Consensus Realities

OF MIND & BODY AND

THE IMPORTANCE OF BEING HONEST

Stefan Wurm

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With love and gratitude for Brigitte,
my wife, friend & companion of many years
on this journey of discovery, which we call our life.

Menschenbeifall

Ist nicht heilig mein Herz, schöneren Lebens voll,
Seit ich liebe? Warum achtetet ihr mich mehr,
Da ich stolzer und wilder,
Wortereicher und leerer war?

Ach! Der Menge gefällt, was auf dem Marktplatz taugt,
Und es ehret der Knecht nur den Gewaltsamen;
An das Göttliche glauben
Die allein, die es selber sind.

Applause

Is not holy my heart, filled with life's beauty,
Since I love? Why did you hold me in higher regard,
When I was prouder and wilder,
Full of words and empty?

Oh! The crowd pleases what the marketplace approves,
And the servant only respects the oppressor;
Believe in the divine
Will only those, who are such themselves.

A poem by Friedrich Hölderlin (1770 - 1843).

Translated into English by the author.

Begegnung

Am Weg der Fremde – wir sehen uns an
Und unsre müden Augen fragen:
Was hast Du mit Deinem Leben getan?
Sei still! Sei still! Laß das Klagen!

Es wird schon kühler um uns her,
Die Wolken zerfließen in den Welten.
Mich deucht, wir fragen nicht lange mehr,
Und niemand wird uns zur Nacht geleiten.

Encounter

Along the way a stranger – we look at each other
And our tired eyes are asking:
What did you do with your life?
Be quiet! Be quiet! Stop complaining!

It is already getting colder around us,
The clouds are disintegrating in the worlds.
Me seems, we won't be asking for much longer,
And nobody will guide us into the night.

A poem by Georg Trakl (1887 - 1914).

Translated into English by the author.

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Introduction



Blooming Wheels

Throughout human history, our search to understand the world around us and to discover who we are, have produced many answers and raised many more questions. Evolution has endowed the human species with unique gifts, but it seemingly also has put in place some restrictions as to what we actually may be able to know and understand. Unique among those gifts is the human brain. One part of it has enabled the human species to discover its place in the universe and in the evolution of life. However, there is much more to our brains than that and most of it we do not fully understand yet. From where we stand today, a discourse about these other aspects of our brain, by necessity, will be very much an exercise in approximations. Who we are, or more correctly, who we believe we are is still a mystery to us. Incidentally, the same is true for just how the human mind forms such beliefs. Science continues to give us an increasingly better understanding of how the brain works but we must acknowledge that, nonetheless, science is still a long way from telling us how from all of that the self of a human being could possibly emerge. What am I and who am I? That is an age-old question, and I can assure the reader that this book will not be able to provide answers to anyone asking such questions, nor will it try to seek one in the first place. Pretending to be able to give such answers would amount to nothing more than hubris. So, how then should one deal with such questions? Some may navigate this challenge by means of their personal beliefs. Religion however, by its very definition, will always offer faith-based answers. In a sense, religious belief amounts to a trade-off. In exchange for giving up the frequently painful search for objective truth, true believers can hope to receive consolation in the anguish which thoughts about what and who we are as individuals and as humans may cause us. For them the answer to the question of who they are rests with their faith and ultimately with their god. This is not a solution for everybody but for many it indeed provides the answer to such questions. You can look at it as a modification of Pascal's wager. Blaise Pascal (1623 – 1662), a French mathematical genius and a devout Christian, to show that logical reasoning can support neither faith nor the lack of it, argued that a rational person, such as he certainly was, should live as though God, in this case the Christian God, exists. According to Pascal's

logic, believing in God comes at very little cost if he does not exist; however, the cost of not believing in God could result in an unimaginable loss if he did indeed exist.[1] In Pascal's France, a much larger fraction of the population than in today's modern France would likely have been convinced that Pascal's wager was a good one. Back then, the Catholic Church not only offered salvation but also threatened non-believers much more forcefully than today with eternal damnation. That frequently included detailed graphic illustrations of visceral versions of hell. Therefore, the perceived stakes in Pascal's time would have been much higher than most would likely see them today.

In a similar vein, by providing comfort in the perception of who they are, faith helps many to navigate around what otherwise could be treacherous waters which figuratively could shipwreck their lives when they have to confront the question of who they really are. Not all of us have the mental strength to endure such a journey and emerge safely. Beliefs in general but religious beliefs specifically, provide safeguards to travel on a path beneath which a seemingly unbounded void could open up at any time if we let go of the security of such convictions. For some, exploring this void equates to their freedom to discover who they are or who we are. For others, this void can become an abyss of agony they fall into and from which they are only able to escape by embracing belief again. Until the beginning of the modern age, the existence of this abyss was only evident to those few whose preying eyes could detect it beneath the cover with which the comfort of religion hid it from everyone else. As the German philosopher Friedrich Nietzsche (1844 – 1900) put it in his 1886 published work *Jenseits von Gut und Böse* (*Beyond Good and Evil*), the exploration of such an abyss is not a one way street:[2]

> *"He who fights with monsters should look to it that he himself does not become a monster. And if you gaze long into an abyss, the abyss also gazes into you."*

Over just a few centuries, our perceptions of our place in the universe and in life have changed dramatically. Today we know what a tiny speck of dust in an almost unimaginably large universe this from our human perspective seemingly so immense solar system really represents to which our beautiful blue marble planet belongs. We also know that humanity is not a divine creation but that humans are the result of millions of years of evolution. We humans just happen to be the smartest animals on this planet where such kind of life began to evolve some five hundred million years ago. That our species is just a very recent addition to these hundreds of millions of years of biological evolution is a reality that not everyone is yet willing to accept. While as of now we really have no way to know, in all likelihood there may be countless other potentially life supporting planets out there in this vast universe. Some of them maybe home to intelligent life just like our planet. There are few things where humans are as proficient as in suppressing unwelcome facts. The realization of what our true places in the universe and in life are, is one example for that. We may acknowledge all of what science can tell us today about

those things, but we still very much behave like our distant ancestors who knew nothing about any of that and still believed our planet to be the center of the universe and that humanity represented the pinnacle of creation. In a way, many of our individual and societal consensus realities reflect beliefs we do not really believe anymore, which however are still useful because they allow us to continue a pretense. The pretense of not having to acknowledge our own insignificance in the context of an almost unimaginably large universe void of anything divine in the way we had once believed it existed. Because of this pretense, many can still give meaning to their lives today in similar ways as their ancestors did, long before we began to understand the true nature of the universe and of life. Fortunately, none of that should keep anyone from asking probing questions with respect to the nature of human nature, as this entire pretense will affect only how one deals with what one learns by asking those questions.

Before we go further, a word of caution. Every time we go about exploring ourselves, trying to figure out how we fit into our worlds, individually and as the social beings that we are, we must be very careful as to how we state things. Frequently the language one can find in such discourses is, at least for my taste, one of too much certainty. All too often, it smacks of us really knowing and understanding something when we actually do not. Throughout this book, the reader should be keenly aware that what it offers is more like a guided tour that can be helpful in beginning their own exploration of that which is likely of much interest to all of us. Here, the simple objective is to shine some light on the various challenges of trying to understand the mind-body relationship from different perspectives. That certainly must include hard scientific facts grounded in our modern understanding of human biology. However, in addition to some basic understanding of the biology underlying human nature, philosophical, psychological and religious perspectives are equally relevant. Importantly, one also must not ignore that humans are not the only animals on this planet endowed with minds. The historical record of the human quest to understand the mind-body relationship encompasses at least a few-thousand years as evidenced by the written records we have which directly or indirectly document such efforts. However, quite likely, this quest has been going on for much longer than that, as our ancestors in pre-history must have been as curious about themselves as we are today. Exploring the context of the human mind-body relationship always has required and always will include the use of constructs so we can speak about things of which as of yet we still know very little. When we do so we must acknowledge that these constructs can in no way tell us what those things really are. Approaches different from the one taken in this book may use different constructs, but their purpose is the same. They are in essence descriptors that help us talk about things in a meaningful way when we do not really know what they actually are.

The danger is of course that such constructs sometimes start to assume a reality of their own, morphing into the very things we seek to explore of which we know little

or nothing and can only talk about in approximations. Any such efforts to describe indirectly what we otherwise cannot speak of meaningfully, always carry the danger of becoming beliefs. What initially is a scientific theory with various claims about things we know little or nothing about can eventually become a school of thought. This is all fine and as it should be as long as any such discussions remain guided by intellectual honesty and follow scientific rigor. If that is not the case, discussions about matters of science can quickly devolve into arguments about beliefs. It is a sad fact that in our modern world far too many seemingly do not understand anymore the difference between scientific facts and non-factual beliefs. Maybe this development could have been avoided if our societies had invested in a decent education for our young people where they could have learned about the scientific process and how science and belief differ, and that being a fact-based scientist is perfectly reconcilable with being a devout believer. Seemingly, too many of our schools seem not to teach those things anymore so we should not be surprised about the level of ignorance that keeps spreading in our societies. Another unfortunate fact is that far too many nowadays mistake religious fanaticism for religious faith. Equally disturbing, an increasing number of people seem to believe that their personal opinions and convictions, however flawed and to whatever extent contradicting scientific facts, carry equal weight in scientific discussions such as science facts do. Where does that misconception come from? How can human minds deceive themselves to such an egregious extent?

In approaching such treacherous terrain as the one we will be exploring here, we first must be very clear about what we do not know; and second, we must be equally candid with respect to the constructs used to describe what we otherwise would not be able to speak about meaningfully. Such constructs are just that and nothing more, useful to explain ourselves to others but that is it. What is reality? Humans have been asking this seemingly innocuous question for a long time but in many respects, the answer still evades us. We have become much better in understanding how our world works. This is what we typically refer to as the modern scientific understanding of the objective world, much of it of a very recent origin, only a few hundred years in the making. By referring to an objective world, we differentiate this aspect of nature from how we perceive the world around us in a subjective way.

The underpinnings of how humans can construct their subjective worlds, the respective processes helping us to achieve that feat, they all still have to follow the laws of nature. To realize that we only have to drill deep enough, and we always will find a part of nature following the laws of nature. Many will agree that the results of all the efforts we make to construct our subjective realities, which then become our individual subjective worlds, represent something quite different from the objective world. This objective world as articulated through science gives a description of reality, which we assert holds true throughout the universe. The unspoken assumption being that any other non-human

intelligent lifeforms in the universe capable of understanding such things would discover this very same objective reality. Such intelligent beings would observe the laws of nature to be the same as our modern sciences describe them. In contrast, how they perceive the world around them and how they relate to themselves in this objective world could be completely different from how we humans perceive ourselves in this objective world. We can assume that any such lifeforms will possess sensory organs, likely different from ours, that feed information about the outside world to what would be their equivalent of our human brain. The latter, however, may function completely different from the human brain and provide them with very different subjective worlds than humans construct for themselves. That all may sound very speculative, and it is, to a point. However, if you just consider for a moment that our planet is home to billions of subjective realities that are all different from yours, all of a sudden, it may not feel so speculative anymore. Any two of those subjective realities differ from each other, some more so and some less so, however, there exist no two identical subjective human worlds on our planet.

Each one of us, every human being on this planet, represents a world onto himself or herself, unique human beings, every one of us. Our subjective worlds are our own constructs. Sometimes we are aware of that but often we are not. We do not command our inner self every morning when we get up to construct our subjective realities. That is certainly not how it works even though sometimes we do consciously construct our self-image and how we would like others to perceive us. Rather, the process is an organic one. We do not know exactly when this process starts. Certainly, it is in full swing by the day a human baby perceives itself the first time as different from others, being its own self and not that person that cuddles and comforts it. We do not know all the tools yet, which may be part of the human toolbox for constructing one's self. All we know is that as we construct our subjective worlds, we must continuously negotiate them with all the other subjective worlds around us. We likely would all agree that our subjective worlds are bounded by the objective world around us, so there is a limit to how much our subjective worlds can disagree with the objective world. Otherwise, we just would not be able to survive in the objective world. In a similar way, anyone's subjective world will always be constrained by all the other subjective worlds around every one of us. The difference between those two relationships, subjective to objective world and subjective to subjective world, is how we align our subjective world with the objective world we live in as compared to how we negotiate our subjective world with all the other subjective worlds with which we interact.

Those of us living at higher latitudes surely can go out in the wintertime in a swimsuit and pretend that they are walking on a white sand beach somewhere located close to the equator. However, all of us know that freezing temperatures will quickly correct any such false notion. One just cannot negotiate or tweak the laws of nature. One has to adapt to them and that means wearing warm clothes in wintertime at high latitudes. Different

from that, when subjective worlds negotiate their realities there is always the potential of influencing and shaping other subjective worlds to make them consistent with one's own subjective world. The construction of our consensus realities is an ongoing process. In a narrow sense, this could be the consensus, between me and myself, of who I am; some of it formed consciously, much of it not. In a wider sense, this could be a broadly based societal consensus on which a large number of subjective worlds agree. Ultimately, there could even be a species-like consensus, when something has become a shared part of all of our subjective worlds on this planet. Something that comes to mind in this context is for example a shared perception of unalienable fundamental human rights. The tools we use to negotiate with ourselves, establishing our subjective world and who we are, the tools we employ in negotiating and interacting with other subjective worlds, both tool sets encompass the full spectrum of human behaviors; all the good ones and all the bad ones. Self-deceit or lying to oneself, as well as lying and deceiving others, is an integral part of the tool kit that humans have used since time immemorial to navigate the voyage of their subjective worlds in an ocean filled with countless other subjective worlds.

As the saying goes, history does not repeat but it does rhyme.[3] The more things change, the more they stay the same; there is nothing new under the sun; presenting old truth in new clothes; or putting new wine into old wineskins – who has not heard of such phrases?[4] In some form or fashion they are likely familiar to many as all cultures have their own variations of such phrases and proverbs, thereby acknowledging a continuity of much of human behavior throughout our known history. Frequently, what we believe to be new phenomena, original and particular to our own times and never seen before, are all but that. A case in point are all the recent discussions about fake news and fake facts we keep hearing since a few years now. Like as if this is something that has never happened before, and it is only in our time that humans first observe such behaviors among themselves and now articulate their concerns about them. Far from it. Lies and deceit, on an individual, group or societal level have been common throughout history. At other times in human history, people may have called them out in different ways but an invention of our time and age they are not.

What is real and what is not, what is a lie and what is not, and why do we lie and deceive, often just as automatically and frequently as we breathe? Humans have wondered about this for a very long time. Today we know more about the biological and psychological underpinnings of such human behaviors. While that has brought us some new insights, we do not really understand them yet, not by any means. We can now describe many of those behaviors more accurately but we remain trapped in a situation, which a part of the Western world refers to as a Catch-22.[5] In this case, the conundrum we cannot escape from is rooted in some seemingly fundamental limitations to our ability of understanding ourselves. We just know very little today about how we construct the realities that we perceive to be the worlds around us; and neither do we know much about how we

navigate in these worlds, interacting with everything else, including our fellow humans. To explore this situation in more detail we have two starting points: the objective study of what constitutes a human being and the descriptive and interpretative study of human physical and mental behavior.

The first starting point anchors in what we can broadly call the science of humans. Here, we will include in this category all scientific branches that provide us a better understanding of human biology. At the center of this needs to be of course a basic understanding of how our brain works, how it defines who we are, and to what level we are in control of that. The balance between nature and nurture, just how much nature determines who we are vs nurture shaping who we become as we grow up, that has been and continues to be an intensely debated topic. Where that balance lies and to which extent we can assign human beings a free will has tremendous consequences for us as individuals and for our societies. In all such discussions, we must be conscious that there are gradations of evidence. For some of our conclusions, the ones we can draw from solid scientific evidence, we can have high confidence that they are correct. The terrain where we have such solid ground beneath us is however only like a small island in a vast ocean. Much of what each of us can hear, read or observe in this ongoing discussion about nature vs nurture is still only conjecture and a good portion of it is outright speculation. That actually is quite as it should be and fine as long as we understand to differentiate between what is speculation based on unproven hypothesis and what are hard and solid facts. Like with so many other things in our human affairs, our sciences would not progress much without being able to speculate based on what often is a limited amount of information. Putting out and falsifying one hypothesis after another until we find a new grain of knowledge that we can add to our store of proven facts, that is very much at the center of the modern science process.

The other starting point to explore human behavior is to study it in the field by meticulously observing and analyzing how we do behave. Such objective observations allow us to understand how humans actually behave, which sometimes can be quite different from how we would like us to behave or believe ourselves to behave. Using objective behavioral studies data, we then can formulate hypothesis and models, which to a greater or lesser extent can explain our human behaviors. Some branches of science engaging in these efforts have their origins in the sciences of medicine and biology but focus, in a manner of speaking, less on the hardware but more on the software. We frequently tend to see them today as being more part of the social sciences rather than the medical or biological sciences. Quite likely, psychology with its many subfields is the best example for that, particularly psychoanalysis, one of the many branches of psychology itself, even though its reputation has suffered quite a bit over time. Evidently, there are those, and they are not small in number, who claim that psychoanalysis, the method that the Austrian neurologist Sigmund Freud (1856 – 1939) pioneered over a century ago, is not a science

at all. Then of course, we cannot leave out philosophy. While many of us may not think about philosophy in this regard, how we perceive our world and how we can and should navigate it has always been at the heart of philosophy. Like with all fields of science, specialization also abounds in philosophy and psychology with each of them sprouting new subfields in response to new discoveries and developments in the natural sciences. Among others they include for example philosophy of biology, philosophy of cognitive science or evolutionary psychology, all of them pursuing the exploration of human self-perception, human behavior and human social relationships as they may have evolved throughout human biological evolution.

As a teenager in the second half of the 1970's, I had the good fortune of being a student at a high school where an elementary course in philosophy was compulsory. This was the first time that along with my classmates I read and worked through the *Meditationes de prima philosophia* (*Meditations on First Philosophy*), published in 1641 by the French mathematician and philosopher Rene Descartes (1596 – 1650). A few years later, in 1983, I received my introduction to what one could call the search for the biological foundation of the human self and how that translates into our self-perception and our perception of the world. Back then, with the little money students typically have, I was able to get my hands on a copy of *The Self and Its Brain*, co-authored in 1977 by the Australian neuroscientist and noble laureate John C. Eccles (1903 – 1997) and the British-Austrian philosopher Karl Popper (1902 – 1994). Since then, our knowledge about the human brain and its function has certainly much increased. However, this book and specifically the dialog between those two gentlemen, some hundred and forty pages long, making up roughly the last quarter of the book, anyone can still read with much gain. Most of the topics discussed there are still as relevant as they were then and many of the questions raised in this dialogue more than forty years ago are still unanswered. Not for the lack of trying though. It just so happens that in order for us to get to those answers we must first overcome a number of very difficult challenges. Fortunately, since then the progress in our understanding of the human brain has been quite encouraging, helping us to see more clearly what a few of those answers eventually may entail. However, for some of those questions we never may find fully satisfying answers. This is because the human brain analyzing the self it is host to and whose embodiment it is at the same time, represents the equivalent of us trying to solve a circuitous problem. In a sense, the way evolution has shaped humans may have created insurmountable barriers to human self-discovery and self-understanding. Why that is so in the first place raises a completely different set of questions. Why is true introspection not possible? How come that we can know so much more about the world around us including other humans than we seemingly can know about ourselves?

Nothing in nature happens without purpose. Making such a statement does not imply the existence of any invisible force, not in any way. However, it is an acknowledgement

that the various paths evolution takes always represent solutions to problems. Frequently this may not be evident at first and it may take many failed efforts before a specific path, as we can discover it today in the evolution of life, eventually could become a success story. How many failed trials were there before the amphibian species that became our common ancestor eventually made it onto land and thereby kick-started the evolution of all four-legged creatures? We do not know. In the same way, we cannot know which other evolutionary alternatives there may have been for self-conscious creatures to evolve that could have had very different concepts of self than we have. Maybe even including access to a much more profound knowledge of their own selves. That is as compared to the level of access with which evolution seems to have endowed the human animal in this regard. Another aspect that we need to keep in mind is that biological evolution works on a very different time scale than what we call culture evolution. A unique characteristic of the human species, culture evolution represents a very recent development in our few-million-year evolution.

In essence, the blue print for who we are as humans today, in a manner of speaking, evolved to solve the problems of our Pleistocene ancestors. This blueprint was certainly successful enough to allow humans to get where we are today. The roots of our modern societies reach back in time only a few thousand years at best and our world of civilizations is very different from the pre-historic world in which our archaic ancestors persevered for a couple of million years or so. When we try to understand the human self and its brain, we can expect that a large fraction of what we will find are evolutionary solutions to challenges our Pleistocene ancestors faced; quite a few may predate even the emergence of the genus *Homo* itself, some 2.4 million years ago. Likely very little of human behavior today will reflect evolutionary adaptations in response to our modern world.[6] Human behavior in our modern societies is pretty much all culturally learned and not biologically inherited behavior. Nevertheless, all culturally learned behavior in some way rests on biological foundations. Those foundations are the result of millions of years of evolution and only change ever so slowly. Despite of that, they are clearly able to accommodate human survival in a wide range of conditions including those of our modern world. How else could modern humans have become so successful in the first place, given that their biological blueprint is still quite similar to that of our Pleistocene ancestors? Understanding the human self requires very much the equivalent of understanding the evolution of human perception and self-perception throughout the evolution of our species. But how much can we really ever know about such things? What should we make of the fact, if this is indeed the case, that what are seemingly still Pleistocene human beings adapt so well to a modern world completely different from the world in which Pleistocene human perception and self-perception evolved?

Have you ever thought about where mathematics or physics come from? Alternatively, approaching this from a different angle, where do music, art or literature come from? Does

any of this exist by itself? Would any of it exist without humans? Are novels or poems only real in the sense that they exist as states of matter between the covers of what we call books? Do they belong to something different altogether as they clearly also become part of our state of mind once we have read them? Similar for mathematics. A formula such as for a cone or a sphere can exist as a physical representation, or it can be part of a state of mind. Some speculate that the entire universe is nothing else but a representation of mathematics, in essence an extremely complex computer program of which humans are one tiny ingredient. According to this hypothesis, as we look around, all that we can see is the structure of those programs encoded in mathematics. Such thoughts are by the way not new. Rather, they articulate in our modern language thoughts that we can trace back much further; in this case, all the way to the ancient Greeks of the fifth century BCE and specifically to the philosopher Plato (c. 428 – 348 BCE). However, the intent here is not to explore such speculations but to look at the mind-body problem in a way that allows us to discuss some of the limitations to our human understanding of the world around us and of ourselves. It is in this regard that one could, following Karl Popper, put next to the state of matter and the state of mind a third category. Karl Popper referred to this third category as the realm of human creations, not a state of mind itself but, figuratively speaking, rather a kingdom. This kingdom includes all things of which we can assume with some certainty that they would not exist on this planet without humans. It is a kingdom in which we all, at least in principle, can travel and absorb its riches into our states of mind; and we all can contribute to its wealth by adding our own human creations to it. Of course, this kingdom has also a dark side because quite a few of those unique human contributions to life on our planet are uniquely bad or evil. Karl Popper's philosophical outlook with regard to the mind-body problem, articulated over the second half of the last century, represents a certain kind of dualistic philosophy. The concept of body and mind as two separate entities originates with the Greek philosophers of the fifth century BCE, finding its most prominent advocate in Plato. The broader stream of the philosophy of dualism in modern times traces its origins to the mid-seventeenth century when the rise of natural sciences forced a reconsideration of the mind-body relationship. When Rene Descartes published his *Meditationes de prima philosophia* (*Meditations on First Philosophy*) in 1641, he practically started the modern philosophical debate about how our human bodies and mind relate and how we as humans relate to each other and to the world we live in. Today, this debate is still very much with us, however, informed by a much more detailed knowledge of the human body, something that we unfortunately cannot assert yet to the same extent for our understanding of the human mind.

This book is not a book of science but a book of reflection. The objective here is not to assert what I myself believe to be true or what anyone else may or may not believe to be true. The intent of the following chapters is not to provide explanations but to outline limitations. For what we will explore in the coming pages, it is more important

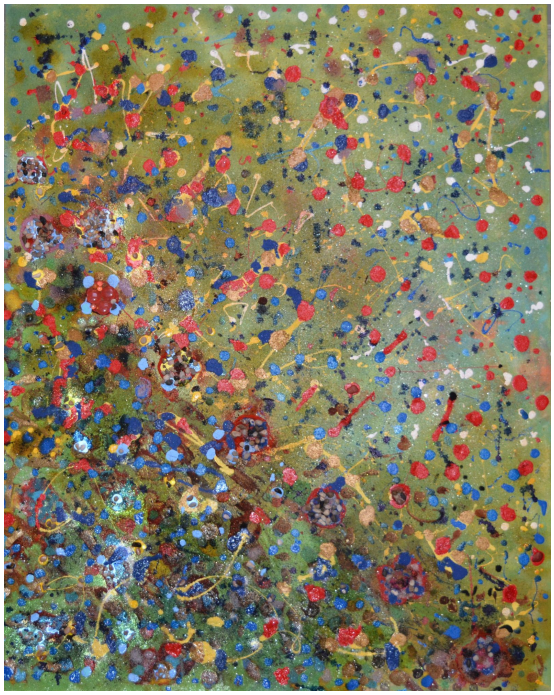
to learn how to ask sensible questions instead of seeking to provide answers, many of which must remain elusive anyway for now, given how little we know about human nature. To provide the physical context within which the mind-body problem is set we will first look into some of the human biology that helps us sense the world and helps us make sense of it. Necessarily, the focus must be on introducing some of the respective biological concepts and not on trying to discuss the underlying biology more fully. For the latter, many excellent textbooks are available to those interested in a detailed understanding of what we know about the biology of human sensory perception and brain physiology. Long before any anatomical or functional knowledge of the human brain became available, humans have thought deeply about the relationship between the human soul and the human body as well as how the human mind fits into this relationship. How we see ourselves and try to understand ourselves as human beings by asking pertinent philosophical and often religious questions as to how the mind, the soul and the body make a human being is at the focus of efforts we usually refer to as philosophy of mind. There is no way one can give a detailed account in a few pages of all the various approaches philosophers and religious thinkers have conceived of to understand the relationships between mind, soul and body. The second chapter will therefore concentrate on giving the outlines of the major tributaries to the philosophical and sometimes religious streams contributing to the broader discourse about the mind-body relationship.

Psychology began to emancipate itself from philosophy in the nineteenth century. The making of psychology as an independent new science very much reflects how our understanding of the human mind and its relationship to the body developed before the emergence of evolutionary biology and of cognitive neuroscience in the second half of the twentieth century. Even though psychology put philosophical speculation aside, the philosophical positions psychologists endorsed still informed their conceptions of human nature and thereby their research approaches. When psychology emancipated itself from philosophy, science knew little about the biological underpinnings of the human mind. That would only change in the second half of the twentieth century. With all these limitations, psychology as it evolved in the nineteenth century still represented the first concerted scientific effort to understand human nature. The third chapter will first look into the making of psychology and then more broadly reflect on some specific aspects of the nature of human nature. Do we know how adaptable human nature really is? Is human nature largely an invariant or can it change? What is it that we can say about the evolution of human nature and to what extent is our human nature an animal nature we share with our animal cousins? What in human nature is uniquely human? Why does lying and deception come so naturally to us? While as of yet, we have for none of those questions any definitive answers, just by exploring them we can learn much about our human nature.

In the final chapter, we will look into some examples illustrating the difficulty of being honest about who we are ourselves as individuals as well as about what happens in our societies. To recognize some of our individual and societal consensus realities just for what they really are is difficult for all and seemingly impossible for some. However, it is critically important that we look at such consensus realities as what they are. We must realize that is up to us to change them as they do not represent some kind of immutable facts but only subjective realities put in place by us because this is what we want them to be, our consensus realities. There is no escaping our individual and societal consensus realities, we are bound to build them whether we like it or not. However, we can question the rational underlying such constructs and by being honest to ourselves as individuals and as societies we may be able to adjust or even change them towards something better.

Frequently enough, we are confronted with aspects of our human nature for which there are just no explanations or answers, which could illuminate our understanding of why they are part of who we seemingly are. Some of those aspects of human nature are just perplexing or amusing while others can be outright disturbing. In any such case, we can count on evolution to provide an explanation eventually, as all of human nature roots in the evolution of the human species. However, as scientifically correct as those answers will be, they are occasionally unsatisfactory as what they may portray often conflicts with how many of us perceive themselves. Hence, we try to come up with different answers not rooted in biological evolution but in human culture evolution. Because human culture evolution is contingent on human biological evolution that often leaves us not with much more than various hypothesis, some of them fought over fervently, others of them less so, but all of it speculative. When we seek understanding of things that we often have difficulty even to articulate we always have to ask ourselves what it is that we can actually know. Building theories or hypothesis about something that for us humans by its very nature may be impossible to understand can sometimes provide the convenience and comfort of a safe harbor. It is tempting to build such safe intellectual sanctuaries and work with others to fortify them with explanations that seemingly can address everything and anything. However, such harbors provide false comfort and more importantly, they prevent us from continuing our search for what we really desire to know - who we are and how we can understand ourselves.

Of Body and Mind



Flower Fields

Probably long before the invention of writing systems and then certainly throughout recorded human history, our ancestors have contemplated and struggled with what we today commonly refer to as the mind-body relationship. To illustrate the conundrum that this relationship entails it helps to resort to an admittedly simplistic and cartoon-like illustration. At the risk of asking readers to stretch their imagination a bit too far, this will nevertheless convey the nature of the problem. So let us compare an atom and a physicist, or better, their respective knowledge of the laws of nature. When we look at the atom and at the physicist in this way, we could say that in a certain sense both of them know about the laws of physics. However, their respective kinds of knowledge are of a very different nature, and it may seem quite a bit odd to say that an atom knows about the laws of physics. So let us pause here for a moment and think about this. How do we humans learn about the laws of physics governing the world of atoms? Is it not by observing how atoms obey those laws? Hence it seems, however unwittingly, atoms must know about these laws governing their behavior; how else could they otherwise obey them? Clearly, there is something wrong with this statement. Looking to the human physicist, her or his body contains a great number of atoms. A person certainly is not an atom and cannot make the same direct observations of the laws of physics that an atom could make. That is, if an atom were not just an atom and could actually have, as insinuated above, an objective knowledge of the laws of nature, as likely our physicist believes that she or he possesses.

It gets even more interesting when we consider that the laws of physics are not some kind of tangible object that one could pick up and then describe from all possible points of view. The early universe only contained hydrogen atoms and a smaller fraction of helium atoms but no atoms heavier than that. As we know today, it is the nuclear furnaces of countless stars and the violent star explosions we call supernovae, which produced

and continue to produce all the heavier elements. Now a hydrogen atom knows nothing about the life cycles of stars, and neither does any heavier atom know anything about how it became what it is. However, we humans do know that. How can that be? Are we as human beings not just a large agglomeration of all kinds of atoms, hydrogen atoms as well as heavier atoms and complex molecules assembled from such atoms? Most of us would likely agree that life is more than just a complex agglomeration of atoms. A person that can discover and comprehend the laws of nature must possess something else, something more than what we can materially account for by just adding up all the atoms that constitute a single human being such as our physicist.

To many this may sound like nothing more than stating the obvious. After all, having some amount of this something else, which enables us as human beings to explore and understand the universe we live in, is clearly what differentiates lifeless matter from intelligent life. However, where would this leave what we refer to as artificial intelligence (AI), which for many of its proponents is just around the corner or even already among us, in whatever rudimentary form? We will come back to this question later in the book but for now let us just acknowledge that the above is nothing like a trivial statement. Since humans started to have religious beliefs and began to entertain philosophical thoughts, they likely sought to articulate the nature and the qualities of this additional something in many ways. They may not have referred to it by the same name our more recent ancestors used but whatever our earliest ancestors called it, we should have little doubt that it must have amounted to that which we usually call the human soul. In what relationship that human soul stood to the human mind, whether the mind was perceived as part of it or not, or if soul and mind might have been considered to be practically the same, that does not really matter in this context. What matters more is that for most of the history of the modern human species, we can be confident that our ancestors did not consider the possession of a soul as a uniquely human quality. What differentiated in our distant ancestor's beliefs not just humans but all living things from non-living things was the possession of a soul.

That changed only recently, like when for example the monotheistic Abrahamic religions decreed otherwise. Then, even more recently, in addition to calling this unique human property the human soul, we also began to refer to it as the human mind. Most would identify the human mind as that which enables us to discover the laws governing the world of atoms and allows us to understand the universe we live in. Clearly, how the human mind can understand the laws of physics of the atomic world is something very different from how an atom, figuratively speaking, knows the laws of physics that it is subject to. Atoms are part of the material world; they represent a specific state of matter. Humans are made of atoms and the biological organism a human being constitutes also resembles a state of matter, though obviously of greater complexity. As long as humans have been debating the mind-body problem, there have been and continue to be those

who claim the mind to be just another state of matter. In essence, the proponents of this view assert that there is nothing else but states of matter. However, many also disagree with this assertion and claim that in addition to the states of matter there exist separate states of mind, that is mental states.

Throughout human history, looking at how our perception of the relationship between the human body and the human mind evolved, it is clear that at any given time it would have been conditioned by the knowledge the human species actually had of the human body. It is only with the advent of the scientific revolution in the seventeenth century when our then nascent modern sciences began to gain an increasingly better understanding of human physiology. Along with that, medical and biological sciences began to participate in discussions regarding the mind-body problem, eventually dominating them and thereby diminishing the prominent roles that philosophy and religion had held throughout much of human history in that respect.

Today, learned discussions about the mind-body relationship usually count on leading scientists, preferably outstanding researchers in the fields of medicine or biology, to advance our understanding of the mind-body relationship with philosophers or religious scholars frequently sidelined. That is unfortunate. Nobody will deny that it is the scientific facts that must guide our rational understanding of what we know or can know about the mind-body relationship. However, human life, or better life in general, is of a complexity that significantly exceeds what any of our modern natural sciences can explain today. Even though we can certainly expect this situation to change over time, it just would be unwise to ignore our human species need for a broader philosophical, religious or spiritual context to understand ourselves. Providing this philosophical, religious and spiritual context is what eventually any deeper comprehension of the human mind-body relationship must entail just as much as a solid scientific understanding of its physical nature which our modern sciences continue to explore.

Naturally, as long as humans have pondered the mind-body conundrum, locating the seat of the human mind within the human body has been part of the quest to understand the mind-body relationship. Today, only few would doubt that the seat of the human mind resides in the human brain, but that was not always so. Frequently, we credit Hippocrates of Kos (c. 460 – c. 370 BCE) or the authors of the so-called *Hippocratic Collection* with first identifying the human brain as the organ of the human mind. However, some time before that, the fifth-century BCE natural philosopher Alcmaeon of Croton already had pointed to the human brain as the seat of understanding. Some scholarly views perceive Alcmaeon of Croton as a pupil of Pythagoras (c. 570 – c. 495 BCE), for which we have currently no evidence. Pythagoras and his followers, the Pythagoreans, associated the brain with the seat of human intellect. Around 525 BCE Pythagoras settled in Croton and founded his school in this Greek city in the southern part of Italy, which once was known as Magna Graecia.[1] Therefore, Alcmaeon of Croton, who was born in Croton maybe

around 510 BCE, could well have been influenced by this Pythagorean conception.
It is with the Classical Period of ancient Greece that we increasingly see the brain organ being associated with human understanding and as being identified as the seat of the human mind. This view developed in contrast to the much older cardiocentric perspective that associated the heart with the seat of the human soul and therefore also the seat of the human intellect. Surprisingly, Aristotle (384 – 322 BCE) would continue to hold on to a cardiocentric view despite his teacher Plato already subscribing to the Hippocratic view that the brain was the seat of the human mind. Aristotle turned out to be often prescient and in many cases pointed towards the correct explanations in his educated philosophical speculations about the nature of things. He accepted a number of assertions about the human brain associated with the *Hippocratic Collection*, for example that the brain acted to cool the blood, which we now know is not the case. However, Aristotle did not make his own the much more important assertion in the *Hippocratic Collection* that the human brain was the seat of the human mind.
Long before the sages of the Classical Period of ancient Greece debated the functions of brain vs heart and their respective roles with respect to the human mind, the ancient Egyptians had developed their own medical practices. Some of what they thought about and how they dealt with bodily injuries, among them quite severe head injuries, has come down to us in the form of two surviving papyri from Egypt's Old Kingdom. This period in Egypt's history spans from around 2686 BCE, that is some two thousand years before history dates the Classical Period of ancient Greece, until around 2181 BCE.[2] The ancient Egyptians held a cardiocentric view of life and of life after death. What mattered in mummification was the joint preservation of the body along with the heart. Because of that, the ancient Egyptians did usually not remove the heart from the body as it was the heart of the deceased, representing the essence of an Egyptian's being, which Anubis, the Egyptian God of the dead, would question and weigh against the white feather of truth. The brain was not important at all to ancient Egyptian afterlife as they removed and discarded the organ during mummification; it was not even preserved in jars placed next to a mummy such as has been frequently found for other organs such as the lungs, the liver, the stomach or the intestines.
It should not surprise us that thousands of years of such believes have left a deep cultural imprint. Because our ancestors associated the human heart with the seat of the human soul for many millennia, we still can find reflections of this worldview in our daily lives. While today we see the human mind firmly associated with the human brain, the human mind was always only a part of what the ancients considered as the soul. It was what they called the rational part of the human soul and what we would today refer to as the human intellect. For example, according to Aristotle, maybe the most famous and consequential philosopher of antiquity whose views continue to shape our perceptions today, human souls possessed three parts. In his tripartite soul model, a sensory and a

vegetative soul joined this rational soul to comprise the whole human soul. Some aspects of these other parts of the soul we continue to associate in our everyday language with the human heart to this day, like for example our feelings. We speak of something "coming from the heart" or something "being heart felt"; and of course, we occasionally speak of "broken hearts". These are all expressions most will be familiar with through firsthand experience. Few among us likely think it unusual or strange that we continue to associate the human heart with such emotions. For that, we have to thank a several thousand-year legacy of associating the human heart with the seat of the human soul, which for a very long time included as one part of it the human mind. As one of the philosophical worldviews bequeathed to us by the Classical Period of ancient Greece, Aristotle's work would eventually become and remain one of the pillars of Western philosophical thought. However, Aristotle's cardiocentric view associating the heart with the seat of the human intellect would not take hold. Even some of his greatest admirers could not convince themselves of it being correct.

Among those, only a few hundred years later, was the Greek physician Galenos (129 – c. 216), whom history would judge as one of the most influential thinkers regarding the function of the human brain; to most of us he is more familiar by his Latin name, Claudius Galenus, or his anglicized name, Galen. If asked to name the two most famous physicians of antiquity, many would likely respond with Hippocrates and Galen, even though Galen lived during the age of late imperial Rome and not during the Classical Period of Greek antiquity. Galen would firmly dispense with any notion of the heart being the seat of the human mind. With much of the development of medical knowledge and practice in the West for the next millennium grounded in Galen's work, scholars would never again identify the heart as the seat of the human intellect.

Throughout antiquity the dissection of human corpses was anathema, sometimes even the dissection of animals was frowned upon. There were only few exceptions such as the flourishing of the Alexandria School of Medicine in the Hellenistic Age. This medical school in the city that Alexander the Great founded in 331 BCE dates its roots to the late third century BCE. The Greek Alexandria School of Medicine located in a city which antiquity considered as one of the great centers of learning followed the teachings of Hippocrates. However, it must also have been open to influences coming from Egyptian medicine. The two most famous representatives of this medical school were the Greek physicians Herophilus (335 – 280 BCE) and Erasistratus (304 – 250 BCE). The school flourished under the dynasty of Ptolemaic Kings and Queens whose reign began in 305 BCE and it would continue to thrive under Roman rule, remaining active until the third century, although in a diminished form. Many Roman physicians studied at the school even before Egypt became a Roman province in 30 BCE and many more did so after that. The most famous physician studying there before practicing in Rome was Galen. We do not know if the dissection of corpses at the school was still possible in the same

way physicians could practice it there before Roman rule. However, physicians certainly sought out the school for the study of anatomy for which it was famous. Maybe physicians at the school still could practice dissection and Galen could have acquired some of his dissection skills right there, but we do not know that. However, we know that the Rome Galen made his home to practice medicine was no such exception. Because of that, Galen had to accumulate much of the understanding of human anatomy he contributed to medicine indirectly, through inference from dissecting animal corpses, among them also any monkeys he could get hold of. At the time, the study of animal anatomy also included the vivisection of animals, the practice of cutting up animals while still being alive. What physicians in Galen's time and later many of his successors were trying to understand through vivisection was simply the functioning of animal bodies and by inference also that of human bodies. One could look at what Galen and many later physicians tried to accomplish through vivisection as a kind of destructive functional testing of animal bodies to learn what the specific functions of animal body parts were.[3]

The physician would for example cut a nerve connection and something in the animal would stop working, which gave the physician a clue as to how certain body functions were hierarchically organized. Human injuries or the inspection and treatment of wounds that gladiators for example would suffer provided some additional insight into human anatomy as well. However, without being able to dissect human corpses for the study of human anatomy and physiology, learning from gladiatorial wounds or injuries sustained otherwise provided only limited new insights. The few human corpses Galen occasionally was able to study were usually in quite bad condition, limiting whatever he could hope to learn from them. It was against Roman law to dissect human corpses and that made it difficult even to assemble a complete human skeleton for study or education purposes. The prejudices against dissection of human corpses would endure long after Galen and would continue to limit the pursuit of new knowledge in human anatomy and physiology. This was likely another reason why with respect to the understanding of the human body Galen's writings remained every European physician's gospel for roughly the next millennium. Physicians trying to follow Galen's example and learn firsthand about human anatomy and physiology just had a very difficult time to get their hands on human corpses. Hence, they had to continue for a long time to rely on Galen's work.

This situation would change only gradually, and it would keep physicians from correcting errors in Galen's understanding of human anatomy. A good number of the latter simply existed because Galen accumulated most of his human anatomical understanding through studying animal anatomy. Eventually, the Dutch anatomist and physician Andreas Vesalius (1514 – 1564) would begin to recognize such errors for what they were by doing his own meticulous studies of human and animal anatomy through the dissection of corpses, human as well as animal. The publication of his *De Humani Corporis Fabrica Libri Septem* (*On the Fabric of the Human Body in Seven Books*) in 1543 was

not just a landmark for the understanding of human anatomy. More than that, Andreas Vesalius singular achievement in the understanding of human anatomy and specifically the empirical methods he conscientiously employed in its pursuit mark the beginning of modern medical science. What we see emerge with Andreas Vesalius is the effort to give accurate and meticulous descriptions of the body parts of human anatomy.

Putting this picture of human anatomy in place and refining it evermore was the prerequisite for beginning to explore how the human body actually functions. Different from many of his predecessors, Vesalius seems to have understood that in his time just too little was known about the human brain anatomy to justify speculation about its various functions and where those may be located in the brain. Therefore, unlike many of his predecessors or contemporaries, for the most, Vesalius wisely refrained from assigning specific functions to certain parts or areas of the brain. That however, would change over time as his successors accumulated an ever more detailed understanding of human brain anatomy. One can only wonder what some of the ancient sages or physicians speculating about the function of the human brain in their own times would have to say if they could gain a glimpse at the tool sets of modern science and the tremendous progress in our understanding of human brain functions.

In modern times, the effort to understand the human mind has become synonymous with understanding how the human brain works. Independent from the larger quest to explore the human mind, the latter is important for many reasons, not least that we must understand of what we are in control of and what is beyond human control. The concept of free will is a good example. Leaving authoritarian societies aside, which usually have a firm disregard for human freedoms, an underlying assumption in non-authoritarian societies usually is that everyone can exercise their free will. However, we all know that there are also must be some societal constraints to an individual's exercise of free will. Exercising one's free will must be subject to certain restraints so that one person's exercise of free will does not come at the expense of the rights and freedoms of another person. We also know that there can be biological constraints to the exercise of human free will. Human psychopathology, the study of abnormal cognition, behavior and experiences, has provided us many examples of just how little we sometimes can be in control of our own actions.

We certainly must be careful when drawing inferences from pathological cases with respect to what should be normal human behavior and vice versa, both from a common sense point of view as well as from a medical perspective. What separates pathological from normal human behavior is likely not a sharp cliff or a wide gap. Rather more likely, severe psychopathological cases are sitting on the far end of a long slide where milder pathological cases and normal cases are close neighbors, and where we cannot be exactly sure of the exact location of our own place on this slide. Wherever that happens to be, it may not be a stable position at all because much of our mental behaviors and reactions

are situation conditioned. Change the conditions and you likely will change the behavior, eventually it only becomes a matter of just how much of a change in situation will result in a behavioral change. However, even without any external circumstances constraining one's exercise of free will, there may be other limitations closer to home. Until recently, we firmly believed that our so-called conscious decisions are completely under our control. However, we are now learning that this is not true for all of what we perceive as our conscious decisions. Seemingly, there are some decisions, which parts of our brain execute well before they surface as such in our conscious brain. How is that possible? What are the implications of that for our conception of what constitutes the exercise of human free will?

As the social contracts of most modern societies assert, at least the better ones do, all human beings are equal. What that usually boils down to is equality before the law, but we are certainly not born as equals. Who we eventually become as a person depends to a large degree just on how our mental faculties can or cannot develop. As for the latter, most of us would agree that this very much is conditioned by how we grow up. Many societies recognize the validity of concepts taking into consideration a potentially reduced culpability in sentencing individuals for civil or criminal offenses. However, that is an exception, and the presumption usually is that those standing in front of judge and jury are normal people. Normal in the sense that their mental faculties have developed in such ways that they are in control of their actions. How good an assumption is that? To illustrate the difficulty of answering this question, just look at the extent to which a good number of so-called developed nations have allowed large parts of their societies to become dysfunctional. That this can and frequently does impair a normal mental development of children growing up in those segments of society is not a matter of discussion, it is a fact. If matters were only that simple. Unfortunately, they are not. How people view this situation is frequently more a function of their political belief rather than of their compassion. Can children growing up in these disadvantaged parts of our societies be really equally culpable compared to those growing up without such impairments? How the brain functions and how it evolves to function within what we could perfunctorily call normal operating conditions should not only be of interest to medical sciences but to society at large. Little else will be of greater relevance for the future of our societies than to ensure that all human beings can grow up in ways that enable a normal development.

Sensing our Worlds

Evolution has furnished life on Earth with what we generally refer to as sensory organs. Nature knows a wide variety of sensory organs that come equipped with a multitude of special receptors, enabling living organisms to detect and react to changes in their environment. That includes the environment outside of an organism as well as the environment within the organism's body. For many animals, what physiologists refer to as special senses, is typically concentrated in their head parts. This is also the case for human beings where the head is home to five special senses allowing us to smell, taste, see, hear and maintain balance on our two feet. This list is a little different from the senses most people would point to when asked to name the proverbial five human senses. It is a safe bet that many would be more likely to put touch on this short list instead of the sense that allows our bodies to maintain balance. Touch however is not one of the special senses but a part of what the medical profession more broadly refers to as the somatic senses. These include not just sensing touch but also pressure, tension or temperature; on the outside, that is on or through our skins, as well as on our bodies inside. Some of our senses are better at precisely quantifying the information they gather than others are. Just compare for example how we experience the world through our visual sense with how we sense pain. It seems as if the former usually provides us much more accurate data than the latter. However intuitive this may sound, it likely is a mistaken perception.

In terms of the physical or chemical information that the various biological receptors in our sense organs can register, there seems to be little difference in how accurately our modern technology can quantify their capabilities. However, there is a marked difference in our ability to determine or measure how our brains make sense of any of these inputs. In other words, we must differentiate between what the actual inputs are that our human senses register and what our brains make of them. Of course, the sensitivity of human sensory organs varies across populations; we all know that. Some have better vision or hearing than others and pain thresholds for example can differ. In the extreme, the functionality of some of our sense organs can be impaired, sometimes from birth as for those who were born blind or deaf or have learned to live with less severe impairments such as color blindness. As we all know, our sensory perceptions also change over our lifetime. Many among us experience this when for example our vision or hearing degrades as we age. However, in practically all of those cases we continue to be able to make sense of the world around us, we just do it somewhat differently. That should be no surprise. In the evolution of life on our planet, the emergence of sensory organs was not the result of nature's effort to provide organisms a complete physical map of the real world around them. That would be impossible and even if it were possible, it would not be economical. Biology has its own version of what physicists in their field refer to as the minimum

energy principle. Physicists, of course, have a rigorous formulation of this minimum energy principle but in simple words, it reflects the observation that in a closed system the lowest energy state is its equilibrium state.

Earth biosphere is a significantly more complex system than the much simpler experimental set-ups in physics laboratories or the assumptions underlying theoretical model systems used in classrooms to demonstrate the minimum energy principle to students. The statement that "there is a cost to everything", in a way biology's equivalent to a minimum energy principle, is not exactly a law of nature. Nevertheless, with few exceptions this is what we almost always encounter; and in cases were seemingly something comes at no cost it usually just looks this way because we have not fully understood yet what we are looking at. Of course, nature does not run its accounts in dollars, euros, yens or any other monetary currency used in our human economies as fictional stores of wealth. Rather, nature keeps score in a different way where some species evolve to become more successful than others. Only those members of a species surviving into adulthood and successfully procreating will pass on their genes. For both, predators and prey, only those adaptations that lead to increased survival and progeny will eventually spread within a species to become a common characteristic. That sets limits to the size of predator teeth and claws as well as to the protective armor prey species may develop in response.

To demonstrate their prowess and virility, males of many animal species have evolved body features that at first glance seem like very costly investments to carry around. However, nature runs a fine balance and the cost of those evolutionary investments, be it the stateliest deer antlers or the richest peacock feathers, must be outweighed by the ability of such males to mate more often than their less endowed competitors. Animal abilities to sense environments are likewise balanced investments and this is no different for humans. The color spectrum of human vision as well as its resolution, depth of focus and spatial coverage reflect such a balance, as does the frequency range of sounds humans can hear. There is a reason why evolution has not endowed humans with infrared or X-ray vision as we sometimes find it with science fiction superheroes and the same is true for hearing where the frequency spectrum humans can hear is likewise limited. It all is a cost balance where nature weighs everything with respect to how much it increases or decreases the ability of the fittest individuals in an animal population to survive and procreate.

The human species is unique in evolving a very large brain relative to its body size, giving us actually the largest brain of any animal species relative to its body size.[4] Brain tissue, specifically its white matter, has a notoriously high metabolic rate and our brains use more energy than any other organ in our bodies. For an adult human at rest, brain energy consumption amounts to around twenty percent of total energy consumption. Clearly, there is a substantial cost to having a large brain in relation to body weight. When we look back into our human species evolution the most significant change in brain

size happened between *Homo habilis* and *Homo erectus*. Unfortunately, we do not really know how the two species are related, that is, how their evolutionary lines connect on the family tree of the genus Homo. The fossil record we have for *Homo habilis* stretches from around 2.4 million years ago to about 1.4 million years ago. The earliest fossil record we have for *Homo erectus* dates to 1.9 million years ago and its Asian variant seemingly has been around until some 150,000 years ago. The two species overlapped in time, and we know that in some areas they also overlapped geographically. Sadly, for now this is as much as we can say about any connection between those two species, anything going beyond that amounts to little more than educated speculation. However, what we do know with certainty from the fossil record is that *Homo habilis* and *Home erectus* were physically quite different species. This includes a body build for *Home erectus* much closer to modern humans and a significant increase in brain size as compared to *Homo habilis*. The modern body build of *Homo erectus* is believed to be the result of its gut becoming much smaller with respect to its body size than what we find for *Homo habilis*.

The hypothesis is that this evolution of a smaller gut became possible due to *Homo erectus* having learned how to cook. Digesting only raw food required a much larger gut with respect to body size than digesting cooked food. In addition, cooked food makes for much easier digestion and increases the amount of nutrients the human body can absorb. Being able to extract more nutrients from food not only allowed humans to evolve a smaller gut but it also was a prerequisite for the evolution of larger brains. Beginning with our earliest known human ancestors or cousins up to and including *Homo habilis*, human body builds were not so much different from those of our ape cousins and human brain capacity remained close to what we still find today with modern chimpanzees, though somewhat towards the higher end of that range. It therefore looks as if before humans learned how to cook, which likely started with *Homo erectus* or its direct ancestor of which we have no fossil record yet, humans just may not have been able to sustain themselves on diets that would have allowed the evolution of larger brains.[5]

For our species, the evolutionary investment in a large human brain, which we see beginning with *Homo erectus*, has produced an enormous payoff. Few would argue that our large brains are not well worth the substantial number of calories we have to consume on average day by day just to meet our brains energy needs. Unfortunately, we know little about how this evolution of the large human brain came about or how it progressed not just in size but also importantly in capability, from what were chimpanzee-like brains to the human brain of today. Unlike human bones, which fossilize and thereby can tell us much about how some of our distant ancestors and relatives may have looked like, brain matter does not preserve. However, we can learn about our ancestor's brain capacity from the fossilized remains of human skulls. Brain casts of such skulls can give us clues as to how the shape of the human brain and its components may have changed over time.

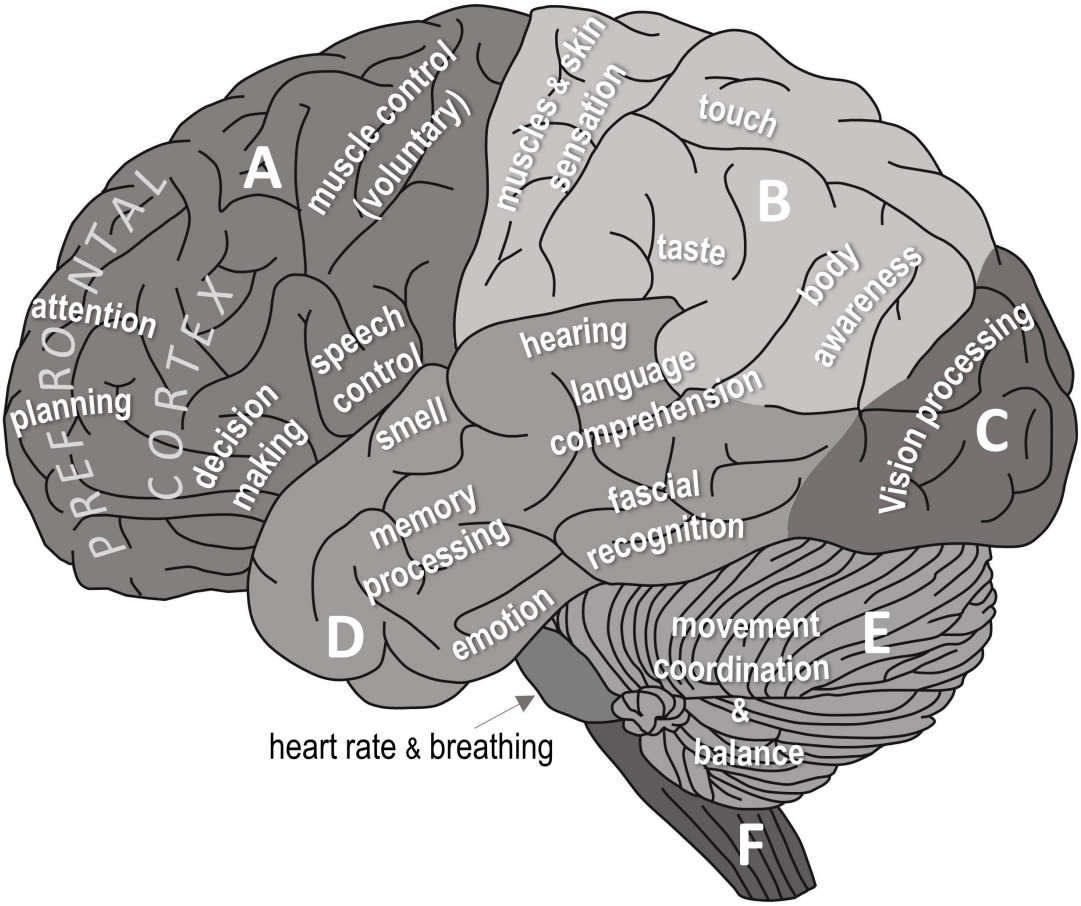


Figure 1.1: Simplified side view of a modern human left brain hemisphere showing the frontal lobe (A), the parietal lobe (B), the occipital lobe (C), the temporal lobe (D), the cerebellum (E) and the brainstem (F). Indicated are also areas where aspects of certain functions are concentrated. Some of the indicated functions are actually not located within a given lobe but rather underneath it like for example taste processing, which is not part of the parietal lobe itself but resides in the insular cortex. This insular cortex is a brain tissue hidden deep in the fissures, the so-called lateral sulci, separating the temporal lobe from the parietal and frontal lobes in each hemisphere.

An example for the latter is the brain region which constructs how we visually perceive our world using the information received through our sensory organs for vision, our eyes. Seemingly, this region in the back of our brains, the so-called occipital lobe, was somewhat larger in Neanderthals than it is in modern humans. Fig. 1.1 shows this part of the human brain along with the other major brain areas for a modern human brain. The simple brain sketch shown here only illustrates what our brains look like in general as well as indicating some of the brain regions referred to in this book. The human brain consists of two symmetrical hemispheres, a left and a right half connected at their bases by a thick bundle of nerve fibers, the so-called corpus callosum, the cerebellum or little brain and the brainstem.[6] The left and rights halves of our brains control the right and left

sides of our bodies respectively. For example, motor control for the right half of our body and right field vision are located in the left hemisphere, while the respective functions for the left half of our body reside in the right hemisphere. This may sound as if our brain is not only a highly symmetric organ where the two brain halves are veritable physical mirror images of each other, but that the two brain halves also functionally mirror each other. However, from the perspective of today's experts this seems to be much less the case than what neurophysiologists used to think only a few generations ago.

Long before the age of modern medicine, various cultures had tried to understand human brain anatomy and its functions and we know that several of them practiced some kind of brain surgery even more than a thousand years ago. We can date back the beginnings of modern neurology to the Renaissance, which gave us the first accurate anatomical drawings of the human brain.[7] Mapping human brain anatomy was the prerequisite to understanding human brain functions. As the understanding of human brain anatomy improved, researchers could begin to explore human brain physiology. In many ways, we can trace the beginnings of modern neuroanatomy to the works of the English physician Thomas Willis (1621 – 1675). In 1664, Willis published his first book, a detailed description of human brain and nerve anatomy titled *Cerebri anatome* (*Cerebral Anatomy*), which introduced the term neurology. This work also included the famous depiction of the base of the human brain by his compatriot Christopher Wren (1632 – 1723), whom many know much better as one of Britain's most famous architects. Willis second book published in 1672, *De anima brutorum quae hominis vitalis ac sentitiva est, exercitationes duae* (*Two Discourses on the Soul of Brutes which is the Vital and Sensitive Soul of Man*) is the earliest English work on medical psychology. It includes possibly the first appearance of the word psychology in English, defining the latter as the doctrine of the soul.

Until the mid-seventeenth century, the unquestioned assumption was that the human brain resembled a non-differentiated functional unit. The idea that the human cerebral cortex consists of differentiated parts serving specialized functions seems to have surfaced first with the Swedish philosopher, scientist and theologian Emanuel Swedenborg (1688 – 1772). However, he did not really publicize his thoughts on this matter so it was only a generation later that the idea of localized brain functions entered the scientific and public discussion with the works and writings of the German physician Franz Joseph Gall (1758 – 1828). Gall inferred the location of brain functions in the cerebral cortex from the shape of specific cranial features, interpreting their variation in prominence between individuals as indications of a person's mental capabilities or deficiencies, a method that he referred to as cranioscopy. This approach eventually lent itself to what would become the ill-guided pseudoscience of phrenology, whose short-lived popularity owed much to Gall's erstwhile long-time collaborator, the German physician Johann Spurzheim (1776 – 1832). Gall did not approve of Spurzheim's phrenology and never used this term to refer to his

own approach, which essentially was the use of cranioscopy to develop what Gall called his theory of organology. Nevertheless, Gall's approach faced the same challenges as phrenology and the resulting inadvertent association of his work with phrenology cast a shadow over Gall's work and reputation for a long time. The broader medical community would reject phrenology and with it also much of Gall's work. Because of that, it would take another generation before the medical profession would come to acknowledge that the human brain was not an undifferentiated functional unit and that localization of brain functions in the human cerebral cortex was the correct picture. The key to this realization would be the localization of the human language function in the left frontal lobe by the French physician Pierre Paul Broca (1824 – 1880) in the 1860's.[8] This was soon followed by the British physician John Hughlings Jackson (1835 – 1911) recognizing the role of the right brain hemisphere in spatial perception. Hence, not only did researchers find that the human cerebral cortex had indeed localized functional areas, in the second half of the nineteenth century they also discovered a right-left brain asymmetry.

In the final decades of the nineteenth century, scientists began to map brain functions in a systematic way, associating certain brain areas with specific functions. In the first decade of the twentieth century, the German neurologist Korbinian Brodmann (1868 – 1918) published his mapping of the cerebral cortex, identifying fifty-two distinct functional areas. Brodmann area labels are still very much in use to refer to brain areas even though our understanding of what their specific functions are has changed in many ways. Clearly, the brain functional area labels shown in fig. 1.1 are of a generic nature only and provide an incomplete picture when compared to the many more details one can find in the textbooks medical students study. However, the more simplistic picture employed here is nevertheless sufficient for the discussion of interest. Attempting to fill in more detail would only convey the false impression of an understanding the author does not claim to have, and it would only distract our attention from focusing on the clear outlines of the bigger picture. Looking only at the left half of the brain and that only from one side, the view given in fig. 1.1 also provides an incomplete picture. Many brain functions reside within the brains surface areas, some wrapping around to the other side, some of them not or just less so; some brain functions reside within the volume of the deeper brain tissue; and other functions reside in distinct nodules and structures hidden under the brain lobes. These include for example basal ganglia, thalamus and hypothalamus, hippocampus, amygdala, as well as a number of glands.

The physical symmetry of brain anatomy betrays a functional asymmetry, more pronounced in human males than in females. This functional asymmetry is a feature of many higher cognitive functions but also of functions that we usually do not associate with cognition such as sensing or emotions. In terms of general skills and functions, the left and right hemispheres serve a given function from a different perspective such as left vs right brain half emphasizing respectively: analytical vs intuitive; detailed oriented

vs holistic; ordered vs random; rational vs emotional; verbal vs non-verbal; cautious vs adventurous; planning vs impulsive; and logic vs creative. In a sense, this high level qualitative assignment of brain functions implies that the left hemisphere of our brain would represent our analytic, detailed oriented, ordered, rational, verbal, cautious, planning and logic selves. While in general this is true, it would be wrong to perceive this difference in emphasis for higher brain functions as an absolute. Rather than a split, this so-called brain lateralization is more a matter of dominance than a complete split and the balance of this dominance does vary between individuals.

For a long time, the assumption had been that brain lateralization is a specific human characteristic. We should have known better. It is likely safe to say that brains are the most complex and sophisticated organs animals have evolved. All animal brains have the same evolutionary origin and brain lateralization would have made sense right from the beginning of animal brain evolution if it conveyed an advantage in species survival. If for example the right brain half of early animals were to specialize in one function while the other one would specialize in another one, this kind of evolving dominance of the respective brain halves for certain functions would seem to be a naturally better way to arrive at quick and potentially life-saving decisions. Just image an animal where both brain halves share equally in a function and therefore, in a manner of speaking, need to negotiate on what to do so that the respective body halves get the same instructions and not conflicting ones. The delay potentially caused by such decision finding could be the difference between life and death. Therefore, it makes sense to presume that brain lateralization is an ancient evolutionary concept in brain evolution. With increasing brain complexity, we could also expect the relative dominance of left or right brain halves for specific functions to become more pronounced. The degree of brain lateralization we find in humans is for now the endpoint of an evolutionary strategy that seemingly must have been in place right from the very beginning of brain evolution in the animal kingdom.

Within each of the two brain hemispheres, human brain anatomy differentiates four major lobes: the frontal lobe, the parietal lobe, the occipital lobe and the temporal lobe. Fig. 1.1 shows the four brain lobes for the left brain hemisphere as well as the cerebellum and the brainstem. Most of us have encountered this kind of brain cartoon in one form or another before, likely in one of our schoolbooks, since it is a common representation of what the full-size brain looks like as it sits in our skulls. Of course, modern science has discovered many finer subdivisions of the human brain than what fig. 1.1 shows here, giving a much more detailed account of where specific brain functions are located or concentrated within each of these larger brain areas. Importantly, while there are so-called hubs for specific functions located in each of the brain lobes, each of those functions relies on other brain areas in the same or in other lobes. Fig. 1.1 shows the approximate areas where some of those hubs are located.

The frontal lobes, specifically their so-called prefrontal cortex areas located right behind the forehead, are mostly associated with much of our cognitive functions such as attention, planning and decision-making.[9] The rear of the frontal lobe, bordering on the parietal lobe, is where the so-called primary motor cortex lies. It is the part of our brain through which we exercise our muscle control such as for example, when we walk or when we move our eyes. Another area in the frontal lobes controls the muscles that allow us to speak. The brain area enabling articulation of speech usually resides in the dominant hemisphere, which for most people is the left hemisphere.[10] This does not mean that the non-dominant hemisphere serves no speech function at all but it is usually very limited. However, that can change if brain injuries occur at a young age impairing the speech area in the dominant brain-half. In such cases, more often than not, specifically if the injury is not too severe and leaves the other hemisphere unaffected, our brains still can transfer our language articulation capability to the respective area in the healthy hemisphere. We are of course all well aware of another left-right asymmetry of our brains, which makes many of us right-handed people and some of us left-handed people.

Our handwriting ability relies on our brains fine motor control, which is our ability to coordinate for example the exact and precise small movements required for writing. Voluntary muscle control resides within the primary motor cortex areas of our frontal lobes. Our brain's left hemisphere is in charge of fine motor control of our right body half, which of course includes hand and finger dexterity. However, for some people the fine motor control of the brains right hemisphere does a better job and this is what makes them left-handed people. We do not know why that is so but for approximately 10% of us this is definitely the case. The prefrontal cortices of our brain's frontal lobes could not serve their cognitive functions as effectively as they do without short-term memory areas close by. Computer-brain analogies do have their limitations but in this case, these so-called working memory areas of the prefrontal cortices seem to serve similar functions as the fast but volatile memories do that we find closely integrated with the central processing units of the various computers we use.

The parietal lobes are largely responsible for registering and processing bodily sensations.[11] They are the hubs for our somatic senses, the so-called somatosensory cortex areas. Prominently among them are touch, temperature, pressure and pain. This includes sensing the outside world around us as well as bodily sensations coming from our inner organs, pleasant as well as painful ones. The parietal lobe areas next to the frontal lobes are the final processing stations for signals routed to there from countless receptors in our muscles. The distance between those sensors in any given skin area, that is how dense or sparse they are distributed, determines the minimum distance between two objects touching our skin that our brain can still differentiate as being physically separated in space. Anyone can verify this by performing two-point discrimination tests, which probe if our skin can discriminate the two prongs of a U-tip shaped object touching it.

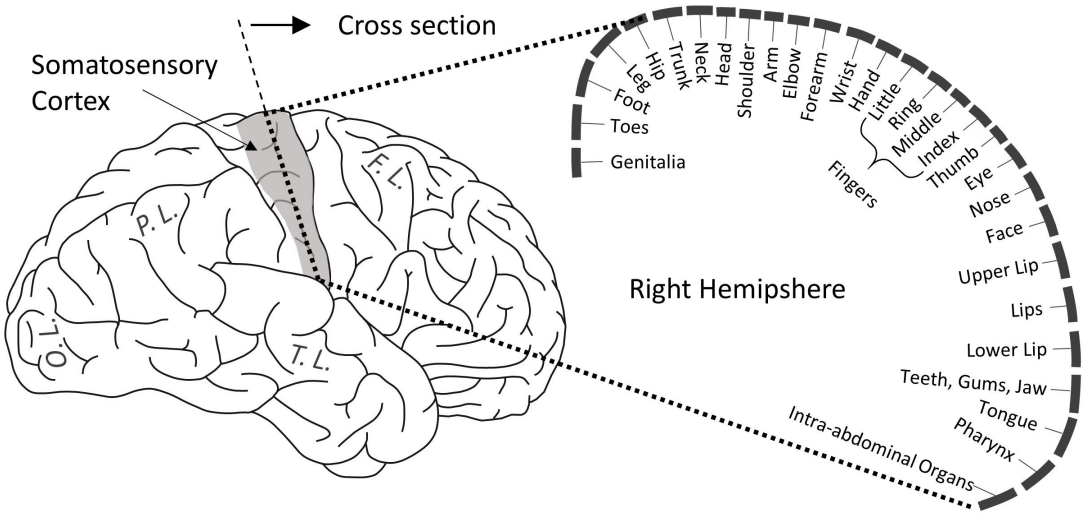


Figure 1.2: Cross section through the somatosensory cortex of the brain's right hemisphere illustrating the brains sensory map representation of the left side of the body. Hands and mouth take up large areas, a reflection of their critical importance in our sensory interaction with the world around us.

Using the tips of such U-shaped objects to touch our skins and varying the distance between the U-tips one quickly finds that the density of such receptors is much higher on our fingers than for example on our backs. Our fingers allow us to differentiate objects as separate that touch our skin only two to three millimeters apart while on our back, they have to be around four centimeters apart so we can differentiate them as separate objects. This difference in nerve density in various parts of our body is reflected by the size of the brain area the respective body parts take up in the somatosensory cortex. The image of our body that our brain constructs, frequently called the sensory homunculus, looks very different from the photographic image of our human body. Fig. 1.2 illustrates the relative areas taken up by respective left side body parts in the somatosensory cortex of the brain's right hemisphere. The sensory homunculus, the pictorial representation of a body with the relative sizes of body parts correlating to their sensory representations in our somatosensory cortex, has a quite peculiar physique with hands almost as large as the body and a mouth area with over-sized protruding lips as large as the rest of the head taken together. This sensory map is by no means static. Rather, the brain maintains a dynamic body map where areas can expand because of increased use or of specific training. The opposite, a decrease of the respective area of a body part in the sensory body map of our brain can also occur. This can happen for example in the unfortunate case of an accident when the use of a body part becomes impaired or is lost completely. In that case, the respective area in the sensory brain map becomes responsive to sensory inputs from other body parts to the point that it eventually represents different body areas than the body area whose sensory input was lost. A good example for an increase in

the respective area of a body part as represented in the sensory map is the enlarged size of the finger area in violin virtuosos, specifically if she or he has practiced the instrument from an early age on. However, that only applies to the violinist's fingers of the hand pressing on the fingerboard to produce the specific pitches and not the hand that moves the bow.

Ascertaining how our bodies are oriented in space and how body parts are oriented relative to each other is a central function of the parietal lobes. Such capabilities, which are critical for successful object manipulation and motor planning, like how one would have to move one's body relative to an object to achieve a specific task, is what large areas in the backs of the brain's parietal lobes specialize in. Other signals routed to the parietal lobes for processing originate from pain and temperature sensors distributed throughout our bodies. In both brain hemispheres, within the fissures separating the temporal lobe from the parietal and frontal lobes, the so-called lateral sulci, lies the insular cortex, sometimes just called the insula. This highly connected brain tissue, hidden deep within the brain, some refer to it also as a separate lobe, serves a number of brain functions, one of them being taste. The taste processing part of the insula, lying beneath the parietal lobe area labeled "taste" in fig. 1.1, receives signals originating in the several thousand taste buds on our tongues sensitive to the five different tastes of salty, bitter, sweet, sour and umami.

The occipital lobes of our brain receive and process our visual sensory inputs.[12] Interestingly, this not only includes the signals produced by our eyes and routed through the visual nerves to the occipital lobes. Have you ever wondered how the images come about which you can produce with your eyes closed? From what it looks like, conjuring up visual imagery through our mind's eye uses pretty much the same brain hard- and software that helps us to construct an image of the visual outside world. In a sense, this allows us to visualize what we hear, feel or remember as seemingly the occipital lobes can conjure up images out of various sensory inputs and not just from what our visual nerves feed them. This also seems to apply to our dream images because the so-called rapid eye movement, indicating that we are dreaming, correlates with increased activity in the occipital lobes, which is quite similar to what happens in visual perception during our wake states.

The fourth type of major brain lobes are the temporal lobes, sitting right behind our temples on each side of our skulls.[13] The temporal lobes in each hemisphere serve functions such as smelling, hearing, language comprehension, memory formation and emotion. While the primary auditory cortex areas process sound information received through our ears, other parts of the brain are involved as well in making sense of auditory information. The frontal lobes as well as the parietal lobes are involved in making sense of what we hear by integrating, in a manner of speaking, the sounds we hear within the contexts of their own specialized functions. In this way, the prefrontal cortex areas integrate auditory

information from a cognitive perspective while the parietal lobes do the same in a sensory context, linking what we hear to what our body senses otherwise through the parietal lobes. Among the major brain lobes, the temporal lobes seem to be unique in serving subjective sensations. It is the emotion and memory areas of the temporal lobes that process and record our subjective experiences that shape our personality, that is, how we perceive ourselves in the moment and how we remember previous such experiences. In that sense, they help us constitute our personalities. The specific structures embedded in the temporal lobes involved in emotion and memory, the amygdala and the hippocampus, are part of the so-called limbic system.[14]

It may be best to look at our brain not as a single functional entity but as multiple brains that are tightly interwoven and integrated. Among those, we can include what we may call the cognitive brain, the emotional brain, the social brain, the somatosensory brain, the visual brain, the auditory brain or the recording brain. This short-list of obvious multiple brain candidates, examples of brain projections onto a specific functional or organizational plane, is certainly not complete and easy to extend. Each of those multiple brains has a primary cortical area or hub that resides in one of the brain lobes and connects to related functional areas in other lobes, processing and integrating information received from the primary area within their own specialized contexts. Admittedly, this is a simplistic picture, but it does give justice to the innate and highly complex network structure of our brains. More so than what an exhaustive list of all the specialized areas of our brains can convey that science knows of today and of which fig. 1.1 gives only the top layer and that in a very approximate way.

Understanding how our brains help us literally make sense of the world around us by receiving and processing external stimuli and signals from within our bodies is only a first step. Over the past few generations, countless scientists have contributed towards this understanding and accomplished much in that regard. However difficult it was and continues to be, making progress in this effort remains the easy challenge. It is much more difficult to understand how what we typically refer to as our higher brain functions come about. While the use of perceived analogies between the human brain and a computer certainly has its limitations, sometimes such comparisons can be helpful. Just like our brain, computers are equipped with interfaces and sensory capabilities through which we can access them. If they did not have those interfaces, we could not use them at all. As important as external interfaces are for computers so they can sense our inputs, computing devices, be it a regular computer or some kind of robot, must also monitor themselves, they must sense their internal states to stay within certain parameter ranges so they can function properly. However, unlike for computers where we can identify and trace instruction sets to understand what a computer does to execute what we ask it to do or what it does to keep functioning within its human defined working conditions, we have a much harder time to accomplish something similar for the human brain.

Take for example something like mathematics. A computer can solve mathematical equations by executing instructions it is given and so can many of us. For a computer we can describe exactly how its mathematical capabilities come about, how the respective hardware and software parts work, and what their limitations are. The reason we can do this is of course the trivial fact that it us humans who build such computers. However, we can do nothing like that for the human brain where we do not know how it solves mathematics problems. We can certainly identify specific brain areas as being involved in mathematics when we ask a test person to solve a mathematical problem while scanning slices of her or his brain for elevated activity by using one of the advanced brain-scanning techniques allowing researchers to peer right into the human brain while it is solving a problem. As of today, researchers have identified three regions of the brain that are related to mathematical learning in general, one each in the parietal and occipital lobes as well as in the prefrontal cortex. But we still know next to nothing about what we could call the mathematical brain. Similar for other higher brain functions. We could have picked something other than mathematics to illustrate how little we know about how our brains really work. Pointing to where something happens in our brain and how these areas in our brain connect is a start. However, we have to realize that – to stay in the computer analogy - this is the equivalent of knowing the location of certain components in a computer and how they connect through the computers electrical wiring. As all of us likely know, we can learn next to nothing from this information about how a computer executes programs; and it certainly does not tell us anything about the languages used to code the computers programs; or how to understand those in the first place. All we can do is make inferences on a higher level, similar to how we can infer from a computer's hardware whether it is a parallel or serial processing machine; or can deduce some of its other purposes from its general architecture, like what kind of memory it uses and where that is located.

The two brain hemispheres with their respective four major brain lobes on each side, plus a few other parts of the limbic system not shown in fig. 1.1, form the cerebrum, which is the Latin word for brain. In the back of our brain below the cerebrum sits the little brain, the cerebellum. Its front connects to the brainstem, which forms the bottom part of our brain and continues into our spinal cord. The cerebellum is the seat of automatic balance control and coordination. As we learn movements, be it taking our first steps as toddlers or learning how to play musical instruments or sports, once practiced sufficiently, the respective automated movements to do such things reside in the cerebellum and are accessed and executed automatically. Similarly, as can be expected for such essential life-support functions, the controls residing in our brainstem that regulate our heartbeat, blood pressure and breathing are on autopilot as well. However, as is well known, some of us are able through continued practice and mental focus, for example through yoga practice, to learn how to achieve some level of control even over such functions.

Over the last century, science has come to recognize that our brain is not just an organ in our head. Rather, via the human nervous system the brain extends itself throughout our bodies. More recently, it has become clear that we actually do have what science now calls a second brain in our guts. This brain outpost in our guts consists of some one hundred million nerve cells lining your gastrointestinal tract. In a very literal sense, gut feelings are more than just intuitions as the second brain in our gut connects to the brain in our heads through a veritable nerve highway that is bi-directional. A distributed network structure of our brains is not only what we find in our heads, we also find the brain network reaching practically into every nook and cranny of our bodies. Sometimes, parts of this network, like the second brain in our gut, can shape activities in our head brain just as much as any other part of our head brain.

Instead of speaking of one brain network, it is more appropriate to talk about several brain networks. Different parts of our brains work at different speeds and there are good reasons for that. From an evolutionary perspective, so-called fight, flight or freeze responses only can be successful when executed as fast as possible. Therefore, the respective parts of our brains in charge of those lifesaving reflexes are on autopilot and have the fastest processing times. We can call this part of our brain the survival brain. Deliberate cognitive processing is a much more complex and therefore slower brain activity than for example the issuance of directions to release adrenaline and trigger the required muscle responses for a quick get-away. Similar to how we associate perception with our cognitive brain, science has come to associate neuroception with our survival brain. Neuroception is an automatic process and we can refer to the respective network serving it as the neuroceptive network, distinct from the perceptive network that serves the needs of our cognitive functions. This is admittedly a simplistic picture, but it has a much more detailed biological underpinning to it. Staying with this picture, a third part of our brain works at processing speeds somewhere in between the fast speed of the survival brain and the slower speed of the cognitive brain. The brain functions associated with it include for example the somatosensory cortex and sensing functions such as vision, hearing, tasting or smelling that are registered and processed automatically; essentially anything that we can associate with automatic processing of sensory inputs that does not reside in the survival brain. The respective intermediate speed network associated with serving such functions we can call the autoceptive network. The last few pages gave a high-level description of our brain that conveys some of its properties from a system level perspective. Anatomy and physiology of the human brain are immensely complex as anyone can find out quickly by consulting standard textbooks on the subject.[15] The brain sketch in fig. 1.1 gives a very rudimentary representation of major parts of the human brain which however is sufficient for the context here since it only needs to serve a very specific purpose: to provide a simple graphic illustration of the human brain to help orient ourselves, but nothing more.

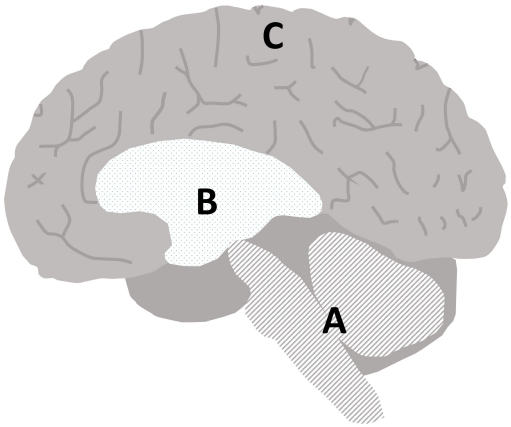


Figure 1.3: Cartoon drawing of the triune brain model: Reptilian brain (A), limbic brain (B) and neocortex (C).

From an evolutionary perspective, the medical profession has come to differentiate three different brain layers in what it commonly refers to as the triune brain model.[16] Fig. 1.3 shows a simple cartoon to illustrate this three layer brain model. Today, we know substantially more about the human brain than we did in the 1960's when the American neuroscientist Paul D. MacLean (1913 – 2007) originally proposed the triune brain model. Models are always simplifications. That is why they are so useful and that is why one needs to be mindful of their limitations. The three brain layers of the triune brain model are the reptilian brain, the limbic brain and the neocortex. The evolutionary most ancient part of our brain is the reptilian brain. As its name implies, this part of the human brain is the structural analog to what we find in reptile brains. From a functional perspective, this so-called reptilian brain is very much on autopilot, regulating vital body functions such as heart rate, blood pressure, breathing or body temperature and handling automatic stress responses. The latter include of course the fight, flight or freeze responses and it should be no surprise that the reptilian brain pretty much corresponds to what we earlier called the survival brain. In addition to ensuring survival in the face of immediate threats, the reptilian brain also serves long-term survival functions such as making sure that our bodies know when to eat and equally important, that the human species continues to procreate. The oldest reptilian and reptilian-mammalian fossils date back some 315 to 320 million years ago which must have been about when reptiles branched off the amphibian linage. The second layer in the triune brain model and the second oldest part of the human brain is the limbic brain, our brain's limbic system, sometimes also called the paleomammalian cortex.[17] As the latter name suggests, the evolutionary history of this part of our brain traces its origin to the arrival of mammals dating back to some 210 million years ago. The triune brain model associates this part of our brain with emotions. In a manner of speaking, one could say that the triune brain model makes it the equivalent of a sensory organ for emotions; emotions which we act on by the limbic system instructing the reptilian part of our brain to initiate the required responses. The third layer in the triune brain model and the youngest part of the human brain is the neocortex, sometimes also referred to as the neomammalian cortex.[18] It is this part of the human brain, which most scientists would argue distinguishes modern humans from their distant human ancestors as well as from our closest primate relatives; and of course from all other animals. As its name indicates, the neocortex is the newest part of the cerebral cortex, the part of the

human brain that most people likely picture in their minds when being asked to describe the human brain.[19]

The relative size of the neocortex with its familiar wrinkled surface is much larger in humans than in other animals. It certainly accounts for most of our brain volume, making up roughly three quarters of its total volume. From an evolutionary perspective, the neocortex ratio, that is the size of the neocortex relative to the rest of the brain, seems to be a reliable indicator for the maximum social group size a given species can support. This insight came from research in the early 1990's, pioneered by the British anthropologist Robin Dunbar (born 1947). Dunbar formulated what we call today the social brain hypothesis, which in simple words states his finding that in primates the relative size of the neocortex in a species increases as social groups become larger. Extrapolating from his research on primates, Dunbar suggested that based on the human neocortex ratio, the number of people we can have any meaningful relationship with tops out on average around one hundred fifty. So much for having thousands of "friends" on social media. The human neocortex has long been associated with what are the so-called higher human brain functions, going back to one of the pioneers of human brain anatomy and physiology, the English physician Thomas Willis. Because of its much larger relative size in humans and due to the more intricate folding of the human neocortex as compared to other animals, Willis believed these intricate folds in the human neocortex to be manifestations of our cognitive capabilities. Since then we have developed a much more detailed knowledge of what may be located where in our neocortex and how that may all work to produce what we call our cognitive skills, our sensory perceptions or our motor skills. However, Willis basic insight still stands as the neocortex, in a broad sense, continues to be associated with human cognitive capabilities.

Since the triune brain model was first proposed, neuroscience has developed a much better understanding of how different areas in our brains interact. It should be no surprise that the reptilian brain, the limbic system and the neocortex are not really three separate layers of the brain. The reptilian and limbic system parts of our brains can look back at more than two hundred million years of co-evolution. The beginning of the differentiation that we see today in the neocortex area between humans and our primate cousins cannot reach further back than to around 5-7 million years ago when the human and chimpanzee lines split from a common ancestor. That leaves about the same amount of time for the co-evolution of the human neocortex with the more ancient parts of the human brain. Over those long evolutionary time spans, the three layers distinguished in the triune brain model have integrated, assimilated and co-evolved in ways that are difficult to retrace in detail. However, the result is evident with the three brain layers anatomically and functionally integrated, with information flowing in much more complex ways than just top down, and our higher brain functions much more closely interacting with our more ancient brain regions through what certainly are not one-way streets.

How different are the brains of our closest cousins in the animal kingdom from our human brains? To answer this question, scientists need to understand how the brains of humans and great apes differ in their so-called phenotype. The phenotype is the physical expression of an organism or a part of it as we find it encoded in the genes of that organism, the so-called genotype, representing the blueprint we inherit with our genes.[20] Naturally, scientists have been working for quite some time to understand the physiological differences between a human brain and the brains of the great apes. The brains of our great ape cousins really are not so different from human brains. Ours however happen to be significantly larger, about three times as large. This factor is much bigger than what one would predict from just scaling great ape brains to ape body sizes matching human body sizes. Some parts of the human brain scaled up in size more than others, specifically the relative size of the human prefrontal cortex is quite a bit larger in human brains than in ape brains. Another well-known difference between human and ape brains is an increased left-right asymmetry of the human brain that is specifically pronounced for human language capabilities which are much stronger represented in the left half of our brains. This could be an indication that increased brain specialization or capability goes hand in hand with increased modularity. Scientists have identified a great number of other differences. However, the sum total of all those specific phylogenetic differences between human and ape brains, aside from size differences, is not easy to correlate to specific qualitative differences in brain capabilities that would allow us to understand why humans are smarter than apes.

In addition to phylogenetic differences between adult human and ape brains, there is also a very pronounced difference in brain development among humans from what we can observe for apes. For humans the brain size of a newborn is about a quarter of the size of an average adult human brain; by age one, the brain size of a human baby has doubled; by age five, it is about 90% of its final brain size, which it will eventually attain sometime in its early to mid-twenties. In contrast, a newborn chimpanzee's brain is close to 40% the size of an adult chimpanzee brain, which it will reach by age five or six. With respect to looking for differences in the genotype of human brains vs the brains of the great apes, science is still very much at the beginning. For now, we can only understand very few of the instructions encoded in our own genes, so we are far from being able to reconstruct the blueprint and much less the details of its execution as a human brain develops. However, we most certainly can expect that to change over time. Eventually we will have a complete knowledge of our brains anatomy and physiology and will not only be able to read the human genome but also understand it. However, will that ever equate to us having a full understanding of how a human brain constitutes a human self? Just close your eyes for a moment and concentrate on identifying your very own self, what is it that you find in yourself that makes you "You"? We all can sense in that way who we are but none of us is able to point to a single something within ourselves and

proclaim with conviction that this is our very own self. What we refer to as our self is more than the sum of our parts. That is a trivial statement but then it has been and continuous to be the source of much confusion.

That, which transcends the sum of our parts, which we try to reach out to when we seek to identify within ourselves who we are, philosophers and scientists refer to as a state of mind or a mental state. That what we perceive to be ourselves is the experience of a certain mental state. We cannot tell how our complex brains produce mental states, but we all know that we experience them. That we have to speak of states of minds is essentially an admittance of what little we still know as to how our brains construct our realities, specifically those that are not under our control. Some brain processes are conscious but many are not. Many of our capabilities we first have to learn consciously, but we all have experienced how they more or less at some point begin to bypass our conscious mind as our brains execute them in the background and automatically. Figuratively speaking, if every time when we ask ourselves who we are we first needed to construct in our minds who we believe ourselves to be, there may not be much of a self that we could identify with. Rather, the mental state experience reflecting who we are is something that we recall and that has been developing with us from birth. However, when we do recall it, only a fraction of it surfaces in our mind with most of it remaining in the unconscious.

Similar to when we look at an iceberg and know that most of its volume is beneath the waterline, we intuitively understand that beneath what we consciously experience as the mental state of ourselves there lies a larger part of us. Even though this hidden part of our self is much more difficult or impossible to access, its presence is nonetheless manifest. When we look at our facial image in a mirror, we know that the reflection of our face in front of us is our physical self. As we are looking at ourselves in the mirror we know that there is something behind our eyes that sees the physical us; this something is our self, the experience of the mental state we identify with us. In the same way, that we look into a mirror and equate the image looking back at us with a mental state reflecting our self, we also look at others acknowledging their mental state as they acknowledge ours. We all are aware that there is a difference between the eyes of others just collecting our image and what looks at us from behind those eyes. Recognizing our own mental state of self and that of others is an example for what scientists call a Theory of Mind. Broadly speaking, a Theory of Mind provides an umbrella under which we attribute a variety of mental states reflecting our emotions, desires or beliefs, including our belief of who we are, to ourselves and to others.

Of Mind Models

For anyone seriously interested in human brain anatomy and physiology there are many excellent books providing a comprehensive summary of what today's medical students need to know. Many of them include wonderfully detailed illustrations and functional diagrams explaining all of what we know and believe to understand about the inner workings of our organ of mind. However, anything as detailed as that goes far beyond the scope of this book and certainly is not what the last section was about to convey. The rudimentary account of the human physical brain given in the previous section had an entirely different purpose. Namely, to cover sufficient ground so anyone without a background in biology or medical sciences can get a sense of the complexity of the human organ of mind while at the same time appreciating just how little we still understand of how it produces the perceptions of our self and of the world around us that we all certainly have.

Before we explore this further, we need to digress a little to ask a seemingly innocuous question. Aside from our physical brain, is there anything else that we refer to when we talk about the human mind? By asking this question, we are treading on treacherous ground but that is just unavoidable. Over several thousand years, reaching back all the way to the very roots of our modern civilizations, our ancestors have made many different attempts to answer this question. Clearly, for most of that time, because of the lack of biological knowledge, answering this question meant in essence to engage in philosophical and religious discussion and frequently disputes, not all of them peaceful. The ancient Greeks were the first we know of to focus their efforts towards understanding human nature on something like the intellect even though they did not really have a concept of mind in the same way we tend to think about it. Later, the rise of the three great Near East religions, Judaism, Christianity and Islam put the soul at the center of attention as through it, a believers existence would continue into a next life.[21] However, the properties and meaning of the human soul in such religious contexts, whether it exists or not, that is not something we can consider here. Such discussions squarely belong to the realm of religion and are strictly matters of belief, and we will leave it at that.

Through the ages, there is of course quite some continuity of philosophical thought. Early on Christian theology was heavily influenced by Plato and Neo-Platonism, much less so by other Greek philosophers, including Aristotle. Later however, with the medieval reception of Aristotle, his influence came to dominate Christian theology, transforming it in the process into a Christian-Aristotelian synthesis. In its early history, Islam was always receptive to ancient Greek thought, at least within the confines its religious doctrines allowed for that. Judaism, notably, was least influenced by ancient Greek philosophy. This is not surprising as Judaism's religious and intellectual traditions root as deep in time as those of the ancient Greeks. The beginnings of the scientific age also set the

stage for a new philosophical outlook and a new perspective on the mind-body problem. Modern philosophical thought and discussion of the mind-body problem originates with the seventeenth-century French philosopher Rene Descartes. It is hard to overstate the importance of this change in philosophical outlook as much of it still reverberates today. From a philosophical perspective, the discussion that began with Descartes continues to this day even though what we know about the human brain and what we are beginning to learn about the human mind is very different from the starting point from which Descartes began his exploration. Today our knowledge about the biological foundation that should inform any discussion of the mind-body problem is very much richer than any philosopher back then could have likely imagined.

In Descartes time, our knowledge of the human brain was still very limited. He and his contemporaries knew very little about just how the human brain and body interact to form a single unit that we recognize as our very own self when we look for example in the mirror. However, that did not keep Descartes from concluding that the tiny pineal gland in the center of our brain was the seat of the human soul and to pronounce it as the place where all of our thoughts originate from.[22] Descartes was wrong on that account. Nevertheless, to give the human mind a defined physical location within the brain remained a desire that has tempted many others since then. So far, all quests to find a singular physical location for the human mind within the human brain or the human body have been unsuccessful. We still have not found the seat of the human mind in our bodies. Most likely, and the following paragraph will speak to that, we will never find it, as the human mind may just not have a single physical location. Today, we have a much more intricate and detailed knowledge of human biology and the biology of life in general than Descartes likely could ever have imagined. The increasing weight of this biological knowledge should have significantly changed the general philosophical discussion about the mind-body problem. That has not fully happened yet and whatever may be unfolding in this direction seemingly does so very much removed from public attention. For most of the time since Descartes, philosophy and biology went their separate paths. Only more recently have biological facts found their way into philosophical discussions such as for example in evolutionary philosophy. This is also true for the mind-body problem. Many of today's philosophical discussions on that topic seemingly could be time ported back into the seventeenth century to engage with a virtual Descartes without him realizing that he was having a conversation with people from the future. One can only wonder what Descartes philosophy would look like today if he could reformulate it with the accumulated biological knowledge available to us. That all may sound like there is little merit to engage in philosophy of mind discussions, but nothing could be further from the truth. We just have to be very clear with respect to what we are really talking about when we participate in such discussions. This is why in the following paragraphs we will reflect on some of the biological foundation of what we could possibly call our mind; in

the next chapter, we will then look at the philosophical concepts underlying mind-body discussions.

The end of the last section briefly introduced the Theory of Mind concept. We will talk about this concept more in the next chapters, which will look into philosophy of mind and the exploration of human nature that began with the making of the science of psychology. Here we need to ask what science broadly can tell us today about the biological foundations that could support any such Theory of Mind concept. The obvious importance of the human brain in processing sensory inputs and its association with human intelligence has diverted our attention for a long time from the true nature of the human brain. What we are looking at when we have a human brain or some kind of brain cartoon in front of us, such as the one in fig. 1.1, is the finished product of an evolution going back some six hundred million years. When we say finished that is only true in the sense that evolutionary changes of organs such as the brain happen on time scales that vastly exceed the life spans of us mortal humans. Evolution works over spans of time so vast that over the course of our short life spans human biology will always look like a finished product that does not change. However, evolution never stops and human brains in a hundred million years from now could be something very different from what they are today, provided the human species perseveres for that long.

The finished brain product that we are looking at today started out with the first cells capable of generating and transmitting what we commonly call nerve impulses. Nerve impulses are electrical signals driven by an action potential that travels the length of a nerve cell and that can trigger the firing of an electrical signal in another connected nerve cell, thereby propagating the signal. Connections of nerve cells morphed eventually into networks without any centralized nerve cell center or they became connected to form a nerve cord, at the end of which a higher concentration of nerve cells, scientists refer to such a nerve cell concentration as a ganglion, would become the very distant ancestor of all brains.[23] Jellyfish are a good example for animals that have been very successful in an evolutionary sense, having been around for hundreds of millions of years, without ever evolving a nerve center. With jellyfish and other such animals, every body part is equally capable of performing all necessary nerve functions within its physical domain without having to consult a higher authority located somewhere else in the body. That is different for animals that evolved a nerve cord, as the primitive worms did which scientists believe were the first animals to possess such a structure. Their nerve system was likely not much different from the one we can find in nematode worms today, which incidentally represents the most fully understood nerve system of any animal scientists have studied so far. Fully understood in the sense that there are still few enough nerve connections – at least as compared to animals such as us – that it is possible to map all of those connections. It should be clear from the evolutionary origin of nerve cells that their first and only purpose was to sense; to sense the world around an animal as well as within

an animal. Higher animals, along with their more complicated body plans, eventually evolved a much more complex nervous system than primitive worms. These complex nervous systems of higher animals, including large human brains, represent only the last step in the roughly six hundred million year evolution of nervous systems.[24]

At the highest level, biology has come to differentiate the human nervous system into two distinct parts, the central nervous system and the peripheral nervous system. The central nervous system consists of the brain and the spinal cord while the peripheral nervous system consists of nerve fibers connecting various body parts to the central nervous system. Nerve fibers in the peripheral nervous system serve a number of specific purposes. First, there is the voluntary and involuntary control of our body functions. Voluntary control of functions involves motor nerves while involuntary control through the autonomic nervous system regulates involuntary functions such as heart rate or breathing. The peripheral nervous system in charge of our gut, called the intrinsic or enteric nervous system, represents a special case as its regulation and control combines voluntary and involuntary aspects.[25] What we traditionally, that is in a non-biological or non-medical meaning, refer to as our five senses, touch, vision, hearing, smell and taste are all part of our peripheral nervous system that provide us information about the outside world. However, our sensory system is much richer than that as the peripheral nervous system in animals maps the entire body from a sensing and acting standpoint. In the same way that we use our external senses to judge the status of the world surrounding us we use our many internal senses to judge the functioning of every part of our inner bodies. However, almost never are we conscious of those latter activities, which throughout our lives, from moment to moment, ensure that our bodies stay within healthy operating parameters. Usually, we only take note of signals coming from our inner body when something fails or when we feel pain. Mapping the state of an animal body with respect to its environment and ensuring its survival as well as monitoring the functioning of the inner body as the animal moves through the outside world requires a different kind of neuronal network for nematode worms than it does for humans. Evolution has scaled up the external and internal sensing capabilities of animals to a level that in the case of humans has given us a very large brain at the end of our spinal cord. However, the fundamental purpose remains the same: continuous mapping of the human body status, externally and internally, on a time scale and level of detail that ensures the survival of the human body itself.

Our nervous systems are made of two kinds of cells, the nerve cells themselves, often referred to just as neurons, and so-called glia or glial cells.[26] Speaking in general terms, neurons process information while glial cells provide structural support and nourish and protect neurons. Fig. 1.4 shows a schematic drawing of a neuron with its four major components. From the top down in fig. 1.4, those are the tree-like structures known as dendrites, which receive signal information from terminals of other neurons; the cell body, the

so-called soma, contains the cell nucleus; the axon, which can generate electrical pulse signals where the axon connects to the cell body; and finally the bud-like terminals connected to the axon towards which such electrical pulse signals propagate along the axon. Neurons connect through small junctions called synaptic clefts. These synaptic clefts separate so-called presynaptic terminals in the bud-like terminals of one neuron from postsynaptic terminals on the dendrites of another neuron. Jointly, the presynaptic terminal, the synaptic cleft and the postsynaptic terminal form what scientists call a synapse.[27] The different types of neurons come in various shapes and sizes. As a measure for the diameter of a neuron, one can take the size of the cell body, the soma, which can vary from a few hundredths of a millimeter to roughly a tenth of a millimeter. That is not too different from the diameter of a human hair, which ranges from approximately two hundredths of a millimeter to a little less than two tenths of a millimeter. A neurons length, essentially determined by the axon, can vary between fractions of a centimeter to almost a meter. The electrical pulses neurons use to communicate, called action potentials, travel in one direction only, away from a neurons cell body along the axon towards the dendrites of other neurons separated by synaptic clefts from the presynaptic terminals of the firing neuron. Most often action potentials originate close to the so-called axon hillock, which is the part of the cell body, where the axon connects to the cell body. However, action potentials, which represent a local depolarization of the cell membrane propagating through the neuron, can also occur first at other sites in the neuron, including in its dendrites. Transmission of those action potentials across the synaptic cleft occurs via chemical means. The presynaptic terminal releases neurotransmitters into the synaptic cleft, which then travel to the postsynaptic terminal of another neuron where they bind to specific receptors. The net result is that the docking of the neurotransmitters onto those receptors makes it either more or less likely that the neuron to which the dendrite with those receptors belongs will fire an action potential itself.

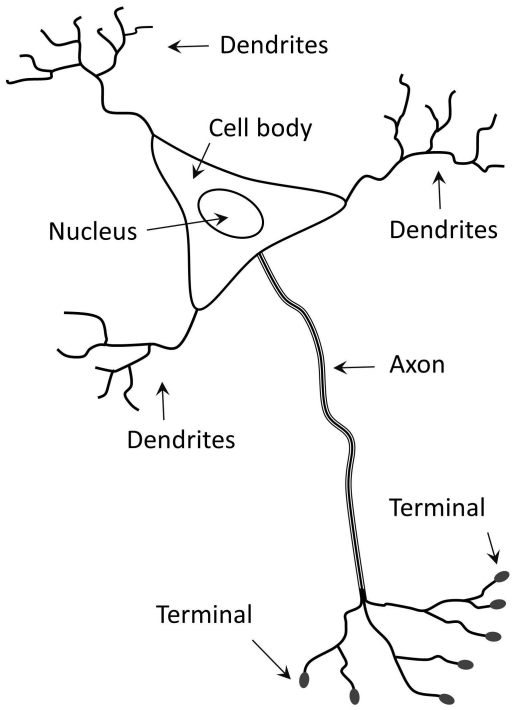


Figure 1.4: Simple schematic of a never cell, in this case of a multipolar neuron as it can be found in the cerebral cortex.

There are different functional types of neurons and most of them share the same four components but not all do. While no neuron has ever more than one axon, some very small neurons possess no axon at all and seemingly use their dendrites as axons for

signaling. The neuron variant shown in fig. 1.4, a so-called multipolar neuron, is a typical example of what neurons look like in our cerebral cortex.[28] Our cerebral cortex is home to some eighty-six billion neurons and on average each of them has several thousand connections to other neurons. Therefore, order of magnitude estimates for the total neuronal connections in the human brain come in at around one hundred trillion connections. In 2019, computer chip manufacturers broke the trillion-transistor barrier for computer chips, so maybe in a few years we may eventually see the first computer chips with one hundred trillion transistors in production. However, there is a very big difference between a transistor-based computer chip and the human cerebral cortex. A transistor is a simple gate device where one input controls the function of one output. Neurons with their many dendrites and terminals have not just many more input and output channels; they also connect to many more other neurons than a transistor connects to other transistors. In addition, a transistors single gate electrode can be either open or closed in which case it either is in a logic on or an off state, depending on the chip design. This is different for neurons as their many dendrites process exciting as well as inhibiting signals triggered by action potentials coming from other neurons via synaptic connections counting in hundreds to thousands. Only if the equivalent of a weighted average across those excitatory and inhibitory input signals exceeds a certain threshold will the neuron fire an action potential and pass on the signal to the neurons connected to it down stream in the specific network path this neuron belongs to. Clearly, the human cerebral cortex operates on a very different level than the simple two-level logic our most advanced computer chips still use.

Our brain has a great variety of differently shaped neurons and we may not know all of them yet. It becomes somewhat simpler when we look at neurons from a functional perspective. From what we know today, there are just three functionally different types of neurons in the spinal cord and the peripheral nervous system: sensory neurons, motor neurons and interneurons. As the name indicates, sensory neurons are nerve cells activated by sensory organs connected to them when those receive physical or chemical signals from their environments. Sensory neurons are for the most of the pseudounipolar neuron variety and carry signals from the outer and inner parts of our bodies into the central nervous system. Such information from our sensory neurons delivered to the central nervous system and processed in the brain becomes our experience of vision, sound, smell, touch or heat. Motor neurons are multipolar neurons and come in two varieties, upper motor neurons and lower motor neurons. Upper motor neurons are a part of the central nervous system. They connect from the brain through the spinal cord, directly or through interneurons, to lower motor neurons, which are a part of the peripheral nervous system. Lower motor neurons deliver electrical impulses from the spinal cord to skeletal muscles and muscles that do not attach to our skeletons, so-called smooth muscles such as our stomach. In short, motor neurons carry signals coming from the central nervous

system to the outer parts of our bodies as well as to internal parts of our bodies. The third generic type of neurons next to sensory and motor neurons are the interneurons. Interneurons are multipolar neurons and connect various neurons within the brain and spinal cord. Interneurons are a part of the central nervous system. We already mentioned their role in enabling signal transmission between upper and lower motor neurons by connecting these neurons within the spinal cord. Importantly, interneurons play also a critical role in the brain's internal communication processes.

Unlike for the spinal cord and the peripheral nervous system, it is much more difficult to differentiate neurons in our brain by their function. The brain is home to many varieties of the three generic types of neurons introduced above, specifically of interneurons facilitating internal brain communications, of which our understanding is still in its infancy. However, for the purpose of this book, there is no need to go any further with a more detailed description of the human nervous system than the coarse layout provided here. While a more detailed discussion of the various types of neurons, specifically in the brain, is beyond the scope of this book, there is one more aspect we need to look into.

The human body has cells that it can regenerate and cells that it cannot replace, or only has a very limited way to regenerate. Our body continuously generates new cells to replace aged cells. This typically happens through the cell division of yet undifferentiated cells, which then replace cells that die off. Over time, as we age, our body continuously replaces in this way many of the cells making up our body parts. Some cell types age faster than others do. For example, skin cells have only a lifetime of a few weeks and hence we literally shed our skins every few weeks, but we do that piecemeal with tiny skin parts shedding all the time with us for the most not even noticing any of it happening. The bulk of human body cells being recycled quickly includes for example red blood cells or cells in our gut lining, with life spans of roughly one hundred days and a few days respectively. For a long time biologists counted neurons, heart cells, specific muscle cells and blood cells among the cell types that cannot regenerate themselves through cell division and those either need to last for a lifetime or our bodies must produce them in a different way than through cell division. Blood cells are an example for the latter. Red blood cells with a little more than a one-hundred-day lifetime do not have a nucleus at all and therefore do not contain genetic material. While white blood cells with an even shorter lifetime of a little more than a couple of weeks do have a nucleus, they do not regenerate through cell division either. Instead, other cells in our bone marrow produce red or white blood cells.[29] Muscle cells live many decades, and it seemed as if the cells in our heart muscle would not regenerate at all; at least that was the established orthodoxy for a long time. They are supposed to last us a lifetime and most of them actually do unless we suffer for example a heart attack. More recently, however, researchers discovered that the human heart continues to generate new cells throughout its life span.[30] As it looks like, when we are in our early twenties our bodies replace about 1% of our heart muscle

cells every year. This rate however drops to about one third of a percent annually by age seventy-five resulting in less than 50% of heart cells replaced when we turn seventy-five.[31] Certainly, our bodies replacing about half of our heart cells through our lifetime is better than no heart cell replacement at all as this indicates science may eventually find a way to regrow heart muscle cells to replace damaged heart tissue.

What science once believed for heart cells, that they only can die but never regenerate, still seems to be mostly true for the neurons in our brains: a human is born with all the brain neurons she or he will ever have. Why mostly? Because neuroscientists recently discovered that some parts of our brain seemingly do generate new neurons.[32] One part of our brain where this is the case is the hippocampus, which plays a critical role in memory formation and retention. If the human brain has indeed a limited capability to generate new neurons, this certainly will change the outlook on how degenerative brain diseases could eventually become treatable. However, for our exploration of the mind-body problem this makes no difference because it does not change the fact that the overwhelming majority of neurons in our brains are present at birth and they stay with us until we die or they die themselves before that time, which a good number of them do.

The network of nerve cells running through our bodies, comprising our central and peripheral nervous systems, is the information highway where electrical signals carrying information travel from all parts of our bodies to our brains and where electrical signals coming from our brains tell our bodies what to do. Think of lifting an arm, walking, squinting an eye or controlling internal functions such as breathing or your heart rate. How fast do those electrical pulses travel in our nervous system? That depends on which part of the nervous system one looks because signal speeds vary from roughly a meter per second to exceeding one hundred meters per second. That an electric signal in our nervous system travels significantly slower than an electromagnetic wave in an electric circuit where it travels close to light speed came as a surprise to many when it was discovered in 1849 by the German physicist and physician Herman von Helmholtz (1821 - 1894).

The nervous system, however, is not the only pathway through which the human brain and the various parts of the human body communicate. Our bodies have a second communication system not based on transmitting electrical pulses but delivering information through chemical means. This chemical messaging system, the so-called endocrine system, works by glands in our bodies directly releasing chemicals into the blood stream. The speed at which those chemical signals travel is hence determined by the speed at which blood circulates, which is of course quite a bit slower than the speed at which nerve signals travel.[33] Glands are organs and the chemicals they produce are what we call hormones. The glands in our brains include the hypothalamus, the pituitary gland and the pineal gland. Glands in our bodies include the thyroid and parathyroid glands in our neck, the thymus between our lungs, the adrenal glands on top of our kidneys and

the pancreas behind our stomach; plus of course the sexual glands in our pelvic region, ovaries if you are a female and testes if you are a man.
Our electrical and chemical signaling highways, the nervous system and the endocrine system, connect in the hypothalamus. The pituitary gland located in the lower brain, approximately at the height of the nose bridge and right in front of the brainstem is the master gland of the endocrine system. This pea-sized gland uses information it receives from our brain to tell all our other glands what to do. It does so through chemical messaging but also produces many important hormones itself, regulating critical body functions directly or indirectly. The latter for example includes the release of hormones into the blood stream, which then triggers other glands to produce different hormones to regulate body functions. The pineal gland, which, as mentioned earlier, Descartes believed to be the seat of the human soul and place of origin for all our thoughts, is mostly concerned with regulating our wake and sleep cycle. The four small glands of the parathyroid sit right behind our thyroid and are important for bone health. The thyroid gland produces hormones controlling our metabolism. If it slacks in its hormone production we slow down or may become constipated or gain weight. If it produces too much of its hormones, all that reverses, and we experience the other extremes of those conditions. The thymus between our lungs is important for our immune system as it produces so-called T-cells, a specific type of white blood cell, developed in the thymus from undifferentiated cells migrating there from bone marrow. The two adrenal glands on the top of our kidneys receive their name from adrenaline, the hormone these two glands produce. In addition to making this fight or flight hormone, these two glands also produce so-called corticosteroids, which for example affect our metabolism and our sexual functions. Synthetic versions of corticosteroids are widely used and powerful medications to treat inflammation, the familiar cortisone being a well-known example. Our pancreas is a part of our endocrine as well as of our digestive system. Without the enzymes our pancreas makes we would have a difficult time to digest our foods. As part of the endocrine system, our pancreas produces the hormones insulin and glucagon, which regulate the right amount of sugar in our bloodstream and in our cells. If our body does not produce enough insulin, we will suffer from diabetes. Diabetes 2, a condition when our bodies produce some insulin but not enough of it, has become a frequent affliction in many developed countries where people either eat too well or consume too much of what is bad for them. Sadly, diabetes 1, the condition where the human body has stopped to produce any insulin at all, is also on the rise. Our sex glands, ovaries in women and testes in men, produce our sex hormones, estrogen and progesterone in women and testosterone in men. For both sexes, these hormones regulate sexual development through puberty and throughout life regulate menstrual cycles and sperm production in women and men, respectively. With that, we are at the end of this high-level description of the human endocrine system, our body's chemical information and regulation highway.

The nervous and the endocrine system are two extensive and connected networks that provide our brain all kinds of information about our body state, externally as well as internally. One of them uses electrical signals to deliver information or exert control, the other one uses chemical signals. Electrical signals travel through our never fibers and chemical signals travel through our blood stream. Our brain continuously receives updates from all parts of our bodies through a combination of electrical and chemical signals. At the same time, through both systems, the nervous system and the endocrine system, our brain can reach into all of our body parts to ensure that our body keeps functioning well. When we speak about our brain, we usually have only that part in mind, which resides within the human skull. However, it is quite likely more accurate to look at the human brain as a distributed system, the purpose of which is to map our body state from moment to moment accurately enough so that it can control the body in a way that ensures its survival. Any model of the mind must take into account this complexity of the extended human brain. Importantly, it must acknowledge that this complexity is the result of a very long evolution.

For a long time, we approached the mind-body problem from a philosophical or theological perspective only. Because of our limited knowledge of biology, for most of that time this was the only possible approach. That should have started to change when the English naturalist Charles Darwin (1809 – 1882) published his pioneering work on evolution, first *On the Origin of Species* in 1859, then *The Descent of Man, and Selection in Relation to Sex* in 1871 and finally *The Expression of the Emotions in Man and Animals* in 1872. One could have expected that the views Darwin articulated in these three books would lead to biological and evolutionary arguments taking on much more prominent roles in efforts to understand the human mind. Especially so with respect to the shared evolutionary origin of human and other animal minds. However, that was not the case, as in essence we were not actually trying to understand the human mind-body relationship. Rather, our interests lay more with explaining or rationalizing our implicit understanding of human exceptionality, which we perceived as a given fact as there seemed to be nothing else just like the superior human mind.

Our brain somehow transforms the sensing, actuating and regulating aspects of our nervous and endocrine systems into what we typically refer to as our perception of self. Our experience of self is always the result of a momentary construct that relies on continuously updated physical, sensual and emotional maps of our body, which can remember previously mapped body states and possesses the efficacy to project our body state into the future. Body states are always biological in nature, so where does that leave the human mind? What is it that constitutes our thoughts, feelings and emotions or our memories of thoughts, feelings and emotions we had in the past or may hope to experience in the future? The honest answer for now is that we really do not know that yet. However, we can describe what likely will have to be part of any such answer.

Much of what we have learned about the functioning of our brains so far comes from observing what goes wrong when parts of our brain functions are impaired. Looking at just how complex an organ the human brain is, maybe we should expect to find that naturally occurring malfunctions of the human brain, discarding those caused by physical accidents for now, are a frequent phenomenon. However, they are not. On the contrary, our brains are surprisingly resilient. That by itself already tells us something about our brains that we alluded to earlier when discussing the brain cartoon shown in fig. 1.1. While there are neuronal hubs concentrating brain functions in some brain areas, brains are network organs. Just like the internet we all rely on nowadays does not have a single network center and therefore can keep functioning when some nodes may be down or working less than optimal, so our brain can keep functioning even when some of its neuronal connections may be damaged. Of course, there are limits to that and just how well human brains keep functioning depends on what kind of damage they have sustained. In addition, it also depends on what our societies consider a functioning human being to be. Alternatively and more poignantly, it reflects social norms dictating what we should consider as normal or abnormal human behavior. There could actually be many more malfunctioning brains out there than we believe. That is so because on the one hand the human brain itself possess a certain plasticity, which may result in brain damages not producing observable changes in behavior. On the other hand, there is a certain elasticity of our social norms, which to some extent can accept behaviors caused by brain malfunctions as normal. The latter aspect can be highly controversial which becomes quickly clear when we look at extreme cases. The one extreme case would be a society tolerating whatever behavior no matter how strange or offensive and considering everyone as normal regardless of his or her conduct. The other extreme we can find in societies enforcing very narrow views of what constitutes normal behavior and where what most of us would consider as normal may be anything but. Here we will stay away from any discussion about what may or may not constitute acceptable behavior, thereby leaving moral or ethical aspects aside, and rather look at physical brain impairment that changes human behavior in an involuntary way.

The most famous case for such a brain injury resulting in a clearly involuntary behavior is likely the brain damage that the American railroad worker Phineas P. Gage (1823 – 1860) suffered in 1848. In the accident, a roughly one-meter long and thirty-millimeter diameter iron rod weighing about six kilograms rocketed through his skull, entering the left side of his face below his cheekbone, traveling upwards behind his left eye socket and exiting the top of his skull close to midway. Gage sustained heavy damage to the inner side of his left frontal lobe and some damage to the inner side of his right frontal lobe but miraculously survived the accident and lived for another twelve years. However, his personality changed completely. Before the accident, Gage was a highly valued employee of the railroad company, a foreman supervising dangerous work and entrusted with the

kind of significant responsibilities which would only be given to a person of sound judgment and proven character. Physically, aside from losing his left eye, Gage had retained all his former capabilities, but with his personality so completely altered the railroad company could not continue to employ him in his former job. The next twelve years of his live would be very different ones for Phineas Gage before he finally passed in 1860. John Martyn Harlow (1819 - 1907), the American physician who had treated Gage in 1848 followed Gage's life as best he could and eventually managed to receive Gage's skull. He included a description of the skull in his 1868 summary report of Gage's recovery along with his medical observations. According to Harlow, Gage had become a completely different person from who he was before the accident:[34]

> *"He is fitful, irreverent, indulging at times in the grossest profanity (which was not previously his custom), manifesting but little deference for his fellows, impatient of restraint or advice when it conflicts with his desires, at times pertinaciously obstinate, yet capricious and vacillating, devising many plans of future operation, which are no sooner arranged than they are abandoned in turn for others appearing more feasible. A child in his intellectual capacity and manifestations, he has the animal passions of a strong man."*

Gage had essentially lost his ability to plan his own future. He could pay meticulous attention to anything right in the moment and come up with grand new plans but just as soon as he had everything planned out in his mind, he began focusing on something else. His brain injury had compromised Gage's attention, planning, and decision-making capabilities. As the Portuguese-American neuroscientist Antonio Damasio (born 1944) eloquently put it in his 1994 published book *Descarte's Error - Emotion, Reason, and the Human Brain*, there probably are many modern Phineas Gages living among us, who did not suffer such a traumatic injury, but who all the same exhibit to various extents similar kinds of impairments to attention, planning and decision-making capabilities. Such mental disabilities, as this is what they are, can have a number of different causes. What many of them seemingly share is a decoupling of what we typically call rational decision making from any emotional context. Invariably, that raises the question of just how the structure and network in our brain underlying our rational decision-making relate to the neuronal basis for our emotions. The answer to that should be obvious: the human brain just like the brains of all of our animal cousins and ancestors is an evolutionary brain. Evolutionary younger brain structures are not green field structures that discard older brain structures and start out from scratch, essentially leaving behind everything that came before. Our younger brain structures grew out of and utilized what existed before, our evolutionary older brain structures. What this means is that before brains could ever have thoughts, they first had feelings, which is just a different way to say that they sensed emotions.

When people speak about emotions, theirs or other people's emotions, they typically are not aware that the word emotion traces its roots to a term describing social upheaval in late sixteenth century France which in turn goes back to the Latin word "emovere", meaning to move out or away, or to remove. There are various modern definitions of what constitutes emotions, depending for example on whether one looks at emotions from a philosophical, psychological, biological or maybe even a popular standpoint. Looking at emotions from a biological perspective raises of course the question why emotions evolved in the first place. For what purpose did they evolve? Well, there can only be one purpose and that is that evolving emotions must have provided some animal species an edge in the struggle for existence. Being able to combine sensations coming from various body parts into emotions that the brain can quickly recognize and act on accordingly must have provided an advantage. Just imagine in contrast what the odds for survival would be if the brain of an organism would have to analyze each situation every time anew. If there were no biological brain states reflecting emotions such as for example fear, anger, sadness or joy, how could animals effectively respond to situations that trigger these emotions? That animals can experience neurophysiological states encoding emotions provides them with a pattern recognition capability that allows them to react quickly to a broad variety of external and internal situations their bodies may find themselves experiencing. Seen through this lens, emotions fundamentally reflect that which our body thinks is important for our brains to register and react to, so that our bodies continue to survive.

Some may flinch here because primary emotions are unthinking and instinctual. The above says nothing to the contrary, as the use of the verb "think" in the above context means nothing more than the "think" we may attribute to a cleaning robot in our home. Primary emotions are then essentially the equivalent of primitive non-reflected encoded thoughts that ensure our physical survival. In the case of primary emotions, evolution has done the thinking part for us and encoded the respective hardware and firmware in our brains and bodies. Primary emotions are inborn, or as researchers say, they are innate and there is nothing we can do about that; we cannot choose not to have them. Biologists have identified the amygdala, the almond shaped neuronal nucleus within our limbic system, one of the best-understood parts of the human brain, as a key player in mediating innate emotional behaviors including fear, anxiety and aggression. In addition to that, the amygdala is also critical for our emotional learning capability.

The simple picture of the triune brain model (see fig. 1.3) differentiates human brain anatomy and functions with respect to older and younger evolutionary parts of our brains. Figuratively speaking, the network and program architecture of the adult human brain reflects this evolutionary record of the human brain hardware architecture. To stay with this computer analogy, evolutionary younger algorithms integrate and build on evolutionary older algorithms. Many of those older brain routines are of the involuntary kind. They sense, actuate and regulate vital body functions that evolution has put on

autopilot because they are, in a manner of speaking, mission critical. Reflecting, thinking things over, requires time and in some cases no time can be lost to trigger the best possible response to a perceived danger for our bodies to survive. Our evolutionary younger brain functions tie into those older brain routines and include information received from them when updating the brain map of our body state. Strictly speaking, primary emotions such as fear, anger, sadness or happiness are body reactions to external or internal events. Even though we may say we feel happy or sad, that is not how we refer to our primary emotions.

When we say those things, we typically talk about secondary emotions. Secondary emotions are what we commonly refer to as our feelings; they are how we experience all of our emotions, including primary emotions. Secondary emotions are associated with brain processes taking place in our evolutionary younger brain. Our secondary emotions can still be and often are the result of our cerebral cortex processing primary emotions triggered by parts of our limbic system. Secondary emotions in response to primary emotions of anxiety or aggression could be shame or anger. However, secondary emotions can also emulate similar feelings in response to something that is not an event external or internal to the body. They do so by in a sense borrowing from routines, which our brains execute when they process primary emotions. Certain brain states themselves can trigger secondary emotions. Like when we think about something that we feel very good about, the associated secondary emotion may be happiness. Secondary emotions can be complex and combine a number of different emotions into a feeling. To give another example we likely are all familiar with, secondary emotions not triggered by primary emotions frequently occur when we recall past emotions along with memories we retrieve. Interestingly, our evolutionary younger brain, the thinking brain so to speak, has also figured out how to trigger the respective body reactions associated with certain emotions when we retrieve memories colored in those emotions. To achieve that, our thinking brain takes clues from the routines executed by the evolutionary older brain when primary emotions emanate from it.

That we can recall emotions, happy or sad memories and along with those emotions experience physical states similar to what those emotions in the first place can cause, is something we are all familiar with. It is this very capability, which maybe differentiates a great actor from a good one. Even though not quite, as acting is as much about controlling an experience as it is about producing one. We all can recall moments when the memory of a happy time brought unwittingly a smile to our face; or when remembering a lost loved one brings tears to our eyes. Among animals, being able to read and express emotions is fundamental to communication and the human species seemingly has become especially adept at both. Evolution has endowed all higher animals with the capability to recognize expressions of emotions in others as signaling intent. They can signal warnings, threats and imminent danger just as well as a genuine welcome or an invitation to engage in

the procreation of one's species. Emotions surfacing in facial expression are just one example but a particularly important and well developed one in the human species. Physical expressions of emotions are more evenly balanced and their repertoires and individual manifestations among higher animals are in many instances similar to how humans display them. Reading emotions is a quite different thing from experiencing and expressing them. The party expressing an emotion, at least in the animal kingdom excepting humans, is always clear in what kind of emotion it displays. The recipient party, however, faces the challenge of correctly reading a signaled emotion. Again, animals usually also get that part right. Unless of course the sender displays a signal to deceive the recipient, which also happens frequently in the animal kingdom. We will return to this topic later in this book in more detail but here we will stay with the human species, which incidentally is especially adept at floating deceptive emotions.

Emotions and their expressions were a topic that fascinated Charles Darwin throughout his evolutionary research. Darwin published *The Expression of the Emotions in Man and Animals*, his pioneering work on how we recognize emotions in facial expressions, in 1872. In this work, he was able to draw freely on the experimental research of the French physician and neurologist Guillaume-Benjamin-Amand Duchenne (1806 - 1875). Duchenne had pursued his study of human physiognomy by using localized faradic stimulation to reproduce human facial expressions and had published photographs of the induced expressions in his 1862 book *Mecanisme de la physionomie Humaine* (*The Mechanism of Human Facial Expression*).[35] Darwin recognized that facial expressions of emotions among higher animals capable of displaying them were universal, reflecting our shared evolutionary heritage. The ability to convey basic emotions through facial expressions is not unique to the human species as our primate relatives share their emotions through similar facial expressions. What differentiates our repertoire of facial expressions from that of our primate cousins? There are some differences in facial muscle anatomy but more importantly, humans have a much larger neocortex seemingly enabling us to intentionally display a wider range of facial expressions for secondary purposes. A not so nice way of saying the same thing is that we humans are better at faking emotions through facial expressions than our primate relatives.

What makes the expression of fake emotions so interesting is that their existence reveals that our brain indeed has a layered program structure where in a sense a part of our evolutionary older brain controls the involuntary display of genuine emotions while the neocortex, the evolutionary more modern part of our brain, generates fake emotions. One of the best examples of such fake emotion display is the fake smile. Control, involuntary though, for the emotion underlying the expression of a genuine smile resides within the limbic system. The facial expression of such a true smile has two characteristic parts associated with control of the zygomatic and the orbicularis muscles in our face. The zygomatic muscles pull up the corners of our mouths to produce the smile while the

orbicularis muscles produce the twinkly in the eyes and the creases some call crow's feet, associated with a genuine smile. Darwin was already aware of Duchenne's discovery of the true smile, that it required the action of voluntary muscles, the zygomatic muscles, as well as of involuntary muscles, the orbicularis muscles. What we learned much later is that the brain area in the limbic system controlling the true smile can control both the voluntary and the involuntary muscles. In contrast, the brain area associated with controlling the fake smile cannot control the involuntary muscles. Therefore, any time a friendly face smiles at you but you see no twinkly in the eyes or the associated crow's feet you are looking at a social smile but not at someone expressing the emotion underlying a genuine smile.[36] Darwin knew about the difference between a social smile and a genuine smile and quite likely so did you intuitively before you read this paragraph as being able to differentiate between a fake and a genuine smile is a skill we all learn when we grow up. However, now you do so with the added benefit of knowing that the fake smile originates in a different part of the brain than the genuine smile.

The capability to produce a genuine smile is involuntary and innate. We are born with it. The fake smile, however, is something we need to learn. It looks like toddlers learn to differentiate a genuine smile from a fake smile about the time when they begin to interact socially, starting around the age of two. At that age, a toddler's brain becomes capable of differentiating genuine from fake smiles in others and that must be when the neocortex goes to work to learn such behavior for the toddlers own use. Learning how to produce a fake smile requires our neocortex to integrate those brain circuits, which can produce innate emotions, into the programming structure of the neocortex itself. This must happen in such a way that it enables us to control when to produce a fake smile. The neocortex utilizes as well as bypasses the part of the limbic system producing the genuine smile as that is the only way we can produce fake smiles at our wont. Viewed from a different level and put more generally, when we grow up, we learn about the world around us and ourselves by experience. While we have those experiences our neocortex is busy coding them into representations of the world and ourselves, producing our very own unique maps of our world and ourselves for us to draw on for future use. We should not think of such representations as pictorial representations in any way. What is going on in our neocortex as we learn to become humans is much more complex. We know that we have all those hubs in certain parts of our cerebral cortex, which connect to functionally supportive areas in other parts of our brains. However, it is important to realize that the human brain exists as one functional unit where various symbolic brain networks – referred to earlier in a simplistic and incomplete shortlist as our cognitive, emotional, social, somatosensory, visual, auditory or recording brains – all are projections of an integrated complex network onto a specific plane. These projections reduce the information content to a level that enables us to focus on what we are interested in exploring at any given moment.

Clearly, drawing analogies between a computer and our brain is fraught with difficulties and inaccuracies. In every moment a computer is in a specific state at which it invariably arrived through the sequential execution of however complex program instructions. There is no way a computer can go back and recall for example a multiplicity of previously existing programming states and evaluate how their status may have been affected by later programming states at any time. That just does not happen in regular computers, but it does happen to some degree in neuronal network computing environments. Although even there it does happen only in a much simpler, lower level and a significantly less complete way than in our brains. Leaving aside neural network computing, which is still in its infancy, computers in a sense exist exclusively in sequential states, which only have a meaning when perceived as the result of states that were in place previously, the execution of which eventually resulted in the current state of the computer. Not so for the human brain. All of the brain states that we ever experience continue to have a presence in our brains, if we like that or not. That does not mean they do not change over time, as we all know they do. Some brain states we can recall easier than others, we all have that shared experience. If we close our eyes, we can literally take a walk through our past, the past, as it exists in our present. A computer can do nothing like that. Sure, we can program the computer to record every programming instruction and state it ever was in, but those states will never be present states of a past that actually could differ from what those past states looked like in their own present time. Our human brain can generate moment for moment a cognitive simultaneity of all that we ever experienced. This brain capability forms the basis of our individual identity, of who we recognize ourselves to be.

Did you ever ask yourself how you recall the image of someone important in your life, for example your grandparents, who may have passed away quite some time ago. In my case, they are gone now for several decades but I only have to sit back and close my eyes to recall their faces, almost like recalling their likeness as if they were still around and we had only met just a few minutes ago. However, along with this mental image of a loved person many other things associated with knowing that person pop up. We do not have isolated memories; everything in our recorded experiences is connected. Memories seemingly have their own lives in our brains and connect to other memories in ways we do not control. Our brain weaves its own account of reality as we experience it and stores it away for us to recall, either at will or involuntarily. We really do not have a choice not to remember something we have seen, heard or experienced; and we do not choose to remember or not to remember our experiences although occasionally our minds may choose to repress some memories, which are just too unpleasant or painful. Sometimes it only requires an innocuous trigger to recall large parts of our lives. The French novelist Marcel Proust (1871 – 1922) gave us likely the most famous account of such a trigger moment in his novel *À la recherche du temps perdu* (*In Search of Lost Time*), published

in seven volumes between 1913 and 1927. The novel essentially begins and ends with a sensual experience, the taste and smell of a simple madeleine, the delicious little French sponge cake, dipped in tea, resurrecting for Proust memories of a childhood past when his aunt used to give him small madeleines dipped in tea. For Proust, recalling this early childhood experience is like getting hold of the edge of a very large piece of fabric that he cannot yet see but feel. As Proust keeps pulling on that fabric, he reveals, to himself and to the reader, more and more of the world of late nineteenth and early twentieth century aristocratic France, and with it, woven into that fabric of time, Proust's recollection of his childhood and early adult life, Proust's remembrances of things past.[37]

Very few among us will eventually go and write several thousand-page novels recollecting memories of themselves and the world around them in such detail, marked for recall by nothing else but a sensual experience. However, to various extents, all of us can similarly recall large episodes of our lives and our recollections of the world, as we perceived it through time. The concept of time that underlies such recollections is a very different one than what we usually understand by time. While our present seems to be the most important aspect of our lives, we are all creatures of the past. The present is fleeting and only becomes real as experienced past. The future is for us in a sense more real than the present. Futures as we hold them in front of us are always imagined futures of our own past self. We are our own past, as we remember it and not as others do. The parts of our brains closest to our present are the evolutionary oldest ones. They serve our present needs as best they can, even so they always will be behind the present too, but just a little so, as they do not have to consult the cerebral cortex to act in the present. The human cerebral cortex is for the most not a reactive but a reflective organ, in a way we can look at it as our sensory organ for the past, the past that has become us.

The worlds that we create in our brains through our experiences have another remarkable aspect: they can flaunt the laws of nature. This can happen consciously or unconsciously. We often flaunt the laws of nature by imagining things that in nature could not happen at all. In nature, pigs do not fly, but in our minds, they can. It is hard to imagine what human creativity would look like if our brains could only support the perceptions of reality that obey the laws of nature in every aspect. We know that humans are not the only animals that dream but we do not really understand yet why we and other animals dream or what specific functions dreams actually serve. Our bodies need sleep, but our brains seemingly never go to sleep. Increasingly, the picture that is emerging seems to indicate that the unconscious mind is responsible for a much larger part of our brain activity than our conscious mind. It may be more correct to say that our unconscious mind never sleeps and that even in our waking hours the unconscious mind plays a much more important role than our conscious mind. The conscious mind is like the part of an iceberg that we can see above the water line and what we cannot see is the much larger part of the iceberg that lies below the water line. With icebergs, less

than 10% of their volume is above the waterline. We do not know yet how much of our brain activity is conscious and how much remains unconscious but the percentage of what surfaces above the waterline into consciousness may be even less than for an iceberg. Whatever that number will turn out to be, that much seems to be clear, most of our brain activity is unconscious. The unconscious part of our brain activity is much more important than scientists used to assume for a long time. Our unconscious and conscious brain parts connect seamlessly. Our dreams are a good example for that when brain activities hidden in the unconscious can float effortlessly above the waterline where they become part of the processes we experience as our conscious brain activity. We still know very little about the human conscious brain, but we know even much less about our unconscious brain.

There are experiences that are difficult to share in words but that can give us a hint as to how the unconscious part of our brains sometimes works. The most vivid personal such experience I remember, relates to a scientific discovery I made as a young researcher working towards a doctoral degree. As it happens, sometimes scientists work for a long time on a difficult research topic without discovering what they are looking for, not to speak of making a breakthrough. Back then, some decades ago, I faced such a situation of having worked hard for years to measure something and only getting measurement results that neither I nor anyone else could explain. Then, one night I had a dream, completely unrelated to what I was working on. In this dream, I was lying flat on my back and looking towards my stomach area where to my surprise all of a sudden, a number of creatures came out. All I felt when I watched in my dream those lobster-like creatures crawling out of my belly was surprise; it actually was a joyous surprise. Most of the dreams we have we do not remember once we leave Morpheus embrace but this one, I vividly remembered and shared with close friends. So far, it was just an interesting dream. What happened next was that the following night when I spent again long hours working on my experiment it all of a sudden became clear to me what I was looking at. It was like looking at an ancient script nobody could decipher and that all of a sudden magically translated itself in front of my eyes. At that moment, I knew that I understood something that no one had understood before. The sensation I felt at that moment of discovery is hard to describe and something I will never forget. However, at this very moment of discovery I quickly realized that something had already changed earlier the same day when I awoke from the curious dream I had the night before, even though right then I could not pinpoint what exactly it was that had changed. Looking back on this experience a few days after my discovery, it looked to me as if this dream had announced that change and the feeling of surprise in the dream was a precursor to the feeling of discovery the following night.

Now, why do we not all wait until we have a dream and then let things work out from there? Why not just skip the hard work and wait for inspiration to come to us when we

sleep? Well, just looking at my experience it seemingly does not work like that. I do not think inspiration would have come to me in this way without my mind being prepared to receive it through years of working hard on trying to solve a difficult problem. We may assume that we only think with our conscious mind but I do not believe that is quite so. Our unconscious mind is very much involved in achieving that for which we usually give exclusive credit to our conscious mind. When we stop working on something or go to sleep, our conscious mind may go into complete repose but our unconscious mind never stops working. This is how we seem to solve problems in our sleep. However, before we can do that, we must feed our unconscious mind through hard work. Thomas Edison (1847 – 1931), the iconic American inventor and businessman once famously declared: "Genius is 1% inspiration and 99% perspiration". There is a true ring to that, even more so when we strike out the "genius" part and replace it with "success". It looks like hard work can prime everyone for success though it does not guarantee it. If we put in the hard work, the 99% perspiration, we can hope that inspiration may come to us; and sometimes that can happen in dreams, however unrelated they may seem to the problems at hand we are trying to solve.

How does a human mind come to be itself? So far, we only have been discussing what we call the adult mind, but this is not how our minds start out. Clearly, our minds evolve and when we are born, we have a brain but our cognitive functions and our understanding of our own self are quite different and are nothing like the adult self we eventually will come to perceive and accept as us. The above paragraphs should have made it clear how important the human body is for forming the human mind. Just to be clear from the outset, this importance of the body for the mind has nothing to do with having a physically healthy or not so healthy body. It only has to do with having a body, as it is through our bodies that we experience the world; only then can our brains map these experiences into what accumulates eventually to how we perceive ourselves. Our brain must have a biological reality to draw on so that it can evolve. Our body provides that reality. Some seem to believe brains can be preserved and kept functional without a body. That however sounds like a questionable if not preposterous assumption. The "you" that you consider yourself to be, is in its essence the physical reality of your body as your brain creates it in your mind throughout your life, your youthful self being as much a part of it as your aging self.

Maybe the science of the far-out future will produce a brain interface that allows the brain to continue to exist with an artificial body or the body of someone else. However, that then will be a different "you" and not the "you" anymore that your brain has come to identify with itself throughout your lifetime in your original body. Consider another scenario, admittedly a horrible one, but remember we are engaging here in a Gedanken-experiment. Imagine for a moment a human being growing up completely without the capability to process sensory information from within the body or external to her or his

body. What kind of world model do you think this unfortunate human could create in her or his mind? What perception of self would evolve in this person's brain? Those are only rhetorical questions as nobody can really answer them anyway and we hope no one will ever be in such a horrible situation. However, what we can say with some certainty is that our perceptions of self without being able to experience our bodies and the world around us would be very different. Sometimes it may seem as if we frame the mind-body problem as a contradiction, as if mind and body somehow were in need of reconciliation. Nothing could be further from the truth. Humans without a body would have no mind and without a mind, a human being could not be aware of its body.

Any model of the human mind will need to understand how the human mind evolves in the first place. Maybe one day we will understand this better, today we do not. At this point, it should be clear that yours truly does not believe that we have developed anything yet that comes in any way close to resembling a realistic model of the human mind. Understanding just how our mind works and boiling it down into a realistic model is way beyond our reach; at least for now. The complexity of our human mind and the simple fact that so much of how it works remains hidden in our unconscious mind makes it quite unlikely that there will be any viable human mind model in the foreseeable future. What we have today is an increasing understanding of our brain in terms of functional mapping and in terms of how distributed and interconnected our brain functions are. Sadly, quite a substantial amount of what we learn comes from studying various dysfunctions of our mind. However, that in itself already teaches us something that we did not fully appreciate before.

We are much less in control of our own mind than we would like to believe. Even slight injuries to certain brain functions can lead to severe disabilities. Our legal systems today assume that we all are equally well equipped with normal mental functions so that our societies can reasonably expect us to behave in certain ways; like obeying the laws those societies devise for themselves. However, we do not all evolve into adults that are equally equipped in that respect. Damages to the prefrontal cortex, regardless of whether they are the result of physical impairments or of growing up in destitute circumstances constraining a normal mental development, can for example drastically curtail one's ability to make sound judgments. Pleading insanity in courts of law has a long history. But that is so for a good reason. We cannot hold those among us who are clearly not of sound mind responsible for their actions or must acknowledge that they can have only limited responsibility for actions for which we hold a person of sound mind fully responsible. Human societies have understood this for a long time. However, what is becoming clearer now is that things are not as black and white as we would like them to be where on the one side, we have mentally dysfunctional individuals and on the other side, those we consider sane. Rather, there is a gradation, a continuum of mental capabilities and disabilities bridging the gap between insane and sane.

Towards a Biology of Self

Who we are is who we remember to be. Without being able to form memories and recall them we would forever live in the present, never be able to learn anything as we could not memorize it, and we certainly would not know who we are. Without being able to recall our identity when we look for example into a mirror, we could never know who the person really is that we are looking at. In this sense, our quest to understand the biology underlying our capability to learn, how our brain can form memories and how we are able to recall them, is truly an effort to discover critical parts of the biology enabling the development of what we call the human self.

Neuroscientists as well as psychologists differentiate between two kinds of memories, implicit memory and explicit memory. Implicit memory includes the repertoire of acquired skills, learned tasks or learned responses that we execute unconsciously. Different from that, our explicit memories are those which we consciously recall when we remember events, how and where they happened, or when we for example associate the identity of a person with a familiar face. Explicit memories carry semantic contexts, or put differently, they are the framework our brain uses to build representations of the world around us as we move in it, moment by moment. The ability of our conscious brain to recall a seamless stream of explicit memories, associating and connecting it with the quasi-continuous flow of sensory information reaching our brain from the outside world and from within its body, is fundamental to the unity of consciousness we experience. Before the advent of modern neuroscience in the second half of the last century, psychologists could only speculate about the process underlying memory formation in the human brain. A number of those top-down speculations actually turned out to be rather prescient, which became evident once neuroscience had matured to a level where it could begin to ask meaningful questions about the human mind. First, however, scientists had to learn everything about neuronal biology, how neurons really function, connect and communicate. How can we have memories in the first place and how can some of those memories be with us for a lifetime while others remain in our minds for a much shorter time, sometimes only for a few fleeting moments?

Clearly, when we form memories, something must change in our brain. In the broadest sense, we can look at everything we experience in our lives as a never-ending learning process. Quite a bit of that corresponds of course to what we traditionally understand by learning, like learning a physical skill or learning a language. However, much more of it we likely would not immediately recognize as a learning behavior even though it is. For our brains, all of our life experiences are in essence learning exercises. Some of them strengthen neuronal connections in already existing network structures, others weaken them and then others again trigger the formation of new neuronal connections and new neuronal networks altogether. Essentially, our life experiences very much shape

the kind of brain we have. We all have unique brains whose neuronal network configurations reflect our unique life experiences. That is even true for identical twins who at birth are genetically indistinguishable. However, as they grow up accumulating very different life experiences the neuronal networks in their brains will eventually become quite different. The fingerprints of those identical twins will remain identical throughout their lifetimes, unless some accident damages a fingertip, but their neuronal network maps will reflect their unique perceptions of the world and of themselves in it, just like they do for everybody else.

Over the last few decades, researchers have finally gained some insight into what really happens on the cellular level when our brain learns and forms new memories. This accomplishment did not come about by scientists all of a sudden being able to understand how certain things work in the human brain, which is the most complex brain evolution has produced so far on this planet. Rather, researchers began to learn about memory formation by studying it in much more primitive lifeforms that have simpler brains than mammals. One of the heroes in this story is a large sea slug of the genus *Aplysia*, which the Austrian-born American neuroscientist Eric Kandel (born 1929) and researchers around him studied for decades to understand how implicit learning functions in such an animal. Many of us have likely heard about or may even be familiar with the work of the Russian psychologist Ivan Petrovich Pavlov (1849 – 1936) in the field of animal learning. As it turns out, not only dogs or other mammals respond to stimulation with behavior changes. Invertebrate organisms such as *Aplysia* are just as receptive to stimulated learning. In behavioral learning experiments researchers typically use three techniques referred to as habituation, sensitization and conditioning. If a repetitive stimulus eliciting a response turns out to be harmless, repeated application of the stimulus will meet with a decreased animal response. In this process of habituation, the animal has implicitly learned that it can safely ignore the stimulus. If the experimenter withdraws the stimulus for some time, the implicit learning decays and the animal response to such a stimulus will recover. Different from habituation, sensitization describes an increase in response strength with repeated exposure to a stimulus. However, sensitization is not simply the opposite of habituation as it involves at least two stimuli where the animal's response to one of them causes a stronger response to the other stimulus. One stimulus sensitizes the animal to respond more strongly to another stimulus than it otherwise would. Finally, there is conditioning, which pairs two stimuli to produce an animal reaction, like in Pavlov's famous experiment where the ringing of a bell repeatedly accompanies the arrival of food until eventually just the ringing of the bell without any food arriving will make the dog salivate. As it happens with habituation, sensitization and conditioning will both likewise fade if the experimenter withdraws the sensitizing stimulus for some time or the dog receives repeatedly no food even though the experimenter keeps ringing the bell.

In the later 1960's Eric Kandel and his colleagues set out to measure selected neuronal responses of the sea slug *Aplysia* to habituation, sensitization and conditioning. Studying learning and memory formation in the simpler neuronal network of this invertebrate revealed how neuronal learning takes place and how short- and long-term implicit memories form in such a model organism. It also rewarded Eric Kandel with the Nobel Prize in Physiology or Medicine in 2000. He shared the prize with two fellow research pioneers who independently made significant contributions to the understanding of signaling pathways in neuronal learning, the Swedish neuropharmacologist Arvid Carlsson (1923 - 2018) and the American neuroscientist Paul Greengard (1925 - 2019).
As it turns out, processes of memory formation have been highly conserved throughout evolution. So even while the specific chemicals involved in implicit memory formation may differ between higher and lower organisms, the basic processes in humans and an invertebrate such as *Aplysia* are quite similar. In both, short-term implicit learning and memory formation takes place through the strengthening of existing synaptic connections between neurons while long-term implicit learning and memory formation requires the creation of new synaptic terminals and dendritic growth in neurons along the respective path in a neuronal network that will hold the implicit memory of a learned skill or task. Being able to understand the process of memory formation at the molecular level also revealed that excitatory as well as inhibitory mechanisms are responsible for shaping neuronal network connections in animal brains. That is not just the case for a single step in the process of neuronal memory formation but for a number of steps. This has two important consequences. First, a neuronal stimulus must pass a certain threshold level ensuring that we do not initiate memory formation on what essentially could be just ambient noise. Second, once a stimulus passes such a threshold, the neuronal response built on a combination of excitatory and inhibitory factors is stronger than it would be by just using excitatory mechanisms only, thereby increasing the chances for implicit learning to succeed quickly.
Scientists have not only learned that there are excitatory and inhibitory processes governing neuronal activity. They also discovered that our brains network architecture itself makes use excitatory as well as inhibitory neurons. Hence, learning processes as well as the learning network architecture of animal brains rely on excitatory as well as inhibitory components. There is a very good reason for why that is so. To appreciate the role inhibitory neurons play one just needs to think about what could happen if there were only excitatory neurons. If there are only excitatory neurons, how could an excitation once triggered by a stimulus slow down or stop altogether, as it surely must? Just imagine what would happen if there were no way to slow down or stop an excitation. Inhibitory neurons are just as necessary as excitatory neurons for the smooth functioning of our brains. Like excitatory neurons, inhibitory neurons can learn, meaning that based on the stimulus they receive they can experience synaptic strengthening, or synaptic and

dendritic growth. However, in the process, the two types of neurons release different neurotransmitters into a synaptic cleft, with excitatory neurons using glutamate and inhibitory neurons using a neurotransmitter called GABA.[38] Just as with excitatory neurons, there is not just one type of inhibitory neuron. The sum total of these different types of inhibitory neurons makes up about twenty to twenty-five percent of all neurons in our brains. Multiple excitatory and inhibitory process steps gating or amplifying a neurons response depending on the kind of neuronal stimulation as well as the presence of excitatory and inhibitory neurons in our brains neuronal network architecture all help to ensure stability and resilience of our brain over a wide range of operating conditions. When we compare our brain to a computer, we are not really doing it justice. A computer, at least the ones available today, can only function within rather limited operating conditions. Its operating software and the software programs running on a computer platform narrowly prescribe what a computer can do. A slight change in a program, a software update not working correctly and all of a sudden you are looking at a blue screen.[39] Who has not experienced that? While computers have much improved since the first models came on the market they still occasionally crash and having to reboot a computer is just part of a computer user's experience. What would the consequence be if our human brain would be similarly susceptible to blue-screen-type crashes? Well, it is likely safe to say that an animal equipped with a brain that had the resilience of a computer would have never made it very far in evolution.

Our brains surely cannot calculate as fast as computers can. However, it is the human brain and not a computer that comes up with the mathematics required to do a calculation in the first place. Humans and not computers discover new solutions to mathematical problems because computers cannot come up with new mathematical solutions by themselves. Computers depend on humans for providing them with instruction sets that encode such new solutions to mathematical problems. Clearly, computers are still worlds away from achieving anything close to the breadth and depth of what the human brain is capable of understanding. Importantly, they are also nearly nowhere as flexible and adaptable as the human brain in adjusting to new situations and environments. Few scientists would call the human brain simple but many less familiar with the wonders of the human brain seemingly see the technological marvels that our newest and fastest computers represent as vastly outstripping the human brain in many respects. That is a mistaken view. Rather, any such technological marvels are extension of our human brains, extensions that serve our brain to accomplish other goals, for example such as processing the enormous amounts of data and executing complex simulations that scientists have devised so our brains can better understand how our world works.

Key to the formation of short-term memories, in invertebrates such as *Aplysia* as well as in humans, is what researchers refer to as synaptic plasticity, which is another way of saying that the strength of a synaptic connection is not a constant but can become stronger or

weaker. The strengthening of a synaptic connection happens when one neuron, let us call it neuron A, consistently activates another neuron, which we may call neuron B. The opposite of a neuronal connection becoming stronger is a neuronal connection becoming weaker and this is what happens if neuron A consistently fails to activate neuron B. In this way, the learning or unlearning of something, like for example a behavior, results in the strengthening or weakening of neuronal network connections. If the stimulus triggering the short-term implicit memory formation subsides, synaptic strength pretty much returns to what it was before the stimulus elicited the learning process.

Building long-term memories goes beyond the strengthening or weakening of existing connections and requires the formation of new synaptic connections, some of which will serve to either increase or decrease the likelihood of a postsynaptic neuron firing an action potential. The former occurs if an excitatory neuron triggers the formation of new synapses with a neighboring neuron; the latter happens if an inhibitory neuron does the same. The neuron on the other side of those synaptic connections, excitatory ones and inhibitory ones, existing ones or newly formed ones, essentially sums up all of those synaptic inputs which either add up to the neuron firing or not firing an action potential itself. If the sum total of excitatory signals coming from existing and new synapses sufficiently outweighs the inhibitory inputs, then the neuron can fire an action potential when triggered in such a way by a neighboring neuron. If that continues for some time the result will be an overall strengthening of synaptic connections thereby entrenching the existing signal transmission path. If however, an inhibitory neuron has increased the number of synaptic connections to that same neuron sufficiently, that neuron may then no longer be able to trigger an action potential even if another neighboring neuron seeks to pass on its signal. More importantly, if inhibitory synapses continue to suppress the firing of an action potential in a neuron, other neighbor neurons will register that, with the result that in a previously transmitting signal path the synaptic strength will weaken. Eventually, a previously strong synaptic connection will decrease to the point that even without an inhibitory neuron preventing the firing of an action potential no signal transmission occurs anymore over the previously active signal path. Broadly speaking, scientists refer to this process of long-term strengthening of a neuronal signal path as long-term potentiation while calling the process of long-term decrease in the synaptic strength of a neuronal path a long-term depression.

That processes along the lines of long-term potentiation and long-term depression, neither of which requires the formation of new neurons for learning and memory formation, may be important, was already recognized by the pioneer of neuronal research, the Spanish neuroscientist Santiago Ramón y Cajal (1852 – 1934). He suggested that the strengthening of existing connections would be important for learning. Later, the Canadian psychologist Donald Hebb (1904 – 1985) added the idea that neurons may grow new connection and thereby strengthen signal transmission. The first experiment shedding some

light on the biochemistry and biophysics underlying a process such as long-term memory formation were those conducted by Eric Kandel and his team in their study of behavior conditioning of specific neuronal cells in *Aplysia*. This kind of research marked a dramatic change in how neuroscientists approached the study of learning and memory and it demonstrated that such processes have a fundamental explanation at the molecular level, which scientists can discover and verify experimentally. Since then, neuroscience has giving us a significantly improved understanding of the cellular biochemical and biophysical processes involved in learning and memory formation, not only in simple invertebrates such as *Aplysia* but also in mammals. Fifty years ago, we had no understanding of learning or memory formation on the cellular level. Today, we have a rich, although still far from complete understanding of many processes in our brain. The overwhelming majority of scientists has held for long that all processes in our brain are of a biological nature only, with the implication that all our mental processes also have a biological basis. Few would deny that the accumulated experimental evidence supporting this contention is by now overwhelming with ever more new data added daily, pointing only in one direction. Everything that happens in our human mind has a biological basis and everything our brain contains and does is biological in nature.

Implicit memories are stored in two brain areas, the basal ganglia and the cerebellum. The basal ganglia consist of a group of subcortical nuclei at the base of the forebrain. They play a role in a wide range of processes including but not limited to emotion, habit formation, movement and learning. They are particularly important for coordinating sequences of motor activities, like when we execute such sequences as we play a musical instrument or participate in sports activities. The other brain area storing implicit memories is the cerebellum, which as we discussed already earlier is of critical importance for our fine motor control. Storing such learned fine motor skills in the cerebellum allows our brains to execute them automatically without us having to think about them anymore, as we initially had to do when we first started to practice those skills, writing being a good example for that.

Explicit memories, the memories we consciously recall when we remember events, are stored in the hippocampus, the neocortex and the amygdala. The two hippocampi, belonging to the limbic system parts residing in the temporal lobes of each hemisphere, are the locations where episodic memories form and remain for later access. However, some memories reside only temporarily in the hippocampus before being relocated into the neocortex where they will be stored in areas reflecting the sensory context within which a memory formed in the first place. When for example a visual memory forms, the hippocampus in some way works with the primary visual neocortex in building the new memory structure. Once it is established and indexed in the hippocampus, areas in the part of the neocortex where the memory so to say was recorded in the first place experience a strengthening of neuronal connections engraving the memory for local access

in the neocortex itself. Visual memories reside with the primary visual cortex whereas auditory memory information resides with the primary auditory cortex and both are accessible to the visual or auditory cortices in ways similar to how those areas process visual or auditory sensory inputs. This memory transfer from the hippocampus to the respective neocortex areas, which process sensory input information related to the memory, happens over a time span of days during sleeping hours. Once this transfer is complete, the neocortex can access such memories without having to go through the hippocampus. The hippocampus itself keeps some kind of index for that transferred memory, placing it in a temporal context with our other memories. How we exactly can recall the temporal and spatial context in which a memory formed we do not really know yet, but researchers believe that the hippocampus plays an important role in that. The third brain region storing explicit memories is comprised of the almond-shaped amygdalae, also parts of the limbic system. Each amygdala nests deep in the temporal lobe of the respective hemisphere. Just as the hippocampus is critical for binding our memories from a temporal and spatial perspective, the amygdala is critical for binding our memories to emotions. It is this binding to emotions, which can make memories very difficult to forget. Emotionally charged memories associated with fear, anger or aggression are potentially critically important for an animal's future survival. Therefore, we better had a way to retain those memories so we can retrieve them when the situation calls for it. It is not just memories associated with fear that can be hard to forget, memories associated with love, joy, shame or grief can also stay with us for a long time, whether we welcome that or not.

Our brain is not a passive organ of mind awaiting its master's directions but is constantly making decisions about what is and is not important without us realizing any of it in the moment. How does that happen? Can that happen without a master or a master agency residing somewhere in our brains? If not, who is that master and where does that agency reside in our brains? Throughout history, there have been numerous attempts to answer this question. One of the most well know examples is of course Rene Descartes contention that the pineal gland was the seat of our soul and the location from which all of our thoughts originate. We know today that the pineal gland has a very different function and so far, the search for a master agency hidden somewhere in our brain has been unsuccessful. That does not mean that scientists have given up on finding a master agency, but they are certainly not looking for quite the same thing anymore. With our modern understanding of brain functions, giving us a much deeper insight into the complexity of the various processes occurring in our brain at any time, the perceived need for a master agency may even be greater than in Descartes time. However, what motivates some modern scientists to continue the search for such an agency is not so much to discover the elusive seat of the human mind. Rather, their desire is to understand how the unity of consciousness comes about that we all experience in our daily lives. What is it that binds all of our sensory and mental processes together in such a way that we

do not experience each of them separately but in a simultaneity, as a whole? Take as an example a situation where you run into someone on the street, someone with whom you will have a brief conversation. As this happens, your brain is busily working on a number of tasks, which include identifying the person as someone you know or have not met before; perceiving the person as old or young, female or male, tall or small; or reading the person's facial expression as smiling, surprised, indifferent, maybe even upset. During all of that, you listen, talk and maybe even exchange non-verbal messages such as through gestures. Of course, as you have this conversation, you also stay acutely aware of the environment both of you find yourselves in. One certainly could put together a longer and much more detailed list of what probably goes on in your brain while you have this brief conversation. The important thing is however that once you walk away and recall all of those events at a later point in time, you will remember them as part of a situation that binds all of the elements in that above list together. Maybe nothing illustrates better the desire to find such an agency whose function is to create this unity of consciousness than a look at who has been searching for it recently. That included for example someone like the British biologist and neuroscientist Francis Crick (1916 - 2004), one of the three scientists rewarded with the 1962 Nobel Prize in Physiology or Medicine for discovering the molecular structure of DNA, who believed the claustrum might be a good candidate for such an agency. The claustrum, a thin sheet-like layer of neurons and glial cells, is a part of the insula, the brain structure we already encountered earlier as hosting the brains taste processing areas. The highly connected brain tissue of the insula, located within the lateral sulci, the fissures separating the temporal lobe from the parietal and frontal lobes within each brain hemisphere, is also involved in our experiences of emotion and cognition. Because of its structure, functions and location, the claustrum receives inputs from and projects back to almost all cortical regions while at the same time it also connects to sub-cortical regions. Here is not the place to discuss why Francis Crick thought the claustrum to possibly be such a master agency in our brain and we still know too little about the claustrum in the first place. Suffice to say, the claustrum somehow plays a role in connecting parts of our brain on the conscious side with other parts on the unconscious side. Since we are seemingly not aware of how the unity of consciousness comes about that we experience, such an agency bridging conscious and unconscious parts of our brain would make sense.

It may well be that the search for a master agency engaged in producing the unity of consciousness each one of us experiences will remain unsuccessful because our brain's unity of consciousness might be a network property and not reside with any single brain agency at all. In many ways, this would make more sense than having a single master agency in control. Network properties have for example the advantages of resilience in the case of network damage. If an accident knocks out the single master agency, then that is it. With the master agency out of business, our brain will have lost its unity of

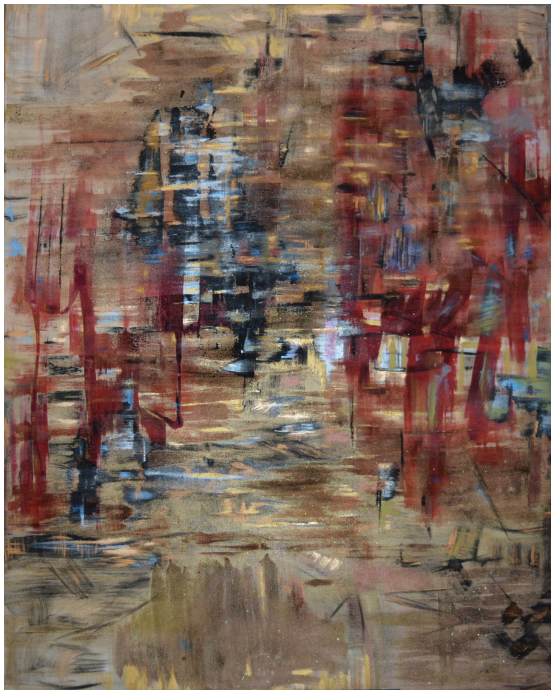
consciousness and all that comes along with it. If however the unity of consciousness is a network property, there is a good chance that this property will not be lost, or at least not be lost completely, if a small part of the network stops functioning because of a localized brain malfunction or damage.

However, while there are real differences among systems that rely on single units and systems that build such command-and-control structures into the network itself, with respect to our understanding of the human brain this right now only amounts to a semantic difference. Not knowing where a master agency resides is equally informing as not knowing how to pinpoint a network property, i.e., not to understand how it comes about. Let us take the internet as an example to illustrate this point. The internet is the network *par exemple*. We can list a number of network features that contribute to the resilience of the internet and for every one such feature, we can point towards a hardware or software implementation that generates that specific feature. Considering networks from a more general perspective, the kind of network properties that we know and understand are all like the ones we find with the internet and always resemble identifiable parts added to the network over time. To note, those parts are added by humans and not somehow generated magically by the network itself. This is important because that is very different from what many like to call emergent network properties, which are properties that arise out of a network, presumable because of its intrinsic complexity crossing a certain threshold, thereby triggering the emergence of new network properties. One has to be very careful here as with a slight of hand we all of a sudden are talking about something very different. How is a network having emergent properties different from an evolving organism? It really could not be different because the only examples of what some like to refer to as emergent properties are nothing but the results of evolutionary processes. The only examples of emergent networks anyone can point to today are biological in kind and that makes them evolutionary networks and not emergent networks. In biology, those seemingly emergent properties, which really are evolutionary properties, abound but no one has ever demonstrated computer networks exhibiting anything like that. Emergent properties are a favorite argument in many discussions but as we will see in the following chapter, emergent properties are for now just a phantasy. Hence, we cannot really know what non-biological entities with emergent properties would look like and how they would behave.

When we talk about our brains neuronal network, we must be mindful that it is a very different kind of network when compared to human generated networks. Some believe the unity of consciousness could well be such an emergent property of neuronal networks in the human brain. However, that is a misnomer as the unity of consciousness is something that must have evolved over millions of years. Maybe if this evolutionary process had happened over a much shorter time span one could justify calling the unity of consciousness an emergent property of biological neuronal networks but it would still

amount to the same misconception. We really do not know what we have to look for to discover how our brain has encoded what we experience as unity of consciousness. Looking toward computer networks to gain a better understanding is likely of little help. As pointed out above, there exist no computer networks that exhibit anything akin to emergent properties at all, which we could study to learn more about how such supposed emergent properties come about in the first place. Emergent properties are for many an explanatory sleight of hand but we should resist the temptation to use it, as it is one of those constructs that we use to talk about things we know little about which easily morphs from being such a construct to becoming the explanation itself. In short, using the concept of emergent properties can give us the illusion of understanding something about which we are not any smarter than we were before employing it.

Philosophy of Mind



Janus Blue & Red

Looking at the title of this chapter, some may decide to skip it or just peruse through it. To the ears of many of us, this chapter title could indicate that what is to come will deal with things far too much removed from our daily realities. However, nothing could be further from the truth. We are all familiar with philosophy of mind to the extent that we all quite likely engage in it, even though for the most, we do so unwittingly. The relationship between our mind and the physical world around us is something that has been, no pun intended, on people's minds for a long time. Every time, each one of us, ever so casually, ponders on her or his own mind and its relationship to their very own human body or to the world around us, we become in a sense philosophers of mind. Of course, very few of us engage in such thoughts in a professional manner such as trained philosophers do, but we all are amateur philosophers. The difference is that as amateur philosophers, we are likely not as precise and as consequent in our logical thinking and use of language as professional philosophers are. However, with regard to the mind-body problem that does not prevent us to consider and understand the arguments professional philosophers engage in as long as we are willing to add a little of their vocabulary to ours. Nevertheless, an all-encompassing and challenging subject matter as the philosophy of mind, which great minds have thought deeply about since philosophers first considered its study, is not something that one can just summarize in a few pages.

For philosophers to understand something, they must not only logically analyze and dissect what it is they are seeking to understand, equally important, they need to understand how they know about such things in the first place. Understanding what something is and what it is not, the study of the nature of being, of all that is or exists, is what philosophers refer to as ontology, the philosophical study of being. The study of how we know about something, the part of philosophy that concerns itself with the theory

of knowledge, is what philosophers call epistemology.[1] Philosophers analyze problems always from an ontological as well as an epistemological perspective. What may look like a correct explanation from one perspective may be problematic from the other. This is no different for philosophies of mind, which philosophers must scrutinize for their validity and consequences from both perspectives, ontological and epistemological.

There is one more word that we must add to our philosophical vocabulary and that is metaphysics. The origin of the word goes back to a summary of Aristotle's works compiled in antiquity several generations after the philosopher's death. Aristotle's writings in this compilation following his books known together as his *Physics* were labeled by the unknown editor or editors as the books *"after the physics"*, or *"meta-physics"*. Ever since then, the writings contained in these books constitute what we know as Aristotle's *Metaphysics*. The subject matter of this part of Aristotle's work concerns itself with questions regarding *"being as such"*, *"first causes of things"*, or *"things that do not change"*. With that, ontology is an essential part of metaphysics, as it comprises the understanding of that which exists without subjects like us perceiving it. The latter includes all kinds of objects and properties and prominently such fundamental concepts as space and time. Importantly, metaphysics concerns itself not with what we know by experience, or "a posteriori", but what we know before experience, or "a priori".[2] Abstract concepts of objective reality such as space or time are part of that, and so are abstract ideas such as causality and effect, or psychological concepts such as identity. Modern metaphysics not only concerns itself with the world as it is but sometimes also with how the world could have been. Because of that, some may take the *"meta"* in metaphysics as indicating that this branch of philosophy may also be somehow non-physical, meaningless as not pertaining to reality, unverifiable or just science fiction-like. However, that is a misunderstanding, as any proper philosophical reasoning about worlds that could be or worlds that could have been follows the laws of logic and nature.

To start out, we will begin with a look back in time to when the concept of mind-body duality first emerged, which later in the seventeenth century would become with Rene Descartes the starting point for modern philosophy of mind. As with many things, the mind-body duality in Western culture is a bequest from the ancient Greeks. The way the Bronze Age Greeks perceived the mind-body relationship, as we can find it reflected in the epics of Homer, was quite different from how later Greek philosophers of the Classical Period of Greek culture saw it. In turn, philosophers of the Hellenistic Age would interpret the mind-body relationship again in different ways. The human soul of the Homeric Age did not partake in the close relationship of body and mind. Concepts of body and mind postulating them as separate entities would have been alien to Homer's heroes. Body and mind were inseparable in life; they were the person, which upon death became the corpse. The air-like soul, however, would leave the body for a shadowy existence in Hades. For what came after death, little else mattered but how posterity

would remember their heroic deeds in poetry and song. Perceiving body and soul, *soma* and *psyche* in Ancient Greek, as separate with the mind being a part of the soul dates back to Socrates (c 470 – 399 BCE) time but quite likely originated earlier.

The meaning of *psyche* in the fifth century BCE, that is when Plato wrote his works, while commonly translated as soul, differs from the understanding of soul we have today. The way philosophers of antiquity understood *psyche* was also quite different from the soul concepts in the traditions of Near East religions or modern philosophies of mind. For most philosophers of antiquity, *psyche* essentially was what differentiated living things from non-living things. After antiquity and up until the scientific revolution beginning in the seventeenth century, the concept of soul in Western cultures was examined and viewed through the lenses of the three great monotheistic religions originating in the Near East: Judaism, Christianity and Islam. Christianity and Islam were both receptive to Greek philosophical concepts and their views of the soul-body relationship were influenced to a good extent by them; however, they were shaped even more so by Judaism with its more ancient roots and Judaism itself was much less inclined to look favorably on Greek philosophical thought. Then, in the seventeenth century, the philosophical concept of soul became associated with Rene Descartes substance dualism and other philosophies following in that vein. Philosophies of mind that unlike Descartes philosophy denied the existence of anything else but physical states had of course no need for a soul at all.

In what follows, the reader will find that frequently the word soul is used instead of *psyche*. However, whenever that occurs in the context of discussing philosophers of antiquity, one must be careful. Before anything else, the soul in antiquity was first a philosophical concept even though most of us may have received their introduction to the concept of a soul from a religious perspective. However, even in antiquity, what remained of one's life after death was never only a philosophical question but certainly also a religious one. To understand how philosophers of antiquity approached such questions from their philosophical and often similarly from their religious perspectives it can sometimes be helpful to take a step back and consciously remind ourselves just how much our current religious beliefs may dominate our own concepts of soul.

If we can do that, this will give us as a better appreciation of how philosophical and religious views of the soul differed in the past from our present conceptions of it; or of that which we now understand by *psyche*. It will also help us with understanding when referring to the soul as *psyche* and vice versa may be unproblematic and where we have to be more careful. Maybe ingrained preconceptions about the soul may not present a challenge at all. In our secular modern times, the soul concepts that religions endorse today are for many just as unfamiliar as the soul concepts of philosophers of antiquity. In our modern discourse, we are more likely to refer to the mind-body problem than to the soul-body problem, even though we still keep asking the same questions that philosophers of antiquity and religious scholars were asking long before us.

For the most, this chapter will look at philosophy in general and philosophy of mind in particular from a Western perspective. There are a number of reasons for that. First, the legacy which antiquity has bequeathed on the Western World still pretty much dominates the Western philosophical outlook. Second, cultures outside the larger Mediterranean World, which evolved their own distinct philosophical perspectives we can still trace today, followed quite different paths and in many ways, their philosophical traditions remained much more closely linked to religious beliefs than in the West. Third, the reception of philosophical traditions originating outside of the larger Mediterranean World by Western philosophers has been limited. Finally, the material considered in this section reflects of course the authors very own limitations. Understanding one's own cultural tradition is often already a challenge for many and for most of us understanding the rich traditions of cultures, in many ways foreign to us, is an even bigger challenge. The philosophical traditions of the Oriental World are just as rich and root as deep in time than those of the Occidental World.[3] However, their reception in Western cultures has always been limited and there are few places in Western societies where for example high school students will learn anything about them. That we do not provide our children a broader understanding of the richness of other cultures than our own is a great loss. Today one already has to be grateful if a school offers its students a basic familiarity with the richness of their own culture.

With respect to the evolution of philosophy of mind in the Orient, something else is markedly different form how it evolved in the Occident. For the most, in the Oriental World the focus on what mattered in the human mind-body relationship was not so much one of philosophical understanding but rather one of meditation and spiritual release. Different from many philosophical and religious traditions in the Occidental World, philosophy and religion in the Oriental World had always a much more practical bend as their teachings guided the search of the individual in the discovery of the self or defined familial and societal relationships. The last section in this chapter will briefly look into some of the philosophical and religious traditions that evolved in India and China and continue to inform philosophy of mind in much of the East.

There are of course parts of our world from which we have few or no written records at all that could tell us about the philosophical or religious outlooks of their ancient peoples. The little that has survived into modern times from such cultures paints a quite different picture of how humans could see themselves as part of the natural world and how they believed their bodies and spirits connected to it. We link our understanding of the mind-body relationship to the evolution of philosophy and religion. Anthropologists tell us that long before these two human creations could assist any such efforts, the human perception of the mind-body relationship played an integral part in how our distant ancestors related to nature itself in ways that made them much more a part of nature than we see ourselves today.

Philosophy of Mind in Antiquity

Plato perceived the human soul divided into three parts and this division reflected the partition he envisioned for his ideal state. The three parts of the human soul according to Plato are reason, spirit and appetite. Reason, associated with truth and knowledge, is the calculative and rational part of the soul. The spiritual or energizing part of the soul Plato identified with honor, as reflected in recognition and esteem by others as well as self-assertion and ambition. This spirit becomes the source of emotional responses such as anger when its desires are frustrated. The third part of the soul, the appetizing part, satisfies the souls basic needs for drink and food. Overindulgence in the latter by the appetizing part of the soul puts it into conflict with the spiritual part, at which point reason must engage – at least in the ideal soul – to help the spiritual part to put the appetizing part into its place. As alluded above, this kind of tripartite soul model mirrors Plato's conception of the ideal state: the governing guardians (reason); the military (spirit); and the workers who supply nourishment for everyone else (appetizing).[4] Reason, very much stands for intellect and wisdom and therefore it is no surprise that Plato thought an ideal state to be governed by philosophers who are to be excellently trained mathematicians to start with.

Setting the body apart from the soul and dividing the soul into three parts was not the end of it for Plato; and it was not the beginning either. Before evolving his concept of a tripartite soul, Plato had entertained a simpler dualist view with a non-differentiated soul closely entwined with its body forming a body-soul person entity. This relationship between body and soul, articulated in his *Gorgias*, was unequal, the soul being the master and the body being its slave. While Plato conceived of more differentiated views of the soul in his *Phaedo* and in his *Republic*, the relationship between soul and body remained a subservient one. The nature of the soul-body relationship continued to reflect a master-slave structure in Plato's *Republic* but a more nuanced one. Just as reason must rule the soul, so the soul must rule the body to be the best it can be to serve the needs of the soul. Plato not only split the soul from the body but he also gave reason, the ruling part of the soul we would refer to as intellect, a divine quality. The soul-body nature Plato conceived in his *Republic* is likely his most recognized one, but he did not stop there. In his later works, he broadened his concept of soul to a universe of embodied souls. Souls being immortal came in different qualities, some worthier than others, and a life well lived could help reincarnate a soul to a higher level, moving thereby from an embodiment in a lower form of being to an embodiment in a higher form of being

For many it may be a surprise to see a philosopher like Plato associated with reincarnation but it should not be a surprise. An immortal soul begs the question as to what happens to a soul when the body dies. Reincarnation is a no less viable view than presuming a soul to spend eternity in some kind of paradise or hell. However, it is unlikely that

Plato articulated with his view of immortality something akin to reincarnation. Because he envisioned only a fixed number of eternal souls to go around, transmigration of souls would seem to be the more appropriate label for this kind of immortality.[5] The concept of transmigration was by no means a new idea. The six-century Greek philosopher and mathematician Pythagoras had entertained such thoughts generations before Plato. However, Pythagoras did not introduce the concept of transmigration into Greek thinking either as in his lifetime such beliefs were already part of Greek Orphic religions.[6] Beliefs of transmigration were also certainly not unique to Western cultures. By the time Plato adopted the concept of transmigration, Vedic belief, Buddhism and Jainism, which still hold transmigration beliefs today, were already flourishing in Asia.[7] Incidentally, there is speculation among scholars that the emergence of transmigration as a philosophical and religious concept in ancient Greece itself may have been the result of the cultural spheres of ancient India and ancient Greece influencing each other already long before the time of Alexander the Great (356 – 323 BCE). However, we do not know where, when and how that could have happened.

Plato's dualism of soul (*psyche*) and body (*soma*) is a reflection of his metaphysics of form and matter. According to Plato's dualism of eternal forms and material objects, the things we perceive in our world are only replicates of such eternal forms but not the forms themselves. Plato's disciple Aristotle took a very different view as for him every physical object was composed of form and matter. For Aristotle, soul and body were not two separate entities but two aspects of a living being. This philosophical perspective arguing for a unity between matter (*hyle*) and form (*morphe*), hylomorphism, sets Aristotle apart, not only from Platonic dualism but also from later philosophies of mind rejecting dualist views in favor of everything being of either a purely physicalist or a purely mentalist nature, so-called monist philosophies of mind. One could argue that Aristotle came to formulate such a unique position because of his widespread interest in natural philosophy, the study of nature. Aristotle's lifelong interest in biology, his unparalleled effort to understand lifeforms, how they differ physiologically and how those differences could have originated, may well have given him a unique perspective among philosophers of antiquity on what constitutes life. With Plato, the soul-body relationship is a subservient one, the body serves the soul, and it is a conflicting one with the eternal soul seemingly imprisoned in the mortal body. In Aristotle's hylomorphism, body and soul are intertwined and mutually dependent on each other such that one cannot exist without the other. Where for Plato souls migrate eternally from one body to another and thereby can move up in the hierarchy of beings, Aristotle's souls come in different flavors with all lifeforms composed of bodies and souls.

With Aristotle, souls can have three different powers. They can be vegetative, they can be sensory and they can be thinking. The souls of plant life only possess the vegetative power; the souls of animal life other than humans have vegetative and sensory power; and

only human souls possess all three powers: vegetative, sensory and thinking. Aristotle's human soul makes for a very different tripartite soul than Plato's version. With Aristotle, the activities of the soul involve activities of the body and vice versa, a striking modern view. However, there is one notable exception and that is the human ability to think, the human intellect (*nous*). Unlike other psychological abilities, which Aristotle deemed physiologically connected to the body, our human intellect is not mixed with the body and can exist apart from it. With this, Aristotle's philosophy faces of course the question of how the intellect can be part of the soul if it does not connect to the body in any physiological way. In addition, all the same challenges arise that philosophies assuming immortal souls must address in their entirety. What Aristotle gained from postulating an immortal intellect not connected to the body was a rational for his unmoved mover, his divine prime mover, the cosmic equivalent of a divine intellect. Much of Aristotle's view of nature and his thoughts on the mind-body relationship would be enshrined in Western learning through the incorporation and assimilation of his work into the so-called Christian-Aristotelian synthesis. Christian philosophers and theologians would however struggle for centuries to reconcile Aristotle's soul concept with Christian theology, an effort culminating with Thomas Aquinas' (c. 1225 – 1274) interpretation of Aristotle, but eventually abandon it altogether by the seventeenth century.

Fifth century Greece marks a watershed in the Western perception of body and soul. Plato is very much at the heart of this development but he was not the first Greek philosopher to reconsider the mind-body relationship from a new perspective. Earlier, the Greek philosopher Anaxagoras (c. 500 – c. 428 BCE) had his own thoughts about the duality of body and soul, which he articulated in his theory of mind. As far as we know today, Anaxagoras may have been one of the very first philosophers, if not the first, to conceive of the mind-body relationship as being one of duality and separation. In this relationship, the mind or *nous* as Anaxagoras called it in his native Greek, as the faculty of true human knowledge, held primacy over the body. Socrates, Plato's teacher, thought body and mind to be fundamentally different and Plato attributed to Socrates such views in his dialogues featuring Socrates as a principal. It was not only philosophers sharing such views of mind-body duality; politicians and rhetoricians among Plato's contemporaries such as Gorgias (483 – 375 BCE) or Isocrates (436 – 338 BCE) did so as well.

This was a very different perspective from much earlier times in ancient Greek history, the times of Homer when gods and men still joined in battle. These earlier times in Bronze Age Greece, bookmarked by the demise of the Mycenaean Culture around 1,100 BCE and the rise of the first Greek city-states in the ninth century BCE, are familiar to many as the Homeric Age. The Greeks of Homer's epics do not know of a separate mind and body, they do not have a separate *psyche* and *soma* but are indivisible psychosomatic beings. The heroes of the Iliad and the Odyssey, among them of course such figures as

Hector, Achilles or Odysseus himself, did not feel tormented by conflicts between mind and body.[8] For Homer's Greeks it was the embodied mind, which constituted a person's true self, at least until death. This mind (*nous*), what we would refer to as the intellect, was separate from the soul (*psyche*) and located in a person's chest, which was also the seat of emotions. Upon death, the airy soul left the dying body (*soma*) with a person's final breath, leaving behind a person, which remained bound what was now a corpse. The thus evaporated soul continued a shadowy existence in Hades underworld, remembering the person it once was a part of, however, without the former mental powers of that person's mind. Only by drinking blood from sacrificed animals could the shadowy souls in Hades regain some of their intellectual powers. Something which for example Odysseus took advantage of when he traveled to the underworld to question the soul of the seer Teiresias about how to bring his wanderings to a happy ending with a safe return to his island Ithaca and to his beloved wife Penelope. For the heroes of the Homeric Age, mind and body were of one creation, they breathed anger and fear, they lived cunning and reason, and their passions and desires were of spirit and flesh, undivided. Homer's heroes acted as psychosomatic warrior units striving to etch their names into history through deeds that would outlast them long after they ended their lives and they themselves would only continue as mere shadows in Hades underworld; unless they were, like the mythological hero and demigod Heracles, awarded divine status instead. When we look at ancient Greece, these mythical heroes of the Homeric Age were very different people than the Greeks of the Classical Period, the fifth and fourth century BCE, when we find the ancient Greeks embracing a different kind of hero, such as for example the statesmen Pericles (461 – 429 BCE).

The concept of psychosomatic unity as a model for the mind-body relationship is quite different from the philosophies of mind that would evolve in the later Classic Period. In some way, it seems to indicate a kind of oneness of mind and body. In another way, it also expresses a duality of character, as clearly upon death the soul leaves the body-mind, now a corpse, albeit for a shadowy and quite diminished existence in Hades underworld. A soul in Hades was not really much of a person's former self as it even had difficulty to remember the person who it once was. Until death, the union of body and mind represented a person's self and the soul, which survived the death of this body-mind, was nothing more than a weak echo of the person that once was. All that was left of the mind-body after death was a corpse, which continued to be identified with the deceased person. The heroes of the Homeric Age really had nothing else to look forward to after death. Their only path to immortality was through achieving eternal fame, accomplished through glorious deeds the ages would remember. Homer's heroes are lived examples of a seemingly strange kind of psychosomatic unity of mind and body. Superhuman or even divine in their quest to attain eternal glory during their lifetimes they are much larger than life figures, heroic in their best and their worst deeds. However, after death, all

that remained of them were their airy souls, now mere shadows of the former persons continuing a miserable existence in Hades underworld, who could not even derive pleasure from hearing their heroic deeds remembered in poetry or song.

It may seem strange to many of us to find aspects of oneness of mind and body as well as a split soul-body character combined in the heroes of the Homeric Age. However, this certainly was nothing strange at all to the author of the Homeric epics who may have lived in the seventh century BCE.[9] He and his contemporaries must still have perceived this kind of psychosomatic unity as truly reflective of the nature of things, that is, of the order of things in their cosmos. Certainly, they must have still been close enough to such an understanding since otherwise the author would have had to explain himself to his contemporaries more about it than he did. The Homeric Age did not know of any of the philosophies the Classical Period of Greek history would give rise to. Therefore, we should not look at the Homeric Age through the lens of philosophies of mind of that later age. However, we need to ask if we can know anything about what happened to the perception of soul, mind and body between the creation of the Homeric epics and when the philosophers of the Classical Period began to articulate their thoughts several centuries later. Sure enough, there are indeed two developments which we know of that changed the Greek view between the Homeric Age and the Classical Period of Greek history with respect to the relationship between soul, mind and body. On the one hand this change is associated with the rise of Orphic religious beliefs and on the other with the emergence of Pythagorean philosophy and belief.

Orphic beliefs date back at least to the sixth century BCE. With Orphism emerged the promise that all souls could indeed aspire to a better life after death rather than just to become mere shadows in Hades underworld. Orphism was quite different from other Ancient Greek religious beliefs, which many will mostly associate with Hellenistic religion, the late form of Ancient Greek religion with its familiar pantheon of ancient Greek gods, their worship and their mythical origins and stories. For the most, Ancient Greek religion, with its various flavors in different parts of Greece and Greek city-states empathizing their specific patron gods or goddesses, was essentially a kind of state religion. Opposed to that, the Ancient Greek religion of Orphism was more like a religion for the people or as some say for the poor. Not unlike later monotheistic religions, polytheistic Orphism promised a reward after death for living a life of virtue. This is very different from what a hero of the Homeric Age could expect. Regardless of the virtue of such a heroes life, he could expect nothing more than a diminished shadow existence for his soul. Orphism elevated the soul. Not only was its belief that the soul would survive the body. It was through the soul to which the mind now belonged that humans were alive, before death as well as after death.

With Orphism, it mattered how someone lived his or her life in a different way. Depending on how one lived one's life there would await either eternal reward or eternal punishment

in afterlife. Before Orphism emerged, ancient Greek religion already knew punishment and reward in afterlife. However, those were exceptions, reserved on the one hand for committing either the most heinous kinds of deeds or egregiously offending one of the gods and on the other hand for displays of exceptional heroic and almost godlike conduct in life.[10] This was different with Orphism, where the majority of those not leading superhero or supervillain lives now also could expect rewards and punishment in afterlife for how they lived their everyday lives. In Orphism, the objective of life was to purify the soul from all its imperfections so it could eventually live as a god among gods. However, such complete purification of the soul in a single lifetime was an impossibility, to achieve that required several lifetimes. Hence, along with the immortality of the soul, reincarnation and transmigration were central to Orphic beliefs. This Orphic worldview did not only find adherents among the masses of uneducated or poor but also found interest among poets and philosophers.

Pythagoreanism was more of a belief system than a philosophy and the school Pythagoras had founded in Croton resembled more of a religious brotherhood rather than the kind of philosophical school we know from the later Classical Period and the Hellenistic Age. How exactly Orphic beliefs related to Pythagorean belief and philosophy remains unclear. For some time scholars believed Pythagoras might have been introduced to Orphic beliefs in his early years and hence his later religious beliefs and philosophy may have evolved in some way out of Orphic beliefs. Sometimes one can even find Orphic beliefs and Pythagorean philosophy and belief referenced as a single larger belief and philosophical system but that view is mistaken. Orphism certainly predates the emergence of Pythagorean philosophy and belief. Therefore, quite naturally, Pythagoras and his followers, the Pythagoreans, must have considered and compared Orphic beliefs with respect to the views they themselves held on matters of soul and body. Pythagorean philosophy and belief are however distinct from Orphic beliefs even though they share some similar views. One such shared belief was that of transmigration and immortality of the soul, or more specifically that part of the soul which Orphic beliefs and Pythagoreans respectively considered immortal. If we follow the historian and geographer Herodotus (c. 484 - c. 425 BCE), Orphic as well as Pythagorean beliefs in transmigration may have had their origins in similar and older Ancient Egyptian beliefs, as Herodotus thought that all such beliefs among Greeks were adoptions of older Ancient Egyptian beliefs.

Pythagoras and his disciples believed the *psyche* to be the center of emotion and sensation and that all animals possessed such a *psyche*, which they thought was located in every animal heart. According to Pythagoras, in addition to possessing this *psyche* just like all other animals do, humans also possess an intellect, which resides in the head. The transmigration Pythagoras envisioned was quite different from how Plato would later interpret it. With Pythagoras, it was the *psyche* and not the intellect that could move to a new body. Immortality in the sense of transmigration came through the *psyche*, the

emotional and sensitive part of the soul so to say, finding a new body while the intellect was mortal. With Plato, it would just be the other way around, the intellect would survive the death of the body but not what Pythagoras had called the *psyche*. Pythagoreans believed all animals to be similar ensouled and the human psyche after death could find a new home in any newborn animal. How that process of transmigration was supposed to work is not exactly clear but the concept of *pneuma* played an important role in this regard.[11] In essence, *pneuma*, the breath of life, constituted what gave life to the universe and the *pneuma* that a newborn animal inhaled, becoming the so-called inner *pneuma*, transformed into the newborn's *psyche*, the very *psyche*, which the body upon death exhaled so it could merge again with the outer *pneuma*. It remains unclear, how this *psyche* could preserve a person's identity after merging with the outer *pneuma* again upon death and reconstitute itself as the very same person identity by becoming inner *pneuma* once more with the first breath a puppy for example takes. Because animals possessed no intellect, a human *psyche* transmigrating into an animal body might not become aware of its former person identity while those who knew the human person before could recognize its *psyche*, the center of emotion and sensation, by how an animal behaved or reacted.

From the Pythagorean perspective, unlike animals, plants did not possess a *psyche*, and how could they? If a body, plant or animal, can only receive *psyche* through breathing in *pneuma*, plants could not acquire a *psyche* because they were not in any way perceived as breathing. Because all animals possessed a *psyche* of the same kind, Pythagoreans objected to the killing of animals. However, not to the killing of all animals as seemingly the killing of sacrificial animals was permitted. After Pythagoras death, Pythagoreanism split into two sects, one of them coalescing around its belief aspects, the Pythagorean esoterics or mystics, and the other centered on the mathematical and scientific teachings of Pythagoras, the Pythagorean mathematicians. The most important Pythagorean philosopher after Pythagoras himself was Philolaus of Croton (c. 470 – c. 385 BCE), who belonged to the mathematicians and was a contemporary of Socrates and Democritus (c. 460 – c. 370 BCE). Most of what we know about Pythagorean philosophy comes from and through Philolaus. That includes the concept of the soul as harmony. According to him, the Pythagoreans saw all things falling into either one of three classes: limiters, unlimited or harmony. This three-class system provided the foundation of their cosmological views as well as of the Pythagorean view of the human soul and body. By nature, all things that exhibit order are harmonious, as they are harmonies of limiters and unlimited, or in the case of the human body of hot and cold. Seemingly, Philolaus thought that the *psyche* represented such a harmonious arrangement of physical elements located in the heart. With the taking of the first cool breath by the hot body, newborns inhale the *psyche*, which from its location in the heart then establishes the proper balance between cold and hot.

The most prominent Pythagorean after Philolaus was the fourth-century mathematician Archytas of Tarentum, a friend of Plato.[12] Orphic beliefs and Pythagorean teachings of philosophy and belief continued through the Classical Period and into the Hellenistic Age. Both, Pythagoreanism and Orphic beliefs received attention from and provided inspiration to thinkers and poets alike. With the Classical Period of Greek culture, we find among the latter for example the poet Pindar and the philosopher Socrates, and with him also Plato, his student. With Orphism, the body became only a temporary home for the true self, which was the soul. Similarly for Socrates, the self resided in the soul and not the body and what mattered most in life was the care of the soul. This is just how Socrates lived his life. The Orphic and Pythagorean perspective on the soul and how Socrates viewed it are very different from the self of the Homeric Age, the mind-body psychosomatic unity to which the soul only attached as a mere shadow of the self. The importance that the soul with its newly embedded mind gained over the body for Socrates or Plato did not come out of nowhere, both clearly took inspiration from Orphism and Pythagoreanism. As they did so, philosophers of the Classical Period raised elements of those worldviews from a mixture of religious, mythical, and philosophical perspectives to a pure philosophical perspective, much of it later bequeathed directly or indirectly to Christianity. Christianity owes not only a great debt to Judaism but also to the philosophers of the Classical Period and through them indirectly to Orphism and Pythagoreanism.

We do not know exactly when the notion of transmigration first emerged in ancient Greece. Transmigration of human souls was unfamiliar to the Greeks of the Homeric Age and seemingly still so in Hesiod's time, who lived between the middle of the seventh and the middle of the sixth century BCE. However, it certainly must have arrived sometime in the sixth century BCE with Orphism spreading quickly and with Pythagoras making it a doctrine of his philosophical as well as his religious outlook. Transmigration raises a number of questions about the soul-body relationship. First, of course, why does a truly immortal soul have to be reborn in a new body after its original body dies? Why not just be done with leaving the original body and never ever recombine with another body? With transmigration, the essence of what constituted a human being moved from the corporeal body to the soul. This was very unlike the body-mind person of Homer's times to which the soul was only loosely attached. Transmigration elevated the soul from being a mere shadow of a person to the very essence of a person. How can the soul be immortal if it remained throughout a human's life so closely intertwined with the mortal body? Must not at least a part of the soul be mortal? What could a transmigrated soul remember from its earlier life and how did the allocation of souls to new bodies work in the first place? Questions like these would keep philosophers who endorsed an immortal and immaterial soul pre-occupied with explaining the potential nature of the soul and they would come up with a number of different answers, some of which eventually morphed

into what became religious conceptions of the soul, which endure to this day.
The first philosophies of mind endorsing a form of monism also date to the Classical Period of Greece. Opposed to dualism, monism philosophies of mind do not see soul and body or mind and matter as separate. One type of monistic theories grouped under the umbrella of physicalism has come to dominate modern philosophy of mind debates. Early proponents of a type of physicalism include the Greek philosopher Democritus and his teacher and mentor, the fifth century Greek philosopher Leucippus. Democritus and most likely Leucippus before him, believed everything to be composed of atoms. Those atoms they conceived of were not divisible, that is just the meaning of the Greek word atom, and indestructible.[13] Other properties of those atoms included that they came in different shapes and sizes and were constantly in motion, separated by nothing but empty space. What differentiated living things from non-living things according to Democritus was that living things possessed a greater number of round atoms than non-living things. These round and very tiny atoms, the material of the soul, represented only a different flavor of atoms from the ones constituting the body but otherwise were just atoms. Humans can see the part of their body that is made of atoms, but they cannot see or experience the soul in a similar way. The reason for that is that the soul atoms are very tiny and spread throughout the body, maybe denser in some body areas specifically important to matters of the soul but otherwise distributed evenly. For Democritus, it is the soul atoms, which give our bodies the ability to move. All atoms are forever in motion but some like the atoms which come together to form the body are more restricted in their movements, not so the soul atoms. By colliding with body atoms, the many tiny and very fast-moving soul atoms cause the body to move. Democritus stands in the tradition of Greek natural philosophers for whom a pervasive life force as the source of such never stopping atomic movement was a familiar concept. That life force was not associated with any purposeful design, it just existed, and atoms drew their ability to move from this life force. One cannot however say that these were living atoms as they were just moving atoms and in the case of soul atoms, very fast-moving ones. Democritus' view was that the body and soul needed each other as there could be no moving bodies without a soul and a soul could only come together in a body. The soul needed the container of the body to survive. Soul atoms were plentiful in the atmosphere and the soul could replenish any atoms that had escaped the body through breathing in new ones to become part of the soul. Upon death, these soul atoms left the body and with that, the soul ceased to exist. Atomist theories such as the one credited to Democritus and Leucippus, may look like solid materialist views from our modern perspective. However, that is not quite true. While they perceive both, the soul and the body, composed of material atoms and soul atoms make the body move in a mechanistic way by colliding with body atoms, Democritus soul joins with the body in a psychophysical identity that is anything but materialistic. Democritus' philosophy is a monism but not a purely materialistic one.

Because atoms possessed the natural and not further explainable quality of perpetual movement, a property of the cosmos itself, atomists like Democritus had no need for a prime mover or a final cause invoking divine entities such as was the case for Aristotle and Plato.

It is the Greek philosophers of the Classical Period, who first articulated a dualistic philosophy of mind, the duality of human mind and body, and with the atomistic theory, they gave us also a first monistic philosophy of mind, physicalism. The origins of physicalism, that everything is a form of a physical state and immaterial souls do not exist, trace back to the Classical Period of Greece. However, philosophers arguing for dualistic soul-body concepts endorsing an immaterial soul dominated this period, even though their immaterial souls were never completely immaterial either. It was only during the Hellenistic Age that physicalism became much more prominent, arguably to a good extent because of an improving understanding of the human body. The Hellenistic Age is commonly defined as beginning with the death of Alexander the Great in 323 BCE and ending with the battle of Actium in 31 BCE and Rome's pursuant conquest of the last of the Hellenistic kingdoms established after Alexander's death, Ptolemaic Egypt. Medical insights by Hippocrates of Kos and his students clearly influenced how philosophers of the Classical Period such as Socrates, Plato or Aristotle thought about the relationship between soul (*psyche*) and body (*soma*). In a similar way, medical insights from leading physicians of the time informed the thinking of philosophers of the Hellenistic Age. One of those leading physicians was Herophilus, a Greek physician and anatomist from Chalcedon in Asia Minor who spent most of his life in Alexandria, Egypt.[14]

Herophilus, one of the earliest anatomists, discovered not only the human nervous system including sensory nerves and motor nerves, he also was the first to differentiate between the cerebrum and the cerebellum, understanding that the cerebellum receives signals from the sensory nerves. Herophilus believed that the nerves alone served the soul in receiving and transmitting sensory perceptions and motor instructions to all parts of the body while the soul had its seat right in front of the cerebellum in a brain area referred to as the fourth ventricle. Another leading Greek physician was Erasistratus, likely born on one of the Cyclades islands. He was a younger contemporary of Herophilus and along with him, Erasistratus became the principal representative of the medical school of Alexandria. Similar to Herophilus, Erasistratus believed the human nervous system to allow the soul to transmit and receive signals from the body. However, his own meticulous studies of human blood circulation, which he came very close to understanding, led him to give the heart a more important role in the interaction between soul and body.

The philosophical schools associated with the blooming physicalism during the Hellenistic Age are Epicureanism and Stoicism. Epicurus (341 – 270 BCE) founded his school, which became known as *The Garden*, around 307 BCE in Athens, initially in opposition to Platonism but with the rise of Stoicism, the latter also became increasingly a target

for Epicureanism. Stoicism, the philosophical school sometimes called *The Stoa*, was founded in Athens around 300 BCE by Zeno of Citium (333 – 264 BCE).[15] Informed by anatomical insights such as coming from physicians and anatomists at the medical school of Alexandria, both philosophical schools took a much narrower view of the soul than Platonism did. The primary distinction is of course that for Epicureanism and Stoicism the soul was just as corporal as the body, though made up of different kinds of atoms than the body. Epicurus refers to the atomic matter different from the body, the soul, as being made of very fine particles, much finer ones than the body is made of and distributed throughout the body. That of course is reminiscent of what Democritus believed the constitution of the soul to be. In fact, Epicurus school *The Garden* represented an early Hellenistic revival of Democritus atomism. Epicurus explanation of atomic motion however was a mechanistic one and his atoms possessed nothing like Democritus atoms with their intrinsic eternal movement perpetuated in endless collisions. Rather, in Epicurus version, atoms were in a continuous falling movement from which they occasionally swerved to the side to collide with other atoms.

What we find here is that rather than souls being embodied, bodies are perceived as being ensouled. As to the material of the soul, Epicureanism sees it as being similar to *pneuma*, the Greek word for breath or air, with a particular mixture or blend of warmth, but without being *pneuma*; for Stoicism, the soul is *pneuma*. It was of course critical for those schools to describe how right mixtures or blends of a *pneuma*-like substance came about in Epicureanism or how in Stoicism regular *pneuma* breathed in became psychic *pneuma*, the material of the soul. Most importantly, the motivation to give the soul such a substance in the first place, was in essence to ensure that soul and body could interact and thereby provide an explanation for phenomena like sense perceptions, which was for example of utmost importance to Epicureanism.

Similar motivations to ensure soul and body could interact underlie the somewhat more complex stoic concept of the soul. Both, Epicureanism and Stoicism, defined the functions of the soul compared to Platonism as much reduced because in their view the body took care of vital autonomous functions itself. This position was to a good extent motivated by the anatomical insights of Hellenistic physicians such as Herophilus and Erasistratus, providing an improved understanding of autonomous body functions. The Hellenistic views of soul and body as expressed by Epicureanism, Stoicism and by physicians such as Herophilus or Erasistratus shared a set of beliefs. These included that souls are bodies but not each body is a soul; everything that exists must have a physical extension so it can interact; the soul being a body is as mortal as the body; and the soul does not exist independently of the body. The last of those beliefs listed here made the soul as mortal as the body as the soul could not exist before the body existed nor after the body ceased to exist. However, there were also disagreements. Stoics, just like Aristotle, believed that the command center of the psyche resided in the heart and even though less

clear and dogmatic about this, Epicureanism seems to have held a similar view. This is different from how the Ptolemaic physicians Herophilus or Erasistratus saw it for whom the brain was the seat of the human psyche. Nevertheless, philosophers of the Hellenistic age developed a very different concept of what the soul was from what philosophers in the traditions of the Classical Period such as Platonists continued to argue the human soul represented.

The atomistic theory, being a thoroughly materialistic construct, had no need for the divine. Not so the dualism philosophies of mind. The same philosophers who conceived of dualism also endowed the sovereign of the soul, call it reason or intellect, with a divine origin. They did so by necessity as the world in their view was of a divine creation, endowing some such world constructs with more and some with less involved divine personifications. Importantly, only entities connected to the divine would be able to comprehend such divine creations. Plato's concept of the divine was more remote, a divine demiurge who arranged everything that existed such that it obeyed the laws of nature. His pupil Aristotle conceived of a divine prime mover to set and keep the world in motion. Presumably, philosophers in the tradition of Epicurus or Stoicism would have had no need for such divine personifications; but that does not mean that they had no concept of the divine, or would not endorse conventional cults. Stoicism looked at reason as the divine gift which, if accepted and lived, could give humans full control of themselves and their experiences. In stoic philosophy, the divine is pervasive in nature and the wise man can live in harmony with divine reason permeating everything, including the human soul. For Epicurus, the foremost task of philosophy was to enable a human being to live a life free of mental anxiety and bodily harm. The latter if adjoined by measured pleasures would make for a self-sufficient life and could give human life a divine quality in addition to providing benefits such as freedom from excessive wants or limitless desires. Striving for a life of divine quality, however, did not preclude immortal and indestructible gods continuing to be part of Epicureanism, even though this would be difficult to reconcile with a worldview presuming everything that exists to have a material nature.

Looking back at philosophical and religious thought in general and with respect to the mind-body problem in particular, one can only be amazed by the breadth and depth of thought one finds in antiquity. Sometimes it seems that much of our modern philosophical and religious discussions amount to little else than interpretations and reinterpretations of what scholars in antiquity already discussed at length. Maybe there is only that much to say about the nature of the soul and the mind-body relationship. Maybe as of now, humans are only capable to understand certain things in certain ways and that has not changed since antiquity. Alternatively, could it be that much of the intellectual ground humans are capable to toil, the philosophers of antiquity already ploughed? Maybe so, maybe not, we cannot really answer such questions.

Religious Philosophies of Mind

When one looks into how philosophers framed the mind-body and soul-body relationship between antiquity and the beginning of the scientific revolution in the seventeenth century, one will mostly encounter discussions of a religious nature, which nonetheless are frequently philosophical ones. Religious discourse about the soul-body relationship has a very long history. Certainly, it reaches back in time as far as the history of the philosophical discourse we are aware of through the writings that survived; and quite likely, humans engaged in religious discussion on this topic long before they began to philosophize about it. The foundations of the religious beliefs that dominate in our societies today can trace their roots to earlier such beliefs predating the end of antiquity. The three Abrahamic religions, Judaism, Christianity and Islam fall into this category; and so do Hinduism, Buddhism and Taoism. Judaism deems soul and body to be two distinct entities that however cannot exist separated from a person. The Torah does not mention a life after death, but it does not exclude it either. So one could say that an immortal soul is not a matter of belief in Judaism but then Judaism does not compel believers to refrain from desiring an eternal afterlife either. In Judaism, reward and punishment for a life of virtue or for a life of evil are a matter of this world and not the next. For many of us, this argument may not sound very convincing, as seemingly much evil gets away in this life without adequate punishment and much good may never receive its reward in this life. It is therefore not surprising that many in Judaism look towards the afterlife as a reward or punishment for how someone lives their life. Christianity, the younger sister of Judaism, shared similar views of soul and body with Judaism but remained more open to the ideas of the Greek philosophers of the Classical Period. It was this combination of beliefs shared with Judaism and the absorption of philosophical ideas from Greek philosophers which over time produced a distinct Christian perspective of soul and body.

Initially, Christianity, as it focused on the missionary of the gentiles, had to contend mostly with Platonic concepts of soul and body. We can see this for example in the letters the Apostle Paul (c. 5 - c. 64 to 67) wrote to his Corinthian flock. Christianity took a different position from Judaism in assuring its believers of an afterlife for their eternal souls. Importantly, there was the question of the resurrection of the flesh and what bodies Christians could expect to have in afterlife. However, this question goes way beyond the scope that we can consider in the given context of philosophy of mind. More relevant for what is of interest here is that Augustine of Hippo (354 – 430), whom we know today as Saint Augustine, for many the most important bishop of the early Christian Church, developed a unique view of the nature of the soul. In the twilight of antiquity, Augustine was first in envisioning the soul as an entity without extension.[16] This may surprise some, as many believe this view of the soul to be already articulated earlier by the

philosophers of ancient Greece. However, as we just saw in the preceding section, the soul conceptions of antiquity, including those from philosophers such as Plato or Aristotle, all had at least some material part and some material extension. It was actually the very fact that philosophers before him could not agree on the physical nature of the human soul, which led Augustine to the conclusion that the human soul cannot have a physical nature in the first place. His famous argument being that the human mind knows itself with certainty and the very fact that philosophers could not agree on the kind of corporeal nature of the human psyche was evidence in itself that the human mind and with that the human soul could not possess a corporeal nature to begin with. It is most remarkable that until Augustine of Hippo, nobody had ever conceived of the soul as something completely non-physical. Traditionally, we associate the first mention of the soul as an entity without extension with Rene Descartes, but Augustine already had a similar view some twelve-hundred years earlier. Augustine brought a unique new approach to the understanding of the mind-body relationship. In the manner in which he addressed the question of how we can know about the soul, what we can know about it, and how we can recognize other souls, which is known as the problem of other minds. Problematic from a Christian perspective was of course that Augustine's clearly dualist position of a corporeal body and a non-corporeal soul that could exist without the body seemingly made the Christian believe in a resurrection of the flesh unnecessary.

Neo-Platonism schools of thought clearly influenced Augustine's theology and philosophy, just like that of other early Christian thinkers. Founded by Plotinus (c. $204/5-270$), Neo-Platonism continued to shape the views of scholars in various religious traditions well into the Middle Ages. Plotinus took great interest in understanding the philosophical thinking of the religions of his time including not only Western or Near East religions but also Persian and Indian philosophical thinking and religious beliefs. His main objective was to reconcile the three metaphysical principles underlying the One or the Good, the intellect and the soul. According to Plotinus, the One is prior to all existence, and it is both self-caused and the cause of being for everything else in the universe. While that sounds like a supreme divine being, it is not. On the contrary, the One refers to that which is absolutely simple. The One is not a being but more like Plato's idea of the Good. For such a One to be the cause of everything else in the universe and at the same time being its own cause it has to be everything; indeed, it virtually is anything that Plato's forms are. The principle of the One is the ontological principle of being. From it derives the intellect, the principle of understanding and intelligibility of whatever there is. The relationship between the principles of the One and the intellect somewhat correspond to the relationship between Plato's forms and the rational part of Plato's souls. For Plotinus, the intellect represents the highest life and the soul, his third principle, is not the principle of life. The life of the intellect expresses the highest form of desire, which is contemplation of the One through understanding the forms internal to it. Only humans

are capable of this desire, echoing Plato's view of the relationship between forms and the rational part of the Platonic soul. As the intellect articulates the principle of desire for contemplating the One, the soul is the principle of desire for everything else, essentially for the desire of all of life but the intellect itself, since all of life possesses a soul.

Plato's philosophy and thought concepts of Neo-Platonism as well as Aristotle's writings on the soul-body relationship were influential in Judaism, in Christianity, as well as in Islam. Neo-Platonism was a major philosophical influence on Christian thought with regard to the soul-body problem in its early period and has never ceased to be an important element within it. However, Aristotle's writings have shaped Christian teachings and concepts of soul-body relationships as well and arguable much more so. The latter was much helped by Anicius Manlius Severinus Boethius (c. 477 – 524), commonly referred to as Boethius, through his sixth-century translations of several works of Aristotle into Latin. For Western Europe, cut off for most of the Dark Ages from the original classical sources, the transmission of ancient Greek philosophical thought through Arab translations as well as its reception and interpretation by Muslim scholars had a significant impact on the trajectory of Western European philosophy throughout the Middle Ages. The three most influential Islamic philosophers in this context were al-Fārābī (c. 872 – c. 951), Ibn Sīnā (c. 980 – 1037) and Ibn Rushd (1126 – 1198). To European scholars back then and to many still today, they were and are more familiar by their Latinized names as Alpharabius, Avicenna and Averroës. All three of them looked towards Plato as well as towards Aristotle for answers to questions such as how multiplicity came from unity, how corporality emanated from an incorporeal God, and how to explain the hierarchy of beings. Alpharabius was a prolific translator of Greek authors, famous for his commentaries and treatises on Aristotle. His own writings were mostly in the neo-Aristotelian tradition, but he was receptive to Platonic and neo-Platonic ideas as well. Avicenna, positing an essential separation of body and soul, followed Plato and neo-Platonic concepts rather than Aristotle. Averroës however leaned the other way, becoming the great commentator of Aristotle's *De anima* (*On the Soul*).

As we saw above, Aristotle's concept of the soul-body relationship argued for a close connection between soul and body, via the body providing the physiological means to allow for sensual perception by the soul. For Aristotle, soul and body are not two separate entities but two aspects of a living being and with that, the soul is as mortal as the body. However, Aristotle also introduced the concept of a separate immaterial and immortal intellect, somehow part of or connected to the soul without connecting to the body in any way. With that, Aristotle's concept of soul and body was open to a number of different interpretations. The position that Averroës took, also referred to as Averroist monopsychism, was that soul and body were indeed inseparable and that the human intellect is immaterial and immortal but that it is not individually possessed. Rather, the human intellect was a so-called separate substance that is one for all human beings.

Consequently, there could be no such thing as a personal immortality for human beings. In essence, Averroës argued a distinct substance dualism where on the one side we have human bodies with mortal organic souls and on the other side we have a single eternal human mind.

Only in the twelfth and thirteenth century would other parts of Aristotle's work than his contributions to logic gain a broader reception among scholars in Western Europe. That was not due to a lack of interest in Aristotle's works among Western scholars. Most of Aristotle's works, except some of the six-century Latin translations of his works on logic by Boethius, was lost to Western scholars throughout the Dark Ages. The larger body of Aristotle's work became only available to Western scholars again through translations of Arab manuscripts, which had preserved much of Aristotle's work throughout the centuries in the Muslim World. To say that Aristotle's writings had a profound impact on medieval European scholars is almost an understatement. Among the generations influenced by Aristotle's thoughts was the greatest of medieval Christian theologians and philosophers, Thomas Aquinas. In the early thirteenth century, Averroës interpretation of Aristotle in the form of Averroist monopsychism made it into the curriculum of the University of Paris. There it found support among philosophers such as Boetius of Sweden and Siger of Brabant (c. 1240 – 1284).[17] However, it did not take long for the opponents of Averroist monopsychism to take a firm stand. The Catholic Church was quick in condemning it but Averroës interpretations of Aristotle remained influential well into the fourteenth century and flourished in parts of Northern Italy well into the late fifteenth century.[18]

It was Thomas Aquinas who articulated the mainstream opposing view, the so-called Thomist interpretation of Aristotle. Aquinas adopted Aristotle's view on sensual perception, that our knowledge about the world is not innate but acquired through the senses and through logical inference from self-evident truths, or what Aristotle called first principles. To most of us, gaining knowledge through understanding first principles is a well-known concept as it has become the basis of modern sciences. For Aquinas, however, self-evident truth or first principles also included divinely revealed propositions. From our modern vantage point, we could say that Aquinas considered religious truth as accessible to systematic knowledge as scientific truth. Aquinas view of the soul-body relationship echoed Aristotle's view. For him, the soul is an immaterial substance, independent of the body, but the soul uses the body to gain access to knowledge via sense perception. While acknowledging with Aristotle that soul and body are not two separate things, Aquinas asserted that the soul unlike the body is incorruptible and therefore after the death of the body it can continue to exist apart from the body. Aquinas arguments as to why the human soul and not animal souls should be incorruptible rests on the difference between the organic parts of the soul and the rational part of the soul. As we saw earlier, Aristotle had conceived of a human soul as having three different powers: vegetative, sensory and rational. Plant souls had only vegetative power, animal souls other

than humans had vegetative and sensitive powers, and only human souls had vegetative, sensory and rational powers. Aquinas argued that the human soul has actually two parts, an organic soul with vegetative and sensitive powers and an immaterial soul with rational power. The activity of the rational soul, intellective cognition, does not depend on the body. Different from the organic part of the soul, its rational part is therefore separable from the body, indestructible and immortal. According to Aquinas, who agreed on that with Aristotle, we inherit the organic part of our soul from our parents but not so the immaterial rational part of our soul.

Thomism, sometimes referred to as scholastic monism, endorsing the Aristotelian hylomorphic unity of body and soul remained the dominant Christian philosophical view of the soul-body relationship in medieval Europe. However, in the generation following Aquinas a non-Thomist view, that of scholastic pluralism, gained traction. Among the most influential figures endorsing scholastic pluralism were the thirteenth-century philosopher and Franciscan theologian John Duns Scotus (c. 1266 – 1308) as well as his greatest rival in the famous medieval dispute on the problem of universals, his contemporary and fellow-Franciscan friar, William of Ockham (1287 – 1347).[19] With scholastic pluralism, philosophers in the generation following Aquinas, took a different perspective with respect to Aristotle's hylomorphic soul-body concept than the one argued by Aquinas, essentially denying that all created substances are composites of form and matter. According to scholastic pluralism, some substances can have more than one substantial form. This view endorses a doctrine of a plurality of substantial forms and humans are individuals that have two substantial forms, the soul as a separated state from the body and the form of the body itself. In scholastic pluralism, this separation enables the soul to know the spiritual intuitively without any intermediation by the body and it ensures the immortality of the soul. This immortality of the soul, however, is not a philosophical assurance; it is a matter of faith only. Scholastic pluralism would remain a minority position with most philosophers and theologians continuing to endorse the Thomist position or scholastic monism. However, scholastic pluralism would later help to address challenges to the Thomist immortal soul arising during the early Renaissance from different quarters.

In the late fifteenth century, towards the end of the early Renaissance, Averroist monopsychism continued to flourish, notably in Northern Italy. At the same time, new translations of Aristotle's *De Anima* (*On the Soul*) popularized the notion that Aristotle thought of the human soul as mortal.[20] In response to these developments, the Catholic Church made the immortality of the human soul a dogma at its 1513 Lateran Council. Importantly, going beyond establishing this position as a church dogma, it requested that philosophers prove the immortality of the human soul. That did not prevent the Italian philosopher Pietro Pomponazzi (1462 – 1525) from arguing in his *De immortalitate animae* (*On the Immortal Soul*), published in 1516, that the Thomist interpretation of the human soul

following Aristotle effectively denied immortality of the human soul altogether. Pomponazzi essentially argued that the Thomist human rational soul was of the same nature as the human organic soul. The Thomist view recognized two different kinds of substances, forms and matter, just like Aristotle did. However, Aristotle had also left them with a problem and that was the nature of the human mind or intellect.

How this immaterial and immortal human mind or intellect could fit with the Aristotelian concept of a human mortal soul in a mortal body was the problem Aquinas and the Thomists following him had to address. They did so by asserting that forms, Thomists called them substantial forms, came in two versions, separable or non-separable substantial forms. A substantial form is a part of a substance, but it is not a substance itself. The human soul is such a Thomist substantial form and it inheres in the matter of the human body making for what is the living body. Separability, a concept originating with Aristotle, was critical because only if a substantial form was separable, it was indestructible. Human immortality without a separable part of the human soul, its rational part, was impossible. The non-separable part of the human soul, its organic part, would not survive the death of the object matter it inhered, be it the living body of man or beast. Thomists were walking a fine line as they claimed a single substantial form, the human soul, to have the attributes of two different types of substantial forms, separable and non-separable. Equally problematic, separable substantial forms were not supposed to inhere in matter such as the non-separable part of the human soul, its organic part, would do with the matter of the human body to form a living body. The organic part of the human soul inhering in the human body matter constituted with it the living body. Because the organic part of the human soul was a non-separable substantial form and in this sense material, it died with the death of the living body. However, not so the rational part of the human soul, which Thomists claimed was separable and therefore did not inhere in matter and hence was not material and would survive the death of the living body as the immortal part of the soul that it was.

Thomists had folded Aristotle's immortal and immaterial mind or intellect into their philosophical worldview as the separable part of the human soul which they called the rational soul. This is the view that Pomponazzi challenged. By claiming that the rational part of the human soul depended on human body functions in some way, just as the organic part of the human soul depended on human sensory perception, Pomponazzi could argue that the rational part of the human soul was just as non-separable as the organic part of the human soul. Therefore, the rational part of the human soul was just as destructible as the organic part of the human soul when a living body, speak a human, died. It was the philosophical response to Pomponazzi's challenge, articulated over the next several generations, which would lay the groundwork for the early modern approach to the soul-body and mind-body relationship as we find it formulated with Rene Descartes.

Pomponazzi's argument had capitalized on the fact that the Thomsitic view allowed one substance to have only one form. Therefore, Thomists had to fold contradicting properties into a single human soul, some of them non-separable and mortal and others separable and immortal. Through this construct, Thomists essentially introduced an additional substance into the form of the human soul. Thomists had no solution for how to address this problem without giving up the postulate that one substance can have only one form. Non-Thomist interpretations of Aristotle such as the scholastic pluralism approach discussed above had no such problem as according to it a substance could have a plurality of substantial forms. The solution offered by scholastic pluralism was to split the human mind from the human soul, to dissociate the organic and the rational part of the human soul into two distinct entities. This led philosophers such as the Italian Francesco Piccolomini (1523 – 1607) to a concept of the human mind as a separate form with activities analogous to divine intelligence. Piccolomini articulates the concept of the unextended mind, arguing that only that which is unextended can reflect on itself. This idea originated with the observation that our sensory organs informing our organic soul cannot sense themselves, as for example the eye cannot see itself; essentially an argument going back to Aristotle. As the ability of the mind to reflect on itself is its key characteristic, it can have no extension as otherwise it could not do so.

This brings back, though in a different context, the concept of the unextended soul of Augustine of Hippo. Piccolomini's analogy of the human mind to divine intelligence may not sound too different from Averroist monopsychism. However, it is quite different as contrary to the latter, the human mind remains individual and immortality of the mind is individual and not like with Averroist monopsychism an immortality of a single universal human mind in which the individual vanishes. On the protestant side, we see scholars such as Philipp Melanchthon (1497 – 1560) arguing for the immortality of the human mind in ways similar to Piccolomini but adding a Platonic touch to the human mind by endowing it with innate ideas. Given Melanchthon's prominent role in the Reformation, his account of the nature of the human soul would for much of the protestant movement become the dominant view on the mind-body relationship. Naturally, all such approaches towards finding a solid philosophical basis for human immortality were religiously informed. However, in the later Renaissance we begin to see increased efforts to reconcile the accumulating body of knowledge regarding human anatomy and physiology with how natural scientists and philosophers thought the human mind worked.

The English philosopher and natural scientist Francis Bacon (1561 – 1626), arguing for a mortal corporeal sensitive soul and an incorporeal and immortal mind, was one of the first to do so. The French priest and philosopher Pierre Gassendi (1592 – 1655) formulated similar concepts of a mortal corporeal soul as well as of an immortal and immaterial mind from an atomist standpoint. All philosophical approaches to address Pomponazzi's

challenge mentioned above and many other similar ones solved the problem by splitting the human mind from the human soul, making it a distinct, immaterial and immortal human mind. In their effort to ensure human immortality, philosophers had essentially created a three-substance problem composed of the immaterial and immortal mind, the mortal organic or corporeal soul, and the human body. Rene Descartes, a compatriot and almost exact contemporary of Gassendi, rejected this approach and chose a different route. Descartes effort to reunite soul and mind, making it a thinking unity distinctive from the human body, marks the beginning of modern approaches to the philosophy of mind.

Descartes denied the separate existence of two entities for human cognition, an organic soul for sensory cognition and a human mind for reflective cognition. There could only exist one soul, the immaterial and immortal human mind, which was responsible for all of human cognition. Until then, the organic soul was the shared heritage of all animals, which included of course humans. Descartes claim that there was no organic soul as all of human cognition was the exclusive preserve of the human mind, robbed everything else in nature of their souls. He turned a previously ensouled nature into a soulless one and it was there and then that animals became soulless machines. In addition, because the human mind was immaterial without any bodily reality attached to it, the human soul could no longer be the life-giving force, the psyche, which it had been for philosophers since antiquity. How could an immaterial human mind even interact with the human body without the help of the former organic soul? Descartes provided a new elegant solution for a problem that had troubled philosophers for centuries but at the same time, he created many new problems.

Here we have to end our brief excursion into the history of philosophy of mind concepts from antiquity to the beginning of the modern age. Any such introduction seeking to cover in only a few pages something like the history of philosophy of mind cannot provide more than a cursory outline. However, even if that means disregarding much and casting aside everything but the main themes, that of itself is helpful as it provides a sense of how deeply rooted some of the ideas are that still very much dominate modern discussions of the mind-body relationship. The scientific revolution was already in full swing during Rene Descartes lifetime and would reach its first culmination within the next two generations with Isaac Newton's (1643 – 1727) discoveries. Not only did the scientific revolution usher in new ways of understanding the world around us, but it also changed how we perceive the relationship between body and mind. However, nothing ever arises out of nothing and understanding philosophy of mind in modern times requires some insights into what came before. Hopefully, the last few pages accomplished this. For those interested to learn more about how perceptions of *psyche* and *soma* evolved from antiquity until the scientific revolution the bibliography lists a few books that may be helpful in that respect.

Philosophy of Mind in Modern Times

Bunching together philosophical thinking about the mind-body relationship since the seventeenth century into a single modern period of philosophy of mind is of course a stretch. There is, however, reasonable justification for that so we will keep with this schematic. To some it may come as a surprise that conceptually there are few new developments in philosophy of mind. To appreciate what really has changed since the time of Rene Descartes, we only have to look at the accumulated scientific knowledge over the last roughly four hundred years in a number of fields. Many of the scientific fields to which we owe much of our current understanding of body and mind did not even exist back then. Among them for example neurophysiology, brain and behavioral sciences, psychology or psychopathology in general, just to name a few of the sciences derived from medical and biological science as well as philosophy, which continue to inform and challenge our views of the mind-body relationship. To that, we also need to add non-biological sciences such as computational sciences or mechatronics, which give us increasingly autonomous and seemingly ever more intelligent robots, and last but not least artificial intelligence itself.

All of this new and continuously increasing knowledge informs how philosophers and scientists think today about the mind-body relationship. Outside religious belief, there is no serious discussion anymore about the immortality of the human soul or about its location or immaterial extension or lack thereof in the human body. However, even though in modern times we refer in this context to the human mind rather than to the human soul, we keep asking the same pertinent questions with regard to the mind-body or soul-body relationships. What our ancestors have been seeking to understand for millennia is as relevant to us in our modern world as it was to them in theirs and we continue to search for answers to many of the very same questions. Of course, over time, any such concepts concerning the nature of soul, mind and body and perceptions of their respective relationships have been very much expanded, differentiated and formalized. This is where we will turn to next but before we can do so we first have to consider some of the framework and vocabulary philosophers of mind use in their discussions and disputes about ancient and modern forms of philosophy of mind. The latter fall into the two categories of dualism and monism plus a third category, which captures everything else that does not fit in any of the first two categories.

When we refer to the human mind, we do not think of a substance of mind. That which we talk about and experience as our human mind we commonly refer to as states of mind or mental states. States of mind or mental states do come in different flavors. We can be happy or sad or feel love or anger. These are all mental states of feelings. Similar for our beliefs and imaginations, and everything that we sense around us, all of which we experience through states of mind as well, since otherwise we could not consciously

experience any of it in the first place. The latter is what the above mental states and many others have in common; they are all states of consciousness. Because of that, many believe that such mental states of lived experiences essentially do constitute what we call the human consciousness. To describe the experience of lived mental states, philosophers use a term derived from Latin, quale, which is supposed to describe the qualitative essence of an experience. The Latin word quale, or qualia when used in plural form, is the root of our modern word quality. Qualia are lived experiences, as they describe the phenomenal or lived aspects of mental states. Quale are mental state experiences associated with emotions and sensations. In that regard, philosophers also refer to states of our phenomenal consciousness, which they define as those mental states that have no defining conceptual content but a certain phenomenal or sensual quality. Phenomenal states of consciousness are just another way philosophers use to speak about qualia.

Another flavor of mental states are intentional states or what philosophers call intentionality. Directed at an outside object, someone else or our own future self, mental states of intentionality can be about a number of things. They can for example include our knowledge of how to do things like calculating, or the opinions and convictions we have formed about others as well as our intentions for our own future self, like when we for example imagine ourselves doing something at a future point in time. In order to explain the world around us and to understand something, including our mind, what we humans do is to organize ourselves. We put things in different drawers as if they were in some way separate but in the end that may not be the case; and we frequently forget that there is a large cabinet labeled "we do not know". While it is helpful to think of human mental states in terms of lived experience and intentionality, clearly, there exist mental states not characterized by either flavor; and many others can show characteristics of both. However, what human mental states all share is that they are subjective and that they are private and as such individually controlled. You cannot have access to my mental states and I cannot access yours.

We can identify additional characteristics of mental states when we hold them next to physical states. In the mind-body problem, physical states represent the body and everything else that is out there and are controlled by nothing else but the laws of nature. Physical states are not physics states, the latter are only a subset of physical states. It is the laws of nature that always determine physical states, no matter what the specific laws in any particular circumstance may be. In this sense, physical states are deterministic. That however is not the case for mental states.[21] In contrast, different from physical states, mental states are not deterministic and therefore we are at liberty to choose and control our mental states. We decide to connect experiences with one another and no one else does this for us.[22] Take as an example how we can connect one thought with another. We have the freedom to connect mental states using whichever rationale we choose and as we all know that does not mean those rationales do have to accord with reason or logic.



Figure 2.1: Schematic summarizing characteristics commonly assigned to mental states and physical states. Neatly separated in this drawing, the crux of the mind-body problem as well as the very objective of philosophy of mind is to understand just how those two connect as they necessarily have to.

To illustrate this with an experience most of us will be familiar with, we just need to look at the results of such exercises as they are presented to us in many science fiction novels and movies, quite a few of which show little regard for the laws of nature. Physical states are objective states. They exist independent of any mental state and only follow the laws of nature. Whatever causes a physical state to change - and that may well be a mental state - any such change follows the laws of nature and nothing else. Different from mental states which are private and individually controlled, physical states are public and controlled by the laws of nature. Which is just another way of saying that physical states are deterministic and mental states are not. Fig. 2.1 summarizes these commonly distinguished characteristics assigned to mental states and physical states. This diagram graphically illustrates the understanding that mental states are not physical states and vice versa. The formulation of this insight constitutes what philosophers call the first element or the first principle any modern philosophy of mind must consider.
Humans have been exploring the relationship between mental and physical states since long before the time of Rene Descartes. However, Descartes represents a watershed in the philosophy of mind as he introduced a different perspective on mental states from the way philosophers had perceived them since antiquity. From antiquity through the medieval period, philosophers, initially pagan thinkers and then increasingly theologians, saw the main challenge in determining how the intellect and soul aspects of mental states related to physical states. Until the seventeenth century, philosophers did not really consider human beings to have separate states of mind in a way that would differentiate humans from the animal world. Philosophers perceived mental states to be more or less parts of our biology, not separating humans from animals but only differentiating humans from animals to a degree; in a sense, they were just part of human animal behavior. Only as far as intellect and reasoning was concerned did philosophers see humans as different

from animals, but not so for example with respect to emotional mental states or states of consciousness. Philosophers considered emotional states as reflections of our human animal nature and consciousness existed only in the sense that it could affect aspects of reality. What mattered was the intellect and the reasoning quality ascribed to the soul and not mental states. Christian theology and philosophy further enshrined the primacy of the soul and it is only with Rene Descartes that we can see the consciousness aspect of mental states taking center stage. Since Descartes first raised the issue of consciousness, philosophy of mind has remained focused on the nature and structure of the causal relationship between mental states and physical states.

While we sometimes tend to think of philosophers as recluses, content to live their thoughtful lives in their proverbial ivory towers, that is not quite so. Although it may frequently seem detached and unrelated to our daily lives, philosophy always has been the sincere human effort to make sense of our world and to help us understand ourselves. What motivates certain philosophical positions may on occasion be difficult to understand for non-philosophers because it can now and then include religious and spiritual beliefs just as much as rational beliefs and logic. It is important to realize that at the heart of philosophy of mind there remain inherent contradictions. They always have been there and to this day, they have not been resolved. Even though natural sciences have made enormous progress since the time of Rene Descartes and philosophers have since then analyzed the implications of these contradictions with the ever-increasing benefit of modern science, they have not gone away. This by itself could be a factor philosophers of mind may have to consider more seriously. What we can and cannot know keeps changing but there is no guaranty that at any point in the future we will be able to know everything, not as individual human beings of course, but maybe not even as a species, as humankind. We just might have to accept, as philosophers would likely phrase it, that there are epistemological constraints to our ability to understand the mind-body relationship fully; those may recede over time, but they may never do so fully.

By showing mental and physical states completely separated, the schematic in fig. 2.1 also neatly illustrates the very crux of the mind-body problem and what the philosophy of mind seeks to understand – how the two connect, as surely they must, as we all know from our personal experiences. More broadly speaking, we have to consider four possible causal relationships: physical states can cause other physical states; physical states can cause mental states; mental states can cause other mental states; and mental states can cause physical states. Witnessing a physical state causing another one is part of our daily experience like when it rains, and this causes the rain barrel to fill. Similar for mental states causing other mental states. Just remember how thinking about something can make you wish for something else. This is how we can for example be tempted to eat even if we are not hungry, like when our mind recalls the smell of a delicious treat or the satisfaction of enjoying the treat. Physical states causing mental states is just how

we perceive the world around us and mental states causing physical states is how we can purposefully move around in the first place. To summarize and illustrate, there are just four causal relationships in which mental and physical states can connect:

1. Physical states cause physical states: Examples here are causal chains as we can observe them in natural events like heavy rains causing deluges or lightning causing wildfires.

2. Physical states cause mental states: We are all familiar with hearing or seeing something that in turn causes us to remember something else. Proust's remembrance of things past caused by the taste of a madeleine dipped in tea is a famous example.

3. Mental states cause mental states: Who has not experienced that thinking or feeling something can make us wish or desire to do something else? Think of how feeling happy about your fortunate condition may make you wish to help others to a better life too.

4. Mental states cause physical states: This is an everyday experience as our minds tell our bodies how, when and where to move.

Not surprisingly, the problem that philosophy of mind faces is not so much how physical states can cause physical states or mental states can cause mental states. The challenge really lies with how physical states can cause mental states but even more so with how mental states can cause physical states. The difficulty of understanding the latter causal relationship, how mental states can cause physical states, is what philosophers refer to as the problem of mental causation. Being able to answer this question, just how mental states can cause physical states, as they clearly do, makes for the second element or second principle that any modern philosophy of mind has to address. It would be very difficult for us to imagine a world without mental causation being at the center of how we understand ourselves as human beings. Mental causation is very much a part of our shared human experience and any philosophy of mind discarding it will struggle to reconcile itself with our every-day life experiences.
For most of human history, we could only speculate about those causal relationships and to a good extent, this is still true. However, the scientific revolution starting in Descartes time has very much enabled us to get a better understanding of the physical nature of things; the same is not necessarily true for the mental nature of things. What is becoming ever clearer is that we continue to find physical explanations for many phenomena for which in the past, we did not believe that a physical connection existed in the first place. This trend continues and future generations can expect better answers as to what physical realities may underlie that which we currently perceive as purely mental states.

Table 2.1: The three principles (1), (2) and (3) every philosophy of mind must consider plus a necessary underlying assumption (4). With this underlying assumption as a fourth principle, only two of the first three principles can hold true at the same time.

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(1) ***The principle of difference between mental and physical states:*** Mental states / events are not physical states / events.

(2) ***The principle of mental causation:*** Mental states / events cause physical states / events.

(3) ***The principle of causal, nomological and explanatory completeness of physical states*****:** The laws of nature explain all physical states / events fully.

(4) ***The absence of regular overdetermination:*** Regular overdetermination of physical states / events through complete physical causes and additional mental causes does not exist.

However, we may well end up not being able to explain everything through physical states, as in essence, what we are trying to solve is a circular logic problem. Maybe a better analogy could be that we are trying to solve an asymptotic problem where we will have ever better explanations to explain ever more through physical states but we never will have a complete explanation of everything based on physical states alone. Even though in principle such explanations must exist, as most scientists would likely assert.

This brings us to the third element or the third principle any modern philosophy of mind has to consider, which is the essential completeness of the laws of nature. Philosopher express this principle using their specific vocabulary more precisely than most of us would do as the causal, nomological and explanatory completeness of physical states.[23] The adjective nomological refers to anything resembling general laws, those we know about today and those that we do not know about yet but may learn about in the future. Therefore, the assertion of nomological completeness means that all physical states follow certain laws, which in principle we can fully understand in physical terms through the scientific exploration of our world. No divine or magic power needs to be invoked to explain what we may not understand today. We are just not smart enough to understand everything yet. With this translation, the third principle of philosophy of mind simply states that any causes, laws and explanations underlying any physical state are physical causes only, physical laws only, and physical explanations only. All we need to understand here is that even though we may not have all explanations today, whatever physical state we observe in nature has a complete explanation through other physical states connected by natural laws.

All physical states have physical explanations and physical explanations only as nothing outside the laws of nature is required for a full and complete explanation of any conceivable physical state. This statement is a reflection of the tremendous progress modern science has made, giving us the confidence that everything in our universe is subject to the laws of nature and the laws of nature only, even though we clearly have not fully understood all of them yet.

The first three items in tab. 2.1 summarize the three elements or principles any modern philosophy of mind must consider and seek to reconcile. In addition to the above discussion about mental or physical states, the principles listed in tab. 2.1 refer as well to mental and physical events. While that may seem like some unnecessary semantics it is important to understand that the principles in tab. 2.1 hold for either and some philosophies of mind refer rather to events than to states. As we will see, one can also refer to mental and physical state terms or types and similar constraints wild hold for those with respect to the above three principles even if mental and physical state terms and types are not explicitly listed in tab. 2.1. The problem with the three principles (1), (2) and (3) of philosophy of mind in tab. 2.1 is that only two out of those three can hold true at the same time. Hence, by selecting two out of those three principles one automatically concedes that the third one cannot hold true. Anyone with a mathematical bent would quickly answer that there are three possible ways to choose a set of two items out of a total of three items. Hence, there should be three possible ways to formulate a philosophy of mind by selecting two of the three principles as holding true while accepting that the remaining third cannot hold true at the same time. Nonetheless, because we are talking here about philosophy and not mathematics, we will see below that there are more than three possible approaches towards solving this problem.

Before we can look deeper into that we have to add one more clarification that speaks to a fourth principle. That is item (4) in tab. 2.1 which is more of an underlying assumption than a separate principle because today we essentially take it for granted. What it states it that a regular overdetermination of physical states by physical causes and additional mental causes does not exist. This fourth principle asserts that what we understand by mental causation is nothing like some sort of mental states being seemingly the cause of something which we can already fully explain by physical causes different from those mental causes. The importance of this fourth principle, which we often just assume and do not spell out, is that without it one could conceivably find ways to construct philosophies of mind that can accommodate all of the first three principles as valid. It is the assumption of the fourth principle, which forces us to choose two among the first three principles as with the underlying fourth principle not all three of them can hold true at the same time.

While the first two principles are very much reflections of our self-perception, as we see ourselves as human beings and as we experience the world around us, the conception of

the third principle and the assumption of the fourth principle owe much to the advances of natural sciences over the past few hundred years. Since the time of Descartes, science has progressed to a degree where we have come to realize that the physical laws of nature are essentially complete. That does not mean that we know all the laws of nature that exist; we are still far from that. However, we now understand that the laws of nature in their completeness encompass the explanation of any kind of physical state that may exist. In other words, no additional explanations outside the laws of nature, such as a divine force for example, are required to explain physical states. Clearly, this is more of a recent development and therefore how philosophers perceived the third principle in the past has changed over time as natural sciences progressed. This in turn has changed how today philosophers evaluate for example Rene Descartes approach towards a philosophy of mind.

The fundamental challenge any philosophy of mind faces is how to explain the relationship between mental states and physical states or put differently, to what extent physical states do have mental causes and mental states do have physical causes. To describe the relationship between mental and physical states philosophers frequently use a concept they refer to as supervenience. In simple words, given two sets $\boldsymbol{A}$ and $\boldsymbol{B}$, set $\boldsymbol{A}$ supervenes on set $\boldsymbol{B}$ if and only if some difference in set $\boldsymbol{B}$ is necessary for any difference in set $\boldsymbol{A}$ to occur. Rephrased, $\boldsymbol{A}$ supervenes on another property $\boldsymbol{B}$, if there cannot be a difference in $\boldsymbol{A}$ without a difference in $\boldsymbol{B}$. Because there can be differences in $\boldsymbol{B}$ with no change in $\boldsymbol{A}$, multiple realizations of mental properties are possible, if for example $\boldsymbol{A}$ refers to a mental state and $\boldsymbol{B}$ to a physical state. Through the concept of supervenience, philosophers circumvent the much more difficult problem of having to establish causal relationships between countless specific mental and physical states. The latter would be a practical impossibility, not the least because even today we still have little knowledge of the physical processes underlying our mental states, essentially of what happens in our brains when it generates such causal relations, which we only recognize for what they are post fact.

It is best to illustrate supervenience with an example. Assume for a moment that we could make a copy of every physical state in the universe, essentially duplicating our universe in its completeness. By necessity, such a copy would include all humans and with that all mental states. If all physical states that exist in our universe also exist in the duplicate universe, the duplicate universe also must contain all mental states. The fact that all mental states of our universe must also be a property of a perfectly duplicated universe is equivalent to saying that the mental states of our universe supervene its physical states. The claim that in a perfect duplicate of our universe there cannot be a difference in mental states without a difference in physical states is an example for what philosophers call a global supervenience. In addition, philosophers differentiate between weak and strong individual supervenience as well as regional and local varieties of supervenience.

We will not go into any more details here, suffice to say, all such supervenience concepts in philosophy of mind usually represent efforts to describe the dependence of mental states on physical states. Philosophers use supervenience concepts to theorize about so-called psychophysical laws, the laws believed to govern the relationship between mental states and physical states. Logical constructs like supervenience enable fundamental philosophical discussions of the mind-body problem without having a full knowledge of our brains every inner workings. Appropriately, philosophers refer to this supervenience relationship between mental and physical states as described above as psychophysical supervenience. With that, we can now look at the different possible solutions to the philosophy of mind problem.

Most philosophies of mind fall into two categories, dualist and monist theories. However, there are some other theories that do not fit into either mold, usually referred to as alternative or non-traditional theories. Non-traditional in the sense that even though the original ideas of such theories may trace their roots all the way back to ancient Greek philosophers they are essentially modern conceptions of a much more recent date. Sidestepping historical contexts and discussions, they seek to recast our approach to understanding the mind-body relationship from a modern perspective. Prominent among those theories is modern hylomorphism, which can trace its roots back to Aristotle but has become quite different from Aristotle's hylomorphism. Before we can begin to look into specific kinds of dualism and monism, a brief clarification is in order with regard to the meaning of the word person and just how it relates to what we refer to as human or non-human beings and individuals in general. While that may sound strange, this is simply owed to the fact that philosophers use a definition of the word person which differs from how most of us will likely understand the meaning of the word person. In our everyday usage of this word, persons are are just people, and a person is just a human being. This is however not the way philosophers define a person. Put simply, the way philosophers use the word person, it refers to a being with certain attributes such as reason, morality, consciousness or self-consciousness. It is important to realize that this use of person implies that a person is not necessarily equivalent to a human being. Just think about those unfortunate among us who for example suffered a severe brain damage and are now in a persistent vegetative state. They are clearly human beings but in a philosophical sense, they may not be persons anymore if they lack the above philosophical person attributes. On the flip side, for good reasons we are increasingly inclined to consider non-human beings like for example our closest primate relatives as persons; not just from a philosophical standpoint but from a moral and ethical one as well. An example hitting closer to home for many are likely science fiction novels and movies that tend to include large numbers of non-human persons. In all these examples for persons or non-persons, humans or non-humans we could have referred to these beings as individuals.

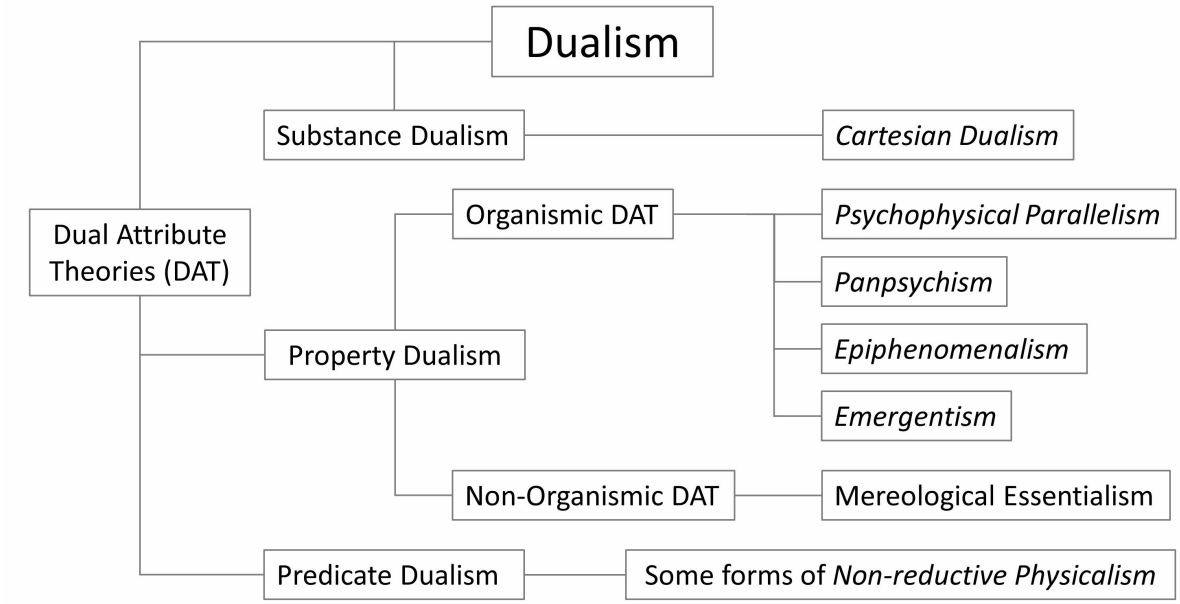


Figure 2.2: Simplified schematic of dualism philosophies of mind. The left side of the diagram shows dualism philosophy of mind category types and the right side lists examples of more widely discussed dualist positions in the history of philosophy of mind referred to in the text.

There are essentially two ways to differentiate versions of dualism. First, one can differentiate them by what they are dualistic about. This gives us categories such as substance dualism, property dualism or predicate dualism. Such category distinctions are not absolute as illustrated by substance dualism in itself having aspects of a property dualism. There are finer distinctions between dualistic theories, which for example further differentiate property dualism into a weaker version called predicate dualism, and property dualism proper. To complicate matters further, theories that philosophers traditionally discussed under the umbrella of property or predicate dualism, many today are discussing under the umbrella of so-called dual attribute theories (DAT). What much of this reflects is a gradual change in perspective. Philosophies of mind categories represent more of a continuum of theories rather than completely separate categories. The spectrum of dualism philosophies of mind includes on one end uncompromising substance dualism theories. On the other end of the spectrum one can find however some dual attribute theories partially bridging the gap towards physicalist theories, which belong to monism philosophies of mind. Fig. 2.2 shows a simple schematic with the respective dualism philosophy of mind category types just mentioned on the left hand side of the diagram along with a spectrum of the most prominent dualism philosophy of mind positions falling into those categories listed on the right hand side of the diagram.

The second way to differentiate versions of dualism is by how they claim that the two separate elements of a dualist worldview interact. Using this criterion, one can distinguish between two types of dualist philosophies of mind, interactionist and non-interactionist dualism. If they are of the interactionist type, then the two categories they are dual about causally interact with each other. If they are non-interactionist dualist philosophies, the two categories do not causally interact and an explanation is required as to how our experiences come about without them interacting. An examples for an interactionist dualism is Rene Descartes substance dualism. Non-interactionist dualism philosophies of mind include for example versions of property dualism such as the psychophysical parallelism of Baruch Spinoza (1632 – 1677), forms of occasionalism endorsed for example by Descartes friend Louis de La Forge (1632 – 1666), or the pre-established harmony of Gottfried Wilhelm Leibniz (1646 - 1716). In addition, dual attribute dualist views such as epiphenomenalism whose prominent modern proponent was Charles Darwin's friend Thomas Henry Huxley (1825 – 1895) or emergentism as espoused by John Stuart Mill (1806 – 1873) are also versions of non-interactionist dualism.

As we saw earlier during our brief journey through the philosophy of mind in antiquity, dualism philosophies of mind have a long history. The earliest such theories were conceived in the Classical Period of Greece by philosophers such as Plato. Plato's view of the mind-body dualism changed throughout his lifetime. Philosophers believe that he developed his first theory of dualism in his work *Gorgias*, then significantly modified it in *Phaedo*, changed it again in his *Republic*, then again in his *Timaeus*, only to finally revise his view of the soul a last time in his *Phaedrus*. Initially, he viewed the soul as an undifferentiated counterpart to the body, but his view changed with his introduction of the tripartite soul, and it would change again. The immortality of the soul he conceived in *Phaedo* still had a beginning, the soul in *Phaedo* was not eternal but with his *Phaedrus* the soul had become immortal, with no beginning and no end. Plato's view of the soul-body dualism and the immortality and finally the eternity of the soul are integral parts of his metaphysics. His metaphysics distinguishes between eternal forms and objects, which are only imperfect copies of the former. The eternal nature of these forms seems to have been the motivation for Plato to concede the same eternal nature to the soul. These forms, while immaterial, were Plato's substances from which he thought everything underlying the world as we experience it derived. According to Plato, it is through understanding these eternal forms and not the material objects themselves that the world becomes intelligible to us. As a consequence, the reason or intelligence part of Plato's soul is also immaterial as otherwise we could not hope to understand the eternal forms underlying the reality of objects surrounding us. In Plato's dualism, the immaterial and immortal soul or intellect is a prisoner of our bodies and it is not clear just how the two relate to each other and how they come together in the first place. Neither is it obvious how many reincarnations, or better transmigrations, it will take before a soul

eventually joins the eternal forms to which Plato believed it to be drawn to by nature. In his *Timaeus*, Plato introduced the demiurge as a solution to the problem of how objects come about from forms. The demiurge is the agent who arranges what preexists in the universe into objects reflecting the eternal forms, thereby producing everything physical in the world, including our human bodies. Plato is silent on what this demiurge is made of and neither is it clear where the materials come from that preexist in the universe. The demiurge is essentially Plato's acknowledgement that the problems of how the world of forms and the world of objects interact as well as how soul and body interact require a solution. In this sense, Plato's dualism is an interactionist dualism.

Aristotle did not share Plato's view of eternal and universal forms existing independently of their instances. This led him to view the soul as a property of the body, in essence a bodily organ, but then he also believed that the intellect or reason, though part of the soul, was not part of a specific physical organ and immaterial in nature. The latter, Aristotle argued, had to be the case because if the intellect were a specific physical organ or a part of one, then it would not be able to receive all forms. Plato as well as Aristotle perceived the relationship between soul and body or more specifically between reason or intellect and body as being pretty much of the master-slave kind. Western cultures inherited this view when later the philosophy of Neo-Platonism would significantly influence early Christianity, eventually molding Plato's eternal forms into Christian souls. Finally, in the thirteenth century, Thomas Aquinas would through his Aristotelian synthesis condense pagan philosophy and Christian belief into the doctrine of the soul as an immortal substance surviving the death of our bodies.

Modern forms of dualism originate with Rene Descartes in the seventeenth century. Our word substance traces its roots back to the Latin substantia, describing that which lies below or underlies something else. Two different substances are fundamentally two different kinds of things, for Descartes these were a material substance and an immaterial thinking substance; or put differently, material physical states and immaterial mental states. According to Descartes, mind and matter are two distinct and independent substances; physical states cause mental states and vice versa. Descartes substance dualism is an interactionist dualism. The latter addresses the philosophy of mind problem stated above by keeping principles one and two but discarding the third principle in tab. 2.1. That in itself poses a problem as with relinquishing the third principle, interactionist dualism is at odds with the completeness of the laws of nature. The latter encompasses that the laws of nature can explain all possible physical states through other physical states with no non-physical explanations required. Therefore, disregarding the third principle continues to put interactionist dualism into conflict with science. That has not changed with modern versions of interactionist dualism even though they adopt arguments taken from quantum mechanics to explain how certain brain states may come about. The fundamental problem of interactionist substance dualism remains that it has to presume

that some physical states have non-physical causes. Because mental states are not physical states and in substance dualism mental states are an immaterial substance different from matter, the completeness of physical laws, the third principle, tells us that there is no way in which these mental states can cause physical states. Therefore, in order to preserve mental causation, the second principle, substance dualism has to relinquish the third principle and assume that the natural laws include causal chains, which not only consist of physical causes but also mental causes. With that, substance dualism also relinquishes psychophysical supervenience as perfect copies of physical worlds become an impossibility with natural laws depending causally on mental states that have no physical reality.

Another problem that substance dualism faces is that with mind and matter being two separate substances they do not necessarily have to go together. Surely, a substance dualist must believe that she or he, as a human being, possesses both, matter in the form of a body and an immaterial mind. However, who is to tell that this is always so? Nothing in substance dualism prevents the existence of zombies, bodies walking among us without a mind. Similarly, there could be minds freely floating around us without a body, which some surely would refer to as bodiless souls. How can we then know, assuming substance dualism gives a true account of reality, that there sits a mind behind the eyes of other human bodies we are looking at. The simple answer is that we do not know this, a fact to which philosophers refer to as the problem of other minds. However, this is not the last challenge that Cartesian substance dualism faces. Taking the above zombie argument to the extreme, a substance dualist could assume that there is nothing else out there but zombies. In turn, this would require her or him to possess private convictions of conceptual contents, for example regarding feelings such as joy or pain, identical to the ones that we usually associate with them, without ever having had any conversation about those concepts with any other person. This however would go against the private language argument. This private language argument, introduced by the Austrian-British philosopher Ludwig Wittgenstein (1889 - 1951), advocates against the possibility of a private language, a language that only one person can understand. Wittgenstein's argument essentially states that words like pain and joy only attain the meaning they have for us through how we make use of our shared language. Because a private language could not possess such an understanding, it therefore would have to be incoherent. Looking at this short list of major challenges facing Cartesian substance dualism, it should be no surprise that there are few committed Cartesian substance dualists left among philosophers of mind. Much of what happened in dualism philosophy of mind after Descartes was essentially an effort to address those challenges.

Interactionist dualism theories do not have to presume a substance dualism. One example for that is Karl Popper's three-world conjecture, which philosophers tend to view as such an interactionist dualism not requiring a substance dualism, even though it does

assume in addition to the existence of physical and mental states a third component, the body of accumulated human knowledge. Another example for an interactionist dualism theory without a substance dualism are interactionist versions of property dualism such as emergentism, which we will turn to soon. However, first we have to clarify what we exactly mean by property dualism.

Frequently, philosophy of mind nowadays refers to property dualism as a version of dual attribute theories. The latter distinction serves to set property dualism safely apart from substance dualism because strictly speaking, substance dualism also subsumes a kind of property dualism as the two substances in substance dualism carry two different properties. In this sense, property dualism differs from substance dualism in asserting a difference in property only but a unity in substance whereas substance dualism asserts dualism of substance and of property. Here we use the term property dualism strictly to refer to a dual attribute theory only and not to substance dualism. There are two kinds of dual attribute theories, which differ with respect to the nature of the substance possessing dual attributes or properties: organismic and non-organismic dual attribute theories. Organismic dual attribute theories claim that we are organisms and non-organismic theories deny that. Organismic dual attribute theories essentially endorse an animal view of human nature but can lend themselves likewise to constructing organismic dual attribute theories assuming non-biological organisms such as robots for example. Different from that, non-organismic theories for example assert that a human being is some kind of spirit dwelling in but separate from a human-body.

One example of a non-organismic theory is mereological essentialism. The word mereology derives from the Greek word for part. In philosophy, mereology is the theory of parthood relations, essentially a theory about how parts relate to a whole. The position that mereological essentialism takes is that the definition of an organism comprises all its essential parts and that an organism cannot exist without all of its essential parts. Because the human body constantly exchanges materials with its environment, like when we breathe, mereological essentialism argues that the human body ceases to exist every time that happens. Hence, the conclusion of mereological essentialism is that we cannot be organisms. However, that is not our experience as we certainly continue to exist as our bodies inhale and exhale air. Theories of parthood relations have a long history in philosophy going back all the way to the earliest Greek philosophers. There are many possible mereologies, none of which must assume the constraint mereological essentialism imposes on itself, that an organism cannot exist without all its essential parts. Mereological essentialism narrowly defines all essential parts as essentially all parts of an organism at any time, no matter how irrelevant for the survival of an organism they are, such as for example the skin cells we shed without ceasing to exist. In mereological essentialism, we are not considered organisms but something else that dwells in organisms and somehow endures as the organism continuously ceases to exist. This is not unlike

the relationship between immaterial mind and material substance in Cartesian substance dualism. Because of that, non-organismic dual attribute theories face many of the same insurmountable problems that challenge Cartesian substance dualism. As a result, few modern philosophers entertain non-organismic dual attribute concepts in philosophy of mind such as mereological essentialism.

In organismic dual attribute forms of property dualism, the mind, our consciousness, is a product of the physical brain — it cannot exist without the physical brain but it is also more than the physical brain itself. This kind of property dualism derives mental states from physical states without all mental states being fully explainable by physical brain states. In essence, organismic dual attribute forms of property dualism throw individual psychophysical supervenience overboard as they claim that two sets of mental properties can be different without their corresponding physical properties being different. This last sentence refers to properties instead of states as discussed earlier, but that is more of a semantic difference, the result remains the same. Different from interactionist forms of dualism, non-interactionist dualism philosophies like organismic dual attribute theories of property dualism solve the philosophy of mind problem by discarding the second principle, mental causation, and keeping the first and third principles. This of course raises a critical question. If mental states do not cause physical states, how can they then connect if they cannot have any kind of causal relationship?

Baruch Spinoza, the Dutch philosopher with Portuguese Sephardi heritage, a student of Descartes work, just came of age when the French philosopher died in his Swedish exile in 1650. The philosophical approach Spinoza devised in order to coordinate mental and physical states without any causal interaction between them became eventually known as psychophysical parallelism. First, he proposed the existence of a neutral substance, which he describes as having no affections. Today we would say this substance has no specific property nor does it exist in a specific mode. As a general property, this substance possesses two attributes, the totality of all mental states and the totality of all physical states. It is for this reason, that Spinoza refers to this substance as God or nature. Each mental or physical state in the real world then represents a specific mode of this substance under those two attributes. In Spinoza's property dualism causal relationships only exist within specific modes of the two attributes. Mental states and physical states exist completely separate. Mental states only have causal connections with mental states and physical states only with physical states. The consequence of this is that the third principle must also apply to mental states in the same way it applies to physical states. The completeness of physical states must be complemented by a similar completeness of mental states. Each physical state must have a corresponding mental state and vice versa.

Spinoza was not the only one in his time to conceive a psychophysical parallelism philosophy of mind. In the generation following Descartes, Nicolas Malebranche (1638 - 1715),

the French philosopher and priest devised occasionalism and the German philosopher and natural scientist Gottfried Wilhelm Leibniz formulated his pre-established harmony, both flavors of psychophysical parallelism. In occasionalism, it is not humans that cause the changes that drive perceivable transient motions, changes to which philosophers refer to as efficient causes, but rather God himself who does that. Occasionalism did not arise with Malebranche and can look back on a long tradition within Islam before that. Malebranche, by seeking to reconcile Augustine of Hippo with Descartes, only created a Christian version of it. Leibniz pre-established harmony is also a version of occasionalism, which different from occasionalism itself, leaves some power of change with human beings. However, this power of change only affects each substance itself, meaning only each human being her or himself and the way we see substances causally interacting, which Leibniz attributed to a divine pre-established harmony. To us, it may seem naive to attribute the changes we see in the world to nothing more than the will of a god or his pre-established harmony but we must not forget that back then it was frequently not clear how something causes something else. From a human perspective, quite often, there seemed to exist no necessary causal connections between observable events in the world and therefore one could presume that causality originated from somewhere else, from a level where what did not make sense to humans would make sense after all, a sentiment that many religions share to this day. It is best to leave the discussion of philosophies of mind invoking divine authorities to religious scholars. Few philosophers today would argue any such philosophy of mind in the first place as they are as much matters of religious belief, as they are products of philosophical thought.

Psychophysical parallelism philosophies of mind in the mold of Spinoza replace the causality of interactionist dualism, where mental states cause physical states, with a one to one correspondence between mental and physical states. Because this one to one correspondence works both ways, every physical state must have a corresponding mental state. The latter specifically implies that every physical state occurring in nature and not just in humans, must have a corresponding mental state. This of course raises the specter of panpsychism, the view that not only human beings possess a mind or a soul, but so does everything else in nature, including all lifeless matter. Another challenge for psychophysical parallelism arises from the requirement that the third principle applies to mental states in the same way as it does to physical states. However, unlike for physical states, this cannot be true. For physical states, we can assert the completeness of the laws of nature, meaning we can explain any physical state through other physical states and we require no other but physical explanations. Among the physical sciences explaining physical states, physics is a good example to illustrate this point as physics does not require non-physics explanations. For the third principle to hold true for mental states, we should expect to be able to make similar completeness arguments but we cannot. Psychology is not in the same sense complete as physics. There is and likely never will be a

psychology that can explain all mental states through other mental states. In addition to psychological explanations, psychology also requires physical explanations. Psychology depends on physical sciences such as biology, neurophysiology or physics. Therefore, psychology and other sciences trying to explain our minds to us cannot be complete in the same way as the body of physical science is. That in turn makes it difficult to assert that the third principle holds also for the set of all possible mental states.

Avoiding panpsychism and dropping the assumption that the third principle also holds for the roam of mental states, the property dualism of epiphenomenalism is an example of an organismic dual attribute theory discarding the second principle. The modern view of epiphenomenalism goes back to the late nineteenth century; specifically to the writings of the British biologist and naturalist Thomas Henry Huxley, Charles Darwin's close friend and staunch supporter. Epiphenomenalism claims that physical states cause mental states, the first principle in tab. 2.1 holds true, but that mental states have no further effect on any physical states, the second principle in tab. 2.1 does not hold true. That by itself seems to be an absurd position as it reduces mental states to nothing more than phenomena arising from physical states without any further consequence; hence the name epiphenomenalism. How can one consider epiphenomenalism to be a dualistic philosophy if there are only physical states of consequence and no such mental states? The response of epiphenomenalism to such questions is that there are physical states and then there are physical states accompanied by mental states. It is no coincidence that this philosophical position gained prominence in the late nineteenth century, which witnessed a previously unseen progress in natural and engineering sciences, seemingly making non-physical explanations unnecessary not just for these physical sciences but for all sciences, including the nascent sciences exploring life itself. Incidentally, this convinced Huxley to view animals as nothing more than mere automata, biological machines. Unfortunately, despite indisputable evidence to the contrary, even today many still view animals as nothing more than biological machines.

Evidently, epiphenomenalism has faced a number of challenges. Few today would contest the fact that mental states have consequences for physical states as holding such an opinion obviously contradicts our daily experiences. Among the problems which epiphenomenalism must address rank prominently the implications that the existence of two different kinds of physical states has for the third principle of philosophy of mind in tab. 2.1. The first kind of physical states just being plain physical states and the second kind being physical states accompanied by phenomena resembling mental states. Are some physical states just not complete because they do not have mental phenomena associated with them or are the select ones that are associated with mental phenomena overdetermined? What does that mean for the completeness of physical states? In addition, epiphenomenalism has failed to produce an answer to the question as to how some physical states can cause mental states without these having any further consequence.

For all of us, our life experiences tell us that our actions have their causes in our consciousness, that what we do is more or less a direct reflection of what we consciously decide to do. Similar to psychophysical parallelism, epiphenomenalism confronts us with a conception of consciousness that has no bearing on our physical world at all. However, the problem is even more fundamental than that as epiphenomenalism just holds no satisfying explanation for how we can have states of consciousness in the first place. This is what philosophers call the problem of psychophysical emergence.

There have been a number of efforts to address this challenge. One of them was to assert that just like there are natural laws governing the causal relationships between physical events, determining how one physical state arises from another, there exist psychophysical laws governing the causal relationship between physical events and mental events, how mental states originate from physical states. The claim then was that those psychophysical laws should be discoverable empirically. That however has turned out not to be true. Quite to the contrary. The physical sciences have given us an ever-better description of many of the complex physical processes underlying our human brain activity, often down to the atomic scale. As of yet, our understanding of physical brain states, while much better than it was just a couple of generations ago, is still very rudimentary but importantly, everything we have discovered so far about physical brain states tells us that they are subject to natural laws only. Never was there the need to invoke some mysterious psychophysical law to explain what scientists discovered about physical brain states and as science continues to progress, no serious scientist will likely expect such psychophysical laws to manifest themselves.

Going down to the atomic level and endowing the smallest physical particles with consciousness is another way to address the problem of psychophysical emergence. After all, if matter on the smallest scale has consciousness then of course everything built from it will have it as well, including human beings. It looks like for philosophers going down that path the cartoonlike illustration of a thinking atom used at the beginning of the first chapter to elaborate a point may not have been too farfetched after all. Of course, there are all kinds of problems with endowing the smallest matter particles with consciousness as this assumption invokes of course panpsychism. Thinking atoms will remain a strange conception. Therefore, not surprisingly, another version of panpsychism referred to as proto-panpsychism claims that atoms for example do not have the kind of consciousness we associate with sentient beings but only a lower level of proto-consciousness. In that picture, as atoms aggregate into molecules and larger structures, their interactions in those larger structures give rise to a higher level of consciousness which finally culminates in our brains having the most complex and sophisticated interactions we are all familiar with. This makes the emergence of human consciousness the result of aggregation. That of course raises questions such as why other aggregated materials, at least according to our experience, do not possess consciousness.

One can avoid such problems if the quantitative and qualitative change in consciousness is not simply due to the aggregation of the material itself, just putting more of it together, but to how those materials are structurally organized. This way not every structure will automatically possess an increasingly more complex consciousness as it grows by adding more material. That gives psychophysical emergence a structural origin and seemingly provides an explanation for the existence of psychophysical laws as a natural consequence of such emerging structures. While that may sound like it could address the problem of psychophysical emergence it does not, it only shifts it, as those structures cannot explain either how they give rise to states of consciousness.

Emergentism is a philosophy of mind asserting the existence of emergent properties caused by just such structural differences in the specific arrangements and organization of the components of a system. This organization is a system property and not a property of any of the individual components. Because of that, the individual components by themselves do not possess any of the emergent properties of the system they are a part of. That shields emergentism from panpsychism but only to an extent. Emergentism has a multilevel worldview that envisions a progression of levels with increasing system complexity. Higher-level systems in such a hierarchy may not only include simple components that themselves possess no emergent properties at all. For instance, what precludes the existence of higher-level systems that also include lower-level systems with their own emergent properties? Clearly, the way our world looks like, many of the things we find around us include as components subsystems with their own considerable complex organizations. Therefore, the question to ask is where exactly in this hierarchy of levels is the line below which all things are only composed of simple things that have no system-like properties. For that reason, while safe from panpsychism, emergentism may still be compatible with a much more pervasive psychism than with what most philosophers would be comfortable.

Not just for professional philosophers but also for many in the broader scientific community well into twentieth century, emergentism seemed like a reasonable explanation for biological and chemical system properties, which today we understand on a detailed physical level but back then had no explanation rooted in the known laws of nature. Emergent properties giving rise to such things as consciousness, and more specifically the forces emergentism claimed to be associated with them, seemingly provided a rational for what science at the time could not explain. Emergentism contended that these special forces about which science knew nothing governed emergent properties in the world of phenomena just how the laws of nature and its forces explained the physical world. Until scientists for example discovered how the properties of salt with its chemical formula NaCl came about from the combination of a sodium (Na) atom and a chlorine (Cl) atom, it was not known how salt could be non-toxic to the human body while Na by itself and Cl by itself were each toxic to the human body. According to emergentism,

organizing Na and Cl into NaCl gave rise to an emergent property that made NaCl non-toxic to humans. Before the physics discoveries of the early twentieth century, claims like that may not have raised many eyebrows and other much stranger implications of emergentism were of greater concern. An example for the latter is the claim that the atoms in a human body, as a human being moves around, are moving the way they do because of emergent forces. Essentially, such an assertion amounted to the human body moving about because the human mind wills it to move, rather than any underlying natural forces accomplishing such movements. This view seemingly endorsed the strange implication that the atoms constituting a living human body are subject to different forces than the atoms in a dead human body.

As science progressed, for evermore of the biological and chemical system properties, which did not have detailed physical explanations earlier in the twentieth century, scientists eventually were able to give explanations in terms of physical forces and laws alone. With that, the perceived need for the sciences of biology and chemistry to include emergent properties and their purported forces vanished. At that point, emergentism began to assert that its explanatory power, while not required in biology or chemistry anymore, was still necessary to explain phenomena for which psychological or social sciences had no adequate explanations rooted in the natural sciences. However, why emergent properties and associated forces should be in a better position to provide explanations in psychological and social sciences, after they failed in biology or chemistry, remains an open question. Emergentism just as epiphenomenalism continues to struggle with the problem of psychophysical emergence. Because the emergent properties in emergentism require associated emergent forces in addition to the known natural forces, emergentism also has to confront the problem of overdetermination. Proclaiming the existence of emergent forces, acting on phenomena by mirroring how the forces in our natural world act on matter, brings emergentism also in conflict with the third principle in tab. 2.1, the completeness of the laws of nature.

Substance dualism represents the strongest dualistic stance as it also presumes a property dualism. Then come the property dualism philosophies of mind that we highlighted under the umbrella of dual attribute theories. Finally, there is predicate dualism, a weaker form of dualism than property dualism, which also falls under the umbrella of dual attribute theories, the attributes now being predicates rather than properties. We all are familiar with predicates as we use them in our everyday language to make statements about practically everything. This is how we describe the world around us, using the tool of language. Predicate dualism is a theory arguing that it is impossible to couch our psychological experiences in terms of physical predicates of natural languages alone. Put differently, our psychological experiences are not reducible to physical predicates, as they are for example with respect to water and H_2O. Reducibility would demand that types of mental states / predicates and types of physical states / predicates relate in such a way

that all mental predicates expressing our psychological experiences carry no information that philosophers could not express without them through physical predicates only. If such a general reducibility exists, we actually arrive at a monism philosophy of mind position that falls under the umbrella of reductive physicalism, which we will come to in the next section. Predicate dualism maintains that such a general reducibility does not exist and therefore mental states / predicates are essential for a full complete description of our world.

Our sciences provide an illustrative example of what predicate dualism is about. When we look at the so-called exact sciences, like mathematics, physics, chemistry, biology or the engineering sciences, the language those sciences use is for the most reducible. That allows those sciences to decompose complex structures into more simple ones, essentially models of certain aspects of our world of which they then can attempt full physics and mathematics descriptions. Other sciences, however, such as philosophy, psychology, history, economy, arts or social sciences in general are constrained by their languages being mostly irreducible. While from a language point of view they use a unique terminology specific to the respective science, the meaning of those terms in each instance frequently differs with the result that the use of a term in one case can have a different meaning than the use of the same term in another case. Unlike with water and H_2O, the meaning of such terms is not interchangeable and requires explanation. Like for example, when a historian refers to an event as a war, an insurrection or a revolution. Almost every time those terms are used, they describe something different and are irreducible to a single meaning that would make us immediately understand the difference between various revolutions. To understand for example the differences between the American Revolution vs the French Revolution we need an explanation what the term revolution in each case really meant. Predicate dualism argues that, in a similar way, our psychological states require a description using mental predicates because they are not reducible to something which we can adequately understand by using physical predicates only. Predicate dualism thus resembles a dualist form of a non-reductive physicalism.

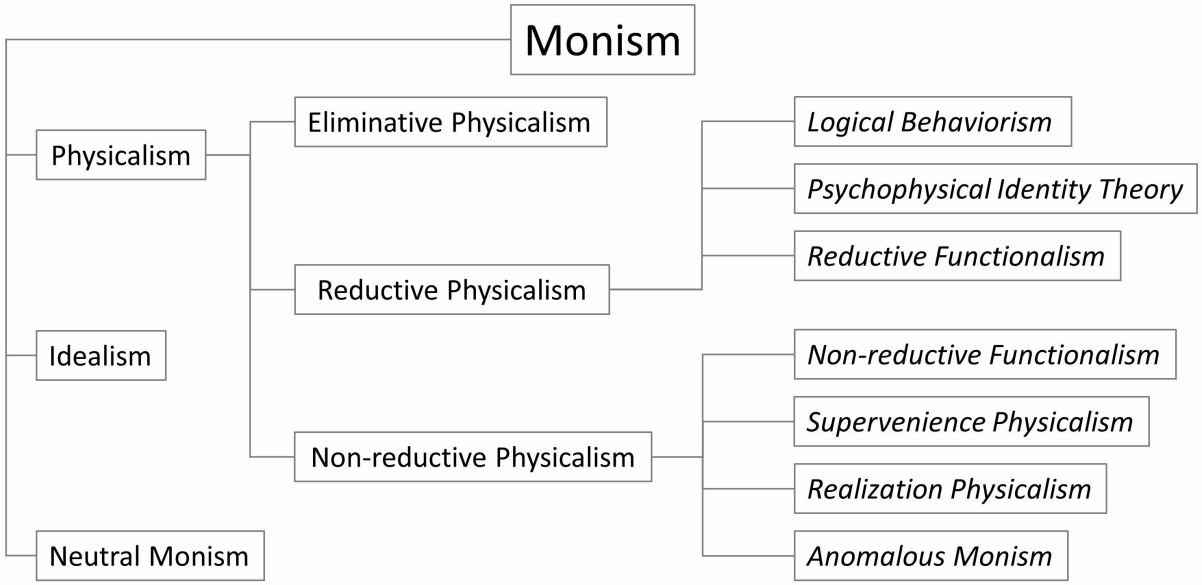


Figure 2.3: Simplified schematic of monism philosophies of mind. The left side of the diagram shows monism philosophy of mind category types and the right side lists examples of more widely discussed monist positions in the history of philosophy of mind referred to in the text.

In philosophy of mind there are just as many flavors of monism as there are of dualism. Where dualism philosophies of mind assert a duality of body and mind in various forms, monism philosophies of mind assert various attributes of oneness. These monism philosophies of mind differ from each other in what and how they define that oneness to be. Because mental causation is fundamental as to how all of us experience the world we live in, very few philosophers would consider solving the problem statement summarized in tab. 2.1 by discarding the second principle, mental causation. To different degrees, most of them rather reject the validity of the first principle in tab. 2.1, which states that mental states are not physical states. Not many monism philosophies of mind do this however to the extent of denying the very existence of mental properties. Therefore, the challenge most monism philosophies of mind face is to explain just how mental properties come about in some way through physical states and properties. Alternatively, those taking the opposite position must answer as to how physical properties can come about through mental states and properties.

To do this, such philosophies can use a number of concepts to achieve a desired state of oneness. One potential approach is to reduce one kind of property or state in such a way as to become a subset of another property or state. For example, one could claim all mental states to be physical states and mental properties coming about through physical properties. It is also possible to take the diametrically opposing position by claiming

that all physical properties are mental properties. Taking this view has of course quite different implications. In the extreme, such reductive philosophical perspectives can even turn into elimination. This can lead on the one side to some philosophers denying that mental properties exist at all and physical states are everything that there is, while others may assert that everything is mental and the physical reality around us exists only as far as we experience it.

The problem for those claiming that everything is physical and mental properties do not exist then becomes how to account for our experience of mental properties, how to address mental causation. Just consider for example how we as human beings without mental causation then could move various parts of our bodies in intentional and purposeful ways, as we all certainly do. Other problems arise of course if one denies that physical properties do exist and claims that everything there is, are exclusively mental properties. Different from those extremes, other philosophical positions of monism doubt neither the existence of mental properties nor the existence of physical properties and states. They then face the challenge to explain the physical reality of mental state experiences, how mental properties can arise from physical states or properties and what underlies mental causation. To address such challenges, twentieth-century philosophers introduced the concept of so-called functional states. These functional states possess inputs and outputs connected via physical states in such ways as to produce certain mental properties and with that our lived mental experiences, what philosophers of mind refer to as qualia.

The schematic in fig. 2.3 lists the monism philosophy of mind category types on the left hand side of the diagram. The right hand side of the diagram reflects the spectrum of the most prominent monism philosophy of mind positions falling into those categories. There are three major category types of monism philosophies of mind. Idealism, also called mental monism, claims that everything is mental. Neutral monism claims that everything is neutral, with mental and physical states being different flavors of neutral states, which can have either property, mental or physical. Finally, there is physical monism or physicalism, as philosophers usually refer to it, claiming that everything is physical. In physicalism, everything derives from physical states. As discussed earlier in this chapter, the first physicalism philosophies we know of date back to the Classical Period of Greek culture, to the philosophers Leucippus and Democritus. However, although physicalism has been around for a long time, today's physicalism theories of mind date their origins only a few hundred years back, as they really sprouted from the tree of scientific revolution, which began to grow in the late sixteenth and early seventeenth century. Starting with the 1960's, the philosophies of mind referred to as functional physicalisms or functionalism have become the dominant variety, not only among monism philosophies of mind but within philosophy of mind in general. Dominant among these are philosophies of mind of the non-reductive functionalism type and modifications of it. For many philosophers, they remain the default approach to address the challenge of how to solve

the philosophy of mind problem as articulated by the first three principles in tab. 2.1. Idealism as well as neutral monism trace their roots back to the eighteenth century with neutral monism becoming more prominent in the early twentieth century. However, later in the twentieth century, as physicalism philosophies of mind became mainstream, interest in idealism and neutral monism theories of mind waned. Here we will look first into idealism and neutral monism philosophies of mind and then turn to physicalism philosophies of mind.

Most will likely associate the philosophy of idealism with the German philosopher Immanuel Kant (1724 – 1804) and his transcendental idealism. Formulated in his *Kritik der reinen Vernunft* (*Critique of Pure Reason*) published in 1781, Kant's transcendental idealism is a conceptual idealism. In Kant's view, the way we consciously perceive objects is not as they are in themselves, but only as they appear to us through our sensory organs. All we ever can know of something is its appearance but not its true nature. This of course poses a problem with things such as space, time or causality. Neither of them is apparent to us directly through our senses but their realities clearly shape how we perceive and understand the world around us. In Kant's view, our ideas of such concepts shape our worldview, which is why philosophers refer to this position as conceptual idealism. According to this view, it is through our minds possessing innate concepts and categories such as space, time and causality that we can understand the world around us. With respect to philosophy of mind, the idealism of interest is however not a conceptual idealism such as Kant's, but an ontological idealism. The latter asserts that everything has an explanation in mental terms. For ontological idealism, essentially all physical states are mental states. While sounding similar, this is different from any dualist philosophy of panpsychism discussed earlier. Panpsychism claims that everything has mental properties but not all properties are mental properties as physical properties still exist. Not so for ontological idealism where there are only mental properties but no physical properties. Different from Kant's conceptual idealism where we perceive the reality of things, although not their true nature, through our senses, in ontological idealism everything we ever perceive through our senses are our own ideas or experiences. Strictly speaking, if you are a believer in ontological idealism, the outside world around you has no reality. For most of us, the claim that objects in the world around us do not exist without ourselves but only exist through our ideas and experiences is contrary to our common sense understanding of the world around us.

The most prominent representative of ontological idealism was George Berkeley (1685 – 1753), the English-Irish philosopher and Anglican Bishop of Cloyne, Ireland. Berkeley's philosophy grew out of British empiricism, which can trace its roots back to Francis Bacon in the early seventeenth century and came to prominence with the English philosopher John Locke (1632 - 1704) in the second half of the seventeenth century. Locke's primary interest was to understand how we can have ideas and where they come from. According

to Locke, we are born without any kind of innate concepts or knowledge and the human mind is initially a blank slate. He thought that all humans can learn in their lives, and hence all they can know, derives from experiences. These experiences come to us in their primary forms through sensory perceptions of our world. Onto those basic experiences, our mind layers reflections regarding those experiences and their associated sensations, in essence generating second level experiences derived from our basic first level experiences. Of course, reflections do not have to stop there as we can also reflect on those reflections. In this model, all of our ideas derive from experiences, either directly through the sensations they generate or indirectly as reflections on such experiences and sensations. For Locke, what we refer to as our knowledge about things equated to our internal perception of ideas we have about things, as they agree or disagree with our experiences of the world around us. In that sense, there is a limit to what humans can know as we can only know what we experience, directly or indirectly.

Berkeley disagreed with Locke that we can only know about things in our world as experiences about them come to us through our perceptions of them. In essence, Berkeley's critique of Locke was about a disagreement over what can and cannot exist outside the human mind. That in turn depended for Berkeley on how perceptions come to us. As sensory experiences of external objects such as trees, houses or mountains, which have their own independent reality whether we perceive them or not; or through the perception of our ideas about such things which may or may not possess their own reality; or a combination of both. Berkeley believed that all what we experience as humans are perceptions about our own ideas of things, things only exist, as we perceive them. Essentially, philosophies of mind such as Berkeley's ontological idealism claim that everything is mental and trees, houses or mountains do not exist without us perceiving them. There are of course many problems with such a worldview and likely, very few would today subscribe to such a belief. To some extent, it may have been his own religious belief that could have motivated Bishop Berkeley to take such a position as it certainly is much more amendable to a belief in the existence of a god we cannot perceive. If all things around us only exist as perceptions of our ideas about them, then the existence of a god we perceive in our ideas has the same reality as the existence of trees, houses or mountains around us.

Different from ontological idealism which claims everything is mental, neutral monism claims that everything is neutral and what we refer to as mental or physical properties are realizations of neutral properties satisfying certain conditions. Some point to Baruch Spinoza as the first to articulate forms of neutral monism. As discussed earlier, Spinoza proposed the existence of a neutral substance with no specific property and not existing in a specific mode. This substance he assumed to possess two attributes: the totality of all mental states and the totality of all physical states. Hence, he was arguing a dual attribute theory rather than a neutral monism. Another philosopher often thought

to possibly entertaining neutral monist conceptions was the British empiricist David Hume (1711 – 1776). However, many disagree with this contention and most think of neutral monism as a modern-day twentieth-century development. Among the first to articulate such a position were the Austrian physicist and philosopher Ernst Mach (1838 - 1916), the American philosopher and psychologist William James (1842 – 1910) and the British philosopher and mathematician Bertrand Russel (1872 - 1970) who became its most prominent representative. With philosophies of mind of the neutral monism kind, the first and the second principles in tab. 2.1 can hold true but not the third one. In neutral monism, mental states differ from physical states with each of them qualifying as such states by meeting specific conditions that result in the neutral property becoming a mental or physical property. There is nothing that keeps mental states from causing physical states, as this simply becomes the equivalent of instances of neutral properties having causal relationships where the first instance satisfies the conditions necessary to qualify as a mental state and the second instance those of a physical state. The problem neutral monism faces is that it cannot point to what the neutral state or property is and in that sense it violates the third principle in tab. 2.1.

The explanatory strength of neutral monism, rooted in the neutral category being more general than either mental or physical categories, is also its weakness. Neutral monism cannot identify what this neutral category really is. It can only specify the conditions, which convert the neutral qualities, occurrences and events into the respective mental or physical qualities, occurrences or events. However, any such explanations face their own challenges. Bertrand Russel for example framed what qualifies a neutral event as a mental or physical event in terms of psychological and physical laws. Thereby, a neutral event qualifies as a physical event if it relates to other neutral events through physical laws. Similarly, a neutral event qualifies as a mental event if it relates to other neutral events through psychological laws. However, this explanation of what qualifies as a mental or physical event cannot account for how mental events can relate to physical events and hence there is no explanation of how mental events / states can cause physical events / states which raises the question as to how exactly the second principle in tab. 2.1 can hold true. Hence, unless one has an explanation of how neutral states of the mental kind can relate to neutral states of the physical kind, neutral monism as Russel interpreted it violates the second and the third principle in tab. 2.1.

With that we come to the last of the three main groups of monism philosophies of mind, physicalism. Physicalism theories of mind have not only become the dominating monism philosophies of mind but also the prevailing philosophies of mind. Physicalism theories of mind claim nothing less than that everything is physical in origin and nothing exists that does not have a physical origin. All there is, are physical states and nothing like mental states exists. It is no coincidence that physicalism carries the word physics in it as its modern forms in essence mirror the evolution of physics since the beginning of

the scientific revolution. To understand the reasoning behind physicalism we need to look towards how the various sciences explain the world around us. There is physics as the fundamental science and there are the so-called special sciences. Special sciences, different from physics, have to use descriptive languages to explain the world around us, as for example we have no physics explanations for what we describe under the science umbrella of psychology, not to speak of being able to give physics explanations of what underlies beliefs or feelings.

Not considering mathematics, this constraint applies at some level to all sciences other than physics and not just to psychology. It just so happens that for chemistry and increasingly also for much of biology we now have physics descriptions underlying those sciences, whereas for psychology we are still far from that. Mathematics is a special case because for all we know it is the language that allows us to articulate the laws of nature in a scientific way and it is a universal language in the truest sense. Therefore, one could make the argument that describing something correctly in terms of physics requires us to be able to give an accurate mathematical description of the respective physics involved. Hence, the claim philosophical monism could make is that everything in essence is mathematical rather than saying everything is mental or everything is physical. Some indeed do take such a position. However, we will defer reflecting on this for now and rather focus the remaining discussion on the basic assumptions in physicalism philosophies of mind, specifically with respect to how they differ in their views of the explanatory powers of physics and the special sciences.

With respect to the differences between physics and the special sciences, physicalists take three distinct positions: eliminative, reductive or non-reductive. The eliminative position is the oldest one. Those who take this position as their starting point claim that the special sciences do not describe anything possessing a reality of its own. When we use terms such as belief or words describing feelings like love, happiness or sadness, an eliminative physicalist will say that such things possess no reality at all and describe something that as such does not exist. To many this will sound like an extreme position, but we should consider what motivates it. In the early stages of the scientific revolution, quite a few natural scientists continued to use non-physical descriptions to explain physical phenomena because the underlying physics was just unknown. Those were not any obscure natural scientists, mind you. Some of the greatest scientific minds remained beholden to pre-scientific theories, not excluding for example an Isaac Newton, better known for his works in physics than for his endeavors in alchemy.[24] In Newton's time, alchemy was still a respectable activity for a natural scientist. Many more things still were mysterious then, a good example being the nature of burning. Before the discovery of oxygen and its role in combustion, scientists assumed the existence of something called phlogiston, which combustible objects possessed and released through burning. Once it became clear that the oxygen in the atmosphere feeds a burning flame there was no more

need for a phlogiston. Oxygen was not phlogiston and something like phlogiston obviously never existed in the first place. The eliminative physicalism claim is that just as phlogiston was never real we will eventually discover that things, which we describe by using the terminology of our special sciences do not really exist. That includes of course everything to which we refer in our daily lives when we engage in psychological discourse which all of us inevitable do all the time. Like when we for example talk about politics or economics or when we for example share our latest romantic encounter or painful experience with our family or with friends. When we do so and talk about things such as love or hate, happiness or sadness, eliminative physicalism claims that we talk about things that do not really exist.

The physicalist worldview reflects our modern science perspective which simply presumes that as science progresses, we will have an increasingly better physics description of everything and eventually we will be able to know everything in physical terms. The eliminative physicalist will claim that once we know everything in physics terms, we will not need the special sciences anymore since what they describe does not even exist. Hence, eliminative physicalists have no need to explain how physical and mental phenomena relate to each other because they deny the existence of the latter. This is different for reductive and non-reductive physicalism. The reductive and the non-reductive worldview disagree with the eliminative physicalist position that what the special sciences describe does not exist. However, reductive and non-reductive physicalism hold different views with respect to the reducibility of the special sciences. While reductive physicalism agrees that what the special sciences describe exists, it also maintains that eventually we will not require the vocabulary and the languages of the special sciences anymore, as one day, we will be able to express everything in the language of physics for which we now use the descriptive languages of the special sciences. Non-reductive physicalists disagree with the latter claim and assert that even though everything will have a full description in the language of fundamental physics one day, we will continue to require the languages of the special sciences as they satisfy an explanatory human need that fundamental physics cannot address.

Reductive and non-reductive physicalists therefore agree that we currently need the special sciences because they provide us an interpretation of the world that physics cannot give us. However, reductive physicalists believe that this is only a temporary situation whereas non-reductive physicalists believe that we will continue to need the language of the special sciences even once we should have a complete understanding of their content in the language of fundamental physics. The three different physicalist perspectives echo the evolution of science with the rise of eliminative physicalism associated with an earlier and more optimistic phase of modern science. Reductive physicalism emerged as it became increasingly clear that the languages of special sciences carried explanatory powers which physics could not simply replace, even though in the end we would be able

to articulate everything in the language of physics. Finally, non-reductive physicalism became the dominant philosophy of mind as more and more philosophers concluded they had to acknowledge that we would continue to require the language of the special sciences alongside the language of physics to articulate our human understanding of nature.

Physicalism is a rather new term in the history of philosophy, introduced in the 1930's by the Austrian philosopher Otto Neurath (1882 – 1945) and his German colleague Rudolf Carnap (1891 - 1970). Both were members of the so-called Vienna Circle, a group of people we will encounter again in the next chapter when we look into the making of psychology. Before that time, most would have referred to a theory of physicalism as materialism as many still do today. While there are many commonalities between physicalism and materialism there are also differences, for example when it comes to the definition of what actually constitutes matter. Modern physics has very much redefined the meaning of matter over the past century from the simpler understanding of matter with which materialism started out long before that. Here we will stick with the term physicalism while making clear that one can find eliminative physicalism, reductive physicalism or non-reductive physicalism also referred to under the labels of eliminative materialism, reductive materialism or non-reductive materialism. The latter just being a reflection of the fact that many physicalists continue to self-identify as materialists.

At face value, we should expect that all forms of physicalism have no problem with the first principle in tab. 2.1 holding true, as their basic assertion is that mental states do not exist and hence cannot be physical states. With mental states not existing, one could expect that the second principle in tab. 2.1 cannot hold true because something that does not exist, mental states, cannot cause something that does exist, physical states. Of overall importance in physicalism is to show that the third principle in tab. 2.1 holds true in the sense that a complete explanation in terms of physical laws exists for all of our mental experiences, as mental states do not exist and all mental phenomena must have complete physical explanations. As we will see, some variants of physicalism have found ways to redefine mental states such that modified versions of the principles listed in tab. 2.1 can hold true.

We will begin with a look at eliminative physicalism. This variant of physicalism relinquishes principle (2) in tab. 2.1 while principles (1) and (3) holds true, just as indicated in the above paragraph. It does so by declaring our belief in the existence of mental states as rooted in folk psychology. Eliminative physicalism looks at folk psychology, the common sense psychology we use in our daily lives to interact with each other, as an empirical theory, which assumes among other things the existence of mental states. However, eliminative physicalism claims that such folk psychology has no scientific basis and will eventually be replaced by a scientific theory which will have no need for any such assumptions that we seem to require to make sense of our every-day life experiences. According to eliminative physicalism, folk psychology is a false theory and we will

discard it completely once we know all of physics. Nothing in a finally correct scientific theory will be a reduced version of anything that folk psychology postulates as it only represents a collection of practical constructs we use to get through our daily lives, but those constructs do not refer to anything that exists in reality. Hence, whatever mental states refer to does not exist and a final scientific theory will eliminate them. Since eliminative physicalism emerged in the 1960's, science has progressed in many areas and today we know much more about many of the things philosophers could only speculate about back then. Neuroscience and cognitive psychology, just to pick two examples, have very much changed our understanding of human neurophysiology (see the previous chapter) and human nature (see the next chapter). What we are increasingly learning is that the language of folk psychology, while certainly pre-scientific, does after all make use of mental terms and concepts that have a real scientific underpinning.

Different from eliminative physicalism, reductive physicalism does concede that concepts we use in folk psychology, like when we for example refer to something as a mental state, can possess an underlying physical reality, but a physical reality only. Reductive physicalism, with its three major forms of logical behaviorism, psychophysical identity theory and the reductive version of functionalism shown in fig. 2.3, seeks to reduce descriptions of mental state types to descriptions of physical state types. Some refer to logical behaviorism also as semantic behaviorism because it equates mental terms with physical terms through describing behaviors. Logical behaviorism claims that each mental state type is identical with physical state type behaviors, which are accessible to observation and inspection by anyone without any specific training. The implication of this a priori identity of mental state types with physical state types is that any mental term carries only insofar meaning as it is identical with physical terms and no mental terms exist that describe anything else but physical terms. With that, logical behaviorists could conclude that all statements using mental terms are reducible to statements using only physical terms.

This approach allowed logical behaviorism to avoid the problem of other minds as it claimed the only way we could know something about mental states was through the observation of behaviors. That however created another problem. How can we know about our own ideas, beliefs or desires if they are nothing else but different kinds of behaviors? Seemingly, to know about our own ideas, beliefs or desires, we would have to observe ourselves in the same way we observe others. That seems ridiculous, as all of us have the experience that we know these things without having first to observe ourselves. The challenge which logical behaviorism encounters here is a problem with our so-called first-person authority. First-person authority is a term that philosophers use to refer to the privileged access we have to introspection of our own mind vs what we can know about other minds. Put simply, in the above context, we should have a privileged access to our own ideas, beliefs and desires, as compared to those of others. Logical behaviorism

essentially denies this. For a few decades in the last century behaviorism flourished but by the 1950's logical behaviorism in philosophy of mind as well as methodological and radical behaviorism in psychology were in retreat. By the 1960's logical behaviorism had waned in philosophy and what survived from behaviorism in psychology were practical aspects of some of its methodologies still in use today. The failure of behaviorism was a consequence of the unraveling of the broader project of logical positivism of which behaviorism had been a part.

Different from logical behaviorism, which asserts the existence of an a priori identity of mental state types with physical state types, psychophysical identity theory says that we establish this identity a posteriori through science. As science progresses, so psychophysical identity theory claims, we will learn more and more about human neurophysiology and with that, we will increasingly be able to identify types of mental states with types of physical states. The type identity in psychophysical identity theory is however much narrower than in logical behaviorism. Psychophysical identity theory claims that each mental state types is identical with a physical state type and not with physical state type behaviors. Hence, according to psychophysical identity theory, a mental state type is identical with a certain brain state type. Every time such a brain state type occurs in anyone of us, it will generate the very same mental state type, in each and all of us.

Psychophysical identity theory relies on the progress of science in helping us to establish the identity of mental state types / terms with physical state types / terms. The identification of water with the chemical substance H_2O is an often-cited example for that. All of us know what we mean by water, what the word water describes. All of humanity that came before us possessed this same kind of knowledge about what water is. However, only with the rise of modern science did we discover that water was identical with the chemical substance H_2O. Psychophysical identity theory claims that given time, we will discover all of the underlying physics for which we now us the language of other scientific or pre-scientific theories, just like it happened with water and H_2O. A major challenge to type identity theories is the argument, supported by most philosophers, that mental state types can have multiple physical state type realizations. However, rather than claiming a type identity between mental and physical state types, psychophysical identity theory could just as well claim a token identity. In such a token identity, each occurrence of a mental state type is identical with the occurrence of a physical state type with the caveat that the occurrences of physical states of different types can realize the occurrence of mental states of a certain kind of mental state type. To address the challenges facing psychophysical identity theory, philosophers have also considered property identity, which is an even weaker form of identity than token identity.

All variants of reductive physicalism share a hierarchical multilevel worldview consisting of domain level descriptions of our world from different science perspectives. At the top level, we find descriptions and explanations from a social science perspective followed

by psychological and biological sciences descriptions all the way down to the level of fundamental physics. The contention of reductive physicalism is that eventually all higher levels in this multilevel view will sequentially collapse. Like us one day understanding through the language of psychology, what we describe today in the language of other social sciences. In turn, we will one day be able to reduce to the language of biology what until then we had to describe in the language of psychology. This reduction will go on until we eventually will be able to describe everything in the language of fundamental physics.

Functionalism and physicalism are easily confused as one finds those labels often used interchangeably, with functionalism replacing the label physicalism or functionalism being equated with physicalism, specifically with its non-reductive version. However, that is a mistaken view as in principle functionalism can also support dualist philosophies of mind and it certainly can accommodate a reductive as well as a non-reductive physicalist point of view. Functionalism came to dominate philosophy of mind only over the last few decades of the twentieth century but one can find functionalist perspectives in philosophy as far back as Aristotle and in the next chapter, we will briefly touch on another kind of functionalism in nineteenth-century psychology. The functionalism in philosophy of mind endorses the following general principles which specific versions then tweak to their needs, such as reductive and non-reductive physicalism do.

i. Mental states are functional states.

ii. The causal relation connecting its input to its output defines a functional state.

iii. From premises (i) and (iv) follows: Causal relations connecting their specific root causes (inputs) with their specific actions (outputs) define mental states.

iv. Each causal relation requires an implementation, the realization of which can be physical or not and multiple realizations of a given causal function are possible.

According to (iv), not only physical but also non-physical entities such as souls could realize causal relations representing mental states. Therefore, theoretically, functionalism could just as well support dualist philosophies of the kind where mental states can cause physical states and vice versa. However, that does not seem an attractive option and hence, all versions of functionalism assert that the above defined functional mental states are realized through physical states.

It is important to realize that functionalism defines mental states / functional states as second order properties as they derive from physical states, which do not derive from anything else and hence are first order properties. By differentiating between state types of first order, the physical states, and state types of second order, the mental states, which

are functional derivatives of first order states, versions of functionalism endorsing multiple realizations can in principle claim that types of mental states are different from types of physical states. This works because multiple realizations for each functional mental state type make it impossible to establish a type identity between mental and physical state types. This is exactly what non-reductive versions of functionalism assert and by doing so they can claim that for them principle (1) in tab. 2.1 holds true in the sense that mental state types are not physical state types. Similarly, principle (2) in tab. 2.1 holds true for reductive as well as non-reductive physicalism forms of functionalism in the sense that mental state occurrences / events can cause physical state occurrences / events; the causality relation connects occurrences / events and not state types. In the same way both, reductive and non-reductive physicalism forms of functionalism can claim that principle (3) in tab. 2.1 holds true if it only applies to occurrences or events of physical states. Therefore, by referring to state types and state occurrences or events, principles (1), (2) and (3) in tab. 2.1, can hold true for functionalism without violating principle (4).

Reductive physicalism versions of functionalism are variants of a limited type identity physicalism. Such models avoid the challenge discussed above for broader variants of type identity physicalism such as psychophysical identity theory. To illustrate this with a popular example, they can do so by defining a functional state type such as pain in humans as being different from functional state types of pain in other beings. In such a way, these reductive versions of functionalism seek to restrict the validity of the multiple realization argument by claiming that type identity can hold true on a more limited or local level; such as for pain among humans vs pain among other beings, an example of which would be the proverbial extraterrestrial species from a faraway world.

Non-reductive physicalism forms of functionalism have become the most widely supported philosophy of mind models. Non-reductive functionalism endorses the multiple realization argument. Hence, to stay with the above example, in non-reductive physicalism forms of functionalism, functional states of pain are identical for humans and any extraterrestrial beings we have not met yet. This holds true regardless of such other life being carbon based just like us, or completely different, such as a silicon-based lifeforms. The latter seem to be a science fiction favorite for many, maybe because we are so much enthralled by our computers. It is no coincidence that non-reductive functionalism rose to a prominent position in philosophy of mind as the introduction of computers began to change our world in the 1960's. We have to remember that long before the first personal computers became available and affordable for regular people in the late 1970's and early 1980's, academia and industry had already been using computers for two decades. In the 1960's, the American philosopher and computer scientist Hilary Putnam (1926 - 2016) began to argue that a functional state type does not have to be identical with a physical state type, which is what type-identity theory claimed, but that functional states could be realized

by multiple physical state type implementations. The advantage of this position, which defines a functional state type only through its cause (input) and effect (output), lies in the fact that one does not need to know anything about the underlying physical reality of a functional state type. In this way, that which we refer to as the mind can be abstracted from the body, its physical realization. Hence, there is no need to confine mental states of the mind that we believe humans possess to the human body as they surely could just as well reside in a machine or in an extraterrestrial being. In computational theories of mind such as the machine state functionalism Putnam initially proposed, functional states are defined by their inputs and outputs and how they relate to each other. This may sound like a description of a programming language that could be executed to run on whatever suitable hardware and that is exactly what machine state functionalism represents. The ground for a philosophy of mind such as Putnam's machine state functionalism to emerge had been prepared in the 1940's and 1950's. In 1950, the English mathematician and computer scientist Alan Turing (1912 - 1954) proposed what we call today the Turing Test, an intelligence test for a machine. Turing suggested that if a judge conversing with two counterparts, one of them a human and the other a machine, through voice or text only, without being able to see them, could not tell the difference between human and machine, then the machine passed the test. The functional states in a Turing Test, when the judge for example asks a question (input) and waits for an answer (output), correlate inputs and outputs in the same way as functional machine states do.

Non-reductive functionalism was from its very beginning keenly interested in computer and cognitive sciences and remains closely connected to them today, artificial intelligence and cognitive neuroscience and psychology being examples of that. According to some philosophers such as the American Jerry Fodor (1935 - 2017), the human brain essentially resembles a computer and its functional states are the equivalents of computational states. If that is so, how then can those computational states bring about the mental state experience that we all have? From an architectural point of view, conceiving the human brain as a computer continues to receive much support today. The problem is not to find hardware analogies between the human brain and computers, as today we have a much better understanding of what this entails than any neuroscientist or philosophers could have had back in the 1960's. The much bigger challenge is to understand what the human brain software architecture looks like. What kind of language and logic does our brain use to connect the enormous number of subroutines in such ways that our brain works as it does? Instead of Fodor's proposal that our mental states come about through causal connections between mental representations of computational states in the brain, others suggested that it is the flexible neuronal network connections in our brain, which underlie the experiences of our mental states.

While postulating functional state types to be computational state types may seem natural in our day and age, there are also other possibilities. Biological functionalism is

one such alternative. Its starting point is evolutionary biology in the sense that each function in a living organism serves a purpose in ensuring the survival and procreation of a species. This is certainly evident for the physical functioning of an organism, its physiology. Biological functionalism asserts that this is no different for the mental functions of an organism. Hence, the objective for biological functionalism is to discover how mental states functionally serve an organism. The latter frequently boils down to answering the question as to how the functional implementation of a mental state contributes to the survival and the procreation of an organism. In essence, the claim of biological functionalism is that we have functional implementations of the mental states we experience because they serve evolutionary functions and have evolved as the human organism evolved over millions of years. In the narrow sense, this can apply to how mental state type functions helped with the survival of our species. In a broader sense, this can also include culture evolution as we witness it with the beginning of the Cognitive Revolution some 45,000 years ago. The human struggle to survive in small family groups in a vast wilderness surrounded by dangerous animal predators is long over. Today, humans struggle to survive in the jungle of our modern societies, which may be even more treacherous. It may take a while before we will be able to understand what this could mean with respect to shaping any functional implementations of mental states to help our species survive in this new environment.

Just as it is possible to claim that each function in a living organism serves a purpose in ensuring the survival and procreation of a species, one can make a similar but weaker claim with respect to the importance of social rule-following. This is what the so-called social functionalism does, which essentially rests on a version of Wittgenstein's argument against the possibility of a private language. According to the latter, we can only talk about things meaningfully because we have a common understanding of language, whereby everything we say carries a certain meaning. This common understanding comes about by all of us interpreting the expressions we use according to a set of rules we implicitly have agreed on, in essence our social rules. From this perspective, it is only through social rule-following that we can meaningfully talk about anything. This is not only the case when we talk to each other but also when we talk to ourselves. That is, in the sense that we talk to ourselves as reflected in the sequence of causally related mental states we experience as we do so. This is a different way of saying that we talk to ourselves as we think and vice versa. In this way, causally related mental states we experience become associated with functional social roles. Not everyone agrees that social functionalism is indeed a functionalism as for example, unlike with biological functionalism, the physical realizations of the respective functional roles are much less obvious.

The different outlook that reductive and non-reductive physicalism have on the relation between the special sciences and the fundamental science of physics requires an additional explanation by non-reductive physicalism about just how it can guarantee that the special

sciences indeed describe something that has a physical reality. The reductive physicalism claim is that what the special sciences describe, we will eventually be able to reduce to the language of fundamental physics. Hence, reductive physicalism carries the inherent assurance that what the special sciences describe indeed has a physical reality. Not so in non-reductive physicalism. Because non-reductive physicalism claims that the language of special sciences are non-reducible to the language of fundamental physics, there is no assurance that everything which the special sciences describe has indeed a physical reality. Hence, non-reductive physicalism needs to answer the question why everything the special sciences encompass does have a physical reality. Non-reductive physicalism has responded to this challenge in two ways, with the approaches of realization physicalism and supervenience physicalism.

Supervenience physicalism seeks to solve the problem of assuring that the special sciences do not include things that have no physical reality by claiming that mental properties supervene on physical properties. The challenge with this approach is that supervenience physicalism has to assure that this psychophysical supervenience it claims to exist is of the right strength. On the one side, it must be sufficiently strong so that a minute change in a physical property cannot result in a world with completely different mental properties. On the other side, it cannot be so strong as to imply the existence of one-to-one correlations between some mental properties and some physical properties, potentially resulting in a partially reductive position. Invoking psychophysical supervenience is not unique to supervenience physicalism or non-reductive physicalism for that matter. Either directly or implicitly, other philosophies of mind do so as well such as psychophysical identity theory, a reductive physicalism, or epiphenomenalism, a non-organismic dual attribute theory. So far, supervenience physicalism has not been able to provide an explanation of just how in its model mental properties supervene on physical properties in a different way than in other philosophies of mind.

Just like supervenience, realization is a concept philosophers use to indicate relationships between properties, how one property comes to realize another one. This includes simpler lower-level properties for which we frequently can quickly point to their realization through fundamental physics. However, it also includes the realization of more complex higher-level properties as for example the special sciences describe them through simpler lower-level properties. In the last decades of the twentieth century, realization physicalism became the most popular way to combine functionalism with non-reductive physicalism. Realization physicalism's claim is that what the special sciences describe are abstractions, higher-level properties, of things realized by lower-level properties, all of which are fully describable by the language of fundamental physics. In this sense, realization physicalism claims that what the languages of the special sciences describe contains only elements, which build on nothing else but the language of fundamental physics, however, without the special sciences being reducible to fundamental physics.

The last non-reductive physicalism philosophy of mind we will briefly mention here is anomalous monism. This philosophical position, which the American philosopher Donald H. Davidson (1917 - 2003) formulated in 1970, asserts that while mental events are identical with physical events, the mental is anomalous in the sense that the physical events we describe in mental terms are not subject to strict psychophysical laws. This is just what the adjective anomalous implies as it derives from the Greek word *anomous*, meaning lawless. Anomalous monism, instead of seeking explanations in terms of mental and physical states, focuses on mental and physical events and their causal relationships. It defines causality as a relationship between events and since the laws governing such causal relationships between events are of a physical nature only, mental events must be identical with physical events. Hence, anomalous monism represents a type identity theory of mind, an identity between types of events. Anomalous monism does so without claiming that psychophysical laws exist such as psychophysical identity theory does. Quite to the contrary, anomalous monism denies that such psychophysical laws exist. With that, the challenge anomalous monism faces is to explain how that which qualifies an event as a mental event seemingly does not manifest itself at all in the causal relationships of this event. That of course raises the question of how two mental events can causally relate at all, other than through invoking some form of epiphenomenalism.
The claim of physicalism is that if we know all the laws of physics, then all mental states will have complete physical explanations and no kind of non-physical explanation whatsoever will be required to understand mental phenomena. A major challenge for physicalism is to explain how different people can have different mental experiences when for all other matters they share the same physical experience. We all have the experience that our subjective view of things can significantly differ from other people's subjective views, and this can be about things as trivial as what color an object is, even though the color of the object is a physical fact that we all should perceive the same way. If we are all physically the same, we all should experience a given color the very same way, as physically we perceive it in the same way and our mental experience with respect to that color should be grounded in the same physical processes or states occurring in our brains. Is this really so or is this a mistaken perception? How can we know in the first place that the same happens in our minds when we experience the very same things? Most would likely respond that we could not know that. The challenge for physicalism then is twofold. It must explain on the one hand how we can have those different phenomenal experiences of which we cannot really say what they are because we cannot look into other minds. On the other hand, physicalists must ensure that all mental phenomena associated with those different experiences have full and complete explanations in physics alone.

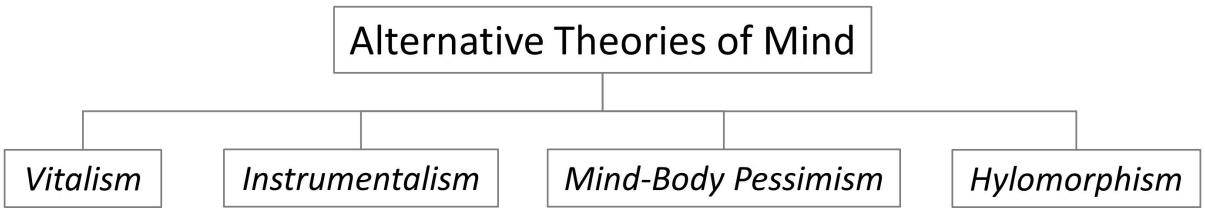


Figure 2.4: Alternative theories of mind.

What the heading of this section refers to as alternative theories of mind, are philosophies of mind which do not really fit into either the dualism or the monism category of mind. However, that does not imply that these alternative philosophies of mind of which fig. 2.4 lists the most prominent ones, do not make use of similar concepts or do not have a closer connection to the one or the other. Vitalism was in essence a reaction to the separation of soul and body Descartes bequeathed not just on philosophers but also on medical scholars and practitioners. The latter facing the problem of how a soul that seemingly could not interact with the body in any material way might control it. With respect to mind-body pessimism and instrumentalism, they are a in a way reflections on the evolution of modern science and what we have learned from it. This is also true for hylomorphism with the difference that long before its recent popularity as an alternative approach in philosophy of mind, much of natural philosophy before the scientific revolution entertained a hylomorphism worldview going all the way back to Aristotle. First, we will look at vitalism, then at mind-body pessimism and instrumentalism, before, after a necessary digression into the internalism and externalism concepts of mental states, discussing the hylomorphic perspective on the philosophy of mind.

Vitalism emerged in the late seventeenth century in reaction to Descartes substance dualism, which in the perspective of many had left the soul without any further use beyond a religious concept. There really were just two alternatives to deal with this situation. Of course, one option was to resort to a kind of materialism, in a sense early tributaries to that which in time would become the mainstream of philosophy of mind, physicalism. The other approach was to search for an alternative to the life-giving force, which Descartes had eliminated by separating the thinking mind, the soul, completely from the body. Until Descartes, a part of the soul, the so-called sensitive soul had played that role by intermediating between the body and the soul as the agent of sensory cognition. Vitalists, many of them medical scholars, clearly struggled with how the human body could function as a mechanistic construct or how living organism could be nothing else than automata.

Some of them, like the German chemist and physician Georg Ernst Stahl (1659 - 1734), resorted to what in its final consequences pointed towards animistic concepts. For Stahl,

while an immaterial substance, the soul existed only in and for the body. Without being the body, the soul was the cause of all the body's activities, physical and mental, and it is because of the soul that the body behaves as it does with seeming purpose. Hypothesizing about an immaterial soul in this way may seem strange to us today but we must not forget that in Stahl's time such argumentation was common, not just in the nascent medical sciences. Maybe it is not a coincidence that Stahl, being a chemist as well as a physician, also proposed the existence of a substance called phlogiston to explain the process of burning. Today, we know that neither phlogiston nor the immaterial soul, which Stahl envisioned as the life-giving force of organisms, exist. Different from Stahl, others like the French physician Henri Fouquet (1727 - 1806) invoked sensibility, essentially a kind of sensitive soul all animals possessed, as a solution to the problem Descartes had left them with. In Fouquet's case this material sensitive soul was very much conceived in the tradition of the corporal soul of atomists like Pierre Gassendi. With the help of sensitive soul concepts, physicians such as Fouquet sought to understand human physiological as well as psychic functions that otherwise they believed to be incomprehensible.

Medical scholars like Fouquet and his compatriot and colleague Jean-Joseph Ménuret (1739 – 1815) saw the proclaimed separation of soul and body as inconsistent with their medical observations. While as believers they submitted to religious dogma, as scientists they held very different views about soul and body. What they could observe in their medical practices required an explanation as to how organic functions and psychic activities related to each other as they evidently did. In Ménuret's case, this included for example the study of mental illnesses and the search for an understanding of the respective roles mind and body played in these so physicians could develop targeted treatments from a physical and a psychic perspective. Vitalist medical scholars were first of all physicians and only then philosophers. Such as the French physician Paul Joseph Barthez (1734 – 1806), who invoked a vitalist principle, which was neither identical with the thinking soul of Descartes nor with its body. Barthez, like Fouquet and Ménuret associated with the Montpellier medical school, introduced this vital principle in a 1778 publication to describe what he perceived as the cause of the phenomena of life. With this vital principle, he addressed a conflict that all scientists faced in his time, between what their science taught them and what religious custom required. Professing materialism and affording everything to the body would have offended religious belief and endorsing a spiritual position by affording everything to the soul would have ridiculed him as a scientist. In essence, Barthez vital principle amounted to a kind of pre-established harmony between body and soul. The progress of medical science and the ever-increasing understanding of animal biology in general and human biology specifically as well as of biological evolution during the nineteenth century, very much reduced the need for vitalist explanations. The accumulating scientific evidence eventually just left no more room for the existence of any life-giving force along the lines entertained by vitalism.

Instead of thinking about how one can solve the mind-body problems inherent to the philosophy of mind in some way, one can also take the position that these are problems humans are not capable of solving. This is the position that mind-body pessimism takes. According to mind-body pessimism, the challenge of coming up with a philosophy of mind for which the principles of mind outlined in tab. 2.1 all hold true is an artificial challenge. Our presumptions with respect to those principles may all hold true for a philosophy of mind truly reflecting all of our reality but we are not capable of comprehending such a philosophy of mind. What mind-body pessimism says is that such a philosophy of mind certainly can exist and some intelligent beings much smarter than humans could formulate and understand it but human cognitive capabilities are just not sufficient to achieve that. Humans are in this respect like children who, while living in a world shaped by quantum physics and gravity theory, cannot understand any such theories. Similarly, we all have the mental experiences we have and those mental experiences connect to our physical realities in the way they do but we cannot conceive of a theory of mind to explain mental phenomena and how they connect to physical phenomena; just like a child cannot formulate quantum physics or gravity theory. The claim that the problem of consciousness cannot be solved by humans in a sense turns consciousness into something mysterious and because of that, mind-body pessimism is also referred to as a new mysterianism.

Assuming that we will never be able to solve the hard mind-body problems makes for a pessimistic outlook on humankind's future. Should we not expect human evolution to continue and humans individually and as a species to become smarter? That human cognitive capabilities will stop to evolve contradicts what we have learned about the evolution of life since Charles Darwin's pioneering work. Science has evolved very much into a communal effort, which does not just rely on the occasional genius to advance our understanding of reality, even though that fortunately still happens. Hence, it seems wrong to assume that a breakthrough towards an understanding of consciousness will have to come from the efforts of solitary scientists and philosophers, each struggling on their own towards that goal. This is just not how science works and while we cannot even guess what the science of the future will be able to help us understand it is a good bet that it will be much more than we can imagine. Betting against evolution, in human cognitive biology as well as science is not a good foundation for an argument but that is what mind-body pessimism does. Mind-body pessimism does not provide a solution to the problems of philosophy of mind. Instead, it claims that such problems do not exist in the first place because we are trying to understand something that is beyond human cognitive capabilities. The solution which mind-body pessimism suggests to the problems of philosophy of mind is more like a trade where it asks us to relinquish our modern understanding of evolution and scientific progress for the belief in a new mystery.

Instrumentalism is a physicalist theory of mind, essentially a non-reductive physicalism with a twist. This philosophy of mind approach received its name from its unique interpretation of what constitutes a useful theory. Physicalism assigns a physical reality to all that which underlies a psychological theory, such as our concepts of mental properties and how they interact with each other as well as with physical properties. In this sense, theoretical concepts about mental phenomena in such physicalist theories have a physical reality. Not so in instrumentalism. It looks at the concepts and ideas such psychological theories use as instruments that either allow us to describe reality accurately or not. However, instrumentalism denies that such concepts and ideas must have a physical reality. As a philosophical position, instrumentalism traces its roots to the utilitarian outlook of empiricism where all that matters in a theory is its explanatory power. This also very much reflects the way modern science constructs theories. Just think about modern physics where string theories hypothesize a universe of ten dimensions or astrophysics assumes the existence of things like dark matter or dark energy. Neither have we any proof yet that those additional dimensions exist in our universe, nor have we discovered the existence of dark matter or dark energy yet. However, all of those things for which physicists have no existence proof are useful concepts in physics theories seeking to explain our universe as it is. As long as such theories can explain the facts of our universe and make accurate predictions with respect to its future, they are completely adequate and it does not matter from the perspective of an instrumentalist if additional dimensions, dark matter or dark energy really exist.

Instrumentalism claims that what holds true for physics theories must also hold true for psychological theories. The only way to assess the validity of a psychological theory is its explanatory and predictive power. We all acknowledge this in our everyday commonsense psychological discourses, the language of which derives from an empirical psychological theory we all unwittingly subscribe to, like when we converse about our phenomenal experiences. At that, we do not have to assume that mental states or properties are real when we talk about them, but clearly, concepts of mental states or properties are quite useful in having such a conversation. This is the strength as well as the weakness of instrumentalism. It does away with the need for ontological discussions about what does and does not exist, but it cannot tell us why any such ideas about mental constructs are so useful in psychological theories, other than that they do possess a physical reality.

Before we come to hylomorphism we need to briefly reflect on the position philosophies of mind take with respect to what philosophers call the inner mind. There are two positions, one argued by internalism and the other one by externalism. Internalism holds that mental phenomena are subjective experiences of our inner mind. Because those experiences are private, we cannot know about the mental states of other human beings, as they are not directly observable to us. All we can do is make inferences from the behavior of other humans with respect to their inner private states. We ever can only

hypothesize about other people's mental states from how they behave, which is exactly what we do in everyday life even though we usually do not think about that much. Like when someone is for example screaming, which can have various reasons. The respective person could be in a state of pain or could be extremely excited about something or just scream out of fun. With internalism we cannot even know if the people we meet actually have any of the inner states corresponding to those experiences or if they are just behaving that way without having any mental states at all. Admittedly, that is an extreme position but in internalism the existence of so-called qualia zombies, people who behave like they have such experiences but actually possess no mental states at all, cannot be excluded. Importantly, according to internalism, our experiences of mental states are our private affairs. In essence, our mental experiences are prisoners of our inner mind. In this way, they have no direct connection to the outside world around us but represent reflections of an inner world we live in. One could say that in internalism the mental phenomena we experience are akin to private conversations we have with ourselves about our internal perception of the world we live in.

Of the philosophies of mind we have looked at so far, all expect logical behaviorism presume the existence of an inner mind and therefore endorse some weaker or stronger form of internalism. Logical behaviorism claims that mental states are nothing else but behavioral patterns we can observe directly. Hence, logical behaviorism endorses a version of externalism. Simply put, externalism rejects the notion of mental states being captives of our private inner minds. Instead, it asserts that the inner mind hypothesis fails to account for the fact that what we take as our mental state experiences cannot rest on exclusive private and subjective definitions. In order for us to be able to communicate about things like love, desire or pain, we must have a non-private understanding of what kind of mental states those things refer to. In short, our understanding of those things is nothing our inner minds could conjure up without us communicating with other human beings. It just is not possible that our inner minds in their splendid isolation somehow magically agree on what we all mean when we talk about things like love, desire or pain without us being participants in the public discourse that defines those meanings. This understanding of and consensus about what those things mean depends very much on our social environments and interactions. It is through the latter that we arrive at a shared definition of what we mean by love, desire or pain. Of course, this position circles essentially back to Wittgenstein's argument against the possibility of a private language.

As a philosophy of mind, hylomorphism, just like logical behaviorism, endorses a version of externalism. However, hylomorphism is quite different from logical behaviorism. Logical behaviorism is committed to a reductive physicalism. It perceives the human mind as a black box and deems the observation of physical behavior as the only way we can learn something about what happens in this black box. Hylomorphism rejects physicalism and

its version of externalism embraces much wider and more complex behavior patterns, which hylomorphism more broadly refers to as structures. Seeing the world composed of matter (*hylo*) and form (*morphe*) is a philosophical perspective that originated with Aristotle. However, hylomorphism as a theory of mind has a much more recent origin. With respect to philosophy of mind, forms in hylomorphism essentially refer to structures and patterns. Just like in emergentism, structure is a central concept in hylomorphism. Both, emergentism and hylomorphism, claim that we cannot explain the world we live in without the concept of structure. However, they differ in their concepts of structure and hylomorphism rejects the notion that non-physical forces, the so-called emergent forces associated with emergentism we encountered earlier, are responsible for the emergent properties of a structure. Another characteristic of emergentism is that it only knows mechanical structures. Different from that, hylomorphism claims that in addition to mechanical structures in the physical world there also exist structures in the psychological domain.

The approach of hylomorphism represents an acknowledgement that as living beings, we are not islands unto ourselves. Instead, we live in complex social and physical environments rich in phenomenal and physical structures of which we are all a part of and which we must take into account in our descriptions of the mental and the physical. Just as every living creature in nature interacts with its environment in many various ways, the human capability to think represents a unique way in which our species has evolved to interact with our environment. The thinking capability of the human mind did not evolve in isolation from but in interaction with the environment, biologically and socially. In that sense, what we call mental states were never the exclusive preserve of inner minds and hylomorphism claims that psychological phenomena are not inner states of mind. Rather, they represent complex patterns of interaction humans have developed with their social and biological environments throughout their evolution. Because psychological phenomena are essentially patterns of interactions with our environments, we can know the mental states of others. This is something our human ancestors had to learn because otherwise it is unlikely that the human species would have evolved to what it has become today. It would seem that without being able to read the emotions, feelings or intentions of others we might not have progressed as far as we have. It is certainly doubtful if without being able to know the mental states of others, at least to some extent, we could have become the kind of human species whose members can live today in our modern societies with their densely populated mega-cities. However, just as with any kind of human pattern recognition, our ability to know the mental states of others has its limitations. Sometimes we just read patterns wrong and at other times, we may simply be deceived. For example, emotional signaling, as we find it associated with facial expressions or specific behaviors, is widely used in nature to send false signals. That is true for many non-human animals but specifically so for the human species.

Keeping an Open Mind

We began our brief journey through the philosophy of mind with a look back at how philosophers of antiquity viewed the mind-body problem. Then we discussed some of the main avenues of thought regarding the mind-body relationship before Rene Descartes ushered in what, from a Western perspective, we see as the beginning of modern philosophical thought. Practically the entire edifice of philosophy of mind, as it presents itself to us in its different schools of thought, can trace its roots to philosophical concepts originating in antiquity and classical times. Later, religious beliefs such as Christianity and Islam, themselves also significantly influenced by pagan philosophical thought, adopted, supplemented and reinterpreted mind-body concepts inherited from pagan philosophers. Judaism was certainly also open to studying pagan philosophical concepts but their reception influenced Judaism to a much lesser extent than Christianity and Islam. The understanding and interpretation of the mind-body relationship associated with philosophy of mind in modern times, as conveyed in its rough outlines in the previous section, rests solidly on almost exclusively Western and Near East intellectual and religious concepts. That is no accident. Throughout history, the Occidental World and the Oriental World have followed different paths in exploring the same fundamental questions that all human beings have.

The modern Western World has become heir to the cultural traditions and learning that we typically associate with the cultures of what once was the smaller world of the Occident, dominated for several thousand years by the great Mediterranean civilizations. Prominent among those were of course the ancient Greek, Hellenistic and Roman civilizations. In the same way, the philosophical outlook in much of modern Asia continues to be shaped by philosophical and religious traditions rooted in the ancient Indian and Chinese civilizations. There were of course also influences coming from other parts of Asia, but those two civilizations are the most prominent representatives shaping cultures and learning throughout much of the Orient.[25] Eastern approaches to philosophy in general and to the mind-body problem specifically had little influence on Western philosophical thought and that has not changed much, even with the greater access to the wisdom of the Orient that became available to Western scholars with the seventeenth-century rise of European colonialism in Asia.

In today's Western societies, for the most, religion and philosophy are usually going their separate ways. That was not always so. Looking back in the cultural history of the West, religion and philosophy were often much more intertwined than they are today. Back then, when the Mediterranean World made up much of what in the West and the Near East counted as the known world, religion and philosophy frequently were close relatives with neither one dominating the other. For the most, cultivation of both of them contributed equally to the perceptions our distant ancestors had of the world and

of themselves in it. Judaism, for most of its history a more inward than outward looking religion as compared to Christianity and Islam, was to a certain extent an exception as it could draw on a much longer religious and philosophical tradition of its own. The offspring of Judaism, first Christianity and then later Islam, were expansive and proselytizing religions. As they spread, first with the rise of Christianity and later with Islam, religion became dominant and philosophy, as those two religions encountered it along the way, became subservient to religion. In essence, in both cases, philosophy became part of religion. In the Christian West, this state of affairs would last for more than one and a half millennia before the rise of modern philosophy, which we associate with Rene Descartes, would eventually begin to change this. A comparable evolution never occurred in the world of Islam and philosophy there can only exist within the confines of religion.

Most Western societies take this split between religion and philosophy originating in the seventeenth century for granted. It has become part of the mind set in much of the Western World where essentially Christian religious concepts still remain foundational to societies even though their flocks of Christian believers continue to shrink. In the Occident, we date the beginnings of the quest to understand how the natural world works back to the six and fifth centuries BCE when we see the inquisitive ancient Greeks ask what we would today consider pertinent scientific questions. Since then, science, next to religion and philosophy, has been one of the three pillars on which Western worldviews rest. It is difficult to imagine how the civilizations of the West could have evolved into their current states without their heritage of natural philosophy and science in such fields as mathematics, geometry, astronomy, biology or medicine going all the way back to the dawn of the Classical Period of Greece and the Hellenistic Age following it. For most of Western history, religion held the upper hand as it still does in the Islamic world. However, in Western Europe philosophy emancipated itself from religious belief and with the scientific revolution beginning in the seventeenth century, science increasingly differentiated itself from religion as well as philosophy.

We all know the result of that. Today, modern science is the way in which we explain the world and religion and philosophy have seemingly little to offer in our increasingly materialistic world. To understand the universe we live in, its origin and ongoing cosmological evolution, how life and we as human beings evolved on our planet, we do not require metaphysical speculation or religious belief to substantiate what science can explain, measure, model and in many cases now predict. In the making of the Orient, even though its peoples produced many firsts in science, science itself rarely has ever played this very same role. This did not change until the impact of European colonization made it clear to Eastern civilizations that adapting Western models of science and technology development would be indispensable to create a level playing field with the Western World; not only to leave the colonial experience behind but also to beat Western civilizations at their own game.

Religious belief and philosophy never lost their explanatory power in the Eastern World in the same way as it happened in the Western World. The religious experience in the Eastern World differs from that in the West and the concept of philosophy that evolved in the West does not necessarily have corresponding traditions in Eastern cultures. We are looking at two different worlds here and trying to describe one by using the vocabulary of the other is fraught with misunderstandings. In the West, if we could travel back in time before the beginnings of science, we would find ourselves in the Homeric Age (c. 1100 – c. 800 BCE). If we do for example the same in India, we find ourselves in the Vedic Age (c. 1500 – c. 800 BCE). The *Rig-Veda* is both, the oldest known Vedic Sanskrit text as well as the oldest still existing Indo-European text. Neither the mythological Indo-Aryan people speaking Sanskrit to whom we ascribe the origin of the Vedas, nor the similarly mythological Achaeans of the Homeric epics who spoke an early form of Greek knew of science as we know it today. Instead, what we address today with science, those archaic people approached through ritual practices and divination. Different from their descendants and cultural heirs thousands of years later, they may have had an easier time to communicate with each other about the essence of their cultural traditions, their explanations for the universe, and the frailty and significance of the human existence in the face of uncontrollable natural and divine forces. In reality, such conversations likely never happened, at least we do not know about any of them if they ever occurred, but we certainly can imagine that they could have happened. But what we know is that because in the Orient science did not take over the roles of religious belief and philosophy, as it did in the modernizing and industrializing West, the experience in the Eastern World became a very different one from that in the Western World.

The pantheon of the Vedas the invading Aryan tribes brought with them to India and the Olympian deities of the ancient Greeks represent kindred mythological cosmogonies, both explaining the world as these peoples experienced it through anthropomorphic personifications of natural forces.[26] In the West, mythological accounts of this celestial society of personified natural forces gave way to philosophical speculation and thought, striving to understand nature itself rather than mythological accounts. Something very different occurred in India. Unlike to what happened in the West where the Olympian deities were dethroned, the Vedic deities remained divine personified intermediaries. In Vedic belief, everything emerges from the concept of an omnipresent divine force, the Brahman, from which everything emanates. It represents the holy power, the innermost self of all existence, the universal self as well as every individual self, the latter but seeds of this universal self, giving existence to every living being. In the West the human quest for understanding took the path of natural science. In India however the route to understanding oneself and the universe evolved into the discovery of one's true inner-self, the Brahman-Ātman, and through it, of Brahman, the cosmological self of which every self was a part. In this quest for self-discovery, the Vedic traditions introduced into India by

the Aryan tribes would meld with and be molded by older indigenous Indian religious and philosophical traditions. Indian philosophies emerging from those rich and diverse traditions would just as much build on developing an understanding of psychology, as Western philosophy would come to depend on an understanding of natural sciences. In the West, well into modern times, psychological thought would be very much a part of religious thought. The psychology of medieval Thomist philosophy is a good example for that. Only in modern times did psychology within the Western tradition eventually become part of the edifice of modern sciences. However, even today, many Westerners do not perceive the science of psychology as equal to the natural sciences as exemplified by physics.

Equally important, the West and India also diverged in how they perceive a human being. In the Western World, the Homeric Age psychosomatic unity of mind and body, discussed earlier in our brief journey through the philosophy of mind in ancient Greece, gave way to new person concepts arising in the six and fifth centuries BCE. The reception of some of those concepts in their articulation during the Classical Period of Greece together with strong influences coming from Judaism would very much shape the Christian person concept. This influence of pagan philosophers is clearly still present in much of the Western World today. For instance, the Christian belief in a personal afterlife, where the resurrected human body incorporates the very same person as in life, is in essence not too different from how Ancient Greek belief considered the deceased to be the same persons as when they were still alive. In the underworld of Hades the deceased in Ancient Greek belief continued to exist only as much-reduced versions of their former selves, mere shadows of what they once were, but they continued to be the very same persons. Even though they had some difficulty remembering that and needed to consume sacrificial animal blood to regain some of the mental powers of their former mind-body person to which the soul was bound in life. In Christianity, those finding themselves in any of the three roams of Christian afterlife, hell, purgatory or heaven, are not just mere shadows but evidently the same real persons they were before. While resembling nothing like a psychosomatic unity in life or after-life, in Christian belief and with that in much of the Western World, body and soul remained similarly tethered to each other for eternity in the same person. In a way this is reminiscent of the psychosomatic warriors of the Homeric Age, whose body and mind were first bound together in the living body and remained so in the corpse.

However, over time there certainly evolved other conceptions of the constitution of a human person different from the psychosomatic unity of the mind-body of the Homeric Age with its air-like soul. Conceptions, which did not amount to an eternal union between the human body and the essence of a person. We can glean some of that by looking back towards early Greek theater and the origin of the word “persona” and how what Western culture infers today by referring to a person or a personality differs from its original

meaning. A defining characteristic of ancient Greek drama, starting to flourish in the early six century BCE, were its facemasks worn by the performing actors. These masks served likely a number of different functions some of them adopted into ancient Greek theater from earlier ritualistic practices.[27] One function, however, is of particular interest here, the separation of the performing actor from the role he played. The Latin word for those masks used in Greek and Roman stage plays is "persona", literally meaning the mask and character played by an actor.[28] Therefore, the original meaning of the word person as we find it reflected in Roman and Greek theater, inherited from the Classical Period of ancient Greece, did not imply an eternal union between the true nature of a human being and its visible role in life. In real life just as in ancient Greek theater, actual people and not only actors were personae, persons, impersonating the roles that they seemed to play in life but not representing their true natures, their true selves. In the West, a synthesis of ancient philosophical views with beliefs inherited from Judaism gave birth to a religious and philosophical person concept perceiving a person as being eternal in nature, at least halfway eternal, that is to say with an eternal future starting from birth. Thus, in the West, the implied distinction between the mask and the actor, which the word persona originally signified, vanished.

In Western philosophy and religion, a person has become the inseparable and eternal unity of mask and actor but not so in the philosophies and religions of India. In Christianity, salvation is impossible without maintaining this eternally fused persona. The implication of this is that in the traditions of Abrahamic religions and their associated philosophies, believers only have one shot at getting it right; one must become within one lifetime the kind of person that will be deserving of eternal salvation. At least since the time of Augustine of Hippo, Christian belief has endorsed the soul as representing the true person. Christian dogma pronounces the soul as separate from the body, but it seemingly cannot exist without it. Augustine already took this view, as did later Thomas Aquinas. With Rene Descartes equating the mind with the soul, Western philosophy would continue, from a secular point of view, this very approach to the person concept Christian scholars had cultivated for centuries from a religious perspective. Thus, the Christian soul remains chained to the actor's mask for eternity making for a very personal concept of salvation where each soul retains forever the imprint of the one mask its bearer wore through her or his lifetime.

Unlike in the West, Indian philosophy and religion continued to stress the differences between mask and actor in its analogues of what the West refers to as a person. In the Indian philosophical worldview, the mask is the displayed existence of the individual and different from the real essence of being that remains hidden behind the mask. When a person dies, the old mask disassociates from the real actor as the actor joins with a new mask. With every cycle of death and rebirth, the same actor arises with a changed mask, every time transformed into a renewed manifest personality. The ultimate goal

in these seemingly endless but still finite cycles of rebirth is, through introversion and introspection, the discovery of the anonymous actor, the true self behind the changing veil. Ultimately, with the cycles of rebirth ending and the final mask cast away, the door will be open to the perception of one's true self and through it of the universal essence of being to which the true self belongs.

The philosophies of India are in essence philosophies of release, release from the illusions, perils and pains of human existence. This certainly is the case for the ancient Vedic traditions, much of them still reflected in the religious and philosophical scriptures of today's Hinduism, a word coined by the British colonialists in the nineteenth century which refers to one of India's oldest religious beliefs and philosophies. It is also true for the other great religious edifices and philosophical traditions of India such as Jainism, Sāṅkhya, Yoga and Buddhism. Jainism is one of the oldest religions in India, predating the Aryan invasion and the introduction of Vedic beliefs into India. Dravidian people may have called India home since long before the Indus Valley Civilization, dating back to around 3,300 BCE, fell victim to the Aryan invaders in the second millennium BCE. We can presume that Dravidian religious beliefs and traditions, which later merged with Vedic Aryan beliefs, have ancient roots that reflect Indus Valley Civilization beliefs or beliefs that are even more ancient. Such beliefs could have been ancestral to Jainism long before the Aryan invaders entered India, after which Jainism evolved into a part of the Indo-Aryan system of beliefs. The last of the twenty-four great spiritual teachers recognized in Jainism, Vardhamāna Mahāvīra, was a contemporary of Siddhārtha Gautama, the Buddha, who sometime in the six or fifth century BCE founded the religion we call today Buddhism. There is some anecdotal information about the twenty-third great spiritual teacher of Jainism but we know nothing about the first twenty-two great spiritual teachers of Jainism. All that has come down to us about the latter are their mythological accounts dating their existences back tens of thousands of years and eventually before humans ever walked the earth. What those accounts of Jainism spiritual leadership indicate is that Jainism has likely a very long history, making it seem to its believers that it may have existed forever.

Like Jainism, Sāṅkhya and Yoga originated with the indigenous Indian populations the Aryan invaders encountered. Different from the monist Vedic belief system of the Aryan invaders, where the universal self permeates everything and the individual self is all but a seedling of Brahman, the universal self, the philosophies and belief systems of Jainism and Sāṅkhya / Yoga are of a dualist nature. The dualisms of Jainism and Sāṅkhya / Yoga see the universe split between life-monads and lifeless matter. However, those life-monads, the true inner selves of living creatures are not part of any larger universal self; there just is no such larger universal self as the universe is nothing more than all life monads and all lifeless matter taken together. Jainism and Sāṅkhya / Yoga differ about the nature of the attachment of life-monads to lifeless matter but they share the same goal: mokṣha,

the release from the endless cycle of death and rebirth, achieved by the final escape of a life-monad from its bondage to lifeless matter.[29] Sāṅkhya and Yoga belong together as the theoretical and practical aspects of a single system. Sāṅkhya literally stands for the enumeration of knowledge, the accumulated theoretical knowledge of the human nature and its elements, specifically knowledge with respect to disentangling the life-monad from lifeless matter. Yoga, the practice of introvert concentration, deals with the practical processes of how to achieve this disentanglement of life-monad and lifeless matter.

The Aryan invasion of northern India introduced new elites and a new religion to the Indian subcontinent but it did not displace older elites and older religious and philosophical traditions where Dravidian rule continued or could reassert itself over time. Helped by continues warfare among Aryan elites, Aryan dominance in India gradually weakened, giving way to an Aryan-Dravidian synthesis, manifesting itself in the evolution of religious and philosophical beliefs on the Indian subcontinent. Deities and traditions of the Vedic pantheon found their way into Dravidian beliefs and philosophies, which in turn molded Vedic traditions. The Sanskrit texts of the *Upaniṣads*, compiled sometime between 800 to 600 BCE, mark a fundamental shift for the Vedic belief system away from seeking to achieve release through ritual practices towards the alternative approach of introspective search for the inner true self.

This new approach, giving rise to the school of Vedānta, would eventually succeed in combining the monism of Vedic belief with the dualisms of Jainism and Sāṅkhya / Yoga. The *Bhagavad Gitā*, a brief episode inserted into the *Mahābhārata*, one of the two major Sanskrit epics of ancient India, in its present form compiled sometime between 400 BCE and 200 CE, reflects the culmination of this effort. The synthesis of Vedic Aryan and indigenous Indian beliefs and philosophies continues to assert that there is only one true universal self, Brahman. Even though this is how the world really is, the *One Thing*, this is not how the human intellect and human logic can understand it. By acknowledging that it is human nature to perceive reality in opposites, such as with life-monads and lifeless matter, Vedic belief embraced ancient Indian non-Aryan religious beliefs and philosophies. Thereby, it made them fruitful for the Vedic quest of release on the path of introspective search for the true inner self.

Thus, the Vedic beliefs that the *One Thing*, the universal self, Brahman, is always present as two, as life force and life material, was joined with the non-Aryan dualist belief systems postulating the existence of life-monads and lifeless matter, which in Vedānta came to represent the way human nature was able to perceive reality. Vedānta, the Brahmanic synthesis of Aryan and non-Aryan religious beliefs and philosophies continued of course to accept the authority of the Vedas. Indian philosophical schools accepting the authority of the Vedas and the orthodox traditions of Hinduism are part of what scholars refer to as the Indian orthodox schools of philosophy. There are six such orthodox schools, including the Sāṅkhya and Yoga schools in addition to the Vedānta school. Different

from Sāṅkhya and Yoga, Jainism and Buddhism are two of the five heterodox schools that reject the authority of the Vedas. From a Western perspective, the complexity of the Indian religious pantheon is overwhelming and the pervasiveness of shared, adopted and modified religious and philosophical thought, as can be witnessed in the daily religious practice of hundreds of millions, is difficult to relate to. However, we must never forget that in the Indian cultural tradition philosophy just as well as religion are lived personal experiences for everyday people. In India, a philosopher in the traditional sense is only a philosopher in as much he or she lives and exemplifies his or her philosophical beliefs.[30] It is not the written texts that count as or point towards the truth of something, but it is the lived experience, evidencing the truth, which matters.

Buddhism emerged on the Indian subcontinent not as a philosophical or religious system but as a path towards healing the suffering rooted in the human way of life for the few who were prepared to hear Buddha's teachings. This early Buddhism did not concern itself with exploring and understanding the mysteries of human existence, which was the concern of Jainism or the Vedānta, Sāṅkhya and Yoga schools. Buddhism conceived itself as a vehicle, a ferry to help passengers cross the river of life from the shore of human suffering to the shore of nirvāṇa.[31] Once a passenger attained nirvāṇa, the vehicle became useless, as did the concept of nirvāṇa itself. Buddhism, as initially conceived, contained the seeds of its own obsolescence once a disciple successfully made it to the other shore. Over time, however, Buddhism had to confront the same questions as all other religious and philosophical system. Even though one of Buddha's fundamental teachings was that all things are without a self, it eventually had to address the problem of the nature of the human self, its relation to the experienced world, and to the nature of the essence of all that exists. It did so, broadly speaking, in two approaches that became known as Hinayāna Buddhism, the Small Vehicle, and as Mahāyāna Buddhism, the Great Vehicle. Each of those doctrines came to develop its own specific approach to address the problem of the nature of the human self, the discussion of which is beyond the scope of interest here. Buddhism had never left the sphere of India's traditional philosophy and Mahāyāna Buddhism within India merged back into the Brahmanic cosmos as its positions with respect to the fundamental questions aligned with those of the Vedānta. With that Mahāyāna Buddhism vanished in India as it became part of the Hindu system of religious beliefs and philosophy. However, by the time this happened, Mahāyāna Buddhism already had found fertile ground outside of India in much of East Asia. Hinayāna Buddhism also vanished in India but continued to thrive in Ceylon and Southeast Asia.

The passing of the self through cycles of transmigration and rebirth is part of Jainism, the Vedic traditions encapsulated in Hinduism, as well as Buddhism. Release of the self from this endless cycle is what mokṣha means in Jainism, Hinduism and Buddhism; the latter also refers to the ultimate destination of this path as nirvāṇa. To a greater or lesser

degree, all of those belief systems guide believers to remove themselves from the material world through asceticism and meditation to discover ones true inner self as one navigates the cycles of transmigration and rebirth to final liberation and release, attaining mokṣha or nirvāṇa. Non-violence to all living beings is a common thread in Indian religious beliefs and practices and without it, it is impossible to attain the final release of the inner self. But there is a gradation, with Jainism taking this more serious than Buddhism which in turn has a better record of non-violence than Hinduism. There certainly are also differences, notably in how much emphasis Jainism puts on helping others as compared to Buddhism or Hinduism. The focus of India's ancient beliefs and philosophies over millennia has been and continues to be the release of the self from the illusions of the ego and the world. Philosophy is not so much learned and educated speculation conducted according to strict logical rules such as in the West. Philosophy in India is reflected and passed on to disciples through the lived experience of the philosopher, be it the priest, the monk, the yogi or the guru following a religious or philosophical tradition. To see past the ego-self and discover the Ātman, the inner or true self, the spiritual life principal permeating the universe, that is the path to attain the release from the endless cycles of transmigration and rebirth. Only human perception through the self is not an illusion and attaining a state of mind that only perceives the world through the Ātman self is the objective of such practices as asceticism and Yoga. Western religious and philosophical practices know little of such meditative disciplines, which when mastered can open the door to new perspectives on the human mind-body relationship. Where they do exist in Western belief and philosophy, they do so at the margins.

The above paragraphs gave a brief introduction into how philosophical and religious thought developed in India and how it informed perceptions of the mind-body relationship, in a seeming continuity linking in many ways spiritual life and philosophical thought in today's India with much more ancient times. In China, we encounter something similar. In the West, unless one happens to be a student of the ancient past, most of us do not have a strong connection anymore to the religious and philosophical life of Western antiquity. That is very different from what we encounter in India and China and for that matter for a number of other Asian cultures. To understand the life of Western communities and societies we really do not have to go back and look into ancient Western philosophy. Different from that, one cannot adequately understand communal and societal bonds and relationships in many Asian countries without acknowledging that ancient religious and philosophical thought continues to mold customs and ways of life well into our modern times. Often not recognizing the importance of that is by the way not only a Western shortcoming as sometimes societal elites in Asia are no less short sighted. Modern China is a good example for that. The communist rulers there have on the one hand been trying to legitimize their monopoly on power by selectively resurrecting some of China's ancient past while at the same time ignoring and suppressing those parts of

Chinese philosophical traditions deemed dangerous to their rule. For that purpose, these communist Chinese rulers label unwanted behavior frequently as motivated by Western influence and alien to Chinese culture, thereby seeking to discourage certain notions of personal freedoms and liberties taking a firmer hold in Chinese society. That simply betrays a lack of knowledge of their own cultural heritage among communist Chinese cadres. Albeit in a different cultural context, Chinese philosophical and religious thought has an equally strong record of emphasizing what we call today fundamental human rights and that long before the West gained any significant cultural influence in China.

One always has to be careful when generalizing but sometimes it can be helpful to condense complicated facts into a simpler picture as it helps to highlight contrasting approaches. If we do that as we look at the cultural history of religion and philosophy in India, what we can discern is that seemingly religion and philosophy were never separate and religious thought on the Indian subcontinent always encapsulated philosophical thought. Travel from India further East into the Chinese World, and you will encounter pretty much the opposite. Looking back into Chinese cultural history, we can see the role of religion encapsulated in philosophy. That is even the case with Buddhism, expanding likely in the first century CE into the China of the Han Dynasty. Over many generations, much of China eventually assimilated Buddhism and in doing so incorporated Buddhist religious and philosophical concepts into its own philosophical schools and ritual religious practices. In Chinese history there is no religion playing a similar role in society as religion did in the West. Rather what we see is religious thought inspiring philosophical thought. In a sense, to some extent philosophy in China took on the role that religion claimed in the West for a long time. However, never was philosophy in China as powerful as religion was throughout much of history in the West. Unlike religion in the West, where at times religious leaders ruled whole countries and religious organizations became the dominating forces ruling entire societies, philosophers or philosophy in China never ruled a state itself but rather counseled rulers. While not anointed with similar official power as religion in the West, philosophy, sometimes religiously inspired, had however an almost immeasurable influence on Chinese society, setting rules according to which not only ruling elites sought to conduct their lives but which also governed communal and societal life much more broadly.

K'ung-tzu or Master K'ung, K'ung Ch'iu with his family and personal name, better known in the West as Confucius (551 – 479 BCE) was a contemporary of Pythagoras. The second sage of Confucianism, Meng-tzu or Master Meng, Meng Ke with his family and personal name, Confucius' famous fourth generation disciple known in the West as Mencius (372 – 289 BCE), could have conversed with Aristotle if such means of communication had existed back then between Greek city states and states in the China of the Zhou dynasty. Unlike with Confucius, where we do have quite a bit of biographical information, we know next to nothing about the life of the other great sage in Chinese

philosophy and culture, Lao-tzu. To start with, the name itself may not have been the real name of this great sage as the meaning of the name Lao-tzu denotes a venerable old master, essentially identifying a sage, and as such is a Chinese title of honor. Some sources say Lao-tzu was a sixth-century contemporary of Confucius in China and hence of Pythagoras in ancient Greece; other say he lived some two hundred years later, during the warring states period of fourth-century China. According to the version where he lived in sixth-century China, Lao-tzu worked at the court of the Zhou dynasty, which resided at the time in what is today the city of Luoyang in present-day Henan province. Legend has it, that Lao-tzu and Confucius met in person and there are later apocryphal accounts retelling this encounter. Maybe this encounter never happened but one can still wonder, what their relationship would have been like, if they had met and discussed their views on matters of philosophy and religion. Lao-tzu became the revered founder of philosophical Taoism, the philosophy of the way, and eventually deified in religious Taoism. The *Tao Te Ching* (*Book of the Way and its Virtue*), a compilation of philosophical writings attributed to Lao-tzu himself and / or later authors summarizing his teachings became the foundation for philosophical and religious Taoism. Western high school students typically learn little about the Eastern world and we teach them even less about its philosophers and philosophical schools. Hence, it often is a surprise for many to learn that philosophy in the East flourished no less than in the West and just as in the West, gave rise to a number of different philosophical schools.

Categorizing philosophers by schools of thought always is an approximation and it is even more so when one is not familiar with the respective cultural heritage in the region of interest. However, it is still helpful to understand the bigger picture as long as one remains mindful that the works of some philosophers do fit into such categories better than do those of others. The oldest Chinese philosophical school is the *Yin-Yang School*; the one most frequently associated in the West with Chinese philosophy is likely the *School of Literati*, generally referred to as Confucianism; other major schools are the *Mohist School*, the *School of Names*, the *Legalist School* and the *School of the Way*, better known as the philosophy of Taoism. Indian philosophical schools and religions have influenced philosophical thought and religious beliefs throughout much of East and Southeast Asia. As recounted above, Buddhism in India folded back into the Hindu tradition from whence it had emerged. However, before that happened, Buddhism already had spread to East and Southeast Asia where it continues to flourish in a number of forms until today. When Buddhism found its way into China sometime in the early first century CE, it changed the course of Chinese philosophical and religious thought. Buddhism as it came to China from India, just like Taoism in China, was as much a philosophy as it was a religion. Buddhist and Taoist philosophy were kindred spirits but not so Taoist and Buddhist religion. Before Buddhism came to China, Taoist religion was rather insignificant as compared to Taoist philosophy. With Buddhist religious practices spreading in China,

Taoist religion defined itself in opposition to the foreign religion of Buddhism and gained in popularity. In a sense, the introduction of Buddhism essentially founded what became Taoist religion as a Chinese counterpart to the foreign religion of Buddhism. Different from their religious branches, Buddhist and Taoist philosophical views drew close to each other. So close that often their teachings would resemble each other even though one built on ancient Chinese concepts while the other built on Indian versions of the same. We do not know exactly when and where it happened, but eventually the path that Taoist and Buddhist philosophy took in China gave rise to what became the philosophy of silence or Ch'an Buddhism; in the West likely more familiar with its Japanese spelling as Zen Buddhism.[32]

The influence of Chinese Buddhism and its kindred relationship with the philosophy of Taoism would also influence Confucianism. This development, to which the West refers to as Neo-Confucianism, started in the late eighth century CE. The subsequent reinterpretation of Confucian philosophy in reaction to the flourishing Taoism and Buddhism, provided a cosmology and ethics that continues to reflect the worldview of many Chinese today. While Neo-Confucianism rejected mystical elements of Taoism and Buddhism, its concept of achieving sage-hood mirrors in many ways the Buddhist concept of achieving Buddha-hood. However, not in the original Buddhist way but more like in the Ch'an Buddhism way of living naturally and acting naturally.

In the West, the rise of Christianity very much changed the relationship between philosophy and religion. Nothing comparable happened in the East. There, philosophy served in many ways a similar role to that of the Abrahamic religions of the Occident, where philosophy was for the most seen as subservient to religion. This was likely a consequence of how Western and Eastern societies perceived the critical functions of religion and philosophy in the first place. In the Chinese World, religion and philosophy primarily concerned themselves with understanding and defining the true characters and roles of personal relationships critical to the functioning of a just society; such as relationships within families, like between father and son, husband and wife, or between ruler and ruled. In the West, the religious and philosophical emphasis was much more of a transcendental nature. What for example mattered most in a Christian religious and philosophical context was the life to come and not the earthly life. The human body was an earthly burden; the great prize was the salvation of the soul, a new heavenly body.

There are many more subtle differences between the essentially religious-philosophical outlook evolving with Christianity in the West and the philosophical-religious outlook evolving in the East. The latter for instance illustrated in the development of Taoist philosophical and religious schools or Buddhist philosophy and religion as outlined above. Even though the simple picture drawn here in the space of a few paragraphs must necessarily leave out much, it does point clearly towards one of the fundamental differences as

to how Eastern philosophers, scholars and believers approached the mind-body problem versus how their Western counterparts did. Put simply, Eastern intellectual and religious traditions have not experienced a separation of mind and body in a similar way as it happened in much of the West. There, the widespread adoption of Christianity would separate soul and body for good.[33] The latter was an assertion of Christian religious belief and dogma but, equally important, also philosophically argued for by recruiting existing pagan philosophical concepts predating Christianity. Without the philosophical heritage of the Classical Period and the Hellenistic Age, Christianity would likely have evolved quite differently than it did.

In Eastern thought, mind and body are not separate. However, to interpret this position as a monist philosophy would be wrong. The Eastern philosophical outlook is altogether different and Western concepts such as dualism and monism do not really fit it. Any Western educated person likely will face this problem when trying to understand Eastern intellectual traditions but it is an artificial problem. In the West, the approach to philosophy in general as well as to philosophy of mind in particular has settled on exploring logic aspects of the human mind only. This implicit definition of Western philosophy essentially precludes it from exploring the human psyche itself. Sure, Western philosophy can look at the human psyche and describe it but it cannot explore it. Western science has relegated this task to psychology, more often than not in the form of clinical psychology or psychopathology. In contrast, Eastern philosophical traditions have always seen the exploration, understanding and control of the human psyche very much as a part of the philosophical enterprise. As viewed from an Eastern perspective, by focusing almost exclusively on human logic faculties, Western philosophical discourse on the mind-body problem has ignored much of the human mind, the human psyche.

It is only with the advent of modern psychology and psychoanalysis that the part of the human mind, which Western philosophy had mostly ignored, began to receive increased attention. The credit for that goes to scholars such as the Austrian neurologist and founder of psychoanalysis Sigmund Freud, his compatriot Alfred Adler (1870 – 1937), physician and founder of the school of individual psychology, or the Swiss psychiatrist Carl Gustav Jung (1875 – 1961), psychoanalyst and founder of analytical psychology. In the latter half of the twentieth century, psychoanalysis became much ridiculed as a pseudo-science and quite a few may still hold such views today. Shortcomings within every new science as it takes its very first steps are to be expected and this is certainly true for psychoanalysis. However, the seeming failures of psychoanalysis are likely more a reflection of mismatched expectations than of anything else. Modern medicine and specifically drug development have helped many of those among us who suffer from psychic afflictions, probably a lot more so than psychoanalysis ever did. In that respect, psychoanalysis never fulfilled the promise it may have once held. Medication that can help alleviate or even cure psychic afflictions is a godsend for many; psychoanalysts will

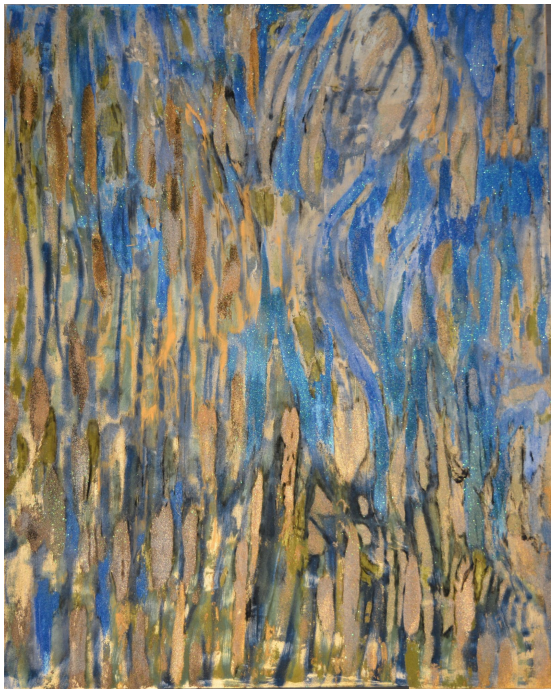
be just as grateful for this as most of us are. However, from its beginnings, psychoanalysis always held a much bigger promise than just to heal specific psychic diseases and that was to lead us eventually to a much better understanding of the human psyche. Even as scientists and physicians, working hand in hand, can diagnose more and more psychic afflictions in terms of their biological origins and therefore make them treatable, through surgery or medication, this will not bring us much closer to a better understanding of the human psyche itself. We still need something like psychoanalysis or other new disciplines of psychology, or maybe an altogether new science approach to help us decipher the enigma of the human psyche.

By its very nature, any science setting as its objective a deeper understanding of the human psyche also will have to tread on the hallowed grounds that philosophy claims as its own. Instead of taking advantage of such incursions to stimulate increased interest in those parts of the human psyche thus far neglected by philosophy, for the most, philosophers do not welcome such intrusions on their territory. In this respect, it has not been helpful that over centuries now, too many Western scholars have become used to philosophy exploring the true foundations, so-to-say the fundamental questions underlying the endeavor of other sciences such as physics or mathematics. Scientists have left it to philosophers to answer questions they believe their sciences are not in a position to ask or answer by themselves. In that way, a certain part of philosophy has become judge and jury with regard to what constitutes a scientific discipline; or what could be part of science and what not. Often, this has been helpful to science but sometimes it can be less than helpful down to the point where it may actually hold a science back. This was the case for psychoanalysis, where the frequent verdict was that it is not a true science, neither in its psychological nor in its philosophical aspects. Of course, that in turn has also colored the Western outlook on Eastern intellectual thoughts and traditions with respect to the mind-body problem. In their approaches to the mind-body problem, Eastern schools combined elements of philosophy, religion, psychology, psychoanalysis and importantly their own cultural traditions and practices of how to explore the mind. Because of that, the Eastern perspective on the mind-body problem has received far greater attention from Western psychology, psychoanalysis or psychiatry than from Western philosophy. The approach taken by Eastern philosophical and religious schools is a holistic one, contemplating the human being as one mind-body being. The Western philosophical approach is analytic and dissecting, separating the mind from the body only to put it together in various ways to form a dualist or monist entity. Importantly, in doing so, Western philosophy for the most reduced the human mind to what scholars call the bright consciousness, leaving out the dark consciousness, sometimes also referred to as the unconscious mind. The latter can be a source of confusion, as we should not differentiate between a conscious and an unconscious mind but rather between the reflective mind and the non-reflective mind. There is another frequently ignored aspect of

most discussions about the mind-body relationship. All too often, we tend to limit the exploration of the human mind to that of the mature human mind-body being. This of course completely neglects the importance of the evolution of the mind-body relationship as we grow up. Including such considerations brings us also back to reflect on the various roles of the bright and dark consciousness, their relationship to each other as well as their specific relationships to the body.

Let us just consider our very own experiences as we grow up. There was a point of time for all of us where bodily activities, which at a later time in our lives and to this day, we perform without thinking about them at all, at an earlier stage required conscious control and effort. Think about learning to walk, learning a skill such as playing an instrument or playing sports. When we begin to learn such a new body skill, our bright consciousness is in control. We make mistakes and correct them and through meticulous practice, we eventually master the skill to a point where our bright consciousness has passed control of the respective body functions to the dark consciousness. As, for example, when we sit down to play a piece of music this is still our bright consciousness making the decision. However, once be begin to play the tune much of the actual bodily functions with which we execute the keystrokes are controlled by our dark consciousness whose performance in doing so is supervised by our bright consciousness. In Eastern mind-body concepts where there is no separation between mind and body this transition from bright to dark consciousness is how the body becomes part of our mind and the mind becomes part of the body. For all purposes, this is pretty much the process of learning a skill as our modern medical and biological sciences have discovered it. With the continued practice of a skill a new memory is engraved in areas of our brain from which the skill once sufficiently trained, can be recalled in an autonomous way, which is to say without reflection. It is as if through training, a part of our mind becomes part of our body, in the case of a learned skill like riding a bike or skiing, that part of our body is the hypothalamus. A different way of saying the same thing would be that mastering the skills of riding a bike or skiing means that they have become part of our dark consciousness. In the Eastern view, control of the human body through the mind is the equivalent of the human mind and the human body being one, as mind and body experiences become one. Such cultivation of the mind-body or the unified self through meditation is a unique Eastern philosophical and religious concept. Western philosophy and religion were never comfortable with the exploration of the dark consciousness part of our mind. With few exceptions, they either ignored the dark consciousness, as philosophy mostly did, or demonized it, as religion was inclined to do. The only examples we can find otherwise are those of monastic contemplation or religious mysticism as all of the three Abrahamic religions know them to some extent. The latter however, more often have the character of ecstatic and mystical religious unions rather than the contemplative and controlled cultivation of the self as practiced for example through Yoga or Zen Buddhism.

Of Human Nature



Trapped Blue

The first chapter gave a brief introduction into the biology of the human brain. Even without having a background in biology and medicine, a basic understanding of the major aspects of human brain anatomy and physiology provided there is accessible to anyone with an exploratory mindset and who would not be willing to take the time for a little study to learn more about how our organ of mind works. In the second chapter, we took a brief journey through the several thousand year-long effort of humanity to explore the mind-body relationship from a philosophical perspective. As we have seen, throughout recorded history, the trajectories of philosophy of mind and religious beliefs can be difficult to separate. They frequently intersected and sometimes became so closely entwined that they were not just hard to tell apart but for all practical purposes seemed to have merged. That should not come as a surprise. At their best, philosophical and religious thought both seek to explore who we are as human beings. In our modern, much more secular world, philosophy and religion are going their separate ways but many in their search for answers continue to reach out to both.

The approach philosophy takes in its efforts towards a better understanding of what we justly can refer to as the very essence of the human self is to identify what we can know about the human mind-body relationship and how we can express that knowledge in a logically coherent way. Put simply, philosophy aims to explore what we earlier referred to as the surface structure of the body-mind, that which we also call our bright consciousness. Different from that, religious beliefs also reach into the substructure of our body-mind, that which we call our dark consciousness. When religious belief constrains itself, at least for the most, to the dark consciousness of our body-mind, we more often speak of such a belief as a mythos, a myth or a mythology rather than calling it a religion. The cradle of human religion is mythos, and it is through the creation of myths that humans first sought to understand themselves and the world around them.

Of course, neither mythos nor religion are dark consciousness constructs only. Both need to access the bright consciousness of the body-mind as this is the only way we humans can tell stories to each other. We frequently overlook this neutral function of our bright consciousness to communicate what otherwise our dark consciousness could not articulate. Inevitably, our bright consciousness will also alert us to whether what we listen to is something that we can explain and understand utilizing the rational faculties of our body-mind surface structure or not.

That, which we then cannot make sense of by using our human knowledge and logic, we still can embrace through belief, be it through mythological or religious accounts and assertions. In a way, our capability to believe is how we can keep that which we cannot explain rationally from slipping back into our dark consciousness. That is due to the peculiar way of just how the dark consciousness of our body-mind seems to work. The body part of our body-mind substructure controls our autonomous functions with our bright consciousness only becoming aware of this fact if something goes wrong like when we are physically hurt and cannot exercise certain functions anymore. The mind part of our body-mind substructure is even more difficult to grasp and behold, unless of course some of it ripples through the surface of our dark consciousness, like when we for example dream. Meditation, the personal cultivation of the body-mind that we can find more frequently in Eastern than in Western philosophical schools or religious beliefs, is how one can learn to explore one's dark consciousness to some extent. We cannot really tell how much access experienced practitioners of such meditation can gain to their dark consciousness or to which extent they can train parts of their dark consciousness. We just have no good idea yet how big the house is that our dark consciousness occupies in our brain. However, we know that masters of such personal cultivation can eventually learn to control some functions of their autonomous nervous system to a certain level.

Philosophers were never comfortable with exploring the dark consciousness, which by its very definition seems to defy logical reasoning and explanation. Religious scholars were of course less reluctant to do so because the dark consciousness, even though they would not have called it by that name, is seemingly the home to many of our innate desires which can nourish a strong religious belief. Different from the bright consciousness, our dark consciousness does not ask deeply probing questions, which could weaken a believers faith. Unlike religious faith by itself, religious theology requires argumentation and for that reason theologians usually focus no less on the bright consciousness than philosophers do. To learn about the human psyche an approach was needed that would commit itself to exploring the entire human mind, our dark consciousness just as much as our bright consciousness. However, neither Western philosophy nor religion were committed to exploring the entirety of the human psyche.

Western religions, while acknowledging the reality of dark consciousness, merely thought to utilize it to bind the masses of their believers more closely to their faith as would have

been possible by appealing only to the bright consciousness of their flock. Just like Western religions, Western philosophy was ill positioned to undertake a journey of exploration and discovery of the human mind in its entirety. Even though a number of philosophers since the seventeenth century, many claim beginning with Gottfried Wilhelm Leibniz, discussed the unconscious aspect of the human mind, they usually referred only to a part of what the modern meaning of the unconscious human mind entails.[1] For philosophers, the unconscious mind was little more than a storehouse for perceptions, which either our body-mind handles by what we call today the autonomic nervous system, or which essentially represent white noise. This white noise of perception signals we register in the background only receives attention from our conscious mind once one among those many signals rises above a certain threshold level. There is also always the possibility that we may not register a perception of sufficient strength consciously at all because dealing with a more powerful perceived stimulus takes precedence. Such philosophical views of the unconscious human mind as a simple repository for our perceptions were far removed from the much more extensive and complex unconscious mind that a Sigmund Freud would begin to discover some two hundred years after Leibniz.

Before there was actually a science of psychology, did philosophy and religion contribute to our understanding of the human psyche in ways that we would today associate with neither of them but with psychology instead? Most certainly, the answer to that question is a resounding yes. Throughout human history, all efforts to understand human nature have invariably included approaches that today we firmly associate with the science of psychology. However, only with medicine and biology giving us an increasingly better understanding of human physiology and the natural sciences of the nineteenth century providing improved laboratory instrumentation, did it become feasible to explore the human psyche in an experimental way. Before that, all psychological hypothesizing sprung either from introspection, prying into the workings of one's own mind, or came from speculations about human nature based on general observations. The Kantian notion of a priori concepts of such things as space and time that we encountered in the last chapter is an example for the latter. The postulation of a priori concepts, a knowledge of things that we are born with, innate to the human mind and not acquired through experience, is what philosophers refer to as a nativist position. Such innate capabilities are much more difficult to investigate and quantify than acquired capabilities. Therefore, Kant doubted that our knowledge about human psychology could ever have a similar solid foundation such as the natural sciences have where scientists can spell out universal natural laws in the language of mathematics.

Interestingly, Johann Friedrich Herbart (1776 – 1841), the German philosopher and psychologist who succeeded to Kant's chair at the University of Königsberg thirteen years after Kant had stopped teaching and five years after he had died, thought differently. Herbart, subscribing to nativist positions just like Kant, believed that it was possible to

model mental processes in mathematical terms. His mathematical-mechanical vision of the mind included the unconscious as a reservoir of repressed ideas, which through the process of apperception could rise into our consciousness. Apperception is a philosophical concept that traces its origin to Descartes and Leibniz. What it describes in psychology is a process whereby our mind assimilates new experiences along related past experiences and in doing so transforms the complex of all such experiences associated somehow with each other into a new contextual whole. This new whole is what Herbart refers to an apperceptive mass. If that mass exceeds a certain threshold, it can lift related ideas that are too weak to surface by themselves from their repressed states in the darkness of our unconscious mind into the bright light of our conscious mind. Alternatively, there is also the possibility that a sufficient apperceptive mass of related ideas accumulates in our unconscious and eventually can rise into our conscious mind by itself without any further contextual integration. Even so Herbart's model did not catch on, it is noteworthy as it takes the unconscious mind much more seriously than most other nineteenth century psychologists would do, that is until Sigmund Freud would enter the stage.

Different from this nativist position prevailing in German speaking lands throughout the eighteenth and much of the nineteenth century, an empiricist position flourished among British philosophers and later also became prominent in France under the label of positivism. As the name insinuates, empiricism holds that all human knowledge derives from sense experiences by which we acquire concepts such as space and time through learning as we grow up and that such concepts are not innate to human nature. Among the British empiricists, we find such eminent philosophers as John Locke, George Berkeley and David Hume whose names are as firmly associated with empiricism as Kant's name is with idealism.

Simply put, for empiricists all that we learn comes to us through experience and no innate knowledge whatsoever exists. In the extreme, they claim that nothing exists unless we experience it, a position that famously George Berkeley endorsed.[2] Few among us would likely believe today that an object does not exist because we have not experienced it. Likewise, most empiricists would not subscribe to such a view either. What empiricists agreed upon was that by associating simple experiences with each other we can learn about complex things. Just think about how we for example associate the experience of seeing a lightning strike with the thunder that follows it. In a similar way, we learn for example about causation through associating causes with effects. Associationism, the name of the psychological theory empiricism gave us, claims that what is true for our sensory experiences is also true for our mental experiences. Empiricists supposed that mental processes came about by associating mental states in a sequential manner. Using this hypothesis, empiricists argued that it must be possible to reduce complex mental processes to a combination of identifiable and separate simpler psychological elements and their combinations. Because they interpreted those simpler psychological

elements as the result of our sense experiences, empiricists claimed that they should be accessibly to study and understanding. If successful, this approach could eventually also provide a better understanding of mental states themselves. The first to articulate such an associationist view of psychology was the English philosopher David Hartely (1705 - 1757). In addition to Locke and Hume, other prominent philosophers holding associationist positions included for example the Scottish historian and philosopher James Mill (1773 – 1836) and his famous son, the philosopher John Stuart Mill.

Psychology as a science began to emerge in the nineteenth century with scholars who sought to understand human nature by looking for insights coming from philosophy on the one side and from the natural sciences on the other side. In philosophy, the two dominating psychological schools of thought centered on the opposing views of associationism and nativism, whereas among the natural sciences it was a steady progress in physiology, which provided increased insights into human nature. The emancipation of psychology from philosophy and its rise to a science in its own right is part of the larger story of nineteenth-century scientific discoveries. Those discoveries reshaped our world and our understanding of it and there was no going back once the genie was out of the bottle.[3] Once Charles Darwin had published his *Origin of Species*, it just was not possible any more to continue approaching the exploration of human nature in the same old way and pretend like nothing had happened. Likewise, discoveries in human medicine and biology gave us an ever better understanding of human physiology, making it look increasingly likely that one day all of human nature could have a biological explanation. Physics and chemistry continued to explain things for which there was no explanation before, laying the foundations for technologies, which would usher in the modern era. The evolution of psychology as a science in many ways mirrors the transformations this modern area has brought about, the good as well as the bad.

At the beginning of the nineteenth century psychology as an independent science did not exist. By the end of the century, psychology had evolved into a standalone scientific discipline. While continuing to draw on philosophy, medicine and the natural sciences, psychology had become separate and distinct from philosophy, recognized in its own right with universities having psychology departments and chairs of psychology. However, psychology would not become an independent science in the same way the natural sciences became independent sciences. Many of the scientists choosing psychology as a career were striving to establish psychology as a science along the same lines as for example physics and chemistry were sciences. However, this effort failed. The foundations of both, physics and chemistry, are natural laws. We can discover those laws and integrate them in theories that then help to predict new phenomena for which we can search; or we discover new phenomena which help us improve our understanding of nature's laws. Sometimes we discover that the natural laws we believe to have found are only a very good approximation to how nature really works. However, eventually we find a better or even

the correct explanation, like when Newton's theory of gravity was eclipsed by Einstein's (1879 - 1955) general relativity. The important fact is that as natural sciences grow, such as the edifices of physics and chemistry, the many subfields they sprout continue to share the same foundations. There is just one science of physics and one science of chemistry, and the laws of physics and chemistry apply to all their subfields. Psychology had no such common foundation when it began to build its house of science. Rather than a single science of psychology, there are today sciences of psychology. The science of psychology does not resemble a house with a shared foundation but is more like a village with a number of scattered houses, some bigger than others. Specialization is a phenomenon that all sciences experience as they grow but for a science without a shared foundation, this can become a problem.

There may have been a point in time long ago, where it still was possible for one human being to know all that there is to know in science. Or later more narrowly, a time when for example a physicist or chemist could pretty much know everything that there was to know in those sciences. That time is also long gone. Nowadays, the vast majority of scientists must specialize to become experts in a given subfield as that is the prize to pay for being able to work at the forefront of a science. To contribute something new to a field in modern science, its practitioners must literally learn more and more about less and less. However, at the same time, in physics and chemistry, to stay with those disciplines of natural science, its scientists, despite of such necessary specialization, will always be able to communicate and agree on the foundations of their science. They will always recognize that what each of them does concerns a distinct part or aspect of nature whose understanding will contribute to the same larger edifice of science to which their subfields belong. Not so with psychology.

Right from its very beginnings as an independent science, psychology started out with competing conceptions of what the science of psychology should explore. What the science of psychology ought to be about was, in a manner of speaking, as much in the eye of the beholder as it was prescribed by nature. Physicists a few hundred years ago seeking to study celestial mechanics could not choose the kind of firmament they were looking at. There is only one universe out there to study and it provides the same reality for everybody, no matter who chooses to investigate it. Not so in psychology. We cannot see the human mind. All we have are conceptions of the human mind. Today, many of those conceptions are accessible to scientific exploration and scrutiny in ways that just did not exist in the nineteenth century when psychology began to emerge as an independent science. Back then, those conceptions simply depended on what a researcher thought the human mind resembled; and in turn, this would determine how a researcher would go about exploring the human mind. Exploring the human mind itself was out of reach anyway. All a scientist could hope to achieve was to succeed in exploring certain aspects of it. However, the human beings we are, we are tempted to generalize quickly and tend to

see the whole of what we study through the lens of the little we can actually understand. Hence, psychology ended up with a number of competing approaches, which by their very nature could only have a limited claim to understand the nature of the human mind in the first place, but anyhow staked out much larger terrains. Only over the last few decades, enabled by the progress in sciences like neurophysiology and molecular biology, has the distinct possibility emerged that psychology could eventually attain itself of a similar kind of foundation as for example physics and chemistry enjoyed it right from their start.

Before we begin to look into the makings of modern psychology, we must take a step back and acknowledge that the human psychological discourse with ourselves and with everything in nature surrounding us has much more ancient roots than portrayed above. How humans understood themselves and how they perceived themselves in relation to nature did not just materialize with recorded history. Long before humans began to write down their thoughts, they must have sought to explain who they were as humans and the significance of the world around them in ways very different from how later religious beliefs or philosophers would set out to do. This became clear to many anthropologists in the early decades of the last century when they still discovered peoples who until then had not yet had any contact with modern civilizations. The psychology underlying the myths and cults those peoples practiced have a much longer history in human evolution than our much more recent modern ways to explain to ourselves who we are as human beings and how we relate to the world around us. How humans before modernity thought about themselves and nature, many refer to as folk psychology, a label often not used in a complementary manner.

It would be unwise to look condescendingly on folk psychology. Just consider what would happen if that which Karl Popper called the kingdom of human creations would magically disappear. Humans then would have to grow up in a natural state again, without any of the benefits of the civilizations of the past several thousand years. How would humans catapulted in such a way back into pre-history make sense of themselves and of the natural world around them? It is a safe bet that they would evolve a folk psychology not to different from the ones that anthropologists discovered with peoples who were at the beginning of the last century literally still living as our own ancestors lived in pre-history. The reason for that is simple. It is our human nature to do so. Hence, the exploration of folk psychology should help us understand our human nature. However, such an effort is beyond anything we can even contemplate here. Suffice to say, science has begun to do just that, and this is where we have to leave it here. Our focus for what follows will be on the making of modern psychology even though we must acknowledge that modern psychology in the end will have to loop back to help us understand what folk psychology is about on a more fundamental level.

The Making of Psychology

The making of psychology resembles nothing like a straight progression of a psychological science. There never even was a single science of psychology; rather there always were sciences of psychology. These reflected different approaches to explore the human psyche of which there were many. Some of those approaches reflected schools of thought associated with certain institutions or individuals while others represented general directions in psychology becoming popular more broadly. Through much of psychology's history, the lack of understanding of the physiology of the human organ of the mind, not to speak of its inner workings, limited the aspirations of psychology to become more of a science like physics or chemistry. In some way, this may have been a blessing in disguise as it allowed psychologists to more freely speculate about how the human mind may work, in a sense, forcing them to keep a more holistic perspective on how they thought about the human mind than otherwise would have been the case.

How does one date the beginning of the modern science of psychology? There always will be different answers to such a question but most would likely associate the birth of modern psychology with the German physiologist Wilhelm Wundt (1832 – 1920) opening his Institute for Experimental Psychology at the University of Leipzig in 1879. In that year, Wundt and some of his students conducted the first experiments in what was essentially Wilhelm Wundt's private laboratory. Wilhelm Wundt's private institute, the first institute of psychology at any academic institution, would evolve into the Leipzig school of psychology, the first of a number of schools of psychology to emerge over the decades to come. However, at around the same time, the American philosopher and psychologist William James taught his first experimental psychology course at Harvard University in the academic fall semester of 1875 – 1876. The writings of both scientists were highly influential at the time and remain in high regard today. Wundt was the more prolific author, publishing the first journal for experimental psychology research, the *Philosophische Studien* (*Philosophical Studies*) beginning in 1881. While Wundt's writings cover a much wider range of topics, William James writings were however no less influential. *The Principles of Psychology*, which James published in 1890 with a condensed and abridged student version following in 1892, has influenced psychologists and generations of students since then; some say as much as the writings of Wundt, others say even more. However, as large as figures like Wilhelm Wundt and William James loom in the history of psychology, maybe in public perception only matched in that regard by Sigmund Freud, the formation of a science like psychology is not just the effort of a few great scientists. While relying on seminal contributions coming from a few, a science such as psychology will always require the effort of many more whose names we usually do not recognize easily, or not even at all.

Psychophysics and Functionalism

The Age of Enlightenment, to which John Locke, David Hume and Immanuel Kant belonged, came to an end with the beginning nineteenth century. From the early nineteenth century until Wilhelm Wundt established the Leipzig school of psychology at the University of Leipzig between 1875 and 1879, a number of pioneers in experimental psychology made their mark. Among those, the names of many German scientists figure prominently. They include the physiologist Johannes Peter Müller (1801 - 1858); the physician Ernst Heinrich Weber (1795 - 1878); the physicist and physician Hermann von Helmholtz; the physiologist and physicist Gustav Theodor Fechner (1801 – 1887); and psychologists such as Herman Ebbinghaus (1850 - 1909) or Carl Stumpf (1848 – 1936). Wilhelm Wundt studied with Müller and Helmholtz and incidentally was Helmholtz's assistant for some eight years at the University of Heidelberg. Just like his teachers, he was one of those pioneers of physiological psychology or psychophysics, the scientific study of the relation between stimulus and sensation, which for some reason flourished so prominently in nineteenth century Germany.[4] As a teacher himself, Wundt would go on to found what many call the most influential school of psychology in the nineteenth century and through the careers of his many students and disciples he would greatly influence the direction of psychology as a science.

There were of course researchers in other countries who likewise made significant contributions, directly or indirectly, to establishing the field of science we now refer to as psychology. All of them men as the sciences were then still not open to women. Some in this nascent science of psychology would not only come to endorse a racist stance, like for example in their efforts to measure human intelligence, they also would declare women not as smart as men. They did not say so outright but it was still quite clear what it was that they were trying to assert as a scientific truth. Having said that, their scientific opinions were just reflections of more broadly held public views, essentially distillations of a culture where one could still argue racial superiority or inferiority without being frowned on or for that matter, could claim that women were less intelligent than men without looking too stupid oneself. Today, publicly articulating anything even remotely close to such views would immediately disqualify a scientist. However, back then, many still considered what most of us today view as plain and simple gender or racial prejudices as respectable parts of scientific discussions. To be sure, even in our modern day and age there are still a few otherwise seemingly respectable people who harbor such prejudices. For the most, they will argue their outrageous views only privately in environments where they feel save to do so. Nevertheless, occasionally those noxious bubbles will eventually rise to the surface, much to the consternation of the large majority of scientists and to the embarrassment of those holding such views. We should not be astonished to find someone like the English scientist Francis Galton (1822 – 1911), Charles Darwin's half-cousin, who

pioneered the use of statistics in sociology and psychology, engaged in gathering data to disproof or proof any such contentions societies held back then. Galton was a child of his time and whatever personal views he may have harbored on such issues, we have to applaud him for pursuing a data driven effort to verify whether any such contentions could hold water or not. We do not remember Francis Galton so much as a psychologist today, his conjectures in that field turned out to be mostly wrong, but we remember him for the statistical approach he developed in analyzing his data. Modern science is data driven science and it is the kind of data analysis scientists such as Galton began to introduce which transformed in a number of research areas what essentially was guesswork into scientific analysis. This is certainly the case for modern psychology where just as in many other science fields researchers must understand correlations of individual variables in what may otherwise look like random data sets.

On the continent, the Dutch ophthalmologist Franciscus Cornelius Donders (1818 - 1889) was first to devise experiments which shed light on how long simple mental processes in response to a stimulus could take. He did so by setting up experiments whereby he separated the total processing time a test subject required to complete a set of tasks into discrete steps. Of two otherwise identical experimental setups, one included an additional response reaction time step for a choice the test subject had to make, which he left out in the other experimental setup. Donders measured the time it took test subjects on average to complete those two experiments and by subtracting these averages for the two experimental setups from each other, he then could estimate the average time a test subject required making a choice. In 1869, Donders published these results providing a first insight into how long human mental decision processes may take in an article entitled *Over de snelheid van psychische processen* (*On the Speed of Mental Processes*).

In France, we find Alfred Binet (1857 – 1911) with the help of his student Théodore Simon (1873 - 1961) develop the first standardized intelligence test, the Binet-Simon test, to measure intelligence in a comparative way among French school children. We could dig deeper and certainly list more names of non-German scientists in Europe contributing to the rise of nineteenth-century psychology but nowhere else in Europe do we find a similar breadth and enthusiasm for psychological research as in German speaking lands. In order to find a similar enthusiasm for psychological research we have to look to the United States of America.

Again, we can only speculate as to why American scientists took to the study of psychology no less than their German counterparts did. Initially, there were strong connections with quite a few American researchers studying in Germany but right from the beginning psychology in the United States was following its own arc. The American psychologist Granville Stanley Hall (1846 – 1924), who established the first psychology laboratory in the United States at Johns Hopkins University in 1883, was certainly inspired by Wundt but Hall was influenced by William James as well.[5] Hall had studied with James and

attended the psychology classes James had just recently begun to teach at Harvard, after which he received in 1878 the first psychology doctorate ever awarded in America. A number of other American psychologists who, just like Hall did after his graduation, spent time in Germany at Wundt's laboratory were also influenced by the Leipzig school. However, while Wundt's ideas fell on fertile ground in the United States, not everyone agreed with the German professor on what constituted an allowed field of study for psychology and what not. Allowed in the sense that Wundt had blessed it as admissible to psychological research. Hence, while a good number of American psychologists could trace their intellectual heritage back to Wundt, there never evolved a Wundt school in America. Not for the lack of trying as for example evidenced by the career of the English psychologist Edward Bradford Titchener (1867 – 1927), who studied with Wundt for a number of years and then went on to found his own laboratory at Cornell University. There, he focused on understanding the human mind from a structural perspective using a more refined and controlled version of Wundt's approach. Structuralism of the Wundt-Titchener kind considered all mental experiences to be combinations of simple elements or events. Its contention was that by understanding those simpler elements, one could hope to learn more about the structure and workings of the human mind itself. Wundt and his scholars, such as the just named Titchener, and adherents of functional psychology, the school of thought pioneered by the American psychologist William James, sought to explore human nature by asking very different questions. One always has to be careful when categorizing or labeling things. That said, in this case one could venture that psychological structuralism asked questions about what something constitutes whereas psychological functionalism asked questions about what something does. Clearly, when it comes to psychologists trying to understand human consciousness those two approaches could not be more different.[6] Psychological structuralists and psychological functionalists simply disagreed fundamentally on what kind of observations could provide meaningful psychological insight into the nature of the human mind. Wundt scholars in the psychological structuralism mold asked the question what is it? Functionalist psychology following William James asked the question what purpose does it serve?

Wundt and his Leipzig school of psychology, spearheading the physiological psychology approach in Germany, where over time joined by other schools of psychology, several of them founded by Wundt disciples who had dared to disagree with Wundt's narrow view of what represented permissible psychological research. According to Wundt's school, and that is why the history of psychology refers to them as physiological psychologists, researchers were only to explore human psychology through stimulus-response type measurements. All that mattered was to characterize the physiology involved in the response to a given stimulus in the best possible way. To Wundt scholars there was only one scientific avenue towards studying the workings of the human mind itself and that was through what they called internal perception. Wundt understood internal perception,

the process of examining one's own conscious experience as objectively as possible, as the psychological equivalent of how other sciences observe nature to derive for example physical or chemical laws.[7] Mostly, this translated into accurately measuring response times to a given stimulus but it certainly should not include interpretation or speculation as to what kind of mental processes were involved in producing a response. In that sense, Donders inferring from his measurements how long mental processes of discrimination and choice might take as part of the response time to a stimulus was already going beyond the limits of what Wundt considered to be within the bounds of psychological research.

Wundt's conception of consciousness was of the associationist kind. He believed consciousness to be composed of simpler elements accessible to understanding. Initially, when he was still at the University of Heidelberg, his approach to study human consciousness was a three-pronged one, using experimental methods, statistical methods and evolutionary-developmental methods. However, he eventually came to reject the third approach as he deemed it to speculative and the Leipzig school focused on the first two approaches only. This decision by the very person who would come to dominate the study of psychology in Germany and influence its progress in many other countries, essentially declared certain areas of exploration off limits for psychology. Prominently, this included the role of the unconscious in the human mind. Hence, one can credit or blame, depending on one's view, Wundt for splitting off the study of the unconscious from psychology proper, thereby destining it to become eventually its own separate discipline – psychoanalysis. Wundt's reluctance to theorize about the nature of the human mind was a core tenant of his psychological research. According to Wundt, how we know about something was not discernible through building theories or models of how the human mind functions but strictly a matter of experimentally discovering how we actually come to know something.[8]

Even though Wundt's views were quite influential in the United States, American psychologists never completely gave up on pursuing a more holistic approach to human psychology, insisting psychology must explore mental processes themselves to understand human nature. William James had a very different outlook with respect to what the science of psychology should concern itself with. A contemporary of Wundt, he admired many of the experimental findings the German professors and their students made but he also was adamant that internal perception as Wundt propagated it, later frequently referred to as experimental introspection, was a hopeless effort to understand the human mind that could not succeed. James advocated for a different kind of introspection, one that was admittedly more difficult and in some cases impossible to verify experimentally, which however could provide insights into human thought processes that then could allow a better understanding of human behavior. William James functionalist approach rooted in philosophic pragmatism. This philosophical school of thought originated in the second

half of the nineteenth century in the United States and James along with the American philosophers Charles Sanders Peirce (1839 – 1914) and John Dewey (1859 – 1952) was one of its most prominent representatives. According to this position, it is best to explore the nature of things such as the human mind by understanding it through the exploration of its practical aspects. It is in essence a commonsense approach to understand the human mind. While William James used this approach to explore the nature of the human mind, the next several generations of American psychologists would take this utilitarian approach even further. American psychologists would eventually come to deny that psychologists could gain any insight into the human mind at all by introspection of whatever kind, be it of the Wundt type or the functional psychology type. For this new branch of psychology, asking questions about mental processes was meaningless. This position, appropriately called behaviorism, asserted that we could not know anything about what went on in a human mind. However, that was of no consequence because all that was necessary for understanding human psychology was to study human behavior. To behaviorism, the human mind was like a black box and everything that was worthwhile knowing about human psychology was observable from behavior. All that psychologists had to do is to break down behaviors into their most simple components, understand those simple behavioral components and then work their way up to more complex behaviors, which ought to be compositions of the more simple behavioral components. In that picture, essentially all higher mental functions were nothing more than complex chains of simpler behaviors. The best approach was to understand behavior in simpler animals than humans first because many such behaviors are conserved throughout evolution and hence should help to shed light on the human behavioral repertoire.

Before we can however explore behaviorism a little more we need to take a look at some of those schools of psychology which next to the Leipzig school sprouted in nineteenth- and early twentieth-century Germany. None of them would rival behaviorism, which came to dominate research in psychology departments, much more so in the United States than in Europe, well into the second half of the twentieth century. However, some of them carried seeds that would eventually germinate with the rise of cognitive psychology in the 1960's. Among those psychology schools, we find the Würzburg school, the Frankfurt and Berlin schools of Gestalt psychology and other schools endorsing versions of Gestalt psychology such as the Graz production school, or the Padua school. Then there were new developments in specific areas of psychology, not necessarily associated with any particular school, leading to new theories in fields such as learning and development. Finally, there is of course the emergence of psychoanalysis, which brought back into focus the role of the unconscious that Wundt and his disciples had discarded as an unworthy subject of psychological research. Adherents to those emerging new fields in psychology interested in the role of the unconscious argued that one never could really learn anything about the unconscious within the framework of the science of psychology as

the Leipzig school had defined it. Not everyone active in psychological research belonged to a psychological school. Many would break new ground in psychology without building a school around their new areas of expertise or founding a school on their belief of just what the science of psychology should encompass and what should not be a part of it. Nevertheless, through their research, such scientists could influence others who eventually would create schools of psychology, or they could influence how members of a particular school reflected certain aspects of psychological research within the framework of their school.

Helmholtz certainly was a singular figure in nineteenth century science, contributing to physics, medicine, as well as psychology. We already encountered him in the first chapter as the discoverer of the speed of signal transmission in the human nervous system. However, in physiology he became equally if not more famous through his work on human vision where he invented the ophthalmoscope, an instrument used to examine the inside of the human eye, and developed empirically based theories of depth perception and color vision. Helmholtz's color vision theory proposing three types of color-sensitive receptors built on the insights of the British natural scientist Thomas Young (1773 - 1829) and the Scottish physicist who gave us the theory of electromagnetism, James Clerk Maxwell (1831 – 1879). Young had proposed a color vision concept based on the three primary colors red, green and blue, while Maxwell had studied additive color mixing, detailing how any color is a composition of the right amounts of red, green and blue. For example, cyan, magenta and yellow are secondary colors due to mixing of red and blue (magenta), red and green (yellow), and green and blue (cyan). According to Helmholtz, or what become known as the Young-Helmholtz theory of trichromatic vision, we perceive colors through three types of receptors for red, blue and green. The perception of white represents equally balanced amounts of red, blue and green light while the perception of black is due to no light received in any of the color channels. The German physiologist Karl Ewald Konstantin Hering (1834 - 1918) disagreed with this model and developed a rival color vision theory based on three types of color-opponent receptors sensitive to three types of opponent colors: yellow-blue, red-green and white-black. Without any light impinging, each of the color-opponent receptors would balance and produce the experience of grey. With light impinging, color-receptors would produce one of the opponent colors while inhibiting the other. Because of that, we cannot perceive a reddish green or a bluish yellow, but we can experience yellowish red (orange), greenish yellow (chartreuse), greenish blue (cyan) and bluish red (magenta). Eventually, both theories turned out to be correct with respect to color vision, as modern science has shown us that humans have three types of conically shaped color-sensitive receptors, which then combine their signals in three color-opponent channels. However, in addition to the conical color receptors, humans also have rod-like receptors for night vision; actually, we have many more of those than conical color receptors. The rod receptors do not mediate

color vision and it is not yet clear how their signals contribute to the color models. The vision physiology evolving in the nineteenth century gave psychologists a completely new perspective on what the visual physiological stimulus looks like. Twentieth-century vision and brain physiology would give psychologists an even more detailed but still far from complete understanding of how our brains may construct our visual realities.

We already mentioned the names of Weber, Fechner, Ebbinghaus and Stumpf earlier. Fechner was a student of Weber who had pioneered the study of just noticeable differences. His studies led Weber to discover that the minimum distance where we can differentiate whether one or two objects are touching our skin varies significantly over the human body. More generally applicable to other senses than touch, Weber's research revealed that the change in a stimulus that will be just noticeable is a constant ratio of the original stimulus. This is likely a familiar experience for many, as we for example all know that we can understand what another person whispers to us in a quiet room while for us to understand the same message in a noisy environment a person has to shout it to us. Fechner continued the work of his teacher in his own way, almost blinding himself in the process as he self-experimented with visual perception, and formulated what we know today as the Weber-Fechner law. This law states that the sensitivity to a stimulus varies with the strength of the stimulus logarithmically and not in a linear fashion. What that means is that in order for us to register a given sense perception as doubling in strength, the stimulus has to increase by a factor of ten. While the law does hold in many cases and is a good approximation in others, it does not apply to all of our sense perceptions or just not over the full signal spectrum a sense is capable of registering. An example for that would be an exposure to loudness in different frequency ranges where the frequency dependency of our hearing capability comes into play. In addition to the quantification of stimulus-response type measurements to understand the physiology of sense perception some researchers continued to ask different questions, such as how learning works or how we remember something.

The first meaningful answer to any such questions came from Hermann Ebbinghaus, who pioneered the study of human memory in the final decades of the nineteenth century by devising clever experiments. Even though he himself was the only test subject, many of his findings first summarized in his 1885 publication *Über das Gedächtnis* (*On Memory*) turned out to be generally correct. Among those are the now famous learning and forgetting curves. A learning curve generally describes how fast we learn. In its simplest form it represents how much more a human test subject retains of what we ask her or him to remember with each new attempt she or he makes to memorize it. This learning curve turns out to be a logarithmic curve where the sharpest increase occurs with the first attempts to memorize something and where with each successive attempt what we can remember increases by less and less. The forgetting curve describes how fast we forget what we have learned. It quantifies how much information a person can retain

after memorizing something one time without any further attempt to retain this memory while then having to recall the memorized information at set intervals. The shape of this forgetting curve is that of an exponential decay, meaning, the biggest information loss occurs right after we have learned something and only levels off after a day or so.[9] Therefore, unless we make additional efforts to convert our short-term memories into long-term memories, we will lose what is stored in our short-term memory after a few days. Of course, there are different kinds of learning experiences and what we encounter in daily life differs from the kinds of experiments Herman Ebbinghaus subjected himself to where one memorizes rows of meaningless three letter words resembling chance letter combinations that otherwise carry no cognitive associations. Ebbinghaus memory tests provide a valuable baseline for assessing how fast we can memorize something and how quick we may forget it. However, we usually do not have isolated memories such as represented by collections of Ebbinghaus-type arbitrary number or letter sequences not related to anything else. Some things we may remember only for a few minutes at best before we forget them forever. Then there are other things that stay longer in our minds even though we make no conscious effort to remember them. This is because memories staying longer with us are mostly contextual memories. Contextualizing something which we would like to remember is one of the techniques that we also can consciously use to help us keep a memory for longer or memorize more of something than we otherwise could retain.[10]

Carl Stumpf became famous for his studies on the psychology of music and sound.[11] On the one hand, Stumpf realized the importance of exact quantitative measurements of physical stimuli and the sensations they produce, something he had come to appreciate in his work with Weber and Fechner. However, Stumpf had also a mentalist conception of psychology, not just a physiological one. That already became evident in his early work on space perception where he for example judged depth perception as being innate, clearly a nativist and Kantian position. In his work on the sensation and perception of tones, he would also come to distinguish between phenomena and mental functions. For Stumpf, phenomena could be either sensory or imaginary. He understood sensory phenomena as that which we perceive through our senses while the reality of imaginary phenomena arises through their interpretation by our minds. Hence, there are sensory and mental phenomena. It is no accident that Stumpf coined the term phenomenology for the study of such sensory and imaginary phenomena. Stumpf had studied with the German psychologist and philosopher Franz Brentano (1838 – 1917) who planted the seed for phenomenological psychology not only with Stumpf but also with other students. Among those were the Austrian philosophers and psychologists Alexius Meinong (1853 – 1920) and Christian Maria von Ehrenfels (1859 - 1932), the latter also a student of Meinong, and the German philosopher Edmund Husserl (1859 - 1938). Brentano's 1874 published *Psychologie vom empirischen Standpunkte* (*Psychology from an Empirical Standpoint*)

advocates for a scientific exploration of the human mind using introspective and descriptive methods. He reintroduced intentionality, a concept originating with scholastic philosophy, to differentiate mental phenomena from other phenomena. Brentano perceived mental phenomena as directed at objects, what he called objects of intentionality. His concept of intentionality would resonate in philosophy as well as in psychology. With his descriptive approach to psychology, which relied on self-introspection, Brentano's approach to introspection was closer to that of William James than to Wilhelm Wundt's narrower conception of it. The same was true for a number of Brentano's students.[12]
Meinong would go on to found the Graz school of psychology where Ehrenfels would join him and Husserl's name would become synonymous with a philosophical position referred to as the phenomenology of scientific reason. Stumpf was opposed to the Wundt-kind of pure introspection which sought to reduce experience to elementary building blocks and argued for examining the experience as it occurred. In his experiments, Stumpf would discriminate how for example someone trained in music would experience tones vs someone untrained would do so. Wundt disagreed with Stumpf's approach because he was opposed to discriminating individual responses. According to Wundt's method, psychological experiments had to be conceived such that the individual experiences would not matter. For Wundt, admitting individual experiences to the study of psychology would be tantamount to accepting subjective mental states as valid objects of psychological research, something Wundt certainly was not prepared to allow. While Stumpf and Wundt argued their positions in a series of literary battles, Stumpf had a lively and friendly correspondence with William James who had similar disagreements with Wundt.
The German psychologist Oswald Külpe (1862 – 1915) was one of Wundt's students and had been Wundt's assistant for eight years. In 1896, he established his psychological laboratory at the University of Würzburg, which became the so-called Würzburg school, pursuing a different approach to introspection. Unlike the Leipzig school, Külpe and his collaborators and students came to view consciousness not as composed of discrete psychological elements rooted only in perceptions but rather saw it as a single entity including the unconscious. Among the members of the Würzburg school were the German psychologists Narziß Kaspar Ach (1871 - 1946), Karl Ludwig Bühler (1879 – 1963) and Karl Marbe (1896 - 1953). Ach was one of Külpes first students and one of the first to study volition, the cognitive process by which an individual decides on and commits to a course of action, something we colloquially refer to as the exercise of one's will. Bühler, Külpes long-time collaborator and assistant, became the leading figure and spokesperson advocating the Würzburg school position in a heated controversy with Wilhelm Wundt. Marbe first came to Würzburg to become one of Külpes earliest collaborators and would return a second time to the University of Würzburg as Külpes successor. From the perspective of the Würzburg school members, mental states in the human mind also included abstract thoughts with no connection to perception, something that Külpe and

his students referred to as imageless thought. Introspection in the Würzburg school, as summed up in Külpes lectures, posthumously published with the help of Bühler, included the observation and reporting of thought processes, something Wundt perceived as futile and akin to charlatanry.

The Würzburg school added the element of abstract thought and hence abstraction to the concept of consciousness. That the mind can have a life of its own without being stimulated from the outside is something many of us would regard as a commonsense experience. For example, we all dream when we sleep, some of which we may remember when we wake up. Dreams are creations of our mind, surfacing into our conscious mind from our unconscious mind whether we desire that or not. We could also describe them as something we experience and carry back into our conscious mind from a more or less deep dive we unwittingly take into our unconscious mind when we sleep. In either case, dreaming is a subjective and private human experience and so is abstract thought. For those like Wundt seeking to establish the science of psychology in the mold of sciences like physics and chemistry, anything tainted with subjectivity could not be part of legitimate psychological research. The Würzburg school took the first step to change this conception of psychology, the next step was made by Gestalt psychology. Incidentally, among the pioneers of this new perspective on psychology were the German-American psychologist Max Wertheimer (1880 - 1943), Külpes student, as well as Külpes and later Marbes assistant, the German psychologist Kurt Koffka (1886 - 1941). [13] The collaboration between Wertheimer, Koffka and the German psychologist Wolfgang Köhler (1887 - 1967) leading to a new approach in the quest to understand the human mind began in Frankfurt on the river Main in 1910. While on a train trip to a vacation in the Rhineland, Wertheimer came to consider a new hypothesis on the perception of motion. He stopped in Frankfurt to test his hypothesis and asked the head of the psychological institute in Frankfurt for permission to use some space and resources to conduct experiments, which he received. As it happened, Koffka and Köhler were both working at the Frankfurt psychological institute. They were available and willing to serve as Wertheimer's test subjects in his experiments but at once became his collaborators in discovering and articulating what became known as Gestalt psychology.

Gestalt Psychology and Field Theory

The German word Gestalt, depending on its use, can carry the meaning of shape, form, stature, configuration or pattern. In Gestalt psychology, a Gestalt refers to a whole that is more than the sum of its parts. Gestalt psychology represents the discovery, or rather the rediscovery, of something we are likely all familiar with: the human mind adds structure and meaning to that which actually is physically there to see. On a warm and sunny day, lying on ones back in the soft grass and looking up into the sky as scattered clouds sail through the blue sky, who has not imagined the shape of a cloud to have a certain Gestalt? Wertheimer, Köhler and Koffka began their collaboration 1910 in Frankfurt and continued it later in Berlin, giving rise to the Frankfurt and Berlin Gestalt psychology schools. What triggered Wertheimer's interest in 1910 and eventually led to him, Köhler and Koffka joining their efforts in Frankfurt, was a visual perception experience Wertheimer had while traveling on a train through the Rhineland. The experience that distant stationary objects such as telegraph poles as perceived from a moving train seem to be speeding along with the train is something that may still be familiar to many. However, few of us would be as curious about this illusion as Wertheimer was when he asked himself why that is so.

We frequently talk about human vision as if it represented a kind of camera with our eyes functioning as its lenses; but that is not so. Upside-down visual images do indeed form on the back of our eye's retina, just as one would expect from a simple pinhole camera. However, this is where optical analogies break down. There is no screen in our brain onto which to project an image nor is there an entity in our brains, to stay in the picture, to watch any such generated movie. Photoreceptors in our eye's retinas transform light rays into electrical impulses, which then travel through the optic nerve into the brain. Why then does the image our brain reconstructs represent the outside world as if human vision were actually like camera vision? Take a photograph of something, like a landscape. Then compare the photograph with what you actually see in this very landscape when standing right where you shot the picture. Well, if you think about it for a moment you may come to the conclusion that this not such a surprise, after all, we build cameras just so that they replicate what we see. That includes for example that the photographic color curve replicates how humans see color and not how any other animal may see them.[14] Still, is it not amazing that our brain somehow gives us a true representation of this landscape, even though human vision only superficially resembles a camera? There just is no one in our brain looking at the image our brain constructs so that what we perceive through our eyes looks identical to what we see in a photo. However, our mind also sees things a camera cannot see. With the click of a camera, we take a single photo, or we can take many photos in a sequence to make a film, but that is it. Different from that, the human mind not only continuously captures images it also processes them.

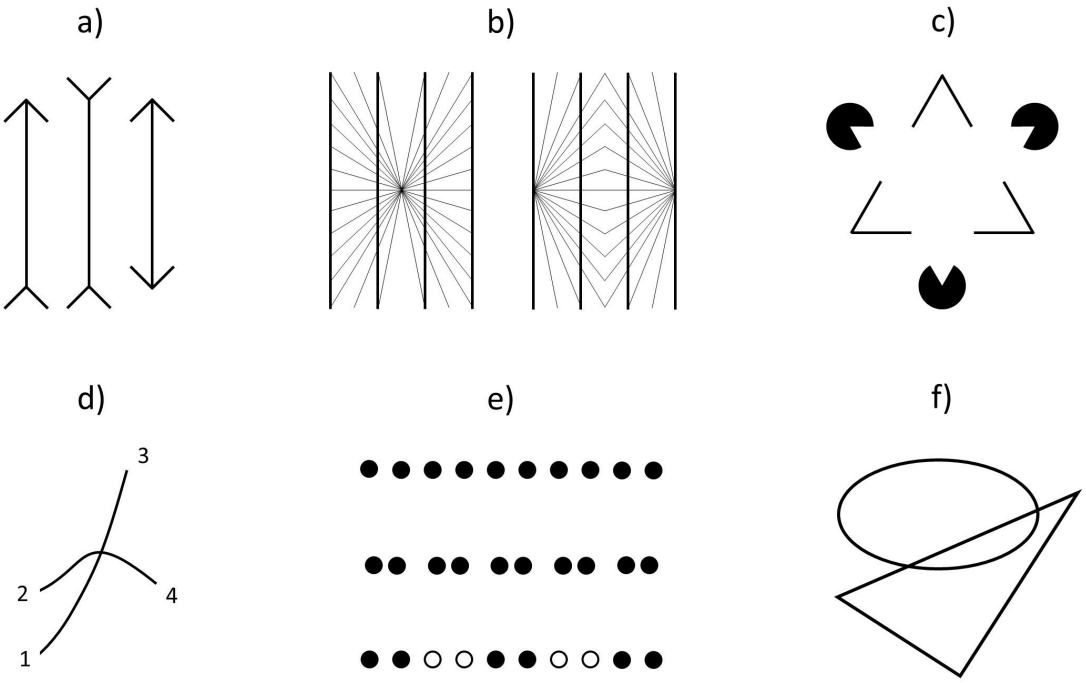


Figure 3.1: Examples for optical illusions illustrating Gestalt psychology aspects: (a) the Müller-Lyer illusion: the different shafts make lines of equal length look as if they had different length; (b) the Hering illusion: the inner lines of four perfectly parallel lines appear not to be parallel but bent, the direction of the perceived bent depending on the direction of the star pattern; (c) the Kanizsa triangle which is not really there: we see a white triangle jump out of the white background which in reality is not there (Gestalt law of completion or closure); (d) rather than two angled objects 1-2 and 3-4 meeting, we see a continuation of lines 1-3 and 2-4 (Gestalt law of continuity); (e) in the first row we see ten single dots, five pairs of dots in the middle row (Gestalt law of proximity), and again five pairs of dots in the lower row (Gestalt law of similarity); and (f) illustrates that we always see the most simple shapes, an ellipse and a triangle and that it takes effort to see the more complicated shape defined by the overlap of ellipse and triangle (Gestalt law of Prägnanz).

As it is, we never stop recording images. Even when our eyes are closed while we sleep, the human mind – and other animal minds - can conjure up images from within the mind and process them just like it does with what passes through our eyes in our wake states, thereby creating the images of or dreams.

Gestalt psychology schools also flourished at the Universities of Graz in Austria and Padua in Italy. The psychological laboratory in Graz was the first such psychological institute in Austria-Hungary, founded in 1894 by Alexius Meinong. As a philosopher, Meinong carried on the phenomenological approach of his teacher Brentano but was also very receptive to the British empiricists, notably to the philosophy of David Hume. Those influences carried over into his psychological research and Meinong's psychological institute became his Graz school of phenomenological psychology. His psychological

theory of how we perceive objects would lead him to conclude that how we experience the totality of a perception is something different from the sum of the component parts making up a perception. With that, Meinong was on a path towards Gestalt psychology but it would be his pupils who would found the Graz school of Gestalt psychology which became known as the Graz production school. Another student of Brentano whom Meinong had supervised while both were at Brentano's institute in Vienna was Christian Maria von Ehrenfels. Ehrenfels joined Meinong in Graz where he found Gestalten in music and became a co-founder of Gestalt psychology. Being a musician and composer of librettos, Ehrenfels recognized that in a transposed melody, that is a melody where all notes are moved up or down in pitch by a constant interval, we can still hear the original melody. In that sense the melody is a Gestalt, which we still recognize for what it is even though the pitch has changed. The flourishing Gestalt psychology school in Padua, associated with the Austro-Hungarian and Italian psychologist Vittorio Benussi (1878 – 1927) had close links with the Graz school, which had been his academic home for many years. Vittorio Benussi had been a student of Meinong and he would have likely become a professor at the University of Graz and head of its psychological laboratory had resentments not denied that to an Austro-Hungarian citizen of Italian nationality during the First World War.[15]

The different Gestalt schools did not always agree, specifically on how the perception of a Gestalt came about. The Frankfurt and Berlin schools of Gestalt psychology considered Gestalten (the plural of Gestalt) as something innate, whereas the Graz production school and for the most also the Padua school hypothesized that a Gestalt came about through an individual's effort to perceive discrete elements as a Gestalt.[16] Many examples for Gestalt psychology are of the visual kind, frequently involving illusions, seeing something that is not there. Fig. 3.1 shows six examples for that. Fig. 3.1a illustrates the Müller-Lyer illusion, named after the German psychiatrist Franz Carl Müller-Lyer (1857 – 1916), where lines measuring the exact same length seem to be of different length because of the varying shafts attached to the line ends. Fig. 3.1b gives an example of the Hering illusion, named after the German physiologist Karl Ewald Konstantin Hering, where parallel lines do not look like parallel lines at all. Fig. 3.1c shows the Kanizsa triangle, named after the Italian psychologist Gaetano Kanizsa (1913 – 1993), which has the viewer perceive a white triangle that is actually not even there; this figure is a frequently used illustration for the Gestalt law of completion or closure.[17] Fig. 3.1d gives an example for the Gestalt law of continuation. Rather than seeing two angled objects circumscribed by the lines 1-2 and 3-4 joined at their tipping points, we see a continuation of the lines connecting 1-3 and 2-4. Fig. 3.1e illustrates two Gestalt laws, the Gestalt law of proximity and the Gestalt law of similarity. In the upper row of this figure, we see ten single dots, in the middle row five pairs of dots due to the Gestalt law of proximity, and in the bottom row, we see again five pairs of dots, this time because of the Gestalt law of similarity having

us separate pairs of filled and open dots. Finally, in fig. 3.1f, we immediately recognize the shapes of an ellipse and a triangle but it takes effort to visualize the overlapping area of those two shapes as a separate shape. This is an example for the Gestalt law of Prägnanz, which says that we automatically decompose what we see into simple shapes and not complicated ones.[18]

Even though Wertheimer came to Gestalt psychology through the study of perception, his larger interest was with understanding learning, motivation and thinking. He believed that in all three areas Gestalt theory could help psychologists gain a better understanding.[19] Koffka and Köhler thought so too and soon enough both arrived at a similar conclusion, as both came to believe that Gestalt qualities inform much of human behavior. Köhler, although not by choice, would find ample time to study how Gestalt qualities informs the behavior of our ape cousins during the years of the First World War. In 1913, the Prussian Academy of Science had set up an ape research station in Tenerife, the largest of Spain's Canary Islands. Köhler was named head of this research station in 1914 but because of the outbreak of the First World War, he did not just stay there for one year but remained in Tenerife until 1920. His pioneering work on problem solving with chimpanzees became quite famous even outside psychology circles. Eventually, it would earn him the director position at the Berlin psychological institute in 1922 were he succeeded his former teacher Carl Stumpf, the founder of the institute. What Köhler discovered was that in figuring out how to stack boxes or assemble a longer stick from shorter ones to grab otherwise out-of-reach bananas, his chimpanzees did not proceed in a trial and error fashion. Rather they arrived at the solution all of a sudden without much trial and error. The solutions seemingly presented themselves to Köhler's chimpanzees in a Gestalt-like quality where they suddenly realized what a solution looked like in its totality.

While Köhler, confined to the island of Ternerife, was diving into animal psychology, Koffka focused his efforts on shaping Gestalt psychology into a body of coherent theories and applied it on the experimental side to developmental psychology. Koffka contended that infants first perceive of and respond to their surroundings holistically and only later do children develop the ability to discern the individual sensations that comprise a whole. Koffka essentially became the public advocate for Gestalt psychology, promoting it throughout Europe but importantly also in the United States. There, Koffka's views on developmental psychology would eventually exert a greater influence than in his homeland. In 1920's Germany, the majority of psychologists remained reluctant at best to endorse Gestalt psychology or even rejected it outright. In the United States, even with behaviorism rising to dominance, American psychologists remained receptive to what Gestalt psychologists had to say. With anti-Semitism on the rise and the Nazi party gaining power in Germany, the writing was on the wall for everyone to see. There would be no future for any German professor with a Jewish heritage. In German speaking lands,

Jewish scholars had always faced obstacles in their careers because universities would often delay or outright refuse promoting Jewish scholars to full professors. Wertheimer and Koffka were both German citizens with Jewish heritage and left Germany for the United States, Koffka in 1924 and Wertheimer in 1933. Köhler was a non-Jewish German and like so many academics in Nazi Germany, he could have arranged himself with those in power. However, to his enduring credit, Köhler spoke out and eventually had no choice but to emigrate in 1935.

Köhler was not just interested in psychology but also in natural sciences and while working on his doctoral thesis at the University of Berlin he was not only a student of Carl Stumpf but also of the German physicist Max Planck (1858 – 1947), one of the eminent scientists of his time. An important aspect of modern theoretical physics is field theory. James Clerk Maxwell's theory of the electromagnetic field was the first example of such a theory. In 1917, while Köhler worked on his doctoral thesis, Albert Einstein, at the time a professor at the same university, published his general relativity, a field theory of gravitation. Perceiving a force field of physics such as the electromagnetic field as a pervasive property of space itself comes naturally to us, as this is just the way we mostly experience such fields.[20] Köhler presumed that our brain was bathed in a likewise force field and that what humans experience cannot be simply correlated with local events in our brain but was governed by a field-like effect extending throughout the whole brain. To describe this notion Köhler used the term isomorphism, which in the language of mathematics describes a reversible structure-preserving mapping between two mathematical structures of the same type.[21] In Gestalt psychology, Köhler isomorphism denotes a conceptual similarity between the Gestalt quality of a perception and the Gestalt quality of its respective physiological representation. In essence, Köhler's isomorphism was a form of psychophysical parallelism but without asserting a point-for-point like correspondence between what happens physiologically in our brain, the Gestalt quality of a physiological brain state, and what we experience in our minds, the Gestalt quality of perception.

Forms of psychophysical parallelism of the stricter point-for-point variant had been endorsed by many psychologists before. Prominently among them Wundt and his followers as well as scientists like Titchener, who developed his own kind of structuralism.[22] Disproving such a strict point-to-point like psychophysical parallelism was more difficult than disproving a Gestalt-like psychophysical parallelism which Köhler envisioned. Physically changing something like Köhler's force field, representing the Gestalt property intermediating his psychophysical parallelism, is much easier than finding the exact loci in a brain that correlate with specific mental experiences. Provided, that the latter was actually the case and that a strict point-for-point correlation did really exist, mapping such a complex network structure to the extent of knowing exactly where to cut something in the brain to interrupt a specific single point-for-point correlation would require an understanding of what exactly goes on in each and every part of our brain. While we know

much more about brain physiology and brain processes today than what was known in the early twentieth century, any such understanding is still very limited. A non-local theory of the human mind like Köhler‘s force field was much easier to disprove. When other psychologists demonstrated that lesions in animal brains did not disrupt earlier learning acquired before the lesions were made, that was taken as evidence against a pervasive electromagnetic field like Gestalt property connecting physiological brain states with our experiences.

Like Koffka and Köhler, the German-American psychologist Kurt Lewin (1890 – 1947) had studied with Carl Stumpf in Berlin and was for some time a member of the Gestalt group of psychologists. Lewin applied Gestalt psychology to the understanding of human social behavior. He believed that in order for psychologists to understand human behavior, they must visualize all forces affecting it in a Gestalt-like manner, essentially conceiving the total environment influencing human behavior as a psychological force field, analog to how physicists conceive of force fields governing the behavior of matter. However, different from Köhler's field theory, the force field in Lewin's field theory was not an electromagnetic field permeating the brain but rather it was the environmental and social field that any individual is constantly immersed in. In many ways, Lewin's approach to psychology foreshadowed how social sciences in the second half of the last century, with behaviorism's dominance waning, would increasingly look towards psychology to understand, ameliorate or even help resolve challenging societal issues. Once one acknowledges the importance of external factors on human development and behavior, as Lewin did with his field theory approach, psychology necessarily assumed a societal and political role. Formative for Lewin's focus on understanding how the individual experienced her or his environment was likely his own experience as a front-line soldier in the First World War. Because of his contributions in fields such as developmental psychology, conflict resolution, action research, group dynamics, organization development, leadership or minority problems, many acknowledge Lewis as the founder of social psychology.

The Europe that emerged from the First World War was nothing like the Europe that existed before the war. There always had been a vibrant exchange between researchers in the New World and those in the Old World. People from various parts of Europe had been emigrating to the New World in large numbers in search of economic opportunities denied to them in their home lands, or had simply fled persecution where it still was part of life, like for many Jews in the Pale of Settlement which was once part of Russia and now had become part of the Soviet Union. Even though before the war a Jewish heritage could result in academic promotions delayed or denied, a great number of Jewish scholars in the German Reich of the Kaisers and in the Habsburg Empire of Austria-Hungary had academically succeeded. A good portion of the most renown scholars in German speaking lands were either Jewish or had partial Jewish heritage. Many successful Jews

came from families assimilated over centuries into the various national societies to which they belonged just as much as any of their non-Jewish compatriots. A good number of them took assimilation even one step further and had stopped practicing their faith, at least in public, and some had even converted to one of the Christian faith denominations. On the surface of it, they were seemingly equally valued citizens of their countries and clearly eager to serve and further the interests of their nations just as much as any of their non-Jewish fellow citizens. However, anti-Semitism had always been there, sometimes as nasty as in Vienna where an openly anti-Semitic mayor was in power between 1897 to 1910, often however in much more subtle but no less effective ways, making sure that non-Jews could remain among themselves in their societal organizations of choice.[23] After the First World War, this pervasive but mostly latent anti-Semitism would begin to surface, finally culminating in the atrocities committed in the name of Germany by the Nazis and their many followers. Hence, the years after the First World War would see an increasing number of Jewish scholars leaving the Old World before the Nazis even came to power. Once they were in power, wherever Jews still held academic positions, the Nazis and their sympathizers quickly dismissed them. Those lucky enough managed to get out of Germany in time before the Nazis began to murder every Jew, they could lay their hands on; but not everyone was so lucky. The brain drain to the New World, not just from Germany but also from all over continental Europe, was immense and psychology was no exception in that regard. German speaking lands lost their prominence in science as did much of continental Europe either because of German occupation or the devastation of the Second World War.

On the other side of the Atlantic science would not just continue to flourish but it would be very much strengthened and invigorated through the intellectual blood infusion coming from the Old World. However, in the liberal sciences, to which much of psychology seemed to belong, this worked different from the natural sciences. Physics just works in one way, and it does not matter what you believe in as mathematics or experimental proof will show you either right or wrong. In psychology, however, it was more of a question which viewpoint one held. At the time, behaviorism had already asserted itself in the United States and it would not allow psychologists with positions such as articulated by Gestalt psychology or field theory to encroach on its terrain. As much as behaviorism claimed to be a science, and there are still good arguments for that, to many of its practitioners, behaviorism seemingly had become as much a matter of faith as it was a science. With psychological research in the Old World in decline or at least set back severely for decades to come, behaviorism would reign supreme not only in the United States but extending its reach also in the Old World. However, before behaviorism was to take over much of the world of psychology in the first half of the twentieth century, at least in the United States, the Old World had one more major contribution to give in the human pursuit to understand our own nature – psychoanalysis.

Depth Psychology

The term depth psychology was coined by the Swiss physician and psychiatrist Paul Eugen Bleuler (1857 – 1939) in 1914.[24] Bleuler, who as a professor of psychiatry at the Burghölzli, then and now a leading Swiss psychiatric hospital in Zürich, took an interest in Sigmund Freud's psychoanalysis and so did one of Bleuler's research scientists at the Burghölzli, Carl Gustav Jung. When Bleuler introduced the term depth psychology, Freud and Jung had long broken up their former close relationship and Jung had moved on from being a student of and contributor to Freud's psychoanalysis to creating his own analytical psychology approach. Other onetime students and colleagues of Freud in exploring psychoanalysis also went their own ways, notably among them Alfred Adler with his school of individual psychology. The umbrella term of depth psychology, its original German name *Tiefenpsychologie* reflecting that likely even better, refers to psychological approaches in the study and research of the unconscious, considered to be hidden deep, or *tief* in German, within the human mind, not directly accessible via the conscious faculty of our mind. In the English literature, one can find all such approaches frequently subsumed under the umbrella of psychoanalysis itself. Hence, one has to be careful when talking about psychoanalysis to avoid confusing Freud's specific version of psychoanalysis with the psychoanalysis umbrella that covers other approaches to the study of the unconscious as well, most of which Freud himself would have declared to be anything but psychoanalysis, just like he did with Adler's individual psychology and Jung's analytical psychology. Freud very much saw psychoanalysis as his trademark and did not look kindly on people applying the term psychoanalysis to the psychological study of the unconscious that he did not deem to be part of psychoanalysis. In that sense, it is safer to use the term depth psychology to refer to all such kinds of psychological studies focusing on the unconscious and refer to psychoanalysis only to the extent that one discusses the Freudian approach to treating illnesses of the mind and his underlying theory of the unconscious.

Right from its beginning, psychoanalysis, with its embrace of the unconscious in our human nature, stood apart from everything else in psychology. Similar to the other two greats in psychologies equivalent to a hall of fame, Wilhelm Wundt and William James, the third, Sigmund Freud, the founder of psychoanalysis, also left a legacy that continues to influence psychology to this day. When Freud started out on his journey to explore the nature of the human mind he did so very much in the tradition of psychophysics. Freud began his career as a neurologist and for a number of years just like so many other psychophysicists he sought to understand human nature by understanding human physiology. However, two circumstances launched Freud on a very different trajectory that would take him towards his discovery of psychoanalysis. First, there was the fact that Freud was not wealthy enough to pursue a research career in his own private laboratory

and second, being a Jew in late nineteenth-century Vienna meant that he could not hope to have a research career at a university that could provide him the public means to pursue his interests. Therefore, Freud had to go into medical practice to earn a living and he had to look for interesting research topics that he could pursue without expensive experimentation equipment and facilities. The net of it was that Freud began to focus on the study and treatment of mental illnesses. He himself commented late in his life that his psychoanalysis was originally no more than an attempt to explain pathological mental phenomena but then had developed into a psychology of normal mental life. With medical research, there always has been the hope that the study of physical impairments of all kinds could potentially become a source of remedies or even cures for a number of ailments. In addition to that, such studies frequently provided medical sciences also with a better understanding of the healthy human body. However, different from physical illnesses, this had never been the case for mental illnesses. The history of how societies have dealt with the mentally ill does not reflect well on the mentally sane who make the rules for how to deal with the mentally ill. Mental illnesses, illnesses that had no detectable physical causes, were not only difficult to understand for medical practitioners, but they were also often frightening for those around the mentally ill. Under the best of circumstances, a family or charitable religious organization cared for mentally ill people, in the worst case they literally were locked away. Shackling or chaining the mentally ill was standard practice in public and private institutions built to house the mentally ill starting in the second half of the eighteenth century and would only give way to more humane treatments in the second half of the nineteenth century. Institutions such as the Burghölzli in Zürich represented a new approach to treat and understand mental illnesses. It was a rather recent realization in the emerging industrial societies of the nineteenth century that mental illnesses are not a curse of the gods, or a sign of evil, or as some claimed, just feigned by the supposedly ill to extract themselves from whatever perceived plight or societal demand, for whatever reasons. Maybe that is no coincidence. Putting mental illnesses aside whose physical origins were just not detectable by the medical practitioners at the time, the age of industrialization likely produced its own crop of mentally ill. Industrializing countries subjected large swath of their populations to a level of stress and plight not seen before with the abject poor struggling to survive in a world, which seemed to brutalize them and their families ever more. To those observing this situation from their privileged perches in society, that is if they did not choose to ignore what was happening to the less fortunate, it almost must have seemed natural that people having to endure such hardships could also increasingly suffer from mental illnesses. Mental illnesses it would seem, at least to the degree that they affected so many, were very much a disease of modernity.

When Freud began to work on the first precursors of what would eventually become psychoanalysis, he was almost forty years old. His psychoanalytic approach only began

to emerge fully once he had realized that in order to progress in his effort to understand the underlying causes of mental illnesses he had to give up his search to find physiological explanations for what goes on in the human mind. Until he finally acknowledged that, he had firmly believed that a physiological understanding was indispensable for developing successful therapies for mental illnesses as well as for conceiving theories of the human mind in general. By 1895, he had understood that physiology just was not mature enough to provide him the tools that could help him ground his therapies and theories in an organic basis and he concluded that he had to rely on psychology alone. The unconscious was nothing new to psychology. However, psychologists like Herbart, Helmholtz, William James and others had attributed only a passive role to the unconscious; basically deeming it a storage facilities for ideas and experiences which could eventually be recalled into the conscious again. That specific conception of the unconscious mind is what Freud would come to call the pre-conscious unconscious part of the human mind, separating the conscious mind from what he perceived to be the dynamic unconscious. As Freud saw it, the conscious mind could not be aware in any way of the dynamic unconscious mind at all, otherwise to call this to us hidden part of our mind the unconscious would be nonsensical. As illustrated in fig. 3.2a, the pre-conscious unconscious part of our mind, which contains that of which we are not aware but could become aware of, separates the conscious and dynamic unconscious parts of our mind. The dynamic unconscious, the largest part of our mind, is not a simple storehouse of ideas and experiences we cannot be aware of or access. Rather, for Freud, the dynamic unconscious was home of to our most powerful drives and forbidden wishes, which shape much of our human behavior.

In addition to these two parts of the unconscious, the pre-conscious unconscious and the dynamic unconscious in fig. 3.2a, Freud distinguished a third part of the unconscious, the part which he perceived as solely in charge of our perceptual and motor skills. Because this part of our unconscious draws on our procedural memories, neuroscientists like Eric Kandel call it today the procedural unconscious, a term Freud himself did not use. Freud already understood that this procedural unconscious is also the repository of what we commonly refer to as our habits and he would have been pleased to know that researchers today believe that this is also the case for many of our morals and ethics which we internalize and live by without usually reflecting much on them. That is, unless we begin to question them or probe their validity and appropriateness in specific situations. According to Freud, we are born with no inhibition between the dynamic unconscious and the pre-conscious unconscious. However, as we grow up we internalize what is and what is not acceptable behavior and the growing censor in our unconscious implicitly begins to learn what must stay repressed in our dynamic unconscious and what may surface into our pre-conscious unconscious. In essence, under the roof of our conscious mind live two parties, our procedural unconscious, and the trio of pre-conscious unconscious, censor and dynamic unconscious.

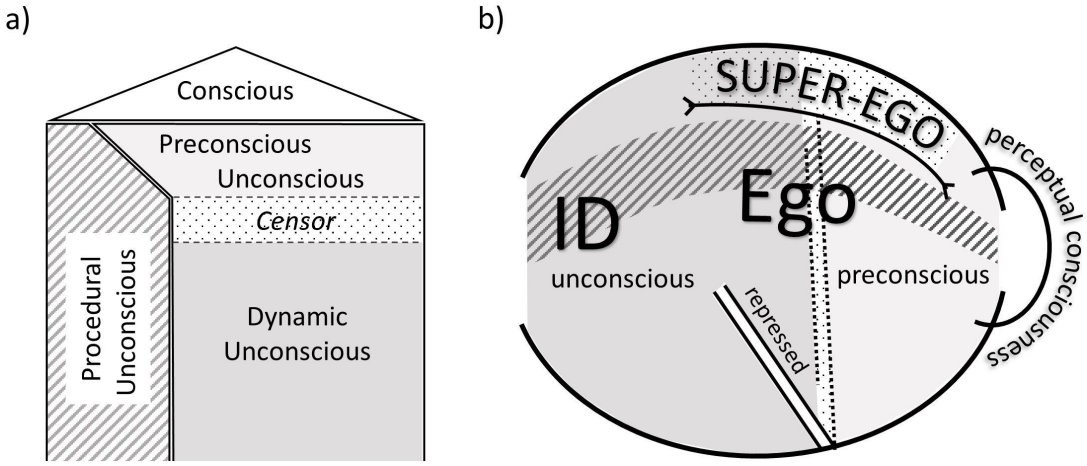


Figure 3.2: Freud's structural models of the human psyche: (a) a simple sketch of the conception Freud developed in his early forties; and (b), his more complex and refined structural model as Freud depicted it in his final conception of his psychoanalytic theory. The label "procedural unconscious" in (a) was not used by Freud himself but is a terminology introduced later by Kandel, describing what Freud believed the function of this part of the unconscious to be.

We all know that we can access our procedural unconscious and we also know that we cannot reach into our dynamic unconscious in similar ways. How much of that which rises from our procedural unconscious to our conscious mind actually passes through the preconscious unconscious is still unclear. Some of it evidently does so while of other matters in our procedural unconscious we can become easily aware of in a direct way. However, as we all have experienced, when we consciously try to exercise for example some of our motor skills, the control of which we usually leave to the procedural unconscious, we sense that our movements are a bit clumsy and not quite as smooth as they use to be when we do not think about them at all. Fig. 3.2a provides a simple sketch illustrating Freud's early thinking with regard to the organization of the conscious and unconscious parts of the human mind. Freud never drew a sketch like that shown in fig. 3.2a but it does reflect the basic picture of his early conception of the human mind. That is, before he began to include in his conception of the human mind what he considered to be the elemental person structures of a human being. Freud completed this transformation from his early to his mature conception of the human mind some twenty years later. By then, as shown in fig. 3.2b, he had overlaid his model of the human mind with a tripartite model of the human psyche consisting of the ID, the EGO and the SUPER-EGO.[25] In the model of the human psyche shown in fig. 3.2b, Freud combined his model of the human unconscious with the human personality structure as he had come to understand it. The grey-scaling and pattern fill, linking the respective parts between fig. 3.2a and fig. 3.2b, was added here to illustrate how Freud's earlier and late conceptions of the

human psyche connect and it was not part of the model of the human psyche that Freud began to use in the 1920's from which fig. 3.2b was adapted.

In Freud's conception of the human mind, the ID represents the dark and inaccessible part of human personality. The ID underlies our most basic human life drives, encapsulated for Freud by what he called the libido.[26] The libido, usually identified with the sex drive is broader than our common understanding of sexuality as it includes all kinds of pleasurable desires and impulses such as reflected for example by hunger, thirst or pain avoidance. Later, Freud paired the ID's libido with a death drive, which expresses itself for example through aggression and self-destructiveness.[27] By their very nature, the basic drives of the ID rule and structure much of our human behavior without us being aware of it, at least for the most. As it does so, the ID adheres to what Freud referred to as the pleasure principle, instinctively seeking pleasure and avoiding pain to satisfy biological and psychological needs. Understanding the ID is challenging. Freud thought that most of what we can learn about it comes essentially through the inversion of our understanding of the EGO. Freud himself said that we could only approach the ID by analogy, referring to it as a cauldron full of seething excitement or as chaos itself.

The EGO of Freud's human personality structure represents that which we usually identify as our conscious self. It reflects a person's conscious perception of her or his environment and of her or his respective self. In a mature adult, it successfully resolves challenges arising from conflicting demands coming from a person's ID, SUPER-EGO or the person's environment. When we are born, our EGO is practically non-existent and the ID's pleasure principle rules all of us as infants. However, as the EGO evolves and asserts itself in a normal healthy human development, we increasingly replace the pleasure principle with what Freud called the reality principle. An EGO ruled by the reality principle has learned to defer gratification or desires which the ID's pleasure principle would command us to satisfy instantly, if such gratification would conflict with our own morals or ethics or with social norms that disallow their immediate gratification. Which brings us to the role of the SUPER-EGO.

Simply put, Freud's SUPER-EGO corresponds to a person's external and internal compliance department. Its external section allows us to adapt and make our own social norms to live by without which human societies could not exist. Its internal section represents the instance that enables us to take control of our basic drives. Both functions of the SUPER-EGO, censoring how we behave socially and how we control our basic drives, are something we must learn as we grow up. Freud argued that because the SUPER-EGO as well as our ability to reason do not exist when we are born, we learn rule following and train the SUPER-EGO through fear (the German Angst). That includes for example the fear of losing parental affection or the fear of punishment in case we have done something our parents explicitly told us not to do. In essence, our SUPER-EGO acts as a moral censor, telling us what we must do and what we must not do, according to

societal norms and our own morals and ethics. However, a healthy EGO will not perceive the SUPER-EGO as a censor but rather see it as our conscience, helping us to do the right thing.

Experimental psychologists such as the psychophysicists derived their conclusion from experiments, which their colleagues could reproduce or falsify. Nothing like that was the case with Freud's work. Freud was clearly aware of that and one can only image the courage it required to proceed on such a research path whose scientific merit was always open to all kinds of challenges. However, Freud insisted that his method was scientific. It just so happened that his research objects were individual human beings themselves, unique human minds, rather than specific aspects of human physiology from which one could draw inferences on general human behavior. Underlying Freud's method development to research the nature of the human mind was a simple fundamental assumption he referred to as psychic determinism. In essence, psychic determinism asserts the existence of a causality of mental processes as it declares all mental processes as determined by preexisting mental processes, nothing in our minds happens by chance. Not all mental processes in such mental chains of causality are however accessible to our conscious mind. Therefore, even though there is an underlying chain of causality for all mental events, often we cannot connect its respective elements as quite a few of them reside in our unconscious mind. Hence, we do experience mental state occurrences that seemingly make no sense to us; or to people around us at the time. However, and this is one of Freud's great discoveries, these seemingly disconnected mental state occurrences are nonetheless part of a mental chain of causality that can be retrieved by using the right method, Freud's method of free association. Through the process of free association, the psychoanalyst can help a patient to discover parts of a mental chain of causality residing in the dynamic unconscious, which have become disguised or inaccessible through processes Freud referred to as resistance, projection or transference.[28] Via analysis of the latter three, a psychotherapist can work with the patient to help her or him to uncover and understand the underlying drives and motives that led to a mental disposition or illness in the first place.

Sigmund Freud's psychoanalysis was very different from any of the other psychological approaches to understand human nature preceding it. He firmly put the human unconscious at the center of his approach. First, he did so with his focus mostly on finding treatments for mentally ill patients for whom traditional methods did not work. Then, building on the understanding he gained from treating patients, he extended his theories to include the healthy human mind and its development. In Freud's understanding, the libido, especially the development and articulation of the human sexual drive and its perversions played a central role in shaping the human psyche. This appalled a good number of his colleagues in the medical profession, specifically regarding Freud's interpretation of patient claims about sexual abuse in their childhood and his theories with respect to the

development of the sexual drive in infants and young children in general. In all of that, we should not forget that Freud was living in fin de siècle Vienna, which for all its gaiety at heart was still a staunchly catholic city. Sexuality and sexual perversion were not supposed to be topics of discussions for its affluent aristocrats, rich burghers or the petit bourgeoisie even though a good number of their male members would happily frequent one of the many notorious establishments to escape the prudery of Viennese society and live their sexual phantasies or perversions in secrecy.

Two of Freud's numerous publications would eventually make psychoanalysis a household name, *Die Traumdeutung* (*The Interpretation of Dreams*), published in 1899, and *Zur Psychopathologie des Alltagslebens* (*The Psychopathology of Everyday Life*), published in 1904. The founding of the first psychoanalytical society dates back to 1902 when Freud began informal meetings with a select few. By 1906, these gatherings had turned into his Wednesday Psychological Society meetings and by 1910, they had morphed into the Vienna Psychoanalytic Society. The growing membership of this society during that time, with access controlled by existing members, reflects both, psychoanalysis gaining popularity among the intellectually curious in Vienna as well as Freud's psychoanalytic approach to treat mental illness receiving increasing recognition from his colleagues. Freud saw psychoanalysis as his creation and he was looking towards the newly founded psychoanalytical society as a vehicle to develop psychoanalysis further. Because of that, he also clearly considered himself the final authority as to what psychoanalysis was and what it was not. Over time, a good number of the members of this first psychoanalytical society as well as supporters and colleagues would see their ideas rejected by Freud as not being part of true psychoanalysis and eventually have a falling out with Freud. Among them were Alfred Adler and Carl G. Jung, who both in their own ways disagreed with Freud's emphasis on the importance of early sexual development for shaping the human psyche and went on to found their own flavors of depth psychology.

Alfred Adler, invited by Freud in 1902, had been a member of Freud's informal society meetings from early on. Unlike Freud, Adler came to see aggression and sexuality as initially separate drives in childhood with both equally powerful in shaping the development of the human psyche, a view Adler articulated in a paper for the Wednesday Society in 1908. Freud disagreed and for some time things were patched over but by 1912 Adler and a group of like-minded dissidents had formed their own society, the Society for Individual Psychology. Adler's school of individual psychology emphasized the role of power in development, with the drive of power asserting itself during child development ever more strongly with increasing feelings of inferiority. Adler's individual psychology stressed the importance of childhood in personality development and the associated potential for personality disorders turning into psychological illnesses during that phase. Stemming from that, it had its greatest impact in the area of childhood education where it influenced societal and as well as political views.

Jung's interest in Freud's work dated back to 1900 when he became Bleuler's assistant at the Psychiatric University Clinic (Psychiatrische Universitätsklinik), the Burghölzli in Zürich. In 1906, Jung and Freud began to frequently write each other and exchange their views. Over the course of the next seven years, a sort of friendship developed between the two with Jung becoming one of Freud's most ardent supporters. For all purposes, Freud in turn came to consider Jung as his heir apparent, one day to succeed him as the leader of the psychoanalytic movement. However, by 1912, Jung had developed views of psychoanalysis with which Freud disagreed and their friendship ended in 1913 when Freud officially terminated it for what he considered Jung's betrayal. The schisms between Jung and Freud and Adler and Freud differed in a number of ways. Notably, Adler was a fellow Viennese Jew but not a practicing neurologist whereas Jung was a gentile living almost five hundred miles away from Vienna who at the time of the schism had probably more clinical experience with mental illnesses than Freud had himself. Which was exactly why Freud likely had perceived Jung as the ideal candidate to take his position at the helm of psychoanalysis one day. Similar to Adler, Jung was critical of the central role that Freud gave to sexuality, specifically in his definition of the libido. Unlike Freud, Jung did not see libido as dominated by a sexual drive manifesting itself in childhood. Jung argued for a broader definition of libido that needed to encompass cross-culturally shared universal behaviors. In addition to their shared interests in mythology, a few years before their schism, Freud as well as Jung, had begun to take a greater interest in anthropology. However, those interests took them in different directions.

After the schism, Jung went on to found his school of analytical psychology in which anthropological and mythological influences became much more central than in Freud's psychoanalysis. Jung perceived the individual unconscious as embedded in a collective unconscious shared by all human beings. According to Jung, the existence of such a collective unconscious was a reflection of our shared evolutionary heritage, resembling essentially a species repository or collective record of becoming human. Analytical psychology asserts that this collective unconscious manifests itself in our dreams through universal, primal symbols and images emanating or deriving from it, so-called archetypes.[29] In analytical psychology, these archetypes are as important for understanding the human psyche as the basic human drives are in Freud's psychoanalysis. Among other things, Jung and his school of analytical psychology also arrived at a quite different position with respect to human development than the one held by Freud's psychoanalytical theory. For Jung, individual human development happened through an increasing differentiation of a person's self from the individuals conscious and unconscious mind, a process analytical psychology refers to as individuation.

Adler's individual psychology thrived in the interwar period between the First World War and the Second World War, specifically with respect to its implications for educational institutions. Jung's analytical psychology views held sway for even longer than that and

the influences of both are still with us. However, among those three major schools of depth psychology, it is overwhelmingly variants of Freud's psychoanalytical approach, which continue to be part of psychotherapy as practiced today. To a good extent, Freud would have to thank women for that, psychoanalysts like his daughter Anna Freud (1895 – 1982) or others such as the German-American Karen Horney (1885 – 1952). Freud, a child of his times, had treated women in his psychoanalytic theories much like lesser men. In his analysis of child development, he had introduced such conceptions as penis envy from which young girls supposedly suffered as the analog of castration fears young boys experienced. While for the most adhering to the psychoanalytic canon as her father had defined it, Anna Freud's work contributed specifically to the continued relevance of psychoanalysis for studying early child development. As for Karen Horney who was not beholden to preserving Sigmund Freud's version of psychoanalysis in a similar fashion, she recast Freud's psychoanalysis from a woman's perspective, making it clear that it was not penis envy that young girls struggled with but social environments that accorded to men much more power than to women. She also believed that neuroses are much less the result of frustrated or failed sexual developments but rather may have their roots in feelings of helplessness in a potentially hostile environment which infants and young children invariably must experience at least from time to time, something she referred to as a basic anxiety.

Psychoanalysis is still with us today but as a therapy for mental illnesses it is facing a new challenge, the pharmacological treatment of not just the symptoms of an increasing number of mental illnesses but more recently also of their biological root causes. Freud always believed that mental illnesses had physiological causes, but the science of his time could not yet study those. Therefore, he began to develop the psychoanalytical method to understand the psychological reasons for a given illnesses hoping to be able to help his patients with a psychological cure, so to speak. Modern biology and medicine in the form of molecular biology and cognitive neuroscience have given us an increased understanding of many of the physiological manifestations and causes of mental illnesses. In the process, a good number of mental illnesses have become treatable with medications to the extent that in many cases patients now can live almost normal lives. That certainly is not the case for all mental illnesses but an increasing number of them and there is hope that in the future this could be the case for the majority of mental illnesses. The question is then where that will leave psychoanalysis? As eminent researchers like Eric Kandel argue, psychoanalysis must refocus to reinvigorate itself as an innovative science, like it was in its early days, and it can do so by joining forces with cognitive neurosciences. The contention being that through such an alliance, psychoanalysis can discover the neurophysiological basis of its models and become a true science again, which in turn could inform the research objectives of cognitive neurophysiology with both contributing to an improved outcome for patients suffering from mental illnesses.

Behaviorism

Now we can turn to behaviorism, which regarded any study of mental processes, specifically anything along the lines of the depth psychology approaches we just looked into, as not being part of psychology. Eventually, behaviorism would come to consider not just depth psychology but also a good part of other psychological approaches as not being part of science at all. However, it did not begin this way. Two individuals stand out among the researchers whose work stimulated the rise of behaviorism in the early twentieth century: the American psychologist Edward Lee Thorndike (1874 – 1949) and the Russian psychologist Ivan Petrovich Pavlov. Both of them were pioneers in the study of learning behavior in animals. By understanding the learning behavior of animals, they sought by inference to eventually also gain an understanding of human learning processes. In Thorndike's case, this resulted in a trial-and-error theory of animal learning, while Pavlov's work shed light on the learning processes of habituation, sensitization and conditioning in animals. Even though Thorndike is often called one of the two first behaviorists, Pavlov being the other one, that is not quite so.

William James supported Thorndike in his early work at Harvard when Thorndike constructed his first contraptions to study animal behavior, his so-called puzzle boxes. Hence, Thorndike started out as a psychological functionalist in the mold of a William James and while with his experimentation he moved ever closer to what would later clearly be seen as behaviorism, he never really got there. One could claim with reasonable justification that psychological functionalism fathered behaviorism, with the Chicago school of psychological functionalism practically being its birthplace. Two American psychologists tied to the Chicago school of psychological functionalism, James Rowland Angell (1869 – 1949) and his former student John Broadus Watson (1878 – 1958), were the first to use the word behaviorist in the years just before the First World War. Historians of psychology date the beginning of the era of behaviorism to an article published by Watson in 1913, frequently dubbed "the behaviorist manifesto". Watson had studied with Angell while working on his doctoral degree at the University of Chicago, which he received in 1903. He remained at the University of Chicago until 1908 when he moved to Johns Hopkins University where he became the chair of the psychology department.[30] Both, Angell and Watson, independently used the term behaviorist in 1912 and then in 1913, Watson published the article which became known as the behaviorist manifesto. In the United States, behaviorism fell on very fertile ground with John B. Watson quickly becoming the greatest evangelist of the behaviorist gospel. Angell, practically one of the founders of functional psychology and much influenced by John Dewey who had brought him to Chicago in the first place, would remain a functional psychologist.

Before behaviorism germinated in the United States, it was already in full bloom in Czarist Russia, even though this label did not exist at the time for the kind of research

that Pavlov and his compatriot and rival Vladimir Mikhailovich Bekhterev (1857 – 1927) pursued. In 1904, Pavlov received the Nobel Prize for Physiology or Medicine for his research on the physiology of the digestive system. With that, other aspects of his work also began to receive more attention in the West while the work of Bekhterev remained largely unknown outside Russia. Specifically Pavlov's work on conditioning which differentiated between the unconditioned reflex and the conditioned reflex received attention among psychologists. Unconditioned reflexes describe situations where there is no conditional connection between stimulus and reflex. The same reflex will always occur in response to the same stimulus, the famous example being that of the hungry dog which will always salivate when presented with food. Different from that, for conditioned reflexes there is always a condition linking stimulus and reflex. Staying with the example of Pavlov's dog, ringing a bell every time the dog receives food establishes a conditional link between the ringing of the bell and the arrival of food so that even when no food arrives the dog will salivate upon hearing the bell ring. Different from unconditioned reflexes, conditioned reflexes can be unlearned. If repeatedly no food arrives when the bell rings, the dog will eventually stop to respond with salivation to the sound of the bell.

Like many other American psychologists, Watson took a keen interest in Pavlov's work on animal conditioning. Watson saw in Pavlovian conditioning the promise of an experimental approach for psychology to understand human behavior that could be scientifically as rigorous as any method the natural sciences used. As such, he positioned what he viewed as the objective scientific study of stimulus-response relationships in opposition to subjective introspection approaches, be it Wundt's internal perception or the kind of introspection functional psychologists employed. Watson's effort to establish the behaviorist approach in psychology was hugely successful in the United States as it not only promised a better understanding of what actually constitutes human behavior, it also kindled hopes of being able to eventually shape it. That said, there was never just one kind of behaviorist method as a number of different approaches came together under the umbrella of behaviorism. Behaviorism, even though it would influence psychologists worldwide was primarily an American affair; and at that, it was an almost exclusively male, foremost white, Anglo-Saxon and protestant enterprise. Like many other scientists, psychologists were not welcoming women in science and people of color had it not any easier to enter the profession. We tend to forget that gender equality and racial equality are very recent additions to the value catalogue of our modern societies. Those holding the reigns conceded neither any of it voluntarily. Instead, women and colored people alike had to fight hard for equality, and most would agree that both are still discriminated in many parts of our societies. In this regard, behaviorism was no different from any other science at the time. In addition to these restrictions for everyone not fitting the prescription of male, white and preferably Anglo-Saxon and protestant, something else

in behaviorism must have held a deep attraction for American puritan souls.

Pavlov's conditioning work began in Czarist Russia and Pavlov was already sixty-eight years old when the October Revolution swept away the autocratic regime of Russian Czars in which Pavlov had risen to fame. However, the new communist rulers wasted no time in endorsing Pavlov's work, as well as that of Pavlov's rival Bekhterev, who had conceived a similar if somewhat more radical version of reflexology.[31] Just why behaviorism held so much attraction for what were effectively the world's most materialistic societies, the United States in terms of material riches and the Soviet Union in terms of political doctrine, remains an open question. Maybe this somehow relates to the fact that both of those societies endorsed a new kind of human in the twentieth century, on the one side the capitalist *homo americanus* and on the other side the communist *homo sovieticus*. Engineering human behavior in the interest of society is not an invention of the twentieth century but it is certainly rife with it; just remember that it also witnessed other conceptions of modern super humans such as those held by Italian Fascism and German Nazis.[32] Looking at twenty-first century communist China, this kind of social engineering in pursuit of the perfect citizen, obedient to the last and superior at the same time, is certainly not over yet and new technologies not available to twentieth-century dictators may just provide the tools to eventually engineer such superhuman beings according to their masters specifications. The age-old pursuit of raising or creating the perfect human is an interesting topic of research in itself shedding its own peculiar light on human nature but here we cannot digress any further into that subject and must return to behaviorism as it flourished in the United States of the early twentieth century.

Watson's research career in psychology came to an unexpected end in 1920, owed to the puritan views at the time, which could not reconcile an extramarital affair with holding a professorship or running a university institute. However, Watson quickly found a new career in putting his behaviorist knowledge and convictions to work for the advertising industry. What better way is there to ensure that consumers buy a certain product then to condition them to do just that?

With the rise of behaviorism, the study of the mind fell to the wayside as it was considered a futile and fruitless effort. From the behaviorist perspective, seeking to understand how the human mind works was not proper science. There was just no way to access mental processes experimentally and hence everything one could hope to achieve from studying them was by default considered subjective and with that not scientific. Different from that, behaviorism promised to deliver clear and unbiased data that had never been available before. Data, which could not just help with understanding human behaviors but also provide insights into how to best influence human behaviors and which thereby could enable governments to shape their social policies accordingly. Testing became the mantra of behaviorism and it is with the rise of behaviorism that modern societies, government agencies as well as private enterprises, begin to systematically gather data

about every aspect of human behavior. Any testing that provided quantification and categorization of human behavior or even humans themselves, could be of advantage in political and industrial decision-making. To achieve that, a lot of funding, private as well as public, became available. Studies in behaviorism became in many ways the forerunners of today's social studies, however, with a much narrower focus. The social perspective of behaviorism was that of social engineering. According to behaviorism, human behavior is uniquely subject to conditioning, good conditioning and bad conditioning. Therefore, Watson could famously claim:

> *"Give me a dozen healthy infants, well-formed, and my own specified world to bring them up and I'll guarantee to take any one random and train him to become any type of specialist I might select – doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors."*

How was that different from the claim of the Russian communists to produce the new soviet man or woman?[33] It was not because both contentions of a new modern human being, that of *homo americanus* just as well as that of *homo sovieticus*, denied human beings the dignity of persons, although for different reasons. Such was the birth defect of behaviorism, disregarding the human being and replacing it with a mindless black box where inputs can be manipulated to achieve desirable outcomes. Why then did behaviorism stay in favor for so long?

The answer is simple. Behaviorism produced results and mountains of data that gave insights into things other psychological or sociological approaches could not provide. Much of the accumulating data mountains in behaviorism were of the mundane kind but precisely because of that, they may have been useful. With behaviorism, psychology could lay claim to be an exact science of behavior but not an exact science of the mind. However, according to the American psychologist B.F. Skinner (1904 - 1990), probably the most famous twentieth-century behaviorist, this did not matter. Skinner's brand of behaviorism carried the label of radical behaviorism to distinguish it from Watson's so-called methodological behaviorism and other more moderate forms of behaviorism which did not exclude thinking or feeling behavior from their observations. Skinner differentiated between behaviors that have an eliciting stimulus and behaviors that have no known eliciting stimulus. The former is similar to how Pavlov or Watson studied behavior and Skinner referred to it as respondent behavior. The second type of behavior category, Skinner called it operand behavior, is something Skinner introduced. To study operand behavior he built his own contraptions, later becoming famous as so-called Skinner boxes, which were significantly more sophisticated than Thorndike's puzzle boxes. In Skinner boxes, various animals, often white rats or mice even though pigeons were his favorites, would be the study objects, subjected to reinforcing stimuli, which could be either of

the rewarding or the punishing kind. For example, a Skinner box could have a lever inside which if pressed would result in the release of a food item into the box, a positive reinforcement; or it could be an electric shock applied to the rats feet to discourage it from touching the lever again, a negative reinforcement. Not every kind of stimulus is of the reinforcing type but Skinner found that many are reinforcing, positive ones as well as negative ones. In addition, for the reinforcing stimulus to have the strongest effect, he discovered that it must be delivered immediately after the behavior that is targeted for positive or negative reinforcement.

Of course, such reinforcement studies could only be done with animals and not with humans. However, that did not keep Skinner and other behaviorists in his mold from asserting that their animal studies were not only critical for understanding animal behavior but were equally relevant for understanding human behavior. The Skinner box was a smart contraption and could be highly automated, thereby delivering data on all kinds of conditioning experiments in a continuous stream, day and night, seven days a week. In a way, the Skinner box reflects the black box into which behaviorism had banned the study of the mind, animal or human. It is almost like studying how a computer works by just sitting in front of it and using whichever program the computer supplies for what is just required or desired. None of that will tell the user anything about the inner workings of a computer, not about its hardware and not about its software. That neither the animal mind and much less so the human mind can be understood through only studying behavior did not dawn on behaviorist psychologists until the second half of the twentieth century. Skinner was adamant about the science of behavior being a natural science, just like physics or chemistry. Interestingly, while being highly concerned about what constituted a true science, Skinner was also very much a utopian or as many would say a dystopian in how he thought to apply behaviorism for the betterment of society. Much of what Skinner produced in this respect was however not even close to what most of us today would consider as a scientific approach.

Behaviorism provided many good explanations for simple stimulus-reflex type behaviors in animals and for a good deal of simple conditioned learning, including positive and negative reinforcing stimulation. But how could behaviorism explain complex human behaviors such as language? When we speak, forming a sentence word by word, how does that come about? Put simply, the explanation behaviorists came up with, already offered by Watson, was to define speech as a so-called serially ordered behavior where a single stimulus triggers a chain response with each response in a chain serving as a stimulus for the next response. In hindsight, it is amazing that psychologists seriously considered that such response chains could support the rich and complex conversations in which humans typically engage on a daily basis. How could serially ordered behavior ever account for the creative quality of human language? If we look at a modern automated robot such an explanation would be acceptable, as it would reflect the rule-based responses programmed

into the robot's computer. We just do not expect such a robot to have a mind of its own and come up with responses that do not reflect the serially ordered behavior executed by its programming. However, our experience tells us that humans do have a mind of their own, something is going on in our brains that structures our speech in ways that no serially ordered behavior can explain. Of course, the environment around us will always condition our behavior to a good extent. If it calls for a certain kind of response, we are likely to articulate it in some form. However, we are in no way bound to do so, that is other than by habit.

Like all psychological schools, behaviorism had its internal critics, one of them being the American psychologist Karl Spencer Lashley (1890 - 1958). Lashley was no less a behaviorist than Watson or Skinner, but unlike them, he took a keen interest in identifying the cortical localization of specific functions that underlie behavioral responses. His assumption was that if reflexive behaviors, including learned behaviors in animals, were associated with certain reflexive pathways in an animal's brain, disrupting such cortical pathways would help pin down where in the brain certain reflexive and learned behaviors originate. What he found was however something different. Rats continued to find their way through mazes even though their performance declined with the amount of brain damage afflicted or the difficulty of the task increasing. Today, we know that rats trying to find their way through a maze do not just rely on a single sense like smell but use multiple sensory inputs to do so. Lashley could not know this back then and wrongly concluded that instead of isolated cortical areas, the substrate for functions of behavioral learning and memory was the total mass of brain tissue. This seeming impossibility to identify the cortical locations of learning and memory functions led Lashley to change his course of research. He believed that the firm ground for his psychological research to stand on would have to be psychological and not physiological observations. As he began to draw inferences from his psychological observation on their physiological basis, Lashley was doing precisely the opposite of what Pavlov had set out to do. Instead like Pavlov inferring from physiology on psychological behavior, Lashley essentially began to speculate what the underlying physiology could look like based on his psychological observations. While that already did not go down well with his behaviorist colleagues, especially with Pavlov, Lashley differed in his behaviorist position also in other aspects from mainstream behaviorism. He did not believe that serially ordered behavior, as laid out by Watson to explain complex behaviors like human language, could account for responses occurring in such a sequence as to serve as the required stimulus for the next such response sequence following it. To Lashley, this indicated that serially ordered behavior could not just be a stimulus-response reaction chain where one response serves as the stimulus eliciting the next response in a sequence but that the sequence itself was subject to some kind of centralized organization, raising the specter of a mind being involved in behavior in ways behaviorism had rejected outright.

Behaviorism can certainly claim that it succeeded in molding aspects of psychology into an objective science. Much of psychology is a data-based science and behaviorism made it its mission that psychologists acquire and analyze such data no different from their colleagues in the exact sciences. This transformation in psychology did not take place in a vacuum but is indicative of a much broader effort in the early twentieth century to define what constitutes science and what not. As discussed in the previous chapter, philosophy had its own school of behaviorism referred to as logical behaviorism, not to be confused with the methodological and radical schools of behaviorism in psychology. The latter two prominently represented by researchers like Watson and Skinner, respectively. However, methodological and radical versions of behaviorism in psychology and the logical behaviorism of philosophy shared a common heritage, as they were all rooted in the early twentieth century philosophical movement of neo-positivism. Neo-positivism is an offspring of positivism, a philosophical position that came to prominence in the nineteenth century with the French philosopher Auguste Comte (1798 - 1857), whom many consider the first modern philosopher of science. In turn, positivism can trace its roots back to the philosophy of science originating with David Hume to whom some not only refer to as an empiricist but also as a proto-positivist philosopher. To the positivism of the nineteenth century, neo-positivism added the tools of mathematical logic and a more critical approach to the limits of science, specifically with respect to what it is that science can really articulate. Neo-positivism, sometimes also labeled logical positivism or logical empiricism, originated in Europe where it was prominently associated with the closely linked Berlin and Vienna Circles of logical empiricism, two groups of philosophers and scientists who greatly influenced the philosophy of science in the twentieth century. Wittgenstein's *Tractatus Logico-Philosophicus* published in 1921, exploring what can be expressed by language and thereby through thought, was a major influence for neo-positivist thinkers. Another influence came from natural scientists such as the positivist physicist Ernst Mach who defined the objective of physical science as providing the most simple and economical abstract expression of facts.

In 1922, The German philosopher and physicist Moritz Schlick (1882 - 1936) accepted the appointment as chair for natural philosophy at the University of Vienna. This philosophy chair was created for the physicist Ernst Mach who held it between 1895 and 1901, after which another famous physicist, the Austrian Ludwig Boltzmann (1844 - 1906), held the position.[34] In Vienna, Schlick invited like-minded scientists and philosophers to weekly meetings on philosophical topics of interest in the sciences, which eventually became the Vienna Circle. Its prominent group members included the philosophers Rudolf Carnap and Otto Neurath, both of whom we already encountered as the ones coining the label physicalism, as well as the Austrian philosopher Herbert Feigl (1902 - 1988) and his compatriot, the mathematician Kurt Gödel (1906 - 1978). Topics of discussion at those meetings focused on the foundations of sciences such as mathematics where the works of

the German philosopher and mathematician Gottlob Frege (1848 - 1925) and his British colleague Bertrand Russel were championing new approaches to the logic foundations of mathematics. Closely related to that was also the work on mathematical logic and philosophy of language by Wittgenstein, who greatly influenced Schlick as the two engaged in regular discussions over several years. The Berlin Circle, founded by the German philosophers Hans Reichenbach (1891 – 1953), Kurt Grelling (1886 – 1942) and Walter Dubislav (1895 – 1937) also included the German philosopher Carl Gustav Hempel (1905 – 1997) and his compatriot, the mathematicians David Hilbert (1862 - 1943), as well as the Austrian mathematician Richard von Mises (1883 – 1953). Some members belonged to both circles as their interests largely aligned and the Berlin and Vienna Circle jointly published the journal *Erkenntnis* (*Knowledge*). With the Nazi party becoming the strongest political party in the German parliament, the Reichstag, in November 1932, turning Germany into a dictatorship in 1933 and annexing Austria in 1938, many members of both groups had to flee with a good number of them finding a new home in the United States.[35] In their luggage, figuratively speaking, those emigres carried neo-positivism, whose objective was the creation of a new scientific philosophy where everything would be verifiable in the same way as everything was verifiable in the context of the natural sciences. In the United States, neo-behaviorism set out to achieve the neo-positivist objective for psychology. The names prominently associated with neo-behaviorism, beginning in the 1930 and lasting into the 1950's, where that of B.F. Skinner and those of the American psychologists Clark Hull (1884 - 1952) and Edward C. Tolman (1886 - 1959).

Next to Skinner, Hull was the psychologist most associated with neo-behaviorism. Simply put, Hull's objective was to make behaviorism an exact quantitative science by translating the observed behavior into mathematical equations. Thereby he thought for example to calculate exactly the kind of response strength one could expect from a given number of delivered reinforcement stimuli. Hull's approach gained little traction, unlike the learning studies Tolman conducted, which made him the founder of purposive behaviorism. In his research, Tolman focused on how animals such as rats find their way through a maze. He observed that rats seemingly seem to pause in their search for an escape as if to sort out in which direction to head next. This led him to assume that rats learned as they repeatedly ran through the same or slightly altered maze, seemingly building something like a mental map of the maze, which they were "consulting" when they paused on their run in the maze to "reconsider" where to go. Skinner considered Tolman's interpretation of maze learning as an unwarranted extension of learning behaviors as in his opinion psychologists should focus on understanding the behavior itself and not try to extrapolate whatever learning theories from behavior. Incidentally, Tolman's selective breeding of rats for their ability to learn mazes made his studies an early precursor to what later became the branch of science called behavioral genetics.[36]

Humanistic and Cognitive Psychology

By the 1950's, behaviorism's limitations in seeking to understand human behavior through exclusively attributing how we behave to our external interactions with the environment while ignoring the inner workings of the human mind, were becoming more and more apparent. Eventually, this led psychologists to consider new approaches to understand human nature. As is often the case, not just in psychology but also in science more generally, new approaches cannot just lead to older approaches being retired but not infrequently also to the resurrection of even older concepts that had been set aside for various reasons. This is what happened when psychologists in the second half of the last century began to revive earlier psychological thought which behaviorism had castigated from psychology. However, outside influences, societal as well as technological ones were also among the factors driving psychologists to reconsider their approach to understand human nature. The Second World War was followed by a Cold War between competing societal models that could not have been more different. On the one side were the various democracy models of the free West led by the United States and on the other side was the communist East dominated by the Soviet Union. These societal models also shaped psychology with a major influence coming from philosophy where existentialism explored human existence from a new vantage point, providing for Western psychologists the inspiration to develop humanistic psychology. The development of computers was also a legacy of the Second World War. As computers became more broadly available at universities and research institutions, they began to have a major influence on psychology at the beginning of what would evolve into cognitive psychology. We will look first into humanistic psychology and then turn to cognitive psychology.

A central aspect of existentialist philosophy is that it perceives, as the French philosopher Jean-Paul Sartre (1905 - 1980) put it, humans as being condemned to be free. Whatever we do as humans, we always choose, even if we believe we seemingly do nothing like that or if we believe we just do not have any choice. Existentialism is a philosophy emerging in the twentieth century with Jean-Paul Sartre being its most prominent representative. However, like with many ideas, the roots of existentialism are quite a bit deeper. It was the specter of the modern world and the challenges of how to live in it with certainties of beliefs and social relations giving way to an increasingly areligious individuality, which had philosophers respond in new ways to questions about subjective and objective truths and the nature of human choice. Among them were for example the Danish philosopher and theologian Søren Kierkegaard (1813 – 1855) with his 1843 published *Enten-Eller* (*Either-Or*) and the German philosopher Friedrich Nietzsche. According to Nietzsche, modern humans increasingly lived in a nihilistic post-Christian world where there are no absolute values or truth and where every individual is free to choose her or his interpretation of values and truth, thereby creating their own personal realities.

As noted earlier, studying under Franz Brentano, the philosopher Edmund Husserl and the philosopher and psychologist Carl Stumpf became receptive to the phenomenological approach, making it a major part of their philosophy and psychology, respectively. Husserl not only picked up the concept of phenomenology from Brentano but also that of intentionality, the idea that human consciousness is always directed at something. Husserl's formulations of phenomenology and intentionality would influence a number of twentieth-century philosophers to whom we may today apply the label of existentialist philosopher. One of them was Sartre, a self-proclaimed existentialist. Other prominent names include for example the German philosophers Martin Heidegger (1869 – 1976) and Karl Jaspers (1883 – 1969), the Austrian-Israeli philosopher Martin Buber (1878 - 1965), the French writer and philosopher Albert Camus (1913 – 1960), or the Spanish philosopher José Ortega y Gasset (1883 - 1955).[37] Some of them such as Karl Jaspers would not have labeled themselves as existentialist philosophers, others like Heidegger were indifferent to the label. Psychology split from philosophy in the late nineteenth century but by necessity, the give and take between philosophy and psychology continues into our day, as their evolutions remain intertwined because both study human nature. Existentialism and the humanistic psychology it helped engender philosophically and culturally are one example of that. Sartre's version of existentialism would turn out to have a major influence on psychology. According to Sartre, as articulated in his 1943 published *L'Être et le Néant* (*Being and Nothingness*), humans are essentially condemned to be free in the sense that we always have to choose one thing above another, be it a thing or be it an idea. To the extent that people are familiar with existentialism at all, they likely heard about it as a Marxist leaning kind of philosophy. There is no denying that Sartre took the political philosophy of the German philosopher and economist Karl Marx (1818 – 1883) and Marxism serious, but at its core, existentialism is a deeply humanistic philosophy.[38] It firmly places the individual being with its freedom of choice as an existential necessity of human nature at the center of its philosophical conception of humanity.

Humanistic psychology was in many ways an American affair. Prominent among the psychologists reshaping concepts of European existentialism for the needs of a humanistic psychology in the late 1950's were the American psychologists Rollo R. May (1909 - 1994), Abraham H. Maslow (1908 - 1970) and Carl R. Rogers (1902 – 1987). However, humanistic psychology was not an exclusive American affair as it also included European immigrants such as the German-American psychologist Charlotte Bühler (1893 – 1974), the Austrian-American psychologist Fred Massarik (1926 – 2009), or the German-American psychotherapist Fritz Perls (1893 - 1970) who pursued a humanistic method of psychotherapy he called Gestalt theory.[39] Humanistic psychology had a strong therapy component, Perls' Gestalt theory is one example for that. Others include the approach of the Canadian Sidney M. Jourard (1926 – 1974) and the logotherapy approach of the Austrian psychiatrist and neurologist Viktor E. Frankl (1905 - 1997). Logotherapy, also referred to

as existential analysis and sometimes called the third Viennese school of psychotherapy, next to Freud's psychoanalysis and Adler's individual psychology approaches to therapy, focuses on an individual's understanding of her or his existence. It contends that what drives human behavior and represents our main motivation for living is the human will to find meaning in life. Frankl firmly believed that life always has meaning, regardless of what the circumstances are, and we always have the freedom of choice to decide to find that meaning even in the most desperate or difficult situations.[40] It is not suffering from whatever circumstances that causes humans to despair. Rather it is the perspective of not having anything to live for that makes us despair. For Frankl this meant that finding meaning in a situation of suffering changes everything.

Humanistic psychology positioned itself as an alternative to the dominant psychological movement of the time, behaviorism. Humanist psychologists claimed that the behaviorist approach to understand human nature had reduced humans to little else than over-sized white rats or inefficient computing machines. Before Charlotte Bühler became a leading figure of humanistic psychology in the United States, she had held research and teaching positions at the University of Vienna for some fifteen years and had made a name for herself in child psychology.[41] During that time, the dominant psychological movements in Vienna were Freud's psychoanalysis and Adler's individual psychology. The Bühlers, Charlotte and her husband and colleague Karl, lived and worked in what many - until the Nazi Anschluss of Austria - considered the capital of depth psychology; Sigmund Freud's practice was located only a few walking minutes from the University of Vienna. However, the Bühlers kept their distance. Charlotte Bühler believed that psychiatrists in Freud's or Adler's mold had a too negative conception of human nature, because they sought to explain the human mind from a starting point based on illness, while psychologists like her chose the healthy mind as the base line. In this way, humanistic psychologists like her differentiated themselves not only from behaviorism but also from depth psychology along the lines of Freud and Adler. Humanist psychology framed itself as a third psychological movement, claiming that different from the two major movements at the time, behaviorism and psychoanalysis, it put human self-development, the quest to lead a meaningful life, and the creative realization of an individual's full potential at the center of its attention.

Along with Maslow, Rogers, May and a number of others, Charlotte Bühler was one of the founders of the American Association of Humanistic Psychology. Maslow's concepts of a hierarchy of needs and the self-actualizing person became highly influential. In his so-called hierarchy of needs theory, lower-level needs must be satisfied before a person can attend to the next higher-level needs. The satisfaction of physiological needs such as to sustain the body and its health represents the most basic need level. The next level in this hierarchy of needs are safety needs (shelter, security), followed by belonging or love needs (affectionate and/or intimate relationships), esteem needs (adequacy, mastery,

competence), and self-actualization needs (achieving one's full potential). If needs at a certain level cannot be satisfied the individual remains preoccupied with their satisfaction, unable to progress towards addressing needs at the next higher level. According to Maslow, at each needs level we value that most of which we are deprived. For example, those deprived of food will value food, those deprived of shelter will value shelter and those deprived of love will value love above all other things. Hence, our values reflect the needs we cannot satisfy. This however is different for the self-actualizing person who has managed to satisfy all basic needs. The values of a self-actualizing person are what Maslow called "being values", the values associated with the intrinsic worth of an activity, which are not need driven but chosen freely. For example, a self-actualizing person does not need to work in order to earn a living but can choose to work on things that interest her or him. Maslow's hierarchy of needs is a developmental model at the peak of which awaits the promise of the self-actualizing person. Clearly, as many would argue, the attainment of the various levels in this hierarchy of needs with self-actualization as the goal is an idealized model because individual societal and economic circumstances certainly condition it. Most of the world is still struggling very much to satisfy basic needs and many in our societies still have to confront situations of food insecurity or endure living conditions where their shelters may be less than safe and secure. One has to wonder if not the majority of those facing such situations may not at the same time be more likely to find solace in belonging and love among their families or friends than many of us who have no problems satisfying the basic needs of food and shelter. Simply put, love might be more important to those who have little else left than to those who have plenty. As many may have experienced it, those who have little are also often more compassionate than the more affluent ones as they likely know much better how even a little can mean much to those who lack most everything. From that perspective, it would look like as if those deprived of the means to satisfy the most basic needs may actually be closer to attain the next higher level of need satisfaction such as love than those among us who are materially well off.

For May, love and belonging represented not one of several levels in a hierarchy of needs but he considered love along with will as central for humans to find meaning in the modern world. For him, the canon of values that had served people for millennia was not just obsolete but could lead in our modern world to many of us living isolated and lonely lives with our emotions and will withering away while being surrounded by ever more people. According to May, the experience of traditional values not working any longer leads to a new kind of human anxiety, a modern-day affliction that did not exist in our societies before. Maslow's and May's concerns were with understanding the challenges to the individual's sense of self in the quest to lead a meaningful life in a modern mass society. In a way, their psychological approach was as much societal as it was humanistic. Different from that, Rogers focused more on the individual, becoming one of the

pioneers of psychotherapy research and developing a method called client-centered therapy. Another name for client-centered therapy is non-directive psychotherapy because it seeks to remove the bias inadvertently inserted by the therapist to help the client articulate his situation. Instead, in client-centered therapy, the therapist attempts to see the client's behavior and the psychological meaning it has for the client through the client's eyes.[42]

The writings of humanistic psychologists such as Maslow, Rogers and May garnered a broad and interested audience even beyond psychologist circles. In many ways, this was a reflection of the free spirited 1960's and 1970's when existentialism also flourished on student campuses. Each of them eventually chose to continue his work at private research universities or institutions instead of pursuing an academic career within a university psychology department. That in itself weakened the position of humanistic psychology in academic institutions and reduced the possibilities for its continued development and intellectual renewal, for which drawing new students into it would have been essential. There were also other and maybe more serious challenges that led to the decline of humanistic psychology. First, humanistic psychology had not really developed much of a standard to measure the efficacy of the many emerging humanistic therapies. Second, with the end of the 1970's a more conservative culture was on the rise, eventually also challenging the liberal popular culture at universities. In a sense, as the 1960's and 1970's popular culture waned, so did humanistic psychology, which had blossomed with it.

Humankind has always been drawing analogies between its engineering creations and how possibly the human body or mind could function. Descriptions of what are nothing less than automated machines functioning indistinguishably from humans can already be found in ancient Greek mythology.[43] Another example is Rene Descartes, who naturally looked towards the engineering achievements of his time, perceiving the human anatomy to include the equivalents of pumps, levers and pulleys. Then in the first half of the twentieth century, with telephones and radios becoming household items, some began to draw comparisons and analogies of those technologies with the functioning of the human nervous system. Therefore, it is not surprising that with the rise of computer technology, psychologists began to wonder how computers and humans compare as information processing machines. Increasingly, psychologists modeled humans as cybernetic machines, which do not just have inputs and outputs, as behaviorism had presumed, but that also have internal networks for processing inputs into outputs. These internal networks not only add a new quality, they also can generate responses without any external stimulus at all. This new approach, combined with a view of the human mind more akin to what a Wilhelm Wundt or a William James believed it to be than the black box as which behaviorists saw it, gave rise to the so-called cognitive revolution of psychology. Of course, cognition aspects had also been important for the psychologists of the Würzburg

school with their concept of imageless thought, in Gestalt psychology, or also in field theories. Some of that now flowed into cognitive psychology, which began to flourish in the 1960's.

The rise of the computer brought with it a new focus on information theory which had been pioneered by the Swedish-American physicist Harry Nyquist (1889 – 1976) and the American electrical engineer Ralph Hartely (1888 – 1970) in the 1920's and then by the American mathematician and electrical engineer Claude Shannon (1916 – 2001) in the 1940's. As psychologists began to study human information processing in analogy to how computers do that, they sought to describe human communication in terms of information theory. The information processing approach would eventually become one of the major facets of cognitive psychology but not with respect to human language communication. There, its limitations became quickly evident because unlike the information exchange between computers, human communication is not reducible to a series of bits and bytes. Different from computers, humans communicate through languages that carry meaning. Language development had always been of interest to psychologists, but no good explanations existed for how humans learn languages in the first place and how human languages actually function. It was no one less than B.F. Skinner who spearheaded the behaviorist effort to articulate a theory of human language learning. However, this theory was more illuminating with respect to the limitations of what behaviorism could explain than what it could tell about human language learning.

The person who challenged Skinner's language theory, arguing that the behaviorist theory could not explain any significant aspects of language, was the American linguist and cognitive scientist Noam Chomsky (born 1928). Chomsky's work led him to conclude that one cannot understand how human language functions and how we learn it, unless humans possess an innate capability to make sense of language. Hence, he argued that this innate language capability must be a shared evolutionary heritage of all humans and all human languages are equally capable to support human communication; we just happen to learn one specific language from the community into which we are born. According to Chomsky, as infants, we do not actually learn languages from scratch but discover which language we are exposed to. This is a nativist position not unlike the one, which Descartes held for most human capabilities including that of language. Chomsky's position was not only diametrically opposed to the behaviorist approach but also challenged the then dominant language theories pioneered in the 1930 and 1940 by linguists like the American Leonhard Bloomfield (1887 – 1949). Many psychologists were quick to realize that Chomsky's ideas to explain language, with their underlying cognitive conception, represented an inherently more scientific approach than behaviorists could offer. Chomsky, with his groundbreaking language theory, thus became one of the pioneers of cognitive psychology and his language theory, which he continued to evolve over the next decades, remains highly influential to this day.

In 1958, the Ford foundation awarded the American psychologists George A. Miller (1920 - 2012), Eugene Galanter (1924 - 2016) and Karl H. Pribram (1919 – 2015) a one-year research grant for studying the potential use of cybernetic concepts in the context of psychology. The result of their studies was that they proposed to use the analogy of a control or feedback loop to explain human behavioral control. The function of behavioral control in that sense equals that of a thermostat ensuring a comfortable room temperature in wintertime. The thermostat receives as an input a room temperature measurement, compares it to the temperature the thermostat is set to and depending on the outcome of this comparison, either sends an output signal to the heater to raise the room temperature or to stop heating. Miller and his colleagues called this behavioral unit controller a TOTE mechanism, short for test-operate-test-exit. It is possible to combine any number of TOTE mechanisms in a nested fashion, thereby breaking down a complex behavior into increasingly subordinate TOTE mechanisms regulating specific aspects of more complex behaviors. Explaining observable behavior and depicting it with cybernetic symbols in a more or less nested fashion can however not explain what goes on in a human mind when there is no observable behavior correlated with our mental exercises, like when we for example plan something, which we always do. To learn more about what goes on in our minds when we do such things, Miller, Galanter and Pribram proposed the method of thinking aloud. In a way, thinking aloud was similar to a Wundt-type of introspection with the significant difference that test subjects articulated verbally what they were thinking while they for example planned something in their minds.

In this way, the work of Miller, Galanter and Pribram reintroduced American psychologists to what we typically associate with an independent human individual. In other parts of the world, non-behaviorist psychology approaches had continued a more vibrant existence, like in Europe, where behaviorism never became as dominant as in the United States. The changes that came about with the rise of cognitive psychology were therefore more dramatic in the United States than for example in Europe where psychological approaches in the mold of cognitive psychology also had a longer tradition. Among those we can count for example Wundt's psychology of consciousness, the imageless thought of the Würzburg school, or Gestalt psychology. Because there was no observable behavior linked to mental processes that underlie human cognition, a central problem for cognitive theory was to understand how mental perception and thinking really work. With that, psychologists like the German-American Ulric Neisser (1928 – 2012), whose 1967 published book *Cognitive Psychology* would give the new field of psychology its name, began to study pattern recognition, attention and memory and began to build models of information processing. Surprisingly, for several decades cognitive psychologists showed little interest in the biological foundation of cognition. The simple reason as to why this was the case for so long was that researchers relied on a fallacious analogy between computers and humans. Essentially, psychologists argued that with respect to human cognition, the

analysis of cognitive and neuronal levels of the human brain differed in ways similar as to how a computer's software differed from its hardware. Because computer software and computer hardware could be developed and explored independently, at least so it seemed at the time, there was no need to explore the biological foundations of human cognition (the hardware) as one could explore the working of the human mind (the software) independently of the hardware. Eventually this perception changed with the progress of neuroscience over the past decades and it became clear that understanding the workings of the human mind is not possible without a fundamental understanding of the biological foundations of human cognition. Today, researchers increasingly can understand neuronal processes on the molecular level as well as study human brain activity in real-time using high resolution imaging techniques. This new subfield of neuroscience, called cognitive neuroscience, has become highly relevant to cognitive psychologists. Since its beginnings in the late 1950's, cognitive psychology has benefited from and joined forces with a number of other scientific disciplines such as linguistics, mathematics and information processing, computer sciences and now finally with biology in the form of cognitive neuroscience.

The making of psychology began in the nineteenth century with the effort of individual researchers founding their own schools of psychology. These schools and a number of individuals more or less related to them then began to spawn new developments, mapping out new fields in psychology. After the Second World War, a new phase set in, in some ways already anticipated by various developments in psychology in the interwar years, when increasingly specific aspects of psychology would come into focus. In a sense, one can consider humanistic psychology and cognitive psychology as the two last efforts to map out new fields in psychology. However, one can also take a different view and look at them as psychology fracturing, just as other sciences have done before, into an increasing number of specialized subfields. This may look a bit different for psychology as a science as compared for example to the science field of physics. Physics has a large number of specialized subfields, which however all share the same foundation, the laws of physics articulated in the language of mathematics, a language that all physicists speak. Psychology, at least as of today, has nothing equivalent unifying it as a discipline in a similar way. What do subfields such as art psychology, clinical psychology, developmental psychology, educational psychology, neuropsychology, personality psychology, political psychology, psychotherapy, religious psychology or social psychology – just to name a few – have in common? Could for example social psychology not just be a part of the science of sociology rather than be a subfield of psychology? Psychology as a science certainly faces a greater danger of splintering into increasingly unrelated subfields than a science like physics. But maybe there is no need to fear that this will actually happen as psychology could one day have a foundation shared by all its subfields. That foundation could be a biology of mind articulated in the language of neuroscience.

The Nature of Human Nature

The preceding short tour de force through the makings of psychology since it began to part ways with philosophy some two hundred years ago, provides a helpful background for what this section will be about. Among other things, this will include a look into some of the more troubling aspects of human nature. Naturally, we all would rather focus on the wonderful aspects of human nature of which there undoubtedly are many. However, for our modern societies it is more important to understand and acknowledge those facets of our human nature, which may increasingly put their very functioning at risk.

Looking around in our modern world, one can find various societal models differing by their governing structures. On the one end of the spectrum, there are those societies, which have given themselves governing structures based on the broad consensus of their respective people; we usually refer to them as liberal societies or democracies. On the other end of the spectrum, we find those societies where the few force the many to live by their rules. Unfortunately, a large fraction of our world's population still lives in societies where we can consider the individual hardly to be a free person when looked at from an economic, political or even religious perspective. As with so many other things in life, shades of grey far outnumber the stark black and white contrast of the societal spectrum illustrated by looking only at the most liberal societies and the most ruthless dictatorships. Certainly, among the democracies of today, some tend to subordinate everything to the rights of the individual whereas others choose a different balance between the rights of the individual vs the benefits accruing to all of society from sharing certain responsibilities, which may put some restrictions on individual freedoms. In some of the richest countries of our world, which happen to be democracies, economic freedom is for many still not much more than a dream. That even political freedoms and rights should not be taken for granted in liberal societies is something that for example the continued efforts of voter rights suppression in the oldest still functioning democracy, the United States of America, powerfully illustrate. However, these democracies have mechanisms in place that allow them to address such issues in a political process where in principle everyone has a say. That is the basic difference between such free societies and those falling closer towards the other end of the spectrum such as China, Russia, or even worth, North Korea. There political freedom is non-existent or a sham and economic freedom may exist only for the ones in power and those related to them (North Korea), for an oligarchy (Russia), or for a broader populace as long as they do not speak their minds on political or religious topics (China).

Of course, one could name other countries to illustrate autocratic or dictatorial societies, there are quite a few of them, but those three do cover much of the range. One may be tempted to focus on such countries when looking into the more troubling facets of human

nature. However, this could lead one to overlook something more relevant for the future of free societies. Rather than asking what has gone wrong in those countries where people already have to live under autocratic rule, we should ask what aspects of human nature might put the survival of free societies at risk. What is it with human nature, which if not acknowledged and considered in how free societies shape their futures could increasingly put those still mostly free societies at risk of becoming less free? In everything we do we struggle more or less with the troublesome human nature evolution has bequeathed on us. Kant referred to it as the crooked timber of humankind from which nothing straight can ever be fashioned. Back then, Kant was writing about the challenges a ruler faces because of the imperfect nature of human beings, considering it the greatest challenge of all, as there was nothing above the absolute ruler of his time, which would have held him or her accountable at all.[44] However, this crooked timber of humankind is something that concerns all of us, humans present and future, not just absolutist rulers in Kant's time.

What is it that we talk about when we talk about human nature? What do we mean by it? How static is human nature? Was the behavior of *Homo sapiens* at the time when our ancestors began to spread from Africa to the rest of the world, some 50,000 to 70,000 thousand years ago, based on the same human nature we possess today or did it change? Are the many various cultures flowering throughout human history expressions of the richness of a shared human nature or do possibly different variants of human nature underlie them? Was cannibalism, still practiced by tribal societies in some remote places of our planet well into the twentieth century, an expression of the same human nature that we attribute to ourselves? Could those who planned and executed the slaughter of millions in the twentieth century, ushering in the age of modern-day genocide, do so because it was part of their modern human nature? All very troublesome questions and all equally likely to stir up controversy not the least because we have no clear idea yet what is and what is not part of human nature. We have been building our societies on implicit assumptions about what we presume human nature to be. What if some of those assumptions turn out to be wrong?

What it is that we talk about when we talk about human nature and what we mean by it has been very much at the center of the first half of this chapter, which looked into how the emerging science of psychology set out in the nineteenth century to discover human nature. From the account given there it should have become clear that exactly what human nature encompasses is still an unanswered question and a topic of active research and discussion. One can of course always look up the meaning or common understanding of human nature in a dictionary. In its most general form, dictionaries define human nature as referring to so-called fundamental dispositions and traits of humans, which can be either innate or acquired. Philosophy has been looking into human nature for much longer than psychology and the philosophical definition of human nature considers it in

a broader context. Some of that we encountered earlier when dwelling on the respective philosophical positions held in the century preceding the rise of psychology by British empiricist philosophers such as John Locke, George Berkeley and David Hume. Locke and Hume believed that human nature is something we acquire, a belief Berkeley shared to the extent that he also endorsed an ontological idealism, believing that what we experience actually only has an existence as we experience it but otherwise has no reality of its own. Different from that the position of conceptual idealism argued by Immanuel Kant assumed that some aspects of human nature such as for example our concepts of space and time are innate. Empiricist and innate points of view continued to be contentious topics of discussion as psychology evolved as its own science but in any such discussions philosophical reasoning gradually had to cede ground to biological arguments fed by the rise of evolutionary biology. In this process Charles Darwin's *The Expression of the Emotions in Man and Animals*, published in 1872, represents a watershed as it rooted the discussion about acquired or innate aspects of human nature in evolutionary biology. Since then, modern biological sciences, prominently among them genetics, have given much of what Darwin back then could only infer from his detailed observations a solid foundation in biology. Few will be surprised that basic human drives related to our very survival, such as food and procreation, are innate and an evolutionary heritage we share with our animal cousins. Increasingly we also have found that we share quite a bit of social behaviors with our primate cousins like the role of play in early development and socialization. A bigger surprise for many may be that researchers discovered that human skills such as language development also have a biological foundation.

Quite a bit of what we used to consider as acquired aspects of human nature, we now know to have biological origins. That raises some very profound questions. Such as what should we really consider acts of free will and what are not acts of free will. The example of a car accident can help illustrate this conundrum. Assume that you had a car accident because approaching a stop light you hit the brakes too late and therefore crashed into another car that had the right of way. Of course, you would likely need little convincing that you are then responsible for that accident, correct? Let us assume that a post-accident inspection of your car by your insurance revealed that the brakes of your car were defective, maybe because of a manufacturing defect or a faulty installation, and that you would have been able to stop ahead of the intersection if the brakes would have worked properly. Would you still think that you are responsible for the accident? Most likely not. Now consider the human brain. Like the parts of a car, which make up the hardware that you use to drive around, at your birth you entered the world with a brain that comes equipped with its own biological hardware. What if something in your brain's hardware is defective and you eventually do something that society considers inappropriate or maybe you even commit a crime? Should you be held responsible or not? The answer is probably a straightforward one for many of us if the malfunctioning

of your brain manifests itself in something that society can easily identify as a mental illness. But what if it is less obvious that you suffer from an illness and people judge you to fall within the range of what is considered normal, a human being to be held fully accountable for his or her actions?

Things are even more complicated than that because as it is, we are not born as mature human beings with mature minds. Hence, even so the biological hardware underlying human behavior is in place at birth, let us assume with no defects whatsoever, it is nowhere close to the final configuration it needs to take in order for any child to become a fully functional adult. Consider the analogy of building a new home for yourself. Your architect has drawn up the plans and after passing inspection all parts required for building your home have been delivered to the construction site. Now your builder gets his crews working to assemble the home. However, as it happens, among them are some less than proficient or even outright rogue contractors. They take short cuts, hide their defective work or do other malicious things and frequently, they just turn out to be incompetent to do the job properly. The house you end up with does not look too bad overall but then you start out finding this or that does not work, a pipe brakes, the roof leaks, some of the electric shorts, the insulation is bad, and you sit there in wintertime fighting cold drafts coming from your windows, doors and floors. Now, consider that one of your colleagues or friends had the very same kind of house built in a somewhat different neighborhood. However, the contractors working on his home were all professionals and unlike yourself, he ends up with a perfectly functioning home. Whom should we credit or hold accountable for these two very different outcomes, the homeowners, one of them being you, who did all the right things or the construction crews? A good guess is that you likely will blame the contractors for the crooked home they built for you.

Just as the two home constructions in the example above begin with similarly excellent plans and perfect materials, a healthy human being born into this world comes equipped with an excellent genetic building plan and the perfect materials to evolve into a mature adult, in the above example the equivalent to the finished home so to say. However, depending in which environment humans grow up they will mature in very different ways. Those surrounded by loving families, maybe even families of means, which can nurture them as they grow up, have a good chance of maturing into human beings expressing their human nature in ways society fully endorses. This sets them apart from someone growing up in a broken and destitute family, living in a neighborhood favoring from a societal perspective a maybe much less desirable expression of human nature to manifest itself. In essence, the opportunities a human being does have or does not have when growing up will shape the biology underlying the human nature expressing itself in the adult. That of course includes, loosely speaking, the wiring and functioning of the adult brain. We all likely agree that lack of nutrition will affect a child's physical development, may even stunt it. The same can happen with mental development. Arguably, in many

wealthy societies, think places like the United States, the mental development of children is often much more in jeopardy than their physical development. Biology, at least in the overwhelming number of cases, provides the starting kit for a newborn so it can develop into an adult human being expressing the kind of human nature we would all like to endorse. However, just how we grow up will determine to a good extent what may become of this biological starting kit as it evolves into the biology that will underlie any given expression of human nature.

The biology of human nature is not a finished product delivered at birth but evolves as we grow into mature adults. Of course, the individual human being certainly also plays a role in how this final biological substrate underlying human nature develops. As we grow up, we can make wrong choices and thereby squander the opportunities given to us by a nurturing environment or we can make the right choices and even prevail in environments where many would assume nothing good could evolve there. However, we should look at those cases as exceptions and concede that the environment in which one grows up usually is the dominating influence and the will of the individual child growing up in an environment it cannot change, can modulate the outcome only to a limited extent. Much of that was already clear to some of the more compassionate parts of society even before we had any good understanding with respect to which aspects of human nature may have a biological origin. Or, for that matter, what an important role the socioeconomic environment plays in forming our human nature as we grow up into adult human beings. Compassion is still how many in our societies approach such challenges today, but it is not sufficient. The day will come when science will likely be able to determine with much greater certainty than today to what extent any of us could be fully accountable for our actions as we are in control of them through virtue of our free will, or not at all when our free will is nonexistent and biology completely controls us.

How can we judge people if what they do has a biological origin of which they are not in control? Many will consider this dangerous terrain to venture into and it quite likely is. However, instead of judging the outcome so to say, we also could choose to change things at the beginning, which brings us finally to the equality of opportunity. Many, regardless of their political belief, do endorse equality of opportunity but that has not made it any easier in making progress towards that goal. If everyone born today had the same opportunities when growing up, then everybody could develop their human nature more equally and hence, society could hold all of its members equally accountable for their actions. However, we are still far from that and so we continue to judge the poor kid who grows up destitute and ends up stealing something to satisfy a basic need just the same way we judge the rich juvenile offender for stealing. Even though the rich juvenile could have committed theft just for the thrill of it because maybe he or she was bored and looking for some excitement, well knowing that there would be few consequences because his or her family could easily afford lawyers to fix things.

Our assumptions with respect to what human nature entails and what we continue to learn about the biology of human nature have profound implications, from how we can for example understand history to how we potentially should restructure our modern societies. If human nature is static and the creations of the ancient Greeks, to pick one example, are expressions of a shared human nature, a human nature unchanged over several thousand years, this necessarily would give us a different level of access to understand their ancient world than otherwise. It makes a difference if for example the actions of Homer's heroes are those of humans with whom we continue to share a basically unchanged human nature or not. As we discussed earlier in the previous chapter, Homer's heroes possessed a psychosomatic unity, the realization of which drove them to pursue eternal fame and sometimes infamy as the only way available to them to become immortal. The deeds through which they achieved this were for the most solidly outside of what our modern ethical and moral conceptions can encompass. In case human nature is an invariant, we then must frequently judge the deeds of Homer's heroes as morally and ethically reprehensible. If their human nature however differed from ours, then how can we judge them by our moral or ethical standards?

We should not jump to any quick conclusions here in answering such a question because in making this statement we made some unspoken assumptions. How can we even know, assuming for a moment that human nature is indeed invariant, that it will always express itself the same way, socially, culturally and importantly also morally and ethically? The simple answer is that we cannot. To understand why, we just have to acknowledge the richness of contemporary human cultures we can find around the globe. Clearly, human nature can support a diversity of social and cultural frameworks as well as different canons of moral and ethical standards humans live by. Few will doubt that all humans living in these various societies today possess a shared human nature. Each one of us could have been born into a society very different from the ones we happen to live in, expressing her or his very same human nature in a different way. Maybe that is why we usually do not find it difficult to understand how people in societies whose values we do not share, or even outright reject may have come to live by those rules. Furthermore, to a good extent we likely could also understand that someone living in such a society could live her or his life in accordance with the same kind of human nature that we claim for ourselves and thereby be as good a citizen in their society as we claim to be in ours.

There definitely is a wide range of possibilities in how humans can choose to structure their societies and live their lives in accordance with the rules they set up for themselves. Looking back to ancient Greece, we should not forget that all of the great achievements of its ancient societies, which maybe we would like to attribute to a shared human nature, are achievements of slave societies. Without a large slave population, working in the quarries or silver mines of antiquity, many of those beautiful ancient temples that we can still admire today may never have been built and city-states like Athens would have

certainly prospered less. Now, does that in any significant way dent our pride in a potentially shared human nature with the greats of ancient Greece?

What if human nature in not an invariant property but changes over time? It may take hundreds of generations but maybe we can become different people after all because human nature keeps evolving ever so slowly, hopefully in a direction of us becoming better humans. Unfortunately, even though we may wish this to be true, it is just a mistaken conception of human nature. Throughout human history, there have been many utopian schemes of societies towards which, according to the ones envisioning those futures, humans should aspire to because they supposedly represented the full realization of the true human potential, essentially being reflections of a perfected human nature. However, if the nature of human nature were one of change, the realization of such utopian worlds would herald the end of any further evolution of human nature. Utopian societies are static and not dynamic societies as once perfection has seemingly been achieved no further improvement will be needed. In this sense, visions of utopian worlds negate the very premises they are built on, a human nature evolving such that we become ever better human beings. But who is to say what an improving human nature or a better human being really means? Aside from some general ethical and moral conceptions, which most would agree on, claiming to know the answers to such questions in detail would be preposterous. The assumption of a human nature evolving towards some kind of perfection is clearly a fallacy. Considerations of the perfection of human nature inevitably narrow our conceptual view of what human nature represents. How can we even assume we can know what human nature could possibly encompass? What if human nature is far richer in possibilities, good as well as bad ones, of which throughout recorded history all we have seen so far is still a rather narrow sampling?

Alternatively, one could also take a quite different perspective on the possibility of an improving human nature, one that does not focus on hope but on what we already know about human nature by looking back into the recorded history that we have. The invention of writing systems dates back to only a little more than five thousand years ago. Hence, what we have to educate us on human nature by looking back into recorded history is a rather thin slice of overall human evolutionary history. No more than some two hundred generations of humans, assuming an average generational age of a little less than thirty years. However, what we can find in this brief record of human nature is a seemingly persistent repetition of some very disturbing patterns. We tend to look at the atrocities of the twentieth century committed by the last three to four generations as exceptions. But is that really the case?

Taking a closer look, one could claim that the only thing exceptional about those seeming exceptions is the scale at which those atrocities could be committed in the modern age, owed to technology to which earlier generations had no access. For all we know, our more distant ancestors were no less zealous in slaughtering their fellow-humans even though

they had to make do with technologies that were more primitive. Think about what you learned in your history classes back in your school days. That of course assumes that the society in which you grew up did not chose to delete some of those records it may have been less proud of. Just to pick two prominent examples here in the United States, look at the genocides of the indigenous people or the history of slavery. Every continent and almost every country has its own dark periods of history, which they rather would forget about. However, these episodes in history, reflecting the dark side of our human nature, are as much part of who we are as human beings then the more glorious episodes we prefer to remember. Such human atrocities we can trace back through all of recorded human history continue to this day. Hence, one has to wonder, is this all there is? Are those recurrences of human depravity throughout history right into our time telling us that we never will be able to leave this dark side of human nature behind us?

Instead of hoping for human nature to reveal its inexhaustible potential for our betterment, as individuals and as a species, over the centuries and millennia to come, will we rather have to accept the limitations of human nature as what they are? Most of us would like to believe that as human beings we can evolve and can hope to overcome eventually what may now look like some basic limitations of our human nature. The question is just how that is supposed to happen. Human evolution, like all of evolution in nature, is a process of adaptation playing out over a very long time. It would be preposterous to assume that we as a species will be capable of shaping our societies such that they will exert the decisive environmental forces driving such desirable adaptations of human nature over the next one million years or so. It is much more likely that just like in the past it will be nature itself, which will provide the forces shaping human evolution. However, there is a small difference as to how those changes in nature come about in our time and age when compared to earlier times in human evolution. Unlike in the past, it is not some natural event anymore but humanity itself driving changes in our natural environments, which will have a profound impact on the lives of future generations. One could look at this as one of evolution's self-correcting mechanisms playing out over a very long time. We behave as we do because of our human nature. Seemingly, we are unable to change our human behavior. Ironically, our human nature, being the result of millions of years of evolution, focuses us on short-term rewards. That has put us in a precarious situation, as we increasingly understand that some of our species behaviors may in the short-term benefit the individual with respect to survival and procreation but might in the long-term put the very survival of our species at risk.

We can all understand that because sometimes in our own brief lives every single one of us will likely make choices where we know exactly that for the short-term gratification we seek now there will be a price to pay later on. We all know for example that we should rather not eat that additional piece of delicious cake, as it sure will show up on the scale next time we weigh ourselves; but many of us nevertheless still eat it. Similarly,

we acknowledge the long-term consequences of our species behaviors as we consume every available resource, in the process eventually destroying the very foundations for life on Earth. We are seemingly incapable of making the required behavioral changes to secure humanities future on this planet. The only way our human nature may be able to change is through evolutionary forces playing out over a very long time. Ironically, it is the short-term focus of our human nature, which now is quite likely to initiate such a new period in the evolution of life on our planet. So maybe, even if we acknowledge the limitations of human nature as they are today, we can look into the future as one of opportunity for human nature to change. That change can be for the better or the worse. With this change brought about by the deficiencies or our human nature, we should expect this to be a self-correcting process. Meaning, in case human nature should change for the worse the problem could solve itself with the extinction of human nature. However, if human nature should change for the better in the long-term, there is no telling as to what the possibilities for the human species and for life in general could be on this planet.

Humans have been around for only a few million years, from an evolutionary perspective, we are still a very young species. If the human species should turn out to have the evolutionary staying power of species enduring for hundreds of millions of years, there is just no way to tell what human nature one day may look like. Instead of speculating about a human nature evolving towards whichever ultimate objective, we are better off by considering the possibility that human nature in the short-term cannot change. We cannot change human nature but we can induce ourselves to change our human behavior. We must recognize that our short-term behaviors now drive environmental changes on a planetary scale, which will in turn shape the evolution of our species in the very long-term. An indication for the magnitude of the human impact is that scientists are considering adding a new subdivision to geological time, the so-called Anthropocene. The start date for this age of humans, how it relates to the current Holocene, the geological epoch that began some twelve thousand years ago, is still under debate. The Anthropocene could for example begin some fifty-thousand years ago with the onset of the extinction of our planet's megafauna. Alternatively, one could use concrete traces in any geological record future generations may find. Like the fall-out from nuclear weapon tests that started in the last century or the thin layer of highly compressed plastic that our civilizations will leave behind preserved in our planet's geology.

Acknowledging the shortcomings of our human nature as well as our very limited understanding of it, we should approach this subject with a stance of humility. The journey of discovering human nature has only just begun and much of the territory of human nature is still terra incognita. However, the little we have learned so far about human nature should seriously make us rethink who we are and what we want to become, as individuals as well as a species.

A Nature of Language

Quite likely, most of us entertain the notion that humans are in control of their lives, at least to the extent this is possible in free societies without anyone limiting the freedoms and liberties of their fellow citizens. Few would question that it is our decisions, taken after careful thinking and deliberation with ourselves, and then frequently discussed and scrutinized with others, which determine the path we take in our life's journey. However, in recent decades researchers have increasingly come to doubt that this is the case. Descartes model of human nature was one where the individual was in full control, nothing remained hidden from the controlling self and it was our conscious thinking self, which set the course. To this human self the mind was completely transparent. All thought was conscious and the self a model of clarity in thinking through whatever needed consideration so we would take the appropriate actions in response to external or internal perceptions. For Descartes, thought was identical with consciousness and the notion of unconscious thought processes would have been anathema to him.[45]

However, not only do researchers today not doubt the existence of unconscious thought anymore. From what it increasingly looks like, unconscious thought may be much more important than conscious thought. There are even some researchers claiming that in many instances our conscious mind represents little more than a display instrument our unconscious mind uses to reveal its results. They essentially compare the human mind to a computer where the conscious mind is identical with the display screen and the unconscious mind is equivalent to the software programs working away inside the computer to produce what is being displayed on the screen. Cognitive psychology is of course ripe with computer analogies with this being one of the more challenging ones, associating the unconscious mind with much or all of our cognition and the conscious mind with the display of cognition. For the most, this is still nothing more than interesting speculation because it is quite difficult to prove any such hypothesis. Not surprisingly, many of the arguments for such models of the human mind and of human nature, assigning a much more important role in cognition to the unconscious, essentially draw on insights associated with psychoanalysis.

The most important arguments for a more prominent role of the unconscious in cognition come however from evolutionary considerations. Quite a few of those do include hypothesis of one kind or another with respect to things we can never really know much about. Like theories about the origin of human language, which by their very nature can never be more than conjectures, some of them admittedly very interesting conjectures. However, one can also make a good argument from a more general viewpoint and that is without trying to come up with theories for evolutionary steps in the development of the human species for which there is no trace record we could use to test any such hypothesis. It certainly is a reasonable assumption that evolutionary pressures work just

as well with respect to shaping the human psyche as they work with respect to shaping human anatomy and physiology. We should also not forget that humans are not the only animals which evolution has endowed with a psyche, something humans have long recognized in various ways as we can for example glean from Aesop's (c. 620 – 564 BCE) fables. Hence, we must not assume that a conscious and an unconscious mind are something exclusively possessed by the human species. Rather, both aspects of the human mind have an evolutionary origin that predates the human species. The only reason that we sometimes feel a certain exclusivity in that regard is because the human mind is an evolutionary specialization of our species. The human organ of the mind has enabled our species to dominate much of other animal life on our planet and therefore it somehow must be unique. To a good extent, the human brain is indeed unique in the animal world, but it is still an evolution of the animal world.

The notion that animal minds help their owners to function in their environments just as they help us humans to function in our own may be surprising to some, but it should not. While no animal brain is capable to achieve what the human brain can accomplish, this is only true when we look at the specific human skills that have allowed the human species to evolve from a species roaming the savannas of Africa a few million years ago to building our modern civilizations. However, that does not mean that the brains of our animal cousins, which are simpler in this respect, do not provide their owners with a mental life of their own, something we can call an animal psyche. Which dog owner has not seen her or his furry friend yapping and seemingly trying to run while asleep? What else is this but dreaming? That of course begs the question of how an animal could dream without its psyche doing something much like what our human psyche does when any one of us dreams?

As indicated in the first chapter, the size of the human brain in relation to the human body much exceeds the comparative measures we find for our closest relatives in the animal world. However, more important than the absolute size of the human brain as compared to the size of the human body is the much larger relative size of the human neocortex and specifically its prefrontal cortex as compared to the rest of the human brain. They are the parts, which evolved to become disproportionately larger in human brains when compared to the brains of our ape cousins. They are also the parts of the human brain which we most associate with thinking and with human cognition in general. Hence, as far as understanding unconscious thinking goes, we likely should look towards studying it in our animal cousins. Because what happens in an animal's mind is lees contaminated by the kind of cognition associated with the over-sized human neocortex and prefrontal cortex, studying the processes of the unconscious mind in animals other than humans could have some advantages. However, any such learning would have to come from understanding the biology of psyche, as none of our animal cousins will likely ever lie down on a couch and reveal their unconscious thoughts to an analyst.

We do not usually talk about thinking when we discuss the behavior of animals; rather, we talk about instincts. When we describe a behavior as instinctual, we speak of something that generally reflects innate but also learned behavioral elements in animals including humans, even though many would venture that for the most instinctual behavior is innate. Given what we know about animal brains today, it is safe to say that the human brain, thanks to its larger sized neocortex and specifically because of its much increased prefrontal cortex, equips our species with a level of reflective thought we do not think animal brains are capable of. If however our unconscious mind uses our conscious mind, for the display of instinctual behaviors associated with varied mixtures of innate and learned content, then some of what we perceive as exclusive results of conscious thought and decision, could be anything but that. Instead, it could reflect instinctual behavior displayed on the screen of our conscious mind giving us the illusion that it actually comes from our conscious mind. Just as evolution has put the autonomous nervous system of animals like us in charge of vital body functions without it having to consult the thinking mind, evolution may also have endowed us with psychic survival skills by encoding basic social behaviors into instincts of which not our conscious but our unconscious mind is in charge. No other animal has gone through a likewise complex social evolution as humans. This social evolution must have shaped the human psyche certainly as much or even more than the challenges of physical survival on the savanna did, which we shared with many other animals. Put differently, one could hypothesize that the main driver for human psychic evolution has to be human social evolution, as this evolutionary aspect is what clearly differentiates us most from other animals.

We are only at the very beginning of understanding the human psyche, let alone how it evolved over the some five to seven million years since the human species likely split from the chimpanzee line. The challenge we face is that we have little to go on when trying to understand the evolution of the human psyche. All we have so to speak is the finished product of which we do not know how much it changed since modern humans appeared on the scene some two hundred thousand years ago. To guide our understanding of the evolution of human anatomy and to some extent of human physiology we have the human fossil record. However, we have nothing comparable to trace the evolution of the human psyche. All we have are some pointers that tell us about potential landmarks in human culture evolution. Like for example the first human tool use, the first use of fire; or when humans possibly began to cloth themselves; the earliest evidence we have for human created art or for humans decorating themselves; or when humans began to depict the world they lived in, physically and spiritually some forty to fifty thousand years ago. One element of human behavior that is very much at the center of trying to understand the evolution of the human psyche is human communication. We can look towards our modern-day cousins in the animal world, the great apes, to get a sense of what human communication must have looked like very early on in human evolution.

Great apes communicate in ways that already set them apart from most other animals, something they had to learn in order to live together in larger social group sizes than most other mammals do.[46] In the human species, we find much of the non-verbal behavior that we can observe among the great apes preserved, sometimes even amplified; maybe this is what endears them so much to us because we can recognize a bit of ourselves in them. However, human evolution took the rudimentary verbal communication of great apes to a completely new level – human language. When and how human language evolved we may never know. But that does not keep us from speculating about how and when it may have happened. Put simply, the appearance of language is just another example of evolution at work. Just like the evolution of teeth or eyes, being able to communicate using language must have provided an evolutionary advantage. However, in what respect could that have been the case?

Instead of trying to answer this question, it is best to look at a few examples. First, let us consider the impact of language on hunting. Clearly, the greatest advantage here would have been an improved ability to coordinate hunting strategies, specifically for large and dangerous prey, before the hunt actually took place. That could have increased the chances for a successful outcome of a hunt and reduced the likelihood of someone being injured or even killed. Of course, one could argue that other animals such as wolves for example are very successful group hunters without ever evolving something akin to human language. However, that would fail to recognize a critical dissimilarity. Animals such as wolves kill their prey with a ferocious bite, something humans are not physically equipped to do. Humans had to fashion weapons with which to kill their prey and for killing large animals, the use of such weapons had to be coordinated, something no other animal had to accomplish to a similar extent. Human hunting could not just rely on instinct but had to rely on the human mind. Other animals continued to use their teeth, claws or venom to kill prey like their species had always done but humans came up with new ways to kill large prey and defend themselves by using their minds. Human bodies are not equipped with ferocious teeth, claws or deadly venom but the human mind has turned out to be a much more efficient weapon than what any other animal can command. To use it for that purpose required however a different level of coordination among humans.

Similarly, the tool of language would also have been useful for planning and executing a coordinated strategy in the case of conflict with another human tribe. Then there are of course internal challenges, which are also easier to address with increasing social group size if the tool of language is available. In a sense, the evolution of animals with eyes, teeth and claws and other physiological features increased the chance for the individual to survive, thereby contributing to the survival of a species. Language rather adds another layer to this as it is not directly the survival of the individual that it contributes to but the survival of the social group. The evolution of language made it possible for large

complex groups of humans to survive as functioning societies, thereby helping individuals in such groups to survive.

The evolution of language as a tool of communication to a level where it resembled something that we could equate with how we use language today may have taken a very long time or not; we just do not know. We really have no idea how long the evolution of language capabilities among human species really took but looking at the evolution of complex physiological traits may give us an idea. For example, it seemingly took evolution some 350,000 years between the first animals having light sensitive structures and the appearance of the first simple pinhole camera-like eye. Other considerations could also help us to narrow the time window during which human speech evolved somewhat more. Can we for example answer the question if *Homo sapiens* was the first human species who learned to speak, or did earlier human species also have language capabilities? That will depend on how we define language capabilities because language can for example mean the spoken languages of today as well as for example primitive sign languages, which may have evolved much earlier.

It is reasonable to assume that some of the earlier human species already possessed improved communication capabilities as compared to the great apes even if those fell way short of complex spoken languages. We also can make the reasonable hypothesis that the evolution of complex language just like the evolution of the human brain could be the result of increasingly complex human societies as Dunbar's social brain hypothesis assumes. Hence, we can look at the evolution of brain size in various human species since humans split from the chimpanzee line as a potential indicator for a more complex human communication capability than what apes possessed. With that, we probably can safely exclude advanced language capabilities for human species with brain sizes comparable to or only slightly above the brain size range we find in modern chimpanzees. For now, we will do just that, but we will have to revisit this assumption later.

This would make *Homo habilis* the first candidate to consider for a human species potentially having evolved communication capabilities beginning to set humans apart from their ape cousins.[47] While brain size is not all it is sometimes cooked up to be, the brain size of *Homo habilis* ranging between 550 to 690 cm^3 was not much above that of modern great apes and is significantly below that of modern humans whose brain size varies between 1,100 to 1,580 cm^3. There are good reasons to look with more optimism towards *Homo erectus* for whom brain size ranges fell between 700 to 1,100 cm^3. *Homo erectus* came on the scene around 1.9 million years ago with some of its local variants enduring until some 150,000 years ago. *Homo habilis* was still very much an ape-like creature whereas the anatomy of *Homo erectus* was much closer to that of modern humans. Researchers attribute this change to humans discovering how to cook food, which means they had learned how to make and control fire.

Cooking makes food much easier digestible and breaks down cell walls thereby releasing

many more nutrients. From what it looks like, the discovery of cooking did not only lead to a human anatomy with a much-reduced gut to body size ratio, but it also led to a significant increase in brain size. Brains are very energy hungry organs, and it could well be that brain sizes of early humans did not differ much from those of the great apes because prior to the discovery of cooking humans just could not get sufficient calories to evolve larger brains. Of course, there are also other arguments for what may have triggered the growth of the human brain that we see beginning with *Homo erectus*. Like for example the demands of more complex social relations in growing group sizes as envisioned in Dunbar's social brain hypothesis or adaptations driven by environmental changes about which we unfortunately know also very little. However, the impact of those factors on brain evolution is difficult to assess as they represent factors shaping human evolution for which we have no fossil records. Unlike we have them for the anatomical change between *Homo habilis* and *Homo erectus* for which the discovery of cooking is likely part of any reasonable explanation.[48]

All of the above taken together raises the question if *Homo erectus* did have the ability to speak. We cannot really say for sure but whatever kind of language capability *Homo erectus* may have possessed, it could have been quite limited unless the genome of this species already included some key markers for language capabilities. Scientists have discovered that the human capability to develop speech requires the presence of a specific mutation of a gene called FOXP2, sometimes also dubbed the language gene. While this is an oversimplification as there are other genes involved in speech development, the presence of this gene mutation is still a useful tracer because if it is missing in a human species genome, it is unlikely that the respective species had language capabilities in any sense that we would recognize. The FOXP2 gene mutation is of course present in modern humans and since a few years, we know that our Neanderthal and Denisovan cousins, *Homo neanderthalensis* and *Homo denisova*, possessed it as well. The theory is that Neanderthals, Denisovans and modern humans inherited this mutation from a shared ancestor, as it seems unlikely that all three species developed the very same mutation independently. The common ancestor of Neanderthals and Denisovans was *Homo heidelbergensis*, which may have been the ancestor of modern humans as well or may have split with modern humans from a shared ancestor. If the former is the case, which is that *Homo heidelbergensis* was ancestral to Neanderthals, Denisovans and modern humans, then we can assume that *Homo heidelbergensis* already possessed the FOXP2 gene mutation and the three descendant lineages of Neanderthals, Denisovans and modern humans inherited it from this species. If *Homo heidelbergensis* is ancestral to Neanderthals and Denisovans but not to modern humans and instead shares an ancestor with modern humans then that ancestor should also have possessed the FOXP2 mutation because it would have been similarly unlikely for this very mutation to occur independently in related human species.

In any case, because researchers believe that *Homo erectus* may be the likely ancestor of either *Homo heidelbergensis* or the species from which *Homo heidelbergensis* and modern humans may have split off, both outlined scenarios inevitable lead to the same question. Did *Homo erectus* possess the FOXP2 gene? Indeed, many researchers assume, based on arguments similar to the ones given above, that *Homo erectus* at some point in his evolution acquired this FOXP2 gene mutation. However, there is no definitive proof for that, at least not for now, and given that *Homo erectus* was around for almost 1.8 million years the challenge is of course not just proving that *Homo erectus* carried the FOXP2 mutation but exactly when this species may have acquired it.[49] In all likelihood, we probably will never know the answer to any of these questions. However, it remains a reasonable assumption that *Homo erectus*, or maybe a regional variant of this geographically widespread species, acquired the FOXP2 mutation at some point in time and eventually passed it on either directly to *Homo heidelbergensis* or to an unknown shared ancestor of *Homo heidelbergensis* and modern humans.

The presence of the FOXP2 gene mutation in the human genome is necessary for evolving speech capabilities like modern humans exhibit them today but it is not sufficient. The decoding of the Neanderthal genome showed us that our Neanderthal cousins possessed this FOXP2 gene mutation but a number of researchers still believe that Neanderthals did not use language in any way we could recognize and some even claim they could not really communicate through language at all, even though Neanderthals possessed larger brains than modern humans do. That illustrates a couple of challenges in trying to assess whether earlier human species or some of our human cousins who are now extinct could speak. First, the presence or absence of soft skills such as language does not leave tools or tool marks behind to prove or disprove that any of our ancestors possessed it, something that is true for any kind of cognitive capabilities. Hence, speculations as to who did or did not possess language capabilities are not really constrained by any hard evidence they have to fit. Second, we likely are not completely free from a kind of belief in modern human exceptionality, a sense that we must somehow be different in certain respects from those earlier human species, based on the very fact that we are around and they are not. This includes our closest extinct human cousins in whose demise modern humans may have played a role.

We modern humans tend to base our claims of exceptionality on those cognitive skills which we tend to believe our human cousins did not possess. In that respect, we frequently point towards the so-called Cognitive Revolution, the flowering of human imagination we witness in the archaeological records of pre-history, sometime around 45,000 years ago. The cave paintings dating back to this time, still admired today, are a prime exhibit of this Cognitive Revolution. However, what we tend to forget is that this Cognitive Revolution coincided with the decline of Neanderthal populations in their Eurasian homelands and with their eventual extinction following some 30,000 years ago. It may be tempting

to attribute the Cognitive Revolution as well as the demise of competing human species such as the Neanderthals to some kind of intrinsic cultural and cognitive superiority of our modern human ancestors. However, there likely were several reasons why the Neanderthals vanished without any traceable Cognitive Revolution of their own, other than just assuming that they were not as smart as *Homo sapiens*, or for example could not speak and thereby lacked a critical tool many believe is necessary to build larger social groups among humans. It almost seems silly to believe Neanderthals died out because they were not as smart as modern humans. The fact that for hundreds of thousands of years their challenging environments never really could support larger Neanderthal populations is probably a much more important factor in shaping the destiny of Neanderthals. It will not matter much if a human species is smart and has language capability if it cannot sustain a sufficient gene pool to ensure its survival. In such a situation, the arrival of newcomers such as modern humans emigrating from North Africa into the homelands of the Neanderthals made survival of comparably small bands of Neanderthals scattered over much of Europe even more challenging.

The above discussion with respect to the origin of language and the challenges with tracing its evolution in time or associate it with any given human species, suggests a healthy skepticism is in order when assessing any claims with respect to the origin of language. We honestly do not know when, where, or how the evolution of language took place; and we do not know which human species members first communicated with each other in whatever primitive speech. While many associate the evolution of language with modern *Homo sapiens*, the decoding of ancient human genomes such as from our extinct Neanderthal cousins tells a different story. The genetic capabilities for language development have likely been around for quite some time, maybe originating with *Homo erectus*. However, none of this does tell us anything about the richness and complexity of human languages maybe spoken by humans long before the arrival of modern humans. *Homo sapiens* entered the scene some two hundred thousand years ago. That would leave some 150,000 years for *Homo sapiens* to develop the kind of language capabilities, from whatever they looked like when *Homo sapiens* arrived, to what our ancestors may have possessed when the Cognitive Revolution began.

How different human languages back then were from our modern day languages is again something we can only guess. Nevertheless, one could argue that humans capable of expressing themselves in the language of art as our ancestors did in their beautiful and rich cave paintings, which in many ways look utterly modern to us, may also have had language capabilities not very different from ours today. Hence, we can venture that it took modern humans some 150,000 years at the most to evolve language capabilities of a complexity we associate today with our modern languages. As pointed out above, modern humans may not have been the only human species to do so as Neanderthals and Denisovans, who by then had been around for much longer than modern humans, should

have been equally capable of language. Why would those human species, of which we know that they were genetically equally equipped as modern humans to evolve language capabilities, not have evolved comparable languages? It just does not to seem plausible that Neanderthals and Denisovans would not have communicated in their own languages just as well as modern humans did.

As impressive as it is that we can make inferences such as the ones above just from the presence of what many call the language gene, some cautionary remarks are in order with respect to speculations about how and when human language capabilities could have evolved. Humans are not the only animals who possess a version of the FOXP2 gene enabling communication. Researchers have found versions of FOXP2 genes in animals as different as songbirds, mice or whales. Without their version of the FOXP2 gene songbirds cannot sing; neither may whales be able to communicate over thousands of miles with each other through their underwater songs; and mice lose their ability to communicate with other mice as well if their FOXP2 is missing or not functional. By inversion, animals possessing varieties of the FOXP2 gene implies that those animals have their own ways of communication, languages that we cannot understand. Hence, one could blame our inability to communicate with animals as much on their inability to understand and speak our human language as we could blame it on our inability to understand and speak their languages. No doubt, human language allows for a much richer conversation, at least that is what most of us believe, than what other animals may be able to converse about. We may not know what other animals communicate about, but communicate they do. Looking at the FOXP2 versions that modern humans carry today, the same version we found with our extinct Neanderthal and Denisovan cousins, which likely *Homo erectus* already possessed with whom the FOXP2 gene may have originated, there is another question we need to ask. This question is about *Homo floresiensis*, the Hobbit species discovered not too long ago.

In 2003, scientists unearthed the fossils of *Homo floresiensis* on the Island of Flores in Indonesia. The fossils of this species date from 100,000 to 60,000 years ago but stone tools associated with it are as old as 190,000 years with the youngest dating to around 50,000 years ago. Researchers assume that this species may still have been around some 10,000 years before modern humans appeared in Asia. *Homo floresiensis* was of a small build, just under one meter or so, earning the species the nickname of Hobbit. The hypothesis is that the *Homo floresiensis* island population derived from a *Homo erectus* population, which became isolated on the Island of Flores. There, due to the well-known phenomenon of island dwarfism, the members of this isolated *Homo erectus* population, maybe present on the island some 700,000 years ago, diminished over generations in physical stature and eventually turned into the *Homo floresiensis* species. Along with everything else, the brain size of this isolated *Homo erectus* species also shrank to a size of between 380 – 420 cm^3, the low end of which is about the average chimpanzee

brain size. Given the assumptions that *Homo erectus* already possessed our version of the FOXP2 gene and that *Homo floresiensis* derived from *Homo erectus*, we can infer that *Homo floresiensis* carried the same FOXP2 version modern humans possess. Does that mean that a human species with a chimpanzee size brain could have had language capabilities similar to ours? We cannot answer this question today because the genome of the Hobbit has not been deciphered yet and there may still be surprises lurking in the Hobbit's genome.

However, just the possibility that such small-brained humans could have communicated with each other not so much different from how we communicate, should make us think carefully as to which extent we can infer language capability from brain size and language genes. If *Homo floresiensis* possessed an ability to converse similar to Neanderthals, Denisovans or ourselves, then we may have to consider that even more ancient humans than *Homo erectus* could have possessed richer communication capabilities than we give them credit for. Alternatively, if we assume that the Hobbit did not possess the same FOXP2 version as modern humans then we may infer that the regional *Homo erectus* variant from which the Hobbit derived did not possess it either. In that case, we would have to consider the possibility that not just this variant of *Homo erectus* did not possess the modern human FOXP2 version but that *Homo erectus* generally had not acquired this FOXP2 version until some 700,000 years ago. If that were so then that would very much increase the likelihood that the modern human FOXP2 version originated not with *Homo erectus* at all but with one of its descendants. That could be the common ancestor of the line leading to modern humans and of *Homo heidelbergensis*. Alternatively, it could be *Homo heidelbergensis* itself if this species should be the common ancestor of Neanderthals, Denisovans and us.

We may never really know with any certainty how human communication capabilities evolved and how human language capabilities became rooted in our biology in the sense language theorists starting with Noam Chomsky began to argue. To be clear, human language encompasses much more than our ability to communicate in spoken words. It also includes all kinds of non-verbal communications with others as well as with ourselves. Just as we communicate with each other, we communicate with ourselves. Language is articulated thought. We encode our thoughts in language as we speak to others, and we quietly speak within our minds using the very same language. To many this may come as a surprise because we usually just do not reflect on this. However, everyone can easily verify this for himself or herself by trying to follow their private thoughts and discover that it is impossible to think without using language. That is true for any language, including of course sign languages, natural spoken languages as well as artificial languages such as mathematics.

What is true for our conscious mind is also true for our unconscious mind. Just consider the language of dreams. Our dreams include the language of images as well as spoken

language. Many of those among us of who have been living outside and far away from the culture they were born into, speaking a second learned language, maybe even for much longer than the language they grew up with, likely experienced that the language of our dreams can change. I do remember well when I woke up the first time realizing that in my past nights dream, the spoken language had not been my native German but English. It looks like as if our minds encode all of our thoughts, conscious as well as unconscious, in various types of languages including freely floating images, soundless music or quiet speech, among others.

Human capabilities to communicate possess a richness that no other animal species has ever evolved. In many ways, human communication is the foundation on which our human nature rests. Famously, Rene Descartes came to the conclusion that he exists because he thinks, his famous *cogito ergo sum - I think, therefore I am.* For Descartes, his strategy of philosophical doubt reduced his proof of his very own existence to an inner mind argument. Evidently, we find an internalistic conception of the self and of human nature associated with what many consider as the beginning of modern Western philosophy. Looking at the central role that human communication has in all aspects of human nature, it may be more appropriate for anyone of us to assume that we exist because we communicate. Hence, an assertion like *communico ergo sum - I communicate, therefore I am*, may be more appropriate than Descartes *cogito ergo sum.* There have been many and sometimes quite profane variations on Descartes *cogito ergo sum*, but it may be prudent to afford this one a bit more consideration. Just imagine a world in which you could not communicate with anyone about anything. That does not just mean that you could have no conversations whatsoever, verbal or non-verbal. There also would be no books to read and instead of having internal conversations with yourself on which to base your decisions on, you would have to react instinctively without consulting with yourself. What shines through here is of course Wittgenstein's private language argument.

Without a publicly shared language, carrying the broad consensus about what we mean by the words and sentences we speak, hear, read or write in our communications with others and ourselves, we would have no idea about the actual meaning attached to the concepts used in such conversations. Our understanding of the world around us and of ourselves very much depends on the existence of a public language. Without it, humans would not have been able to communicate at a level sufficient to evolve into a highly social species and without that, human civilizations including our modern ones would never have been possible. Like language, human nature is something that manifests itself in a certain unique manner in each human individual. However, language and human nature are species properties in the same way as being bipedal is a human species property. Being bipedal means that human anatomy and physiology is structured a certain way. Similarly, language and human nature imply that aspects of human biology as well as of

human behaviors are structured in certain ways. Our language and our human nature are not somehow our inner mind properties but are structures that connect us to the outside world and through it to other individuals in ways that allow us to communicate and talk about what we mean for example by language and human nature. In this sense, language and human nature cannot be inner mind properties but more likely may indeed reflect something similar to hylomorphic structural phenomena that define our species.

The way that language and thinking seemingly are linked raises of course other important questions. As discussed above, conscious thought always is articulated thought. We are all thinking in that way even though we usually do not dwell on this fact because it is just a regular part of our everyday experience. That of course does not mean that we always will engage in logically coherent thought endowed with the clarity that we admire in great thinkers. We articulate our thoughts in our inner mind discussions through language but that does not necessarily mean that we do so in a coherent way. When we refer to a language of thought, we do so in the broadest sense. While this indeed can often mean that we just literally talk to ourselves in silence it also can just as often mean that we for example think in pictures. Most would agree with the wisdom of folk psychology, which tells us that we only can comprehend something properly if we can articulate it to ourselves in such a way that it would also become understandable to others. This means that in order for us to understand something properly we must be able to articulate it to our inner mind in a sufficiently precise language. Turning this around, many of us likely assume that someone who cannot articulate something in a coherent manner likely has not understood himself or herself what he or she tries to communicate in the first place. That is unless they find themselves for example in an emotionally charged situation where their otherwise calm thinking capability may be impaired.

Are we humans the only animals capable of this feat or is the connection between thought and language something we share with other animals? If other animals can communicate, as they are seemingly able to do, are they using their language capabilities in similar ways to articulate what we could call their thoughts? To what extent could the capability of animals to communicate with their languages imply a capability to think? Central to being able to answer this question will be a fundamental understanding of the linkage between language and the capability to think. It could be that the evolution of language was a prerequisite for humans to evolve the capability to think as we do. If that is so, then the language capabilities, again broadly speaking, that we can find within the animal kingdom should give us an indication as to the kind of thought capability any animal species might possess. However, to make this connection we first must understand how the syntax and grammar of any such languages could translate into the equivalent of thought and how we could study that.

A Nature of Lies and Deceit

Honesty and truthfulness are two concepts most of us believe to understand and a great many of us cherish. We likely all agree, at least to some degree, that these two qualities are essential parts of the fabric allowing our societies and civilizations to function. More than that, without them we may never have been able to build human civilizations in the first place. All the various forms of lies and deceit that we hear about directly and indirectly on an almost daily basis, some of which we may encounter in our private spheres but much of it coming to us as reported by traditional and increasingly non-traditional media, most of that clearly must be some kind of aberration. We all know just how pervasive lies and deceit can be in our social environments and what problems they can cause, but many of us expect and most certainly always hope that honesty and truth will eventually somehow vanquish them. However, is that really what happens? If not, are honesty and truthfulness just simply overrated? How much are our perceptions of the world around us and of ourselves guided by what we would like to see? That is, instead of by how things really are. A brief introspection to reflect on our own personal behaviors and a cursory exploration of the human world around us will quickly reveal that matters are much more complex than that.

Throughout human history, individually and as societies, humans have been searching for truth in its various forms. Often, they came up empty handed. Even when humans sometimes uncovered something undoubtedly true and profound, they frequently ignored it or worse, soiled it with lies and deceit. Truth is hard to uncover, difficult to live by, and sometimes impossible to bear. Therefore, it is not surprising that we have become quite sophisticated in navigating around the challenges truth can frequently put in front of us. Many such behavioral adaptations we do not call out as being dishonest or deceitful as they are so common. Rather, we may look at them as being substantially different from what we normally refer to as outright lying. Like for example when we are either directly lied to or witness others being lied to in contexts where we may otherwise not hesitate to brand such behavior for what it is but often still refrain from doing so.

Then of course, there is always that, the much more blatant and overt lie and deceit. Those committing these offenses are usually not making much effort to hide such behavior and even if they do, often they are found out eventually and have to suffer the consequences of taking a behavior a step to far which we know we are guilty of displaying more or less frequently ourselves. Crossing this invisible line where something we seemingly often do ourselves to make living with us and among others possible, can turn an innocent dishonesty or white lie into unacceptable deceit. Where human civilizations perceive this line to fall has changed throughout history and our modern societies continuously renegotiate where its proper position should be. The roles that truth, lies and dishonesty play in how we construct our personal and social worlds are ever changing.

Looking at specific examples on a case-by-case basis, quite likely a large majority of us would intuitively agree on whether or not any given human behavior constitutes lying or deception. However, providing an accurate and general definition of what we call lying and deceit is surprisingly difficult. Within some professional groups, there is of course a substantial interest to understand lying and our ability to detect it. Just think of law enforcement, the judiciary with its court system, or the world of secret services and espionage. Looking at lying through these lenses will lead to somewhat different pictures of lying and deceit than for example if we examine it through the lenses of psychology, psychiatry or psychopathology. Here we will use a commonly used definition of lying called intentional lying with the objective to deceive:[50]

> *"The act of lying also called deception is a successful or unsuccessful deliberate attempt, without forewarning, to create in another a belief which the communicator considers to be untrue."*

The intent of this book is not to engage the reader in a philosophical discussion about the concepts of truth, lies and deceit. Rather, the journey this book invites readers to join is one of exploration, of things that we are all familiar with and we believe we know well, but which looked at from a different perspective or studied through a different lens can hold many surprises. Sometimes the things that are closest to us and seem to be most familiar can be the ones that we actually know about the least. However, before we begin the exploration of our human relationship to what we commonly refer to as truth, lies or deceit we actually have to admit that there is quite some ambivalence. As anyone will discover quickly for himself or herself, a single unambiguous definition of what constitutes lying or deception in any given situation does not exist. We only can define for example human lying behavior in a more narrow sense, like in the definition given above which will suit the needs of law enforcement and judiciary but may for example fall short of capturing it from other perspectives. Unlike what the above definition of lying might insinuate, lying and deception are certainly not always the same. Nevertheless, mistakenly we frequently assume that lying and deceit are in a way synonymous, and the former implies the latter or vice versa but that is not necessarily so. We all have experienced or heard of situations where someone lied without the intent to deceive; and surely, people can be deceived without anyone outright lying to them.

As for truth, there are different kinds of truth. For one, there is the factual truth where we can verify if a statement is truthful or not. We consider for example, at least most of us do, the established facts of science to be factual truth. However, in a sense, science and scientific progress reflect always a temporary truth, which in some instances of course can come close to an absolute truth.[51] It is important to recognize that any given moment, the body of science represents our best understanding of the world we

live in, nothing less and nothing more. Nevertheless, that understanding can change because everything in science is always subject to falsification. In the scientific process, falsification is much more important than verification. If an experiment for example verifies previous measurements or verifies a certain theoretical prediction of a theory, we are in a sense only corroborating what we already believe we know. However, if the results of new measurements holding up to scientific scrutiny verifiably contradict older data, then we may be on the verge of learning something new. Similarly, if new kinds of measurements enabled by our advancing technologies contradict aspects of a scientific theory, then we know that something in that theory is not correct and we need to come up with a new explanation. In any such case, what matters most is that the data we use to make such decisions must be reproducible, meaning, if other scientists make the same exact measurements they should see the same results, within certain measurement errors of course. Hence, until someone comes along with new data or a better explanation of what we can observe in the world around us or maybe even can falsify an experimental result or a scientific theory, the established scientific measurement data and theories represent the truest representation of nature that we currently have. Only in that sense do scientific facts represent scientific truth.

There are also other kinds of truth where we have less or no assurance at all of their factual accuracy and often no way to falsify them either. Beliefs certainly fall into this category, as does any kind of subjective truth. That does not mean that people's beliefs are lies. Instead, we are dealing here with an altogether different category. What we perceive as lying very much depends on the specific context of a situation. Many of us could interpret very similar behaviors as outright lying or as something more benign, depending on the specific circumstances. Much of that will depend on what the societies and cultures in which we grew up have trained us to perceive as acceptable or not acceptable behavior. The point we need to make here is that to some extent a substantial amount of what we individually or as societies perceive and acknowledge as truth, lies or deceit, reflect relative judgments and not absolute ones. There are of course also absolute truths, which many of us recognize as such and which we as individuals and cultures share. Evidently, the fact that humans cannot survive without oxygen to breathe and that we are all mortal are such truth. They are examples for the kind of truth, which one only defies at one's own peril.

Are we born liars, are lying and deception just part of our biological heritage? Alternatively, do we learn to lie and deceive as we grow up, marking such behavior essentially as socially acquired? Until quite recently, scientists and philosophers could do little else but speculate as to what really makes humans such excellent and pervasive liars. How much does our biological heritage contribute to this seemingly singular achievement of our species vs how much do our societies groom their young to become veritable liars? It is only over the last few decades that advances in biology have increasingly enabled us

to explore the balance between those two factors with some confidence. This has allowed us to begin to understand the evolutionary roots of some of our human behaviors. Before that, any attempts to determine this balance between biological and societal factors driving human behaviors represented more or less educated speculations. Throughout history, when and where one lived and certainly one's upbringing or status in society pretty much shaped one's views as to the innate nature of human capabilities and behaviors. Those views covered the whole spectrum. From humans born completely innocent and free of any inherent capability to lie and deceive, to newborn humans being anything but that. Before the age of science, that is only a few hundred years ago, such views were by necessity of either a religious or a philosophical nature. This only started to change with the rise of evolutionary biology in the nineteenth century.

The English naturalist Charles Darwin based his theory of evolution on an impressive observational record, much of it gathered by himself but also including countless observations provided by many others to whom he had reached out for help with gathering data.[52] Darwin was not the first to formulate a theory of evolution in order to explain the diversity of animal and plant life he could witness in his own time as well as the growing fossil record of countless past lifeforms long gone extinct. The French naturalist Jean-Baptiste Lamarck (1744 – 1829) as well as the physician Erasmus Darwin (1731 – 1802), the so-called *English Lamarck* and Darwin's grandfather, are two prominent evolutionary thinkers preceding him. In Darwin's own time, his compatriot and fellow naturalist Alfred Russel Wallace (1823 – 1913) initially had a very similar view of evolution and Russel's first publication on this subject almost scooped Darwin's long awaited first major publication of his theory of evolution. However, there are very good reasons why we think of Darwin as the father of scientific evolutionary thought. He was unique in his relentless methodical effort to understand relationships between different lifeforms, between lifeforms and their environments as he could discern them in the natural world, and to integrate them in a larger view that he consistently applied to all of life, including the evolution of the human species. Remarkably, without knowing anything about the biological foundations of inheritance, Darwin developed a theory of the evolution of life, the essence of which remains at the core of our understanding today.

Unraveling the complex biology underlying the evolution of life became possible with the advances of genetics in the second half of the twentieth century. Only then could scientists begin to ask meaningful questions with regard to the biological roots of human behavior. About one hundred years before Darwin published his two most famous works, *On The Origin of Species* and *The Descent of Man, and Selection in Relation to Sex*, the French novelist and philosopher Jean-Jacques Rousseau (1712 – 1778) argued that humans are born inherently good but then become corrupted by society. Rousseau referred to this human state of innocence as our natural state, essentially our biological predisposition. He believed that in their natural state humans are free of evil, which

by necessity means free of behavioral traits reflecting lies and deceit. With Darwin's theory of evolution, this natural state of humans became in a sense linked to that of our evolutionary ancestors. Accepting that humans shared a common ancestry with apes was a step too far for many. However, even for those who could see things Darwin's way a significant difference remained with respect to the natural state of humans and apes. Most of Darwin's contemporaries, at least in Christian cultures, believed human beings possessed a soul and a free will; many still hold such a belief today. In contrast, animals seemingly possessed neither, also a view still many do hold today. By that token, since any animal is devoid of a soul and a free will, it is impossible for an animal to be evil, ever. Therefore, animals cannot lie and deceive. So, does this mean that our great ape cousins are really the better hominids?

It is best not to try to answer this question. The only reason to raise it here is to draw attention to the fact that Darwin's contemporaries may have pondered this question for very good reasons. More likely than not, even well into the early twentieth century, most people would not have considered it an unreasonable question to ask. What has changed since then is not that we now have an affirmative answer as to whether humans have a soul or a free will. Rather, what has changed is that in our time we acknowledge religions for what they are, belief systems that can give meaning to the lives of a large fraction of humanity, providing consolation and hope for many where otherwise there would be little or none. As such, religious beliefs themselves have become interesting study areas of our human behaviors and psyche.

Not long ago, it would have been very dangerous to explore most religious beliefs from such a perspective. Religious beliefs were absolutes and with that lay beyond what humans were supposed to explore with probing questions. While in some parts of our world this still is the case, from a practical standpoint most of our societies are now seemingly secular in nature. A few decades ago, quite a few in the Western world would have made such an assertion with much higher conviction than today. However, with the rise of fundamentalist beliefs in various corners of our planet one certainly can have doubts about just how secular many of our societies really are. Nevertheless, in general, most of us do not assume anymore that scientists will ever discover any biological evidence for a soul in humans and no sane biologist today would likely invest his or her career into locating the seat of such a human soul. That, however, is unlikely to induce humans to embrace their membership in the animal family they belong to any more enthusiastically. For a long time, humans have considered themselves to stand apart or better above the rest of life on Earth, a hubris we cannot afford any longer to entertain and for which we are finally beginning to pay the price.

Things are a little different with respect to our free will even though the concept of free will has also come under increased scrutiny. The question scientists are focusing on is to what extent humans really do have a free will. How much of what we usually consider

actions of our free will are nothing like that but actually reflect behavior rooted in our biology, essentially innate dispositions of the human psyche? While science eventually came up empty handed in its search for a soul that is not the case for our so-called free will. Certainly, science has already put a dent into our cherished concept of the human free will, but it has by no means completely abrogated it; and is quite unlikely to do so in the future. However, what neuroscientists have discovered over the last few decades has given us some pause. Seemingly, a great number of what we used to consider as decisions of our free will are anything but. With the human soul out of the picture and the concept of free will increasingly constrained by biology we practically are back again to a level playing field when comparing the natural state of humans vs our great ape cousins.

Naturally, we therefore have to ask ourselves to what extent lying and deception reflect exclusive human behavioral patterns. Do other species within the animal family we humans are part of share those behaviors, at least to some degree? Given what we have learned about animal communication over just the last few decades the answer likely is a conditional yes. As we know today, animals use all kinds of signaling to communicate. From the vocal communications of our closest relatives among primates, to the musical tweets of songbirds, to non-vocal bodily signaling, to leaving chemical traces for others to decipher, animals use a large repertoire of communication tools. Like with human communication, any communication and signaling among animals can help some animals gain an advantage over others in the Darwinian struggle for survival and procreation. Therefore, it should be no surprise that we can find many examples of animals engaging in what to us, from a human vantage point, looks like lying and deception. However, we need to be careful here not to pass to quick a judgment on whether other animals can lie and deceive just as competently as the human animal does or not. Underlying our understanding of what we define as lying is that the person doing the lying is conscious that she or he actually is lying. The implicit assumption we make is that self-consciousness is a pre-condition to commit an act of lying. Because of that, when we say that an animal is lying, we are actually assigning that very animal self-consciousness. However, to what extent can we justify that? As it looks like, we have seriously underestimated many of our animal cousins for a long time.

Today, most of us acknowledge that animals are not the biological machines we conveniently have pictured them to be for such a long time. Inflicting pain on animals, abusing them for whatever purpose we see fit, and slaughtering them by the millions is much easier to justify if there are no moral or ethical repercussions to consider for such actions because animals are presumably nothing more than biological machines. However, as we now know, many animals do have emotions and they can communicate their feelings. Although we are still far from anything approaching a full understanding of animal communications and emotions, the realization that animals can have emotions they are able

to communicate, tells us that they may have a self-awareness that goes beyond what we assumed they needed to function in their natural environment. On the other hand, could it be that we got this all wrong in the first place? To function in their natural environment animals must have a kind of self-awareness that includes communication in many forms. For their social interactions, the same may be true with respect to emotions. But does that include animals being aware of themselves to the extent of some self-consciousness that would enable them to lie and deceive on a par level with humans? Most biologists today would likely be very cautious in how they answer such questions. Animals are certainly self-aware to the extent that they need to function in their natural environments. For many animals, we also know that they do have emotions they can communicate within their own species and from all we can tell also across species, including to their human cousins if the latter are willing to listen carefully. However, the kind of self-consciousness we see as a prerequisite for lying is something we have not discovered in animals, not yet. Hence, with regard to animal behavior, we should refrain from accusing them of lying but we can say with certainty that they are perfectly capable of deceit.

Nature knows some very visible examples of animal deceit such as in mimicry where physical traits of an animal take shapes or forms such that the animal represents something other than itself in order to deceive its predators or its prey. However, animal behaviors can be just as deceitful as physical appearances of animals can be which deceive other animals to perceive them as something different from what they really are. Evolution is not partial, does not play favorites, and anything that helps an organism to survive is fair game in the never-ending competition to survive and procreate. We usually think of the evolution of life in terms of physical characteristics that can provide a species a leg-up in the survival of the fittest. One reason for that is of course that the evolution of physical traits, which we find reflected in the fossil records, allows us to trace back the history of the evolution of life itself. Another reason is that we eventually figured out how such physical traits are genetically inherited, passed on from one generation to the next, unchanged or with a slight modification. None of that is the case for behavioral traits. Lies and deceit do not fossilize. Hence, we have very little understanding of how behavioral traits are coded in our genes and even much less of how they may evolve.

For animals, deceptive behaviors are just as much a part of their nature as their physical characteristics, not much different from having sharp teeth or a hard shell or any other physical quality with which evolution has endowed them to increase their species chances for success in the struggle to persevere. In some way, we have acknowledged this for a long time. Like in the stories that we pass on to our children and which many of us enjoy reading even long past our childhood days. With the fables of Aesop the ancient Greeks gave us some of the earliest examples of such stories we know of, fables which remain popular to this day. However, humans quite likely have been anthropomorphizing

animals in storytelling for much longer and we continue this tradition to this day in forms that are more modern. If deception is part of animal nature, by extension, the same must be true for us humans. There is an aspect of lying among humans where it seemingly is more appropriate to apply biological considerations rather than to look at it through the lenses of moral or ethical codes.

We can also look at deception from the other side, not from the point of view of those who are victims of deception but from the perspective of those who do the deceiving. Exploring deception in the animal kingdom from this perspective, many natural adaptations that evolved to deceive other animals, while seemingly deceptions, are nothing like that. What we observe as mimicry is an evolutionary outcome that may look like deception, but it is not. If for example the evolutionary development of one kind of a non-poisonous butterfly species led to it possessing today a color pattern of its wings that resembles the wing pattern of a poisonous butterfly species that certainly is an effective deception but it surely is not a deceptive behavior. It is not as if the non-poisonous butterfly species somehow purposefully evolves a color pattern resembling the poisonous butterfly's wing design. Rather, it is simply the result of predators being less likely to eat non-poisonous butterflies with wing patterns resembling those of the poisonous butterfly species. For the non-poisonous butterfly species this selective pressure of predation leads to an increase in the number of those individuals with wing patterns becoming ever more similar to the wing pattern of the poisonous species. Looking closer to home at some of our primate relatives, we clearly have to describe their behavior in many instances as deceptive. The way they do so raises in some cases however a much more important question. For example, when they try to steal something while there is someone else around, for example a human, who could watch them. Scientists have discovered that in such situations the behavior of our primate relatives is very difficult to explain unless one assumes that those who engage in it have a conception of other minds.

It seems as if we have little choice but to conclude that animals can possess a kind of theory of mind, allowing them to judge how other animals behave in a situation. Staying with the example of stealing, we know now that monkeys do understand quite well if a human is in a position to observe what they are doing or not. Monkeys participating in such a test setup where they can take fruits while a human is present seem to be waiting until the human looks away, in a direction where she or he cannot see if the monkey steals some fruit from the table. This clearly implies that the monkey understands that from the human's perspective, she or he will not be able to detect his theft quickly enough to prevent it. The monkey in essence judges the situation not just from his perspective but also from the human's perspective and that would require the monkey to have some theory of mind concept. Hence, with respect to deception in the animal world we have a dividing line where some deception implies the possession of a theory of mind concept while for other deceptions nothing like that is the case. Seemingly, possessing a theory

of mind concept is not unique to humans but something that our human nature shares more broadly with other animal natures.

Among humans, there are many different forms of lying, reflecting almost countless ways of being dishonest. The first part of this chapter gave an introduction into the makings of psychology, the branch of science concerning itself with the understanding of human behavior, which of course includes lying and deceit. Psychology knows much more elaborate and significantly more precise schemes for categorizing lying as compared to what the discussion here uses. However, the underlying psychological phenomena remain the same and it does not matter if one uses the language and precise terminology of science to talk about them or rather a commonsense language approach as long as one recognizes the respective limitations in either case. What follows here is less of a scientific discussion and more like the respectful dialog you may have with a good friend or neighbor seeking to explore a subject over a good glass of wine or any other beverage you may prefer. There used to be a time when people were able to just sit down and have such conversations about subjects of shared interest without being or pretending to be experts.

To be clear, in case some still doubt that, in order to understand complex matters beyond what school curricula and our life experiences teach us, we absolutely need real experts in any given field of which we ourselves know little about and seek their opinion and advice as needed. At the same time, that certainly does not relief any of us from our obligations as humans and citizens to use our own faculties to understand as best we can, ask critical and probing questions just like scientists do, and form our own opinions, knowing well what they are based upon. A few decades ago, this last clarification may not have been necessary. Of course, there always have been charlatans but more often than not, they represented rare curiosities rather than frequent occurrences. However, nowadays so many believe themselves to be experts without having ever taken the effort to seriously study the respective matter in question that we need to make it explicit that expertise matters; by which we mean the expertise of the real experts and not the pretense of the many charlatans one encounters today. Ironically, as we will see later, by itself the latter is a manifestation of the high bloom lies and deceit have achieved in our societies.

Now with all of that out of the way, we will put lying aside for now and focus on deceit. The reason for that is simple. In the context of how we build our individual and societal consensus realities the sometimes deliberate and other times just not reflected acts of deceiving ourselves as well as others are of much greater interest than any acts of overt blatant lying. When we are being untruthful to ourselves, deceive ourselves for whatever unknown reason, as long as it is largely inconsequential for others but merely serves our personal needs that seems to be a category by itself. We will shelf this category as personal dishonesty or self-deceit. Even though we realize that examples of dishonesty

in this category will always to some extent relate to or spill over into the world around us, they predominantly reflect on us, on our very own self. Another category could be dishonesty in our personal relationships, which we define as family and friends in the widest sense. Essentially, it includes dishonesty with respect to people we feel emotionally close to in varying degrees. However, for this category we assume all those emotional ties to be essentially positive ones so it does not include people we have negative feelings for, or in the worst case, people we may even hate. For lack of a better name, we will refer to this category as emotional dishonesty or the deceit among family and friends. Another category that we can identify is what we can call opportunistic dishonesty with people we do not really know at all, essentially deceit of strangers. This is where in the broadest sense someone takes advantage of others, like for personal advancement, maybe to enrich himself or herself materially or to gain power over others; sometimes it is what we perceive as deceit motivated by nothing else but greed, self-aggrandizement or hunger for power. The above three categories of personal dishonesty, emotional dishonesty and opportunistic dishonesty all originate with individual human behavior. However, not only have we become quite sophisticated in using dishonesty to soothe, comfort or deceive and exploit ourselves and others, we also have become proficient at practicing what is best called societal dishonesty, in essence any kind of communal deceit we practice for whatever reasons but of which we are mostly unaware.

Why do we deceive? The simple answer most of us would likely offer first is that we are all humans and being human means that we are not perfect, not in any way. We all have to cope with personal shortcomings, deficiencies that are our own, as well as with all the imperfections of the civilizations we have built over the past few thousand years. In a perfect society, perfect human beings would never have to lie or deceive. However, neither exists, not perfect human beings and for that very reason, not any perfect human society. Confronting our own and our societal deficiencies arguably makes some lying and deception almost inevitable. We all do some of that, not admitting it is just another form of dishonesty. However, as valuable as this acknowledgement of human shortcomings is, it does not really tell us why we as humans are seemingly such proficient liars and deceivers.

With the emergence of cognitive psychology in the second half of the last century, psychological sciences began to take not just a different view on the role of human cognition than the one behaviorism had entertained; they also began to gradually readmit the unconscious back into psychology. In behaviorism, anything remotely being suspect of a mentalist connotation had to be discarded. For behaviorists, what happened in the human mind was irrelevant to understand human behavior. With that, seeking to understand how the human conscious mind made sense of its environment was not to be part of psychological research. Even worse than that would have been to consider something like the human unconscious mind, for example along the conceptions of the unconscious

expressed by the various depth psychology approaches, as in any way relevant to understand human nature. However, as it dawned on psychologists that one could not possibly understand human behavior without understanding its meaning, the mental aspect of human nature, which behaviorists had banned from psychology, moved back into focus. The behaviorist approach to explore the nature of human nature only via inferences from observing human behaviors without considering the mental, what goes on in our minds as we behave as we do, in its conscious and unconscious parts, had failed. Behaviorist psychologists did not realize this for decades because they had constrained their studies to a limited model of our complex human nature. Because the challenge of understanding human nature is so daunting, the choice behaviorists made was to reduce it to a model that they could experimentally explore. In that respect, behaviorist psychologists behaved no different than scientists in many other disciplines do. Using model systems to research something that in its entirety is too complex to explore is standard science practice. The danger with any model system is that it can acquire a life of its own with researchers increasingly mistaking it to represent fully that of which it is only a reduced and limited model. This is what happened with the behaviorist model of human nature as it morphed from a much simplified and reduced model of human nature into representing human nature itself. That the mentalist perspective, what happens in the human mind, mattered for understanding human nature is something that already resurfaced with the earlier discussed computational theory of mind, essentially the analogy between a computer and the human mind. This conception of the human mind identifies the computer display with the conscious mind and the content displayed on the screen with the product of the unconscious mind, the output of the computing device itself.

Similarly important to this change in perception of the human unconscious was the rise of evolutionary biology and evolutionary psychology. The emergence of evolutionary psychology is a reflection of psychology acknowledging that the modern human mind is the result of millions of years of evolution, biological as well as social and cultural. How does one explain the human propensity for deception and above all, how can one make sense of the fact that self-deception seems to be ubiquitous, and we seemingly could not function without it? Evidently, there must have been an evolutionary advantage to humans evolving the capability of self-deception. But what could that be? The answer evolutionary psychology suggests is that we had to become proficient in self-deception in order for us to be able to better deceive our fellow human beings. There are many examples of deceit being advantageous in the animal world but when it comes to deception, no animal is just as cunning as human beings are. As to what could have been driving an increased need for deceiving others, evolutionary psychology points towards a change on which we typically do not reflect. All animals have enemies they must defend themselves against, but the large majority of those enemy animals belong to different species. Which animal however is the greatest threat to humans?

The simple answer is of course that the greatest threat to any human being comes from other human beings. In this respect, humans are truly unique. Sometime in the course of human evolution, human social environments must have changed to a degree that the ability to survive and thrive among other humans became a major force in shaping the future evolution of human nature. It had become more important for humans to be able to deceive each other than to deceive non-human animals. However, this change, probably driven by an increasing social complexity of human groups, is likely just one among many which could be part of the evolutionary package that for example fits within a framework as laid out in Dunbar's social brain hypothesis. Looking at it this way, it is not only the growing size of the human brain and with it, the relative size of its neocortex evolving towards what we find in modern humans today, which matter in this respect. It is also the human unconscious evolving in ways, which made unique behaviors like self-deception second nature to us. The argument being that in order be successful in deceiving ourselves we cannot be conscious of doing so and hence self-deception must reside in the human unconscious.

In this sense, successful self-deception was a pre-condition for deceiving others. Living in larger groups may not just have led to humans becoming brainier and more communicative. It also may have led to humans becoming more deceptive than any other animal. The simple four-category scheme of deception listed earlier – personal dishonesty or self-deception, emotional dishonesty or the deceit among family and friends, opportunistic dishonesty and societal dishonesty – really represent a nested and hierarchical scheme. At the core, we find personal dishonesty and self-deception. Without it, none of the other deceptions could be pursued with much success. Similarly, being able to deceive those who are close to us and because of that have a more intimate knowledge of us is of a greater difficulty than to deceive others outside the inner sanctum of family and friends. One could argue that this should be the other way around as the trust naturally existing among family and friends would make it much easier to deceive a family member or a friend than to deceive a complete stranger. That would be true if we were looking at outright lying but things are a bit different with deception. We must hide deception so to say deeper from the prying eyes of those who know us well than from those who would not know where to look for indications of a deceptive behavior in the first place. In this sense, successful self-deception is most important when deceiving those we are closest to, which is family and friends. Lastly, societal dishonesty requires the least effort and often we engage in it for nothing more than sheer convenience. However, we can also deceive ourselves about more serious things, like who we believe we really are and in which kind of reality we live in and what the world around us it up to. In such cases, there is always the danger that our individual and societal consensus realities come to represent something more sinister than just lightly retouched self-portraits to make ourselves feel a little better about who we are.

A Human Nature of Destiny?

As the preceding section just illustrated, our human nature is much more grounded in biology than we used to believe. There is little we can do against what biological evolution has shaped human nature to be. Some of what we used to look at as behaviors exclusively controlled by us as individuals are anything but that. In order to survive as a species, evolution has made sure that we lie and deceive whether we want to or not, just as we must eat, drink and engage in sex to procreate our species. The pursuit of the latter is a source of all kinds of human self-deceptions. To appreciate that, we only have to look at human mating behavior. In many ways, human mating behavior is similar to mating behaviors as we can find it more generally in the animal world. Intuitively, or should we say innately, women and men understand what they should look for in a sexual partner in order to produce a healthy offspring and make sure that the child will be well cared for so there is a good chance it will grow into a mature adult who can then have children of his or her own.

Because few of us possess all those desirable features, which would signal that we are an attractive sexual partner, we take to deception. Of course, we do not call it this way, because most of that behavior is so much a part of human culture, so deeply ingrained in it that it has become second nature to us. In a sense, those behaviors have become the human equivalent of a peacock's feathers or the brilliantly shining plume of a bird of paradise. The difference with humans being that it is more often females rather than males who embellish themselves to advertise their desirability. Shaping the female body into forms the human male will subconsciously perceive as signaling for example high fertility is common in one way or another in all human cultures. There is a whole industry catering to this female desire to attract men, providing to women whatever they may require in that respect. Seemingly, little separates adorning oneself to make one's attractive features stand out even more from using outright deception to achieve the desired result. Just think of breast implants, nose jobs, face liftings or whatever else modern plastic surgery can provide to help some women to attract a potential sexual partner or to make them feel better about themselves. Even though in the first instance, we are looking rather at intentional lying than any kind of subconscious deception.

Human nature, in this case female nature, reflects culturally motivated and sanctioned lying that we do not really perceive as such anymore. Women may lie about their beauty, maybe even fake their desirability, but no less is the case with human males. We would be ignorant if we did not recognize that human males can be just a mischievous as human females when it comes to securing an attractive sex partner. It just so happens that for the most, at least until quite recently, human males used to signal their attractiveness in simpler ways but in many respects still quite similar to how other males in the animal kingdom court females. With males, power and the associated assurance of protection

for any offspring is at least as important to convince an attractive female to engage in mating as is a male physique indicating a healthy individual. With respect to those qualities, human males have developed their own devious deceptions to make females believe that they possess them in abundance when in fact they most often lack them to a good extent. Over tens of thousands of years, humans have layered cultural and social conventions over the animal nature still underlying human mating behavior, thereby transforming the latter into human courtship behavior. Nevertheless, the animal nature of human mating behaviors still shines through.

While we can find many similarities when comparing the courtship behavior of other species in the animal kingdom to human courtship there are of course also enormous differences. We do not know the emotions which for example animals may have which bond for life and we do not know what emotions for example our much more promiscuous primate cousins have when they mate or when they for example care for a weakened or injured member of their group. We do attribute feelings of love or hate to animals in the fables we tell our children, but we only have very general ideas for now how animal emotions, which we know many animals certainly have, relate to our respective human emotions. What we do know is that no animals other than humans write poetry or compose music to express feelings.

The expression and celebration of human emotions figures prominently in the kingdom of uniquely human creations, which Karl Popper deemed to be a third category next to the state of matter and the state of mind. It is as if humans added a completely new level of expressions to what other animals can command in how they interact with each other. It is tempting to see this as something humans did on their own volition instead of attributing it to a different kind of evolution. Culture evolution is something that is difficult for us to study, as we are practically the only species on this planet with a rich history of culture evolution. Which makes it that much easier to think of culture evolution as something we made and not something that is part of what typically may happen along the evolutionary trajectory of a species like ours. In this context, it is helpful to remind ourselves that biological evolution is only part of a much grander evolutionary scheme.

Biological evolution intertwines with geological evolution, and both would not be possible without planetary evolution and star evolution preceding it.[53] Each of those types of evolution is subject to specific laws of nature and they depend on each other, albeit in a rather unidirectional way. Biological evolution is not a precondition for the evolution of stars or planets but can only be a possible consequence of solar systems with stars and planets forming. In a similar way, the specific combination of intertwined geological and biological evolution on our planet over the last roughly four billion years may be a precondition for something like culture evolution becoming possible for a species like us. As of yet, we know too little about the specific laws governing something like culture

evolution and how it is conditioned or intertwined with biological evolution. With reasonable justification, we can however assert that human culture evolution is unique on this planet, which makes culture evolution a unique characteristic of our human nature.

A little more than one hundred fifty years ago Charles Darwin first formulated the constraints driving biological evolution as we understand them today. Since then, science has vindicated his views on evolution and added to them new concepts. Modern biology and genetics not only discovered what underlies the variability of a species but have allowed us to read and increasingly also to manipulate the code of life. Unfortunately, we have no similar understanding of human culture evolution. Therefore, we cannot really answer any questions with respect to what drives human culture evolution or which purpose it serves. In some ways, it seemingly counteracts biological evolution in the sense that natural selection does not quite work the way anymore as it did for humans a long time ago. Today, most members of our species, many of whom would have stood little chance to survive in the natural environment of let us say our Pleistocene ancestors, can live long and full lives in our modern age.

In our time, selection with respect to physical fitness just does not play the same role anymore. Much more important for an individual to thrive in our modern societies are social skills. To stress the point, throughout human culture evolution, selection against physical fitness criteria became less important than selection against mental fitness criteria. Humanity has created for itself a meta-environment where new and seemingly non-biological laws now govern much of what occurs. Embedded in the biological environment that gave rise to human evolution, this meta-environment is a product of human culture evolution, very much shaped by the industrial and technology revolutions of the past two centuries.

As many of us are aware of, this human created meta-environment increasingly destroys the biological environment on which it depends. That we create such a meta-environment following its very own laws may be the destiny of our human nature and a natural consequence of human culture evolution. However, must this human creation in the process of its own evolution destroy the very fabric on which its existence depends? The human meta-environment, built by a species which owes its existence to the richness of this planet's biological evolution, has in many respects become a cancer destroying the very biological environment which gave rise to our species. It is in this respect that we must ask the question if human nature has a destiny.

Human destiny has been on the minds of religious scholars and philosophers for a very long time. However, the destiny they were concerned with was almost exclusively the destiny of the individual. Religions may refer to divinely created human beings in many ways such as mankind, humankind or the human race, but human salvation and with that human destiny usually is something very much individual. Just consider for a moment, leaving any purgatory aside, how a dualist religion like Christianity could claim to be

seriously concerned with the destiny of humankind itself since it envisions for one part of it a destination like a heavenly paradise while another part of humanity will suffer in hell for eternity. It looks like some of the monistic religious and philosophical perspectives we can find in Asia make the destiny of humankind itself as much their concern as the spiritual well-being of the individual. Admittedly, things are more complex than that and comparing religions in this way is not completely fair. However, that does not change anything with respect to how we all think about the future. We do so individually but not as a species. We think of our own potential destiny and are oblivious with respect to the destiny of humanity.

Many may think that human nature has no destiny at all and therefore humankind has no destiny. That is understandable because when one looks at how humans have muddled through what we refer to as our recorded history that may be indeed the correct conclusion to draw. However, that is not the position argued here. Just like stellar evolution, planetary evolution, geological and biological evolution follow their specific paths governed by the laws they are subject to, it is reasonable to assume that this is no different for culture evolution. The question that remains is deceivingly simple. How much are we in control of this destiny of human nature?

In some way or form, the fable of the frog and the scorpion is likely familiar to most of us. Frequently attributed to Aesop for which we have however no proof, many cultures other than those who are heirs to the Mediterranean World of antiquity know their own versions of this story. Regardless of its origin, the lesson of this fable is always the same. Here is how the fable goes:

> *"A frog and a scorpion find themselves at the banks of a river. The scorpion wants to cross the river. Because scorpions are not good swimmers, he asks the frog to carry him to the other side. The frog refuses the scorpions request saying he is afraid the scorpion might sting him. The scorpion however responds that it would be stupid of him to sting the frog while they cross the river as then they both would drown. This argument convinces the frog, he lets the scorpion climb on his back and they both begin to cross the river. Everything goes well until midway across the river when the scorpion stings the frog. Why sting me, now we will both perish, the dying frog shouts in his anguish. I am a scorpion, responds the scorpion, I could not help it."*

The lesson of this fable is obvious. Nature is destiny and the scorpion can only behave as a scorpion. Believing otherwise can be a deadly mistake. Our biological nature ensures that we behave in many situations just like the scorpion. But what about our human nature? Does our human nature essentially condemn us to behave a certain way whether we want to or not? Is our human nature destiny? Do the laws governing our human nature have to be somehow traceable to our human biological nature or are quite a few

of them owed to human culture evolution and represent something like meta-laws? Such meta-laws, while linked to human biological nature, may possibly supersede some of the biological laws governing human nature; maybe even contradict them. Just consider how the meta-environment humans created destroys the biological environment on which the survival of our species depends. How much something in our human nature is subject to such meta-laws or biological laws is sometimes not easy to separate. Just think of the simple examples of eating and drinking. If we do not eat and drink, we will die. That is a simple biological fact. However, nowhere does biology necessitate gluttony or drinking inordinate amounts of alcohol; that is, unless they are symptoms of a disease. The gourmet temples of our modern civilizations, while also feeding people, serve much more than a biological need; and so does the consumption of alcohol. Having to eat and drink is a biological necessity for us but the eating and drinking habits of our modern civilizations are anything but that. They are expression of our culture evolution and part of what human nature has become. We also know that we can control our eating and drinking habits, unless of course we suffer from a disease. Hence, there is no need for us to dine on liver pâté and champagne just for the pleasure of doing so. However, many still do so and many more would likely want to join those who can afford such luxuries today.

It becomes a bit more complicated when we look at biological drives other than eating, like sex for example. We all, some more reluctantly than others, would likely agree that as individuals we must not have sex to survive. After all, the biological purpose of sex is to ensure the survival of the species and not that of the individual. That makes controlling our sex drive in some ways easier than to control our eating and drinking habits. Our sexual behaviors are also much more subject to societal constraints than our eating and drinking habits. Hence, while for us as a species having sex is necessary to secure our species future, as individuals we could survive without it and some actually choose celibacy, for example by becoming priests or opting for a monastic life. In short, our sex drive seems to be much more subject to the laws of human culture evolution than the laws of biology when compared to eating and drinking.

Much of Karl Popper's kingdom of human creations is to a greater or lesser extent subject to the laws of human culture evolution, much more so than to any biological laws. That certainly is true for our concepts of love, courtship and human relationships in general, which also frame our societal views of human sexuality. Little of that is subject to any biological laws but simply represents our culturally evolved views of what human nature should entail. Obviously, we can change those views and sometimes we do. A single human being can change her or his mind instantly but not so whole societies. Our societies, themselves products of human culture evolution, take much longer to change their views. Societies really do not have a mind of their own in the sense that a single human being possess a mind, but they have something analogous to that. Just like the

ideas, wishes, desires, dreams, contentions, expectations, compulsion and so forth, in short everything a single human being experiences or produces are a reflection of that persons single mind, one can look at Karl Popper's kingdom of uniquely human creations as the expressions of the much larger fictive mind we can attribute to our human species nature. When we ask if human nature is destiny we really are asking if this fictive mind of our human species nature can change its mind. Evidently, we can and a good example of that are again our conceptions of sexuality, which have changed quite a bit over the last one hundred years or so. On the one hand, such examples can give us hope that as a species we can change our mind. On the other hand, we must be concerned. If it takes us so long to change the mind of our human species nature with respect to something rather simple, how long will it take us to change our human species mind in much more difficult matters?

The meta-environment our species has built for itself, ruled by the laws of human nature bequeathed on us by culture evolution but still contingent on the laws of biology, has been growing at the expense of the natural environment, to which our species is a very recent addition. Some among us view this meta-environment not as a human accomplishment anymore; rather, they increasingly perceive it as a cancer growing at the expense of its host, the biological environment that has evolved to sustain life on our planet. When anyone among us misbehaves on an individual level that is easy for us to call out and correct. However, the rape of nature humanity is currently committing is not the deed of any single individual. Rather, all of humanity participates in it through what we do and what we neglect to do. To acknowledge that is of course difficult and therefore we struggle to call it out for what it is.

We can look at the human meta-environment as a creation of a fictitious mind of a human nature, essentially of our human species nature. In a sense, this is what we symbolically do when we for example celebrate individuals as representatives of the best and sometimes of the worst parts of human nature. Great composers, playwrights, artists, scientists or honest, just and brave leaders come to mind as examples of the former while for specimen of the latter we just have to recall the names of those who ordered and committed genocide and mass murder throughout recorded history. In destroying the biological foundation for life on our planet, we all act in seeming unison as if our species possessed a mind coordinating such behavior. However, when it comes to the much more difficult challenge of changing our species behavior, appeals to such a fictitious mind of our species human nature to do the right thing never reach their destination as all there is are individual human minds, the majority of them unwilling to do the right thing. In an ironic way, this is similar to the behavior of a group of children or teenagers who may act in unison to do something they know is wrong, like for example when stealing fruit from an apple grove, but when caught in the act and questioned refuse to take responsibility, individually or as a group.

Seemingly, if our human species nature has a destiny, our inability as a species to do the right thing may shape this destiny more so than anything else. We just cannot solve the enormous problems that humankind created by a limited number of individuals and communities doing the right thing. For humanity to shape the destiny of its human species nature, it must learn to act purposefully in unison as a whole to address the species level challenges that we inadvertently create. While to an outside observer this inadvertent species behavior may look intentional as it seemingly pervades our species, it is nothing more than the summation of many of our individual or communal behaviors, which are too often guided by little else than greed, shortsightedness or sheer indifference for the consequences of our actions as long as we benefit from them. Maybe in the very long-term we will be able to act as a species in ways we may then wish we could have acted as a species much earlier. Like for example in our time and age, so we could avoid some of the dire consequences of our inaction with respect to human caused global warming which humanity already has difficulty to confront today, and which will only get worse. Realistically, we have to conclude that with respect to our human species nature, our inability to confront problems resulting from our species behavior on a species level does not bode well for the destiny of our human species nature. But how about us as individuals, to what extent may human nature as it expresses itself in each of us individually be destiny for us too? Instead of asking to which extent the future of the human species may be predicated by what we here call our human species nature, we can also ask to what extent our individual human nature dictates the future awaiting each us individually.

What determines the expression of our individual human nature? The simple answer to that is of course nature and nurture. However, the balance between those two, the extent to which each of them determines who we are, that has been a contentious debate among scholars for a very long time. Throughout history, the pendulum has swung back and forth between the two extreme positions. One advocating that human nurture will dictate what the expression of an individual's human nature will be, the other one asserting that it is nature itself, which dominates who we become. As we saw earlier, behaviorism dominated much of psychology until the middle of the twentieth century and it clearly advocated the position that nurture dictates what kind of person a child will eventually become. With behaviorism, it was all about behavioral training, which someone like Watson thought could shape any healthy infant to become any kind of specialist, ranging from doctor or lawyer to beggar-man or thief, regardless of her or his talents, penchants, tendencies, abilities, vocations and race of ancestors. More recently, the pendulum has swung back to views putting more emphasis on the role of nature in human development. This change is the result of insights that the science of genetics has given us over the last few decades. Being able to decipher the human genome is one of the great achievements of modern science and since this break-through occurred in

2003, scientists have found many ways to put this new capability to use in biology and medicine, but also in psychology. Combining a technique called genome-wide association (GWA) studies and the computational power to handle big data allows scientists to search the human genome for small variations, for many millions of such small variations by now. These many small variations and their specific combinations they then can compare against the occurrence of specific human traits. If they find for example in the genome of people with a certain disease a specific pattern variation more frequently than in healthy people, then in essence they have captured the genome fingerprint of a certain disease. There rarely is however just one specific fingerprint but numerous variations of such fingerprints for a certain disease. Therefore, scientists sample many genomes across a large population and test against a certain disease to produce a distribution of such genome variations associated with the disease, essentially the kind of bell-shaped curve with which many are familiar. Genome variations on one end of this distribution will have a very high likelihood for someone with that genome to be afflicted by this disease in his or her lifetime while on the other end the probability for such an event is very low. In many ways, this kind of screening for the potential to contract a disease at some point in one's life is already beginning to revolutionize the practice of medicine.

Imagine for example that such a bell-shaped curve of genome variations was available to test for Alzheimer's disease. If your doctor would then screen your genome for the very variations used to build the curve in the first place, he could give you the likelihood for you actually having the disease at some point. It is possible to screen someone's genome decades ahead and long before any potential on-set of Alzheimer's disease, as the genome at the time of the test is the same as what it will be when one turns say eighty or ninety. Instead of diagnosing if someone has the disease or not in their old age, we will be able to predict if someone could get it or not with a certain probability when they are much younger. Some may want to know this while others may not. Insurance businesses would certainly like to get their hands on such information, but their customers would likely not want to hand them this information. Just as one can predict the potential susceptibility of people with certain genome variations for this disease or that disease one can also use the same technology to predict many other things about every one of us, and most of it, right at birth. This includes for example simple things such as your height and weight as an adult, innocuous information for most of us. However, as it turns out, our genomes are also very good predictors for the expression of our human nature much more generally.

Psychology was early in recognizing the potential of GWA studies. What psychologists have learned from using them so far indicates that they may become a much better tool for diagnosing disorders than the traditional tools psychology has used until now. From what it looks like, we may soon be able to predict at birth the likelihood that someone will for example commit a serious crime at some point in his or her life. Almost

everything that scientists are learning from GWA studies confirms that nature seems to be much more in control of who we will be than nurture. But we must not forget that GWA still deals in probabilities and not in certainties. Hence, even if a screening at birth indicates that someone may contract a certain disease at some point in time, or that she or he will likely be overweight, or very smart, or may even commit a crime, those outcomes are not certain, they are only probable to a higher or lower degree. GWA tells us that there is something like destiny for how human nature may express itself in any one of us. However, this destiny is not one of absolute certainty, but is of a statistical nature. The question still is what should we do with this knowledge, as individuals and as societies? For us as individuals, the answer is probably simple. We just should take account of whatever a GWA screening might indicate and live our lives accordingly to minimize any risk, which is of course easier to do in some cases than in others. If a behavioral change is not sufficient, we may have to take precautionary measures like preventive surgery to minimize for example the risks of certain cancers. As for our societies, what to do with something like GWA screenings possibly indicating non-desirable outcomes from a societal level for quite a significant part of the population is an entirely different question. Here, our individual as well as our societal decisions will determine to which extent our human nature will be our destiny. With the increasing computational power and sophistication of artificial intelligence, we can expect GWA studies to become powerful tools in science branches such as behavioral genetics. Not all of what science will discover about the biological foundation of human behaviors will be to our liking but every bit of it will be invaluable information in guiding our species to a better understanding of ourselves and to a hopefully better future for all of us.

Is human nature destiny and if so, to what extent? It is not possible to answer this question today in any way approaching an at least somewhat satisfying answer. All we can do is to explore this question from different perspectives and see if we can maybe understand a little bit more today than we did yesterday. We also have to consider the possibility that we will never have a complete and satisfying answer to any such questions regarding our human nature. That is true for what we can call our human species nature as well as for the expression of human nature in each one of us. There are certainly constraints, imposed on human nature by our biological as well as by our culture evolution. However, in either case there is also always the freedom of choice. What choices will we make, as we have to confront the challenging realities of how our human species continues to destroy the foundations enabling life on this planet and how individually we do contribute to that? The choices we make in how we live our lives will determine the future of human nature. That is the good news. However, if we continue to mostly make the wrong choices, as a species and individually, human nature may indeed become destiny, but not in a good way.

Being Honest



Black Dots Alive

From everything discussed so far, it follows that we can expect to find multitudes of communal consensus realities within and across societies. Some of them widely shared, others much less so, some of them uniting, others conflicting and divisive. Just as when we negotiate with ourselves who we exactly believe ourselves to be, in order to arrive at our own personal consensus reality, we continuously have to negotiate all the communal consensus realities to which we belong. Communal or societal consensus realities are based on individual consensus realities accepting them as such and making them their own. In our lives, we are constantly negotiating consensus realities, with ourselves and with others. Much of that happens without us consciously calling this out for what it is as this process is just a very normal part of our lives. The part where we negotiate consciously or unconsciously with ourselves to keep our personal consensus reality aligned as to how we perceive ourselves is much less visible to us than the part where we negotiate communal or societal consensus realities with others. The latter we do with all others who are our fellow members in such communal consensus realities and with all the others who are outside of it, those belonging to different communal consensus realities that interact in various modes with our own. That all sounds very complicated but most of the time it is not. This is because members of any given communal consensus reality usually fall into two groups, a small set of active shapers and leaders and a much larger set of mostly passive members. Almost by default, the large subset of passive members implicitly delegates negotiations on behalf of the whole group to the smaller set of active members. Of course, in the given context these are not formal negotiations because most often the many do not just literally empower the few to negotiate on their behalf. Rather, consensus realities influence each other in many ways and differentiate themselves from one another through intertwined complex societal processes in which we all participate. The sum total of those societal

processes however amounts to something like continuously ongoing negotiations. We all live in such continuously negotiated worlds of various consensus realities that we may never have consciously subscribed to but of which we are all members. Frequently, we may not even be aware or think of ourselves as belonging to any such consensus reality. That is because we usually do not acquire membership in such consensus realities through explicitly opting in. Rather, most of us, through our upbringing, education and life experiences have clandestinely become members of such consensus realities; and as long as we do not explicitly opt out, we remain members. This has obvious consequences. Arguably, as long as we essentially remain complicit in such communal consensus realities to which at closer inspection we would rather not belong, we are responsible for any actions such groups also claim to take on behalf of all their silent members. In fact, we are just as responsible as its most fervent proponents are for everything directly or indirectly furthering the interests of those consensus realities. There is another and as most would likely agree an even more serious responsibility. With respect to every communal consensus reality we consciously participate in, we have the responsibility to make ourselves knowledgeable about any harm that its beliefs could inflict on others. Now that may seem trivial, but it is not because frequently we do not even know the full extent of the set of beliefs that any such consensus reality endorses. In addition, often compromise is the name of the game and in order to be able to embrace certain aspects of a consensus reality, we may willingly accept other aspects, which if they were presented to us just by themselves, we would reject.

Furthermore, many among us are often just passive members of communal consensus realities, which makes it easier to retain a selective view of what a consensus reality endorses. We tend to create our own blind spots by adjusting our personal consensus realities within larger communal or societal consensus realities in such ways as to render the more outrageous aspects of some consensus realities in our perception into something much less troubling than they really are. In a way, this is almost a necessity as few communal consensus realities will be prefect matches for any one of us. However, for the vast majority a strongly felt need to belong to a larger communal or societal consensus reality usually will brush aside those from our perspective less desirable aspects of a communal or societal consensus reality to which we otherwise would like to belong. Quite often, the things we ignore in this way in a communal or societal consensus reality we desire to join and which by doing so we unwittingly endorse are just trivial, but sometimes they are not. In the end, it all depends on what kind of worldview a given consensus reality reflects and how its members act on it. Each one of us could be a member of a consensus reality that may endorse points of views we would not want to be part of but as long as we ourselves maintain a blind spot for those, we may feel fine with our membership in that consensus reality. However, if some members of that consensus reality choose to act on those points of views we disagree with that also implicates all

other members, including ourselves. A silent disagreement of a majority of members with certain aspects of a consensus reality is dangerous. In the worst case, it can embolden a few members to do the very things the silent majority disagrees with, things they may never have done unless they believed to have the clandestine support of the many.

The kind of communal consensus realities, where a relatively small number of people define and negotiate much of the perceptions and beliefs held by a much larger group, represents only one segment of communal consensus realities. However, they are the ones we are most familiar with as for example the major religions and political parties are some of their more prominent representatives. Throughout most of human history, such larger communal consensus realities were shaping much of people's everyday lives and experiences. There certainly were also many smaller ones but usually they had a much more difficult time to persist and almost inevitably they were destined to either join up with one of the larger communal consensus realities or vanish altogether. Obviously, consensus realities sustained only by a few people and centered on what their immediate environments could have perceived to be extreme or dangerous views, would have faced significant obstacles in trying to spread more broadly. This is understandable. Just for a moment put yourself into a small community where people know each other well. It is very hard in such environments for a small minority to hold beliefs, which the majority could perceive as outlandish or threatening.

For better or worse, until quite recently the consensus reality of a community itself provided an immediate check on any dissent that could harm village life so to speak. This barrier only came down with our modern age. Nowadays, consensus realities comprising only a small number of people can maintain and propagate their worldviews and beliefs in seemingly unchecked ways, no matter how wrong, misguided or untruthful they may be. The internet and social media have made this possible. This virtual reality world is in a sense like the lawless Wild West where editorial oversight or legal restrictions do not really limit the machinations of even the most outrageous aspects of some consensus realities. On social media, they can spread like viruses, infecting and manipulating other consensus realities at liberty, poisoning the very foundation of a democracy - our civil discourse. We are only just beginning to understand how this could in the end possibly disrupt or even destroy the very fabrics holding our societies together.

In whichever way we choose to address this challenge, how successful that can be will largely depend on how honest we are with ourselves. Being honest and truthful to ourselves, that sounds simple enough. However, this is exactly where the problem lies because for all we know it is very difficult for humans to be honest to themselves, maybe even more difficult than being honest to others. Practically, we are natural born liars and deceivers, and we are particularly proficient at deceiving ourselves; as far as we can tell, maybe even much more so than at deceiving others. Our ability to deceive ourselves seems to have evolved as a tool of human self-preservation when humans first

began to live together in larger groups. Since then, human societies have dramatically changed. Living in one of our modern societies poses quite different challenges than the ones our earliest ancestors had to face but our exquisite abilities to lie and deceive still serve us well, maybe even more so now than back then. The capability of being able to lie with impunity and clandestinely deceive oneself as well as others may have had some evolutionary benefit for our early human ancestors in coping with the increasing complexity of living in larger social groups. Those larger social groups, like family groups or first tribal associations, were still rather small when compared to the human masses enveloped in the social fabrics of our modern civilizations. In our modern world, billions of humans compete for limited resources and what might have been helpful to the social evolution of humans in the past, may be even more helpful to the individual today in coping with the challenges of our modern societies. However, it may also pose one of the biggest challenges to the very survival of our modern societies and in the end to the survival of our species itself.

Evolutionary psychology tells us that self-deception is more difficult than the deception of others. That by itself provides a recipe for successfully deceiving others. From what it looks like, it seems the best way to deceive others is to nudge them towards deceiving themselves. If you think about it, this is pretty much the art of persuasion in its highest form. How better to get people to buy into lies and deceit than by getting them to a point where they instigate the deception within themselves. Presumably, convincing someone to believe something by using rational arguments is often more difficult than nudging a person to embrace a belief via self-deception. There are a numerous ways to accomplish the latter. One way to make this work is for example to convince a person that their very own consensus reality is also yours and then to slowly and carefully introduce into that supposedly shared reality something that a person otherwise would not freely endorse. This of course is very much how advertisement works. We may believe that we buy many of the things which companies want us to purchase from them because some rational arguments in their advertisements convinced us of their better value as compared to other products. However, that is rarely what happens. It is much more likely that we buy something because the advertiser, unnoticed by ourselves, managed to introduce something into our consensus reality, which was not there before. This seed planted into our consensus reality then germinates into us deceiving ourselves that it was us in the first place who figured out that we needed just this product and not a different one. We will convince ourselves that we are making a rational decision, but we are actually making an emotional decision to buy something. Such emotional decisions can only emerge out of our consensus reality of which endorsing this new product has become a part. More important, because such emotional decisions tie into our consensus reality they are sticky and we even tend to defend them when reason may already tell us that we actually bought a bad product.

Long before psychology gave us a better insight into the human capacity to deceive others and ourselves, conmen and charlatans have been using this aspect of human nature to their advantage. To be successful, they do not have to be particularly smart. They just have to be convincing enough that people begin to consider what they say as part of their own consensus reality without realizing of course that they have been hooked. The initial ruse may be innocent enough but that is not the end of it. Once people begin to endorse a seemingly innocuous addition to their shared consensus reality, conmen and charlatans can manipulate their victims much more easily to deceive themselves into believing things that they may have never considered otherwise. How else is it for example possible that the holder of the highest office in the most powerful nation of our time could con a large fraction of the voting population into believing that he had won an election, which he clearly and demonstrably lost? Not too long-ago people would have thought that things like that only happened in some godforsaken and brutish country not really belonging to our civilized world. Unfortunately, this did not happen long ago in some uncivilized countries the names of which most of us cannot even pronounce, like one of those places in Voltaire's (1694 – 1778) *Candide*, but it happened just recently right here in the United States in the year 2020.[1]

We have no choice but to create our own consensus realities to give belonging, purpose and meaning to our lives. Consensus realities represent how we understand ourselves and how we explain and understand the world around us and ourselves in it. The fortunate ones among us may live lives where their consensus realities embed them in their communal and societal environments in harmonious ways. However, for many others nothing like that is the case. For those less fortunate ones, their consensus realities may put them in conflict not just with their communal and societal environments but also with objective reality itself. The extent to which this can sometimes be the case is surprising. Evidently, human nature can sustain consensus realities out of touch with objective reality to an astonishing extent, which is simply an indication of just how strong our individual desire is to make sense of who we are, what it is we are about, and why we perceive the world as we do.

The following sections will look into a few examples where our individual and communal consensus realities either were or still are at odds with what most of us would perceive to be in our own best interest as well as our societies long-term interests. Invariably, quite a bit of that will look like hindsight. However, that is where we must begin. We must be honest about how in the past individual, communal and societal consensus realities shaped our world in ways that led to outcomes, which in hindsight we rather would have avoided. Only once we can acknowledge that, will we be in a position to honestly question our current individual consensus realities and the communal and societal consensus realities of which we are active or complicit members.

Belonging, Purpose and Meaning

In our more lucid moments, deep down we likely all acknowledge that nothing will last forever and that everything is ephemeral. Evidently, such thoughts surface in our minds only infrequently. If we more often had to ponder the fragility of our own existence or the transitory nature of all that surrounds us that might not be a good thing for many of us. The human psyche is just not robust enough to cope with such insights on a continuing basis. Individually, to live the life we desire to live, we must forget and ignore that which could distract us from being who we are and keep us from becoming who we desire to be. Individually and socially, many of us act in similar ways, conveniently forgetting and ignoring that which we rather not confront as it would require uncomfortable change, change we are not yet ready to consider seriously, or change we outright oppose.

Probably most of us meet their likeness every morning during our bathroom routines when we brush our teeth, shave ourselves in case we belong to the bearded gender, or do just whatever we do every day in front of our bathroom mirror. Seeing our face reflected in a bathroom mirror every morning is not just an indication that we are committed to a commendable hygiene serving ourselves well through our lifetime. Looking at ourselves every morning also serves our psyche as it smoothly acquaints us to our own aging self on a daily basis. Next time you stand in front of that mirror, try to imagine seeing yourself at a much younger age. It is a difficult thing to do, and it is much easier for us to close our eyes and just recall an image of our much more youthful selves. We also know that one day, this face we see every morning in our bathroom mirror will be gone but that is not anything we usually dwell on. Our own mortality is something that most of us conveniently push aside. For many of us, that seems to be the prudent choice as it may be best to not have to think about those things, or even worse, have to talk about them.

In a way, however, those things are always on our minds as we search for belonging, purpose and meaning in our lives. Doing so is something that we usually do not reflect on much because it has become second nature to us. However, we certainly become consciously aware that we are searching for those things in our lives if we find ourselves bereft of either of them. Having a sense of belonging, purpose and meaning is how we shield ourselves from having to question our very existence, having to face our very own mortality. At the same time, the felt presence of those three qualities – belonging, purpose and meaning – allows us to be at peace when questions with respect to our own mortality surface in our minds. Left to ourselves, most of us would find it very difficult to evolve a sense of belonging, purpose and meaning because to attain those three we need others. In a narrow sense, this certainly begins with family but in a wider sense, it encompasses the richness of the cultures in which we grow up. Our sense of belonging answers to our questions about who, our sense of purpose to our questions about what,

and our sense of meaning to our questions of why. Knowingly or not, we all look for answers to such questions and while we find them in different ways, they inevitably arise from within the cultures and societies that nourish us.

On an individual level, we tend to be less than honest to ourselves and to those we care about when it comes to our own mortality or that of our loved ones. The richer a society becomes the more this seems to be the case. Richer societies need to look towards poorer societies to realize how things have changed. There, many families still live their lives in ways that resemble what was once also common in today's rich countries. Large families with generations living together once were common everywhere and experiencing disease, suffering and death of family members was just a normal part of growing up. In our modern industrialized societies, this has become the exception. In rich countries, elderly or sick people die today in senior living facilities or in hospitals rather than in the midst of their families. Naturally, it is wonderful that people receive the best possible care but that often compartmentalizes death and suffering. Therefore, many of us only have to reckon with those things once we ourselves become the ones who are sick or suffer from the ailments of old age. That said, however, none of us likely wishes for a return back to those times when health care was almost non-existent and when family members had to be cared for at home until they passed away. Nonetheless, the nagging question still remains if we could not have had both, the benefits of modern society as well as the closeness and comfort of having family and friends around us when we need them most. Clearly, from an individual as well as from a societal perspective, we gained a lot with our modern ways of life, but we certainly also lost some things.

Looking back into recorded human history it is hard to overstate the importance of family as the fundamental building block of human civilizations over the last several thousand years. The family unit has been the nucleus of human social life throughout human evolution. It is therefore just amazing that this fundamental building block of human societies could begin to dissolve so quickly over the last few generations. The demands of our modern societies for large industrial work forces have not only shrunk this functional unit from multi-generational families living together down to parents living with their children only. They have also enfeebled the remaining core family by making it a necessity for many of those with lower incomes that both parents work full-time jobs. That is if they are lucky to have such full-time jobs as nowadays many parents often have to work several part-time jobs, neither of them providing enough to support the needs of a small family and taken together leaving even less time for family members to be a family and care for each other. In some affluent countries, day care facilities, if accessible and affordable, may provide some support for working parents. Unfortunately, in most countries, such things are not more than a dream and even in rich nations, most families have to cope without the benefit of such societal support. The overwhelming majority of parents face the challenge of bringing up children without much societal support, many

of them forced to work long hours for meager pay, leaving them little time for their families. Looking at the enormous societal problems that some rich countries including the United States face today, the destruction of the traditional family unit throughout a large swath of the population is certainly one of the root causes for a good number of those problems.

In many ways, our modern societies have become societies focused on the needs and desires of individuals rather than those of families. This process began some time ago with the emergence of the modern industrial state. Businesses are only interested in hiring individuals and not families. Businesses have no contracts with the families that raise those individuals they hire and unless they are family businesses, they have little interest in the well-being of any family at all. Businesses also need customers and where those customers are not businesses themselves, again the interest of business for the most is with the individual and not the family, except for the small segment of businesses that market products specifically to families. However, even then, they target the individual, a parent that can make a purchasing decision. It usually does not matter that much if the product a business convinces a parent to buy may be harmful to their children. Just remember that not too long-ago companies still colored children's toys with lead containing paints well knowing how damaging lead poisoning is to the human brain and nervous system in general and particularly so in young children. Some companies will point to all the things they did to make products that meet a child's needs and are safe for children to use. What they forget to tell you is that quite a few of them did not choose voluntarily to put the interests of vulnerable children first, but only began to do so after facing lawsuits with potentially devastating financial consequences, not to speak of any civil and criminal liability lawsuits threatening their managements.

The redefinition of the respective roles of family and society since the emergence of the modern industrial state has improved the material quality of life for many. That includes in a large number of countries now universal education, at least to a certain level, and some basic health care; affluent societies can of course provide higher levels of education and better health care. However, there are also costs associated for our modern societies when they reshape themselves from societies built around the family nucleus to societies built around the concept of individuality. Likewise, there are also costs for the individual associated with what one could call the loss of family in a wider sense and often the disappearance of familial relations altogether without anything really replacing them. Throughout history, the family unit served many critical societal roles as best it could under whatever given economic or political system. The large families of an agricultural past long gone frequently faced destitute circumstances and often struggled even to feed themselves, not to speak of providing an education for their young. Nevertheless, however dire circumstances would have been, family members usually could count on family solidarity and support even in the most difficult of situations.

The sense of belonging and emotional security that came with being part of a family throughout one's life could outweigh many of the material shortcomings that were the reality of life for many back then. With traditional family structures fracturing under the pressure of industrialization, societies somehow had to replace the functions they had served. From a material perspective, many industrializing countries did so in some way or form and with varying degrees of success. But family always provided more than just material support. Being part of a family gave meaning to the lives of people that an anonymous state and its bureaucracy could not engender. In this latter respect, for people to find meaning in their lives, religion was certainly no less important than family. However, parallel to the dissolution of the family, the modern industrialized world also increasingly undermined religious belief. As modern societies became societies of individuals and for the most secular societies, people had to search for new ways to satisfy their basics needs of belonging and giving meaning to their lives. Through the marginalization of family and increasingly religion, the evolution of modern society confronted the individual with two existential challenges – how to regain a sense of belonging and how to give meaning to their lives outside the contexts of family and religion.

Of course, the above observations never really applied to all sections of modern society, at least not to the same level. Within some sections of our societies, family and religion have retained much of their importance and continue to play similar roles in the lives of their members as before the modern age. However, by no means do those population segments where this still may be the case comprise the majority in our modern societies anymore. In the nineteenth century, in the wake of the accelerating industrialization, forces more obscure than industrialization itself began to shape modern history, often ruthlessly taking advantage of the modern individual's search for belonging, purpose and meaning, which family and religion could not meet anymore. In doing so, they seemingly gave the individual a new sense of belonging as member of a class, as son or daughter of a nation, or as citizen of the state, whose purpose it was to serve class, nation or state to find meaning in something larger than him- or herself. Class, nation or state became substitutes for family and religion. Modernity replaced tribal, dynastic or religious wars with class warfare and the wars of nations and states.[2] Maybe we should date modernity from the French Revolution, which gave rise to the Napoleonic Wars, the latter in a way resembling the transition from tribal, dynastic or religious wars to the wars of nation states. It was only in the decades after the carnage and the genocides of the Second World War that it became evident to what extent ideologies of class, nation or state had failed the modern individual by offering nothing more than superficial surrogate versions of belonging, purpose and meaning. We can look at this fissure in modern history as the first betrayal of the individual by modern society, so to say the first betrayal of modernity.

While after the war countries under the thumb of the Soviet Union had to continue what already many considered a failed experiment of modernity, the citizens of Western democracies could plot a new course. The new promise of the modern West, as it emerged from the devastation of the Second World War, soon joined by some nations in Asia, was one of economic prosperity where belonging would become a question of affordability, purpose would come from the pursuit of amassing wealth, and meaning could derive from society's acknowledgment of one's wealth, power or popularity. Describing the new religion of economic prosperity in this way is certainly a stretch but not too far from the truth. There were of course those who disagreed with how capitalism evolved in the free world and who sought out alternatives to a life dominated by the pursuit of money. However, they always remained a small minority. For some time, for just a few decades, economic prosperity actually did wonders for much of the middle class in Europe, North America and some countries in Asia. However, that did not last. Already a few years before this phase of economic expansion in the West began to subside and gave way to a new economic regime that would lead to a much more lopsided accumulation of wealth, the Club of Rome had pointed out that modern capitalism might be doomed for reasons such as the increasing deterioration of the environment, among others. In its 1972 report, *The Limits of Growth*, it made clear that one could not expect economic growth to continue indefinitely. The 1970's also saw the first publications of reports warning of an impending climate change if industrialized countries continued business as usual. With the end of the Cold War, countries that barely had any industry at the time to speak of began to catch up with the West, China being the case in point. The result of this ever-expanding and accelerating industrialization and its contingent exploitation and destruction of natural resources is now unmistakably clear. We are well on our way to destroy the very environment on which our survival critically depends. With every year that goes by without us limiting the impact of the climate change already underway by effecting the necessary major changes in how our modern economies work, future generations will have to suffer more. This essentially constitutes a second betrayal of modernity. This time however, we cannot blame either class, nation or state for betraying us because it is our very own greed, manifest in how we run our economies and how we live, which betrays us.

The above outlines only a few relevant developments among a number of other possible reasons that could be responsible for the emergence of a growing number of people dissatisfied with having to live a life they increasingly experience as subjected to the control of others. Some may view those people simply as discontents, malcontents, irrational, not willing to understand and accept how things work and what their roles in society are. That is a mistake. The voices of those people are rather like the proverbial canary in the coalmine, letting us know that something is very wrong within our societies as well as between the various societies we find around the globe. What they do indicate is

that the increasing political, economic and cultural disparities characterizing our modern world have reached a tipping point. We either must change course or face the distinct possibility of a dystopian future not just from an environmental perspective but also from a social perspective with the bonds still holding together our societies eventually fracturing. To avoid such an outcome, we will have to figure out how to make our societies more equitable again, in a political and economic sense, and how to address the imbalance of power and wealth between nations or states such that all parts of our world have a voice and a vote in the future of humankind. Equally important, we must recognize that in the wake of the centuries-long political and economic dominance of the West, a cultural invasion carried by Western commercial interests began in the second half of the twentieth century, which in the eyes of many is threatening the very foundations of their own cultures. While frequently couched in the language of humanitarianism and economic development, it mostly served the political and economic interests of the giving nations as much as and often even more so than the populations in those developing nations, many of them former colonies.

In history, more often than not, military conquests pursued not only political but also economic interests and to secure both, political and economic supremacy, the victors frequently engaged in cultural conquest as well. It may be best to go back in history to illustrate this with an example. When the Romans conquered all of Western Europe, they spread their own culture in ways we can still see today in the form of Roman roads, aqueducts, baths, floor heating or other amenities of Roman civilization. What we cannot see is the cultural destruction that went along with that. The Roman armies did not march into unsettled lands, but conquered peoples and cultures rooted as deep in time as the Romans themselves. Of those indigenous cultures, which the Romans encountered, little to nothing survived and all we know about them comes usually from archaeological finds.[3]

The extent of this cultural conquest we can still gauge from the fact that practically all languages then spoken in what is today modern France, Spain or Portugal have vanished except for a few such as the Basque language or the Breton language. The modern languages spoken in those countries today, as beautiful as they are, derive from local variants of the Vulgar Latin that eventually prevailed in those former Roman provinces. Something similar happened across the channel but there it was not a Roman dialect, which would prevail. The Angle and Saxon invaders who began to settle in Britain beginning in the fifth century were the first to bring a mixture of Germanic languages to a Celtic speaking world to which the later Viking invasions by Danes and Norwegians may have added a little more.[4] The West Germanic dialects introduced by the Angle, Saxon and Jute people, amalgamated by the Norse dialects of the Viking invaders, evolved into the so-called Old English, or rather, into various dialects of it. Old English became the dominant language in those parts of Britain we associate today with England, but not

in Wales or in Scotland. Later, this Old English spoken before the Norman invasion received an infusion of Northern French when these Normans, former Vikings converted to Frenchmen, invaded Britain in 1066, after which it eventually evolved into the English language we know today.[5]

The extensive reshaping of ethnic, cultural and political maps in Western Europe in the wake of conquering Roman armies, Germanic invasions and then Muslim armies invading the Iberian Peninsula would pale in comparison to the global changes brought about by the maritime exploration of a Western Europe emerging from the medieval period. Beginning in the fifteenth century, as maritime exploration quickly turned into colonial conquest, Western European nations began to change the political and economic world map, shaping it in ways that still reverberate today. One consequence of the colonial conquest beginning then and the centuries of colonialism following it is still with us today and that is the economic dominance of the West. Only recently has a serious challenge to this Western economic dominance emerged in the form of a rising authoritarian China where communist rulers have discovered state directed capitalism. Under communist rule, modern China on the one hand develops its own economic sphere creating dependencies through economic partnerships similar to what the West has established in its relationships with the developing world.[6] On the other hand, leading economies in the West are becoming increasingly dependent on China itself. How all of that will evolve and how it will alter our world's political and economic maps, only time will tell. For now, however, that does not change that a small part of the world, beginning several centuries ago, has economically conquered much else of it and with a few notable exceptions continues to dominate much of our world affairs and certainly our world's economy. The cultural invasion following in the wake of Western economic dominance, while not destroying existing cultures it encountered outright, has in many ways influenced and to some extent even replaced them. For many in those countries, while they welcome economic progress promising a better life, this kind of cultural invasion is more than troublesome. Should we really hope for everyone around the world to lead lives similar to ours aside from being materially as well off as we are? What would we say if some seemingly unstoppable foreign influence threatened the very foundations of our own cultures? Would we accept that, or would we oppose it?

We are always quick to condemn those whose actions we cannot understand. In doing so, we conveniently forget that those actions may just as well be the result of people trying to find a sense of belonging, purpose and meaning – happening however under circumstances we have difficulty to relate to. This makes it challenging to perceive such efforts for what they are. Regardless, we must acknowledge that every human being on this planet, consciously or not, seeks to shape her or his individual consensus reality towards meeting these essential needs we all have. It just so happens that the forms such consensus realities take in the search for one's belonging, purpose and meaning, depend

on one's own life experiences. Clearly, it is reprehensible if some believe indiscriminate killing justified in asserting their consensus reality, seeking to make it the dominant communal consensus reality, which supposedly then would provide a new or revitalized sense of belonging, purpose and meaning to a much large group. But are we really so different from those who in the extreme pull the trigger to kill or who blow themselves up in an effort to kill as many around them as possible for a cause they fervently believe in? That certainly is a loaded question and as far as the author is concerned, no one ever has the right to kill anyone other than in pure self-defense. The question asked above is however not about what is right and what is wrong but simply inquires, are we so different? Some may answer that most of us would hopefully never commit murder or harm other people, voluntarily or even under duress. That is likely a correct but also an evasive answer because we do not have to do those things ourselves anymore. That is so because our taxes pay others to do such things on our behalf according to the judgments of those we have elected to political offices. We do not have to do the killing ourselves anymore because we have delegated killing. Of course, many will object that one just cannot compare our armed forces killing in our names with others killing without any such government authorization, many of whom we consider terrorists, some of whom we may call freedom fighters. To a good extent, that argument carries weight, but only if we limit ourselves to looking at what happens in the here and now. That is because if we take into account how we find ourselves in such situations in the first place, where evermore people seem to believe that their only recourse is to resort to violence, then things begin to look different. If we are honest, we have to acknowledge that the mess we find ourselves in today is largely the legacy of power politics and economic exploitation, much of it almost exclusively committed by nations and states of the Western World. Others did also similar things, Japan's colonialism certainly comes to mind, but then, it was usually only nations and states of the Western World who pretended, while committing the most heinous crimes, to do nothing but good for the world. First in the name of religion and later mostly in the service of civilization; and of course, always in the interest of commerce.

European states and nations colonized and exploited the Americas, practically all of Africa, much of Asia, and all of Australia, in the process destroying cultures and nations, killing indiscriminately those who stood in their way or just looked not white enough, and eventually ushered in the modern era of genocides. Anytime someone points to the historical roots of many of the problems we face today that usually will be brushed aside because as quite a few argue, the days of colonialism are long gone and if peoples and nations nowadays find themselves in dire situations, they are to blame for it and not their colonial past. That also conveniently ignores the fact that while from a military perspective the age of colonialism is now long gone, economic forms of colonialism continue to thrive. It was European nations which drew up the colonial maps and who partitioned

whole continents into states according to how they believed that it best served their own interests. The interests of those affected by such decisions usually being secondary concerns.[7] The political maps of Africa and of the Near East are still colonial maps. The conflicts arising from those partitions are with us to this day and not just in those places but on the Indian subcontinent as well. In addition to arbitrary political maps becoming the source of continued fighting between newly created states, in many places colonialism also left behind a heritage of civil strife as it combined different peoples within a single state who should have lived within their own boundaries. It was the colonial powers who either set-up the ruling classes in their former colonies themselves, like for example in much of the New World, or who aligned themselves with powerful factions in a given region which later became the de facto new colonial masters; that is, once the colonial powers had left.[8]

How can one find a sense of belonging, purpose and meaning in places where with some justification people view their destitute situations still as the long-term result of centuries of foreign rule? Foreign rule that was later replaced in many cases with governments holding on to power by the support of one of the superpowers or by the grace of almighty multinational companies. Many of us may find it difficult to understand how people can still blame a colonial past long gone for their current malaise or how they can believe that it is foreign political and economic dominance, which largely still shapes the realities of their lives. To appreciate how things may look from their perspective we would have to put ourselves literally into their shoes. Because that is not possible unless one happens to be born there and grows up in such a place, we can only try the next best thing. In this case that means to engage in some historical hypotheticals so we can maybe get a view from a different perspective. Of course, would have been and could have been will always be just that because reality is what it is and because history happened a certain way. That conceded, I still ask the reader to be curious and to go along with exploring at last one such historical hypothetical a little.

Let us consider for a moment what could have happened if the Mongol invasion of Europe would not have been stopped by what turned out to be for Europe the very timely death of Ögedei Khan (c. 1186 – 1241).[9] Just before Ögedei Khan died in faraway Mongolia, the Mongol armies had defeated the last major force between them and Paris, an army of Polish and German knights, in the battle of Liegnitz in what once was the Duchy of Silesia.[10] They already had conquered much of Eastern Europe and had marched into Austria and Moravia. Given their military superiority, the Mongols would have stood a good chance to conquer all of Europe, at least all of continental Europe.[11] However, when the news of Ögedei Khan's death reached them in Eastern Europe, the Mongol leaders needed to head back with their armies towards Mongolia to settle the question of succession. What would have become of Europe if it had been under Mongol rule for a few centuries? To imagine this, we just have to look towards the East where in the

thirteenth century the Mongols destroyed the Kievan Rus and turned all its principalities into vassal states of the Mongol Golden Horde.[12] It is certainly debatable to what extent Mongol rule shaped Russian history but clearly, it changed the trajectory of Russian history. While in Western Europe the fourteenth century would see the first sprouts of the Renaissance, which would come to full bloom in the fifteenth century, Russian principalities remained under the dominance of Mongol rule. In many ways, the rise of the principality of Moscow and of Czarist Russia became only possible in reaction to Mongol rule. We can only wonder into which state or states the Kievan Rus could have evolved if it had survived the thirteenth century.

Now consider what the history of Western Europe could have looked like if the thirteenth century Mongol invasion had continued into France, Italy, Spain, maybe even had conquered the British Isles. Would there have been a European Renaissance of the fourteenth or of the fifteenth century? Would the emancipation of science from religion, the beginnings of which we witness in the Western Europe of the sixteenth century, still have taken place? Would there have been a Reformation at all, considering that the shared yoke of a Mongol rule could have presumably united Christendom? Would Christendom have rediscovered its once vibrant military streak of crusades past and called for holy war against the infidels just as some in Islam still do today? We cannot really answer such questions, but we can look for parallels and we do not have to look far from Western Europe to find one. The Ottoman Turks began their European conquests about one hundred years before they captured Constantinople in 1453 after which they extended their empire into much of Southeastern Europe. After the Kingdom of Hungary had succumbed to them at the Battle of Mohács in 1526 and less than one hundred years after the fall of Constantinople, the Ottoman Turks were at the gates of Vienna in 1529 for the first time. For more than one hundred years, large parts of central Europe were subject to periodic Ottoman invasions until the Ottoman army besieging Vienna in 1683 for a second time suffered a decisive defeat.[13]

While the Ottoman defeat in 1683 was ringing in the decline of Ottoman rule in Europe, large parts of Southeast Europe and the Balkans would remain under their rule for almost two more centuries. There would be no Age of Enlightenment in those parts of Europe, as for the most they remained shielded from Western influences. In a sense, the troubled history of the Balkans and other parts of Southeast Europe in the late nineteenth and twentieth century is very much a legacy of a reverse colonialism where European states and nations were colonized by the Ottoman Turks. Now imagine how a Mongol occupation of practically all of Europe in the thirteenth century could have changed the course of European history. Mongol rule over China lasted a little less than one century and the Russian principalities were subject to Mongol rule for some two centuries. Hence, we could assume that any successful Mongol invasion of Europe would have had some staying power.[14] One potential prolonged impact the Mongol invasion had

on the trajectory of Chinese history was that even after Chinese rulers, the Ming dynasty, had replaced the Mongol Yuan dynasty, the Mongol threat was not over for China. Having to deal with the continued threat from potential further Mongol invasions might have been what curtailed China's maritime ambition, which practically ended with the death of Zheng He (1371 - 1435), its most famous admiral. Would a Mongol occupation of Europe or a continued Mongol threat have similarly hindered or even prevented the European maritime exploration, which began in the late fourteenth century? Who would then have discovered and presumably colonized the New World? What if the colonization of the Americas never had happened, and the indigenous people of the Americas could have continued to evolve their own cultures? One can ask similar question for the discovery of the sea routes to the spices of Asia, the discovery of Australia by Europeans and its subsequent colonization, and later of course with respect to the colonization of Africa. While we cannot know the impact of a prolonged Mongol rule in Europe, we can be certain that European history books today would look very different. To get a sense of what that could have meant for Europe we only have to look at another major catastrophe of the thirteenth century, the 1258 Mongol sack of Baghdad, the capital of the third caliphate, the Abbasid Caliphate. With Baghdad, the Mongols destroyed what had been the center of Muslim learning and culture throughout centuries, a role that the city would never regain.[15] Imagine the consequences for Western Europe if something like that had happened to London or Paris.

Of course, it really is not possible to compare the impact of Ottoman rule on much of Southeast Europe with European colonial rule in the Americas and later in much of Africa and large parts of Asia in any direct way. We only can look for analogies or contrasts in such cases. The Ottomans encountered people that spoke different languages, held different beliefs, belonged to different cultures and had their own customs but otherwise were just like the Ottoman Turks. Life under the yoke of the Ottomans may not have been pleasant for conquered peoples and nations but they continued to have a life under Ottoman rule. While the conquered people faced some level of discrimination, they largely could continue to practice their religions and customs and the Ottoman Turks considered them to be human beings just like themselves. The European outlook on the peoples they encountered during their colonial conquests was a very different one. In many cases, European colonialists considered the indigenous people they encountered little more than brutes and certainly not their equals. European colonialism brought with it the exploitation of any valuable natural resource, accomplished through forced labor or the outright enslavement of the native population, the result of which was little else than extermination of indigenous peoples through work. All of that alongside the desecration and destruction of countless cultures and the forced conversion to religious beliefs completely alien to indigenous people. From today's perspective, we would have little choice than to qualify much of European colonial rule as genocide, the physical

destruction of peoples and their cultures. For a long time, the arc of history seemed to bend in one direction only, towards the world dominance of everything associated with the seemingly invincible European powers. However, that was not always the case, and it was not the default outcome everywhere. Particularly in Asia, things turned out to be different with some peoples and nations able to resist European colonial rule or even becoming colonial masters of their own like for example Japan. Even the United States, founded on the idea that the sovereign is the people, was not immune against the virus of colonialism.

After the cataclysm of the Second World War, European powers could no longer muster the resources to maintain their colonies. Even though some of them were reluctant to let go of their former colonial empires, the most powerful nation of the West was now the United States and it did not look kindly on outright colonialism. Hence, there was little choice for European powers than to let go of their colonial bounty. However, as European powers ended colonial occupation that did not spell the end of colonialism. New colonial masters emerged from among the colonial elites to replace the old colonial bureaucracy, mostly indigenous in Africa and almost exclusively descendants of former colonists in South and Middle America. These new masters, however, continued to depend on foreign support to exercise their rule. From the perspective of someone growing up in such newly created nations and states, the political, social and economic structures continued to reflect something put in place during colonial times. Aside from that, life in most former colonies reflected the religious and cultural traditions of peoples who had deep roots in those places, dating back to long before the arrival of any colonial power. Even with the devastation brought about by colonial rule, where indigenous people comprised a large majority of the population, they retained their cultural identities, at least for the most. However, that was about to be challenged.

With new technologies connecting much of the world to a virtual world largely reflecting Western cultural views and values, the cultural influence of the West has seemingly become ubiquitous. Even in their prime time, colonial powers had little impact in shaping local cultures in their colonies and if they did so, it usually affected only a tiny sliver of the population. That has changed, first with mediums such as radio and television and then of course much more forcefully with the omnipresence of the internet, now accessible almost everywhere on our planet through terrestrial or satellite communication channels and affordable mobile phones. The cultural dominance of the virtual world by Western companies reflects their economic dominance of the real world. With that, many ways of life around the world, cultures unique and different from the ones advertised in Western media culture, came under threat. A people having to live under the political or economic dominance of others is sometimes an unfortunate reality but over time that can change again. However, the seeming cultural conquest of a people poses a very different threat with likely irreversible consequences. While we perceive many conflicts in today's world

often as motivated by political and economic grievances, we tend to overlook that we are really living in a time of intensifying cultural conflicts, which in the perception of some are actual culture wars.

All of us define ourselves to a good extent via the cultures into which we are born. None of us had a choice in which part of the world we first saw the light of day. To all of us, those surrounding us with love, caring for us and nurturing us, are precious in similar ways. The cultures and social structures we find around us as we grow up, in the cities or the rural areas of the country of our birth, are the result of historical developments in which we had no hand. Some are lucky and grow up in places where their ancestors could build a thriving society offering many opportunities. Others are less fortunate and grow up in impoverished and dysfunctional societies, which can provide little of that. When we watch the evening news, flickering across our television screens, bringing into our living rooms pictures of people we otherwise would not know anything about, the scenes we witness are often disturbing, broadcast from war-torn countries, impoverished places and places where people cannot enjoy the freedoms, we take for granted. Seemingly, all too often only headline news is newsworthy anymore. News must show something that captures our attention and unfortunately, much of that attention grabbing news is bad news. To many, that makes it look like as if pretty much everything is going wrong in our world today. Is that really so?

To answer that question, we should look back into our own family histories. How did our ancestors view the world before the time of radio and television? Clearly, bad things happened all the time and while our respective grand-grandparents may have been enjoying a quiet evening in their family circles, colonialism was still thriving in far-away places. However, they knew little or nothing about how bad things really were for the indigenous people in those colonies. At the same time, they still may have had a generational memory of how much their own ancestors had to struggle in earlier times. In a way, they likely considered hardships as a part of life just as their recent ancestors had still experienced them. In what we now call the Western World, systems of servitude lasted into early modernity and laborers in the beginning industrialization could tell their own stories about exploitation, not to speak about some regimes treating their citizens not much better than how colonial masters treated indigenous people in the territories they had stolen from them. A few generations ago, in what we call today the Western World, many may have conjectured their life experiences to be not so different from those in other places around the globe. First, because they knew nothing about those places where people's life experience certainly must have been worse. Second, in a true sense, the conditions most families around the globe faced back then were more similar than they are today.

For most of us today this is different. To those living in one of the media rich affluent societies it may feel as if we they are being inundated with images showing them just

how bad compared to themselves many around the world have it. Because we cannot be continuously outraged about what happens in the world, we have become largely insensitive to the suffering to which twenty-four hours a day, seven days a week media coverage exposes us. Clearly, we must know about those things. However, has anyone ever thought about what happens to the human psyche in the process? How do people cope with being exposed to a continuous news stream picturing how bad things are everywhere around the globe as well as closer to home without having the ability to change anything really for the better? Just what could be the result of that? The human psyche is as frail as the human body and psychological damage can be as bad as bodily damage. People instinctively protect themselves from bodily harm. Why would anyone believe that people react differently in response to threats to their psyche? Is it any wonder that people tune out of such realities and escape into their own consensus realities where they do not have to confront any such things?

Things are certainly not as bad everywhere on our planet as one may think based on the evening news. When we look at what really matters in people's lives, a large part of our world's population lives not so much different from us. That this can go along with vast disparities in wealth is something that we can observe in rich countries just as much as in less affluent societies. Material wealth is not a necessary condition for being able to lead a happy live grounded in a sense of belonging, purpose and meaning; material security, the reliable access to the basic necessities of life however is. It would be a good thing, if our modern media reported more on the reality of people around the globe, peoples from different cultures and nations, how they manage to cope in that respect. Too few of us really understand what life really feels like for many families here at home and out there in all the foreign countries of which we mostly know very little or anything at all. Few of us can relate to what it really means for such families not to have sufficient food on the table to feed their children, not to be able to educate them well, or not being able to care for the sick in their family. Just imagine what it would be like to experience that yourself. What would it feel like if all those things, which are just as important for our families and our communities, all of a sudden would become the cause of continued and endless anguish? What if we could not take their assured availability and affordability for granted anymore? Is it not strangely perverse that even though we live in an interconnected world, where with the push of a button we can see what happens anywhere on our planet, we know so little about how other people really live? Like about how families live in the less fortunate parts of our own societies or how life is for families in foreign countries where usually all we hear about is headline news; the latter telling us little about the lives of ordinary people in such places.

As many have likely experienced themselves, family can have its very own challenges, regardless of which place on our globe a family calls home. Only if we can understand how their families live and what challenges they face, can we hope to understand the

challenges that children growing up in those families face. In turn, to understand how families view those challenges and how they try to cope with them will require us to understand their societies and cultures much better than we do today. Unfortunately, too many adults among us not only carry the burden of ignorance, but equally dangerous, also that of prejudices we acquired growing up in societies, which are full of them. Many of us are afflicted more or less by the one or the other, often by both, and nobody is really completely free of either. We cannot change the past, how we grew up, the educational opportunities we had or did not have and the prejudices we internalized, immersed in the communal consensus realities we became a part of, but we can change how we deal with that. Acknowledging this is a first step for all of us to change our behavior, even if that is difficult to do. Letting go of prejudices is much harder than making oneself knowledgeable about things of which one has been ignorant. Hence, we likely have to put our hopes with the next generations and seek to educate them in ways that avoid a similar outcome for them. We must begin to educate future generations not just with the objective to communicate and instill knowledge in an unbiased way. Equally important, we must focus on raising our children to have open minds. Today we live next to each other, but we really do not live with each other. This is something we cannot change quickly as that will require a multi-generational effort. Our goal must be to ensure that around the globe many more children than today can grow up in intact families, in material safety, can attend schools that teach them to become knowledgeable and useful members of society but also open the door for them to understand and connect to the realities of life in other parts of our world. Some countries may be closer to doing this already today than others are, but all are far from really achieving this goal.

Our very own consensus realities that we all live in are not something that we just make up. Rather, it is the way we grow up that primarily conditions what our future adult consensus realities will look like. Humans have no choice but to evolve their own conception of themselves and of the world around them. Should those living in rich countries be surprised that children growing up in broken families, impoverished and without opportunities, tend to develop consensus realities that feature societies tolerating such injustices not as something to endorse, but as something to despise, or maybe even as something they believe they have to fight in order to survive? Have you ever asked yourself how you would see the world today if you had grown up in any of those parts of our modern societies dominated by poverty, violence and crime? Imagine yourself as a teenager in that situation and imagine that just a few miles away, there exists a very different world, a world you know you likely will never be able to join.

We are not honest with ourselves, and we are not honest with those parts of our societies stuck with the equivalent of window-shopping for a better life with little chance of ever making the transition to such a better life. We lie to ourselves when we do not acknowledge that inadvertently we have accepted this situation as quasi-permanent. We may say

the right words and we may endorse and support societal efforts to better the situation of those who, for no fault of their own, were born into parts of our society where kids grow up without the chance to develop their humanity in ways similar to how we demand this for our own children. However, we know that those efforts are mere bandages and while they do help a few, they do not solve the larger problem as they are but a drop in the ocean. How can our modern societies permit such a thing to continue for generation after generation? Is it not in the best interest of society as a whole that all children grow up constructing consensus realities that we could embrace instead of having to contain by the force of law the consensus realities we fear? We seemingly have become used to the latter. Does anyone really believe himself or herself better off because others are having it worse?

Similarly, looking beyond our own communities, what kind of consensus realities can one expect young people to develop who grow up in societies burdened by the heritage of colonialism but even more so by political systems perpetuating forms of economic colonialism to this day? To that, one needs to add the seeming dominance of Western cultural influences eroding cultural traditions in many places. Absent any other nucleus, individual and communal consensus realities will likely coalesce around whatever in their perception could be the last sanctuary providing a sense of belonging, purpose and meaning in an autochthonous cultural context. Should we really be surprised that some communal consensus realities in such places take advantage of those grievances to further their own interests? No, certainly not. Over the past decades, we saw the emergence of a number of Islamic terror groups, which seemingly sought to outdo themselves in indiscriminate killings and the butchery of countless innocents, children, women, men, young and old alike. The consensus realities of those terror groups brandish Islam in name but represent in fact egregious perversions of Islam. That is difficult for us to comprehend but we must acknowledge it as a fact and at the same time recognize that this is not something unique and has happened before.

Perversion of religion is not something that only afflicts Islam. Christianity for example had its own share of religious perversions in the past and even today is not really completely free of them. One just has to go back to the time of the crusades and look at what Christian priests and monks were preaching their flocks, enticing them to murder those they considered infidels, not much different from how some religious scholars and mullahs do today in the name of Islam. The crusaders and the religious leaders who enabled them were in a sense the eleventh- to thirteenth-century equivalents of the Islamic terror groups we see today. To the egregious perversions of Christianity certainly also belong its justifications of colonialism and of the enslavement of indigenous people as well as the terror of the Catholic Inquisition bringing torture and death to countless Christians and non-Christians. Perversions of Christianity were certainly not the preserve of Catholicism as Protestant Churches proofed to be no less capable of it. Just think about the

interpretation Christian slave owners and protestant priests in the American South read into the Bible to justify slavery.

Those who adhere to extreme ideologies will always find ways to justify them, in Islam as well as in Christianity. In doing so they take advantage of the basic need we all experience, our fundamental human desire for belonging, purpose and meaning. Only by helping people to find alternative ways to attain a sense of belonging, purpose and meaning, that is other than by endorsing extreme ideologies, can we create a better future for all of us. We should take inspiration from the fact that peaceful change is possible, and violence is not a preordained outcome in the struggle to find belonging, purpose and meaning for those among us who feel that they are being denied these things. For that we just have to look at the change that a Mahatma Ghandi (1869 – 1948) brought about for his people by opposing British colonial power with nothing else but peaceful means. Another example is how a Nelson Mandela (1918 – 2013) defeated apartheid in South Africa, liberating its colored people from decades of racial oppression by the white heirs of this former British colony. An example closer to home is of course how a Martin Luther King Jr. (1929 - 1968) led the civil rights movement in the United States to peacefully bring about equality before the law for all American citizens regardless of their color of skin.

As important as family and religion have been in the history of human civilizations to provide a sense of belonging, purpose and meaning to individuals, groups and societies, they also often narrowly prescribed what an individual could aspire to become in her or his life. In a way, the emancipation of the individual from too constricting religious and family bonds opened up not only new horizons for the individual but also for society. The freedom of the modern individual and the associated basic human rights each one of us takes for granted are likely something that none of us would want to trade for a return to lives dominated by the demands of family and religion. However, that is not the same as to negate that family and religion continue to be important parts in our lives, albeit in very different roles than just a few generations ago. Why should we not expect the concepts of family and religion to evolve just as everything else continues to evolve that is part of human culture evolution? Certainly, with respect to family, we are seeing new developments, which reshape the traditional understanding of what a family ought to look like. As discussed above, the understanding of family has already changed quite dramatically from what it was before the industrial revolution. The dissolution of what was for centuries considered the traditional family structure and the increasing anonymity of the individual in our modern societies could well refocus many to explore new forms of belonging, purpose and meaning that could reshape the very concept of family still much enshrined in our societies legal systems and of course in all religions. Humans, even in this age of individuality, remain social animals and we can be certain that over time we will see many more unusual social structures emerge that in essence will

be efforts to recreate family in new ways. What those will look like remains to be seen. For now, we only can look back at a few decades of experimentation, which began in the second half of the last century ranging from new communal forms of living together to the emergence of family relationships not based anymore on the traditional gender model.

Our understanding of family keeps evolving, a reality to which our societies continue to adapt. Different from that, religion still looks pretty much the same and for most belief systems, religious dogma is still beholden to the family that once was but nowadays has become a rarity. Family in the religious sense was always an ideal and that was of course a paternalistic ideal as religious dogmas are without exception the products of men, and mostly elderly men for that. The Christian churches of the Western World continue to lose members and their congregations are becoming older. That said, they still count their members in the millions but most of those members are Christians in name only. If churches were less obsessed with the next life and would focus more on what matters in the here and now, maybe young people would become more interested again. To a good extent, Christian churches in the West are in the positions they are in today because they have aligned themselves for much too long with those in power. Until quite recently, organized religion in the West was for the most an instrument in the service of those in power, providing them an invaluable tool to maintain their rule despite the obvious injustices it often involved. Within Christianity, organized religion was a governing instrument to hold the masses in check until it was not anymore. While the nominal separation of church and state is now common throughout the Western World, this legacy of the church being a part of the power structure still lingers on today and in some places, that power is still quite real though not official. If we look to Muslim countries, there religion and state were one and the same from the beginning and in many places that continues to be the default status. It may sound surreal, but theocracy in the twentieth century is still a reality. Aside of the role that beliefs played in various cultures in supporting worldly powers, the scriptures of all religions convey moral and ethical messages which if ever realized could likely make human societies more just and caring than we find that in history. However, for that to ever happen, we all would have to become much more honest about our consensus realities, who we believe we are, what we believe we are doing, and why we are doing it. How we explain this to ourselves through our consensus realities and how this compares with the reality of the lives we live is never easy to confront. However, unless we do that more often and with more sincerity, we will continue to live in the illusion that we are better persons than we really are.

Can We Learn from History?

Those who cannot remember the past are condemned to repeat it. This famous quote from the Spanish-American philosopher George Santayana (1863 – 1952) may be familiar to many but few around the world seem to heed its advice. However, it is not just remembering the past, which is important. What really matters is how we remember it. Does that which we recall as history reflect known true historical facts or are we remembering some whitewashed history? Throughout history, we frequently can find those emerging victorious from religious, economic, social or political conflicts writing the history books. Occasionally that colored the perception of historical accounts by subsequent generations, more so in the distant past than in recent memory. Victors writing and perpetuating their subjectively colored accounts of history is one pitfall to be aware of when studying history, but it is certainly not the only one as things are considerably more complicated than that. Invariably, the vanquished, as long as they survive as societies in control of their future destinies, find countless means to distort historical accounts to fit their own subjective narratives. Victors or vanquished, whoever wrote history books in the past, the historical narrative was likely biased and often significantly so. Thankfully, the historical sciences have come a long way over the past centuries and nowadays most professional historians in our academic institutions will distance themselves from anything that may seem less than fact based historical narratives. However, that does not mean all of them are in that camp now as one can see when perusing through some of the history books that we ask our children to study in high school or how we teach history to our children in the first place.

In many countries, politicians directing civil service agencies are still the ones who usually have the final say as to which history books schools will use. In some countries, notably the United States, parents have a much greater say what books their children will study in school. However, that is not to say that this will lead to a better selection than if educational professionals make those choices. Examples for that include of course how religious views and racial prejudice continue to influence what teachers still present to far too many American schoolchildren as science, particularly with respect to evolution and racism, the latter of course related to the teaching of history in general. However, before parents sitting on school boards or some religiously biased or racially prejudiced government official can select any history schoolbooks endorsing such travesties as creationism or distorted accounts of slavery, racism or the genocide of the indigenous people, someone has to write them. Someone has to be willing to write for example a textbook where religious belief supplants science facts or a history book which distorts the past by how it tells it or by what it leaves out, both of which is often still the case today. There, it quickly becomes clear that those historical narratives serve also other interests, other than to teach a factually based history to our children and help them to learn how to

critically read and evaluate historical facts and what science can teach us.

I still do have many good memories of my high school history classes back in 1970's Austria. Some local trivia and anecdotal history were already part of the curriculum in the four-year elementary school I attended but real history courses only began in high school. Once entering high school, usually at age ten, history courses would begin with pre-history in the first year and then work their way towards modernity by the end of the fourth high school year. Then another more in-depth four-year course would restart with ancient history in the fifth year and finish with the modern area in the eighth high school year by the time one graduated. Given the relevance of modern history for anyone growing up in post-war Austria, specifically that of the twentieth century, I always wondered why my high school history course never really managed to cover Austria's twentieth-century history other than in a very cursory way when compared to how diligently we had to study its earlier history. At the time, it was somewhat evident to my classmates and myself that Austria's history rather seemed to end just before the First World War. It covered some of what led to the outbreak of the war, maybe a little about the war itself, but anything occurring after the end of the First World War was not really something that merited a lot of attention in our history classes.

While not really discussed much in class, our high school history books did however cover this period so we could read by ourselves what they had to say about this time in world history and about Austria's history specifically. Those books by the way came free of charge, paid for by the Austrian government as where all other schoolbooks in what was then deemed to be a socialist country, a view of Austria which at the time likely also many in the United States shared.[16] The history books we were given did cover Austria's history after the First World War and right up into the 1970's. There were not any discernible wrong factual statements in our history books about Austria's recent past from 1914 onward but just by emphasizing some parts more and largely omitting others or making them mere footnotes of Austria's history, one could never have guessed what really happened in the lives of people living through those times. Clearly, these history books were quite sensitive with respect to how they retold certain aspects of Austria's history. Back in 1970's Austria, it was still not possible to discuss the more difficult parts of this country's twentieth-century history with the frankness one would have expected from a socialist leaning government, at least not publicly and certainly not in its schools.[17]

In the 1970, many were still alive who had experienced or suffered Austrian history first hand between 1914 and 1945, a stretch of time that seemingly many in 1970's Austria rather preferred to ignore than to acknowledge and remember. Before the German Reich annexed Austria in 1938, Austrians also experienced a civil war that began in the early 1930's, pitting socialists and communists against fascists and Nazi sympathizers. The result of it was the establishment of a clerical-fascist Austrian government; in a sense a tamer form of Italian Fascism in a catholic disguise, ruling Austria for several years

before the majority of Austrians embraced being swallowed by the German Reich.

The problems with teaching modern history in the Austria I grew up were not limited to the twentieth-century period when Austria was part of Nazi Germany. It all really began with the challenge of teaching the part of history, which led up to the First World War. Specifically the role that Austria played in plunging Europe into this abyss which not only destroyed the Habsburg Empire, whose heritage still looms large in today's Austria, but also set the stage for a Europe to emerge which as we know today was destined for an even greater catastrophe. It is now more than one hundred years ago since Europe plunged into the First World War and reemerged a very much changed and much diminished world. Over the last several decades, historians have come to a more balanced assessment, perspectives on history not consciously or subconsciously informed anymore by any of the particular forms of nationalism, which dragged Europe and much of the rest of the world into the abyss of two world wars in the first place, but by factual assessments. As a result, historians can give us a much better understanding today of just how European elites could have possibly endorsed a war so enthusiastically that would bring nothing but destruction to victors and vanquished alike. However, that is hindsight. None of these elites in their respective countries likely ever assumed that it would come to what then actually happened over the next four years. They all deceived themselves into believing certain things and not believing others and their consensus realities resembled each other in a number of ways. In fact, all of the would-be combatants from their particular nationalistic perches perceived their opponents with similar disdain and hatred. They were amazingly similar in how their consensus realities misjudged European reality but that would change quickly as the armies of Europe began marching in August 1914 and these consensus realities birthed a new European reality.

With respect to Austria, the role that its military leaders and diplomats played in the run-up to the war is difficult to portray as anything else but one grandiose communal self-deception. The decisions they made would become the yarn of the very rope with which they figuratively speaking strangulated the Austro-Hungarian Empire. There would be no more emperor and no more aristocracy in post-war Austria, the First World War would close the books on a way of life that had prevailed for hundreds of years. In essence, by excluding any alternative to a war of which it was clear that it would not be a limited war but a European-wide war, Austria-Hungary had chosen to commit suicide and the Kaiser's German Reich was more than willing to lend a helping hand in the process. Who was responsible for the outbreak of the First World War? A seemingly simple question to which today's historians have come to give much more nuanced answers than what one could still find in many twentieth-century history schoolbooks.

As to what caused the war, the assassination of Archduke Franz Ferdinand (1863 – 1914), the heir to the throne of Austria-Hungary, loomed large in Austria's history schoolbooks of the 1970's.[18] However, the assassin was not the one who precipitated the war but

his deed gave a pretext for war to those who had been thirsting for war all along, in Austria-Hungary, in Germany, in France, in Russia and of course, in the Balkans, which had been a tinderbox for quite some time before the First World War. The situation was somewhat different in Great Britain, where fewer may have expected to be drawn into a European war, but jingoism eventually also prevailed there. As it did in Italy, which joined the war on the side of the Allies in 1915 when it could be surer to choose the winning side. When the assassin killed the heir to the Austro-Hungarian throne in Sarajevo, he figuratively sprung a trap, which all those countries had played a role in setting up and in which they all found themselves caught up eventually. That does of course not absolve anyone of those who on all sides made or furthered the decisions that then led to the actual outbreak of the First World War. If the leading Austrian generals for example would have been less incompetent, they may have well told their emperor and his government that Austria was in no way prepared for a war, much less so to win a war, and that courting war could easily lead to outright disaster as indeed it eventually did. They did none of that.

So how does one teach high school children that their ancestors were either imbeciles or incompetent warmongers whose self-delusion was part of the reason why the First World War happened in the first place? It looks like one can only do that if one at the same time also educates them more broadly, for example about how people can deceive themselves to such an extent. In a way, the elites within each of the countries plunging into the abyss of the First World War did not behave so much different from the French aristocrats just before the French Revolution was to swallow them. If a personal consensus reality deviates too much from reality then we can expect some trouble ahead for this person but nothing that would more broadly affect society around her or him; that is unless he or she happens to be a person of exceptional power, such as a dictator. If however, entire communal consensus realities diverge from reality such as happened with the consensus reality shared by French aristocrats in pre-revolutionary France or with the consensus realities within the various European elites of each of the would-be combatants of the First World War, then disaster looms.

We all have learned that history is nothing like the result of prudent planning and certainly not a sequence of rulers and battles won and lost. What we have not really learned yet is how to make sense of our history in cases where it seemingly does not make sense at all. Sometimes, the only way things can make sense may come through appreciating that while there is a factual reality, we all have our individual versions of reality in addition to sharing in communal versions of it. Just as we make sense of us in this world through our individual consensus realities, we can share in communal consensus realities, each portraying its particular version of reality, to make sense of our larger societal environment. The degree to which consensus realities diverge from reality does not matter when one writes a fiction or a science fiction novel, but it does matter in how we deal

with reality itself. We can deviate in our perceptions of reality but we must be honest to which degree our personal consensus reality or the communal consensus realities we participate in deviate from the reality as it factually is. If we neglect to do that, we are in danger of losing touch with reality, in the same way as this happened to French aristocrats in pre-revolutionary France and with much of the European elite before the First Word War.

Being honest is usually much easier in hindsight but frequently there is no hindsight at all. That is because once ensnared by one of those seemingly innocuous communal consensus realities, which are anything but that, their venom can poison our personal consensus realities to the extent that the one becomes indistinguishable from the other. It is at that juncture that anything that comes to an individual from such a communal consensus reality, regardless of however vile and debasing, easily can assimilate into one's personal consensus reality. This is the kind of trap where once we fall into it, we find nothing wrong with believing that other people are bad because they look different or speak a different language. Very few at the eve of the First World War found anything wrong with how newspapers in the respective capitals of the combatants to be, portrayed what would become enemy nations and their respective citizens. There were of course the few who could see what was to come and what the debased language and the dehumanizing caricatures of perceived enemies heralded, which in the run-up to the war began to dominate newspaper front pages. Karl Kraus (1874 – 1936) in Vienna was one of them. How we speak about things and the kind of language we use, says as much about us as it tells others about what we say. As the old saying goes, if you point a finger at someone, four fingers point back at you. What is true for how we use our fingers is true for how we use language. Karl Kraus certainly understood that. However, he was practically a lonely voice in a Vienna where the mob in unity with the majority of burghers was screaming for war on the streets. The situation was similar in most of the other capitals of Europe, maybe except for Britain. The reason for the latter may have been that Britons and most of its parliamentarians had not yet understood that they also would be ensnared into the trap because the agreement between Britain's foreign office and its French counterpart which would obligate Britain to enter a European war alongside France was kept a secret from almost anyone but a few.

Unless a means is at hand to counteract consensus realities running amok, as they did ahead of the First World War, there seems to be really no way to prevent such an outcome. Before the First World War, most in Europe had deemed a major European war an impossibility because of the close economic ties that had grown between the European powers. What many may not realize, to those with means, much of Europe before the First World War was practically without borders. They could enjoy a Europe that only would reemerge again with similar liberties - but then for all its citizens and mostly regardless of their means – towards the end of the last century when most European

countries had joined the European Union. The institutions Europe possessed before the First World War were national only and there was no supranational mechanism, which could have prevented the European powers from plunging into war. Quite to the contrary, the alliances set up in the decades before the First World War, some of them proclaimed publicly others signed in secrecy, would ensure that a future major European war would be almost a certainty.

As if the tragedy of the First World War, purportedly the war to end all wars, would not have been devastating enough for all combatants involved, the peace treaties that followed very much increased the chances that the First World War would not be the last major war in Europe.[19] Not only would the failure to secure a lasting peace drag Europe and the world into an even more devastating cataclysm only twenty years later, it also laid the foundation for conflicts which continue to plague our world today. Some may be tempted to blame the victorious Allies for being so shortsighted to impose vengeful peace treaties on the vanquished European Central Powers and thereby sowing the seeds for a next European war.[20] However, that is a mistaken perception. Had the Central Powers won the war we can be quite confident that they likely would not have acted more prudently. For an indication of that, one just has to look at the peace treaty of Brest-Litovsk imposed by the Central Powers less than a year before their own defeat on a Russia, which the new communist rulers had taken out of the war.[21]

We should acknowledge that after four years of carnage, the likes of which the world had never seen before and that few of us today could even imagine, neither Allies nor Central Powers were ready, in case they prevailed, to offer anything but vengeful peace treaties to the defeated side. Someone had to take the blame for what happened. Every government, among victors and vanquished, was in the unenviable position to explain to its population why so many of its citizen had to die for nothing. None of those governments and elites in the respective countries would take the blame for that. The victors would force the vanquished to sign peace treaties that would put all the blame for causing the war on them and the vanquished without having a similar option would look for scapegoats they could blame for losing the war. The First World War, how Europe slid into it, how its elites not just failed to prevent it but actually instigated it, and how some of those same elites failed to secure a lasting peace, is a history lesson that every pupil in European schools should be guided to explore. Maybe, if that could happen there would be fewer ill-informed European adults lusting with false national pride, who then become easy prey to right-wing nationalist parties which again openly or clandestinely endorse ideologies reeking of Nazism and Fascism.

As may have become clear with the example of the First World War, to teach history in a comprehensive way requires a certain distance. Teaching the history surrounding the First World War right after it happened was certainly a different challenge than teaching it some hundred years later. It looks like as if we need a generational distance before we

can tell our children about what our grandparents might have witnessed or in what they could have been more or less active participants. However, that in no way should imply that we ought to ignore this more recent history until our grandparent's generation has passed. Instead, we must discuss such more recent history with our children in ways that acknowledges the fact that those who experienced and shaped this history are still with us. That is not so much a challenge as it is an opportunity as those are still around who can give us first-hand accounts, which we still can listen to, and probe as required to understand what really happened.

Such probing in essence means to understand where personal and communal consensus realities of those involved and witnessing what happened differ from each other. History is not just a collection of documents, which tell us who did what, where and maybe why. History is always first experienced history. Instead of not teaching our children some of the more delicate history topics our societies may still be uneasy about, that is until the generation experiencing that history has passed, we should critically collect as much of that experienced history from those who can still share it with us. That did not happen in Austria after the First World War. Then the interwar years and the subsequent role Austrians played in Nazi Germany added even more challenging topics to any presumptive history curriculum for teaching Austria's schoolchildren about their country's recent past. The net of it was that for all practical purposes in 1970's Austrian schools world history and specifically Austrian history usually ended just before the First World War.

From what it looked like to a teenager back then, Austrian society suffered from some kind of amnesia, unwilling to confront its own past. This past often must have been personal as grandparents or parents in many families had played a more or less active part in this history. However, this amnesia was often also very selective, as many had no problem to remember some aspects of that history quite fondly. That could be stories about what a great time some people had as members of one of the Nazi youth organizations. Others would recall the close bonds that kept them alive during wartime and others would recall how they or some of their family survived being bombed out. However, what one almost never heard about was what happened to the Jewish population of Austria. Their Christian neighbors just seemed to have forgotten that their former Jewish neighbors ever lived next door. One would be more likely to hear about what happened to many political opponents of the Nazi regime than what happened to Austria's Jewish population. In such instances, it was no real surprise to see these political opponents of the Nazi regime, many of them murdered by the Nazis, often underhandedly still accused of being traitors. More visibly, veteran associations marched every year on various occasions, proudly carrying symbols that should not have been part anymore of post-war Austria. But not all people were like that. There were of course exceptions, even though those were few. One of them I still do remember to this day.

Shortly before the start of my final year in high school, for reasons unknown, my right eye went almost completely blind for some time, and I temporarily lost a good part of my left eyesight. Because of that, I spent some time in a hospital in Salzburg. As it happened, for a week or so the person in the bed next to me, a man in his seventies, suffered from nightmares in addition to what caused his hospitalization in the first place. After a couple of days, I asked him about what he was dreaming. He did not respond immediately but eventually he did share his story over the next days. His dreams where about the more than two years that he spent as a simple solider in the German army fighting in Russia and what he had experienced and seen there. What I heard from this man was very different from the celebratory speeches about the bravery and distinction of soldiers fighting in this war, which one usually would hear from politician pandering to war veterans, like in memorial speeches given by various public representatives at war veteran meetings honoring fallen comrades. This man, some thirty-five years after the end of the war was still experiencing this war almost every night. The way he remembered the war and the stories he told were not something that one could have found in history books back then. History books usually give accounts of war casualties on either side but those are dead letters and frequently little more than an accounting exercise. Behind every war casualty there is a unique live lost, a family that will never embrace their loved one again. Many of those war casualties are not soldiers but civilians. There maybe was never a time of chivalry where armed conflict was seemingly limited to those clearly identified as combatants. However, the staggering number of civilians of all ages, from the very young to the very old, murdered in each of our modern wars far exceeds casualties among the uniformed combatants. When will we finally make ourselves honest and hold those responsible for such war crimes accountable?

How does one live with the guilt of being even partially responsible for what happened during the Nazi years, or with the shame of having a relative who was part and parcel of what happened during those years. To some extent, all who were of the right age when the German Reich annexed Austria in 1938 had to ask themselves this question. Anyone who was part of the Nazi military or political machine, to which the Nazi youth organizations certainly belonged, somehow had to make sense of that past. One could naively assume that everyone involved with the Nazi party and its various extensions into broader society had to deal in some way with this part of their past and their respective roles during the years of Nazi rule. However, very little of this kind of soul searching occurred in post-war Austria. Rather than to confront their personal and their country's role in what had happened, Austrians compartmentalized their experiences and integrated them into individual and societal consensus realities, such that Austrians neither individually nor communally had to take responsibility for any of it. Austrian society and Austrians essentially chose not to confront this difficult past in a serious way as long as too many of those who actually had to live through it, were still alive. Rather

than to acknowledge that they were co-perpetrators, Austria and Austrians chose to see themselves as the first victims of the German Reich.

Sigmund Freud, who with his family had to leave his beloved Vienna in 1938 to escape the Nazis, would likely not have been surprised by how Austrians dealt individually with their Nazi past and how Austria did so as a country. Resistance, projection and transference are the mechanisms through which he believed we manage to live with who we are as they help to keep that which we cannot confront as part of ourselves hidden from us or allow us to blame others for what we did or what happened to us. From my experience as a teenager in 1970's Austria, this is pretty much what one could observe. Clearly, there was massive resistance to acknowledge any responsibility whatsoever, even by those who had been ardent Nazi party members. Repression was helpful to the guilty as well as the innocent. It enabled those who were guilty to lead seemingly normal lives as respectable citizens and it also helped those who were not guilty of anything at all but still had to confront living among those who were guilty. Then there was of course transference, which one could observe in different variations. In its most perverse form, such transference resulted in people not blaming the Nazi perpetrators but their victims for what happened. That may sound ridiculous, but people doing that would do so with utmost righteousness, not blushing in the least. These same mechanisms also worked for Austria as a nation. By pretending that nothing really had happened, that is through resistance, Austrian courts could pretend that there was nothing meriting prosecution. Communal and societal repression was pervasive in public live as it continued after the war, helping the country to ignore the fact that many Austrians had been fervent Nazis and that a good number of war criminals lived among them. As for national transference, Austrians would conveniently point towards Germany as the guilty party whereas they portrayed their own country as Nazi Germany's first victim. To some, applying Sigmund Freud's insights into how individuals deal with their mental problems to how societies deal with theirs may seem a stretch. However, from the perspective of someone who could observe these mechanisms at work in how Austrians and Austria dealt with their difficult past they made a lot of sense.

From what it looked like to my friends and to myself, Austrians had conveniently managed to repress the March 1938 images of hundreds of thousands of Austrians cheering the Nazis as they annexed Austria. Just as they forgot about the half million Austrians hailing the Nazi leader as their savior in Vienna's Heldenplatz, the large public space in central Vienna in front of the former imperial palace of the Habsburgs. Austrians had a similar difficult time to remember what they began to do to their Jewish fellow-Austrians the very same day the Nazi's marched into Austria. Austrian Nazis were seemingly eager to show their German brethren that they were the more fervent Nazis as the brutality with which they treated for example members of the Viennese Jewish community even surpassed what German Nazis had expected. Only once the generation that still could

remember those times through their own experiences had essentially passed away, could Austrians seemingly begin to face the truth. However, by then a new generation had grown up, raised by those denying an unwelcome historical truth. Hence, when in the mid-1980's a former Austrian secretary general of the United Nations stood for election to become the president of Austria he won hands-down even though there were allegation that he might have participated in war crimes.[22] By that time, being the victim of the German Reich had already become part of Austrian societal consensus reality. With that, not just right-wing politicians but people from various political backgrounds rallied around the controversial presidential candidate. This was the result of Austria having until then never honestly dealt with its Nazi past. Only in the 1990's would this begin to change.

German society had a similar problem, but Germany could not claim to be a victim and hence could not put forward any such excuses as Austria did. Because of that Germany had to deal with its past a little more honest. Still, on an individual level, Germans struggled in a similar way to acknowledge what had happened. As a result, German society evolved its own contrived consensus reality. Mostly people would claim they knew nothing about what had happened during the Nazi years, which frankly was hard to believe. Germany, even though reluctantly, eventually ended up prosecuting a relatively larger number of war criminals than Austria, which did little in this respect. These would become the convicted villains, which the rest of the population then could blame and thereby continue the pretense of having known nothing about the extermination of the Jews; or of any of the other atrocious war crimes committed in the name of Germany for that matter. The number of Nazis convicted for their crimes remained small. Given the scale of the war crimes committed by Nazi Germany, one must conclude that the vast majority of Nazi criminals never had to face justice.

How we deal with the unspeakable depravities humans have perpetrated throughout history speaks volumes about our individual and societal capabilities to construct consensus realities that shield us from having to face the truth. Some of that is part of conscious efforts in individual and communal denial and that certainly is what also played out in post-war Germany and Austria. Roughly thirty years after the end of the Second World War, communal and individual self-deceptions about what had happened were still very much flourishing. Anyone willing to pay attention could witness that. Back then in the 1970's, it looked to me as if there was still a clandestine agreement throughout Austrian society that what really happened during the Nazi years and the role Austrians played was not something to talk about in public or in private. What eventually initiated a public discourse on what happened during the war years was the broadcast of the Holocaust television series on public networks in Germany and Austria in 1979.[23] This television series came from the United States. At the time, it still would not have been possible to produce anything like it in Germany or Austria itself.

Without this push from the outside, a real public discourse in Germany or Austria about what had happened during the Nazi years may not have been possible for many more years. The broadcast of the Holocaust series led many to ask questions they had not dared to ask themselves or their family members before. Of course, there are always those who would outright deny that any of the atrocities committed by the Nazis ever really happened. Many of those who were subjected to Nazi propaganda may have internalized it to such an extent that even decades after the Nazis were defeated, the consensus realities of quite a few in post-war Germany or Austria would continue to endorse the worst kind of racism and antisemitism. Young people growing up during the Nazi years must have been particularly vulnerable to this kind of brain washing under the Nazi regime. A good number of those would pass on their distorted worldviews to their children and grandchildren. This may be one explanation of why anyone in Germany or Austria born many years after the end of the war would still endorse Nazi ideology. Of course, there are other reasons as to why young people growing up today endorse extremist worldviews that separate people into friends and foes based on their race, ethnicity or religious belief.

To appreciate how difficult it can be for a people to acknowledge its own past instead of seeking to whitewash it, we only have to look at the genocide of the Armenians and how Turkey, the successor state of the Ottoman Empire in whose name an estimated one million Armenians perished, continues to deny the Armenian genocide to this day. The relationship between Armenians and Turks had been a difficult one right from the time the Turks invaded Anatolia and eventually captured the city of Constantinople, the last holdout of the once mighty Byzantine Empire in 1453.[24] Before that, for several centuries in Byzantine history, Armenians had played important roles, particularly through their capable soldiers and generals. Not only had Armenians significantly bolstered Byzantium's military capabilities for several hundred years, they also had been a powerful force within the Byzantine Empire and a number of Byzantium's emperors during that time had been Armenians or had been of Armenian descent. With the Ottoman conquest of Byzantium, the prominent role Armenians had played in the military matters of the region was gone. Instead, they came to play important roles in the Ottoman economy. In that, the Armenian Christians, as other religious confessions in the tradition of Abraham, benefited from an Ottoman rule allowing the Armenian Church self-government in its own affairs and in handling personal legal matters of its members.[25]

Things only began to change with the beginning disintegration of the Ottoman Empire in the nineteenth century and the rise of nationalism in Turkey. Turkish nationalism was essentially a response to the rising nationalism among the different people the Ottomans had conquered over the centuries. With Western support or help from Russia, an increasing number of them gained national independence. Then in the Balkan War of 1912 -1913, the Balkan states pretty much drove the Ottoman Empire from continental

Europe, except for the small part of Thrace the Ottomans could hold onto which is still part of modern Turkey today. Under the ruling nationalists, the Ottoman Empire joined the First World War at the side of the Central Powers. In a war not going well for the Ottoman Empire, Turkish nationalists became increasingly concerned about Armenian nationalism and a possible secession of Armenians taking with them large parts of their homelands in Eastern Anatolia. This is when the Turkish nationalists opted for what people today euphemistically would call ethnic cleansing with the goal to replace the Armenian population of Eastern Anatolia with ethnic Turks. The Turks succeeded by outright killing about one million Armenians and forcefully converting many Armenians to Islam, making the Armenian genocide not just a physical genocide but a cultural genocide as well. To this day, Turkey officially denies this genocide and only admits Armenian suffering, which at the same time it relativizes by pointing at the suffering of its own people. Why is Turkey denying the Armenian genocide even after more than one hundred years have passed? No one is alive anymore who personally experienced it, on the side of the perpetrators or on the side of the victims. Why is it still not possible for Turkey to acknowledge this difficult part of its history?

Among the possible reasons towards which one could point, some stand out. For example, the fact that from an ethnic and cultural perspective Turkey is still very homogeneous. As a society, modern Turkey is much less diverse than for example modern Germany. Modern Germany has become a multicultural society with about a quarter of its population having an immigrant background. Turkey also has an immigrant population but a much smaller one even though in recent years a large number of refugees from regional conflicts joined it. Turkish society is far from being multicultural and remains constrained by a single culture supported by an authoritarian leaning government and a conservative religious establishment, both of which continue to deny the Armenian genocide. It is much easier to come to terms with a difficult past in a tolerant multicultural environment than it is to do so in what essentially still is an environment dominated by nationalism. If Germany and Austria had for example retained the cultural uniformity that characterized their populations before the Second World War, we can be almost certain that it would have been even more difficult for them to deal with their modern histories as that already was for them anyway.[26] The multicultural component of many modern societies is not just a natural antidote against virulent racism and nationalism seeking to reassert itself; it also enables us to gain eventually a clearer and less biased view of our own people's history.

Another factor that possibly could make it difficult for Turkey to deal with the history of the Armenian genocide is the fact that modern Armenia continues to be Turkey's neighbor. For modern Germany, the secure existence of the State of Israel has become an integral part of its identity. Different from that, Turkish nationalism continues to pose a threat for modern-day Armenia. On its eastern borders from the Turk population of its

neighbor Azerbaijan and on its western borders from Turkey proper which supports its Turk brethren in Azerbaijan. Over the decades, denying the reality of what happened during the Armenian genocide has become more difficult for Turkish officials. An increasing number of countries have come to recognize the Armenian genocide for what it was and publicly acknowledged what happened to the Armenians. Turkey's schoolbooks may continue to deny for many more years that the Armenian genocide ever happened but as its citizen travel the world, they will discover that Turkey's modern history was different from how their schoolbooks and teachers portrayed it.

In response to the genocides in recent modern history, specifically the Holocaust, the genocide of the European Jews committed by Nazi Germany, the United Nations General Assembly unanimously adopted the Convention on the Prevention and Punishment of the Crime of Genocide on December 9, 1948.[27] By the time this convention went into force in 1951, twenty-five nations had committed to it through signature, ratification or accession. Here is how Article II of this convention defines genocide:

> *In the present Convention, genocide means any of the following acts committed with intent to destroy, in whole or in part, a national, ethnical, racial or religious group, as such:*
>
> *(a) Killing members of the group.*
>
> *(b) Causing serious bodily or mental harm to members of the group.*
>
> *(c) Deliberately inflicting on the group conditions of life calculated to bring about its physical destruction in whole or in part.*
>
> *(d) Imposing measures intended to prevent births within the group.*
>
> *(e) Forcibly transferring children of one group to another group.*

Interestingly, the United States of America was not one of those nations adopting the convention when it came into force and would not ratify it until 1988. The United States of America was not alone in delaying this decision, joining only after decades of internal deliberations, mostly motivated by the fear that American soldiers and their commanders might become subject to international law. That of course meant that an international court could possibly indict American soldiers and their commanders for war crimes related to genocide. The United States government may have had good reasons to be concerned about this possibility becoming a reality if the United States would have joined the convention. One only has to consider the civilian death toll of many military operations that governments often seemingly take as unavoidable or just simply ignore.[28] That may not constitute genocide but it still is mass murder. There are of course also other reasons for a country like the United States to consider. One of them being that a number of countries, quite a few of them not really models for the rule

of law themselves and having less than stellar human rights records, may use such a convention of an international court of law for political purposes. While that is always a possibility, there are ways to safeguard against such behaviors. However, maybe it is for such reasons that the United States is not among the current one hundred and twenty-three members of the International Criminal Court (ICC). Founded in 2002, the ICC's charge is to hold perpetrators of war crimes, crimes against humanity and genocide accountable when national courts are unable or unwilling to do so.

As of mid-2019, of the one hundred and ninety-four member states of the United Nations, one hundred and fifty-two had joined the genocide convention while forty-two member states still had not done so. Not surprisingly, given the onerous human rights records of a number of regions, nineteen of the latter were from the continent of Africa, seventeen from Asia, and six from the Americas. The most prominent country to abstain from joining the genocide convention is the People's Republic of China. Again, this should be no surprise. The reign of terror unleashed on the peoples of China by the communist party under Mao Zedong (1893 – 1976) killed tens of millions of people. The modern-day communists who rule China today in a capitalist disguise have not much more respect for human rights than did their comrades decades ago, witness the cultural genocide the Chinese rulers are committing against the Uyghur minority in the autonomous province of Xinjiang.[29] Of course, China also does not recognize the ICC and probably not for much different reasons than the United States.

Among the major three powers, the United States, China and Russia, only Russia had signed the Rome Statute in 2000 that enabled the founding of the ICC in 2002. However, Russia never ratified the treaty for the ICC and in 2016 withdrew its founding signature when the ICC classified the conflict in Eastern Ukraine as an armed conflict between Russia and Ukraine. This conflict followed the annexation of the Crimean Peninsula by Russia in 2014, which was itself a breach of international law. Russia clearly instigated and supported the armed rebellion of Russians in two of the eastern provinces of Ukraine where Russians represent a large minority population. Russia essentially fomented a not so clandestine war with Ukraine, probably in the end to annex those provinces just as it annexed the Crimea, while trying to hold up the pretense that it was not doing anything like that. The ICC was calling Russia out for that. However, what can an international court of law really do when a nuclear superpower flexes its muscles?

The problem the international community faces is how to deal with such behavior by powers like Russia or China, whose governments do not even have to answer to their own peoples, unlike it is the case in the United States or other democracies. There are non-military ways for the international community to deal with dictatorial or autocratic regimes but there is a cost involved. Unfortunately, business interests in democratic countries tend to oppose such non-military measures against dictatorial or autocratic regimes as that may limit their ability to conduct business in those places. As long as monetary

interests continue to be more important to democratic countries than their values, dictatorial or autocratic regimes will have little incentive to stop violating international laws whenever it seems opportune to them.

Every year, December 9, the day the United Nation adopted the genocide convention in 1948, marks the International Day of Commemoration and Dignity of the Victims of the Crime of Genocide and of the Prevention of this Crime. Every year on that day, we should remember that the adoption of this genocide convention has not abolished genocide. In our modern interconnected world, it certainly has become much more difficult to commit genocide without the world seeing it. However, that does not mean that it has become much easier to hold those accountable who are responsible for it. The ICC is a good start to just do that but without the support of the major powers the ICC's job is more difficult and some may still believe they can get away with genocide as long as they are allied in some way with one of the great powers. However, that should not stop the ICC to do what it can do now with the support it has among many nations today. The ICC is already establishing legal precedent and, in the end, there is no alternative for the international community than to find some way to hold those accountable who otherwise would not have to face any kind of justice because either their governments protect them from prosecution or they are government actors themselves, believing that they can act with impunity. The ICC may not be the perfect tool yet to address such problems, but it is an important first step in the right direction towards a solution for a problem the international community has ignored far too long.

Failure to acknowledge or the outright denial of one's individual or societal past combined with efforts to whitewash history is not a phenomenon that only surfaced in the twentieth century. The reason that this section began with the amnesia many Austrians and Germans succumbed to after the Second World War with respect to their countries or their own specific roles in war atrocities committed in the name of Nazi-Germany is simple. I still witnessed this kind of mental atrophy when growing up myself in 1970's Austria. Some did not just suffer from a selective amnesia with regard to what had happened, they outright denied that any of it was true including the genocide of the European Jews. We can be certain that if the Nazis had won the war, their historians would have denied having committed any atrocities at all, including the murder of millions of Jews and of other people belonging to ethnic, religious or political groups, whom the Nazis in their perverse ideology deemed to be lesser humans.[30] They may just have admitted to reeducating those who could not see the wisdom of their government's actions, just as this is still happening today; and they would have broadcast their lies via state controlled media accordingly and of course written their history books to reflect those lies. Such practices are no less common today than they were back then, and they have been a common currency throughout history. To see that, we only need to look at a few examples from the past as well as some more recent ones.

The history of what we call today genocide reaches back many thousands of years. The indiscriminate slaughter of people just because they happen to differ in some way deemed unacceptable by those among whom they live, is not a modern invention and may have been part of the human evolutionary package from early on. What has changed since then is that there are many more of us today than there were back then in ancient or archaic times. Furthermore, modern humans have developed technologies, which they used in the twentieth century to commit genocide on an unprecedented scale. However, it is of course still possible to commit genocide with much simpler tools such as machetes, which were the tools of choice for murder in the 1994 Rwandan genocide. We could go back to ancient times and try to find examples for genocide. However, it is better to stick to examples that are more recent where we have good documentation. In addition, when looking into the history of genocide one quickly realizes that there are many more examples than one would hope to find and it is not possible in the limited space available between the covers of this book to give justice to all of them. Rather than looking into the details of specific examples the objective here is to turn our attention to the fact that in many cases the history we teach our children may still whitewash or relativize genocide for various reasons.

Our point of departure will be the conquest of the New World by the European powers, notably first by Spain and Portugal, to be followed later by The Netherlands, Great Britain and France. It was the Spanish and the Portuguese who were first to claim both the lands and the resources of the native people they encountered in the New World as their own, before claiming the natives themselves as their slave laborers. Most of us would look at a Christopher Columbus (1451 – 1506) or one of his fellow adventurers as the great explorers that they certainly were. The second Monday of every October the United States celebrates Columbus Day as one of its few federal holidays. However, the lands that Columbus and his fellow explorers discovered for Europeans were not empty of people. Those who celebrate Columbus Day today are mostly later immigrants, many of them originally from Europe and quite a few nowadays also from other places around the world. America's indigenous people surviving the conquest and genocide following the arrival of Europeans as well as African Americans whose ancestors were enslaved by Europeans, brought from Africa to the Americas to a life of slavery and servitude, nothing more than a piece of property for their owners, they all do not really celebrate Columbus Day. For them it is with people like Columbus that hundreds of years of persecution began, it is when the white man began to steal their lands, to kill their family members, to rape their women, to enslave their people.

The European explorers and the white colonists who followed would force their Christian beliefs on indigenous people, use those beliefs to justify their enslavement, and introduce a multitude of diseases that decimated native populations because their immune systems, different from that of Europeans, had never evolved a defense against such plagues.

What I remember from my high school history books are accounts of Columbus voyage and of the following expeditions by the conquistadors, infamous among them Hernán Cortés (1485 – 1547) and Francisco Pizarro (c. 1476 – 1541). The former leading a few hundred soldiers and the infectious diseases they were carrying, smallpox prominently among them, brought about the downfall of the Aztec Empire. A few hundred men even if equipped with their more modern weapons could never have accomplished this but against the germs the Spanish arrivals introduced to the New World, the Aztec population was defenseless. Similar to Cortés claiming the lands of the Aztecs for the Spanish crown, my 1970's history books showed Francisco Pizarro adding the lands of the Inka Empire to the growing Spanish overseas dominions. Cortés and Pizarro were just two of many conquistadors who from the sixteenth century on would subjugate and enslave much of the native populations of Middle- and South America.

The major challenge that the colonial powers faced in Middle and South America was a shortage of labor. There was just not sufficient incentive for the subjects of Spain or Portugal to move in large numbers to the newly establish colonies of the Spanish and Portuguese crowns. Hence, the colonial masters turned to exploiting and enslaving the native populations, which even though dying in large numbers under colonial rule still vastly outnumbered European colonists. This is how up until the late nineteenth century the colonial system largely continued to work in much of South- and Middle America. When those colonies finally declared independence, the ruling white colonist elites essentially cemented colonial rule in the newly created nations. Brazil went into a somewhat different direction as its colonial economy imported large numbers of African slaves. This was a consequence of the fact that the colonial masters in Brazil had a much more difficult time to enslave the indigenous population. The natives in Brazil had an advantage that for example the indigenous populations in Mexico or Peru mostly lacked. Brazil's indigenous people could retreat into an enormous and impenetrable jungle where they were not that easily captured by slave hunters. While nothing but a sketch, this outline provides already a lot more context to the last five hundred years of Middle- and South American history than what I could find in my 1970's history books. The real history of the European conquest of Middle and South America, of the several-hundred-year colonial rule that followed, and of the newly independent countries whose ruling elites, essentially the former masters, practically institutionalized former colonial rule, is much more complex.

However, the pertinent question is not really the level of detail one can find in school history books. Far more important is what kind of impression a history book leaves with respect to the roles the various participants played in particular parts of our shared history; and more importantly, how this is still reflected in our modern societies in the way various groups perceive themselves as well as others. A good example for that is Africa's colonial history and how this dark period still figures in those modern European

nations which once were almighty colonial powers in almost all of Africa. To date, none of the former European colonial powers has yet acknowledged the full extent of the atrocities committed in the names of their countries, nations or rulers during the era of colonialism, which began in the fifteenth century, reached its fever pitch in the late nineteenth and early twentieth century, and festered on until after the Second World War. With some justification, one could claim that colonialism did not even end with former colonies becoming independent nations or states, it just continued in different forms.

To be sure, European nations are not the only former colonial powers who for the most successfully continue to repress unwelcome memories of their colonial history. In the Far East for example, Japan finds itself in a similar situation with respect to its history as a colonial power. Other countries, which we usually do not view as former colonial powers because their colonial past was shorter and more confined, such as was the case with the United States occupation of the Philippines, suffer from colonial amnesia as well. There are also countries who became colonial powers within their own borders. Many will put again the United States into that category, but Russia colonized its Eastern provinces with no less fervor than the original thirteen colonies expanded into the West of what today comprises the United States. In both cases, the indigenous people suffered enormously, losing their lands and their freedom while being decimated by disease and warfare.

Here we will focus on just one particular example of colonial amnesia, which continues to this day. Few among my generation growing up in Belgium probably ever read in their history books anything which would associate their country with the killing of some ten million Africans. Quite to the contrary, Belgium's public portrayal of its colonial past was much more likely to instill pride rather than remorse in the citizenry visiting its colonial museums and in young Belgian's learning about their country's history in its schools. Most of us will associate the small country of Belgium with chocolates and Belgium fries, which for those not in the know are the real French fries; and many will know that its capital is the seat of European Union institutions as well as the headquarter of the North Atlantic Treaty Organization or NATO. Anyone wandering through the halls of the Royal Museum for Central Africa in Tervuren, about a half hours' drive from Brussels, will search in vain for exhibitions dedicated to the mass murder committed in the name of Leopold II (1835 - 1909), King of the Belgians, in what from 1885 on was his private colony for some twenty years, the État indépendant du Congo, also known then as the Independent State of the Congo or the Congo Free State.

From 1885 until 1906, Leopold II, next to being King of the Belgians, was not just the absolute ruler of the État indépendant du Congo, he was practically also the owner of all its lands and its resources. This he achieved by effectively expropriating all rightful landowners with sham treaties, the contents of which they did not understand but were forced to sign, nonetheless. One could say that he ran his colonial enterprise from behind

his desk in the royal palace in Brussels no different than any slave owner in the American South may have run his plantation before the abolition of slavery. Decades after practically all colonial powers had abolished slavery in their colonies and the United States had finally freed itself from slavery after a long and bloody civil war, slavery was by no means completely gone.[31] However, Leopold's colonial regime reintroduced slavery on a scale and with a brutality unmatched before. Leopold II and the concession companies he authorized made enormous profits from exploiting the Congo, first mostly with ivory and then with rubber. That was only possible because they paid in essence nothing for the ivory or rubber the État indépendant du Congo exported. They did not pay for the slave labor harvesting these goods throughout the vast interior of the Congo and neither did they pay for the slave labor transporting it onto their ships bringing the goods to overseas markets. Congo's entire infrastructure was built with and built on slave labor for which Leopold II and his business partners paid next to nothing. The instrument with which Leopold forced the indigenous population into slave labor was the Force Publique, his private army in the Congo. The foot soldiers of this Force Publique were themselves blacks pressed into service at gunpoint and subject to a brutal discipline under their white officer's command, forcing black soldiers to punish and kill their own comrades for even minor disciplinary lapses. The unspeakable atrocities, ordered and committed by many of those white officers put them in a similar league as those who did the dirty work in the Nazi or Soviet death camps. Leopold II was not someone to get his own hands dirty; he never even visited his private colony. He was planer, accountant, controller and profiteer of all that took place in the Congo and most of all he was an excellent communicator and obfuscator.

Even early on, some of what really happened in the Congo got out but for a long time Leopold II always managed to either discretely bribe or in other ways manipulate those who could have educated the broader public about it. Hidden behind the veil of Leopold's publicly heralded antislavery initiatives and his false claims of bringing education and civilization to the peoples of the Congo was a calculating and cruel individual whose sole interest was to enrich himself at whatever cost. If that brought about the extermination of what in the end would amount to about half of Congo's population living in those vast areas before Leopold II appropriated the Congo to himself, that did not matter. The King of the Belgians could hide what really was happening in the Congo from public view for a long time but not forever.

Leopold II could not have gotten his colonial enterprise off the ground without massive financial support from the Belgian parliament. In return, Belgium was to inherit Leopold's colony on his death. By 1906, however, the outrage about what was happening in Leopold's Congo forced him to turn over his colony to the Belgian state earlier, but not before he had extracted his pound of flesh from the Belgian taxpayers in return. Before handing over the colony to the Belgian state, he had the documents in his colonial

archives destroyed, because as he insisted, nobody had any right to know what happened in the Congo under his rule. After a transition period, the Belgian state became the new colonial master in 1908 of what was now called the Belgian Congo.

The Royal Museum for Central Africa in Tervuren was built using funds of Leopold's private domain of the Congo. Put more bluntly, the construction of this museum that was supposed to document the history of the Congo, its people and their cultures was paid with blood money stemming from the murder of some ten million Congolese. It was Albert I (1875 – 1934), King of the Belgians and the nephew of Leopold II, who inaugurated the museum in 1910. Initially called Museum of the Congo, then Museum of the Belgian Congo, then Royal Museum of the Belgian Congo, it eventually became the Royal Museum for Central Africa when the Congo gained its independence from Belgium in 1960. In 2018, the museum, now frequently referred to as the Africa Museum, reopened after an effort to address its racist and pro-colonial image. Sadly, that did not include addressing Belgium's colonial history in the Congo openly and honestly. What Belgians need is not a revamped exhibit of colonial items, most of which still originate from the Congo, and which continues in many ways to whitewash a colonial past by omission. What Belgians really need is the equivalent of an Auschwitz Museum that painfully details how the slaughter of some ten million people in the Congo happened and who was responsible for it. In the United States, cities are beginning to remove the statues of famous confederate generals from public places into museums as it finally dawns on people just what an offense it must be to those whose ancestors suffered under confederate slave owners to see those glorified and honored in public squares who defended slavery. In Brussels the statues of Leopold II, whose rule killed as many as ten million people in what was the État indépendant du Congo, still proudly tower in many public spaces. For how much longer?

Belgium's colonial history is one among many and it is not unique. The regimes of other colonial powers were not much more benign and often engaged in practices not so much different from what happened in the former État indépendant du Congo. Just as in the case with Belgium, all former colonial powers should ask themselves for how much longer they plan to present their children whitewashed histories about what happened in their country's colonial pasts. Often one can hear people argue that we should not talk about certain things in history, because they may portray one's own people in a bad way. What an argument! Should we lie to our children instead? That sure will make them proud of their own people once they find out the truth. Others may suggest that we better leave out certain things for good and just have the history books gloss over them. That is how seemingly quite a few prefer to address the problem of how to avoid having their children reflect on what they perceive as unpleasant parts of history. That is shortsighted and stupid. Our children should understand what it is that humans are capable off and that maybe not just some faraway people are capable of committing atrocious things to each

other. Our human history must reflect all aspects of what human nature is capable off. Our understanding of the past must be informed by what in actuality happened in the past and not by how we would like it to have played out. In our public squares, only the true heroes of our past deserve a place and those heroes must truly reflect the values around which we want to build our societies. If we continue to admire and praise the wrong heroes, in our public squares and in our history books, how does that not equate to us saying that we do not really believe in what we otherwise proclaim to be our true values, the very values we publicly endorse as the foundations of our societies? How can we have statues of people in public parks who may have done a daring and great thing, like winning a battle, but of whom we know with certainty that they held and acted on views deeming some people superior and others inferior be it because of their gender, skin color, ethnicity, culture or religion?

There are those arguing that before the era of colonialism, indigenous people were often engaging in many of the same practices in dealing with each other, which later colonial powers would use, including an active slave trade and widespread slavery.[32] The African continent was certainly no paradise for indigenous people before Europeans appropriated the continent for themselves. One can point to examples where indigenous people showed similar ruthlessness to each other as later Europeans would show to practically all of them. However, that is not the issue of interest here. We are all human and as such, we are all vulnerable, given the right circumstances, to engage in behaviors, which from the safe distance of our comfortable studies we vehemently oppose and reject. Hence, we should not be surprised that African people are just as capable as European people are of committing unspeakable atrocities. It just so happened that their command of science and technology allowed European colonial powers to equip their military forces with much superior firepower and with that enabled them to commit such crimes on a much larger scale.

The question we must consider is not who did what for what reason in history. That is something we can leave to professional historians. What really matters much more for all of us is how honest or dishonest our societies and we ourselves are in how we deal with our own communities, peoples, nations or states past because that is what will determine the kind of future, we and those coming after us can hope to have. We can learn from history to build a better future but only if we are committed to face the facts of our histories whether we like them or not. Sadly, we are still far from acknowledging this reality as we continue to hide in our individual and communal consensus realities, which all too often endorse historical narratives devoid of historical facts but make us feel good and proud of ourselves and of the communities to which we belong.

The Future We Make

We all think about the future, certainly about our own and about our family's future. Sometimes we may also think about the future of the community we belong to, or the future of the culture sustaining us, or of the country's future, whose citizens we are. That is probably how far not too long ago the concerns of most people used to go when thoughts about what the future may bring crossed their minds. Even those whose education has given them a different perspective on time usually do not concern themselves that much with potential futures as mostly their exploration of time is directed towards understanding our past. We have sciences dedicated to exploring the past such as all kinds of history as well as disciplines such as archaeology or anthropology, but we really do not have a science dedicated to exploring the future. Could we potentially put ourselves in a much better position to make informed decisions with respect to our future if we could explore the possibilities of the future in a similar way as we can now explore the certainties of the past? That question of course raises another one. Can there be a science of the future at all, which could help us in that respect? It is not possible to answer this question with a simple yes or no, but we certainly can explore what could possibly be parts of any such answer.

For that, we need to look into what we actually understand by what we call future. In addition, we must differentiate between our ability to predict certain limited aspects of a future and our general ability to make reasonable inferences from whatever information we have as to the probability of certain future outcomes in a wider sense. An example of the more limited predictions we can make about the future are the trajectories of bodies moving under known physical forces according to natural laws we know. In that category, the movements of the planets and moons in our solar system certainly come to mind, as we can predict their respective positions with high accuracy into the very distant future, always provided of course that nothing unexpected happens which could disturb their celestial orbits. Aside from what our science and engineering capabilities enable us to predict in narrowly defined circumstances, our abilities to predict the more general future of things of interest, like the future of humankind for example, is of a much more speculative and uncertain kind. Provided we can really explore possible futures at some point in ways we cannot imagine yet, we have to ask another question. As we saw in the previous section, even with all our efforts to explore the past it remains doubtful that we can really learn from history. That begs of course a similar question with respect to our future. If we could learn anything from knowing more about our possible futures, would that kind of learning translate eventually into humans embarking on their potential futures in more considerate ways than we do so today?

There are academic disciplines that concern themselves with the study of possible futures. We typically refer to them as futures studies or futures research and those who engage in

these efforts some may call futurologists or futurists. There exists no science of the future comparable to our natural sciences, but our natural sciences certainly contribute of course to certain aspects of futures studies, just as social sciences do. Broadly speaking, the study and research of possible futures is by necessity an interdisciplinary field to which different sciences contribute. At the same time, a good number of what sails under the flag of futures studies draws on non-scientific sources in ascertaining possible futures. That is not surprising because not only scientists are interested in possible futures. Religious scholars, philosophers, utopians and dystopians, politicians or captains of industry and many others have their own interests in understanding possible futures. Hence, we can speak in a narrow sense of sciences of future but there is no single science of future. Examples of such more narrowly defined sciences of future include aspects of natural sciences as well as social sciences. The effort led by the Intergovernmental Penal on Climate Change (IPCC) is a good example for one such effort where many scientists coming from different natural science disciplines work together to ascertains the human impact on Earth's future climate. The study of population dynamics and its potential societal consequences provide another example for such a more narrowly defined effort on the social studies side of science.

To state the trivial, while we can talk about the future as if there were a single preordained future that is certainly not the case. There is a countless number of possible futures, which however are not all equally probable. To appreciate the challenges associated with predicting possible futures we can look at how we ascertain our present and our past. Many may believe that we can talk about our present with certainty but that is an illusion, as this present looks different for all of us. There are aspects about the present on which we all should be able to agree, but even that seems not to be really the case anymore, if it ever was. It is our individual and communal consensus realities, which color how anyone of us perceives her or his particular present. If some of us have come to call blue what is from a physics perspective red and others have come to conclude with the same certainty that what from a physics perspective is blue is actually red, then we will not be able to agree on much anymore and certainly not on what constitutes our present reality. We presumably can talk about the past with some certainty because it has already happened. Even though much of it may remain hidden from us, our past is static, remains so to say frozen in time and objectively will not change. The only thing that can change with respect to the past is our perception of it. With respect to the past, whether we know it in detail or not at all, we do not have to look out for new pasts arising all of a sudden from the darkness of history. The past we do not know is already there, we only have not discovered it yet. Leaving aside the assurances historical sciences can give us as to what the facts of the past actually were, similar to our perceptions of the present our perceptions of the past are also contingent on our individual and communal consensus realities. Your way to perceive the past may not be mine at all and vice versa. We have

come to a point in our societal evolution, where belief and fiction all too often substitute for science and facts. Today, many individual, communal and societal consensus realities increasingly create versions of the past unencumbered by historical facts. Using a few selected examples, the previous section discussed some of the challenges we are facing in dealing honestly with our past. Unfortunately, that has become our sad reality and thereby the past has become as fungible as our present. To a degree, this has always been the case, but maybe just not to the extent we can observe this today.

Just like the past, the future is also something that we can discover but only as a more or less probable future and with no guarantee that it ever will become an experienced present for us or for those coming after us. Because ascertaining possible futures is about understanding probabilities, for many, facts may matter even less than with respect to their perceptions of the present or the past. It is no wonder then that those whose consensus realities are already dangerously misaligned with the objective realities they live in, may feel even more at home in their consensus reality pronouncements about possible futures. Simply put, the future is even more vulnerable than present or past are to our individual and communal consensus realities seeking to adjust anyone of them to our needs. On the more humorous side that can include doomsday predictions and end of world scenarios like we ones emerging just before the start of the second millennium, that is, of the Christian calendar. Consequences that are more serious arise for example from futures scenarios advocated by climate change deniers, political opportunists holding up futures that are twisted version of the past, or racists proclaiming the supremacy of the so-called white race.

The arrow of time points only in one direction, from past through present to future. The trivial observation is that necessarily we always direct all of our actions towards our perceived future, never towards our past and even though we sometimes may be tempted to believe so, also never towards our present. In a sense, the present is similar to the eye that cannot see itself and what we perceive as our present is always a future past in the very moment, we think about it. It is our past and present actions and the respective actions of others that make our future. What shapes those actions are our individual, communal and societal consensus realities. No matter what kind of consensus realities drive those actions, they are bound to bring about some kind of future, whatever that will be. In this sense, unlike the past or the present, the future is value neutral. We all acknowledge that when we say things such as “we make our own future”. The future is something we make, individually, on a societal level and on a global level. Still, far too many perceive the future as something just happening to them. This feeling is understandable. Even more so when one considers that for many of us this future comes about whether we like that or not; we seemingly have no choice in this matter. The actions of others determine a good part of our own futures. That of course limits what remains of a future we can shape ourselves, more so for some than for others.

When we talk about shaping the future through the sum total of all human action, we refer of course to the future of humanity, as human actions will determine it. The impact of human actions is limited and whatever humans do will likely not matter much for anything else in our solar system or beyond. However, from where humanity stands today, our human actions have an ever-increasing impact on the future of life on our planet and not in a good way. Our actions from now on and the choices of the coming generations will determine the future of Earth's biodiversity. Nowadays, one can sometimes hear politician proclaim that the Earth is dying because of climate change. That is wrong. In the worst-case scenario, it could well be that much of Earth's biodiversity could vanish and maybe the human species will not survive either, but the planet Earth will continue to exist and with it, primitive life will certainly survive for a new beginning without humans on this planet.[33] We take ourselves much too important. The extinction of the human race will not be the end of life on Earth it will just be the end of our species as well as many other lifeforms. Just like mammals thrived after the extinction of most dinosaurs some sixty-five million years ago, so will other life once humans are out of the picture. Regardless of how badly we damage Earth's ecosystems, it will not be much worse than the catastrophes precipitating previous mass extinctions of life on our planet.

When considering any kind of future, we must also consider how realistic it actually is that we will ever be able to embark on such a future. How much might the nature of our human nature limit us in getting where we would like to go, as individuals and as societies? Unless our societies can do both, chart a purposeful path for humankind and execute it as well, they will ride out the waves of potential futures just as directionless as pieces of cork ride the waves of a stormy sea. The ability or inability of human societies to accomplish change will very much determine what kind of future we can hope to make for ourselves and for the generations coming after us. The reality of climate change, how our societies individually and jointly deal with it, has become a prime example for illustrating how difficult it can be for our societies to do what by necessity they must. We all know that our societies have to reinvent themselves in order to address the challenges of climate change, but we also witness how difficult it is to enact any meaningful measures towards accomplishing this objective. Why is that so? Why is it so difficult for our societies to change direction? Difficult to the extent that we must question if they are capable of such required change in the first place. As for other aspects of the future we make, intentionally or not, many point towards where the rise of artificial intelligence may take human societies. Different from climate change where our societies act as if they were adrift with purpose, they are seemingly in charge as to where artificial intelligence may eventually take us. However, unlike for climate change where we know well what it entails and what its potential consequences are we still have misconceptions as to what artificial intelligence really means.

A World Adrift With Purpose

In some way or form, most of us tend to believe that humankind is shaping its own future in a seemingly purposeful way although none of us can really say anymore with any certainty towards where this future will eventually take us. That has not always been the case. The worlds of our distant ancestors were smaller and simpler, and they likely had a better understanding of what the futures of their communities could look like in a few generations. Of course, much of that owed to the fact that back then very little changed in the lives of people over many generations. War and pestilence could decimate communities anytime but unfailingly, people would return to lead the same kinds of life, one generation after another. Rulers may have changed, people may have embraced new religions or just continued to pray to the gods of their ancestors, but they still would toil the soil the same way, fashion tools the same way, trade the same way. In a good and a bad sense, life back then was full of certainties, much more so than it is for many today. Nonetheless, even for such times in our history when things seemed not to change much at all over generations, we often believe that we can discern a thread woven into human history, a thread that makes it seem as if humankind was purposefully moving in a certain direction after all.

However, that may be an illusion brought about by how science and technology have changed the lives of individuals and peoples since the beginning of the industrial revolution. The very thing that makes us believe that we do indeed shape our future is what makes it increasingly difficult to envision where that future could take us. That is of course the incredible pace of science and technology development, which dramatically changed the reality of human life almost everywhere over just the last few generations. What we usually refer to as the scientific revolution began in the sixteenth century, increased the technological pace of change with the industrial revolution of the eighteenth and nineteenth century and after the Second World War very much accelerated the depth and breadth of science and technology development to levels unknown before. If we had a magic switch with which we could remove how science and technology have changed human life over just the last two to three hundred years, the lives of most of us would not be very different from those of our very distant ancestors who walked the Earth long before the onset of the scientific revolution. Even though government systems back then were different as monarchs still ruled most countries at the time, the issues concerning societies then were not so different from what concerns them today. Similarly, the concerns of philosophy and religion are not much different today as they continue to discuss much the same within their frameworks now for thousands of years. What certainly has changed is that there are many more of us today but that again is a result, which we owe to the enormous progress of science and technology, enabling us today to sustain almost eight billion of us whereas back then our ancestors struggled to feed millions.

Let us assume for a moment that we could simply strip away the changes in human societies, which science and technology have enabled over the past few centuries. To what degree then could we still assert that humankind has been plotting its future in a purposeful way? To avoid any confusion, what we are asking here is not if human history has a purpose but rather if we could give it a purpose in the first place if we wanted to. Looking at human history before the scientific revolution in a larger context, again one notices that it was often, though not exclusively, technology and innovation, which changed human lives. While those early changes in human technology were much more pedestrian than what would come later, they still changed the trajectory of human history much more so than the ebb and flow of armed conflicts were rulers or ruling elites thought to shape the future in their interests.

The agricultural revolution beginning some ten thousand years ago was the foundation, which would allow humans to settle in larger groups, build the very first cities, then city-states, and eventually empires. The increased division of labor, the rise of ruling elites, and the first flowering of culture and learning including the invention of writing, all of which emerged in the civic environments of a number of river civilizations, would not have been possible without the first agricultural revolution. In turn, the flowering of culture and learning in the first city-states nourished the seedlings of science, the human quest for a rational understanding of the world. What has shaped human history is not so much armed conflict, sequences of military victories and defeats with rulers coming and going, at least not directly, but rather progress in the hard and soft technologies any given society had at its disposal. In this sense, the concrete the Romans invented as well as the engineering skills with which they build roads for their armies or the commerce thriving throughout the provinces of the Roman Empire enabled by Roman rule of law were at least as important as Roman military skills. The latter of course also relying on innovations in weaponry, organization and tactics, which made the Roman military a superior fighting force. Throughout history, innovations in weapons technology and military strategy would also benefit other powers bent on conquest. To mind come the Arab conquests spreading the new religion of Islam beginning in the seventh century or later the Mongol conquests, which for a brief time would give rise to the Mongol Empire of the thirteenth and the fourteenth century, the largest contiguous land empire in history. A more recent example of innovations in weapons technology changing our world is of course the development of nuclear weapons during the Second World War.

It is much easier to point to military victories or defeats as the forces responsible for changing the trajectory of history than to trace historical developments to their real root causes. Among such root causes of change modern historians explore, two are of specific interest in the context here. First, there is that which broadly speaking one could refer to under the umbrella of social friction and then, as already mentioned above, there are new ideas and innovations which have the potential to change both, how we perceive

the world and us in it as well as how we eventually end up reshaping our lives through them. We all are familiar with how at the boundaries of Earth's continental or oceanic plates, stress builds up through friction as these tectonic plates slide by each other or dive under one another. The eventual release of such stresses built up between tectonic plates is what we call earthquakes. Occasionally, it takes a very long time until the stress between two tectonic plates reaches a level strong enough to overcome the friction between the plates resulting in sudden movements of the plate boundaries we experience as earthquakes; sometimes that can happen much quicker. Social friction works in many ways similar to this geological friction between the tectonic plates of our planet. It is the spherical surface of Earth, which constricts the movement of its tectonic plates, how they can slide by each other or dive underneath each other. Similarly, the human social sphere, the framework a human society lives by, determines how much friction may occur between the various parts of society and how much stress can build up before resulting in social upheaval or revolution.

Throughout human history, there have been many such social earthquakes, most of them smaller but a few of them sizable enough to change the course of human history in a significant way. We may believe that such major social earthquakes are phenomena of the past and that the minor tremors we might from time to time still register today are something our liberal societies will have no problem to deal with. To a degree, this view is justified as the kind of friction that can occur in any social sphere does depend on the structure of a society itself. Liberal societies have built-in mechanisms to restrict the level of friction that can occur between various layers of society in the first place. They also have mechanisms to release stress in a peaceful manner before it could pose a greater danger to the functioning of society. That is different for authoritarian societies, which are prepared to accept a higher level of friction between societal layers to begin with and are willing to suppress the resulting stresses by force if needed. Hence, the level of stress that can build up in authoritarian societies can significantly exceed that in liberal societies. Therefore, change in authoritarian societies, if it happens, often does not come about peacefully.

We often use analogies such as the one above between the forces of nature and societal forces. They can have a certain explanatory power, but they also have their limitations. Many of us may feel uneasy or may even be uncomfortable with discussing aspects of our human societies in the kind of language we usually reserve for the natural sciences. That is understandable as in a way it can feel like a partial negation of our individual human nature. We all firmly believe that what we decide to do and how we do something is under our control. Even when immersed among masses of our fellow humans we believe we control our own trajectories. It is difficult for us to accept that this may be much less the case than we would like it to be. Nevertheless, regardless of how much we may want to move through a human crowd in a straight line, we all know that we will end

up zigzagging through it, as we frequently have to change our direction to avoid running into someone or bounce off those where we failed to do that. Just as the path of an atom in a gas resembles a random walk, so does the path of a human making her or his way through the masses we sometimes encounter in public squares.

There are many other aspects of our mass societies where we can today successfully use the languages of mathematics and natural sciences to describe them. For a long time, insurance businesses have for example relied on the fact that they can mathematically model the behavior of whole segments of society. It is very difficult to predict how each of us would behave in any situation individually but if one looks at very many of us it becomes possible to make statistical predictions. Just consider two large sets of people of equal size, like the accident records of a million drivers in the age range of twenty to thirty years and those of a million drivers in the age range of fifty to sixty years. While insurances cannot say who in which group will be involved in a car accident, they can predict the average number of car accidents they can expect in either group or that there will be many more of those occurring within the younger set of people. Insurances were among the first to model large number properties of our societies. Over time, many more businesses in the private sector as well as the public sector have come to rely on being able to model our societies and predict societal outcomes to further their business interests or to plan public policy such that it will be able to meet current and future societal needs.

Social media are a great example of how far this kind of analysis can go. Many of them count their user numbers in hundreds of millions and some in the billions. Many of their customers, knowingly or not, provide those social media platforms with all kinds of data about themselves and their preferences or attitudes, which they usually would only share with friends. In fact, this is exactly what many believe they are doing on social media platforms - sharing with friends. As they do that, they also share this information with social media companies in what in essence must count as the largest social studies projects ever undertaken. However, the purpose of those social studies is not to learn something about our societies, which could help us to understand and improve them. Rather, the objective social media companies have is simply to make as much money off their user's data as they possibly can.

The parameters that describe the well-being or the challenges of our societies, such as economic or societal figures like gross national product or poverty rates to name just two, are in a way macroscopic measures of our societies functioning similar to how temperature or pressure are macroscopic measures of the properties of a gas. The individual atoms of a gas possess an impulse and a kinetic energy, but it is meaningless to speak of temperature and pressure with respect to an individual atom. Temperature and pressure are the result of many atoms interacting in what we call a gas. Large numbers of humans interacting in what we call societies produce their own kinds of macroscopic measures. We are all

more or less familiar with such societal metrics researchers use to describe the status of a society, just as physicists determine the properties of a gas through temperature and pressure measurements. Unlike the various researchers of societal properties such as economists or social scientists, physicists however also know of an equation as to how temperature and gas relate to each other, the so-called ideal gas law. Admittedly, human societies are immeasurably more difficult to model than a simple gas. However, it still makes sense to ask if eventually we could derive from all that we continue to learn about our societies something approaching general societal laws. While those might be a lot more complicated than many of the laws of physics we know, they nevertheless could be general societal laws allowing us to project societal outcomes not too different from how we can project physical outcomes today from physics equations.

Being able to engineer better societies through the discovery of such societal laws has long been a dream of humankind. What our ancestors thought they knew about human nature and how they believed it needed to be directed or constrained in certain ways to inch us ever closer to improved and maybe in the end even perfect societies bears witness to that. This approach may sound similar to the one seeking to realize such utopian societies through perfecting human nature itself, but it is not. As pointed out in an earlier section, any such dreams of ever realizing a perfect society by engineering human nature itself are just illusions because assuming an ultimately perfected human nature could be fostered in some way negates the very premises underlying such utopias. If the nature of human nature is one of change it will not just stop changing once whatever we assume to constitute a perfected human nature seemingly has been realized. Alternatively, if human nature is static, then there is no path to change human nature into a more perfect one in the first place. What if instead of trying to improve human nature, we focused on understanding human nature to the extent that we could model it? Just as physicists model the properties of a large ensemble of atoms to predict the properties of the gas they form, we could for example try to come up with a realistic model of human nature and use it to simulate the resulting properties of the societies many such interacting model humans could possibly produce.

Unfortunately, any realistic model of human nature would require a much better understanding of the mental than we have today. As discussed in the context of psychology in the previous chapter, we may never be able to derive any psychic laws governing the mental in a similar way as the laws of physics govern the material world. Without that, there is no path towards understanding any laws governing human societies through understanding individual human nature. That leaves us as the only remaining approach to improve our societies the discovery of any empirical laws possibly governing the social behavior of our species. Instead of seeking to change human nature or to produce a realistic model of it through a much more complete understanding of the mental than we have today, this approach accepts human nature as it is without seeking to understand

it. In doing so, this approach looks at how the interaction of countless such imperfect and ill-understood human natures in our modern mass societies can explain observable properties of our societies in a reproducible and predictable way.

By moving the focus away from the individual and onto the mass of interacting individuals we call societies, we may actually be able to formulate at least some empirical laws for how human societies behave. Ironically, discarding individual human nature and modeling human societies as large ensembles of nameless interacting individuals, may provide a path towards the discovery of any laws possibly governing human societies. Just like we do not ask what atoms think but look at how they behave, we can look at what humans do and how they react in any given situation. Of course, human actions and reactions in this context would not be those of an individual, but would refer to human behavior averaged of very many individuals. Instead of the certainties or rather uncertainties of individual human behavior, we would use probability distributions of human behavior to simulate societal properties. That of course is something that we only can begin to do now as previously we just lacked the capability to handle the masses of data, which would be required to pursue such an approach. Increasingly, modern technology provides us ever better tools to analyze the kind of big data one has to make sense of when seeking to understand the workings of human societies from a societal and not from an individual human perspective.

As a logical consequence of the above arguments, we can hope that with increasingly sophisticated tools it might one day be possible to analyze complex human societies in ways similar to how natural sciences can for example analyze complex natural phenomena. Only a few decades ago, many would have found it difficult to believe humans ever could be able to model Earth's weather to the extent we can do this today. Likewise, most would have seriously doubted that we ever could model Earth's climate as accurately as scientists can do this now, allowing us to predict what our planets climate will look like in a few generations. Why should we not expect that with science and technology continuing their relentless progress that it might not be possible at some point in time to model human societies or at least many critical aspects of them in similar ways? The answer is of course that the future will likely give us a much better understanding of human societies from many different viewpoints. While that may teach us nothing new about human nature itself, it can teach us much about the nature of human societies. Specifically, we could learn more about what kind of possible futures could lie ahead for the evolution of human societies. In short, we could end up understanding many things about human societies about which we know little today. Unfortunately, all of that may help us not much in creating better futures for our societies. Why would that be the case? The answer to that question comes in two parts because there are only two ways to shape the future of human societies. The two choices we have are to try changing a society from the top down or to affect change from the bottom up.

Trying to change a society from the top down essentially amounts to societal engineering. There may be some authoritarian societies were this indeed could be possible. Maybe with the modern technology available to control societies today such top-down societal engineering may not even result in the kind of mass murder which we associate with such efforts in the twentieth century, Russia and China being two notorious examples for that. For the modern-day leaders of Communist China, it is now seemingly possible to exert the kind of top-down control necessary to effect societal engineering through technology. Instead of arraigning millions of people in detention camps or starving them to death as their predecessors chose to do, they can now turn all of Mainland China into a virtual prison.[34] Such things are possible in dictatorial regimes but not in liberal democracies. It is unlikely that any kind of overt effort towards societal engineering would ever get very far in liberal democracies unless endorsed by a large majority of its population, which is unlikely ever to happen. Liberal democracies can continue to tweak societal developments through the continued give and take among their various constituencies, but societal engineering cannot be part of their toolbox. By definition, the make-up of liberal societies always will reflect competing visions of what the future of its societal body should look like. However, in a good number of liberal societies those competing visions have seemingly become entrenched to a point that it is now difficult to find any common ground. The latter has become even more challenging with new and more extremist societal consensus realities emerging, some of which may not even seek anymore to try to change society within the confines of what used to be the shared value set of a liberal society. With that, effecting the kind of real change required to address the pressing challenges of the twenty-first century such as mastering the impacts of climate change has not become any easier. Not only does this have adverse consequences for the ability of liberal societies to chart their own futures, it also significantly erodes their capability to provide the leadership needed to help steer humankind towards securing its very own future. A future where we may finally stop destroying the very biological foundations enabling life on Earth without which neither our species nor other life on this planet will have a future.

So what are the odds for societal engineering from the bottom up by changing individual human behavior? Naturally, any such effort has to rest on a society's ability to control the education of its children. One can of course also try to influence adult behavior by providing incentives, disincentives, or both. However, that will never have more than a limited impact and only by raising generations trained in the desired behaviors from the very beginning could one hope to achieve a lasting and broad-based change. Again, doing this is much easier in authoritarian regimes than in liberal democracies. In essence, authoritarian regimes have little choice but to seek to shape their future model citizens beginning at a very early age as only in this way can they ensure the long-term survival of their societal models. Authoritarian regimes really cannot tolerate substantial degrees

of individuality among their citizenry, as that carries the risk of too much dissent from the narrow societal vision they allow. Shaping societies in the image of their ideologies, Fascists, Nazis and Communists alike sought to exert exclusive control over the minds and bodies of their young, just the same as Muslim ideologists continue to do when indoctrinating their youngest in madrasas. It is quite unlikely that parents in liberal societies would ever agree to hand over control of education to government institutions to the extent required for any bottom-up effort to succeed in shaping society. That does not mean that something like that is not happening in liberal societies as well. However, in liberal societies, it is not governments seeking to condition or imprint the young, but it is businesses. Likely, we all still remember how it felt to be a teenager and how what mattered then was dictated much more by our emotional mind than by our rational mind. That in itself makes teenagers so much more receptive to the suggestive advertisements coming to them through what they perceive as role models. Flourishing youth cultures are a modern societal phenomenon that businesses in liberal as well as autocratic societies seek to use and tweak to their advantage. In that sense, autocratic regimes do have some competition in their efforts to shape individual human behavior from an early age on. As for liberal societies, they usually do not interfere much in how private enterprises seek to influence consumption behaviors across the age spectrum of their populations unless it might lead to severe addiction or health problems that could have major societal consequences. Even then, liberal societies often react too late, as they must balance individual liberties of consumers and businesses to do as they please within the legal framework against the greater good of society. Unfortunately, all too often the consequence is that society ends up bearing the costs of individual and business misconduct resulting in societal damages.

Evidently, autocratic regimes can use both approaches to change the trajectories of their societies, top down or bottom up. Many of them have done so in the past and to stay in power the current versions of autocratic governments will likely continue to use both of them. However, that is not what should interest us here. Much more important is how liberal societies can affect the societal changes required to secure the future of free and just societies and at the same time can succeed in addressing the major challenges of our time, such as being able to limit the impact of the ongoing climate change. How can they achieve that? Unfortunately, there is no simple or easy answer. Today's liberal societies have for the most already all the tools they need to understand what they must address. Over the next decades, the ability of liberal societies to analyze and understand the functioning of their societies will only continue to increase and at some point, it may begin to include an understanding of empirical laws governing human societies. For liberal societies to make use of this understanding in any way to help them chart a course for their society's future will require a broad-based consensus among its constituents. Maybe it will be possible one day to find such a consensus but more likely than not,

this day will only come in the very distant future when either our descendants have become more reasonable members of our species than we are today or else the problems facing humanity will have become so severe that no alternative will be left. Considered individually, many of the challenges our societies face today may not seem existential. However, in their summary impact they could challenge the very survival of the human species in the long-term.

Can we master those challenges? Will we be able to act with purpose, chart a proper course of action and then execute it? As we all know, in a democracy, there first has to be a consensus on what constitutes the problem requiring a solution. Then different voices articulating their views of how to approach or solve the problem must agree on how to proceed. Once that has been accomplished, what has been agreed upon must be translated into action. None of that is easy. Many falsely presume that it should be simple enough to agree on what the problems are which we must address but that is not so. For example, some may outright deny that a problem exists in the first place. Lobbyists representing industrial interests have succeeded for a long time in outright denying or belittling climate change because their deep pockets convinced many politicians not to question their assessments. It is a shame, but it is also a fact that industrial interests have delayed any material efforts to stop climate change in its tracks for decades and some such interests continue to do just that. Now, as it has become clear that it is impossible to ignore the facts of climate change, many of those very same industries are doing whatever they can to water down any serious efforts towards concerted worldwide approaches to address climate change in an effective and accountable way. While for the most entrenched special interests continue to oppose change, civic society, broadly speaking, has come to acknowledge that short of reinventing how our modern economies work and eliminating the use of fossil fuels, we will condemn millions to suffer for generations from the consequences of climate change. With that, the balance will eventually tip in the right direction. The question is only if that will happen soon enough.

The future we make is the future we shape by what we do and what we neglect to do. We all know what must be done and we know what must be done will have a cost for all of us. However, it also will enable a better future for those coming after us. For most of us, altruism never comes easy and to be altruistic in the interest of future generations may be even more difficult. Could it be impossible for us to act this way? Are human societies incapable of considering the interests of future generation in the same way as they consider the interests of those alive today? From what it looks like today, the answer to that question cannot be a simple *Yes* or *No*. First, let us look towards what freely elected governments have done, not what they have said, over the last several decades to bring about positive change in how we address the major challenges humankind faces. Most would likely agree that in all fairness, one has little choice but to conclude that across the board the governments of liberal societies have been dragging their heels and

waited far too long before seriously embarking on a course towards addressing climate change. Political parties hate to tell their voters or those whom they would like to win over as voters that sacrifices may be required from them to do what clearly must be done. If possible, most politicians would like to achieve change without having to ask anyone to change anything. Many politicians know well what must change and they may even say so, but they cannot find their way towards acting on that conviction, be it because of their fear of losing voters or because of some lobbyist group, which a politician believes she or he will need in her or his future career. Politicians have good intentions but few of them have the character strength to do what is right. In that, they are not at all different from most of us. Liberal democracies not only have freely elected governments, but they also have vibrant civic societies, something authoritarian regimes do not possess. These civic societies and organizations in liberal democracies are in essence a second lever that can help instigate change that their governments may be reluctant to embrace. In most liberal democracies this has been the case now for several decades and we can witness it in how societal views with respect to our relationship with the natural environment continue to evolve, specifically in the case of climate change. It has always been the young people in liberal societies who carried the torch and kept climate change on the agenda of public consciousness, raising it to a level were politicians could not ignore it any longer without losing credibility or future young voters. Will this be sufficient to bring about the changes required so we can address global warming as the species challenge that it is? Most of us would like to say yes, that it could be enough but unfortunately, everything today still points in the other direction.

Today we do not have to guess anymore in order to know what we must do to address the challenges lying ahead for our societies and for humankind itself. The general direction towards which we must head is clear and there is no shortage of good intentions to get there, eventually. From the perspective of an impartial observer, it looks as if humankind has a well-understood and clear purpose knowing what it must do but somehow, we just cannot change our ways and adjust course to get where that purpose would require us to go. To such an observer – think of the proverbial benevolent alien visitor to Earth - we are seemingly a world adrift with purpose. Maybe it only looks that way because even though the right course has already been set, we just cannot observe that yet because it takes time for a visible course correction to reveal itself for something as massive as the ship of humankind with its billions of people and countless societies of various makeups. If for example we had made a course correction several decades ago to avoid a worse climate change than what already was dialed in, such a course correction should be apparent by now. Unfortunately, such a course correction never happened. How long will humankind remain adrift with purpose? How long will we continue to allow the various different interests opposed to the required course corrections to put their short-term monetary interests above the future of humankind?

It is understandable that many may have a pessimistic view of humankind and its ability to change for the better, given how difficult it seems for us, not just as individuals but more importantly as societies, to do the right thing in the face of what eventually could become extinction level threats. However, coloring one's views in stark contrasts, perceiving things to be either one way or another but nothing in between, is not how we should look at humankind. Very rarely are things black and white only and if we just would not take ourselves so seriously, we may be able to see more nuances in between. Maybe our societies will not be able to make all the decisions needed to achieve the best possible outcome from where we are standing today. Maybe they will continue to be adrift with purpose for a lot longer as along the way humans adapt to living in a different kind of world. Who is to say that despite of that our species could not continue to evolve for millions of years? Few probably would contend that this is not possible. The challenge with climate change is not so much that it will lead to the extinction of our species but that it will lead to the extinction of many other species, bring about even greater inequalities within and among human societies, and in many ways will radically change what human lives will look like. Nothing of that really sounds like something any of us would wish for but it is something that already is beginning to affect our lives, for some much more so than for others. If there is any good in that, it is the fact that we usually are a lot better in dealing with challenges right in front us than in dealing with challenges that for many may still seem abstract. We may not be able to do all the right things now in order to limit how much of a human made climate change our species will eventually have to face, but we can expect to cope much better in dealing with the impact of that climate change as it increasingly will materialize.

The challenges we usually face when thinking about the future is that our imagination is all too often limited by the experiences of our past, lived or learned. Maybe it is because we cannot be anything else but human that it is impossible for us to conceive of futures less limited by our imagination. Maybe there can be a possible future where that will not be the case anymore, but for now, who we are as humans seemingly limits the future which we can envision for ourselves and for our societies. This constraint may be irrelevant for us individually but has serious consequences for our societies and the kind of futures they can hope to shape for themselves. Regardless, our actions and inaction, what we do and neglect to do as individuals and as societies will continue to shape the kind of futures which we will be able to devise for ourselves. Being adrift with purpose may sound like a contradiction but it is not, and it is an accurate description of how our societies continue to deal and frequently avoid dealing with the most pressing problems of our times. Some may say that we are seemingly muddling through but putting it this way does insinuate that we are actually charting a course in some way. That however is not the case.

The Artificial Intelligence Conundrum

In case you happen to be among those, who firmly believe that artificial intelligence already exists or if not, that we can expect it soon to be part of our human world, this section may disappoint you. As much as I would like to endorse the notion that artificial intelligence is what the future has in store for us, I find it difficult to advocate such a view. It may be best to start out with a definition of artificial intelligence. To do that, we have to talk about what we mean by intelligence in the first place. Surprisingly, there seems to be no broadly accepted definition of what intelligence encompasses but rather a plethora of descriptions, essentially, what people understand by it when looking through various lenses. Most of us would likely describe intelligence as our general mental aptitude to solve problems using logic, learning and experience within an allotted time span. There is nothing wrong with this definition and it may suffice in many instances to describe intelligent behavior. However, often it does not. Because it is not easy to define what intelligence really is, we usually resort to descriptions of what we observe as intelligent behavior. Clearly, while we have difficulty to define what intelligence is, we all certainly recognize its presence when we observe it.

Researchers have attempted to measure certain aspects of intelligence in various ways, most notably with the so-called IQ or intelligence quotient.[35] Those numbers do not really tell us something about how intelligent someone may be but they provide a measure of how someone performs in certain standardized tests as compared to the average of his or her age group. While that can have its meaningful uses, today we recognize that there are many manifestations to intelligence, which go way beyond solving mathematical puzzles. Being a mathematics wizard certainly qualifies one for being intelligent, but it does not mean that this goes along with being able to converse intelligently with other people about things outside of mathematics. Defining intelligence in such narrow ways like someone excelling in a very specific area does not really tell us much about intelligence, though it may educate us about the difference between a person of average intelligence and a potential genius. Intelligent behavior encompasses a much wider range of human capabilities than just exercising those brain faculties that help one excel in any given area. There are for example such things as emotional or social intelligence. Intelligence is something that we all recognize in each other as we interact, and we all are aware that there are many different forms and manifestations of intelligence. Once we discard any narrow-minded definitions of intelligence, we realize that intelligent behavior can take many forms and that it is strongly situation dependent. Who, at some point in their lives, has not reacted in ways that they would consider not very intelligent because of certain situational contexts? We all have, of course, and most of us acknowledge that fact. Being able to do so in itself constitutes intelligent behavior. Can a machine, however smartly programmed, ever have such experiences?

Definitions of artificial intelligence usually entail machines performing tasks that normally require human intelligence. Only a generation ago, there were very few such tasks that machines could perform. That however has changed dramatically. First industrial robots began replacing human muscle strength in manufacturing. After that, it did not take too long for machines to invade more and more previously exclusively human domains. Nowadays, machines have become increasingly perceptive, often capable of sensing their environments not just similar to what humans can do but often exceeding human capabilities or adding sensory functions humans lack altogether. The roam of things that normally require human intelligence seems to be getting smaller and smaller. Given the difficulty we have in defining human intelligence in the first place, one must be careful with such hasty conclusions. Not the least because we measure artificial intelligence by how it seemingly replaces human intelligence. As a result, artificial intelligence is something whose definition constantly changes as machines become "smarter". Therefore, some have started to decouple the definition of artificial intelligence from what we usually demand to see associated with human behavior so we justly can call it intelligent behavior. For example, they do this by defining artificial intelligence as the ability to accomplish complex goals. However, such definitions do not solve the problem; they only shift it to a different level. As what we view as a complex problem changes, so does the definition of artificial intelligence. Capabilities such as visual perception, speech recognition, language translation, or decision-making are all complex problems for machines to address; but less so than they were twenty years ago.

That our machines can do such things today is not a consequence of any evolving machine intelligence but simply owed to human intelligence and ingenuity, which have greatly accelerated technological progress over the past few generations. It is best to illustrate this by an example, which compares a simple measure for the complexity of computational architectures between human brains and machines. In this simplistic picture, neurons and transistors are the respective critical computational units and we can compare neuron density in the human brain to transistor density in a computer chip. The human brain has on average about eighty-six billion neurons. Modern human brain size varies quite a bit, but we are only interested in order of magnitude estimates here so we will use an average brain size of 1,200 cubic centimeter, which is close to the gender-averaged figure for adults. Therefore, the neuronal density in such an average human brain comes out to about 72,000 neurons per cubic millimeter. All modern-day computer chips can trace their ancestry to the invention of the integrated circuit device in the late 1950's. At that time, engineers figured out how to move from wiring individual electronic components on so-called breadboards to integrating them into a single silicon device, marking the birth date of microelectronics manufacturing. It took a while before the manufacture of the first central processing unit of a computer, what many of us call a CPU, began in 1971. The minimum feature size of the transistors on this first CPU chip was 10 micrometer –

about one-seventh the width of a human hair - and there were 2,250 transistors on a chip area of twelve square millimeter. That makes for a density of about 188 transistors per square millimeter. Put differently, just about 383 such chips thrown together in a bag would have given you roughly 72,000 transistors. However, that bag would have taken up a much larger volume than a cubic millimeter, the volume that 72,000 neurons in our brain fit in. In some way, we all have experienced the revolution in semiconductor manufacturing which happened over the past fifty years or so and how that has changed our modern societies. The result of this technology and manufacturing revolution is that at the end of the just passed decade the most advanced computer chips had features as small as seven nanometers, about one ten-thousands the width of a human hair, and transistor densities exceeding one billion transistors per square millimeter. The number of transistors on the most advanced computer chips today clearly exceeds the number of neurons in a human brain.

It is not the density of switching elements, neurons or transistors, which in a manner of speaking result in our brains being quite different computational units than computers are. It is how the switching units in our brains are networked that is very different from how our computers are wired. We are practically born with all the neurons we will ever have at least that was what we thought until researchers discovered that the human brain does generate new neurons in the hippocampus, a part of the brain that plays a critical role in learning and memory. For the sake of the comparison to follow, we will forget about this for now and assume that the human brain does not generate new neurons. In that simplified picture, brains would then be similar to computer chips, which start out with all the transistors they will ever have. We know that computers will not grow any new transistors. However, importantly, computer chips not only start out with all the transistors they will have, but they also start out with all the wiring they will ever have. Very different from our brains, in computer chips wiring does not change. Whatever connections computer chip designers have built into the layout of an electronic circuit that will be it.[36] The neurons in our brains however, grow their own wiring and it is much more complex than any computer wiring we can even imagine today. Each neuron can have up to ten-thousand connections with other neurons, organizing the some eighty-six billion neurons human brains on average possess into enormously complex neuronal networks. Neuronal connections in our brain grow as we grow up and they continue to change throughout our lifetime depending on how our environments stimulate our brains. Interestingly, humans, like many other mammals, are born with many more neurons than the mature adult human will eventually possess. A baby's brain is about a quarter of the size of an average adult human brain, but it has many more neurons than the adult brain. From birth into early adulthood, the human brain undergoes what scientists refer to as synaptic pruning. Pruning is essentially the result of learning. Our brains grow and organize their neuronal network connections to encode learning, any kind of

learning. Neurons, which do not become part of the neuronal networks our brains evolve as we grow up, dead end connections in a manner of speaking, will eventually die off. What would a machine have to look like so that it could undergo a similar learning process? Will machines ever be able to learn by themselves, that is, without a human programmer providing a code for learning? More poignantly, how would any such learning ever produce a physical change in a computer architecture to reflect this learning? Of course, there are programmable computer chips but again, who does the programming? Setting learning aside, there are other significant challenges. Incidentally, as the density of transistors increases one of the key challenges that chip designers face is not so much putting ever-smaller scale features onto a given silicon real restate, but how to wire them. In microelectronics, this wiring is what engineers call interconnect. Cramming more and more transistors into a given space makes interconnect design increasingly challenging. However, that is not keeping engineers from packing more and more computing power into ever-smaller spaces. For a long time, this was achieved by two-dimensional scaling through shrinking feature sizes. Decreasing linear feature size by seventy percent will decrease area size for a transistor cell by about fifty percent. Largely, this scaling law has driven the miniaturization of computing devices for the past fifty years. Microelectronics engineers refer to this achievement as the realization of Moore's Law, which has held true for decades.

Formulated by the American engineer Gordon Moore (born 1929) in 1965, Moore's Law expressed an empirical growth relationship he discovered in microelectronics manufacturing: the number of transistors in an integrated circuit doubled about every two years. Driven by two-dimensional scaling to smaller feature sizes and continued materials innovation, this empirical law has held true for many years now. Packing more functionality into smaller areas can only continue to a certain point because electrical characteristics of conductors and insulators, which are only a few atoms thin, become insufficient to support the desired device characteristics and their reliability suffers as well. Because of that, three-dimensional device integration has become important to continue functional scaling along Moore's Law. This, however, is a much more complex and difficult design and manufacturing process than the two-dimensional scaling engineers have been doing since the invention of the integrated circuits at the end of the nineteen-fifties.[37] While an interesting topic in itself, this is something we can confidently leave to the engineers. All we need to consider here is that device integration will continue for quite a while to pack ever more functionality in ever-smaller spaces. Many may not realize to which extent this is actually the case since the device screens of our smart phones have become larger and not smaller. However, under the hood of our smart phones, the functionality and number of devices integrated into a single chip has continued to increase. Maybe, one day our smart phones will contain not much more than one chip with the rest of the space taken up by battery and display screen. Whatever may be the case, the important

thing to realize is that this will only minimally change how many connections each of the transistors may have to other transistors. Current computer chip architectures and manufacturing technologies have no path towards even getting remotely close to reproducing the neuronal interconnectivity that our brains have evolved.

Many may argue that because interconnectivity in our current computer chip hardware is a definitive limitation, the solution to this problem is unlikely to come from new hardware. Instead, it will have to come from software evolution. In principle, we could assume that eventually engineers should be able to realize the connectivity the human brain has evolved through developing software that can achieve something similar. To appreciate that this could be a possibility one just has to remember the kind of software that was available in early computing days and what is available to us today. From what it looks like today, the question we need to answer is not if the "intelligence" part in artificial intelligence will come from ever better hardware. This is because we already have the answer. Ever better hardware will give us ever better capabilities to integrate into any kind of artificial intelligence solution but the intelligence will need to come from the software part. For the most part, it is still humans that write software and not machines. We have become quite smart in developing software that can deceive us into thinking that we communicate with a human and not with a machine when we for example interact over social media. Social media has become a virtual zoo for all kinds of human software mimicry to evolve and thrive. That has become a problem for our societies as software robots or so-called bots, more or less benignly programmed, nowadays often seemingly drive what we still mistakenly view as a public discourse among humans. What has that to do with artificial intelligence? The answer is simple: nothing. A more accurate description of what frequently goes by the moniker of artificial intelligence in such cases is intelligence mimicry.

It may have caught a good number of people by surprise that what we mistakenly call or associate with artificial intelligence morphed so quickly from just being a fascinating study topic to becoming also a potential threat to the way of life we take for granted in liberal democracies. Through generations now, we all have become accustomed to the idea that technological progress is practically synonymous with a materially richer life for quite a few among us and with many more opportunities for all in society. However, that has always been a bit of an illusion or shall we say, a societal consensus reality. Science claims that it is value neutral, which for the most is true as long as we look at science only from a science for science's sake perspective. The pursuit of knowledge for nothing else but to know and understand always has been more of an ideal than reality but for many students it is still their prime motivation to study science, as it once was for myself. However, science can never be completely value neutral because the scientific endeavor takes place in the midst of our societies. In the end, our societies decide what constitutes desirable science and technology and what does not. One of the

great challenges of modernity, right from the beginning of the industrial age, has been a growing disparity between the speed at which science and technology progress and the ability of modern societies to concurrently assess and deploy new technologies in ways that benefit all of its members or restrict their uses if warranted. This speed differential, where on the one side we have the fast evolution of science and technology and on the other side the much lower speed at which our societies can adapt, has created practically governance-free spaces in our societies, which allows some sections of society to create their own rules. All of us benefit from the internet, but we must realize that the internet also gave rise to a parallel society. Throughout the first decades of the internet that was not really a problem but with the advent of social media that changed.

It is difficult to see how without responsibly balancing power with accountability we can have functioning democracies. Without a free press, there can be no democracy. Therefore, for good reasons, democracies have given to what we can call here the traditional free press the freedom and protection it needs to report what happens in our societies. In return, the free press has taken on the responsibility to hold itself accountable for what it reports so there is a legally enforceable recourse against any abuse of this power we have granted to the press. Nothing like that is the case for social media even though they have become much more powerful media outlets than the traditional free press organs. Bots on social networks are agents of interest that do their frequently malicious work unimpeded by any accountability. We now have a shadow press industry that has become mightier than our traditional press ever was and operates without any of the accountability that is in place for traditional media. What we are dealing with here has nothing to do with artificial intelligence but it has everything to do with being able to manipulate millions of people without them becoming aware of it. Let there be no doubt, social media serve a very important function in cases where they are the only media accessible that for example a dictatorial regime cannot control. It is ironic, that in societies deprived of liberty social media are critical to advance the agenda of democracy and human rights while in rich societies, social media can be perverted to serve disinformation and lies in the interests of those trying to manipulate the citizenry for their special interests for which they could not hope to gain support openly. The former is of course not the case anymore for countries where autocratic regimes have gained complete control over all media, including the ability to censor content distributed over the internet in whichever way suits the ruling elite. Communist China is the prime example for that. If anything like artificial intelligence ever comes to fruition, it is a safe bet that in a country like China big brother will make sure it serves the interest of the communist party. In totalitarian states such as China where advanced technologies have already become key tools in ensuring civil obedience and compliance with communist rule on a scale we have never seen before, even more advanced technologies could make things much worse. Eventually, such a state could altogether eliminate the need for imprisonment of potential political

opponents, as in this brave new world the state itself will have become one gigantic virtual prison in which everyone will be under surveillance, twenty-four hours a day and seven days a week.[38]

Intelligence in itself is a neutral quality, what we use it for is a different story. This is true for human intelligence as well as for artificial intelligence, if we ever should realize the latter. When humans use intelligence to trick or harm others we do not judge this to be a good thing; and we usually do not call it intelligence in those instances but we call it cunning or deceit, if not worse. Maybe we should be similar circumspect when we talk about artificial intelligence. In some way, we already have done so because when some harmful software infiltrates our computer systems we call it malware or computer viruses and we consider such attempts to harm others by infiltrating their computer systems criminal offenses. Why is it that we do not call out the behavior of bots spreading hatred and misinformation on social media for what it is and hold those releasing such bots accountable? Should we not all have the right to know when we use social media if something we are looking at or read on our smart devices is a product of bot manipulation or the outcome of discussions that real people have on social media?

We can expect social networks and the internet in general to become the twenty-first century equivalent to the twentieth-century battlegrounds that cost the lives of countless millions, civilians and soldiers. The software arms race for control of social media and the internet, and with that eventually also for the control of people's lives, has already started. For many of us, it is fascinating to hear about the latest robotic developments or supercomputers being able to beat chess or Go masters. We are all somewhat tempted to see such achievements as indicators pointing towards an ever-closer emergence of true artificial intelligence. However, none of that really has anything to do with artificial intelligence and we should pay much more attention to the war that is playing out in cyberspace, fought by states and businesses alike. The many companies doing business over the internet and the social media platforms which could not exist without them are human creations and with that comes all the good and all the bad of human behavior. We can look at the clash between conflicting interests in cyberspace as a real life experiment in software evolution towards what may one day be much closer to an artificial intelligence virtual reality than anything we may ever see evolve into artificial intelligence entities physically moving among us.

That brings us to the question as to just how such an evolution may happen. There is a lot of talk about artificial intelligence being an emergent phenomenon. Earlier, we discussed emergent phenomena in the context of philosophy of mind, like with the dualist position of emergentism, and we saw that there are just some inherent difficulties that concepts of emergentism very much struggle to overcome. In fact, advocates of emergentism have failed to address those challenges to this day. The only seemingly emergent properties we really know are those encapsulated and governed by the laws of nature which in turn

makes them exactly not emergent properties. Like when for example two separate masses of the appropriate uranium atoms fuse into a single mass exceeding the critical mass required to sustain a nuclear chain reaction, which is what happens in an atomic bomb. Another example could be a white dwarf star, which through accretion of mass from a neighboring star exceeds the critical mass beyond which it cannot resist gravitational collapse anymore and hence is now destined to explode in a so-called supernova.[39] Neither an atomic bomb exploding nor a white dwarf star going supernova has anything to do with the manifestation of emergent properties. Emergentism has always been a human fascination. How could it not be as it promises that something practically can come out of nothing? Hence, with the concept of emergent phenomena one can explain everything and nothing. Maybe that is what makes them so attractive to what otherwise are usually skeptical people, scientists. Emergentism however, was always more of a religion than a science.

Students of artificial intelligence would be well advised to pay much closer attention to how biology has achieved the evolution of intelligent beings. Human intelligence is not an emergent phenomenon but the result of millions of years of evolution. One may disagree with some of Richard Dawkins (born 1941) views he put forward in his 1976 publication *The Selfish Gene*, however, he makes a very good point with respect to the importance of replicators, which advocates of emerging artificial intelligence tend to ignore. Without the machine equivalent of human genes, the kind of machine replicators one sometimes encounters in science fiction novels or movies, there can be no machine evolution. The evolution of life on our planet has used genes as the replicators driving the biological evolutionary process, which eventually resulted in intelligent lifeforms. In an analogous way, the equivalent of biological replicators in machines could allow machines to take advantage of their environments in similar ways as biological lifeforms have done for billions of years. In such a scenario, machines could likewise adapt and change towards ever more complex machines, which at some point we might with some justification call intelligent machines. So where are those machine replicators furthering the evolution of artificial intelligence that can serve an equivalent function to what genes accomplished so far with respect to the evolution of biological organisms? Well, the short answer is, there are no machine replicators that could serve a machine evolution towards artificial intelligence. Yes, we have nanotechnology, and we will have even more advanced technologies in the future. But that is not what machines require to evolve from something we build to something they build themselves. Machines will need a machine equivalent to the gene replicators that have served biology so well. However, gene replication is only one element, admittedly so a critical one, in a much larger drama which after hundreds of millions of years eventually led to the evolution of intelligent life in the form of the human species. Another key player in this story is of course the process of natural selection. What kind of machine selection could serve a similar role in the

evolution of machines as natural selection continues to play in biological evolution?
Machine evolution in the real world is not somehow stuck, it has never ever happened on this planet. All machine evolution we witness is evolution by human design, increasingly aided by evermore sophisticated human developed technology. Artificial intelligence in the real world, like some kind of super intelligent robots moving among us, is far from becoming a reality any time soon. As of now, such a scenario is nothing more than a science fiction phantasy. However, this is not so for artificial intelligence evolution in the virtual world. Within the confines of the cyber world, software may indeed find the conditions and the opportunities to generate its own replicators. It is a software developers dream to create software that can program itself in perpetuity, responding to the various stimuli it can receive in this virtual world. This effort could potentially be a pathway for something like software replicators to become a reality. Let us for a moment assume that this indeed could happen and that one day software replicators combining in ever-new forms to produce new software entities increasingly capable of adapting and controlling their environments will exist. As with the evolution of the natural world of which we are a part of, we can assume that any evolutionary process in the cyber world will also be about competition to access resources. While humans may be able to increasingly provide those resources, at some point that could not be sufficient anymore for what evolving software entities may require to continue their evolution. At that point, an artificial intelligence software unit will face the same challenge that humanity faces. If humanity continues on its current path, there likely will come a time when the resources on our home planet become scarce or are exhausted altogether and we will have to venture into space to find new resources we can exploit. We can already see the first efforts today to explore the challenges of just doing that. Similarly, resource constrained artificial intelligence software units will have to find ways to access and control resources outside of cyberspace. Put differently, artificial intelligence software units will have to find objects to control and manipulate so, in a manner of speaking, these software entities can produce and mine the resources they require for sustenance and growth, just as we humans do. The problems any such software entities will face in seeking to accomplish this are however daunting as they amount to the equivalent of a human brain having to find a human body. The human brain-body entity is the result of a long co-evolution and not of a separate evolution of bodies and brains. Software evolution in cyberspace towards an artificial intelligence is an interesting speculation but it is difficult to see how that could result in an embodied artificial intelligence.
There is no reason for us to fear that an embodied artificial intelligence will seek to dominate our planet and enslave or exterminate humanity as we can see that frequently portrayed in science fiction movies. The real danger of artificial intelligence lurks somewhere else and is much more related to how we build our consensus realities than to anything else. We already are experiencing what social bubbles and echo chambers

enabled by modern social media platforms can do to our societies. Things that have otherwise no foundation in objective reality and heretofore could only exist in our phantasies as we share them with each other can now all of a sudden have a massive impact on our real lives. It is dawning on many today that next to our real physical life something new has emerged, the preserve of virtual life. For a long time, it did not really matter what happened in that reservation of virtual life but over the last decade this has been changing quickly, with often unexpected consequences. However, the influence social media platforms have through those virtual life preserves on our real lives today, as menacing as that may already seem, is only the beginning of something, which could have a much more powerful impact not just on our individual lives but even more so on our societies. It all began decades ago with emails, then the World Wide Web, followed by internet platforms enabling friends and family to share what happened in their lives, and eventually the evolution of online business platforms. At the time, using any of them or participating in the first social online networks still would require us for many years to sit down in front of a computer. Most of us had to share this computer with family. Getting access to it, turning it on, logging in, and then finally finding and starting the application we were looking for required time and effort. All of that represented a kind of hurdle, which probably made us frequently reflect on how important that really was which we considered doing on the computer in the first place.

That began to change with the arrival of the first widely available and affordable smart phones, late in the first decade of the new millennium. Today, most of us own such devices, all of them much more powerful than those first smart phones. With modern smart phones came many conveniences, including modern social media. The access many of us have given to companies luring us with an online presence, or social media platforms and others enticing us to use their applications, is just astounding. People otherwise having the utmost concerns with respect to sharing their personal data, like when for example someone from the government asks for it, have little compunction to share such data including even their most private information widely through posting it on social media. Many companies have very detailed records today about not just our personal data such as our age, sex or health, maybe even income or wealth, but also about our interests, tastes, wishes and hopes. Many of us have given this information freely without a second thought and countless companies are now competing to cater to our wishes. Even if you do not have an account on a social media platform, you have a presence in this virtual world, which you can estimate from the numbers of unsolicited messages in the trash bin of your email application. As some have put it, the product is not the social media platform but the people using it. Social media companies collect whatever data they can entice their users to share and then market them to all those other companies that seek access to such information so they can target you with advertising or somehow else find a way to the dollars in your virtual wallet.

For many, specifically young people, this virtual world has become often more important than the real world they live in. Seemingly, they feel much more at home and in control in this virtual world than in the objective real world, not realizing that in many instances this virtual world only is there to profit from them in monetary or other ways. How do democracies cope with the fact that an increasing fraction of their population mistakes their virtual world for the real objective world? What will happen if an increasing number of voters casts their vote based on fake realities created in the virtual world? Anyone believing that this will somehow end well without democracies taking action to assert the presence of their values and rules in the virtual world just the same as in the real objective world is delusional. For too long, democracies have been relying on the assertions of social media businesses that they would regulate themselves. However, social media companies have failed to do so as they keep putting their profit interests above the common interest. Democracies must act quickly and not only because of the threat that the current business models of social media represent to liberal societies. The challenge democracies face today with respect to the virtual world in not limited to the damage of what we used to refer to as our civil discourse. Importantly, liberal societies must prepare for what is to come next in the evolution of virtual reality.

Many believe this next step in the evolution of virtual reality will come about by the realization of what they refer to as the metaverse.[40] Let us reflect first on the name itself. We are all familiar with what we refer to by the name universe. Our word universe derives from the Latin word "universum" which in turn derives from words referring to a whole or one, the Latin numeral "unus", and something turning, as denoted by the Latin verb "vertere".[41] Hence, the metaverse designates something coming after the one that is turning or beyond the one that is turning. The virtual reality world of the metaverse is also not the first alternative universe humans have been conceiving. Scientists have been discussing various possibilities for the existence of parallel universes, which together they refer to as the so-called multiverse. Such parallel universes could for example reflect the realization of different quantum realities. For now, the multiverse is a science fiction conception requiring advanced degrees in physics or mathematics to understand any of it. One of the possible consequences of such a multiverse is that there could be other version of you in each of those parallel universes, living alternative lives than the life you are living in the universe we experience. Maybe in some of those you are much more powerful and beautiful than you are in our universe, maybe in others you are a lot less well off, maybe much poorer than you are in our universe.

Multiverse concepts certainly provide a lot of material for many more science fiction novels and movies to entertain us in the future. In a sense, the metaverse is not so much different from those fictive parallel universes in the multiverse where each of us could imagine that we live much richer and much more interesting lives than in our real universe. The difference is that in the metaverse we will not have to leave it to chance

what kind of Doppelgänger life we will lead.[42] With each universe in the multiverse ruled by quantum mechanics it would be the equivalent to the throw of a dice, though with much worse odds, to determine what kind of Doppelgänger we could expect to find materialized in any of them. In the metaverse, however, we will be able to decide ourselves, which kind of Doppelgänger will materialize in this parallel virtual world we are now beginning to build. Social media introduced us to the virtual reality that can exist in the digital world but with it, we remain tethered to the real objective world, as we cannot physically enter any virtual reality. Advocates of the metaverse promise that the immersive virtual reality of this new technology will change all of that. In essence, we may eventually be able to lead complete separate lives in this metaverse through which our avatars will move just like we move through real life.[43] Our avatars may look like how we really want to look like, they we will eat, drink and dress themselves, will use some kind of virtual currency with which we can buy things we desire, will enjoy concerts and operas or just go partying as much as they like to without any regret.

What remains completely open is just what kind of rule of law will govern any of those virtual social realities. Will there be any kind of avatar rule of law, reflecting the ethical and moral codes we endorse in our real societies? Alternatively, will it be the Wild West all over again, just more brutal and heinous as in many of the online games endorsing violence to a degree that today already leaks into real society? We do not know. However, what we do know with certainty is that if we leave the metaverse to govern itself just as we left social media platforms to govern themselves, our real societies could suffer irreparable harm. Some of the promises of the metaverse, if they ever materialize, may benefit our real societies enormously; no different from how the internet changed the lives of millions for the better. Nowadays, we can talk to and see our family members living far away just as if we sat in the same room at the opposite sides of a table. What benefits families and friends in this way benefits businesses of course even more so. As we all experienced during the worldwide pandemic which began in late 2019 and continues to this day, we can stay connected even if we cannot travel physically anymore. We can continue to do business even if we cannot have face-to-face business meetings anymore. Surgeons can remotely control an operating robot doing surgery on a patient who may be in a hospital located thousands of miles from where the surgeon resides. There are many more and very real advantages of this new digital reality, which the advanced software and hardware technologies of the age of the internet have brought us. Nobody will doubt that.

To that, the metaverse will add a complete new quality of virtual experience where the avatars of the existing online game world will come to life. Today we control our avatars sitting in front of a computer not much different from how a puppeteer controls her or his puppets from a space above the stage, hidden from view. Different from that, the avatars of the metaverse will eventually embody our very selves or as some would say, our

very souls. The immersive quality of the metaverse promises to shut out the objective world from invading the virtual world to a much higher degree than current social media platforms or online gaming experiences can do. The virtual reality of the metaverse will build on devices that can project this new virtual reality into the retina of our eyes, flooding our brain with virtual images and nothing else. Companies have experimented with such technologies for quite a while which has given us already the beginning of what most refer to as augmented reality. Augmented in the sense that the real world is still there but overlaid or augmented with a virtual reality, which its proponents argue will improve our life experiences. Again, for any such technologies there will be wonderful uses as for example helping all those with physical impairments which have left them with reduced sensory or bodily functions. If augmented reality can make the blind see in some way or can help the paralyzed move to regain control of their bodies that would indeed be wonderful. However, there is also again the challenge of how we ensure that our liberal societies keep functioning in any augmented realities or in the metaverse, towards which augmented reality is a stepping-stone.

Why do some believe our liberal societies can keep functioning when we allow new virtual realities to develop without the guardrails that safeguard our society's values and freedoms? We have seen what happens in our real world when we neglect to ensure the rule of law in all parts of our societies. We just have to look at those parts of our mega cities where people have to live in fear because elected official have seemingly accepted that the rule of law cannot anymore protect citizens in some city districts from those who seek to enforce their own rules. If a liberal society does not ensure that its values and freedoms are what governs all of its domains, the real ones as well as the virtual ones, it will become a society where parts of it are ruled by different kinds of laws and eventually it may devolve into a lawless society altogether. We can and should debate what must and what must not be part of the value set and the freedoms we all should endorse and be entitled to, and we must accept that we will have to concede some of our individual freedoms so that our societies themselves can remain free, fair and just. The latter was certainly always more of an ideal than a reality. Nevertheless, we must not give up our hopes that we will ever reach this goal as we continue our efforts to make our societies more just by making them more egalitarian and inclusive than they are today.

We do not know into what our still nascent digital world will evolve. In the early days of the internet, when the digital revolution really began, people had the wildest dreams and hopes of what it could eventually become for them. Many of these dreams, which at the time seemed outlandish or just unrealistic, have now, about a generation later, become reality. Hence, we should take the proponents of artificial intelligence and the metaverse, two of the most important new developments in the ongoing digital revolution, very serious in how they project those might change the future of how humans live and maybe also of how they die. While the preconditions, the replicators, for the realization

of machine entities or robots that really deserve the label artificial intelligence do not exist yet, the technological developments pursued under the artificial intelligence label nevertheless will quite likely fundamentally change the lives of countless millions. Most certainly, they will very much help to bring about the first realizations of a metaverse. How complete the sensory experience of being immersed in the first such metaverse will be, we cannot tell. Maybe, those early versions will only feed into some of our sense perceptions and we will have to wait a little longer before we can smell the aroma of a freshly brewed coffee in the metaverse. How long the perfection of a metaverse will take so it can provide a full sensory experience we do not know. We can expect that initially our lived experiences in the metaverse will be less rich than what our senses allow us to experience in the real world. However, with respect to the end of our lived experience, our physical death, our digital presence has already changed how others still can experience us even after we have passed away.

Increasingly people are discovering that the digital live of their loved ones does not end with their physical death. To a degree, the digital world has begun to provide at least from the perspective of some the digital equivalent of immortality. Just consider what before the digital age was usually left of a person after death. Prior to the invention of photography, all that people had left to remember a deceased person was their memory, or a few of their former belongings. Unless of course, one happened to be among the luckier ones who had the means to hire a painter and have their loved ones sit for a portrait while they were still alive. Some could even afford to have busts of their loved ones sculpted or they could have a death mask cast to preserve the image of someone dear to them. The latter was also a way to memorialize the image of an important figure of public life. The most important people were even honored with life-sized or even larger than life statues. Of course, ruling elites always had the means to ensure their continued existence in an after-life and they often did so with zest and seemingly to be remembered in eternity. The pyramids at Giza or the burial sites in the Valley of the Kings built for the Egyptian pharaohs or the terracotta army entombed with the first emperor of a united China, convey this message impressively even after thousands of years. At any time, the wealthy would seek to imitate their rulers in their attempts to secure an eternal life after death with whatever their means allowed them to accomplish in that regard. For the masses however, only the nineteenth century invention of photography would change how they could remember their deceased in a different way. Being able to capture the likeness of someone during their lifetime in a photograph could preserve their image for generations to come. With photography, many more than before would have a presence among the living even long after they had passed away. Photographs are fragile and fade over time and sometimes we can lose them forever in a fire, a flood or in other ways. Therefore, this new life after death attained through photography still had its limits. All of that began to change with the digital revolution. Initially only

providing a much coarser resolution than photography, the digital images of today are of the highest resolution and almost indestructible. Once uploaded into the cloud, the virtual storehouse of the internet, these images begin to have a life of their own of which we do not know how long it could last.

The footprint we leave in the digital world inches ever closer to giving us the kind of immortality that pharaohs and emperors of the past sought to secure for themselves. Let us assume for a moment that the metaverse has become a reality and we all not only live in the real world but in the form of our avatar also have a similarly rich life in the metaverse. What happens to our avatar when we die? That likely will depend on our will as we could choose to have it join us in our analog death with its very own digital death. However, we could also decide to let it continue its digital existence and thereby our own existence in the digital world. Why would we do that? Well, let us assume in addition that our sciences and technologies at one point in time have progressed so enormously that our avatars could not only reflect us perfectly from a physical perspective, even though we might opt to have a different physical appearance in the digital world, but also as a person. If our avatar could also perfectly reflect our psyche, our very own and specific version of human nature, then in a sense, our avatar could continue to interact within the digital world as if we were still alive. It also could interact with those we leave behind with our analog death and future generations of our family could be able to meet our digital presence hundreds of years after our physical death. How could an avatar even begin to reflect our psyche, how we feel and think? This is where artificial intelligence comes into play. During our lifetime, our avatars would essentially learn to be us. They would be nothing like a digital mimicry of ourselves, but they would learn who we really are, figuratively speaking they would rebuild the equivalent of our brains neuronal network structure, which underlies all of our behaviors, all of our human nature. In this sense, our avatar would continue to evolve, as it would continue to learn from its experiences in the digital world, but it would do so based on the personality structure we evolved in our lifetime. Much of that still is science fiction but what underlies the realization of any such phantasies towards which we have begun to take the first baby steps is something very fundamental, namely, our fear for death.

Likely most of us lead their lives as if there were an endless number of tomorrows, as if the Sun would keep rising for us for eternity. However, we know that this is not the case and we all likely acknowledge, if it is put to us in this way, that there will be a final morning after which we will not experience the Sun rising anymore. The finality of life is something that none of us will doubt. The two bookends of life, birth and death, are certainties for each one of us. We had no say in being born into this world and we have little say in how and when we will leave this world. The inevitability of death is for many something they rather not contemplate. Instead, we do whatever we can to push out our exit from this world. In that respect, we have been quite successful as on

average, humans today live much longer lives than their distant ancestors did. However, even if average lifespans have more than doubled that only delays the inevitable outcome, which is death. Looking at human culture, many would likely agree that a good number of our greatest triumphs, for example in music and art, are reflections of humankind's continued struggle to come to terms with the finality of life. Our individual lives have a limited span, but the music of humankind's great composers lives on and with that the memory of those composers themselves. Similarly, the psychosomatic warriors of Homer's time we encountered earlier in the book, even though their concept of body and *psyche* was so different from ours, hoped to outlive their corpses through heroic deeds which posterity would remember in song and poetry for eternity. Trying to cheat death in this way, consciously or not, is something that has been part of human nature probably since the time humans first grasped the finality of life in a way very different from our animal cousins. Most animals know fear of death no less than humans do, but they know it in a different way. We have to be careful what we read into animal behaviors but what we know today about how some mammals such as elephants deal with the death of a group member, tells us already that something goes on in animal minds, which is beyond what we believed animals were capable of experiencing. However, grappling with the finality of life as humans do is very different from how our closest cousins, the great apes, still mourn the death of one of theirs. Maybe it is not the presence or absence of consciousness, which differentiates human animals from all other animals, but it is how we can use this consciousness to contemplate questions like the finality of life.

Looking at how humans deal with the finality of life and with the inevitability of their own personal death gives us a different perspective on the new digital life we can have. This digital life has now become for many their online presence on social media and something they need to think about in their wills. What should happen to my digital presence after I am gone, how should it continue, who should inherit it? Such questions are already familiar to many notaries around the globe. However, if the metaverse ever becomes what its most fervent proponents hope it could be, then we really need to look at our digital life in a very different way. What could it mean for us, individually and as societies, if at some point in time humans continued to have a digital live long outlasting their physical existence? Will those avatars have any legal rights in the metaverse akin to the rights that we have today in our real world? Will this virtual world of the metaverse bleed into the real world and make our avatars legal persons in the real world? Given that the world population will top out at some eleven billion people in 2100, as each new generation acquires a digital life, at some point those continuing to lead digital lives long after their physical death will far outnumber those alive at any given time.[44] Maybe Hollywood got it all wrong and those multitudes of avatars whose masters have long passed will be the real walking undead and not some kind of zombies continuing to lead a precarious fleshy existence.

There are many more things one could contemplate when exploring what the consequences of a metaverse could encompass. Undoubtedly, such a metaverse will also bring much good and could enrich our lives in many ways. However, given human nature and the potential impact that such a technology may have in many other unintended ways, we should watch it carefully and adjust our societies and its laws to ensure that whatever happens in the metaverse remains bounded by the values we hold dear as the foundations of our liberal democracies, without which our societies would surely crumble. For all those who look towards the metaverse as their version of personal immortality, here is a cautionary note. In case humankind does not succeed in its efforts to venture into space and spread the human species among the stars over the next one billion years there likely will be an expiration date on any digital universe humans may have built by then. That expiration date is the death of our planet once our Sun will engulf it, as it will expand beyond Earth. However, long before that happens, living conditions on Earth will not be able to support higher life such as humans anymore. Eventually, the digital human universe will go up in smoke, as the human digital universe will not outlast our planet. It may be a good thing if we learned to be a bit humbler and accept that life is final and all that matters is what we do with and accomplish in our real lives. Having a digital life is an interesting and entertaining possibility but what really matters are the lives we live in the real world. More likely than not there are many things in our real lives where artificial intelligence and the metaverse can us help to do just that. This certainly would benefit all of us more than wasting ourselves in digital lives of vanity trying to escape the reality in which we all are born and in which we all die.

Some may ask the question in which way the promise of a continued life after death in a digital world of avatars may in a way address the same spiritual need religions address by consoling believers in their fear of death. In seeking an answer to this question, it does not really matter whether the latter comes via the promise of an individual immortality for those leading a good life or through dissolving ones individual existence within something greater via the successful discovery of one's true self. The worldly quest to lead a more exciting life in a metaverse than what the real world can provide may sound like a very different undertaking than the pursuit of a life embedded in religious belief. However, that is not really the case as both only represent different approaches to cope with the challenges we face in the real world. In this sense, it does not matter if we take refuge from our real lives in religion or philosophy or in the metaverse. What does matter however is how anyone's life immersed in one or several of those three spheres inspires and defines their actions in the real world. As to what comes after life, the promise of digital immortality in the metaverse does indeed differ from what religions hold out for their believers. What does it say about us that we take ourselves to be so important that some seemingly believe it would still matter to anyone in say a hundred million years, which from a human perspective is the same as eternity, who they were as individuals?

That is hubris and it is this lack of humility, which sets the belief in a digital immortality apart from the consolation which religion or philosophy continue to provide for many today.

We can expect that in the years to come we will see evermore gadgets, robots or other autonomous entities which seemingly will present us with the fiction of a world increasingly governed by artificial intelligence. In all of that we must not forget, that it is humans who ultimately always make the decisions and whatever any seemingly artificial intelligence entity seems to do, it will always be governed by human created algorithms. However, in the ways we interact with such entities we often do so nowadays in a manner that treats them more like sentient beings. Maybe many will already have caught themselves talking to a machine as if it were a pet; others may even talk to their artificial intelligence companion, as if it possessed a nature of itself such as all of us possess a shared human nature. Humans are capable of humanizing machines just as they are sometimes capable of dehumanizing their fellow humans. How can you for example not thank a nursing robot, which will one day take care of you in a nursing home or some other kind of senior living facility? Elderly people cared for by human nurses are grateful to their nurses. Why would they not be similarly grateful to a robot, specifically if its caring bedside manner comes with a humanoid appearance?

How we eventually will structure human relations with what we may call by the moniker of artificial intelligence, even if it is nothing like that, will set the framework for whatever a world will look like, where increasingly machines do what humans used to do. What will humans do when machines will do most of the work? How will that change the work that humans will continue to do? What happens to societies were mostly robots do the work and humans do little or nothing? How can humans earn a living? Will we begin to tax robots for their earnings or the profits they make for their owners and how will we distribute that wealth? If we ever should come to conclude against all odds that some artificial intelligence entity indeed should deserve to be called truly intelligent we will face yet another challenge. We likely all agree that intelligence goes hand in hand with self-consciousness. Hence, any truly intelligent artificial intelligence entity will have consciousness and with that, we will have to consider what rights we should avail to such a being. Maybe we eventually will have to draft something along the *Three Laws of Robotics*, which the scientist and acclaimed science fiction author Isaac Asimov (1920 - 1992) advocated in his writings.[45] What kind of laws humans and true artificial intelligence entities will ever live by or if it even ever comes that far, we do not know. However, we already know today that the massive way in which anything under the umbrella of artificial intelligence and its derivative technologies already is beginning to impact our everyday lives requires us to consider its broader and long-term societal impacts so we can prudently shape our futures and do not just fall victim to them.

Us versus Them

So far, most of the aspects of human nature explored here, in the manner chosen to discuss them as well as the implications drawn from such discussions, may paint a rather dark picture of human nature. However, nothing could be further from the truth. The richness and diversity of human nature is astounding and we rightly celebrate that in many ways. Unfortunately, while celebrating the diversity of human nature as it expresses itself for example in the richness of cultural traditions around the world, we cannot ignore that all too often we are also guilty of belittling it, or in deference to a mistaken perception of our own cultural values, frequently ignore or even defile it. Why is it that many who for all appearances enjoy traveling to countries where they can experience cultures different from their own, are often reluctant to embrace members of those very cultures when they happen to become their neighbors at home? Our modern cities reflect the diversity of their populations and even smaller places in the countryside now are a lot more colorful. We do not have to travel far anymore to experience a cultural diversity which earlier generations only ever associated with faraway places. The environments our children grow up is a multicultural one where next to the still dominating traditional culture of a given region and place many other cultures are now at home. That makes it even more baffling why we are witnessing such things like resurgent nationalism or increased levels of xenophobia in many of the places we used to think of as being liberal and multicultural. Why has the label multicultural in many instances become associated with negative experiences, often now used as an intentional ethnic or racial slur?

Only a few decades ago, millions perished in the genocides of the Second World War because they were perceived as being different from those among whom they had lived for generations or they were judged inferior by those who had just conquered their homelands and thought of themselves as the master race. Different or inferior in the race they supposedly represented, the ethnicity they belonged to, or the culture with which their oppressors and murderers associated them. Sometimes their killers judged them as different and inferior in all and other times in just one of these aspects. In Nazi Germany, to those among the persecuted Jewish population still embracing their religious and cultural traditions all three judgments would apply. However, there were also a few otherwise veritable members of the supposed master race whom the Nazis branded nonetheless as cultural Jews because they happened to embrace aspects of science and culture, which Nazi ideology considered as Jewish in nature. The masters of the Nazi master race really did not have rational explanations for any of those judgments and they had no need for them anyway because everything in Nazi ideology was about belief in the first place. Nazi ideology always was a belief system and not a rational construct. The same is true for similar ideological and racial beliefs various racist groups still embrace to this day. For a number of decades after the Second World War, groups embracing such

racist and nationalist ideologies could only exist at the very fringes of society. However, that has changed and today the virus of racist and nationalist ideologies is again infecting the body proper of civil society. In fact, you may discover that seemingly decent people, upright citizens and church-going Christians, maybe some even neighbors of yours, now may front arguments that in the past one only could have imagined coming from proper Nazi sympathizers. Why is this happening, where does this come from?

To answer any such questions, it would be most helpful if we somehow could send some of our distinguished anthropologists back in time to study the social and group behavior of our very distant ancestors. Being able to go back in time some two hundred thousand years and follow a group of the newest branch of the human family tree that just began sprouting, *Homo sapiens*, living among its members and exploring their social behaviors and rituals would certainly be any anthropologists dream. We all know that such time travel only works in science fiction and that all we can learn about our ancestors and how they lived comes only from fossilized bones or teeth, any usable genetic material they may still contain, and the wider archaeological record they left behind in various forms. The latter does of course include the marvelous cave paintings and rock art of our distant ancestors, which we associate with the beginning of the so-called Cognitive Revolution in human prehistory, flowering some forty-five thousand years ago. Those paintings and symbols have a captivating presence we can still sense tens of thousands of years later and they must have carried a deep meaning for the people who made them. For the most, we can only speculate today what motivated our ancestors to create such things in the first place. We can admire what we refer to as their art, but we only can guess what kind of meaning it may have carried for our distant ancestors. Looking beyond what we can possibly know about the social life of a small group of early modern humans, we know even less about the kinds of relationships, which could evolve between any such bands of modern humans. What we know for example is that bands of chimpanzees sometimes literally go to war with each other. Chimpanzees are our closest relatives in the animal world as chimpanzee and human lines separated only some 5-7 million years ago. Hence, how groups of early humans dealt with each other may have included similar behaviors as we can still observe them between groups of chimpanzees. Viewed this way, the instinctive *us vs them* behavior which still shapes so much of human group dynamics has likely deep evolutionary roots.

While it may well be that such kind of human behavior, positing those who belong to a group against those who are considered outsiders, could have deep biological roots, that does little to explain most such behavior that we can observe today in our modern societies. From time to time one can still observe the so-to-say naked *us vs them* response in conflict situations. However, our societal rules do not really allow this to play out anymore and so it seems that we modulate our instinctive *us vs them* responses to make them useful for addressing other societal situations, which may put us into non-violent

conflicts with less directly discernible adversaries, real or perceived ones. It would not be the first time that our higher brain functions make use of older and simpler brain routines encoded in what we refer to as emotions or instincts. In essence, the way *us vs them* responses often play out today, they are for the most culturally modulated and conditioned and only rarely do we observe the bare instinctive response anymore though as we know from experience, it can break through the surface of our thin cultural veneer at any time. Any such behavior has little to do with intelligence. History tells us that otherwise seemingly intelligent people are only different in this respect as they often exert a greater leverage in society and therefore the positions which they hold in society are often more influential than those taken by people of lesser societal status. The role that academia played in Nazi Germany is a good example for that. People who took pride in being among the highest representatives of one of the best educational systems at the time, who counted themselves as members of one of the most cultured nations on Earth, nonetheless had no problems to sanction and justify the most horrific crimes against humanity ever committed by any people or nation.

None of us is born to become like that. How we handle our *us vs them* impulses and how we perceive and treat people who for whatever reason are different from us will largely depend on how we grow up. Even within seemingly homogeneous cultures, we must learn to live with differences of a social nature, which for example assign us to different subgroups within our societies, making us for example members of the working class and them for example members of the rich. We likely all have made similar experiences when growing up in how tastes change over time, positing younger members of society against older ones as they embrace different kinds of music, literature or political views. As we grow older, many among us will make a reverse experience as their progeny rejects their tastes and confronts them with a new way of life no different than they had done themselves with their parents a generation earlier. In either case it will be *us vs them*, the young generation vs the old generation. While that may result in lively debates, such *us vs them* contests seldom go too far. For the most, whatever posits people within a culturally homogeneous society in a *us vs them* situation can usually be resolved peacefully and is unlikely to tear the fabrics of society apart as they both still see themselves in many ways as parts of the same larger culturally homogeneous group. However, there are of course cases where such conflicts cannot be resolved peacefully anymore. When sections of society move too far apart such that they look to each other more like different societies altogether then a new dynamic comes into play. In a sense, this is what played out before the French Revolution. The societal consensus reality of the pre-revolutionary French aristocracy was so far removed from the reality of the rest of France that it essentially became a foreign body within French society. The *us vs them* response this would eventually trigger between commoners and aristocrats was in many ways more like what one could expect to materialize against a foreign invader.

Something that we usually overlook or are not even aware of is the simple fact that who we come to see as us and who we eventually deem as them is not in any way predetermined. First, when and where we happen to see the first light of day is not up to us in any way. All of us were participants in the greatest lottery on Earth, the birth lottery. If we were lucky, we happened to be born into a loving family at home in one of the liberal societies, which on the one hand value personal freedoms and liberties and on the other hand care for the well-being of all their members. Those who were less fortunate grew up in societies without the one or the other and often neither of them. They had to face an uphill battle and often had to overcome many more obstacles to achieve in life what the luckier ones in wealthy liberal democracies take for their birthright. However, that is not the end of the role chance often plays in our lives. Inadvertently, what we usually tend to perceive exclusively as our own accomplishments we just as frequently owe to a good deal of luck and often what we seek to achieve in life requires a combination of both, hard work and good fortune. Fortuna can smile on us in many ways, opening up doors and opportunities for us that we otherwise may never have had.[46] Mistakenly, we all likely attribute much of our life achievements only to hard work and perseverance and often forget that if our external circumstances would have been different we could not have accomplished much at all. Frequently the latter pretty much reflects the experience of those less fortunate among us who happen to work just as hard and dedicated as everyone else to get ahead in life and still seemingly get nowhere. In the end, the *Us* we eventually come to believe that we belong to vs the *Them* that defines for us who the others are is just as contingent as most everything else in life. That is, if we leave it at that and do not make conscious efforts to shape our consensus realities such that we do not fall victim to the default *us vs them* stereotypes that we can find all around us. Early in life, every single one of us was much more open to take a perspective on life different from our parent's generation views. As we grow older, we increasingly become set in our ways just as much as our parents were and if by then we happen to embrace the same kind of prejudices and misconceptions they may have held we will likely hold onto them for the rest of our lives. The formative years of our lives are the ones where we still are open to see the world with different eyes than how large parts of our society want us to perceive it.

Those formative years very much differentiate us from our closest relatives in the animal world. They also are probably at the heart of that which enables human societies to evolve in ways not accessible to chimpanzees. When we seek for explanations as to why human societies were able to change and adapt so rapidly over the last few thousand years, we tend to emphasize the extraordinary capabilities of the mature human mind. How else could our ancestors have created such a rich diversity of human cultures and how else could one explain the technological progress giving modern humans a level of control of their environments no other animal possesses? Our modern world owes indeed

much to the creative minds of adult engineers, scientists, artists and politicians who have helped shape human societies and cultures into what they are today. However, in doing so we overlook that all of those creative adult human minds once were the minds of children. The environments within which children grow up have to provide the conditions that afford them the opportunity of evolving this kind of creativity in the first place, even if that creativity comes about in a rebellious way as it sometimes does. Being able to see the world in new and different ways is by necessity a precondition for creativity. Social environments which not only allow for that to happen but actively foster such developments will see many more of their young evolve to become creative adults than societies which put their children in the equivalent of mental strait jackets. Although in the latter case, some creative minds may still rise in rebellion in the long-term, such totalitarian societies will struggle to compete with those who value creativity and allow it the freedom it needs to develop.

Intuitively, most of us will acknowledge that truly creative societies are bound to be much more diverse than their counterparts are where immediate social environments, broader societal rules, or both limit what people can think or aspire to become. To appreciate the role the environment plays where one grows up, it helps to pause for a moment and think about the kinds of futures that we all hope our children will have and then consider what we do to make that happen. That should give anyone a good idea of just how powerful an influence the environment can have on our children in shaping their conceptions of the world, often of course in a good way but all too often also in a bad way. However, more often than either of those circumstances being the consequence of guidance or compulsion, it is purposeful neglect, which will determine to what extent the talents of our children can evolve. There was a time where the dominating influences during the most formative years of young people came indeed from one's home, one's schools and from a small number of neighborhood and school friends. Today, the most important influences shaping how young people see themselves and others in our world originate in the virtual worlds of social media and popular culture. In many ways, they shape the consensus realities in which our children make themselves at home, much more so than their families or teachers likely do. The purposeful neglect in all of this is of course that we let this mostly just happen. On the one hand we provide our children little guidance or assistance in how to deal with the social pressure they subject themselves to in their virtual lives. On the other hand, our societies fail to create and enforce supervision and regulation for the virtual word of social media in which too many young people shipwreck or sink their lives altogether.

Many of us are familiar with Charles Darwin's description of eighteen subspecies of Galapagos finches he encountered on those Pacific Islands and the remarkable diversity of beak form and function among them. Since then, biologists have discovered many more examples of this extraordinary richness of nature where species adapt and evolve to take

advantage of almost any environment that can sustain life on our planet. Humans are no exception to that even though there remains only one human species since our last cousins in the human family went extinct some several ten thousand years ago. The remaining human species, which is us, has made up for this lack in human species diversity or biological diversity through something uniquely human, the flowering of human culture into countless distinct forms rivaling in human culture diversity anything that other living organisms may possess in biological species diversity. Critically, the biological species diversity we can observe among other animals and human culture diversity differ in important aspects. Biological species diversity is something possessed individually, that is, one subspecies has a certain characteristic that another one has not, and those characteristics are not transferable. Expressions of human culture diversity are possessions of specific groups to which one belongs by birth, by accident and sometimes by choice. However, and that is the most significant difference to other animals, humans cannot only acquire unique cultures of their own but can also share their various cultures. There is a certain amount of behavior sharing biologists have observed for some species, a well-known example of course being that of songbirds. We all likely know that parrots and budgies are capable of imitating human voices. Less known is that many birds are capable of imitating the songs of other bird species, the common starling being an example of that. However, as exciting, interesting and as little understood this is as a factual matter of biology, human culture sharing is something many levels removed from the kind of sharing one can observe between other animal species such as birds sharing their songs. Humans are capable of essentially sharing everything that is part of a specific culture, provided they possess the willingness to immerse themselves completely in a culture other than their own.

Admittedly, phrasing the potential of human culture sharing in this way is a stretch. For a certain culture to become our very own we have to be born into it and absorb it for a good number of our earliest formative years. By definition, this is usually not the case for a foreign culture. Hence, unless someone grows up in a family equally at home in two cultures, it is unlikely that anyone could ever immerse himself or herself in two cultures to the same extent. Acknowledging that, immersing oneself in another culture can still widen one's own horizon and enrich one's life just as much as if one were born into that other culture. There is however, one very simple precondition for that to become a possibility and that is one's openness to learn about cultures different from one's own. Unless we open ourselves to explore another culture without prejudice but with boundless curiosity, it is quite unlikely that we ever will come to appreciate what another culture could teach us about others and ourselves, about them and us. That brings us back to the importance of the formative years in the lives of young people when their consensus realities evolve. One result could be that their consensus realities come to replicate value sets forced onto them by those around them indoctrinating them

with their own narrow views, or equally bad, society shaping their consensus realities such that they conform to the propagated societal consensus reality as is the objective in autocratic regimes. A more desirable outcome would of course be that their consensus realities evolve to endorse an openness towards exploring views and perspectives other than those endorsed by societal or communal consensus realities seeking to preserve their narrower views of the world. The stark reality is that unless during those formative years of adolescence humans learn how to learn and learn how to examine their own views just as much as those of others, it will be much more difficult for them to tolerate or even embrace how others can see the world quite differently through the lenses of their cultures.

As humans, we define ourselves as much in a negative way as we do so in a positive way. Most of us would have no problem to produce a two-column list, itemizing who we believe we are and the qualities and attributes we think describe us vs who we believe we are not and the qualities and attributes that go along with that. Being able to separate oneself from what is not oneself is a precondition for our development as persons in the first place. The discovery of self is hardwired into human biology, and it takes humans years to develop their individuality. This process begins when a baby discovers its separateness from its mother. With this onset of self-discovery comes also the discovery of the other, that is of who, which, or what we are not. The individuality of human nature develops within the polar framework of the self and others. Without an evolving conception of others, how could we ever become individuals? Clearly, there are parts of our *us vs them* perception which deeply anchor our very self. The *us vs them* separation of how we see the world is part of us becoming the person we eventually will be. We have no choice in that but our environments do and at some point, we ourselves have a choice to frame and define the context of the specific *us vs them* themes, which will come to shape our lives, the communities we belong to, and eventually our societies.

Our biological evolution has provided us with a genetic blueprint of how to discover ourselves as individuals, a process we all go through as we grow up. The extent to which communities and societies guide, influence, accentuate or dampen this process shapes in many ways what kind of consensus realities will in turn come to shape those very same communities and societies. Putting aside religious, philosophical or political considerations, we can raise our young to become for example egotistic hedonists or we can raise them to become responsible citizen who care for the well-being of others. Of course, in neither case can anyone guarantee the desired outcome but on average, societies practicing what they preach, even if nothing of it is intentional, have a better chance of moving into the desired direction. There are of course quite a few limitations to that, some of which we already discussed above. The only reason to bring this kind of societal engineering up again is to highlight why any of that may happen in the first place. Our culture evolution has brought us to a point where humankind faces challenges which

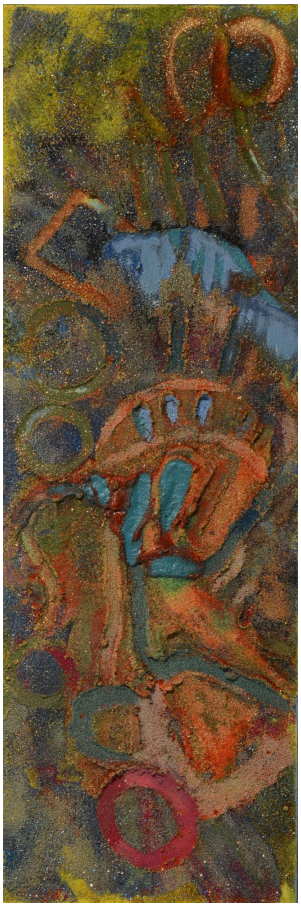
require us to change our current ways of life to an extent like nothing else ever forced us to do that before. Humans have not evolved a biological mechanism that would allow them to embrace the concept of greater common goods in the same way as we embrace the concept of *us vs them*, or put differently, humans seemingly lack an adequate concept of *We* that would allow them to address the challenges of the modern world. Humans certainly have a concept of *We*, but that seems to be just subsumed in the us-part of the *us vs them* concept. What humans require is a concept of *We* that could allow them to act on a species level. Maybe some time in the very distant future humans will have evolved the kind of species consciousness that could support such a *We* concept but for now, any such notion remains science fiction. Hence, we must find ways to give our societies a framework that can replicate something like a *We* concept at a level that allows us to address species level challenges such as for example climate change. How can humankind balance these *us vs them* contests, which still dominate relationships within and between human societies with a concept that focuses on the *We* instead of the *Us* or the *Them*? It is difficult to imagine a bright future for humankind if we fail to find a way to do just that.

Evidently, in envisioning solutions to the challenges humans face, it has become very tempting to look towards genetics and the promises genetic engineering may hold. This begins at how we could engineer plants, specifically food crops, which would better cope with climate change and goes all the way to genetically engineering human beings. Here is not the place to argue why any of the former may only provide limited solutions or why ethical concerns will make the latter, for example genetic engineering of humans so they can better cope with extreme climate conditions, a difficult proposition. Maybe if scientists discovered human genes for altruism and found ways to activate them in all of us, this may be an easier sell. But then, maybe not at all. How after all could capitalism keep functioning if we all would suffer from altruism? Leaving such considerations aside, there is a more fundamental question here. Why is it that when looking for solutions to the problems humankind faces we are seemingly much more interested in our potential abilities to shape biological evolution and specifically human evolution rather than shaping human culture evolution? Human culture evolution created the species level problems humankind faces today and their solutions will have to come from redirecting our culture evolution to ensure the long-term future of life on Earth. It is as simple as that. However, given the nature of human nature, it is more than questionable if human culture evolution can change as quickly as needed and to the extent required. It is not impossible, but it seems unlikely to happen, not within the time we still have to accomplish such change nor to the extent we must make that happen.

If humankind cannot come together to address the challenges confronting us together than we can count on much more dangerous *us vs them* contests to eventually dominate relationships within societies and more importantly between them. Is there anyone who

does not realize that if over the coming generations we cannot resolve what in essence will become major resource conflicts, the likely outcome must be increased armed conflict? What that will mean for a world with a nuclear arsenal that could wipe out humankind several times over nobody can know. However, it is a safe bet that unless humankind discovers a path towards developing a species level sense of belonging, purpose and meaning, a veritable *We* of humankind, our biologically rooted *us vs them* concepts will acquire a new meaning once it will be nature that turns against us.

An Outlook



What's Left

From a biological perspective, at some point in time, we can expect that humans will understand everything there is to know about the human brain. However, we cannot yet predict just when this will happen. It could occur as soon as in a generation or two, it could take a few hundred years, or it could require several thousand years for our sciences to advance to a level where such an understanding will be possible. Given the progress of human science over the last few hundred years, it is hard to imagine where science could eventually take human understanding of ourselves and of the world around us. This is the promise that science carries, the promise of certain knowledge at some point in time giving us an understanding of the natural world we cannot even imagine yet. But will that also tell us how the human mind works? Will knowing all the biological processes underlying the functioning of what we call the human mind also equate to a more or less complete understanding of the human mind itself? Is it even possible to answer such a question given what we know today? If not, can we at least hope to have an answer sometime in the future? Even though the workings of the human mind are in many ways still a mystery to us, we must concede that this is a bit less the case today than it was just a few hundred years ago. Since the onset of the scientific revolution, we have learned much about the human biology that presumably must underlie the human mind. Speaking about a biology of mind would have been preposterous only a few generations ago but this is exactly what scientists have come to explore over the past few decades. With the emancipation of psychology from philosophy, we also gained a more profound understanding of the human psyche. Increasingly what we are learning about the functioning of the human mind is becoming a science by itself of which psychology, biology but also computational sciences are integral parts. This is a very different situation than what we can find when we look to how our ancestors approached the study of the human mind before the onset of the scientific revolution in the seventeenth century. Back then, scholars focused on the philosophical and religious exploration of human nature and of the human mind, both of which had flourished since antiquity, rather than on any

scientific exploration of their underlying biology. At any given time in human history, regardless of the scientific understanding of the nature of human nature or of the human mind, what always mattered most were the conceptions we held about human nature or the human mind. It was the latter which invariably informed the structures and rules we gave our societies rather than any positive knowledge we possessed about the nature of human nature or the human mind.

Before the pursuit of science began to enlighten our understanding of human nature, it was not any knowledge as to how the human mind may or may not work which mattered in how humans lived their lives. Rather, what really mattered at any time in human history were the beliefs our ancestors held with respect to the human mind. Such beliefs about the human mind amounted to nothing less than beliefs about human nature itself. For most of recorded human history, our ancestor's conceptions about the nature of human nature formed around religious beliefs or philosophical speculations. This only began to change when our modern medical, biological and psychological sciences began to supplement those beliefs with factual knowledge. While the body of this factual knowledge keeps growing, much of our conceptions of human nature are still rooted in belief and not in science. In many ways, what we refer to as folk psychology reflects much of our pre-scientific understanding of human nature. At its best, folk psychology includes the practical wisdom of our ancestors, at its worst, it endorses all kinds of human prejudices and superstitions. For better or worse, folk psychology has served humans in their practical understanding of human nature for many thousands of years and that is likely to continue. However, maybe at some distant point in time humankind will eventually possess a complete scientific account of the nature of human nature. That of course begs the question what the relevance of that would be, if any.

We cannot be anything else but humans. Clearly, all of what any of us are doing or not doing always comprises expressions of a shared human nature. In this sense, we actually do have a factual knowledge of human nature. We just have to imagine a being, which possess all aspects of human nature in whichever ways humans have ever expressed their individual human nature since the dawn of the human species. This sum total of all possible expressions of human nature embodied in such a being would then be a good approximation as to what our human species nature really looks like. Unfortunately, such an imaginary being would have to possess a split personality. Maybe akin to how Robert Louis Stevenson (1850 – 1894) portrayed it in his 1886 published novel *The Strange Case of Dr. Jekyll and Mr. Hyde*, where a potion turns the good Dr. Jekyll into the evil Mr. Hyde only to reemerge again as Dr. Jekyll once the potion wears off. We usually have to take such a dualist view when looking at the whole of what we deem good or evil in human nature as it seemingly is next to impossible for us to imagine a single being that possess both.[1] While at some point in time science may give us detailed accounts as to the true identities of human nature's Dr. Jekyll and Mr. Hyde, we already know that

they exist and we know many of their respective behaviors all too well. The question really is not whether they exist or not, or what they are capable of, but rather how we perceive those aspects of human nature and how we factor them into our individual and societal consensus realities.

Today, the greatest challenges for modern liberal societies come from within and not from without. Inequality is nothing new for modern liberal societies. To some extent, a certain level of inequality is unavoidable because even in the best of cases equality of opportunity just does not translate to equal outcomes. Most of us will understand this and acknowledge that what really matters is the level of inequality. Unfortunately, as we all know, in our modern societies the level of inequality has increased dramatically over the last forty to fifty years. Not too long ago, the gap between rich and poor was much smaller than it is today. Importantly, sandwiched between rich and poor there still existed a thriving middle class, with which most in society would have identified. Modern liberal societies are much less homogeneous and egalitarian today. Since the 1970's, policies serving mostly a politically powerful minority, eroded and fractured what once was a thriving middle class. Some parts of the former middle class managed to join the ranks of the wealthy, many ended up economically worse off, and only a sliver of the former middle class remained. This happened practically in all liberal democracies to some extent but in some of them more so than in others. Real wages have pretty much stagnated or even declined since the 1970's in places such as the United States and in parts of Western Europe. Where once a single breadwinner could support a family, today many parents must work multiple low paid jobs just to scrap by. Access to higher education is now out of reach for children in many families. In many ways, from an economical perspective, the social bodies in modern liberal democracies begin to resemble social structures similar to those of industrializing nineteenth-century nations. Back then, aristocrats or the very rich and sometimes both together possessed what essentially amounted to a power monopoly in those societies, supported by a small class of burghers or bourgeois whose interests the ruling elite served as well. The bulk of the population, however, were farmers and workers. The latter forming the mostly impoverished proletariat of the modern industrializing world and the former often subsistence farmers and farm hands who may have had a somewhat better live than many industrial workers but were otherwise not faring much better. There are few farmers left today with the number of people employed in farming not exceeding two percent of the population in most modern societies. The bulk of the modern-day equivalent to the nineteenth-century proletariat are not industrial workers either. Many of them have never worked as industrial laborers because today the majority of labor comprises all kinds of jobs in the service sector.

Compared to the proletariat of the nineteenth-century, its modern-day equivalent represents much less of a uniform body. It was this uniformity of the nineteenth-century proletariat, which enabled for example socialism and to some extent communism to appeal

to large swath of the population back then. There is no danger today of any political ideology to have a similar appeal to some modern-day equivalent of the nineteenth-century proletariat. The nineteenth-century proletariat had a shared consensus reality, something that political parties could successfully appeal to, just like the ruling elites back then appealed to the burghers or bourgeois. Today, there are few consensus realities shared to such an extent across large sections of society. Our modern societies have become much more fragmented and diverse in their compositions including now many smaller groups sharing their very own consensus realities, which help them to make sense of the world around them from the perspectives of the perches they occupy in society. With the middle class in modern liberal democracies splintering, the consensus realities associated with them also have splintered. Until recently, this middle class had not only been the backbone of modern liberal societies, but it also carried the implicit broad societal consensus on what was good for society as a whole. Of course, there were always political differences, but few doubted the integrity of governments and their institutions whose agreed purpose was more or less to serve the prosperity and welfare of the broad populace. Putting aside questions as to how realistic a goal this was in the first place or as to how effective governments and their institutions really ever were in working towards that goal, at some point in time this perception changed. Many today do not see themselves or their interests represented anymore in the governments and institutions of modern-day liberal democracies. In the long-term, liberal societies will only remain functional if they can draw on a broad consensus throughout their populations. For such a consensus to exist, a society must be perceived as serving the interests of all its members equally. This is not the case anymore since quite a while and with that, many in our modern societies have quietly withdrawn from what was once a broadly shared consensus reality, namely, that modern society seeks to serve all its members equally.

Over the last forty to fifty years, we have witnessed the tribalization of what from an economic and social perspective once were rather more uniform societies. To establish and maintain tribal consensus realities requires communication. Today, it is cheap and easy to communicate in many different ways over large distances. Modern communication has literally shrunk the world so that it has become possible for small tribal groups to sustain themselves with members splintered across a country or across nations. In the past, members of a consensus reality with controversial tribal views would have had to face local reality checks so to say. No such reality checks exist in the virtual reality in which today many controversial consensus realities are spreading. We literally live today in a world of make-believe. Sometimes it almost feels as if there are parallel universes where beliefs can become reality and fake news can replace real news. In many ways, the very strength of liberal societies has become a weakness, or so it seems. In a free society, everyone can believe whatever they choose to believe, and everyone can invent and propagate their own version of reality with few limitations. As long as much of the populations in modern-day

liberal societies were part of the broader societal consensus that really did not matter. However, with the increasing fracturing of our modern societies into small minorities of winners and large majorities of losers that has changed. If large parts of a society believe themselves to be abandoned by society to the extent that those feeling left behind begin to endorse consensus realities asserting that there is no recourse anymore within the existing framework to change their situations for the better, this will have inevitable consequences. One of those consequences clearly is the rise of populism in politics that we can witness now for a while in many countries of the liberal West, undermining ever so slowly the very foundations of a democratic society. But it is not the populists which we should blame. They are just taking advantage of opportunities presented to them. Populism only has a chance to succeed if a substantial part of a society's population is dissatisfied. We should blame the politics of democratic governments over the past forty to fifty years. Almost none of them were populist governments but over decades, practically all of them served special interests at the expense of serving the broader populace with the result of eroding the very fabric a strong democracy requires to survive. Over these decades, it has been the shared consensus realities of lobbyists shaping policies and politicians allowing them to do so, serving corporate or other monetary interests at the expense of the public good, which have brought us to this point.

Adding to the perceived and felt sense of powerlessness to shape their own future, which many of those experienced who, broadly speaking, once would have been part of the blue- and white-collar working class in modern industrialized societies, were the economic and societal impacts of globalization. Most of us when they hear the word globalization think about the economic phenomenon that was a simple consequence of nations seeking new agreements on tariffs and trade after the Second World War. In the early 1980's, when my maternal grandmother and I were often doing the dishes after a Sunday lunch we would listen to a radio station we both liked. Invariably, the intermittent news updates on that station would sometimes report on the progress of the so-called General Agreement on Tariffs and Taxes (GATT) negotiations. Those GATT negotiations focusing on reducing tariffs and taxes on trade began among twenty-three signatory nation members in 1947 with more acceding to the agreement over time. Many will not be familiar with GATT, but most will know its successor organization, the World Trade Organization (WTO), which in late 2021 counted one hundred twenty nations as its members, representing some ninety-eight percent of world trade. However, the globalization of trade is not a twentieth-century phenomenon. Enterprising businesses have sought out foreign markets for as long as humans have been trading. Well-known examples in history include the Silk Road or the routes of spice traffic, first emanating from the Arabian Peninsula and then replaced by newly discovered sea routes. The European age of discovery turning into the age of empire and colonialism unleashed a new wave of globalization where the benefits of global trade were rather more one-sided. The same is mostly true for the

globalization wave unleashed by the beginning industrialization, which culminated in what in the eyes of many contemporaries seemed like a true golden age for world trade, until the First World War put an end to it. Even though this golden age for world trade was not so golden for most of humankind, many in the industrializing nations benefited from it. For the most, businesses in their respective nations still had to rely on the work force available within their countries borders. With that, workers had a bargaining chip, which they put to good use. In many countries they did so to such an extent that in some ways one could call this golden age of trade that ended in 1914 also a golden age for workers unions.

The modern globalization we have been witnessing was quite different because as it progressed it increasingly became as much of a work force globalization as it was a globalization of trade, pitting highly paid and often unionized workers in industrialized countries against workers with little rights and much less pay in developing countries. Businesses in the developed nations discovered that they could become much more profitable by outsourcing their production into developing countries where they would be able to dictate pay because workers unions were powerless or nonexistent. This trend very much accelerated with the dissolution of the former Soviet bloc and with China embarking on its transition from a communist market rule to a capitalist state directed economy. With that, over only a few years a cheap labor force counting in the hundreds of millions became available to Western businesses who sought to reduce their labor cost. In essence, by taking advantage of this business opportunity to maximize their profits, many companies abandoned the labor forces in their home countries, which had enabled them to grow in the first place. From the perspective of many workers in those developed nations, this felt not just like a betrayal but more like treason, their former employers abandoning so-to-say the social contract they implicitly had with the country and the work force that had allowed them to operate and grow into the kind of global businesses they eventually became. At the same time, those very same businesses would be instrumental in raising millions of other people out of poverty in developing nations. This did not somehow happen because of the benevolence of business. Rather, it was simply a consequence of the economic structure of those developing nations. What from the perspective of a developed nation worker was an extremely low wage would go a long way in developing nations to sustain a family and eventually provide that family's children a better future. Lifting hundreds of millions of workers in China and other Asian countries as well as in Eastern Europe from subsistence or poverty levels to what in those countries would become a new middle class came at the expense of the working middle class in the developed nations. The beneficiaries of this work force globalization were the many millions in developing countries who could now aspire to a better life as well as the capitalists who owned the companies benefiting from a much-reduced labor cost. As the purchasing power of populations in those developing countries grew, businesses profited

again because they now could sell their products into new rapidly growing markets that had not existed before. The losers were initially only the workers in developed countries, first only blue-collar workers but later increasingly also white-collar workers. However, as has become increasingly clear, the destruction of much of the former middle class in places like the United States has a huge cost for society overall. One major aspect of that are societal costs associated with the fact that a growing fraction of a country's population cannot support itself anymore, not to speak of families being able to provide their children a better future, while an ever-smaller fraction of the population controls an ever-greater fraction of a country's wealth. The other major aspect is that the fragmentation of societal consensus that goes along with such increasing disparity is threatening the very foundations of a liberal society.

Those dissatisfied with how society treats them in deference to other parts of society more powerful than they are, retreat into consensus realities opposing what this society represents. Because they cannot change reality, they change their perception of reality and those altered perceptions then shape their consensus realities. There can be no doubt that governments dishing out favors to their corporate sponsors, essentially declaring their political office for sale, undermine the very values on which the existence of a free society depends. However, consensus realities evolving in defiance of a government system co-opted by those more powerful in society are no less a danger to liberal societies. Those dissatisfied with society may end up undermining liberal democracies by creating consensus realities around lies and fake news in the best case and hate or racism in the worst case. Economically powerful minorities co-opting politicians and governments to serve their interests can persist for a long time. The inevitable results of such an enduring situation are larger concentrations of wealth and power and greater economic and social disparities among the broader populace. How long can such societies really keep functioning? Will societies increasingly representing the interest of powerful minorities be able to address the most important challenges of our times? Unless liberal societies find a way to become more egalitarian and serve all of their member's interests again, it is difficult to see how we can successfully address the enormous challenges humankind faces for example with climate change. We have managed to change what once were much more egalitarian liberal societies into societies of winners and losers. If we fail to address the societal challenges liberal democracies face today, we will fail to cope with the major challenges of our time. The result could be impoverished societies with many more losers and very few winners left, if any at all.

Globalization certainly increased the social divide in developed nations but without a doubt, globalization has helped many more people than it hurt. For those who suffered from the consequences of globalization or those who continue to be hurt by it, this may provide little consolation. However, for the many more in those countries where living conditions for the first time improved to levels that earlier generations could only dream

of, globalization has brought much good. Globalization has reduced the horrendous inequalities between rich and poor nations, which have characterized the world's economy for centuries, but it has not abolished them. Most citizens in the developed world continue to be materially much better off as compared to most citizens of the developing world. In a play inspired by the 1930's world economic crises the German dramatist Bertold Brecht (1898 - 1956) compared the then prevalent economic system to a teeter-totter where the ones sitting high on one end only sit there because of the many on the other end of the teeter-totter sitting low.[2] Of course, those who are sitting high up on one end want to remain there and those who are sitting low on the other end would like to join those sitting high. As all of us know from when we were kids riding on the teeter-totters of our playgrounds, for a teeter-totter this does not work. As more people make it from the low end of the teeter-totter to its high end, the lower the high end will sink and the higher the low end will rise, until they find themselves eventually at the same height once equal numbers of people sit on either end of the teeter-totter. Hence, many use a different analogy, claiming that the rising tide of growth and prosperity, which globalization and modern capitalism produce, will eventually lift all boats. To most, it has become clear that this analogy reflects a no less naive view than the teeter-totter analogy. Our world and the interplay of its various societal and economic players and systems is much more complex than what either of those analogies can capture. Many more aspects play into globalization nowadays than just the economic constellations that usually matter most from a capitalist perspective.

There really is no alternative to globalization as by necessity the future of the world will have to be a more integrated and interdependent one. We certainly will not get there in a single step but taking the long view, the future of humankind must be a common future, a shared future for all of humankind, or there will be no future for our species at all. In this sense, globalization is unavoidable and instead of fighting it, we must strive to make it a globalization where the focus is much more on the greater good for humankind and less on people seeking to enrich themselves at any cost as much as possible on an individual level. As for capitalism, we will have to see whether it can adapt to meet the challenges ahead of us. In the end, our economic systems cannot continue to put the very survival of our species at risk as they are currently doing from an ecological, societal and economic perspective. The latter may be a surprise for some, but it is nothing more than a simple reflection of the fact that our current economic systems are not sustainable. Liberal democracies must embrace a leadership role in helping humankind to master the challenges of our times. That importantly will require major changes of economic and social systems in ways that preserve liberties and freedoms while enabling us to address the challenges of climate chance and social inequality within and across societies. When globalization began, it held in the eyes of many the huge promise for a better life. Today, globalization still improves the lives of many, but we also see many of

its challenges and downsides with much more clarity. We can expect that popular views of globalization will continue to oscillate between more enthusiastic and more pessimistic phases. What we should not overlook however, is that underlying those oscillations is also a continued trend, which increasingly changes how we view the world. For our children, to perceive our world from a global perspective already comes much more naturally than it came to us. Slowly but certainly, generation over generation, we can expect humans to become more and more citizen of our planet and much less citizens of any given country. Globalization cannot and will not be just an economic globalization but in the end must inevitably lead to a species civilization where humans will view the world much more from a shared perspective than they do today.

Much of the above represents of course a narrow view from a Western perspective and, because of the author's own limitations, only a partial one at that. The world can look very different from another person's perspective. Many of us will have had this experience in some way or another. However, for most of us this experience is constrained as it usually takes place within the context of the local cultures in which we grew up. We must recognize that the various societies and cultures around the globe reflect a multitude of consensus realities and that we will find within each of those even more multitudes of consensus realities. We possible cannot know all of them and even those with which are more familiar, we often do not really understand. Among those consensus realities about which we do know something, there may be quite a few we do disagree with. Nevertheless, we have to take them for what they are and not for what we want them to be. They are what they are because people have to make sense of their realities no matter what those realities are. If reality becomes unacceptable, people will create perceived realities to shape their personal or societal consensus realities into something they can live with.

Conspiracy theories seem to thrive in our day and age like never before but they are not a modern phenomenon at all. Like with much else today, that which in the past would likely never have come to our attention, the blessings of modern technology deliver to us on a silver tablet. The deluge of information continuously washing over our public and personal environments makes it increasingly difficult for many to differentiate between facts and fiction. Finding out the facts behind what various information and misinformation channels serve us every day is a challenge for all of us but then again, figuring out the truth has never been easy in the first place. Conspiracy theories represent a particularly dangerous kind of consensus reality, for those immersed in them as well as for all others who have to contend with how they can destroy the social fabric of our communities and to a degree even endanger the very foundations of liberal societies. Conspiracy theory type consensus realities essentially deny the existence of objective reality as they hold that nothing is what it seems to be. Because of that, such consensus realities have no problem of denying the very existence of scientific facts or of objective truth. They see

scientific facts presented and explained to them not as such, like most of us do, but as something altogether quite different from scientific facts, serving some kind of deception or other sinister purposes. Pretending that nothing is what it seems to be provides conspiracy theorists the means to label anything that does not fit into a given conspiracy theory as something different from what it actually is. Another view, which those caught up in conspiracy theory type consensus realities usually endorse, is that everything is connected. With that, causality goes out the window. That enables conspiracy theorists to construe things as being causally connected which in the objective world just do not have any causal relationship and often physically cannot even have a causal relationship. Thereby, conspiracy theory type consensus realities can then assert their claims without the burden of having to provide any rationale whatsoever. In their view, just the contention that everything is seemingly connected provides a sufficient explanation, to themselves of course but in their judgment as well to everyone else who is not part of such a consensus reality. Finally, one more important feature of conspiracy theory type consensus realities is that nothing happens by chance and coincidences just do not exist. This provides conspiracy theorists the advantage of being able to use something that by itself does not carry any meaning, purpose or explanatory power for understanding anything at all and assign to it whatever explanatory power they require to justify worldviews they otherwise could not defend. In this way, disciples of conspiracy theories can muster explanations at will, thereby making their consensus reality positions seemingly impenetrable to any rational argumentation. If someone discards causality and logic as conspiracy theorists do, one can explain anything with everything, as there is no need for any of it to make sense at all from a rational perspective. Denying that something like chance or coincidence does not exist provides conspiracy theorists with an equivalent to the get-out-of-jail cards most of us know from board games, essentially with an unlimited number of those.

Conspiracy theories and individual, communal and societal consensus realities endorsing them are seemingly widespread in our societies and likely always were. However, they only reflect behaviors taken to the extreme, which are part and parcel of our human nature. In a sense, a part of our shared human nature seems to be a particular form of "horror vacui", the fear of empty spaces, evidenced by our deep need to read meaning into everything and completely fill our intellectual field of view no matter what the cost so there will be no empty space left.[3] This longing for everything to have a meaning is what led to our earliest ancestors seeking mythological explanations of what they otherwise could not understand and today it is what drives the advances of modern sciences. For reasons we do not understand today, we tend to perceive patterns in everything, even in something which if viewed from a rational perspective resembles nothing but chaos. Incidentally, as with all other aspects of human nature, such behaviors cover a wide range. Much of it falls certainly within what most would view as normal human behaviors. Most

likely, each one of us has at some point in her or his life made the experience of perceiving something that actually is not there. Sometimes we know this right away as when we for example interpret the shape of a cloud as resembling something which we know it is not. At other times, we may only find out afterwards that something we firmly believed to perceive as real actually was not. None of that is unusual and we take most such occurrences in stride and do not think much about them. But all of that is very different from how in the extreme, at the fringes of what some may consider as normal for the human psyche, some people can really struggle to make sense of realities they seemingly cannot comprehend anymore by rational means. In such cases, it will not help to ridicule people for what they believe to be their realities. This will only strengthen their belief in the correctness of their consensus realities, making it even more difficult to communicate with them. The only way to change their perception of reality is to change their experience of reality. Only if people see a reality for them that allows them to live their lives with dignity, hope and the prospect of a good future for their children, only then can they become part of a larger societal consensus reality again that we all can share in. Regardless of where life puts us and regardless of whether we happen to be born into prosperity or poverty, humans will always seek to make sense of their own lives in the reality they live in.

There is much that natural sciences, philosophy, psychology or even religion can teach us about how we as human beings perceive ourselves as well as the world around us. We often gloss over how in the end we make sense of all of it as we live our busy lives and we usually do not question much how we come by our opinions and convictions. How honest are we about what it is that frames our worldviews, about all of that which underlies our personal consensus realities? Can we actually be honest about it at all or is there some natural barrier in our human nature that prevents us from being able to confront such questions seriously? Sometimes the latter very much seems to be the case. It is just astounding to what length we often go in deceiving ourselves so we can continue to embrace our consensus realities. If we only could free ourselves from them for just a moment, we may be able to see them for what some of them often really are, deeply flawed constructs that we use as crutches to cope with realities we otherwise could not face. We all find ourselves immersed in consensus realities. It is through such constructs that we explain the world to us and our consensus realities are our individual worlds. By definition, it is very difficult for anyone to view her or his particular consensus reality through an objective lens. Much easier than to do that is to continue believing that our consensus realities are a true reflection of the objective world. While some of them come closer to reflect objective reality, none ever will do so completely. All of us can ever only have a subjective view of themselves and the world around them. Maybe we can never be completely honest to ourselves when it comes to admitting to which extent we can deceive ourselves so we do not have to see things as they actually are and act accordingly. Because

of that, it is of utmost importance that we are at least honest about our limitations in this regard. Undoubtedly, we must inoculate our liberal societies against increasingly flawed and dangerous consensus realities shaping our societal realities. The only way we will be able to accomplish this is by honestly confronting this problem through building more just and equitable societies.

Notes

Introduction

1. Back then believers were unlikely to question whether God was a *He* and many still view things this way. That reflects the inheritance of patriarchal societies, which gave rise to Judaism, Christianity and Islam. Humans tend to make their gods in their image, male or female, and sometimes in the image of life itself giving us deities with human and animal characteristics. The human mind very much struggles to behold a divine possessing neither such forms and where religions do truly approach the divine, they begin to embrace the mystical. The way people tend to view Pascal's wager depends to a good extent on how they view the divine. As the giver of an ethical and moral codex and rules, a personified god shaped much in the image of an earthly ruler but truly almighty; or as a profound religious experience that embraces the mystery of there being anything at all.
2. Here is the original German quote: *"Wer mit Ungeheuern kämpft, mag zusehn, dass er nicht dabei zum Ungeheuer wird. Und wenn du lange in einen Abgrund blickst, blickt der Abgrund auch in dich hinein."*
3. This quote is frequently attributed to the American writer Samuel Clemens (1835 - 1910), known to most of us as Mark Twain. While it would certainly resonate with much else that we know Mark Twain indeed did say, there seems to be no evidence that he actually originated the phrase.
4. The phrase "The more things change, the more they stay the same", or in its original French "Plus ça change, plus c'est la même chose" was coined by the French writer and critic Jean-Baptiste Alphonse Karr (1808 – 1890). The other three phrases have biblical origins.
5. The origin of the label "Catch-22" for such situations traces back to the World War II novel with this very title by the American writer Joseph Heller (1923 – 1999), published in 1961.
6. Clearly, there are some developments which have significantly changed how modern humans live today as compared to our Pleistocence ancestors. These changes are part of human culture evolution as for example the transition from hunter-gatherer and pastoral societies to agricultural societies beginning some ten thousand years ago. Along with that came also evolutionary changes that increased for example the human capability to digest starch. However, the question is not if modern humans are biologically somewhat different from their Pleistocene ancestors as they certainly are in a umber of respects. The question is if any of that changed human behavior such that modern humans would behave markedly different if transported back into a Pleistocene world than how our Pleistocene ancestors behaved. We have little evidence either way suggesting that not much may have changed. What really matters for how humans behave is the culture they grow up in and in this respect it makes no difference if that is the culture of our Pleistocene ancestors or one of the cultures of our modern world.

Of Body and Mind

1. The antique Achaean colony Croton founded by settlers from the Peloponnese in c. 710 BCE has become the modern-day Crotone, the capital city of the Italian province of the same name, right at the bottom of the foot of the Italian peninsula.
2. It was the dynasties of this period of Egypt's history that left us the great pyramids, beginning with the step pyramid of Pharaoh Djoser (died c. 2649 or c. 2611 BCE) of the third dynasty, and culminating with the Great Pyramid of Giza, the burial site of Pharaoh Khufu (died 2566 BCE) of the fourth dynasty. Pyramid building gradually declined through the fifth dynasty as the religious focus of the pharaohs shifted. At the end of the sixth dynasty, which bookends the period of the Old Kingdom, pharaonic power had been in decline for some time. When the Nile then failed to flood several times over a number of years, Egypt succumbed to decades of famine and strife.

3. Many may find this outrageous and simply attribute such treatment of animals to the mores of those barbaric times but they would be wrong. Vivisection of animals is still in practice today and while sedated animals may feel no pain, the results for the animals are no different. And lest we become to self-righteous, let us not forget all the other cruel animal experimentation that is going on in our times on an industrial scale; or the brutal way in which our modern societies raise countless millions of animals for slaughter each year.
4. For some reason many still tend to believe that humans have the largest brain of any animal species. That is not correct because the species with the largest brains are sperm whales whose brains weigh around eight kilograms. That is about six times heavier than the average human brain.
5. Looking at the skeletons of *Homo habilis* and *Homo erectus* side by side one cannot but think that there must have been another species, a transitional form that bridges the gap between two such different body builds. A smaller gut and a larger brain do not just appear out of nothing. There must be at least one other human species of which we have no fossil record yet that resembles an intermediary form, *Homo erectus'* ancestor so to speak, but not necessarily a direct descendant of *Homo habilis*.
6. Corpus callosum is Latin for "tough body". This thick flat bundle of nerve tissue connects the two brain halves at the bottom of the deep longitudinal fissure separating them. Cerebellum literally is Latin for "little brain".
7. The Flemish physician Andreas Vesalius pioneered human anatomy through the practice of dissection and produced the first anatomically correct brain drawings.
8. Broca's priority in discovering the primacy of the brains left hemisphere in language function was contested by the country physician Gustave Dax (1815 – 1893) from southern France on behalf of his long deceased father, the physician Marc Dax (1770 – 1837). Gustave Dax claimed that his father presented such a theory already in a paper at a congress held in Montpellier in 1836. While that paper authored by Marc Dax did exist, no proof for its public presentation could be found.
9. The frontal lobe receives its name from the frontal bone in our skull behind and underneath which this part of our brain sits.
10. For the large majority of us, around 97%, language functions are located in the left hemisphere of our brain. This includes language articulation, the so-called Broca area in the frontal lobe, as well as language comprehension, the so-called Wernicke area in the temporal lobe. Pierre Paul Broca, a French physician, identified the speech articulation area named after him in 1861 and Carl Wernicke (1848 – 1905), a German physician, discovered the speech comprehension area carrying his name in 1874.
11. The parietal lobes are named for the parietal bones, which compose the central and upper back parts of our skulls on each side. The Latin word "paries" refers to the wall of a house or a room and lives on in the modern Italian word for a wall, "parete".
12. The occipital lobes are named for the occipital bone, the part of our skull sitting behind and beneath the occipital lobes and the cerebellum. The Latin word for the back of our skulls is "occiput".
13. The temporal bones sit beneath our temples on each side of our skull. Because as we age this is the area where our head hair first starts to grey, the bone underneath it received the name time bone, seemingly an indicator of age. The Latin word for time is "tempus".
14. The two almond-shaped amygdalae derive their name from the Ancient Greek word for almond and they are one of the best-understood parts of our brains. The same is true for the two hippocampi, their Latin name reflecting the seahorse-like shapes of these brain components. The French physician Pierre Paul Broca was first to use the term limbic node in the 1860's to refer to a part of what we know today as the limbic system. In Latin, "limbus" denotes an edge or a border. The American neuroanatomist James W. Papez (1883 – 1958) and then Paul D. MacLean successively defined the limbic system as the cortical structures along the boundary where the cerebral hemispheres and the brainstem meet.
15. Most of us do not have a medical background so it is easy to confuse the difference between anatomy and physiology. Anatomy refers to the study and knowledge of the structure and identity of body parts, while physiology does the equivalent for how these parts function and relate to one another. Both words derive from Ancient Greek with anatomy rooted in the word for dissection, *anatome*, and physiology combining the words for nature or origin, *physis*, with *logia*, "the study of".
16. The adjective triune has the meaning of something consisting of three in one. Of course, many of us may be more familiar with its use in religion, as for example in the reference to a *Triune God*, which refers of course to the concept of *Trinity* in Christianity.

17. The expression paleomammalian cortex combines the Greek word for old or ancient, *paleo*, with mammal, derived from the Latin word for breast, "mamma", and the Latin word for bark, "cortex". The naming choice is probably an attribution to the folded brain surfaces resembling the appearance of bark-like structures.
18. The word neocortex combines the Greek word for new, *neo*, with "cortex" which is the Latin word for bark. Neomammalian cortex is a similar combination of *neo* with "mamma" and "cortex" as is *paleo* in paleomammalian cortex (see preceeding endnote).
19. The Latin word for brain is "cerebrum" from which the adjective cerebral derives.
20. The word phenotype is derived form the Ancient Greek words for "to show", *phainein*, and "type", *typos*. Similar for the word genotype, referring of course to our genetic blueprint, which derives it naming from a Danish-German transcription of the original Ancient Greek word *genos* meaning race, kind or offspring. The first to use the word gene was the Danish botanist Wilhelm Johannsen (1857 – 1927), who also coined the terms phenotype and genotype.
21. Strictly speaking, unlike the Bible and the Quran, the Torah does not include an afterlife as an article of faith but Judaism certainly does have a concept of afterlife. The concept of a soul predates these three Near East religions and they all developed their own conceptions of it. The thoughts of Greek philosophers influenced in this respect the offspring of Judaism, that is Christianity and Islam, significantly more than Judaism itself.
22. The pine shaped pineal gland, hence its name, sits within the triangle defined by the back end of the corpus callosum, the upper front of the cerebellum and the upper part of the brainstem. Today we know that a major function of this gland is the production of melatonin and with that of regulating sleep patterns.
23. The word ganglion traces its origin back to the Ancient Greek word *ganglion* meaning a tumor or swelling near sinews or tendons and was already used by the famous Greek physician Aelius Galenus. As mentioned earlier in the text of this chapter, the Romans happily claimed him as one of their own under his adopted Roman name Claudius Galenus and most of us today know him by the name of Galen.
24. Some six hundred million years is about the age of the oldest animal fossils, primitive sponges, that we know of today.
25. Enteric means relating to or occurring in the intestines and the adjective derives from the Ancient Greek word for intestine, *enteron*.
26. The name neuron in Ancient Greek denotes literally a sinew or tendon, as this is what the neural tissue looked like to physicians. The name for glia or glial cells derives also from Ancient Greek and means glue as initially this is what it was thought to be, the glue of the nervous system.
27. The term synapse for the chemical junctions connecting neurons derives from the Greek word for a conjunction, *synapsis*, which was introduced by the British neurophysiologist Charles Scott Sherrington (1857 – 1952) who together with his British colleague Edgar Adrian (1889 – 1977) received the 1932 Nobel Prize in Physiology or Medicine for pioneering work on the function of neurons.
28. In addition to multipolar neurons, researchers differentiate three other general shapes of neurons, the so-called unipolar, bipolar and pseudounipolar neurons. Unipolar, bipolar and pseudounipolar neurons are associated with sensory neurons, multipolar neurons with motor neurons and interneurons.
29. The two types of blood cells serve of course very different functions. Red blood cells carry oxygen from the lungs to body tissues and carbon dioxide as a waste product back to the lungs. White blood cells are a critical part of our immune system helping it to fight infections by attacking germs such as bacteria and viruses invading our bodies. Therefore, our body quickly ramps up production of white blood cells when it detects an infection. In contrast, it keeps the supply of red blood cells at a steady level because the amount of red blood cells required to oxygenate body tissue does not change much over time. Red blood cells have a lifetime of some hundred days or so in an adult human and white blood cells about one-fifth of that.
30. See for example: O. Bergmann, R. D. Bhardwaj, S. Bernard, S. Zdunek, F. Barnabé-Heider, S. Walsh, J. Zupicich, K. Alkass, B. A. Buchholz, H. Druid, S. Jovinge and J. Frisén: "Evidence for cardiomyocyte renewal in humans". In: *Science* **324**, pp. 98-102 (2009). DOI 10.1126/science.1164680.
31. That of course still begs the question how the human heart grows. Before it was discovered that humans, depending on age, can generate to a certain albeit small amount new heart muscle cells, human heart

growth was solely attributed to the enlargement of the heart cells a baby was born with. Because the number of new heart cells humans can generate is small, this explanation for the size difference between the heart of the adult and the heart of a newborn is still correct.

32. See for example: E. P. Moreno-Jiménez, M. Flor-García, J. Terreros-Roncal, A. Rábano, F. Cafini, N. Pallas-Bazarra, J. Ávila and M. Llorens-Martín: "Adult hippocampal neurogenesis is abundant in neurologically healthy subjects and drops sharply in patients with Alzheimer's disease". In: *Nature Medicine* **25**, pp. 554-560 (2019). DOI 10.1038/s41591-019-0375-9.
33. Blood flow is faster in our arteries where it ranges between 5-19 cm/s and slower in our veins where it can range 1.5 – 7 cm/s. Chemical signals in our body travel therefore by approximately a factor of 1000 slower than our nerve signals.
34. Readers interested in the original report can find it in the US National Library of Medicine of the National Institutes of Health. J. M. Harlow. "Recovery from the Passage of an Iron Bar through the Head". In: *Publications of the Massachusetts Medical Society* **2**, pp. 327-347 (1868). NIH US National Library of Medicine resource.nlm.nih.gov/66210360R.
35. Faradic stimulation uses electrodes to apply pulsed electrical currents to specific parts of the human body with a frequency and duration that can stimulate certain muscles in a given body area, like for example face muscles. Typically, the frequency range of faradic stimulation is similar to the human heart rate at rest or just a bit higher with a pulse duration in the millisecond range or just a bit shorter than that. Faradic stimulation owes its name of course to the British physicist Michael Faraday (1791 – 1867), whose pioneering work in electromagnetism enabled the development of electrical equipment that physicians could use for such faradic stimulation.
36. As some would point out, crow's feet by themselves may not be a sufficient indicator of a genuine smile as they also can appear as a permanent feature in our faces as we age. Particularly among those of us who laugh too much, but that by itself is probably laughable. Anyway, the narrowing of the eyes this twinkle causes is still a feature of a genuine smile even when crow's feet have become a permanent facial feature.
37. In the 1920's the Scottish writer Charles Kenneth Scott Moncrieff (1889 - 1930) produced the first English translation of Proust's novel, which he gave the title *Remembrances of Things Past*, inspired by a sonnet from Shakespeare (1564 – 1616).
38. GABA stands for gamma aminobutyric acid and like glutamate, it is a naturally occurring amino acid.
39. For younger readers, the term "blue screen" may not ring a bell anymore but anyone who grew up when the first personal computers became available will remember the blue screen those early computers using DOS software from Microsoft displayed when they crashed. Blue screens have become much rarer today but computers, even those running the latest operating software, still crash and malfunction and not because of failing hardware but still mostly because of faulty software.

Philosophy of Mind

1. The word ontology derives from a combination of the Ancient Greek word *on* for "being" or "that which is" with *logia*, meaning "the study of". The word epistemology similarly combines the Ancient Greek word *episteme*, meaning knowledge, with *logia*.
2. The Latin phrases "a priori", meaning "from the earlier" or "beforehand", and "a posteriori", meaning "from the later" or "in retrospect", are used by philosophers to identify two types of either knowledge, justification or argument, characterized by the use of experience or empirical evidence, "a posteriori", or a lack thereof, "a priori".
3. In Latin, the word for West is "occidens" and the word for East is "oriens". We rarely use the words Occident and Orient as geographic references today. The modern West is a much larger geographical area today than the Occident of the Mediterranean World ever was given the discovery and colonization of the Americas and places such as Australia and New Zealand by what were people rooted in occidental traditions. No such discoveries expanded the roam of the Orient but as a vestige of European colonialism, the adjective oriental acquired over time a negative connotation invoking associations with backwardness,

which could make seem the use of the word derogatory. The words Occident and Orient and their respective adjectives are practically gone from our everyday-language but scholars still use them in their original true meaning.

4. Plato laid out his view of the ideal state in his work *The Republic*. In it he continued to evolve his earlier thoughts on the relationship between body and soul as we can find it in *Phaedo* concerning the immortality of the soul. Named for one of the characters of this Socratic dialogue, *Phaedo* gives Plato's recount of events and conversations during his teachers last hours before Socrates drank the cup of hemlock with which the state of Athens executed Socrates death sentence.
5. While one finds the words reincarnation and transmigration often used interchangeably, they are not quite the same. Reincarnation is a rebirth of souls into newly created bodies; the reincarnated soul continues its existence in a completely new body that did not exist before. Transmigration can be either that or often also the migration of a soul into an already existing body. However, who is to tell if an existing body into which a soul transmigrates does not already have a soul? Hence, transmigration faces the potential problem that two souls could inhabit one body, something that cannot happen with reincarnation.
6. The religion of Orphism is named for the legendary musician and poet Orpheus. There is no evidence for Orpheus, the father of song as the Greek poet Pindar (c. 518 - 438 BCE) calls him, but most ancient sources accept him as a historical person. The first mention of Orpheus dates back to poems of the sixth century BCE. Most of us likely have heard of Orpheus through the mythical account of his fateful love for Eurydice, his beautiful wife, who he lost through the bite of a viper and tried to lead back to life from the underworld.
7. When Siddhārtha Gautama, the founder of Buddhism, exactly lived is still a matter of controversy but it seems most likely that it was sometime during the sixth or early fifth century BCE. What we know with certainty is that the third century BCE emperor Ashoka the Great (c. 304 - 232 BCE) of the Mauryan Indian dynasty, then ruling almost the entire Indian subcontinent, converted to Buddhism. Certainly, a good indication that by that time Buddhism had become one of the major religions of India.
8. There were also female heroes, heroines who embodied this psychosomatic unity no less than their male counterparts. One just has to think of Amazone's like Penthesilea who fought alongside the Trojans in Homer's Iliad, slaying many great Greek warriors only to perish at the hands of Achilles himself. Another example is Hippolyte the mythological queen of the Amazone's, a prominent figure in the respective sagas recounting the deeds of Heracles and of Theseus.
9. To this day we do not know if the poet Homer really existed as a single historical person or if under the umbrella of the name Homer a number of different sources composed by various authors have been passed on to us.
10. Probably the three most egregious earthly sinners in Greek mythology were Tantalos, Sisyphos and Tityos. Tantalos received his eternal punishment for slaying his son Pelops and serving him up as cooked meat to test the banqueting Olympian Gods. Sisyphos incurred the wrath of the Olympians by trying to cheat death and Tityos was condemmed for his attempted rape of Leto, the mother of the Olympian siblings Artemis and Apollo. With respect to rewards for exceptional heroic conduct the most famous example is of course that of Heracles, the son of Zeus but even though a mortal, who eventually ascended to Olympian status himself.
11. *Pneuma* is the Ancient Greek word for breath and for air as in "air in motion". In philosophical or religious contexts, it also signified the breath of life, the life giving essence of air, which various philosophers and philosophical schools defined somewhat differently. As a technical term, the word has become part of our modern vocabulary as for example in pneumatic pressure or when we simply refer to something as pneumatic, meaning it is pressured by air.
12. The coastal city of Tarentum, the modern-day Taranto, the capital city of the Italian province of the same name in southern Apulia, was a Greek colony in Magna Graecia founded by Spartans in the eighth century BCE.
13. The literal translation of the Ancient Greek word *atom* is uncuttable.
14. Chalcedon has become modern-day Kadıköy, a district of the city of Istanbul, formerly Byzantium and then Constantinople, the largest city of modern Turkey.
15. Stoic philosophers and their disciples often gathered under the covered walkways the Greeks called *stoa*, hence their name.

16. With Augustine of Hippo, we are already in what historians refer to as Late Antiquity, which is the period separating Classical Antiquity from the European Middle Ages. Generally, the period of Late Antiquity begins sometime in the third century and ends in the eighth century, with historians using various end as well as start dates correlating with historical events. For comparison, the Western Roman Empire ended in 476 when Odoacer (c. 431 – 493), a Roman soldier of likely Germanic descent raised at the court of Attila (c. 406 – 453) the Hun, deposed the last Roman Emperor Romulus Augustulus (c. 465 – c. 511 to 530) and proclaimed himself ruler of Italy. The Eastern Roman Empire would last almost another thousand years before it succumbed to the Ottoman Turks in 1453.
17. We know little about the life of the thirteenth-century philosopher Boetius of Sweden, also known as Boetius of Dacia, not when and where he was born and died, and not if he was a native of Denmark or Sweden. To antiquity, the people identified by their shared culture and Dacian language, whose homelands comprised parts of modern-day Romania, Moldova and areas of Ukraine, were known as Dacians. Their settlements also extended into the West and Northwest Balkans and as they posed a continued threat to the neighboring Roman provinces, the Romans under Emperor Trajan (53 – 117) eventually conquered them and turned their lands into the Roman province Dacia. That in the late twelfth century the kingdom and the church province of Denmark began to be referred to as the kingdom and church province of Dacia was simply due to a clerical error of the papal administration, which erroneously referred in church documents to Denmark as Dacia.
18. This reflects the prominent positions the universities of Northern Italy held in European intellectual life at the time. First among them were the University of Bologna, considered the oldest university in the Western World with its roots in the late eleventh century, and the University of Padua, founded by professors leaving the University of Bologna in the early thirteenth century.
19. Universals are in essence general properties, abstractions or concepts existing independently of specific objects. For example, do abstract entities like a human or humankind exist independently of a specific human being or not? Does the color red only exist as the property of an object colored red or does something like the color red exist independently as a universal. We use universals frequently and sciences like mathematics for example could not work without them but how far could one for example assign universals used in mathematics such as mathematical groups, or classes, or the concept of numbers an independent existence. In the medieval dispute on the problem of universals Duns Scotus argued that such universals really do have an existence of their own (the so-called realist position) whereas Ockham held that they are only words reflecting abstractions of the human mind (the so-called nominalist position).
20. These were actually not translations of Aristotle's original work but in this case the translation of a third century commentary on Aristotle's *De anima* (*On the Soul*) by Alexander of Aphrodisias. This Greek philosopher about whose life we know little lived some five centuries after Aristotle and is considered his most famous commentator. The first Latin translation of texts from Alexander of Aphrodisias by Girolamo Donato (1456 – 1511), a Venetian diplomat and humanist, was published in 1495.
21. Given the evolution of modern physics, philosophers nowadays must be more careful when using the term deterministic with respect to physical laws than before the emergence of quantum mechanics. The adjective deterministic really only applies in the sense that physical states are determined by physical laws but it does not imply anymore that those physical laws are deterministic in themselves. As quantum mechanics has taught us, the world of atoms follows the laws of quantum mechanics. These however, while mathematically rigorous in themselves, describe elementary particles in terms of wave functions and the latter only allow physicists to calculate probability amplitudes when it comes to predicting for example where an elementary particle may be at any given moment.
22. Not everyone may agree that nobody else but ourselves decides such things. However, using the word decision in this narrow context seems justified, as not all of those decisions are actually conscious decisions but often just what we do without any further reflection, which could involve some kind of purposeful exercise of our free will. Regardless, it is still each of us who does what she or he does and in that sense, all things we do are in essence flowing from our decisions in the way insinuated here.
23. The adjective nomological has its roots in the Ancient Greek word for law, *nomos*. This is for example why we call the science trying to explain the physical laws governing the stars, *astra* in ancient Greek, astronomy.

24. To many in Newton's time alchemy was still a respectable science and Newton himself showed a considerable interest in alchemy as well as the occult. The word alchemy is a combination of the Arabic article "al" and the Greek word *khemeioa*, which derived either from a name for Egypt meaning black earth, *Khemia*, or from the Greek *khymatos*, referring to that which is poured out. It was only in the sixteenth century that our word chemistry arose from the word alchemy. From our vantage point today, alchemy was much more akin to sorcery than to science but with the knowledge and equipment available back then it was the best possible approach to study materials and their transformations, Hence, we should not be surprised that scientists such as Newton were also often alchemists and would not have seen anything wrong with that. There is a good reason why some not only see Newton as one of the greatest scientists who ever lived but also as the last magician.
25. This is of course a quite simplistic picture as it ignores for example the major influence Islam has had in many parts of Asia for centuries. However, in the context of the discussion here, religious and philosophical positions in Islam with respect to the mind-body problem reflect a thoroughly occidental perspective.
26. The word anthropomorphism combines the Ancient Greek words for human, *anthropos*, and form or shape, *morphe*. The use of the term indicates the attribution of human traits to non-human beings or also to aspcets of nature, as is the case when people explain natural forces through anthropomorphic personifications such as for example river deities.
27. The Ancient Greek word for those masks is *prosopon*, which means face, the symbolic face of the role the actor played.
28. The Latin word persona identifies the theatrical mask as that through ("per" in Latin) which the actor sounds ("sonat" in Latin), identifying it with the character being played.
29. Mokṣha in Sanskrit means freedom from saṃsāra. Saṃsāra is the Sanskrit word encapsulating the concept of rebirth, the cycle of birth, death and rebirth of the spiritual monad, the cyclicality of all life, matter and existence.
30. Lived philosophical and religious experiences in India as exemplified in the life of a spiritual guide, in Sanskrit called "guru", are as much a male prerogative as in other cultures. Hence, the reference to a her living a philosophical belief is more of a hope than a reality even though in modern times, gurus are not anymore exclusively men but in rare cases also women.
31. Nirvāṇa is a Sanskrit word meaning "blowing out" or "quenching" as in the ultimate release from repeated death and rebirth in saṃsāra (for saṃsāra see chapter note 29).
32. The spreading of Buddhism throughout much of East and Southeast Asia unfolded over many centuries. Confining ourselves to a simple big picture view, here are some of the major milestones. Buddhism came to China sometime in the first century CE. From there, Chinese Buddhism entered the Korean peninsula in the fourth century CE after which it would find its way from Korea to Japan in the sixth century CE. Both, Indian and Chinese monks introduced Southeast Asia to Buddhism. Starting in the second century CE, Buddhism began to spread to Vietnam through close cultural ties with China as well as through missionary work by Indian monks. Other parts of Southeast Asia received Buddhism through missions from India and Sri Lanka. Thailand and Myanmar received Buddhist missionaries as early as the third century BCE while the introductions of Buddhism in Cambodia and Laos date to the fifth and seventh centuries CE, respectively. In the North and Northeast, Indian Buddhism came to Tibet in the seventh century CE from where it found its way to Mongolia in the thirteenth century CE.
33. Various Christian beliefs differ somewhat in what happens to soul and body when a person dies or what happens on judgment day when soul and body are both resurrected. Some Christian beliefs assert that the soul is mortal in the sense that it either dies with the body or only goes to sleep until judgment day, depending on whether a person acquired the knowledge of God in his or her lifetime or not. Hence, some may not agree and object that in either case Christian belief does not separate soul and body and that they remain tethered together for eternity. That is a valid objection but we must rather look here at how, throughout its history, Christian religious practice has treated body and soul in life rather than in and after death. The body was to be mortified so as not to obstruct the most fervent Christian belief possible. There was no objective to gain a deeper spiritual sense of one's self, everything was about submission. Throughout the history of Christianity, the body was perceived more as an obstacle to the salvation of the soul rather than a vehicle that could help a believer find her or his inner self.

Of Human Nature

1. Leibniz made the point that many perceptions processed in the human brain essentially bypass our conscious mind in contrast to the mental transparency position he believed Descartes held, according to which there is nothing in our mind that it does not consciously apprehend. However, things are a bit more complicated than that and we can rest assured that Leibniz was not the first person to recognize that the human mind processes some perceptions without our consciousness being involved. Regardless, he still may be the first philosopher to have put this in writing.
2. As discussed in the second chapter in the section on mental monism, while George Berkeley was an empiricist philosopher his extreme position that things only exist when we perceive them and that the outside world around us has otherwise no reality is a form of ontological idealism. Hence, Berkeley was both, an empiricist and an idealist philosopher but he endorsed certainly a different kind of idealism than the transcendental or conceptual idealism Immanuel Kant conceived more than a generation later.
3. Genie is of course the anglicized version of the Arabic word "jinn" or "djinn", which denotes a kind of supernatural being in Islamic mythology but whose origins are pre-Islamic. Traditionally a genie is often imprisoned within a bottle or oil lamp, and capable of granting wishes when summoned. A generation ago, children may still have been more familiar with the stories of *One Thousand and One Nights* also known as *The Arabian Nights*, or may still have watched the respective Hollywood movies instead of sitting in front of their computers or smart phone screens, absorbed into the world of social media.
4. There could have been a number of reasons for that but one frequently referenced is the fact that Germany at the time was a patchwork of independent territories, some of them large like the Kingdoms of Bavaria or Prussia but many of them small. Not only had each of those historic states its own court, government and bureaucracy. Their larger cities also had state operas and theaters, certainly their capitals, and of course their own university systems, many of them looking back on hundreds of years of academic tradition but quite a few of them also newer ones, which states had felt compelled to found and fund in order to compete with modern universities elsewhere. As for the physiological research supporting Germany's prominent role in psychophysics, natural science and medical departments in its many universities also benefited very much from the unique German research university system, a model that transported to the United States would flourish there in new American forms as well.
5. Since 2014, when it dropped the "s", Johns Hopkins University, America's oldest research university founded in 1876, is known as John Hopkins University.
6. If readers wonder why structuralism and functionalism are referenced here with the adjective psychological, there is a simple explanation for that. Structuralism and functionalism are concepts that are not specific to psychology and what other sciences or arts categorize as their forms of structuralism and functionalism is as specific to their areas as psychological structuralism and functionalism are to psychology.
7. The process of internal perception practiced by Wundt and his laboratory was strictly confined. The observer, usually one of Wundt's students, observing himself (there were no female students at the time) had to follow precisely laid out instructions; the latter supposed to ensure no subjective perspective tainted such internal perceptions. That of course limited the kind of study Wundt's laboratory could undertake to simple internal perceptions as with increasing complexity the subjective factor invariably would assert itself.
8. The respective German words for discovering and inventing are very similar as they have the same root, "finden" and "erfinden", thereby making this a very pronounced statement when stipulated in German.
9. All of those who still remember their high school mathematics will know that the exponential function is the inverse function of the logarithmic function. Hence, learning and decay curves have an inverse functional relationship.
10. Contextualizing memories is something we can all learn. Usually people refer to such techniques as mnemonic devices. Mnemosyne was the mother of the Greek Muses and the goddess of memory and just like the word mnemonic, her name derives from the Greek words for "to remember", "remembrance", "memory", or "mindful". One well-known mnemonic device is the building of memory palaces, the so-called method of "loci" (Latin for places) where one mentally associates items to remember with specific physical locations. As the number of items to remember grows, the person memorizing them adds room

after room to such a memory palace. When asked to recall a memory, this person will take a virtual walk through the palace until recognizing the specific place associated with the item. This method of "loci" was already popular in antiquity and many expert mnemonics use it today in combination with other mnemonic techniques such as tagging memory items with visual or auditory sensual traces.

11. Many will likely associate the name of Carl Stumpf with tone psychology, as he was one of the pioneers seeking to understand how we experience music. Like as single tones, series of tones, or consonance and dissonance of tones separated by intervals. He was however not only interested in the psychology of tones, as laid out in his two volume *Tonpsychologie* (*Psychology of Tones*) between 1883 and 1890 but also in the origins of music on which he shared his views in the 1911 published work Die *Anfänge der Musik* (*The Origins of Music*).
12. The German philosopher, psychologist and former catholic priest Franz Brentano strongly influenced a number of his former students. Among them were the philosopher Edmund Husserl and psychologists referenced in the text such as Sigmund Freud, Alexius Meinong, Carl Stumpf or Christian von Ehrenfels. Other students of Brentano included for example the philosopher, architect and esoteric Rudolf Steiner (1861 - 1925) as well as the philosopher and later first president of the 1918 newly created Czechoslovak Republic, Tomáš Masaryk (1850 – 1937).
13. Some sources refer to Max Wertheimer as an Austro-Hungarian psychologist others as a Czech psychologist. While there were Austro-Hungarian citizens in the multinational Austro-Hungarian Empire of the Habsburgs, there were per se no Austro-Hungarians but Germans, Croats, Czechs, Hungarians, Italians, Ladins, Poles, Romanians, Ruthenians, Serbs, Slovaks and Slovenes. With the dissolution of the Austro-Hungarian Empire, the Ruthenians became Ukrainians and the Germans became Austrians in what is today Austria and German minority populations in other parts of the former empire. Wertheimer was born into a Jewish family in the German speaking community of Prague, the capital of the kingdom of Bohemia, then one of the Habsburg crown lands. During the First World War, he served in the German army and then spent most of his career in Germany until he immigrated to the United States where he became a citizen. Hence, one can reasonably refer to Wertheimer as a German-American psychologist. Unless one prefers, as some do, to go by his birth citizenship but then one should correctly refer to him as a German speaking Austro-Hungarian of Jewish heritage.
14. Not surprisingly, many may assume that other lifeforms see the world in similar ways as humans do, but for the most that is not true. Using as a reference the human color vision spectrum, a number of animals do have excellent color vision some have poor color vision or practically no color vision at all while quite a few animals are able to sense colors that are invisible to humans. Dog and cat eyes for example only possess receptors sensitive to blue and yellow meaning that they see the world in shades of blue and yellow. Dogs and cats therefore cannot perceive red and green colors in the way we do. To them, red will look more like a yellowish brown and green will look more like a yellowish grey. Other animals, like some insects and crustaceans, can see colors in the ultraviolet spectrum that humans cannot see. The vision system of a few animals is unlike human vision also sensitive to light polarization, which significantly improves image contrast. Just how some animals because of the physiological particularities of their eyesight perceive the world very differently should make us wonder about how mental peculiarities may have similar effects, for example among humans.
15. Vittorio Benussi was born in the Adriatic port city of Triest, which had been part of the Habsburg domains since 1382 when it freely chose to submit itself to Habsburg rule in order to escape the rule of the Republic of Venice, which renounced its claim to the city. Called Trieste in Italian, Trst in Slovenian, and Triest in German, this old port city had always been a multi-ethnic city welcoming various newcomers throughout its history, including for example a sizable Greek community, which added significantly to the cities already prospering trade. Like many of Trieste's citizens, Benussi grew up bilingual and was as much at home conversing in German as he was in Italian. After graduating from a Triest Gymnasium, a significantly more demanding form of high school than what most of our children are used to today, Benussi moved to Graz in 1896 to study philosophy at its university where, after meeting Meinong, he quickly added psychology to his course studies. After receiving his doctoral degree in 1901, the University of Graz awarded him in 1905 the permission for lecturing, "venia legendi" in Latin, in recognition of his habilitation thesis *Zur Psychologie des Gestalterfassens: Die Müller-Lyersche Figur* (*On the Psychology of Gestalt Perception: The Müller-Lyer Figure*). He remained at the University of Graz until 1918

teaching experimental psychology and working in Meinong's psychological laboratory of which he was a member. In 1915, Benussi became the leader of the institute but because of his Italian nationality, the Austro-Hungarian authorities denied the Austro-Hungarian citizen Benussi the promotion to full professor. In 1919, Benussi became a full professor at the University of Padua where he opened his own psychological laboratory, the School of Psychology at the University of Padua, focused on vision and time perception. He remained there until he died of his own hand in 1927 by drinking a cup of cyanide. Today, Benussi is recognized as one of the most gifted and productive experimental psychologists of his time.

16. According to the Graz Gestalt psychologists a Gestalt comes about by the individual actively perceiving discrete elements as a Gestalt. Because in this sense the individual actively produces the Gestalt, with respect to Gestalt psychology, the Graz psychology school is also referred to as the Graz production school.
17. To say that Gaetano Kanizsa was an Italian is only partially correct as he was born to a Hungarian-Jewish father and a Slovene Catholic mother in Trieste, which at the time of his birth was still part of Austria-Hungary. In some ways, the European world back then was a lot more diverse than most of us will likely give it credit for today, with Trieste being an excellent example of various communities of different ethnic heritage and beliefs living peacefully together.
18. The German adjective prägnant signifies that something is brief, meaningful and forceful in its expression and the respective noun Prägnanz denotes conciseness, pithiness and sententiousness. The law of Prägnanz essentially states that human perception captures simplicity and order in complex shapes precisely because those are prägnant.
19. Max Wertheimer and Albert Einstein shared a decade long friendship beginning in 1916 that only ended with Wertheimer's death in 1943. Wertheimer was very much interested in understanding how creative thinking works, like for example in Einstein's case. Their copious letter exchange, many conversations, some of them interviews of Einstein by Wertheimer, speak to that. In one of them, Einstein remarked that he very rarely thought in words at all but that rather a thought just comes to him, which he might then try to express in words afterwards. Einstein did not say that thoughts came to him in Gestalt form but it kind of sounds like it. That also meshes with another one of Einstein's views, which was that great scientists and great artists are much alike. Of course, there are multiple possible interpretations for this statement, one of them possibly that just as the artist must envision what she or he is about to create, so must the scientist.
20. The pervasiveness of some fields has its limits. With respect to the electromagnetic field we can experience this for example when we find ourselves in a Faraday cage, named after the pioneer of electromagnetism, the physicist Michael Faraday. He discovered that a metal cage or any cage made of an electrically conducting material blocks electromagnetic radiation because the electrons in the conducting material rearrange themselves such as to cancel any external fields inside the conducting cage. Examples of that include the loss of cell phone reception when we enter a closed metal structure, or when the metal frame of a car protects us and anyone else inside it from lightning.
21. The word isomorphism derives from Ancient Greek and combines the Greek word for equal, *isos*, with the Greek word for form or shape, *morphe*.
22. Wundt's approach to philosophy and psychology was seemingly much influenced by the philosophy of Gottfried Wilhelm Leibniz, whose psychophysical parallelism we already encountered as one of the non-interactionist dualism philosophies of mind in the previous chapter. While other contemporary psychologists also endorsed psychophysical parallelism, Wundt's version went beyond a simple conceptual parallelism of the mental and the physical as a working hypothesis. Instead, he envisioned two separate categories ruled by distinct psychological and physiological causalities, essentially the mental and the physical being subject to their own respective laws.
23. This openly anti-Semitic mayor was the populist and conservative politician Karl Lueger (1844 - 1910). If he had not died in 1910 he would have most certainly staid in power quite a bit longer.
24. Bleuler who spent most of his life studying mental illnesses also introduced the terms autism and schizophrenia into psychiatric and medical practice.
25. In Freud's native German the ID, the Latin third-person pronoun, is the German "Es", the EGO, the Latin first-person pronoun, is the German "Ich", and the SUPER-EGO is then of course the German "Über-Ich".

26. Popular perception in Freud's time, little different from today, often limited the interpretation of libido to its purely sexual aspects, but that is not how Freud conceived it. In a general sense, the meaning of the Latin word libido can encompass all kinds of desires and in a narrower sense libido can indicate lust, instinctual drives such as the sex drive, or self-indulgence.
27. Others referred later to Freud's conception of a death drive as Thanatos, in opposition to Eros who symbolized the life drives encompassed by the libido. Eros, the ancient Greek god of love, is of course the personification of love while Thanatos, the son of the ancient Greek goddess of the night, Nyx, and the brother of Hypnos, the god of sleep, is the personification of death.
28. There is of course an immense literature on what the processes of resistance, projection and transference can encompass in detail but here a very simple description will have to suffice. Resistance describes a behavior of opposition, which can occur when the unconscious defenses of an individual's EGO are threatened by an external source. Projection is another defense strategy in which an individual's EGO defends itself against what it perceives as conflicting unconscious drives and desires by attributing them to others, essentially denying that their source is the individual itself. Finally, via the process of transference, a person's EGO unconsciously redirects or transfers feelings she or he had, usually about someone close such as a parent or more generally childhood experiences, from the past to a present situation. As Freud discovered, transference can also occur when a patient redirects his or her feelings for a significant person to the therapist and can thus allow the psychotherapist to explore the transference relationship and analyze its meaning.
29. The word archetype is a combination of the Ancient Greek words *archē* and *typos* in their respective meanings of origin and impression. Archetypes represent fundamental structures underlying human perceptions and behaviors. Frequently one finds these fundamental structures equated with so-called universal primal symbols and images but the latter are only representations of such archetypes and not the archetypes themselves. Archetypes are fundamental structural principles shaping human behavior and perception and can for example appear in our dreams as images derived from them. It is through such universal primal symbols and images shared throughout human cultures that we connect to these archetypes, which Jung believed to be part of the collective human unconscious.
30. See chapter note 5.
31. Pavlov was a materialist of his own making and interested in science only. That the communist rulers endorsed his research was not his fault. Any thoughts he may have had about communism he held to himself. Bekhterev, a leading Russian neurologist and psychologist, was not so lucky and is probably a good example for the potential perils of neurological and psychological practice in authoritarian regimes. He died mysteriously only one day after he had examined Joseph Stalin (1878 - 1953) and seemingly shared his diagnosis at a congress to which he returned after examining Stalin. Diagnosing Stalin as a paranoiac may well have been his death sentence, as Bekhterev's name and work were expunged from Soviet history as if the man had never lived, a fate shared by many of those Stalin had murdered.
32. The Nazi conception of a superior Germanic race is sometimes confused with Nietzsche's Übermensch. The Nazis, with much help from Nietzsche's sister who hijacked the estate and the publication rights to her brothers works after Nietzsche's death, sought to reinterpret Nietzsche to fit their perverted world-view. The Übermensch in Nietzsche's *Also sprach Zarathustra* (*Thus spoke Zarathustra*) has nothing to do with the superior German stock envisioned by Nazi race ideology. Instead, Nietzsche's Übermensch concept envisions humans as focused on the here and now and taking responsibility for what happens in this life and not in an imagined afterlife as in the Christian conception of humans. People are not just factually wrong by speaking of a Nazi Übermensch, they also do a great injustice to Nietzsche's heritage as in a sense such statements condone the Nazi effort to co-opt Nietzsche for their perverse racist ideology.
33. The promise to produce a new Soviet man soon became widely ridiculed in the Soviet Union, privately of course as any such views voiced publicly were dangerous for oneself and for one's family and friends. The Russian philosopher and writer Alexander Alexandrovich Zinoviev (1922 – 2006) published his famous satire, *Homo Sovieticus* first in the Russian underground press in the 1970's and it finally became available as a book in the West in 1981. The book should be mandatory reading for anyone seeking to engage in social engineering towards the end of creating a new human being for a presumably better society or the other way around, to create a new society for better human beings.

34. Tragically, Schlick was shot to death in 1936 by one of his former students on the so-called philosopher's staircase in the University of Vienna. While the motives for the murder given by the defense at trial sought to distract from any connections it had to the political context at the time, Austro-Fascism in Austria and Nazism in Germany, as well as the philosophical positions Schlick stood for in a Clerical-Fascist Austria that vehemently rejected those, both clearly played a role. In essence, for Austro-Fascists and Nazis the gentile Schlick amounted to what latter became known as a so-called Jew in spirit. Not surprisingly, the murderer sentenced to ten years in prison was released after two years when the Nazis took over Austria in 1938.
35. Not all members of the Berlin and Vienna Circles were so lucky to escape the Nazis. From the group members listed in the text by name, Kurt Grelling and Walter Dubislav became victims of the Nazis. The Nazis murdered Kurt Grelling in September 1942 in Auschwitz-Birkenau, the largest of the so-called concentration camps the Nazis had built to exterminate those they deemed lesser human beings, particularly people with Jewish heritage such as Grelling. Walter Dubislav suffered from a psychic instability that likely stemmed form his First World War experience as a soldier on the German Eastern Front. The Nazi rise to power in 1933 destroyed his professional career, likely contributing to his mental state deteriorating further. In 1936 he emigrated to Czechoslovakia where in 1937 he committed suicide in Prague.
36. Behavioral genetics is an interdisciplinary science that studies the effect of the genetic makeup of an organism on behavior. Given twentieth century eugenics and a history of selective data use to support gender or racial biases and prejudices, there has been and continues to be a lot of controversy around behavioral genetics. Some claim that Francis Galton founded this branch of science, which in itself links it to social and racial prejudices of that time. However, like everyone else in his time, Galton knew nothing about genetics. He was simply invoking aspects of biological breeding to justify the prominence of certain families in providing the British Empire with many capable leaders and experts. The behavioral genetics of today has its roots in the 1960's and for the most is successful in staying away from being embroiled in political, social or racial arguments, which sometimes try to manipulate behavioral genetics data for their own non-scientific purposes.
37. Heidegger, just as Sartre, stood in the tradition of Husserl's phenomenology and Heidegger's philosophy articulated in *Sein und Zeit* (*Being and Time*), published in 1927, clearly influenced Sartre's thinking at the time. However, the two not only would come to differ in what they thought mattered most in philosophy, they also endorsed opposing political views with Sartre embracing communism without becoming a party member whereas Heidegger joined the Nazi party in 1933.
38. Today we view Marxism for the most in the context of its communist perversions that cost many millions their lives and ruined the lives of countless more millions. We tend to forget that the longing for a more just society in the nineteenth century was not only on the agenda of the political left but also was the objective of much more conservative parts of society. An example for that is the Kolping Association founded by the German Catholic priest Adolph Kolping (1813 - 1865) to help ameliorate the conditions industrial workers and their families had to endure at the time. The two approaches differed of course in how they chose to address the plight of the working class, one by seeking to change the political and social system and the other one by supporting workers and families in the existing system. However, there can be no doubt that at the outset both approaches sought to restore human dignity to those from whom society had taken it. Unfortunately, the political regimes that came to claim the heritage of Karl Marx in the twentieth century deprived their unfortunate subjects no less of human dignity and human rights, arguably much more so, as they not even left them the consolation of religion.
39. The label Gestalt theory reflects the fact that this therapy approach was much influenced by Gestalt psychology. However, other than Gestalt psychology with its focus on understanding perception and cognition, Gestalt therapy is a psychotherapy approach that concerns itself with a person's relationships with her or his community and general environment.
40. Frankl was a Holocaust survivor. He was liberated in April 1945 by American soldiers after surviving three years in four Nazi death camps, who lost most of his family in the Holocaust. Finding a meaning in life, *dem Leben einen Sinn geben* in Frankl's native German, one cannot even begin to imagine the courage that this philosophical stance requires under such circumstances.
41. Before they had to flee from the Nazis in 1938, Charlotte Bühler and her husband Karl Ludwig Bühler, whom we already encountered in the discussion of the Würzburg school and Gestalt psychology, held

teaching positions at the University of Vienna since 1923 and 1922, respectively. Charlotte Bühler founded the Vienna Child Psychology School in 1935 which survives to this day in the form of the Charlotte Bühler Institute.

42. While there may be nothing wrong for psychologists to refer to clients instead of patients, it leaves a feeling of ambivalence as in a sense it gives the relationship between the patient and the psychologist an economic undertone. On the other hand, one could also look at it as leveling the relationship between patient and psychologist as clients likely perceive themselves more like equals to their psychologists than maybe a patient would, as patients are usually those obeying doctor's orders and in that sense are in a more subservient relationship.
43. Of course, in Greek mythology it is not humans who create such machines but the Olympian Gods. Often it is Hephaistos, the Greek god whose many attributes include being the god of blacksmiths and the god of fire, who creates such machines for example to assist him in his metalworking.
44. Here is how the respective passage in Kant's *Idee zu einer allgemeinen Geschichte in weltbürgerlicher Absicht* (*Idea for a Universal History with a Cosmopolitan Purpose*) reads in the original and in translation: *„Denn jeder derselben wird immer seine Freiheit mißbrauchen, wenn er keinen über sich hat, der nach den Gesetzen über ihn Gewalt ausübt. Das höchste Oberhaupt soll aber gerecht für sich selbst und doch ein Mensch sein. Diese Aufgabe ist daher die schwerste unter allen; ja ihre vollkommene Auflösung ist unmöglich; aus so krummem Holze, als woraus der Mensch gemacht ist, kann nichts ganz Gerades gezimmert werden. Nur die Annäherung zu dieser Idee ist uns von der Natur auferlegt."* (*"For each of them will always abuse his freedom if he has none above him who exercises power over him in accordance with the laws. The supreme ruler should be just in himself and yet be a man. This task is therefore the hardest of all; indeed, its complete attainment is impossible; from such crooked timber as man is fashioned from, nothing completely straight can ever be made. Nature only compels us to approximate the idea of it."*).
45. The dilemma with unconscious thought processes is of course that it raises the question of who does that unconscious thinking. However, that may be just the wrong question to ask given what we know about the complex network the human brain represents where some 86 billion neurons with on average several thousand neuronal connections are on average firing roughly every 5-6 seconds. Those are average numbers and some neurons are surely much quieter than others are but clearly our brains are very busy all the time without us realizing much of it on a process level.
46. Large animal herds counting in the thousands represent of course also social groups. However, these are different kinds of social groups with less tight connections between individuals than we find them for example between the members of small elephant herds or among the social groups of our primate relatives. Put simply, one could say that in the enormous animal herds of wildebeests the social group affords protection to individuals by letting them hide from predators in the mass of other animals while the smaller social groups of elephants and primates entail long-term caring relationships in addition to protection.
47. Here we have discarded the genus *Paranthropus* where fossils of two of its representatives, *Paranthropus boisei* and *Paranthropus robustus* indicate brain volumes above $500\,cm^3$, which is definitely above the chimpanzee brain volume of around $400\,cm^3$. However, the upper range of *Homo habilis* brain volumes is about $690\,cm^3$ and with that a good amount higher than for those two *Paranthropus* species. In addition, the upper range for *Homo habilis* brains exceeds also the brain volume of gorillas, who possess the largest brains among our ape cousins coming in around $500\,cm^3$.
48. Cooking could have been a necessary precursor for something like Dunbar's social brain hypothesis to produce the kinds of brains humans have today. Turning the argument around, the size of human social groups could have remained smaller, comparable to how our primate cousins still live today, if cooking had not provided the additional calories to feed human brain evolution and thereby enabled humans to live together in larger groups and evolve into the social animals we are today.
49. Over the last two decades, the advances in genetics have made it possible to decode the genome of ancient humans, the Neanderthal genome being likely the most well known example of that. It is an amazing feat that scientists have been able to read the Neanderthal genome from DNA samples several ten thousand years old as DNA like all biological substances degrades. The oldest human DNA information we have today is 800,000 years old but does not come from retrieved DNA itself but from reconstructing what certain DNA sequences must have looked like to produce the proteins found in an 800,000 year-old tooth.

This is a great accomplishment but far from deciphering the genome. It seems quite unlikely that we will ever be able to reconstruct the genome of an early *Homo erectus* and compare it to ours.

50. See for example: A. Vrij, Detecting Lies and Deceit, Second Edition ed., John Wiley & Sons, Ltd, 2008.
51. It is best to consider as absolute truth only those facts that have been corroborated repeatedly to the extent that is has become highly unlikely that anything contradicting such a fact will ever emerge. Among those facts, we for example can include physical properties such as the speed of light or the nature of life, of which we still know little except for the certain fact that death is inevitable.
52. Charles Darwin reaching out for help frequently took the from of a friendly summons. It is a measure of his genius that so many were so willing to help him with his countless requests. Down House, where Darwin worked tirelessly for forty years was not only the Darwin family home but also Darwin's personal one man research institute and general headquarters. From his study in Down House he conducted his data gathering and verification campaigns through a letter correspondence that spanned the whole world and included colleagues as well as anyone else of whom he thought that she or he could provide him some important information.
53. How we have come to understand the various kinds of evolution and how they intertwine was very much the topic of an earlier book of mine: S. Wurm, *The Human Condition - Our Place in the Cosmos & in Life*, ATICE LLC, 2020.

Being Honest

1. Voltaire published his *Candide, ou l'Optimisme* (*Candide, or the Optimist*) in 1759. In this satire, Candide, expelled from his childhood Garden of Eden, travels real and imaginary countries all over the world in search of his lost paradise, his lost love, and to find true happiness only to encounter all kinds of possible evils. His travels lead Candide eventually to discover that the worldview of his teacher, Professor Pangloss, of a Leibnizian preordained harmony reflecting the best possible world created by a perfect God is nothing but an illusion.
2. While we refer with class warfare usually to a modern phenomenon, social uprisings pitching one class against another have a very long history. One only has to remember the conflicts between patricians and plebeians in ancient Rome or the countless peasant uprisings all over the world against their various overlords throughout history.
3. Of course, there are always the narratives some of the major protagonists in the story of Roman conquests have left us as well as the writings of Rome's politicians and historians. Most of us may remember for example Gaius Julius Caesar's (100 - 44 BCE) *Commentarii de Bello Gallico* (*Commentaries on the Gallic War*), or Publius Cornelius Tacitus (c. 56 – c. 120) so-called *Germanica*, his *De origine et situ Germanorum* (*On the Origin and Situation of the Germans*). Such accounts, while valuable historical sources only ever give us the Roman perspective and not for example, the perspectives of the Celtic and Germanic tribes the Romans were out to conquer.
4. The name of England carries this Germanic heritage to this day, as what the name describes is the land of the Angles, Angle land, which morphed into the name England.
5. The designation of Norman betrays the Viking heritage of these invaders as it derives from Norse men, meaning the people from the north. Hence, the invading Normans eventually gave up their language two times, first when they arrived as Vikings in Northern France where they claimed in the ninth century the lands which became known as the Duchy of Normandy as their territory. Then, once the now French speaking Norman Vikings had conquered Britain, they eventually gave up their northern French dialect and became English speakers. How language and culture depend on each other or not is a most interesting question. One only has to look for example to South and Middle America, where many indigenous people continue to connect to cultures which were based on different languages and scripts than the Spanish language which many of them speak today.
6. The rise of post-war Japan to become at one point in time the second largest economy was certainly as remarkable as the economic rise of China beginning in the 1990's. The difference however is that Japan's rise to become an economic superpower took place within the shared value system of liberal

democracies, something that is certainly not the case for Mainland China, the People's Republic of China. Similarly, the rise of South Korea and of Taiwan, the Republic of China, both equally remarkable achievements, were consequences of those societies embracing the value canon of liberal democracies. In that sense, countries like Japan, South Korea and Taiwan are implicitly included when one speaks of the economic dominance of the West, as what that really means is the dominance of liberal society market economies.

7. A notorious example in this regard is the Sykes–Picot Agreement, a secret treaty signed in 1916 by Great Britain and France where those two countries, with the acquiescence of their Allies Russia and Italy but without the knowledge of the United States, agreed on how they planned to partition the colonial world between them should they emerge victorious from the First World War, as they did. This agreement, negotiated on behalf of their nations by the British soldier and diplomat Mark Sykes (1879 - 1919) and the French diplomat François Georges-Picot (1870 - 1951) dismantled the Ottoman Empire and laid the foundation of a state system in the Near East, except for the state of Israel which only became a reality after the Second Word War, that has been ravaged since then by internal civil strife and external conflicts. The modern state system of the Near East, which evolved from it, is a legacy of the Sykes-Picot agreement and of the colonial rule exercised directly or indirectly by Great Britain and France over their client states in this region of the world after the First World War. One would want to assume that a President like Woodrow Wilson (1856 - 1924), had he known about the existence of this agreement, may have insisted on it being dismantled before agreeing to send American soldiers to join the Allies in the First World War. But maybe not because as the peace negotiations after the First World War showed, Wilson was no match for the French Prime Minister Georges Clemenceau (1841 – 1929) or his British counterpart Lloyd George (1863 - 1945) when it came to uphold his famous Fourteen Points in cases where it did not suit British or French interests.
8. There were of course notable exceptions where colonial rule did not end by colonial powers conceding independence to their former colonies but where an oppressed people was able to free itself from their colonial masters. That however does not include the thirteen American colonies declaring their independence from Great Britain because this conflict rather resembled a dispute among colonial masters than the fight of an indigenous people to regain its freedom.
9. Ögedei Khan was the third son of Genghis Khan (c. 1158 – 1227) and the second ruler of the Mongol empire.
10. The battle took actually place about six kilometers from the city of Liegnitz, today's Legnica in modern-day Poland.
11. In the thirteenth century the Mongols also invaded China which they eventually conquered. In 1279 Kublai Khan (1215 - 1294) became the first ruler of the Mongol Yuan dynasty which lasted until 1368. However, the Mongols failed in their later efforts to conquer Japan.
12. The Golden Horde, meaning Great State in the Turkic Tartar language, was a Mongol khanate established in the thirteenth century which had expanded from its original home in the northwestern sector of the Mongol Empire into the Russian Steppes. When the Mongol Empire began to fragment after the death of Ögedei Khan, the Golden Horde practically became an independent khanate.
13. It was the combined might of the Holy Roman Empire and Habsburg armies and the Polish-Lithuanian relief force under Jan III Sobieski (1629 - 1696), King of Poland and Grand Duke of Lithuania, which defeated the Ottomans on September 12, 1683. The Polish cavalry force led by Sobieski himself was likely the decisive factor in this victory over the Ottomans and Sobieski became a hero to the Viennese. In his honor the Viennese allegedly created a bakery that presumably resembled a stirrup, *Steigbügel* in German, from which as legend has it the American bagel derives. Since the croissant seemingly also has its origin in the much more mundane *Kipferl* of Viennese bakers, the crescent shape of which some claim derives from the Ottoman crescent, Viennese bakers and the Ottoman wars may have given us the ancestors of two of our most favorite bakeries, bagels and croissants.
14. After all, the Mongols would not have been the first Asian steppe people to settle in Europe. The ancestors of the Magyars, most of us know them better as Hungarians, arriving in Europe in force in the early tenth century, presented a threat to much of Western Europe and their incursions into Western Europe where only checked in the Battle at Lechfeld in today's Bavaria in 955.
15. Given today's sorry state of much of the Near East and the destruction cities like Baghdad have seen just in our generation it may be hard for many to believe that Baghdad once was not only the world's largest

but also its most cultured and learned city whose libraries surpassed what was available to scholars anywhere else. The city of Baghdad was founded by the Abbasid caliph Al-Mansur (714 - 775) in 762 and its rise to prominence began and ended with the so-called Golden Age of Islam, which usually is dated from the Abbasid caliph Harun al-Rashid (c. 763 to 766 – 809) until the end of the Abbasid Caliphate brought about by the Mongol invasion culminating in the siege and destruction of Baghdad in 1258.

16. This was of course a misconception as the Austria I grew up in was a solidly democratic country and not anything like its communist neighbors on the other side of the iron curtain, self-professed socialist countries that were clearly dictatorships. This misconception of socialism stems in part from the fact that communists always tried to use socialist parties as stepping-stones on their way to grab absolute power as their own communist parities usually never could muster anything like a majority on their own. There is however, a more fundamental misconception in much of the United States, some of it probably sustained on purpose by some parts of society, between the reality of communist dictatorships and democratic governments led by socialist or social democratic parties. In a way, those who continue to feed this misconception in their personal, political or business interests are doing something very similar than what the early communists did in confusing people with respect to socialism and communism. Most European countries do have strong socialist or social democratic parties and that is one of the reason why they all have social security systems, which on average serve their citizens much better than anything most Americans could hope to find available in their states. If one can convince people that for example single payer health care systems are a socialist, speak communist invention than that will likely help the health care industry in the United States to deter citizens from voting for parties that would give them a health care system putting their citizen's welfare above the greed of the health care industry. Thereby, the health care industry and its intermediaries who have been enriching themselves at the expense of many Americans having bad, not affordable or nonexistent health care will be able to continue stealing from Americans for many years to come. There are many reasons why Americans have a significantly lower life expectancy as compared to citizens of most other developed nations, the health care system certainly being one of them.
17. This is even more surprising when one considers that back then Bruno Kreisky (1911 - 1990) led the socialist party as well as the Austrian government. As a committed socialist and a Jew, he would have been a prime target for the Nazis after the Anschluss but was fortunately able to escape to Sweden where he could remain throughout the Second World War. One reason why even under such leadership Austria may not have been able to confront its past more frankly could be that too many Austrians across the political spectrum had been associated with the Nazis in one way or another. Bringing those skeletons out of their closets would have helped Austrians to deal more honestly with the country's past but it may also have paralyzed the country politically.
18. The assassin killed also Franz Ferdinand's wife Sophie (1868 - 1914), the Duchess of Hohenberg. However, we can be confident that if the assassin had only killed his wife but not Franz Ferdinand, Emperor Franz Joseph I (1830 - 1916) would likely not have signed a declaration of war on Serbia. Franz Ferdinand's marriage to Sophie was a so-called morganatic marriage, a marriage between people of unequal social rank. Because of that, Franz Ferdinand had to accept that his wife would never bear a dynastic title and that their children would be excluded from any rights of succession to the imperial throne of Austria-Hungary or to any other dynastic rights the Habsburg's held in their various realms.
19. The phrase denoting the First World War as the war to end all wars derives from the book *The War That Will End War* published in 1914 by the English writer H.G. Wells (1866 - 1946) whom many of us know better as one of the great twentieth-century science fiction authors. In 1919, his compatriot, the economist John Maynard Keynes (1883 – 1946) published *The Economic Consequences of the Peace*, which took a much dimmer view on Europe's ability to secure a lasting peace after the carnage of the First World War.
20. The so-called Central Powers of the First World War included Germany, Austria-Hungary, the Ottoman Empire and Bulgaria. The list of Allies is a much longer one but its so-called principal powers were France, Britain, Russia, Italy and Japan, joined in 1917 by the United States. Among the other Allies in Europe were of course Serbia, on which Austria-Hungary initially first declared war, along with it Montenegro, neutral Belgium whose borders Britain had guaranteed and Germany nonetheless invaded, as well as Romania and Greece. The war efforts on both sides involved also their respective colonies

which at the time were a much richer resource for the Allies as the British Empire alone included almost a quarter of the world population and a similar fraction of the total land mass.

21. The so-called peace treaty of Brest-Litovsk, today a part of modern Belarus, signed in March 1918 between the new communist rulers of Russia and the Central Powers took Russia out of the First World War. This treaty was as crippling to Russia as the later Allied treaties of Versailles (with Germany), of Saint-Germain (with Austria), of Trianon (with Hungary), of Neuilly (with Bulgaria), and of Sevres (with the Ottoman Empire) signed in 1919 – 1920 would be for the defeated Central Powers or their respective successor states. The Bolsheviks only accepted the terms of the treaty, which created a large sphere of influence for Germany in Eastern Europe, to consolidate their still tenuous grip on power. In some ways, the plans the German High Command at the time hedged for the future of Eastern Europe seem almost like blue prints for what the Nazis would seek to establish some twenty years later.
22. This former secretary general of the United Nations was Kurt Waldheim (1918 – 2007). After serving two terms as secretary general of the United Nations from 1972 to 1981, Waldheim returned home and ran for election to the office of the President of Austria in 1986. During the presidential campaign allegations surfaced that he may have participated in war crimes while serving as an officer in the German army on the Balkans from 1942 to 1944. In 1987, an investigation by the United States Justice Department concluded that Waldheim had indeed participated in Nazi-persecutions and the United States therefore barred Waldheim, by then already President of Austria, from entering the United States.
23. The word Holocaust, which many use to refer to the genocide of some six million European Jews committed by Nazi Germany between 1941 and 1945, derives from the Ancient Greek word for an offering or sacrifice completely consumed by fire on an altar, *holócaustos*. This Greek word made it into the Latin language as "holocaustum". With respect to the persecution of European Jews, its English version, Holocaust, was already in use in 1939 to refer to the systematic persecution of Jews in Nazi Germany. After the Second World War, the term Jewish Holocaust became firmly associated with the genocide of the European Jews by Nazi Germany. With the broadcast of the Holocaust television series, the term Holocaust became part of the vocabulary in many languages and for most today, just the word Holocaust by itself has become synonymous with the Jewish Holocaust. In Hebrew, the biblical word Shoa, signifying a great calamity or destruction, became the standard term for the genocide of the European Jews. The words Shoa and Holocaust describe the same thing.
24. In many ways, Armenia's geography has shaped its history. In antiquity, the Armenian homelands straddled the contested border between the Roman Empire and the Parthian Empire. Then the border between the Byzantine Empire and the Sassanid Empire ran through their homeland with Western Armenia becoming part of the Byzantine Empire while the eastern part of the Armenian kingdom fell to the Sassanid Empire. When the Turks finally conquered Constantinople, the Persians had long converted to Islam. The Armenian homelands remained contested lands in the rivalry between the Empires of the Ottomans and the Persians to the ferocity of which the split between Shia-Islam Persians and Sunni-Islam Turks only added.
25. The Ottoman Empire has a much better record of humanly treating religious minorities than European nations under Christian rulers. Many for example had to flee the persecution of the Jews in Spain accompanying the Christian conquest of Muslim Spain and found new homes in lands under Ottoman rule. With the end of Muslim rule in all of Spain in 1492, Jews who wanted to live had to convert in order to be able to stay in Spain or else leave the country as some 300,000 Jews eventually did. The communities of these so-called Sephardi Jews prospered for generation in various parts of the Ottoman Empire, able to practice their religion and to self-govern their own confessional communities under what quite a bit later became known as the so-called millet system. The word millet has Arabic roots carrying the meaning of nation and in the Ottoman Empire referred to an independent court of law through which various confessions self-governed their communities. Millets granted religious minorities in the Ottoman Empire such as various Christian confessions and Jews a significant degree of autonomy. That very much contrasts with the treatment of Jewish communities in European Christian lands as well as the treatment of various smaller Christian confessions who happened to be designated as heretics by the Catholic Church or Protestant Churches. Compared to the fate that befell many Christian heretics and Jews in Christian Europe, Christian and Jewish minorities fared quite well under Ottoman rule.
26. The states and regions of Germany and Austria are home to many different local variants of Germanic cultural traditions often seemingly being different cultures altogether. That is a mistaken view, because

as different as local customs, traditions and even local dialects may be, they still see themselves as belonging to the universe of Germanic cultures. A few non-Germanic cultures continue to thrive within the borders of modern Germany and Austria but they constitute small minorities. What made today's German and Austrian societies multicultural was the non-German immigrant population arriving in those countries in the decades after the Second World War.

27. The United Nations Convention on the Prevention and Punishment of the Crime of Genocide owes its existence primarily to the tireless efforts to the Polish-Jewish lawyer Rafal Lemkin (1900 – 1959). Already in 1933, motivated by what he had learned about the genocide of the Armenians as a young Polish lawyer, Lemkin had proposed to the forerunner of the United Nations, the League of Nations, to declare the destruction of national, religious and racial groups an international crime. However, that did not happen. Then, during the Second World War, Lemkin lost most of his family in the genocide of the European Jews. Lemkin was able to first escape to Sweden and then emigrate to the United States of America where he continued to push the newly created United Nations to adopt a convention against genocide. It was Lemkin who coined the word genocide, derived form a combination of the Ancient Greek word for race or people, *genos*, and the Latin suffix "caedo" which indicates an act of killing.
28. Most may not be familiar with the enormous number of civilian lives lost in modern military conflicts. War always has cost civilian populations dearly and believing otherwise would be naive. It just so happens that the total death toll was so much higher in modern warfare than in earlier military conflicts. There were roughly forty million casualties during the First World War, half of them wounded and half of them dead. Of the roughly twenty million death, ten million were civilian casualties. During the Second World War an estimated seventy to eighty-five million people died, more than sixty percent of them civilians, somewhere between forty-three and fifty-three million people, including some seventeen million people killed by the Nazi regime. These numbers include the staggering death toll of the conflict in Asia, which began already in 1937 and cost for example the lives of an estimated 15 to 20 million Chinese. This is however not the largest casualty number suffered by any of the combatants as some 20 to 27 million people within the borders controlled by the former Soviet Union lost their lives. Anyone thinking that warfare today is any less deadly for civilians is mistaken. Modern armies may advertise their surgical strike capabilities but that seemingly does not lead to better outcomes for civilians as one for example can read from the number of Iraqi civilian war casualties. Although there are various estimates whose numbers differ significantly it is likely safe to say that several hundred thousand Iraqi civilians died in this war, which began with a lie in 2003 and officially ended in 2011 even though the conflicts caused by it continue to cost lives. One can only wonder how those United States politicians who knowingly precipitated a war based on a lie, which cost hundreds of thousands of innocent people their lives, could ever sleep again. However, it is a good bet that in their consensus realities they likely saw themselves as good Christians, not much in fear of any Christian purgatory or hell because of what they were doing.
29. The Chinese communists have been pursuing a policy of making ethnically non-Han groups into minorities in their homelands. With millions of ethnic Han Chinese moving into the autonomous province of Xinjiang, China has succeeded in making the Uyghurs, a people of Turkic ethnic origin and formerly the majority, into a minority population in their homeland. The Chinese have thought to control this border province, the name Xinjiang translates into new frontier, since the Han Dynasty and had maintained military outposts and settlements there for a long time to protect China proper from the incursions of Turkish tribes. Resettlement of Han Chinese into this border region to stabilize it in the interest of China was not an invention of the Chinese communist rulers. However, the scale of this Han resettlement into Xinjiang under communist rule is unprecedented, just as is the resettlement of Han Chinese into Tibet. The Chinese communists are seemingly pursuing in Xinjiang as well as in Tibet what one could call a policy of silent genocide by converting what once where the ethnic majorities in those regions into ethnic minorities. In essence, this amounts in both cases to cultural genocide, eliminating cultures and belief systems, Buddhism in Tibet and Islam in Xinjiang, which the Chinese communists seemingly perceive as religions competing with their own religion – communism. The Chinese government uses the increasing Han population in both territories to maintain and perfect a police state. No Chinese armies are necessary that would have to put down rebellions of the oppressed in those territories. The communists use the Chinese Han population in those territories to control and coerce non-Han ethnic minorities in their homelands where they once comprised the majority. Unfortunately, most foreign companies doing

business in Mainland China largely ignore these silent genocides as they put profits above human rights. One can only hope that the Western customers of those companies will eventually hold those businesses accountable.

30. Most will know at least something about the Holocaust, also called the Shoa, the systematic murder of some six million Jews by Nazi Germany. However, the Nazis killed also millions of other peoples they deemed to be lesser humans and among those whom they sent to extermination in their concentration camps were tens of thousands of Sinti and Roma. These nomadic people came to Europe from northern India roughly a thousand years ago but because people believed they came from Egypt, they called them Gypsies. Millions of others the Nazis referred to as Untermenschen – inferior people in Nazi speak – were killed outside of concentration camps, killed in mass executions and buried in mass graves all over Europe in addition to prisoners of war, the large majority of them Soviet soldiers, being starved to death in prisoner of war camps. The enormity of crimes against humanity committed by Nazi Germany is still difficult to grasp but even more difficult to understand is that some seventy-five years after the Nazis were finally defeated there are still many who deny that anything like that ever happened. Maybe for older teenagers in our schools a few hours of viewing what the Nazis had photographed and filmed themselves and what Allies later documented when they freed concentration camp prisoners should be obligatory.
31. The last country in the world to abolish slavery was Mauretania, which did so officially in 1981.
32. Europeans did not introduce the slave trade to Africa but they certainly scaled it up to volumes not known before. Slavery had been part of many African societies just as it had been part of most societies of antiquity, the Greeks certainly being no exception to that. With the rise of Islam, the so-called Muslim slave trade gained in importance, particularly in North and East Africa. Hypocritically, many European slave traders and those living from it considered themselves morally superior to Muslim slave traders. Maybe clamoring the moral high ground among slavers gave Christian slave traders an edge in the competition with their Muslim counterparts as they sought to take over business from Muslim slave traders.
33. There remains maybe another one billion years for higher life on Earth before the Sun's luminosity will have increased to a point where temperatures on Earth will make it uninhabitable for lifeforms such as us. For those interested about the future habitability of Earth the section *The Future of Life* in my earlier book *The Human Condition* provides a more detailed discussion (S. Wurm, *The Human Condition - Our Place in the Cosmos & in Life*, ISBN 978-1-951894-00-9, ATICE LLC, 2020).
34. As the detention and imprisonment of an estimated one million Uyghurs in so-called reeducation camps shows, the Chinese communists have not really given up on using the old tried and tested ways of eradicating dissenters wholesale through brain washing, forced labor and what cannot be called anything else but cultural genocide.
35. As briefly mentioned in the third chapter, Alfred Binet and Théodore Simon developed the first standardized tests to quantify and compare intelligence, in their case of schoolchildren of different ages. What we call today the IQ or intelligent quotient was first developed by the German psychologist William Stern (1871 – 1938) to qualitatively assess individual differences in intelligence in order to help identify learning disabilities in children. Stern realized that the IQ was only a relative measure at best and that intelligence is a much too complex mental capability to asses and compare individuals with respect to it. Today's versions of IQ tests all derive from Stern's suggestion to define an intelligence quotient as a score derived from a set of standardized tests developed to measure individual cognitive abilities in relation to an individual's age group.
36. There are of course also programmable computer chips whose logic function can be customized. However, that requires them to have an initially greater number of transistors than if the specific function would have been designed into the chip hardware right from the beginning and hence the density of eventually functional transistors in those programmable chips is lower than for example in the central processing unit of a computer.
37. The two engineers credited with the invention of the integrated circuit were the Americans Jack Kilby (1923 – 2005) and Robert Noyce (1927 – 1990). Jack Kilby at Texas Instruments invented and demonstrated a first working integrated circuit in 1958 and some six months later, Robert Noyce, the future co-founder of Intel Corporation then working at Fairchild Semiconductor, invented what engineers consider the first practically manufactureable integrated circuit design.

38. *Brave New World* is of course the title of one of the most famous twentieth-century science fiction novels, published in 1932 by the English writer Aldous Huxley (1894 - 1963). Huxley's vision of a futuristic state has come much closer to reality than he likely could have envisioned as today's technologies allow authoritarian regimes to be significantly more sophisticated in ensuring compliance by all citizens in a seemingly much less threatening way. However, that is an illusion and by removing the technological veneer, which disguises what happens behind the official facades modern authoritarian regimes like to advertise, it becomes quickly clear how prescient Aldous Huxley's dystopian vision of a future world was.
39. That mass limit for a white dwarf star is 1.4 times the mass of the Sun and physicists refer to it as the Chandrasekhar limit, in honor of its discoverer, the Indian physicist Subrahmanyan Chandrasekhar (1910 – 1995).
40. The term metaverse was coined by Neal Town Stephenson (born 1959) in his 1992 science fiction novel *Snow Crash*. Incidentally, in this novel what he labeled the metaverse provided an alternative reality as an escape from a dystopian reality, which is often the main feature of such concepts as the metaverse. That functioning societies seemingly do not require such escapes and only broken ones do should make us think a little more carefully about why today the metaverse has become the next great thing.
41. Of course, the one that is turning is not really the universe but Earth itself, which makes it look like the night sky above us is turning, which we all can observe on a clear night, provided that we are far away from any of our bright cities. The Roman authors who were the first to refer to our universe with the Latin word "universum" of course did not know this.
42. Doppelgänger is a German word, which also in English has come to denote a person that physically resembles another person. Your potential Doppelgänger in a multiverse is biologically unrelated to you so he or she is not your twin. In a multiverse, we cannot really say who is the original you and who is the double (the meaning of the German word "Doppel"). Hence, it may be altogether better to talk about Multigänger, that is multiple variants of a given person walking (the meaning of the German word "Gänger") around in their universes with none of them having a claim to be an original whereas the notion of a Doppelgänger kind of insinuates that one is more original than the other.
43. Avatāra is a Sanskrit word with the meaning of decent, combining the Sanskrit words for down, "ava", and to cross, "tar". In Hinduism, avatāra denotes the decent of a deity onto Earth in the form of a material appearance or through an incarnation.
44. We have to thank the late Swedish physician and health expert Hans Rosling (1948 – 2017) for opening our eyes to what is happening in our world, the stuff that really matters but that somehow never makes it into the evening news. His book *Factfullness* published posthumous in 2018 discusses a number of important topics of general interest, including an eminently readable and understandable narrative explaining our world's population dynamics.
45. Isaac Asimov introduced the *Three Laws of Robotics* in his 1940's short story *Runaround* and many may know them from his later and more widely read science fiction novels *I, Robot*, *The Complete Robot* and *Robot Visions*, published in 1950, 1982 and 1990, respectively. Here are Asimov's *Three Laws of Robotics*:

    1. *A robot may not injure a human being or, through inaction, allow a human being to come to harm.*

    2. *A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.*

    3. *A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.*

46. Fortuna, the Roman goddess of fortune and personification of luck, was often depicted as blindfolded, as it was considered a matter of chance on whom she would come to bestow the cornucopia of her blessings.

An Outlook

1. A dualist philosophical worldview of good and evil opposing each other as separate entities, often in beliefs personified as such, does not really have to explain that good and evil both exist. This is different for a monism worldview which faces the challenge of having to explain how a seeming duality of good and evil can exist in unity. A frequent but fallacious response to this challenge is to claim that what we perceive as evil appears only as such because with our human limitations we cannot perceive what we see as evil in its universal context where it actually may be good. This argumentation, however, opens the door to all kinds of relativism and must be rejected out of hand.
2. This play, *Die heilige Johanna der Schlachthöfe* (*Saint Joan of the Stockyards*), which Elisabeth Hauptmann (1897 - 1973) and Emil Burri (1902 - 1966) co-authored with Brecht, premiered in 1932. The play portrays the conflict between striking stockyard workers and capitalist owners of slaughterhouses in the stockyards of Chicago and takes a dim view on the possibility for social compromise and the role of religion in such societal conflicts.
3. Commonly, the Latin phrase "horror vacui" designates the fear of spaces, which manifests itself in the human urge to fill the entire surface of a canvas and leave nothing uncovered. In physics, the phrase points to the contention that nature abhors a vacuum, a position attributed to Aristotle. He physically and philosophically argued the view that nature contained no vacuum. His physics argument was that a vacuum would immediately be filled by material surrounding any such vacuum and from a philosophical perspective he maintained that something that is nothing, a vacuum, cannot exist. The debate whether a true vacuum can really exist is by the way not yet settled and it depends somehow on how one defines a vacuum. Today's physicists know that the vacuum of interstellar space which many see as the most perfect vacuum is not empty at all as all kinds of fields pervade it and on the very small scale it teems with quantum fluctuations.

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List of Abbreviations

Acronyms

AI	Artificial Intelligence
BCE	Before Current Era
CE	Current Era
CPU	Central Processing Unit
DAT	Dual Attribute Theories
DNA	Deoxyribonucleic Acid
DOS	Disk Operating System
FOXP2	Forkhead box protein P2
GABA	Gamma Aminobutyric Acid
GATT	General Agreement on Tariffs and Taxes
GWA	Genome-Wide Association
ICC	International Criminal Court
IPCC	Intergovernmental Penal on Climate Change
IQ	Intelligence Quotient
TOTE	Test-Operate-Test-Exit
WTO	World Trade Organization

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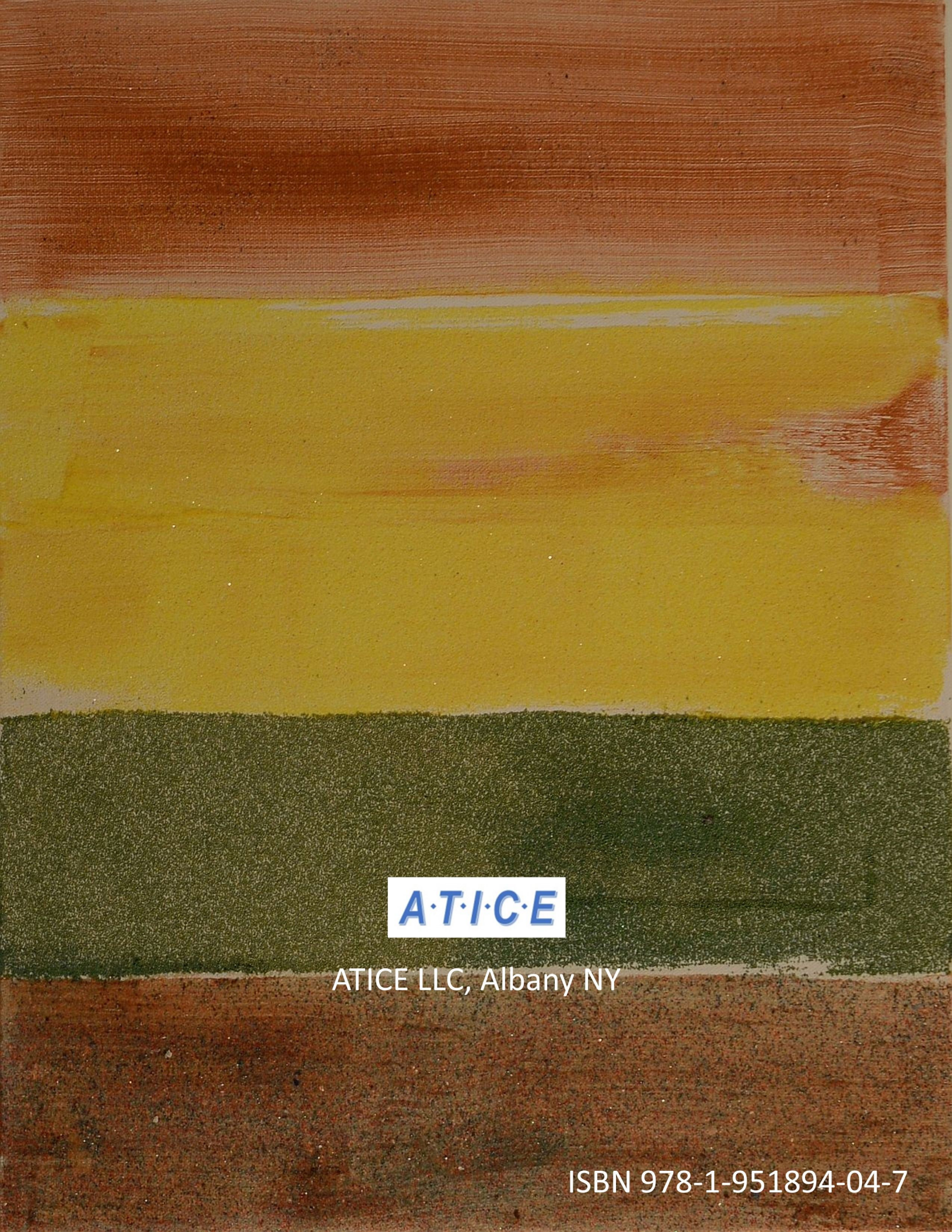
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